

Introduction to Complex Analysis

Part I: The Complex Plane, Mappings, Analyticity, and the Elementary Functions

Edition with animated figures

Lecture Notes

A. Lorent

How to read these notes

These notes are written to be *self-contained*. Every object that appears is defined before it is used, every symbol is given a type and a range, and every assertion is either proved, or explicitly flagged as a forward reference to a later part of the course, or quoted from real analysis (and then we say so). Nothing is left to “it is clear that”; where a computation is routine it is still carried out, because in complex analysis the routine computations are exactly where the interesting phenomena (multivaluedness, branch cuts, failure of familiar real identities) hide.

The material is organised in eight parts and an appendix.

- **Part I (Chapters 1–6)** builds the field $\mathbb{C}$, its geometry, its polar description, roots and powers, the point-set vocabulary we need, and two immediate payoffs: quadratics and constant-coefficient ODEs.
- **Part II (Chapters 7–11)** treats complex functions as *mappings* of the plane: linear maps, z^n , $z^{1/n}$, and $1/z$.
- **Part III (Chapters 12–15)** is the analytic core: limits, continuity, differentiability, analyticity, the Cauchy–Riemann equations, harmonic functions.
- **Part IV (Chapters 16–18)** constructs the elementary transcendental functions e^z , $\log z$, $\sin z$, $\cos z$, $\sinh z$, $\cosh z$, z^α and describes their mapping behaviour.
- **Part V (Chapters 19–22)** is integration: contour integrals, the Cauchy–Goursat theorem, Cauchy’s integral formulas and their consequences.
- **Part VI (Chapters 23–26)** treats series: sequences and series, Taylor and Laurent series, zeros and poles.
- **Part VII (Chapters 27–32)** develops residues and their applications: the residue theorem, real integrals, the argument principle, and summing series.
- **Part VIII (Chapters 33–34)** is conformal mapping and the linear fractional transformations.
- **Appendix A** collects the background from multivariable calculus that the notes use.

Exercises appear at the end of each chapter. Exercises marked $(\star)$ are used later in the notes.

Contents

How to read these notes	i
How to watch the animated figures	viii
I The Complex Plane	1
1 Complex Numbers and Their Properties	2
1.1 Why the real numbers are not enough	2
1.2 The field $\mathbb{C}$	2
1.3 Conjugation	5
1.4 Modulus	7
1.5 Complex numbers as conformal matrices	10
1.6 Exercises	14
2 The Complex Plane	16
2.1 Complex numbers as points and as vectors	16
2.2 Distance, circles, discs, and lines	16
2.3 Inequalities read geometrically	18
2.4 Exercises	19
3 Polar Form of Complex Numbers	21
3.1 The argument	21
3.2 Polar form and Euler's formula	22
3.3 The geometry of multiplication	23
3.4 Exercises	24
4 Powers and Roots	26
4.1 Integer powers and De Moivre's formula	26
4.2 n -th roots	26
4.3 Roots of unity	28
4.4 Exercises	29
5 Sets of Points in the Complex Plane	31
5.1 Neighbourhoods	31
5.2 Connectedness and regions	32
5.3 Reading a set from an inequality	34
5.4 Exercises	35

6 Applications: Quadratics and Differential Equations	36
6.1 Quadratic equations over $\mathbb{C}$	36
6.2 Linear ODEs with constant coefficients	37
6.3 Exercises	40
 II Complex Functions as Mappings	 42
7 Complex Functions as Mappings	43
7.1 What a complex function is, and why we cannot graph it	43
7.2 Mapping curves, mapping regions	44
7.3 Injectivity, surjectivity, and inverse maps	47
7.4 Exercises	47
 8 Linear Mappings	 49
8.1 The three elementary motions	49
8.2 The general linear map	49
8.3 Exercises	53
 9 The Power Function z^n	 55
9.1 Action on the polar grid	55
9.2 Action on the Cartesian grid; the case $n = 2$	58
9.3 Exercises	61
 10 The Power Function $z^{1/n}$	 63
10.1 Branches	63
10.2 Exercises	68
 11 The Reciprocal Function	 69
11.1 Inversion and reflection	69
11.2 Circles and lines are preserved	71
11.3 The extended plane and the Riemann sphere	73
11.4 Exercises	76
 III Limits, Continuity, and Analyticity	 78
 12 Limits and Continuity	 79
12.1 Limits	79
12.2 Showing a limit does not exist	81
12.3 Continuity	82
12.4 Exercises	83
 13 Differentiability and Analyticity	 84
13.1 The complex derivative	84
13.2 Analyticity	86
13.3 Conformality	86
13.4 Exercises	88

14 The Cauchy–Riemann Equations	90
14.1 Necessity	90
14.2 Sufficiency	91
14.3 The Cauchy–Riemann equations as a differential inclusion	93
14.4 Consequences: rigidity	96
14.5 Exercises	96
15 Harmonic Functions	98
15.1 Laplace’s equation	98
15.2 Harmonic conjugates	99
15.3 Level curves and physical interpretation	101
15.4 Bonus: harmonic functions as steady states of heat flow	103
15.5 Exercises	106
IV The Elementary Functions	108
16 Exponential and Logarithmic Functions	109
16.1 The complex exponential	109
16.2 The complex logarithm	112
16.3 Complex powers	115
16.4 Exercises	116
17 Trigonometric and Hyperbolic Functions	117
17.1 Definitions	117
17.2 Real and imaginary parts; size	119
17.3 Mapping properties of $\sin$	120
17.4 Exercises	125
18 Complex Powers, Continued	126
18.1 All branches of z^α	126
18.2 Power functions as mappings	127
18.3 The binomial series	130
18.4 Exercises	131
V Integration	132
19 Real Integrals and Contour Integrals	133
19.1 Integrals of a complex-valued function of a real variable	133
19.2 Contours	134
19.3 The contour integral	135
19.4 Two computations that decide the theory	136
19.5 Exercises	137

20 The Cauchy–Goursat Theorem and Independence of Path	138
20.1 Antiderivatives and the fundamental theorem	138
20.2 Goursat’s lemma	140
20.3 The Cauchy–Goursat theorem	143
20.4 Consequences for harmonic conjugates	149
20.5 Exercises	150
21 Cauchy’s Integral Formulas	151
21.1 The first formula	151
21.2 The second formula: derivatives	154
21.3 Morera’s theorem	155
21.4 Exercises	156
22 Consequences of the Integral Formulas	158
22.1 Cauchy’s inequalities and Liouville’s theorem	158
22.2 The Fundamental Theorem of Algebra	159
22.3 Exercises	159
VI Series	160
23 Sequences and Series	161
23.1 Sequences	161
23.2 Series	161
23.3 Power series	162
23.4 Exercises	164
24 Taylor Series	166
24.1 Taylor’s theorem	166
24.2 The standard expansions	168
24.3 The identity theorem	170
24.4 The maximum modulus principle	172
24.5 Underneath: the maximum principle for harmonic functions	173
24.6 Exercises	176
25 Laurent Series	178
25.1 Laurent’s theorem	178
25.2 Computing Laurent series	180
25.3 Exercises	182
26 Zeros and Poles	183
26.1 Isolated singularities and their classification	183
26.2 Removable singularities	184
26.3 Poles	184
26.4 Essential singularities	185
26.5 Exercises	188

VII	Residues	189
27	Residues and the Residue Theorem	190
27.1	The residue	190
27.2	The residue theorem	191
27.3	The winding number and the general form	193
27.4	Exercises	195
28	Evaluation of Real Trigonometric Integrals	197
28.1	The substitution	197
28.2	Worked examples	198
28.3	A general formula	200
28.4	Exercises	200
29	Evaluation of Real Improper Integrals	202
29.1	Principal values and the semicircular contour	202
29.2	Rational integrands	203
29.3	Fourier-type integrands and Jordan's lemma	207
29.4	Indented contours and simple poles on the real axis	209
29.5	Exercises	212
30	Integration along a Branch Cut	213
30.1	The keyhole contour	213
30.2	The model integral	214
30.3	Logarithms in the integrand	216
30.4	Exercises	216
31	The Argument Principle and Rouché's Theorem	218
31.1	The argument principle	218
31.2	Rouché's theorem	221
31.3	The open mapping theorem	224
31.4	Exercises	224
32	Summing Infinite Series	225
32.1	The summation kernels	225
32.2	The summation theorems	226
32.3	The Basel problem and its relatives	227
32.4	Exercises	229
VIII	Conformal Mapping	231
33	Conformal Mapping	232
33.1	Conformal maps	232
33.2	A catalogue of standard maps	233
33.3	Transporting harmonic functions	236
33.4	The Riemann mapping theorem	238
33.5	Exercises	239

34 Linear Fractional Transformations	240
34.1 Definition and basic structure	240
34.2 Three points determine the map	242
34.3 The disc and the half-plane	244
34.4 Symmetry and fixed points	248
34.5 Exercises	251
 IX Appendix	 252
A Background from Multivariable Calculus	253
A.1 Open sets, limits, continuity	253
A.2 Partial derivatives, the gradient, the Jacobian	253
A.3 Curves	258
A.4 Line integrals of scalar fields	258
A.5 Line integrals of vector fields	259
A.6 Dictionary: contour integrals in these terms	262
A.7 Exercises	263

How to watch the animated figures

Wherever these notes introduce a function and explain what it does to the plane, an *animated figure* follows (its caption says “animated”). It shows the function f as a *deformation of the plane*: a piece of the domain, drawn as a grid or as coloured tiles, is carried continuously onto its image under f .

Playing them. Open this PDF in the PyFig Reader and click a picture: it plays in place on the page, runs forward to f , pauses, runs back, and loops; click it again to pause. In any other PDF viewer the pictures show the finished map, and clicking one opens its animation in your web browser. On paper you see the finished map.

What you are watching. Each panel is drawn from a *homotopy* from the identity to f : a continuous function $H(z, s)$ of the point z and a time $0 \leq s \leq 1$, with

$$H(z, 0) = z \quad \text{and} \quad H(z, 1) = f(z).$$

Frame s of the animation is the picture of $z \mapsto H(z, s)$. You will see two different homotopies for every function:

- A the *straight-line* homotopy $H(z, s) = (1 - s)z + sf(z)$: every point travels along a straight segment from z to $f(z)$;
- B a *path of maps* chosen for the function at hand, in which every stage $z \mapsto H(z, s)$ is itself a map of the same kind: a rotation followed by a magnification and a translation for a linear map; the powers z^t for z^n and the roots, the exponent t running from 1 to n (from 1 to $1/n$ for the roots); a rotation of the Riemann sphere for $1/z$; for another linear fractional map M , a one-parameter path M^s of linear fractional maps (not in general rotations of the sphere); $\frac{e^{sz}-1}{s} + s$ for e^z and $\frac{\sin(sz)}{s}$ for $\sin z$ (both extended by continuity to $s = 0$, where they equal z);
- C the same path as B, applied to coloured tiles, with the undeformed tiles alongside for comparison.

Neither homotopy is canonical. There are infinitely many homotopies from the identity to f , and every one of them ends in the same picture, the image of the domain under f . The two shown here are choices, and you should feel free to prefer one of them, or to imagine a third that makes more sense to you. The path is not the point; the destination is. What matters is to understand what f *does*: where lines, circles and regions go, which angles are kept and which are doubled, where the map folds the plane over itself or tears it along a cut.

As your understanding grows, the animations themselves will start to make sense: why the straight-line homotopy folds the square over itself for z^2 while z^{1+s} opens the plane like a fan,

why a rotation of the sphere keeps every circle a circle, why no path of one-to-one maps of the plane onto itself (in particular no rigid motion) can turn it into its mirror image; the straight-line homotopy for $\bar{z}$ manages only by crushing the plane onto a line halfway. When that happens, the homotopies have done their job.

Part I

The Complex Plane

Chapter 1

Complex Numbers and Their Properties

1.1 Why the real numbers are not enough

The real field $\mathbb{R}$ is complete and ordered, and yet the polynomial equation

$$x^2 + 1 = 0 \tag{1.1}$$

has no solution in $\mathbb{R}$: for every $x \in \mathbb{R}$ we have $x^2 \geq 0$, hence $x^2 + 1 \geq 1 > 0$. Any *ordered* field has this defect, since in an ordered field squares are non-negative. So enlarging $\mathbb{R}$ to a field in which (1.1) is solvable necessarily costs us the order relation. That trade — give up “ $<$ ”, gain the roots of every polynomial — is the bargain struck by the complex numbers, and it is an extraordinarily good one.

We construct $\mathbb{C}$ carefully rather than by declaring “let i be a symbol with $i^2 = -1$ ”. The declaration is harmless once one knows that a consistent object exists; the construction is what shows that it does.

1.2 The field $\mathbb{C}$

Definition 1.1 (The complex numbers). Let $\mathbb{C}$ denote the set $\mathbb{R} \times \mathbb{R}$ of ordered pairs of real numbers,

$$\mathbb{C} := \{ (a, b) : a \in \mathbb{R}, b \in \mathbb{R} \}, \tag{1.2}$$

equipped with the two binary operations $+$: $\mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ and $\cdot$: $\mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ defined for all $a, b, c, d \in \mathbb{R}$ by

$$(a, b) + (c, d) := (a + c, b + d), \tag{1.3}$$

$$(a, b) \cdot (c, d) := (ac - bd, ad + bc). \tag{1.4}$$

Two complex numbers are *equal* exactly when they are equal as ordered pairs:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d. \tag{1.5}$$

Definition (1.3) is nothing but vector addition in $\mathbb{R}^2$. Definition (1.4) is the whole content of the subject: it is up to isomorphism, the only multiplication that makes $\mathbb{R}^2$ (with its vector addition,

and with multiplication by reals being the scalar multiplication of $\mathbb{R}^2$) into a field extending $\mathbb{R}$. It is not unique as a formula: for instance $(a, b)(c, d) := (ac - 4bd, ad + bc)$ also gives a field, but it is just (1.4) written in the basis $(1, 0), (0, 2)$; all other choices amount to a different choice of the second basis vector.

Theorem 1.2 ($\mathbb{C}$ is a field). *The set $\mathbb{C}$ of (1.2) with the operations (1.3) and (1.4) is a field. Explicitly, for all $z, w, \zeta \in \mathbb{C}$:*

$$(F1) \quad z + w = w + z \text{ and } zw = wz \quad (\text{commutativity});$$

$$(F2) \quad (z + w) + \zeta = z + (w + \zeta) \text{ and } (zw)\zeta = z(w\zeta) \quad (\text{associativity});$$

$$(F3) \quad z(w + \zeta) = zw + z\zeta \quad (\text{distributivity});$$

$$(F4) \quad (0, 0) \text{ is an additive identity and } (1, 0) \text{ a multiplicative identity};$$

$$(F5) \quad \text{every } z \in \mathbb{C} \text{ has an additive inverse } -z;$$

$$(F6) \quad \text{every } z \in \mathbb{C} \text{ with } z \neq (0, 0) \text{ has a multiplicative inverse } z^{-1}.$$

Proof. Write $z = (a, b)$, $w = (c, d)$, $\zeta = (e, f)$ with $a, b, c, d, e, f \in \mathbb{R}$. Throughout we use only the field axioms of $\mathbb{R}$.

(F1). Additive commutativity is immediate from (1.3) and commutativity of $+$ in $\mathbb{R}$. For multiplication,

$$zw = (ac - bd, ad + bc), \quad wz = (ca - db, cb + da),$$

and $ac - bd = ca - db$, $ad + bc = cb + da$ in $\mathbb{R}$; by (1.5), $zw = wz$.

(F2). Additive associativity is componentwise associativity in $\mathbb{R}$. For multiplication we compute both sides. First

$$(zw)\zeta = (ac - bd, ad + bc) \cdot (e, f) = ((ac - bd)e - (ad + bc)f, (ac - bd)f + (ad + bc)e),$$

whose components expand to

$$ace - bde - adf - bcf, \quad acf - bdf + ade + bce.$$

Second,

$$z(w\zeta) = (a, b) \cdot (ce - df, cf + de) = (a(ce - df) - b(cf + de), a(cf + de) + b(ce - df)),$$

whose components expand to

$$ace - adf - bcf - bde, \quad acf + ade + bce - bdf.$$

The two first components agree and the two second components agree, so $(zw)\zeta = z(w\zeta)$.

(F3).

$$z(w + \zeta) = (a, b) \cdot (c + e, d + f) = (a(c + e) - b(d + f), a(d + f) + b(c + e)),$$

with components $ac + ae - bd - bf$ and $ad + af + bc + be$. On the other hand

$$zw + z\zeta = (ac - bd, ad + bc) + (ae - bf, af + be) = (ac - bd + ae - bf, ad + bc + af + be),$$

which agrees componentwise.

(F4). $(a, b) + (0, 0) = (a, b)$ is clear. Also $(a, b) \cdot (1, 0) = (a \cdot 1 - b \cdot 0, a \cdot 0 + b \cdot 1) = (a, b)$.

(F5). Put $-z := (-a, -b)$; then $z + (-z) = (0, 0)$.

(F6). Suppose $z = (a, b) \neq (0, 0)$, i.e. not both $a = 0$ and $b = 0$. Then $a^2 + b^2 > 0$ in $\mathbb{R}$, so we may define

$$z^{-1} := \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right). \quad (1.6)$$

Using (1.4),

$$z \cdot z^{-1} = \left(\frac{a \cdot a - b \cdot (-b)}{a^2 + b^2}, \frac{a \cdot (-b) + b \cdot a}{a^2 + b^2} \right) = \left(\frac{a^2 + b^2}{a^2 + b^2}, 0 \right) = (1, 0). \quad \square$$

Proposition 1.3 ($\mathbb{R}$ sits inside $\mathbb{C}$). *The map $\iota : \mathbb{R} \rightarrow \mathbb{C}$ defined by $\iota(a) := (a, 0)$ is injective and satisfies*

$$\iota(a + c) = \iota(a) + \iota(c), \quad \iota(ac) = \iota(a)\iota(c) \quad \text{for all } a, c \in \mathbb{R}. \quad (1.7)$$

Proof. Injectivity is (1.5). The first identity in (1.7) is $(a, 0) + (c, 0) = (a + c, 0)$, which is (1.3). The second is $(a, 0) \cdot (c, 0) = (ac - 0, 0 + 0) = (ac, 0)$, which is (1.4). $\square$

Because of Proposition 1.3 we *identify* the real number a with the complex number $(a, 0)$ and write a for both. This identification is harmless: all arithmetic performed in $\mathbb{R}$ agrees with the arithmetic performed in $\mathbb{C}$ on the image of ι .

Definition 1.4 (The imaginary unit). Set

$$i := (0, 1) \in \mathbb{C}. \quad (1.8)$$

Proposition 1.5. $i^2 = -1$; that is, $(0, 1) \cdot (0, 1) = (-1, 0)$.

Proof. By (1.4), $(0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0)$, which is the number we have identified with $-1 \in \mathbb{R}$. $\square$

Proposition 1.6 (Standard form). *Every $z \in \mathbb{C}$ can be written in exactly one way as*

$$z = x + iy, \quad x, y \in \mathbb{R}, \quad (1.9)$$

where the right-hand side means $\iota(x) + i \cdot \iota(y)$ computed in $\mathbb{C}$. Indeed $z = (x, y)$ if and only if (1.9) holds.

Proof. Compute $\iota(x) + i \iota(y) = (x, 0) + (0, 1)(y, 0)$. Now $(0, 1)(y, 0) = (0 \cdot y - 1 \cdot 0, 0 \cdot 0 + 1 \cdot y) = (0, y)$, so the sum is $(x, 0) + (0, y) = (x, y)$. Uniqueness is (1.5). $\square$

From now on we always write complex numbers in the standard form (1.9), and (1.3)–(1.4) become the familiar rules

$$(x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2), \quad (1.10)$$

$$(x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2). \quad (1.11)$$

Equation (1.11) need not be memorised: it is what one gets by expanding the product formally and replacing i^2 by -1 .

Definition 1.7 (Real and imaginary parts). For $z = x + iy$ with $x, y \in \mathbb{R}$ we set

$$\operatorname{Re} z := x \in \mathbb{R}, \quad \operatorname{Im} z := y \in \mathbb{R}. \quad (1.12)$$

We call $\operatorname{Re} z$ the *real part* and $\operatorname{Im} z$ the *imaginary part* of z . Note that $\operatorname{Im} z$ is a *real* number: the imaginary part of $3 + 4i$ is 4, not $4i$. A number z with $\operatorname{Re} z = 0$ and $z \neq 0$ is called *purely imaginary*.

Warning 1.8. $\mathbb{C}$ cannot be made into an ordered field. Suppose $<$ were an order on $\mathbb{C}$ compatible with the field operations. In any ordered field $s^2 > 0$ for every $s \neq 0$. Taking $s = i$ gives $-1 = i^2 > 0$; taking $s = 1$ gives $1 = 1^2 > 0$. Adding, $0 > 0$, a contradiction. Consequently an expression such as “ $z_1 < z_2$ ” is *meaningless* for general complex numbers, and it is illegal to write $z > 0$ unless one has first established $z \in \mathbb{R}$. Inequalities in these notes are always between *real* quantities, typically moduli or real parts.

1.3 Conjugation

Definition 1.9 (Complex conjugate). For $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$, the *complex conjugate* of z is

$$\bar{z} := x - iy. \quad (1.13)$$

Theorem 1.10 (Algebra of conjugation). For all $z, w \in \mathbb{C}$:

$$\overline{z + w} = \bar{z} + \bar{w}, \quad (1.14)$$

$$\overline{z\bar{w}} = \bar{z}w, \quad (1.15)$$

$$\overline{(z/w)} = \bar{z}/\bar{w} \quad (w \neq 0), \quad (1.16)$$

$$\bar{\bar{z}} = z, \quad (1.17)$$

$$\operatorname{Re} z = \frac{1}{2}(z + \bar{z}), \quad \operatorname{Im} z = \frac{1}{2i}(z - \bar{z}), \quad (1.18)$$

$$z\bar{z} = (\operatorname{Re} z)^2 + (\operatorname{Im} z)^2 \in [0, \infty). \quad (1.19)$$

Moreover $z \in \mathbb{R}$ if and only if $\bar{z} = z$, and z is purely imaginary or zero if and only if $\bar{z} = -z$.

Proof. Write $z = x + iy$, $w = u + iv$ with $x, y, u, v \in \mathbb{R}$.

(1.14): $z + w = (x + u) + i(y + v)$, so $\overline{z + w} = (x + u) - i(y + v) = (x - iy) + (u - iv) = \bar{z} + \bar{w}$.

(1.15): by (1.11), $z\bar{w} = (xu - yv) + i(xv + yu)$, hence $\overline{z\bar{w}} = (xu - yv) - i(xv + yu)$. On the other hand

$$\bar{z}\bar{w} = (x - iy)(u - iv) = (xu - (-y)(-v)) + i(x(-v) + (-y)u) = (xu - yv) - i(xv + yu),$$

the same number.

(1.16): let $q := z/w$, so $z = qw$. Applying (1.15), $\bar{z} = \bar{q}\bar{w}$. Since $w \neq 0$ we have $\bar{w} \neq 0$ (its components are $u, -v$, not both zero), so $\bar{q} = \bar{z}/\bar{w}$.

(1.17): $\overline{x - iy} = x + iy$.

(1.18): $z + \bar{z} = (x + iy) + (x - iy) = 2x = 2\operatorname{Re} z$ and $z - \bar{z} = 2iy = 2i\operatorname{Im} z$.

(1.19): $z\bar{z} = (x + iy)(x - iy) = x^2 - (iy)(iy) + ixy - ixy = x^2 + y^2$, a sum of two real squares, hence real and ≥ 0 .

Finally $\bar{z} = z$ means $x - iy = x + iy$, i.e. $2iy = 0$, i.e. $y = 0$, i.e. $z \in \mathbb{R}$; and $\bar{z} = -z$ means $2x = 0$, i.e. $\operatorname{Re} z = 0$. $\square$

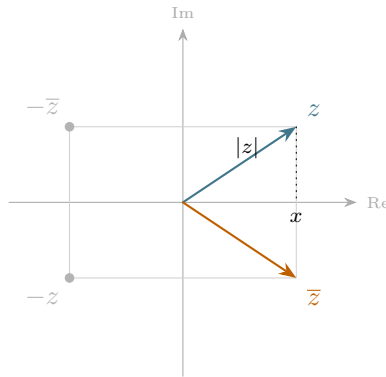


FIGURE 1.1. Conjugation is reflection in the real axis, and negation is the half turn about the origin. The four points z , $\bar{z}$, $-z$ and $-\bar{z}$ form a rectangle symmetric about both axes; identity (1.19) says that the product $z\bar{z}$ of z with its mirror image $\bar{z}$ is the squared distance to the origin.

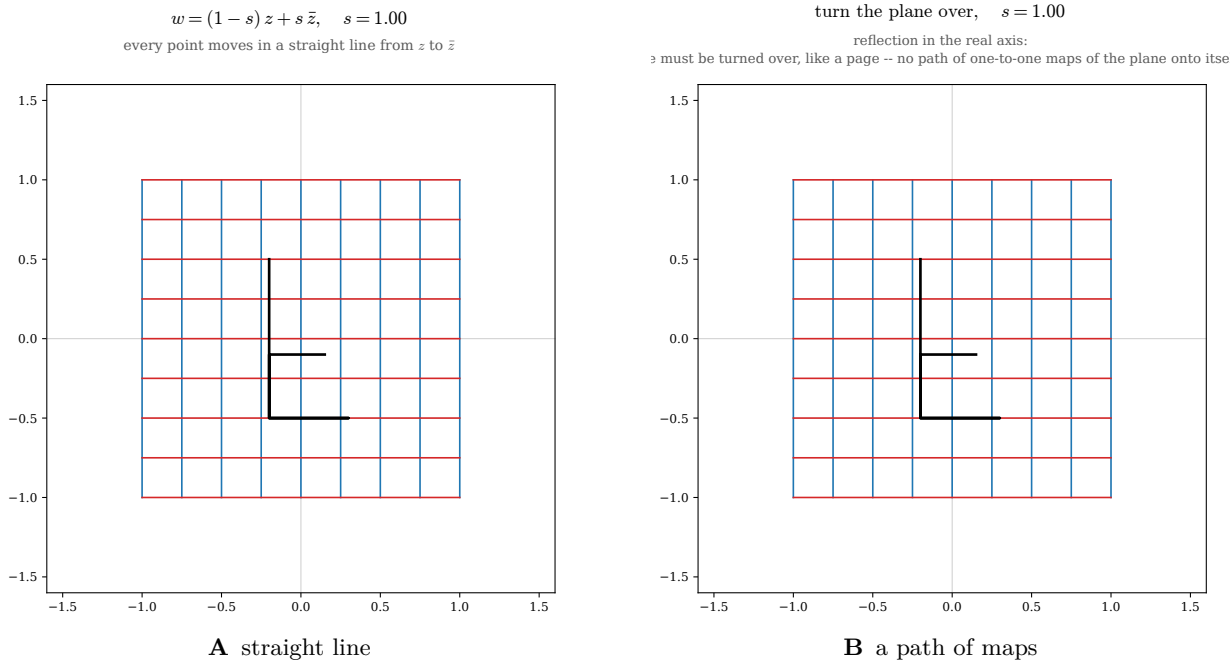
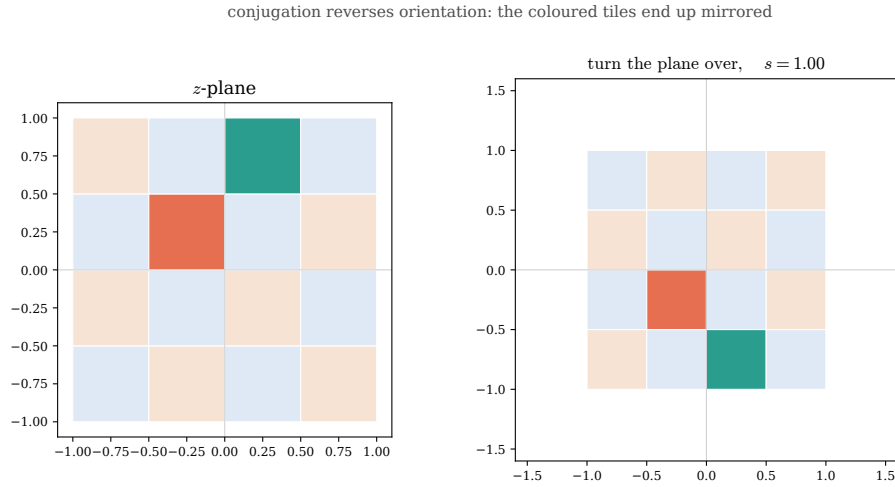


FIGURE 1.2 (animated). Conjugation $z \mapsto \bar{z}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: reflection in the real axis: the plane must be turned over, like a page -- no path of one-to-one maps of the plane onto itself does it. **C**: the tiles on the left deform along the path of B.



Identity (1.19) is the engine behind division: to compute z/w one multiplies numerator and denominator by $\bar{w}$,

$$\frac{z}{w} = \frac{z \bar{w}}{w \bar{w}} = \frac{z \bar{w}}{(\operatorname{Re} w)^2 + (\operatorname{Im} w)^2}, \quad (1.20)$$

because the new denominator is a positive *real* number and dividing by a real number is done componentwise.

Example 1.11. Compute $\frac{2+3i}{4-i}$. Here $w = 4 - i$, $\bar{w} = 4 + i$ and $w\bar{w} = 4^2 + (-1)^2 = 17$. Hence

$$\frac{2+3i}{4-i} = \frac{(2+3i)(4+i)}{17} = \frac{8+2i+12i+3i^2}{17} = \frac{8-3+14i}{17} = \frac{5}{17} + \frac{14}{17}i.$$

One checks the answer by multiplying back: $(\frac{5}{17} + \frac{14}{17}i)(4-i) = \frac{20+14}{17} + \frac{56-5}{17}i = 2+3i$.

1.4 Modulus

Definition 1.12 (Modulus). For $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$, the *modulus* (or *absolute value*) of z is the non-negative real number

$$|z| := \sqrt{x^2 + y^2} = \sqrt{z\bar{z}} \geq 0, \quad (1.21)$$

where $\sqrt{}$ denotes the non-negative real square root.

Note that for $z \in \mathbb{R}$ the definition (1.21) reduces to the real absolute value, so the notation is consistent.

Theorem 1.13 (Properties of the modulus). *For all $z, w \in \mathbb{C}$:*

$$|z| \geq 0, \quad \text{with } |z| = 0 \iff z = 0; \quad (1.22)$$

$$|\bar{z}| = |z|; \quad (1.23)$$

$$|zw| = |z| |w|; \quad (1.24)$$

$$\left| \frac{z}{w} \right| = \frac{|z|}{|w|} \quad (w \neq 0); \quad (1.25)$$

$$|\operatorname{Re} z| \leq |z|, \quad |\operatorname{Im} z| \leq |z|; \quad (1.26)$$

$$|z + w| \leq |z| + |w| \quad (\text{triangle inequality}); \quad (1.27)$$

$$||z| - |w|| \leq |z - w| \quad (\text{reverse triangle inequality}). \quad (1.28)$$

Proof. Write $z = x + iy$, $w = u + iv$, $x, y, u, v \in \mathbb{R}$.

(1.22): $x^2 + y^2 \geq 0$ always, and equals 0 iff $x = y = 0$.

(1.23): $\bar{z} = x - iy$ has the same $x^2 + (-y)^2$.

(1.24): by (1.19) and (1.15),

$$|zw|^2 = (zw)(\overline{zw}) = zw\bar{z}\bar{w} = (z\bar{z})(w\bar{w}) = |z|^2 |w|^2,$$

using commutativity. Taking non-negative square roots of both sides of an identity between non-negative reals gives (1.24).

(1.25): apply (1.24) to $z = (z/w) \cdot w$ and divide by $|w| \neq 0$.

(1.26): $|\operatorname{Re} z| = |x| = \sqrt{x^2} \leq \sqrt{x^2 + y^2} = |z|$, and similarly for Im .

(1.27): expand using (1.19),

$$\begin{aligned} |z + w|^2 &= (z + w)(\overline{z + w}) = (z + w)(\bar{z} + \bar{w}) \\ &= z\bar{z} + w\bar{w} + z\bar{w} + w\bar{z} = |z|^2 + |w|^2 + (z\bar{w} + \bar{z}w) \\ &= |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w}), \end{aligned}$$

where the last step used (1.18) and $\overline{z\bar{w}} = \bar{z}w$. Now by (1.26) and (1.24),

$$\operatorname{Re}(z\bar{w}) \leq |z\bar{w}| = |z| |\bar{w}| = |z| |w|.$$

Therefore $|z + w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2$, and taking non-negative square roots gives (1.27).

(1.28): from (1.27) applied to $z = (z - w) + w$ we get $|z| \leq |z - w| + |w|$, i.e. $|z| - |w| \leq |z - w|$. Interchanging the roles of z and w and using $|w - z| = |-(z - w)| = |z - w|$ gives $|w| - |z| \leq |z - w|$. The two inequalities together are exactly (1.28). $\square$

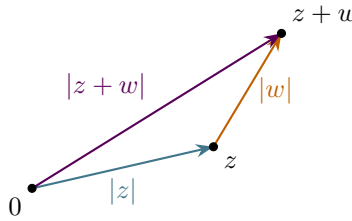


FIGURE 1.3. The triangle inequality (1.27), drawn. The direct route from 0 to $z + w$ is the third side of a triangle whose other two sides have lengths $|z|$ and $|w|$; it can be no longer than the detour, and it is exactly as long only when z and w point the same way (or one of them is 0).

Remark 1.14 (Equality in the triangle inequality). Tracing the proof, equality holds in (1.27) if and only if $\operatorname{Re}(z\bar{w}) = |z\bar{w}|$, i.e. if and only if $z\bar{w}$ is a non-negative real number. When $w \neq 0$ this says $z = tw$ for some real $t \geq 0$: the two vectors point in the same direction. We will meet this condition again in Chapter 3.

Proposition 1.15 (Powers of i). *For every $n \in \mathbb{Z}$,*

$$i^n = \begin{cases} 1, & n \equiv 0 \pmod{4}, \\ i, & n \equiv 1 \pmod{4}, \\ -1, & n \equiv 2 \pmod{4}, \\ -i, & n \equiv 3 \pmod{4}. \end{cases} \quad (1.29)$$

Proof. By Proposition 1.5, $i^2 = -1$, hence $i^3 = i^2 i = -i$ and $i^4 = (i^2)^2 = 1$. Writing $n = 4q + r$ with $q \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3\}$ gives $i^n = (i^4)^q i^r = 1^q i^r = i^r$. (For $n < 0$ the identity $i^{-1} = -i$, verified by $i(-i) = -i^2 = 1$, justifies the same computation.) $\square$

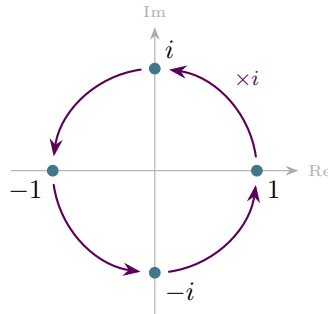


FIGURE 1.4. The powers of i cycle with period four. Multiplication by i is a quarter turn counterclockwise, so four of them return every point to where it started — which is the statement $i^4 = 1$ read as geometry rather than algebra.

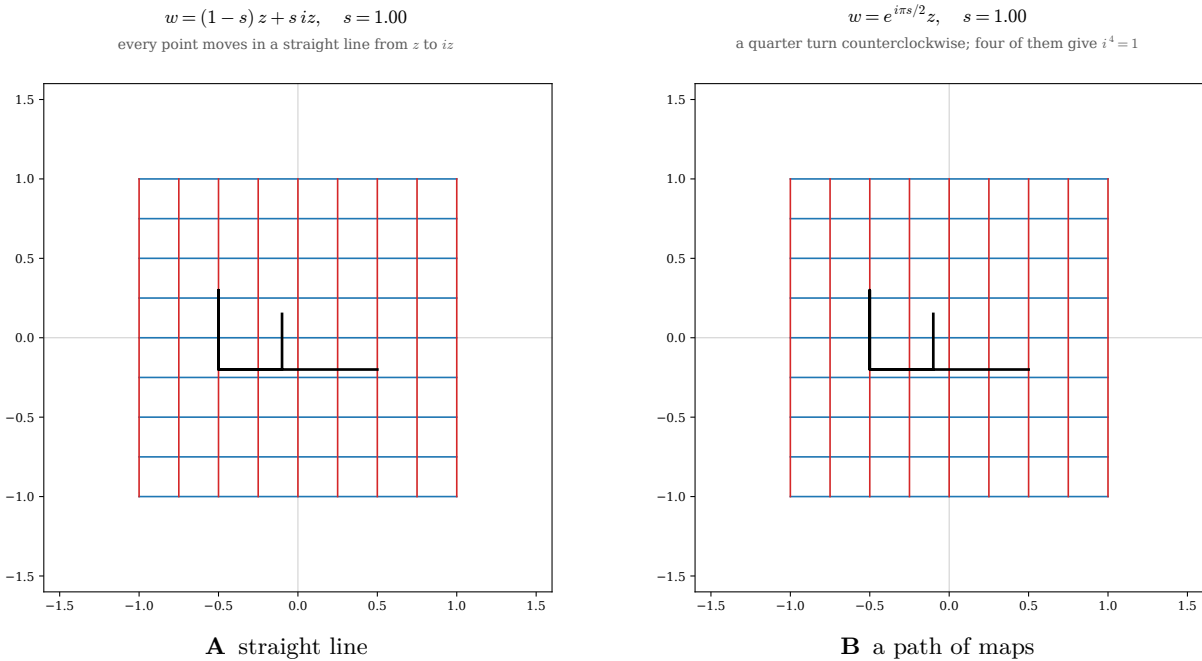
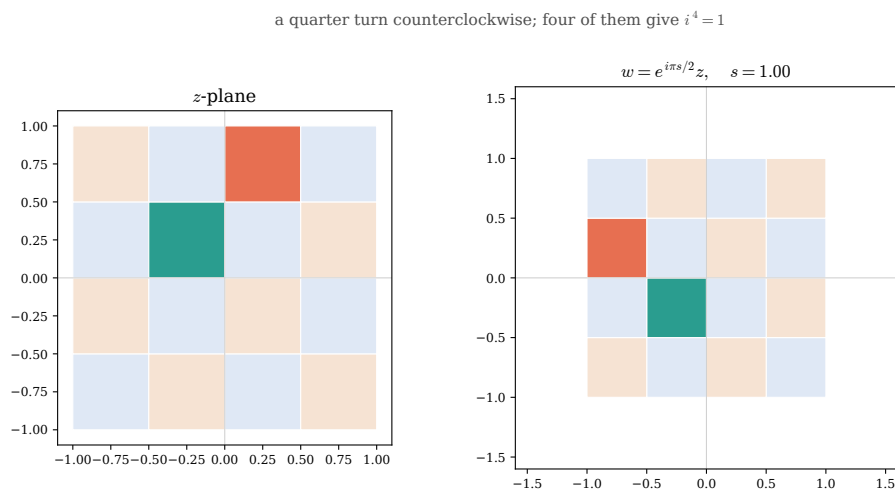


FIGURE 1.5 (animated). Multiplication by i . **A**: every point moves in a straight line from z to $f(z)$. **B**: a quarter turn counterclockwise; four of them give $i^4 = 1$. **C**: the tiles on the left deform along the path of B.



Warning 1.16 (The symbol $\sqrt{\cdot}$). It is common to write “ $i = \sqrt{-1}$ ”. This is at best a mnemonic. The symbol $\sqrt{\cdot}$ was defined only for non-negative reals, and extending it naively to $\mathbb{C}$ breaks the rule $\sqrt{a}\sqrt{b} = \sqrt{ab}$:

$$-1 = i^2 = \sqrt{-1}\sqrt{-1} \stackrel{?}{=} \sqrt{(-1)(-1)} = \sqrt{1} = 1.$$

The error is in the step marked “?”. The equation $w^2 = -1$ has *two* solutions, i and $-i$, and no canonical way to prefer one of them arises from the algebra. In Chapter 10 we will introduce a genuine square-root *function* on $\mathbb{C}$ by making a deliberate choice (a *branch*), and we will see precisely which identities survive that choice and which do not.

1.5 Complex numbers as conformal matrices

There is a second model of $\mathbb{C}$, inside the 2×2 real matrices, which will matter a great deal later: it converts the Cauchy–Riemann equations of Chapter 14 into a single membership statement, and it explains why the whole subject has no analogue in dimension three or higher.

Definition 1.17 (Rotations, similarities, and $CO_+(2)$). Write $\mathbb{R}^{2 \times 2}$ for the real 2×2 matrices and

$$SO(2) := \left\{ R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} : \theta \in \mathbb{R} \right\} \quad (1.30)$$

for the rotation matrices. Define the *conformal orientation-preserving matrices*

$$CO_+(2) := \{ \lambda R : \lambda > 0, R \in SO(2) \} \subseteq \mathbb{R}^{2 \times 2}, \quad (1.31)$$

and write

$$\overline{CO_+(2)} := CO_+(2) \cup \{0\} \quad (1.32)$$

for the set obtained by adjoining the zero matrix. The bar here does *not* mean complex conjugation: $\overline{CO_+(2)}$ is the closure of $CO_+(2)$ in $\mathbb{R}^{2 \times 2}$ (identified with $\mathbb{R}^4$). Indeed $\lambda R \rightarrow 0$ as $\lambda \rightarrow 0^+$; conversely, if $\lambda_n R_n \rightarrow M$ then $\lambda_n = |\lambda_n R_n \mathbf{e}_1| \rightarrow |M \mathbf{e}_1| =: \lambda$, and if $\lambda > 0$ then $R_n \rightarrow M/\lambda$,

which again satisfies $R^T R = I$ and $\det R = 1$, so $M \in CO_+(2)$, while if $\lambda = 0$ then $M = 0$ (each R_n has entries of modulus ≤ 1). Finally let

$$\mathcal{C} := \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\} \subseteq \mathbb{R}^{2 \times 2}. \quad (1.33)$$

The point of the section is that these three descriptions — complex numbers, similarities of the plane, and the special matrix shape (1.33) — are the same object.

Lemma 1.18 (What conformal means for a matrix). *Let $M \in \mathbb{R}^{2 \times 2}$ be invertible, and regard it as a linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. The following are equivalent:*

- (i) *M preserves angles and orientation: for all nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ the angle from $M\mathbf{u}$ to $M\mathbf{v}$ equals the angle from $\mathbf{u}$ to $\mathbf{v}$, including sign;*
- (ii) *there is $\lambda > 0$ with $\langle M\mathbf{u}, M\mathbf{v} \rangle = \lambda^2 \langle \mathbf{u}, \mathbf{v} \rangle$ for all $\mathbf{u}, \mathbf{v}$, and $\det M > 0$;*
- (iii) *$M^T M = \lambda^2 I$ for some $\lambda > 0$ and $\det M > 0$;*
- (iv) *$M \in CO_+(2)$;*
- (v) *$M = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ for some $(a, b) \neq (0, 0)$.*

Proof. We prove (i) $\Rightarrow$ (iii), (ii) $\Leftrightarrow$ (iii), (iii) $\Rightarrow$ (v) $\Rightarrow$ (iv) $\Rightarrow$ (iii), and (iv) $\Rightarrow$ (i). Throughout, the signed angle θ (modulo 2π) from a nonzero vector $\mathbf{u}$ to a nonzero vector $\mathbf{v}$ is determined by

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{|\mathbf{u}| |\mathbf{v}|}, \quad \sin \theta = \frac{\det[\mathbf{u} \ \mathbf{v}]}{|\mathbf{u}| |\mathbf{v}|}.$$

(i) $\Rightarrow$ (iii). Let $\mathbf{e}_1, \mathbf{e}_2$ be the standard basis; the angle from $\mathbf{e}_1$ to $\mathbf{e}_2$ is $\pi/2$. By (i) the angle from $M\mathbf{e}_1$ to $M\mathbf{e}_2$ is $\pi/2$ as well, so $\langle M\mathbf{e}_1, M\mathbf{e}_2 \rangle = 0$ and $\det M = \det[M\mathbf{e}_1 \ M\mathbf{e}_2] > 0$. Next, the angle from $\mathbf{e}_1 + \mathbf{e}_2$ to $\mathbf{e}_1 - \mathbf{e}_2$ is $-\pi/2$, so by (i) these images are orthogonal too:

$$0 = \langle M\mathbf{e}_1 + M\mathbf{e}_2, M\mathbf{e}_1 - M\mathbf{e}_2 \rangle = |M\mathbf{e}_1|^2 - |M\mathbf{e}_2|^2.$$

Hence the two columns $M\mathbf{e}_1, M\mathbf{e}_2$ of M are orthogonal and of the same length $\lambda := |M\mathbf{e}_1|$, which is > 0 because M is invertible. The entries of $M^T M$ are the inner products of the columns of M , so $M^T M = \lambda^2 I$.

(ii) $\Leftrightarrow$ (iii) is the definition of the transpose: $\langle M\mathbf{u}, M\mathbf{v} \rangle = \langle M^T M\mathbf{u}, \mathbf{v} \rangle$, and two symmetric matrices agree iff their bilinear forms do.

(iii) $\Rightarrow$ (v). Write $M = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. Then $M^T M = \lambda^2 I$ says

$$p^2 + r^2 = \lambda^2, \quad q^2 + s^2 = \lambda^2, \quad pq + rs = 0. \quad (1.34)$$

So the columns (p, r) and (q, s) are orthogonal and of equal length $\lambda > 0$. In the plane, the vectors orthogonal to (p, r) of length λ are exactly $\pm(-r, p)$. Hence either $(q, s) = (-r, p)$, giving $\det M = ps - qr = p^2 + r^2 = \lambda^2 > 0$, or $(q, s) = (r, -p)$, giving $\det M = -p^2 - r^2 < 0$, which is excluded. Therefore $M = \begin{pmatrix} p & -r \\ r & p \end{pmatrix}$, which is (v) with $a = p, b = r$.

(v) $\Rightarrow$ (iv). Here we use the existence of an argument, which is proved in Theorem 3.2 independently of this section. Given $(a, b) \neq (0, 0)$ put $\lambda := \sqrt{a^2 + b^2} > 0$ and let θ be an argument of $a + ib$ in the sense of Definition 3.1, so $a = \lambda \cos \theta$ and $b = \lambda \sin \theta$. Then

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \lambda \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \lambda R_\theta \in CO_+(2).$$

(iv) $\Rightarrow$ (iii). If $M = \lambda R$ with $R^\top R = I$ and $\det R = 1$, then $M^\top M = \lambda^2 I$ and $\det M = \lambda^2 > 0$.

(iv) $\Rightarrow$ (i). Let $M = \lambda R$ as before and let $\mathbf{u}, \mathbf{v}$ be nonzero. Then $\langle M\mathbf{u}, M\mathbf{v} \rangle = \lambda^2 \langle R^\top R \mathbf{u}, \mathbf{v} \rangle = \lambda^2 \langle \mathbf{u}, \mathbf{v} \rangle$, in particular $|M\mathbf{u}| = \lambda |\mathbf{u}|$ and $|M\mathbf{v}| = \lambda |\mathbf{v}|$; and $\det[M\mathbf{u} \ M\mathbf{v}] = \det M \det[\mathbf{u} \ \mathbf{v}] = \lambda^2 \det[\mathbf{u} \ \mathbf{v}]$. Dividing by $|M\mathbf{u}| |M\mathbf{v}| = \lambda^2 |\mathbf{u}| |\mathbf{v}|$, the cosine and the sine of the angle from $M\mathbf{u}$ to $M\mathbf{v}$ equal those of the angle from $\mathbf{u}$ to $\mathbf{v}$, so the two angles coincide (modulo 2π), sign included. $\square$

Theorem 1.19 ($\mathbb{C}$ is the field of conformal matrices). *Define $\Phi : \mathbb{C} \rightarrow \mathbb{R}^{2 \times 2}$ by*

$$\Phi(a + ib) := \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad a, b \in \mathbb{R}. \quad (1.35)$$

Then:

(i) Φ is injective with image $\mathcal{C}$, and

$$\mathcal{C} = CO_+(2) \cup \{0\} = \overline{CO_+(2)}; \quad (1.36)$$

(ii) Φ is a field isomorphism of $\mathbb{C}$ onto $\mathcal{C}$, where $\mathcal{C}$ carries matrix addition and matrix multiplication:

$$\Phi(z + w) = \Phi(z) + \Phi(w), \quad \Phi(zw) = \Phi(z) \Phi(w), \quad (1.37)$$

with $\Phi(0) = 0$, $\Phi(1) = I$, and $\Phi(z)^{-1} = \Phi(z^{-1})$ for $z \neq 0$. In particular $\mathcal{C}$ is a commutative field, even though $\mathbb{R}^{2 \times 2}$ is not commutative;

(iii) the dictionary of invariants is

$$\Phi(i) = J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I, \quad \det \Phi(z) = |z|^2, \quad \Phi(\bar{z}) = \Phi(z)^\top, \quad \frac{1}{2} \operatorname{tr} \Phi(z) = \operatorname{Re} z; \quad (1.38)$$

(iv) under the identification of $\mathbb{C}$ with $\mathbb{R}^2$ by $x + iy \leftrightarrow (x, y)^\top$, the matrix $\Phi(z)$ is exactly the matrix of the $\mathbb{R}$ -linear map

$$\mu_z : \mathbb{C} \rightarrow \mathbb{C}, \quad \mu_z(w) := zw. \quad (1.39)$$

Proof. (i) Injectivity and the description of the image are immediate from (1.35) and (1.33). Equation (1.36) is Lemma 1.18 (iv) $\Leftrightarrow$ (v) for the nonzero matrices, together with the observation that $\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = 0$ exactly when $a = b = 0$.

(iv) Let $z = a + ib$ and $w = x + iy$. By (1.11),

$$zw = (ax - by) + i(bx + ay), \quad \text{i.e.} \quad \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax - by \\ bx + ay \end{pmatrix},$$

which is precisely the claim.

(ii) Additivity is clear from (1.35). For multiplicativity, note that by (iv) the map μ_{zw} equals $\mu_z \circ \mu_w$, because $(zw)u = z(wu)$ by associativity in $\mathbb{C}$ (Theorem 1.2); and the matrix of a composition of linear maps is the product of the matrices. Alternatively, multiply out:

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} x & -y \\ y & x \end{pmatrix} = \begin{pmatrix} ax - by & -(ay + bx) \\ ay + bx & ax - by \end{pmatrix} = \Phi((ax - by) + i(ay + bx)) = \Phi(zw).$$

Since Φ is an injective ring homomorphism onto $\mathcal{C}$ and $\mathbb{C}$ is a field, $\mathcal{C}$ is a field isomorphic to it; in particular $\Phi(z)\Phi(w) = \Phi(w)\Phi(z)$, so these matrices commute with one another even though general 2×2 matrices do not. Inverses correspond because $\Phi(z)\Phi(z^{-1}) = \Phi(1) = I$.

(iii) $\Phi(i) = \Phi(0+1 \cdot i) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = J$, and $J^2 = \Phi(i^2) = \Phi(-1) = -I$ by (ii) and Proposition 1.5. Next $\det \Phi(z) = a^2 + b^2 = |z|^2$ by (1.21). Transposing $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ gives $\begin{pmatrix} a & b \\ -b & a \end{pmatrix} = \Phi(a - ib) = \Phi(\bar{z})$. Finally $\text{tr } \Phi(z) = 2a = 2 \text{Re } z$. $\square$

Remark 1.20 (What each fact means geometrically). Theorem 1.19 repackages several things already proved.

- (a) $\det \Phi(z) = |z|^2$ says that multiplication by z scales *areas* by $|z|^2$; since $\Phi(z)$ is a similarity (Lemma 1.18), it therefore scales lengths by $|z|$ — which is (1.24).
- (b) The factorisation $\Phi(z) = \lambda R_\theta$ with $\lambda = |z|$ is exactly the polar decomposition (3.7), and the statement “multiplying by z rotates by $\arg z$ and scales by $|z|$ ” of Theorem 3.11 is the statement that $\Phi(z)$ is a similarity.
- (c) $J^2 = -I$ is the equation $i^2 = -1$: the quarter turn done twice is the half turn. Any real matrix J with $J^2 = -I$ turns $\mathbb{R}^2$ into a copy of $\mathbb{C}$ by $(a + ib) \cdot \mathbf{v} := a\mathbf{v} + bJ\mathbf{v}$; this is the beginning of the theory of complex structures.
- (d) $\Phi(\bar{z}) = \Phi(z)^T$ pairs conjugation with transposition. Note that reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, which is the matrix of $w \mapsto \bar{w}$, is *not* in $\mathcal{C}$: it has determinant -1 . Conjugation is conformal but orientation-reversing, and that single fact is why $\bar{z}$ is not differentiable anywhere (Example 13.4).

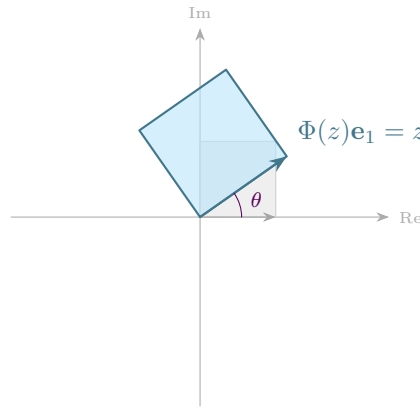


FIGURE 1.6. The matrix $\Phi(z) = \lambda R_\theta$ acting on the plane, for $z = \lambda e^{i\theta}$ with $\lambda = 1.4$ and $\theta = 35^\circ$. A similarity carries the unit square to a square: angles and shape are preserved, only the size and the direction of the frame change. Every element of $CO_+(2)$ looks like this, and by Theorem 1.19 every element of $CO_+(2)$ is a nonzero complex number.

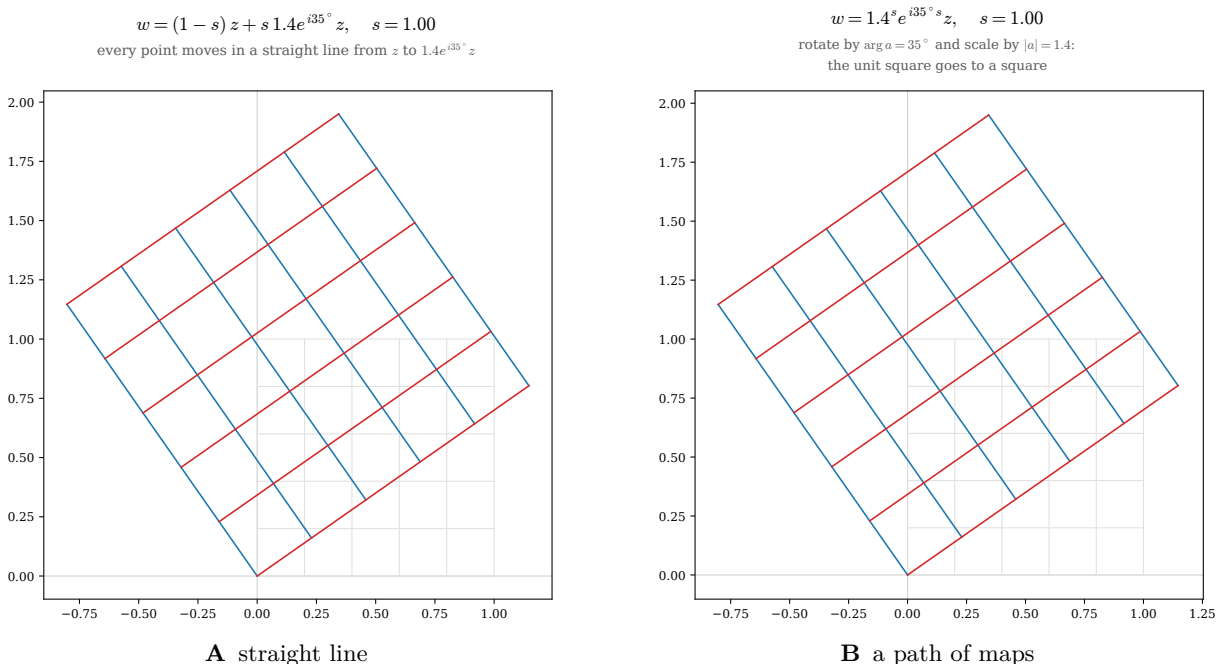
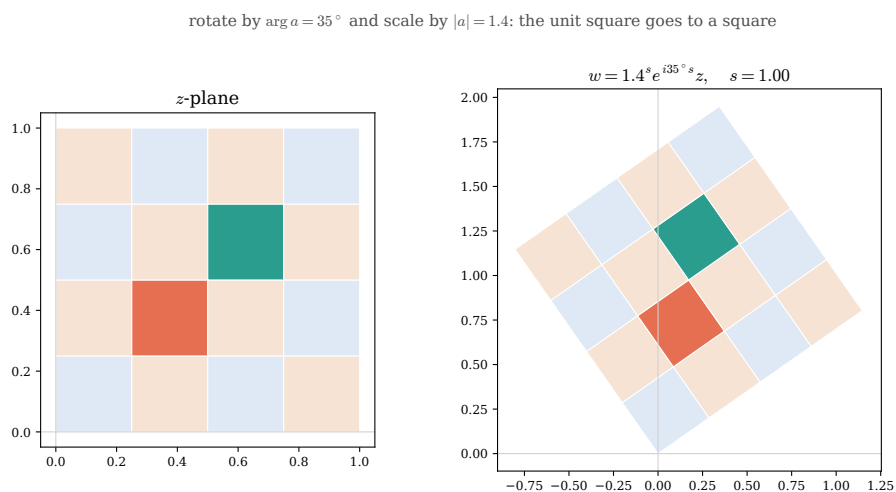


FIGURE 1.7 (animated). Multiplication by $a = 1.4e^{i35^\circ}$, $f(z) = az$. **A**: every point moves in a straight line from z to $f(z)$. **B**: rotate by $\arg a = 35^\circ$ and scale by $|a| = 1.4$: the unit square goes to a square. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 1.7, continued)

1.6 Exercises

Exercise 1.1. Put each of the following in standard form $x + iy$ with $x, y \in \mathbb{R}$:

$$(3 - 2i)(1 + 4i), \quad \frac{1}{2 - 5i}, \quad \frac{(1 + i)^3}{1 - i}, \quad i^{2026}, \quad \frac{3 + i}{i}.$$

Exercise 1.2. Prove that $\operatorname{Re}(iz) = -\operatorname{Im} z$ and $\operatorname{Im}(iz) = \operatorname{Re} z$ for every $z \in \mathbb{C}$, and interpret the result geometrically after reading Chapter 2.

Exercise 1.3. Let $z, w \in \mathbb{C}$. Prove the *parallelogram law*

$$|z + w|^2 + |z - w|^2 = 2|z|^2 + 2|w|^2,$$

and explain the name by drawing the parallelogram with vertices $0, z, z + w, w$.

Exercise 1.4. (★) Prove *Lagrange's identity*: for $z, w \in \mathbb{C}$,

$$|z|^2 |w|^2 - (\operatorname{Re}(z\bar{w}))^2 = (\operatorname{Im}(z\bar{w}))^2 \geq 0,$$

and deduce the Cauchy–Schwarz inequality $|\operatorname{Re}(z\bar{w})| \leq |z| |w|$ directly from it.

Exercise 1.5. Show that if $|z| = 1$ then $\bar{z} = 1/z$. Use this to prove that for $|z| = |w| = 1$ with $zw \neq -1$, the number $\frac{z+w}{1+zw}$ is real.

Exercise 1.6. Determine all $z \in \mathbb{C}$ satisfying $z^2 = \bar{z}$. (There are four.)

Exercise 1.7. Suppose $p(z) = a_n z^n + \cdots + a_1 z + a_0$ with all $a_k \in \mathbb{R}$. Prove $\overline{p(z)} = p(\bar{z})$ for every $z \in \mathbb{C}$, and conclude that the non-real roots of a real polynomial occur in conjugate pairs.

Exercise 1.8. Prove that there is no function $f : \mathbb{C} \rightarrow \mathbb{C}$ satisfying $f(z)^2 = z$ for all z and $f(zw) = f(z)f(w)$ for all z, w and $f(1) = 1$. (*Hint: evaluate at $z = w = -1$ after determining $f(-1)$.*)

Exercise 1.9. Compute $\Phi(3 + 4i)$, $\Phi(3 + 4i)^{-1}$ and $\det \Phi(3 + 4i)$ using (1.35) and (1.38), and check the inverse against (1.6). Then write $\Phi(3 + 4i)$ in the form λR_θ of (1.31), giving λ exactly and θ to four decimals.

Exercise 1.10. (★) Let $M \in \mathbb{R}^{2 \times 2}$ satisfy $MJ = JM$, where $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Prove $M \in \mathcal{C}$. (So $\mathcal{C}$ is exactly the commutant of J : the complex numbers are the real matrices that commute with “multiplication by i ”.) Deduce that $\mathcal{C}$ is a two-dimensional real subalgebra of $\mathbb{R}^{2 \times 2}$ and that it contains no nonzero nilpotent and no zero divisors.

Exercise 1.11. Show that $\mathbb{R}^{2 \times 2}$ decomposes as a direct sum of vector spaces

$$\mathbb{R}^{2 \times 2} = \mathcal{C} \oplus \mathcal{C}^-, \quad \mathcal{C}^- := \left\{ \begin{pmatrix} c & d \\ d & -c \end{pmatrix} : c, d \in \mathbb{R} \right\},$$

and that $\mathcal{C}^-$ consists of the conformal *orientation-reversing* matrices together with 0. Given $M \in \mathbb{R}^{2 \times 2}$, identify the two components explicitly in terms of M and the matrix J of (1.38) (*Hint: compute JMJ*). (This splitting is the matrix form of the Wirtinger decomposition of Exercise 14.3, and it reappears in Theorem 14.10.)

Chapter 2

The Complex Plane

2.1 Complex numbers as points and as vectors

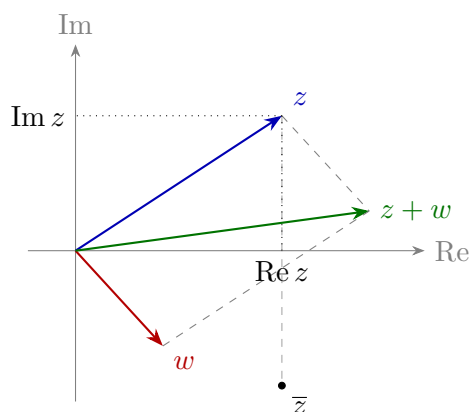
Definition 1.1 built $\mathbb{C}$ out of $\mathbb{R} \times \mathbb{R}$, so a complex number *is* an ordered pair of reals and can be drawn as a point in a Cartesian plane.

Definition 2.1 (Complex plane). The *complex plane* (or *Argand diagram*) is the Cartesian plane $\mathbb{R}^2$ in which the point with coordinates (x, y) is labelled by the complex number $z = x + iy$. The horizontal axis $\{x + i \cdot 0 : x \in \mathbb{R}\}$ is the *real axis*; the vertical axis $\{0 + iy : y \in \mathbb{R}\}$ is the *imaginary axis*.

Two pictures of the same object will be used constantly and it is worth naming them.

- *Point picture*: z is the point $(\operatorname{Re} z, \operatorname{Im} z)$.
- *Vector picture*: z is the arrow from 0 to that point.

In the vector picture, addition (1.10) is exactly the parallelogram rule of vector addition in $\mathbb{R}^2$, and $z \mapsto -z$ is the point reflection through the origin. Conjugation $z \mapsto \bar{z}$ is the reflection across the real axis, since it flips the sign of the second coordinate.



2.2 Distance, circles, discs, and lines

Proposition 2.2 (Modulus is Euclidean distance). For $z, w \in \mathbb{C}$ the number $|z - w|$ equals the Euclidean distance in $\mathbb{R}^2$ between the points z and w .

Proof. Write $z = x_1 + iy_1$ and $w = x_2 + iy_2$ with all four coordinates real. Then $z - w = (x_1 - x_2) + i(y_1 - y_2)$ and by (1.21),

$$|z - w| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2},$$

which is the Euclidean distance formula. $\square$

Proposition 2.2 converts every statement about $|\cdot|$ into a statement of plane geometry. The three basic translations are collected next.

Definition 2.3 (Circles and discs). Let $z_0 \in \mathbb{C}$ and let $\rho > 0$ be real. Define

$$C(z_0, \rho) := \{z \in \mathbb{C} : |z - z_0| = \rho\} \quad (\text{circle of centre } z_0, \text{ radius } \rho), \quad (2.1)$$

$$D(z_0, \rho) := \{z \in \mathbb{C} : |z - z_0| < \rho\} \quad (\text{open disc}), \quad (2.2)$$

$$\overline{D}(z_0, \rho) := \{z \in \mathbb{C} : |z - z_0| \leq \rho\} \quad (\text{closed disc}). \quad (2.3)$$

We write $\mathbb{D} := D(0, 1)$ for the *open unit disc* and $\mathbb{T} := C(0, 1)$ for the *unit circle*.

Example 2.4 (Reading a locus). The set $\{z : |z - 2i| = 3\}$ is the circle of radius 3 centred at $2i$. The set $\{z : |z + 1 - i| \leq 2\}$ is the closed disc of radius 2 centred at $-1 + i$, because $|z + 1 - i| = |z - (-1 + i)|$. Rewriting a condition so that it reads $|z - (\text{centre})|$ is the standard first move.

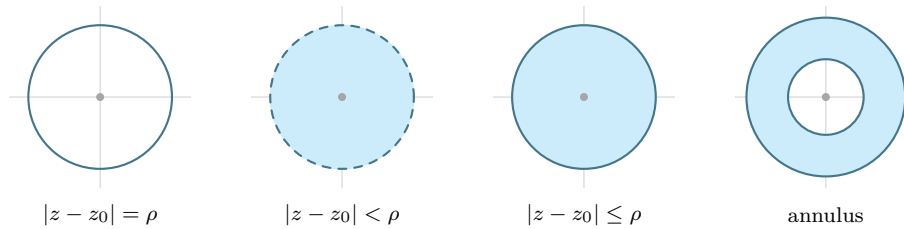


FIGURE 2.1. Three loci of Definition 2.3 and an annulus, all built from one idea: $|z - z_0|$ is the distance from z to z_0 . From the left: the circle $C(z_0, \rho)$, the open disc $D(z_0, \rho)$ (boundary excluded, drawn dashed), the closed disc, and an annulus $\rho_1 < |z - z_0| < \rho_2$. (Annuli appear in Example 2.8.)

Proposition 2.5 (Perpendicular bisector). Let $a, b \in \mathbb{C}$ with $a \neq b$. Then

$$\{z \in \mathbb{C} : |z - a| = |z - b|\} \quad (2.4)$$

is the straight line through the midpoint $\frac{1}{2}(a + b)$ perpendicular to the segment from a to b .

Proof. Square both sides (legitimate, as both are ≥ 0) and use (1.19):

$$|z - a|^2 = |z - b|^2 \iff |z|^2 - 2\operatorname{Re}(z\bar{a}) + |a|^2 = |z|^2 - 2\operatorname{Re}(z\bar{b}) + |b|^2,$$

where we expanded exactly as in the proof of (1.27). Cancelling $|z|^2$ and rearranging,

$$2\operatorname{Re}(z(\overline{b - a})) = |b|^2 - |a|^2. \quad (2.5)$$

Writing $z = x + iy$ and $b - a = p + iq$ with $x, y, p, q \in \mathbb{R}$, the left side is $2(px + qy)$, so (2.5) is the real linear equation $2px + 2qy = |b|^2 - |a|^2$ with $(p, q) \neq (0, 0)$; this is a straight line with normal vector (p, q) , i.e. perpendicular to $b - a$. Substituting $z = \frac{1}{2}(a + b)$ into (2.4) gives $\left|\frac{1}{2}(b - a)\right| = \left|\frac{1}{2}(a - b)\right|$, which is true, so the midpoint lies on the line. $\square$

Proposition 2.6 (The general circle-or-line equation). *Let $A, D \in \mathbb{R}$ and $B, C \in \mathbb{R}$, not all zero, and consider the set*

$$S = \{x + iy : A(x^2 + y^2) + Bx + Cy + D = 0\}. \quad (2.6)$$

- (i) *If $A = 0$ and $(B, C) \neq (0, 0)$ then S is a straight line.*
- (ii) *If $A \neq 0$ then S is a circle, a single point, or empty according as $B^2 + C^2 - 4AD$ is positive, zero, or negative.*
- (iii) *If $A = B = C = 0$ then $D \neq 0$ and $S = \emptyset$.*

Equivalently, in complex notation (2.6) reads

$$Az\bar{z} + \bar{\beta}z + \beta\bar{z} + D = 0, \quad \beta := \frac{1}{2}(B + iC) \in \mathbb{C}. \quad (2.7)$$

Proof. Statements (i) and (iii) are immediate. For (ii), divide by $A \neq 0$ and complete squares:

$$x^2 + y^2 + \frac{B}{A}x + \frac{C}{A}y + \frac{D}{A} = \left(x + \frac{B}{2A}\right)^2 + \left(y + \frac{C}{2A}\right)^2 - \frac{B^2 + C^2 - 4AD}{4A^2}.$$

So S is the set of points at squared distance $(B^2 + C^2 - 4AD)/(4A^2)$ from $(-\frac{B}{2A}, -\frac{C}{2A})$, and the trichotomy follows.

For (2.7): with $z = x + iy$ and $\beta = \frac{1}{2}(B + iC)$ we have $z\bar{z} = x^2 + y^2$ and

$$\bar{\beta}z + \beta\bar{z} = 2\operatorname{Re}(\bar{\beta}z) = 2\operatorname{Re}\left(\frac{1}{2}(B - iC)(x + iy)\right) = Bx + Cy. \quad \square$$

Proposition 2.6 is the reason a single family of curves — “circles and lines” — keeps appearing in this subject: it is the family preserved by the reciprocal map, as we prove in Chapter 11. A line is best thought of as a circle through the point at infinity.

2.3 Inequalities read geometrically

The triangle inequality (1.27) says the direct route from 0 to $z + w$ is no longer than the detour through z . Two standard consequences are used constantly when estimating.

Proposition 2.7 (Estimating quotients). *Let $p(z) = \sum_{k=0}^n a_k z^k$ and $q(z) = \sum_{k=0}^m b_k z^k$ be polynomials with complex coefficients, $b_m \neq 0$. There is $R_0 \geq 1$, depending only on $b_0, \dots, b_m$, such that, writing $R := |z|$,*

$$|p(z)| \leq \left(\sum_{k=0}^n |a_k|\right) R^n \quad \text{for all } |z| \geq 1, \quad |q(z)| \geq \frac{1}{2} |b_m| R^m \quad \text{for all } |z| \geq R_0. \quad (2.8)$$

Consequently, for all z with $|z| = R \geq R_0$ we have $q(z) \neq 0$ and

$$\left|\frac{p(z)}{q(z)}\right| \leq \frac{2\sum_{k=0}^n |a_k|}{|b_m|} R^{n-m}.$$

Proof. For the first estimate, (1.27) extended to $n + 1$ summands (induction) and (1.24) give $|p(z)| \leq \sum_k |a_k| R^k \leq \sum_k |a_k| R^n$ whenever $R \geq 1$, since then $R^k \leq R^n$ for $k \leq n$.

For the second, write $q(z) = b_m z^m (1 + r(z))$ with $r(z) = \sum_{k=0}^{m-1} \frac{b_k}{b_m} z^{k-m}$ (for $z \neq 0$). By the triangle inequality and (1.24), $|r(z)| \leq \frac{1}{|b_m|} \sum_{k < m} |b_k| R^{k-m} \rightarrow 0$ as $R \rightarrow \infty$, and the bound depends only on R and the b_k ; so there is $R_0 \geq 1$, depending only on $b_0, \dots, b_m$, with $|r(z)| \leq \frac{1}{2}$ for $|z| = R \geq R_0$. By the reverse triangle inequality (1.28), $|1 + r(z)| \geq 1 - |r(z)| \geq \frac{1}{2}$, and (1.24) gives the second estimate. Finally, for $R \geq R_0$ the second estimate gives $|q(z)| > 0$, and dividing the first estimate by the second gives the bound on the quotient. $\square$

Example 2.8. Describe $\{z : 1 < |z - i| \leq 3\}$. By Proposition 2.2 this is the set of points whose distance to i is greater than 1 and at most 3: an *annulus* centred at i , with the inner boundary circle removed and the outer boundary circle included.

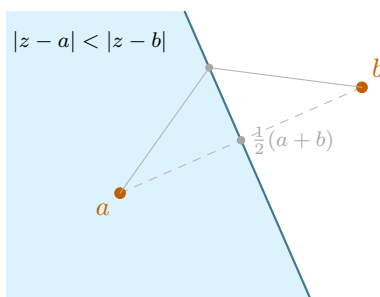


FIGURE 2.2. Proposition 2.5: the set where $|z - a| = |z - b|$ is the perpendicular bisector of the segment $[a, b]$. Every point of the line, such as the marked point joined to a and b , is equidistant from the two points a, b ; the shaded half-plane is $|z - a| < |z - b|$, and the unshaded one is $|z - a| > |z - b|$.

Example 2.9 (A locus that is not a circle). Describe $\{z : |z - 1| + |z + 1| = 4\}$. By Proposition 2.2 the condition says the sum of the distances from z to the two points ± 1 equals 4; this is the ellipse with foci ± 1 and major semi-axis 2. Explicitly, with $z = x + iy$ one obtains after two squarings $\frac{x^2}{4} + \frac{y^2}{3} = 1$.

2.4 Exercises

Exercise 2.1. Sketch each set and say whether it is a circle, a disc, a line, a half-plane, or none of these:

$$|z - 3| = |z + i|, \quad |2z - 1| = 4, \quad \operatorname{Re}(z) \geq \operatorname{Im}(z), \quad |z|^2 = \operatorname{Re}(z), \quad \operatorname{Re}\left(\frac{1}{z}\right) = 1.$$

Exercise 2.2. Show that $\{z : |z - a| = k|z - b|\}$ with $a \neq b$ and $k > 0$, $k \neq 1$, is a circle (an *Apollonius circle*). Find its centre and radius in terms of a, b, k . What happens when $k = 1$?

Exercise 2.3. Prove that three distinct points z_1, z_2, z_3 are collinear if and only if $\frac{z_3 - z_1}{z_2 - z_1} \in \mathbb{R}$.

Exercise 2.4. ($\star$) Let $a \in \mathbb{D}$. Prove that the Möbius map $\varphi_a(z) = \frac{z - a}{1 - \bar{a}z}$ maps $\mathbb{T}$ into $\mathbb{T}$. (*Hint:* for $|z| = 1$, use $\bar{z} = 1/z$.)

Exercise 2.5. Use (2.7) to write the equation of the circle through 1, i , and -1 in the form $Az\bar{z} + \bar{\beta}z + \beta\bar{z} + D = 0$.

Exercise 2.6. Sketch $\{z : |z - 1| + |z + 1| \leq 4\}$ and $\{z : ||z - 1| - |z + 1|| = 1\}$. Identify each conic and give its equation in x, y .

Exercise 2.7. Let a, b, c be the vertices of a triangle. Prove that its centroid is $\frac{1}{3}(a + b + c)$, and that the triangle is equilateral if and only if $a^2 + b^2 + c^2 = ab + bc + ca$. (For the second part, factor $a + \omega b + \omega^2 c$ with $\omega = e^{2\pi i/3}$.)

Exercise 2.8. (★) Prove *Ptolemy's inequality*: for all z_1, z_2, z_3, z_4 ,

$$|z_1 - z_3||z_2 - z_4| \leq |z_1 - z_2||z_3 - z_4| + |z_2 - z_3||z_1 - z_4|.$$

(Hint: verify the algebraic identity $(z_1 - z_3)(z_2 - z_4) = (z_1 - z_2)(z_3 - z_4) + (z_2 - z_3)(z_1 - z_4)$ and apply (1.27).) When is it an equality?

Chapter 3

Polar Form of Complex Numbers

Rectangular form $z = x + iy$ makes addition transparent and multiplication opaque. Polar form does the opposite, and since analysis is largely about multiplying, dividing, and iterating, polar form is the more important of the two.

3.1 The argument

Definition 3.1 (Argument). Let $z \in \mathbb{C}$ with $z \neq 0$. A real number θ is called an *argument* of z if

$$\operatorname{Re} z = |z| \cos \theta \quad \text{and} \quad \operatorname{Im} z = |z| \sin \theta. \quad (3.1)$$

The set of all such θ is denoted $\arg z \subseteq \mathbb{R}$. The number 0 has *no* argument, and $\arg 0$ is undefined.

Theorem 3.2 (Existence and structure of the argument). *Let $z \in \mathbb{C} \setminus \{0\}$. Then $\arg z \neq \emptyset$, and there is exactly one $\theta_0 \in \arg z$ with $\theta_0 \in (-\pi, \pi]$. Moreover*

$$\arg z = \{ \theta_0 + 2k\pi : k \in \mathbb{Z} \}. \quad (3.2)$$

Proof. Put $r := |z| > 0$ and $(c, s) := (\operatorname{Re} z/r, \operatorname{Im} z/r)$. Then $c^2 + s^2 = (\operatorname{Re} z)^2/r^2 + (\operatorname{Im} z)^2/r^2 = 1$ by (1.21), so (c, s) lies on the unit circle of $\mathbb{R}^2$. It is a standard fact from real analysis that $t \mapsto (\cos t, \sin t)$ maps $(-\pi, \pi]$ bijectively onto that circle; let θ_0 be the unique preimage of (c, s) . Then (3.1) holds with $\theta = \theta_0$, so $\arg z \neq \emptyset$.

If $\theta \in \arg z$ then $\cos \theta = \cos \theta_0$ and $\sin \theta = \sin \theta_0$, hence $\cos(\theta - \theta_0) = \cos \theta \cos \theta_0 + \sin \theta \sin \theta_0 = c^2 + s^2 = 1$, which forces $\theta - \theta_0 \in 2\pi\mathbb{Z}$. Conversely $\cos$ and $\sin$ are 2π -periodic, so every $\theta_0 + 2k\pi$ is an argument. This proves (3.2), and uniqueness in $(-\pi, \pi]$ follows because a half-open interval of length 2π contains exactly one element of any coset of $2\pi\mathbb{Z}$. $\square$

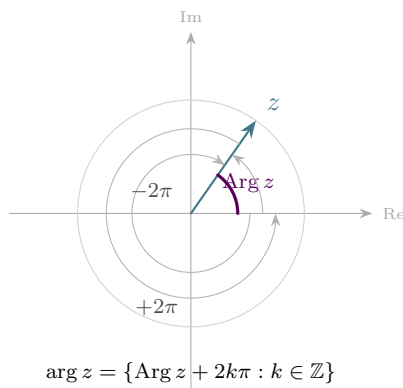


FIGURE 3.1. The argument is a *set*. Every ray from the origin meets the circle once, but the angle naming it is determined only modulo 2π : the same point z is reached by θ_0 , by $\theta_0 + 2\pi$, and by $\theta_0 - 2\pi$. The principal value $\text{Arg } z$ is the unique representative in the half-open interval $(-\pi, \pi]$, drawn heavy.

Definition 3.3 (Principal argument). For $z \in \mathbb{C} \setminus \{0\}$ the unique element of $\arg z \cap (-\pi, \pi]$ is called the *principal argument* of z and denoted $\text{Arg } z$. Thus

$$\text{Arg} : \mathbb{C} \setminus \{0\} \longrightarrow (-\pi, \pi], \quad \arg z = \{\text{Arg } z + 2k\pi : k \in \mathbb{Z}\}. \quad (3.3)$$

Convention 3.4. Because $\arg z$ is a *set*, an equation such as $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ must be read as an equality of sets, where $A + B := \{a + b : a \in A, b \in B\}$. Equations involving Arg are equations between real numbers and are generally *false* without correction terms; see Theorem 3.11 (iii).

Proposition 3.5 (Computing Arg from coordinates). *Let $z = x + iy \neq 0$ with $x, y \in \mathbb{R}$. Then*

$$\text{Arg } z = \begin{cases} \arctan(y/x), & x > 0, \\ \arctan(y/x) + \pi, & x < 0, y \geq 0, \\ \arctan(y/x) - \pi, & x < 0, y < 0, \\ \pi/2, & x = 0, y > 0, \\ -\pi/2, & x = 0, y < 0. \end{cases} \quad (3.4)$$

Proof. The function $\arctan$ takes values in $(-\pi/2, \pi/2)$, and $\tan \theta = \sin \theta / \cos \theta = y/x$ whenever $x \neq 0$ and $\theta \in \arg z$. If $x > 0$ then $\cos(\text{Arg } z) > 0$ by (3.1), so $\text{Arg } z \in (-\pi/2, \pi/2)$ and $\text{Arg } z = \arctan(y/x)$. If $x < 0$ then $\text{Arg } z \in (\pi/2, \pi]$ when $y \geq 0$ and $\text{Arg } z \in (-\pi, -\pi/2)$ when $y < 0$; in either case adding or subtracting π moves $\arctan(y/x)$ into the correct interval and preserves the tangent. The two remaining cases are read off directly from (3.1). $\square$

Warning 3.6. Formula (3.4) shows that “ $\text{Arg } z = \arctan(\text{Im } z / \text{Re } z)$ ” is wrong in the left half-plane. For instance $z = -1 - i$ has $\arctan(1) = \pi/4$, but z lies in the third quadrant and $\text{Arg } z = -3\pi/4$. Always locate the quadrant first.

3.2 Polar form and Euler’s formula

Theorem 3.7 (Polar form). *Every $z \in \mathbb{C} \setminus \{0\}$ can be written as*

$$z = r(\cos \theta + i \sin \theta), \quad r = |z| > 0, \quad \theta \in \arg z, \quad (3.5)$$

and conversely if $z = r(\cos \theta + i \sin \theta)$ with $r > 0$ and $\theta \in \mathbb{R}$ then $r = |z|$ and $\theta \in \arg z$. Two such representations $r_1(\cos \theta_1 + i \sin \theta_1)$ and $r_2(\cos \theta_2 + i \sin \theta_2)$ describe the same complex number if and only if $r_1 = r_2$ and $\theta_1 - \theta_2 \in 2\pi\mathbb{Z}$.

Proof. The representation (3.5) is just (3.1) rewritten, using Theorem 3.2 for the existence of θ . Conversely, if $z = r(\cos \theta + i \sin \theta)$ with $r > 0$ then $|z| = r\sqrt{\cos^2 \theta + \sin^2 \theta} = r$ and (3.1) holds. The uniqueness statement is Theorem 3.2. $\square$

Definition 3.8 (Euler’s notation). For $\theta \in \mathbb{R}$ we *define*

$$e^{i\theta} := \cos \theta + i \sin \theta. \quad (3.6)$$

With this notation (3.5) becomes the *exponential polar form*

$$z = re^{i\theta}, \quad r = |z|, \quad \theta \in \arg z. \quad (3.7)$$

At this stage (3.6) is a definition — an abbreviation for the right-hand side — and nothing is being asserted about exponentials. Chapter 16 defines e^z for all $z \in \mathbb{C}$ and proves that the definition given there agrees with (3.6) when $z = i\theta$, so the notation is consistent. The next proposition shows that the notation is already justified in the only way that matters: it turns multiplication into addition of exponents.

Proposition 3.9 (Law of exponents on the unit circle). *For all $\theta, \varphi \in \mathbb{R}$,*

$$e^{i\theta} e^{i\varphi} = e^{i(\theta+\varphi)}, \quad \overline{e^{i\theta}} = e^{-i\theta} = \frac{1}{e^{i\theta}}, \quad |e^{i\theta}| = 1. \quad (3.8)$$

Proof. Using (1.11) and (3.6),

$$\begin{aligned} e^{i\theta} e^{i\varphi} &= (\cos \theta + i \sin \theta)(\cos \varphi + i \sin \varphi) \\ &= (\cos \theta \cos \varphi - \sin \theta \sin \varphi) + i(\cos \theta \sin \varphi + \sin \theta \cos \varphi) \\ &= \cos(\theta + \varphi) + i \sin(\theta + \varphi) = e^{i(\theta+\varphi)}, \end{aligned}$$

by the real addition formulas for cosine and sine. Next, $\overline{e^{i\theta}} = \cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta) = e^{-i\theta}$, and $e^{i\theta} e^{-i\theta} = e^{i0} = 1$ by what we just proved, so $e^{-i\theta} = 1/e^{i\theta}$. Finally $|e^{i\theta}|^2 = \cos^2 \theta + \sin^2 \theta = 1$. $\square$

Example 3.10. Convert $z = -1 + i\sqrt{3}$ to polar form. Here $|z| = \sqrt{1+3} = 2$. Since $\operatorname{Re} z = -1 < 0$ and $\operatorname{Im} z = \sqrt{3} \geq 0$, (3.4) gives $\operatorname{Arg} z = \arctan(-\sqrt{3}) + \pi = -\frac{\pi}{3} + \pi = \frac{2\pi}{3}$. Hence $z = 2e^{2\pi i/3} = 2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$. Conversely, $4e^{-i\pi/4} = 4(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}) = 2\sqrt{2} - 2\sqrt{2}i$.

3.3 The geometry of multiplication

Theorem 3.11 (Multiplication multiplies moduli and adds arguments). *Let $z_1, z_2 \in \mathbb{C} \setminus \{0\}$, say $z_j = r_j e^{i\theta_j}$ with $r_j > 0$ and $\theta_j \in \arg z_j$. Then:*

- (i) $z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$ and $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$;
- (ii) as sets, $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ and $\arg(z_1/z_2) = \arg z_1 - \arg z_2$;
- (iii) there exists $k \in \{-1, 0, 1\}$, depending on z_1, z_2 , with

$$\operatorname{Arg}(z_1 z_2) = \operatorname{Arg} z_1 + \operatorname{Arg} z_2 + 2k\pi, \quad (3.9)$$

and in general no smaller set of k values suffices.

Proof. (i) By Proposition 3.9, $z_1 z_2 = r_1 r_2 e^{i\theta_1} e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)}$, and since $r_1 r_2 > 0$, Theorem 3.7 identifies $r_1 r_2 = |z_1 z_2|$ and $\theta_1 + \theta_2 \in \arg(z_1 z_2)$. The quotient statement follows from $1/z_2 = r_2^{-1} e^{-i\theta_2}$, itself a consequence of (3.8).

(ii) “ $\supseteq$ ” is (i). For “ $\subseteq$ ”, let $\psi \in \arg(z_1 z_2)$. By (i), $\theta_1 + \theta_2$ is also an argument of $z_1 z_2$, so by (3.2) $\psi = \theta_1 + \theta_2 + 2m\pi$ for some $m \in \mathbb{Z}$; then $\psi = (\theta_1 + 2m\pi) + \theta_2 \in \arg z_1 + \arg z_2$. The quotient case is the same argument with the second formula of (i): “ $\supseteq$ ” is (i), and if $\psi \in \arg(z_1/z_2)$ then $\psi = \theta_1 - \theta_2 + 2m\pi$ for some $m \in \mathbb{Z}$, so $\psi = (\theta_1 + 2m\pi) - \theta_2 \in \arg z_1 - \arg z_2$.

(iii) Both $\operatorname{Arg}(z_1 z_2)$ and $\operatorname{Arg} z_1 + \operatorname{Arg} z_2$ are arguments of $z_1 z_2$, so they differ by $2k\pi$ for some $k \in \mathbb{Z}$ by (3.2). Since $\operatorname{Arg} z_j \in (-\pi, \pi]$ we have $\operatorname{Arg} z_1 + \operatorname{Arg} z_2 \in (-2\pi, 2\pi]$, while $\operatorname{Arg}(z_1 z_2) \in$

$(-\pi, \pi]$; their difference therefore lies in $(-3\pi, 3\pi)$, forcing $k \in \{-1, 0, 1\}$. All three values occur: for $z_1 = z_2 = -1$ we get $\text{Arg}(1) = 0$ but $\text{Arg}(-1) + \text{Arg}(-1) = 2\pi$, so $k = -1$; for $z_1 = z_2 = i$ we get $\text{Arg}(-1) = \pi = \frac{\pi}{2} + \frac{\pi}{2}$, so $k = 0$; for $z_1 = z_2 = e^{-3\pi i/4}$ we get $\text{Arg } z_1 + \text{Arg } z_2 = -\frac{3\pi}{2}$ while $\text{Arg}(z_1 z_2) = \text{Arg}(e^{-3\pi i/2}) = \text{Arg}(i) = \frac{\pi}{2}$, so $k = 1$. $\square$

Theorem 3.11(i) is the single most useful fact in the subject:

Multiplying by $re^{i\alpha}$ is the same as rotating the plane through the angle α about 0 and then scaling by the factor r .

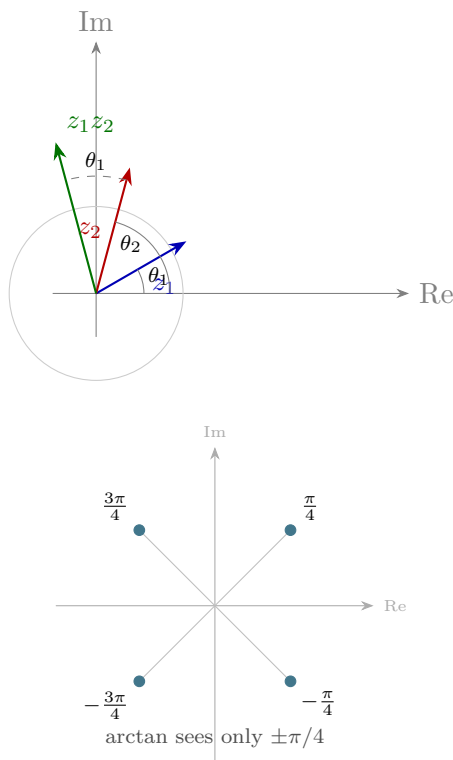


FIGURE 3.2. The quadrant trap of Warning 3.6. The four points $\pm 1 \pm i$ all have $|\text{Im } z / \text{Re } z| = 1$, so arctan cannot tell them apart — it returns only $\pm\pi/4$. The principal arguments are in fact $\pi/4$, $3\pi/4$, $-\pi/4$, $-3\pi/4$: one must locate the quadrant before trusting the arctangent.

Example 3.12 (Rotation as multiplication). Multiplication by $i = e^{i\pi/2}$ rotates by a quarter turn counterclockwise. Indeed $i(x + iy) = -y + ix$, which is the vector (x, y) rotated by 90° . Multiplication by $-1 = e^{i\pi}$ is a half turn; multiplication by the positive real r is a pure dilation.

Example 3.13 (Trigonometry for free). Take $\theta = \varphi$ in the computation proving Proposition 3.9. Comparing real and imaginary parts of $e^{2i\theta} = (e^{i\theta})^2$ gives, in one line,

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad \sin 2\theta = 2 \sin \theta \cos \theta.$$

Every multiple-angle identity arises the same way; see Exercise 3.3.

3.4 Exercises

Exercise 3.1. Write in the form $re^{i\theta}$ with $r > 0$ and $\theta = \text{Arg } z$:

$$1 + i, \quad -3, \quad -2i, \quad \sqrt{3} - i, \quad \frac{1 + i}{\sqrt{3} - i}.$$

Exercise 3.2. Write in the form $x + iy$ with $x, y \in \mathbb{R}$: $5e^{i\pi}$, $\sqrt{2}e^{3\pi i/4}$, $e^{-i\pi/6}$, $2e^{i\pi/2} \cdot 3e^{i\pi/3}$.

Exercise 3.3. Use $e^{3i\theta} = (e^{i\theta})^3$ to prove $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ and $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$.

Exercise 3.4. Give explicit z_1, z_2 for which $\text{Arg}(z_1 z_2) \neq \text{Arg } z_1 + \text{Arg } z_2$, and verify (3.9) for your example by exhibiting k .

Exercise 3.5. (★) Prove that for $z \neq 0$, $\text{Arg } \bar{z} = -\text{Arg } z$ unless z is a negative real number, in which case $\text{Arg } \bar{z} = \text{Arg } z = \pi$. Where does the exception come from?

Exercise 3.6. Let $\theta \in \mathbb{R}$ with $e^{i\theta} \neq 1$. Prove the finite geometric sum

$$\sum_{k=0}^{n-1} e^{ik\theta} = \frac{e^{in\theta} - 1}{e^{i\theta} - 1},$$

and by taking real parts derive *Lagrange's trigonometric identity*

$$\sum_{k=0}^{n-1} \cos(k\theta) = \frac{1}{2} + \frac{\sin((n - \frac{1}{2})\theta)}{2\sin(\theta/2)}.$$

Exercise 3.7. Show that the map $z \mapsto e^{i\alpha}z$ preserves distances, i.e. $|e^{i\alpha}z - e^{i\alpha}w| = |z - w|$ for all z, w . Which complex numbers a have the property that $z \mapsto az$ preserves distances?

Exercise 3.8. Using (3.8), prove that for all $\theta, \varphi \in \mathbb{R}$

$$e^{i\theta} + e^{i\varphi} = 2\cos\left(\frac{\theta - \varphi}{2}\right)e^{i(\theta+\varphi)/2},$$

and read off both $|e^{i\theta} + e^{i\varphi}|$ and an argument of the sum. Interpret the formula as the statement that the diagonal of a rhombus bisects its angle.

Exercise 3.9. Let $z = re^{i\theta}$ with $r > 0$. Express $\text{Re}(z^n)$ and $\text{Im}(z^n)$ in terms of r and θ , and use the result to sketch the curves $\text{Re}(z^3) = 0$ and $\text{Im}(z^3) = 0$. How many rays does each consist of?

Exercise 3.10. (★) Prove that $|e^{i\theta} - 1| = 2|\sin(\theta/2)|$ for all real θ , and deduce that the perimeter of the regular n -gon inscribed in the unit circle is $2n\sin(\pi/n)$. Show this tends to 2π as $n \rightarrow \infty$, and say which standard limit you have used.

Chapter 4

Powers and Roots

4.1 Integer powers and De Moivre's formula

Definition 4.1 (Integer powers). For $z \in \mathbb{C}$ and $n \in \mathbb{N} = \{1, 2, 3, \dots\}$ define $z^1 := z$ and $z^{n+1} := z^n z$. Set $z^0 := 1$ for every $z \in \mathbb{C}$, including $0^0 := 1$ (the usual convention, which makes a polynomial $\sum_{k=0}^n a_k z^k$ take the value a_0 at $z = 0$), and for $z \neq 0$ and $n \in \mathbb{N}$ set $z^{-n} := (z^{-1})^n$, where z^{-1} is the multiplicative inverse of (1.6).

Theorem 4.2 (De Moivre). For every $\theta \in \mathbb{R}$ and every $n \in \mathbb{Z}$,

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta), \quad \text{that is,} \quad (e^{i\theta})^n = e^{in\theta}. \quad (4.1)$$

Consequently, if $z = re^{i\theta}$ with $r > 0$ then

$$z^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta), \quad n \in \mathbb{Z}. \quad (4.2)$$

Proof. Case $n \geq 0$, by induction. For $n = 0$ both sides equal 1. Assume (4.1) for some $n \geq 0$. Then, using Proposition 3.9,

$$(e^{i\theta})^{n+1} = (e^{i\theta})^n e^{i\theta} = e^{in\theta} e^{i\theta} = e^{i(n+1)\theta},$$

completing the induction.

Case $n < 0$. Write $n = -m$ with $m \in \mathbb{N}$. By (3.8) the inverse of $e^{i\theta}$ is $e^{-i\theta}$, hence $(e^{i\theta})^{-m} = (e^{-i\theta})^m = e^{-im\theta} = e^{in\theta}$ by the case already proved applied to the angle $-\theta$.

Finally (4.2) follows because $|\cdot|$ is multiplicative (1.24), so $|z^n| = r^n$, and $(re^{i\theta})^n = r^n (e^{i\theta})^n$ by commutativity. $\square$

Example 4.3. Compute $(1 + i)^{10}$. Since $1 + i = \sqrt{2} e^{i\pi/4}$, (4.2) gives

$$(1 + i)^{10} = (\sqrt{2})^{10} e^{10i\pi/4} = 32 e^{5i\pi/2} = 32 e^{i\pi/2} = 32i,$$

where we reduced $5\pi/2$ modulo 2π . Doing this by the binomial theorem is possible but far more work; this is the typical situation.

4.2 n -th roots

Definition 4.4 (n -th root). Let $w \in \mathbb{C}$ and $n \in \mathbb{N}$. A number $z \in \mathbb{C}$ is an n -th root of w if $z^n = w$. The set of all n -th roots of w is written $w^{1/n}$; it is a set, not a number.

Theorem 4.5 (The n -th roots of a nonzero complex number). *Let $n \in \mathbb{N}$ and let $w \in \mathbb{C} \setminus \{0\}$, written in polar form as $w = \rho e^{i\varphi}$ with $\rho = |w| > 0$ and $\varphi \in \arg w$. Then w has exactly n distinct n -th roots, namely*

$$z_k = \rho^{1/n} \exp\left(i \frac{\varphi + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1, \quad (4.3)$$

where $\exp(i\theta)$ is another way of writing $e^{i\theta}$ of Definition 3.8, and $\rho^{1/n} > 0$ denotes the unique positive real n -th root of the positive real number ρ . These n points are the vertices of a regular n -gon inscribed in the circle $C(0, \rho^{1/n})$ of (2.1). The number 0 has exactly one n -th root, namely 0.

Proof. Each z_k is a root. By (4.2),

$$z_k^n = (\rho^{1/n})^n \exp\left(in \cdot \frac{\varphi + 2k\pi}{n}\right) = \rho e^{i(\varphi + 2k\pi)} = \rho e^{i\varphi} = w,$$

using 2π -periodicity (3.2).

They are distinct. The moduli all equal $\rho^{1/n}$, so by Theorem 3.7 $z_k = z_l$ forces $\frac{\varphi + 2k\pi}{n} - \frac{\varphi + 2l\pi}{n} \in 2\pi\mathbb{Z}$, i.e. $\frac{2\pi(k-l)}{n} \in 2\pi\mathbb{Z}$, i.e. $n \mid k-l$. For $k, l \in \{0, \dots, n-1\}$ this means $k = l$.

There are no others. Suppose $z^n = w$. Since $w \neq 0$ we have $z \neq 0$, so we may write $z = se^{i\psi}$ with $s > 0$. Then $s^n e^{in\psi} = \rho e^{i\varphi}$, and Theorem 3.7 gives $s^n = \rho$ and $n\psi - \varphi \in 2\pi\mathbb{Z}$. The first equation has the unique positive solution $s = \rho^{1/n}$ (strict monotonicity of $t \mapsto t^n$ on $(0, \infty)$). The second gives $\psi = \frac{\varphi + 2m\pi}{n}$ for some $m \in \mathbb{Z}$; writing $m = qn + k$ with $k \in \{0, \dots, n-1\}$ yields $\psi = \frac{\varphi + 2k\pi}{n} + 2q\pi$, so $z = z_k$.

Geometry. The z_k lie on the circle of radius $\rho^{1/n}$ and consecutive arguments differ by exactly $2\pi/n$; that is the definition of a regular n -gon inscribed in that circle.

The case $w = 0$. If $z^n = 0$ then $|z|^n = 0$ by (1.24), so $|z| = 0$ and $z = 0$. $\square$

Definition 4.6 (Principal n -th root). For $w \in \mathbb{C} \setminus \{0\}$ and $n \in \mathbb{N}$, the *principal n -th root* of w is the root obtained in (4.3) by taking $\varphi = \operatorname{Arg} w$ and $k = 0$:

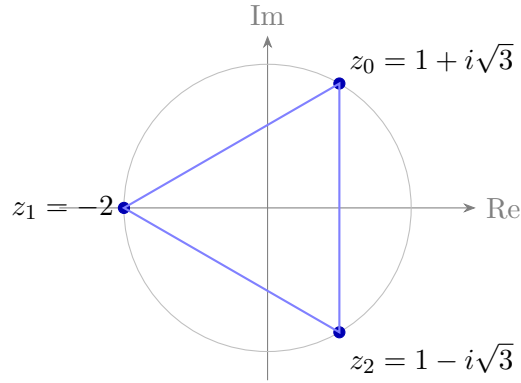
$$w_{\text{pr}}^{1/n} := |w|^{1/n} \exp\left(i \frac{\operatorname{Arg} w}{n}\right). \quad (4.4)$$

Its argument lies in $(-\pi/n, \pi/n]$. When w is a positive real, (4.4) returns the usual positive real root, so the notation is consistent with the real convention.

Example 4.7 (Cube roots of -8). Here $\rho = 8$, $\operatorname{Arg}(-8) = \pi$, $n = 3$, so $\rho^{1/3} = 2$ and the roots have arguments $\frac{\pi + 2k\pi}{3}$ for $k = 0, 1, 2$, i.e. $\frac{\pi}{3}$, π , $\frac{5\pi}{3}$. Thus

$$z_0 = 2e^{i\pi/3} = 1 + i\sqrt{3}, \quad z_1 = 2e^{i\pi} = -2, \quad z_2 = 2e^{5i\pi/3} = 1 - i\sqrt{3}.$$

The principal cube root is $z_0 = 1 + i\sqrt{3}$, *not* the real root -2 . Note the roots form an equilateral triangle on $C(0, 2)$ and that the two non-real ones are conjugates, as they must be by Exercise 1.7 since $z^3 + 8$ has real coefficients.



4.3 Roots of unity

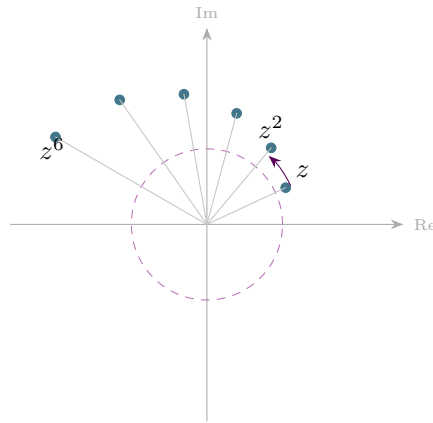


FIGURE 4.1. De Moivre's formula (4.2) as a picture: raising to the n -th power multiplies the argument by n and raises the modulus to the n -th power. Here z has $|z| = 1.15$ and $\text{Arg } z = 25^\circ$; the successive powers spiral outward, turning 25° each time. Had $|z|$ been less than 1 the spiral would wind inward, and on $|z| = 1$ it would stay on the circle; these three cases reappear when we study the map z^n in Chapter 9.

Definition 4.8 (Roots of unity). Let $n \in \mathbb{N}$. The n -th roots of unity are the solutions of $z^n = 1$. By Theorem 4.5 with $w = 1$, $\rho = 1$, $\varphi = 0$, they are

$$\omega_k := e^{2\pi i k/n} = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}, \quad k = 0, 1, \dots, n-1. \quad (4.5)$$

We write $\omega := \omega_1 = e^{2\pi i/n}$, so that $\omega_k = \omega^k$.

Proposition 4.9 (Structure of the roots of unity). Let $n \in \mathbb{N}$ and $\omega = e^{2\pi i/n}$.

- (i) $\mu_n := \{1, \omega, \omega^2, \dots, \omega^{n-1}\}$ is a group under multiplication, cyclic of order n .
- (ii) If $n \geq 2$ then

$$\sum_{k=0}^{n-1} \omega^k = 0. \quad (4.6)$$

- (iii) $z^n - 1 = \prod_{k=0}^{n-1} (z - \omega^k)$ for all $z \in \mathbb{C}$.

(iv) If $w \in \mathbb{C} \setminus \{0\}$ and z_* is any one n -th root of w , then the full set of n -th roots is $\{z_*\omega^k : k = 0, \dots, n-1\}$.

Proof. (i) $\omega^n = e^{2\pi i} = 1$ by (4.1), so products and inverses of powers of ω are again powers of ω (reduce the exponent modulo n); the elements listed are distinct by Theorem 4.5, so the group is cyclic of order n generated by ω .

(ii) Since $n \geq 2$ we have $\omega \neq 1$, so the finite geometric series sums to $\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega-1} = 0$.

(iii) The polynomial $z^n - 1$ has degree n , leading coefficient 1, and by Theorem 4.5 the n distinct roots $\omega^0, \dots, \omega^{n-1}$. Dividing successively by $(z - \omega^k)$ (polynomial division over the field $\mathbb{C}$) exhausts the degree, giving the stated factorisation.

(iv) Each $z_*\omega^k$ satisfies $(z_*\omega^k)^n = z_*^n(\omega^n)^k = w$; they are n distinct numbers because $z_* \neq 0$ and the ω^k are distinct; and by Theorem 4.5 there are no others. $\square$

Part (iv) is worth restating: *the solution set of $z^n = w$ is a single solution times the group μ_n* . Geometrically, once you have found one root, the rest are obtained by rotating it repeatedly through the angle $2\pi/n$.

Example 4.10 (Solving a quartic by roots). Solve $z^4 = -16$. Write $-16 = 16e^{i\pi}$. By (4.3) with $n = 4$, $\rho^{1/4} = 2$ and arguments $\frac{\pi+2k\pi}{4}$ for $k = 0, 1, 2, 3$, i.e. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$. Hence

$$z \in \{\sqrt{2}(1+i), \sqrt{2}(-1+i), \sqrt{2}(-1-i), \sqrt{2}(1-i)\}.$$

Equivalently, one root is $\sqrt{2}(1+i)$ and the others are that root times $i, -1, -i$ — the fourth roots of unity, in accordance with Proposition 4.9(iv).

4.4 Exercises

Exercise 4.1. Compute, using polar form: $(\sqrt{3} + i)^6$, $(1 - i)^{-8}$, $(\frac{-1+i\sqrt{3}}{2})^{2026}$.

Exercise 4.2. Find all solutions and plot them: $z^3 = 8i$, $z^4 = 1$, $z^6 = -1$, $z^5 = 32$, $z^2 = 3 - 4i$.

Exercise 4.3. Show that the principal square root of $a + bi$ ($b \neq 0$) is

$$\sqrt{\frac{|a+bi|+a}{2}} + i \operatorname{sgn}(b) \sqrt{\frac{|a+bi|-a}{2}}.$$

(Set $(x + iy)^2 = a + bi$, solve $x^2 - y^2 = a$, $2xy = b$, and use $x^2 + y^2 = |a + bi|$.) This formula is used in Chapter 6.

Exercise 4.4. ($\star$) Let $n \geq 2$ and $\omega = e^{2\pi i/n}$. Prove that for $j \in \mathbb{Z}$,

$$\sum_{k=0}^{n-1} \omega^{jk} = \begin{cases} n, & n \mid j, \\ 0, & \text{otherwise.} \end{cases}$$

This orthogonality relation is the whole of the discrete Fourier transform.

Exercise 4.5. Prove that the product of all n -th roots of unity is $(-1)^{n+1}$.

Exercise 4.6. Let P be a regular n -gon inscribed in the unit circle with one vertex at 1. Prove that the product of the distances from 1 to the other $n-1$ vertices equals n . (Hint: factor $\frac{z^n-1}{z-1}$ and let $z \rightarrow 1$.)

Exercise 4.7. Explain why $(w^{1/n})^n = \{w\}$ but $(w^n)^{1/n}$ has n elements in general. Which containment always holds?

Exercise 4.8. Draw the n -th roots of unity for $n = 3, 4, 6, 8$ and verify (4.6) geometrically in each case by adding the vectors head to tail. Why must the resulting polygon close up?

Exercise 4.9. Factor $z^6 - 1$ completely over $\mathbb{C}$, then group the conjugate pairs to factor it over $\mathbb{R}$ into linear and irreducible quadratic factors. Do the same for $z^4 + 1$, and note that this one has *no* real linear factor.

Exercise 4.10. (★) Let $\omega = e^{2\pi i/n}$ and let ζ be any n -th root of unity. Prove that ζ generates all of μ_n — i.e. that $\{\zeta^k : k \in \mathbb{Z}\} = \mu_n$ — if and only if $\zeta = \omega^j$ with $\gcd(j, n) = 1$. Count such ζ for $n = 12$. (*These are the primitive n -th roots of unity, and the count is Euler's $\varphi(n)$.*)

Chapter 5

Sets of Points in the Complex Plane

Every later definition — limit, continuity, derivative, analyticity — quantifies over small discs. This chapter fixes the vocabulary. All of it is the standard topology of $\mathbb{R}^2$ transcribed into complex notation; nothing here is special to $\mathbb{C}$, but the notation must be fluent before Part III.

5.1 Neighbourhoods

Definition 5.1 (Neighbourhood, punctured neighbourhood). Let $z_0 \in \mathbb{C}$ and let $\varepsilon > 0$ be real. The ε -neighbourhood of z_0 is the open disc

$$D(z_0, \varepsilon) = \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}, \quad (5.1)$$

and the *punctured* (or *deleted*) ε -neighbourhood is

$$D'(z_0, \varepsilon) := D(z_0, \varepsilon) \setminus \{z_0\} = \{z \in \mathbb{C} : 0 < |z - z_0| < \varepsilon\}. \quad (5.2)$$

Definition 5.2 (Interior, exterior, boundary). Let $S \subseteq \mathbb{C}$ and $z_0 \in \mathbb{C}$. We call z_0

- an *interior point* of S if there is $\varepsilon > 0$ with $D(z_0, \varepsilon) \subseteq S$;
- an *exterior point* of S if there is $\varepsilon > 0$ with $D(z_0, \varepsilon) \subseteq \mathbb{C} \setminus S$;
- a *boundary point* of S if it is neither, i.e. if for every $\varepsilon > 0$ the disc $D(z_0, \varepsilon)$ meets both S and $\mathbb{C} \setminus S$.

The sets of such points are written $\text{int } S$, $\text{ext } S$, ∂S respectively. By construction

$$\mathbb{C} = \text{int } S \sqcup \partial S \sqcup \text{ext } S \quad (\text{disjoint union}). \quad (5.3)$$

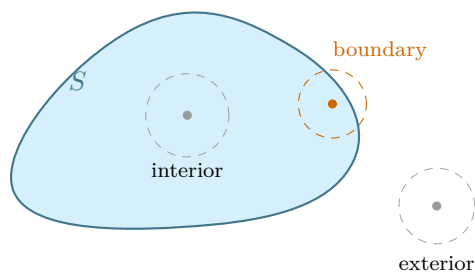


FIGURE 5.1. The trichotomy (5.3). An interior point of S carries a whole disc inside S ; an exterior point carries a whole disc inside the complement; a boundary point carries neither — every disc about it, however small, meets both. The three classes partition the plane, whatever S is.

Definition 5.3 (Open, closed, closure). A set $S \subseteq \mathbb{C}$ is *open* if every point of S is an interior point of S , i.e. $S = \text{int } S$. It is *closed* if its complement $\mathbb{C} \setminus S$ is open. The *closure* of S is $\bar{S} := S \cup \partial S$.

Warning 5.4. “Open” and “closed” are not opposites. The set $\mathbb{C}$ is both open and closed; so is $\emptyset$. The set $\{z : 0 < |z| \leq 1\}$ is neither.

Proposition 5.5 (Discs are open). *For every $z_0 \in \mathbb{C}$ and $\rho > 0$, the disc $D(z_0, \rho)$ of (2.2) is open, the closed disc $\bar{D}(z_0, \rho)$ of (2.3) is closed, and $\partial D(z_0, \rho) = \partial \bar{D}(z_0, \rho) = C(z_0, \rho)$.*

Proof. Let $a \in D(z_0, \rho)$, so $|a - z_0| < \rho$. Put $\varepsilon := \rho - |a - z_0| > 0$. If $|z - a| < \varepsilon$ then by the triangle inequality (1.27)

$$|z - z_0| \leq |z - a| + |a - z_0| < \varepsilon + |a - z_0| = \rho,$$

so $D(a, \varepsilon) \subseteq D(z_0, \rho)$; hence a is interior and $D(z_0, \rho)$ is open.

For the closed disc, let $b \notin \bar{D}(z_0, \rho)$, i.e. $|b - z_0| > \rho$, and put $\delta := |b - z_0| - \rho > 0$. If $|z - b| < \delta$ then by the reverse triangle inequality (1.28), $|z - z_0| \geq |b - z_0| - |z - b| > \rho$, so $D(b, \delta) \subseteq \mathbb{C} \setminus \bar{D}(z_0, \rho)$. Hence the complement is open and $\bar{D}(z_0, \rho)$ is closed.

Finally let $|c - z_0| = \rho$ and $\varepsilon > 0$. The points $c_t := z_0 + t(c - z_0)$ satisfy $|c_t - z_0| = t\rho$ and $|c_t - c| = |1 - t|\rho$; choosing t slightly less than 1 and slightly greater than 1 produces points of $D(z_0, \rho)$ and of its complement inside $D(c, \varepsilon)$. So c is a boundary point of both discs. Conversely no point with $|c - z_0| \neq \rho$ is a boundary point, by the two paragraphs above. $\square$

Example 5.6. (a) $S = \{z : \text{Re } z > 0\}$ (open right half-plane) is open; ∂S is the imaginary axis; S is not closed.

(b) $S = \{z : 1 \leq |z| < 2\}$ is neither open nor closed; $\text{int } S = \{1 < |z| < 2\}$, $\partial S = C(0, 1) \cup C(0, 2)$, $\bar{S} = \{1 \leq |z| \leq 2\}$.

(c) $S = \mathbb{Z} \subseteq \mathbb{C}$ is closed with empty interior; $\partial S = S$.

(d) $S = \{z : \text{Im } z \neq 0\}$ is open and disconnected (see below).

5.2 Connectedness and regions

Definition 5.7 (Polygonal path). Let $z_0, z_1, \dots, z_m \in \mathbb{C}$. The *polygonal path* with these vertices is the union of the m segments

$$[z_{j-1}, z_j] := \{(1 - t)z_{j-1} + tz_j : t \in [0, 1]\}, \quad j = 1, \dots, m. \quad (5.4)$$

Definition 5.8 (Connected open set; region). An open set $S \subseteq \mathbb{C}$ is *connected* if for every pair $a, b \in S$ there is a polygonal path with $z_0 = a$, $z_m = b$ and all segments (5.4) contained in S . A nonempty connected open subset of $\mathbb{C}$ is called a *region*.

Remark 5.9. For *open* subsets of $\mathbb{C}$ the polygonal criterion of Definition 5.8 is equivalent to the general topological notion of connectedness (no partition into two disjoint nonempty relatively open pieces). We take the polygonal version as the definition because it is the one used in proofs such as Theorem 14.13, where one integrates or estimates along a path.

Here is the proof of the equivalence. Let $S \subseteq \mathbb{C}$ be open; for an open S , “relatively open in S ” just means “open”.

Polygonally connected $\Rightarrow$ no partition. Suppose $S = U \cup V$ with U, V open, disjoint and nonempty; pick $a \in U$, $b \in V$ and a polygonal path in S with vertices $a = z_0, \dots, z_m = b$. Along the path some segment $[z_{j-1}, z_j]$ has $z_{j-1} \in U$ and $z_j \in V$. Put $\gamma(t) := (1-t)z_{j-1} + tz_j$ for $t \in [0, 1]$ and $t^* := \sup\{t \in [0, 1] : \gamma(t) \in U\}$. If $\gamma(t^*) \in U$ then $t^* < 1$ and, U being open and γ continuous, $\gamma(t) \in U$ for all t slightly larger than t^* , contradicting the definition of t^* . If $\gamma(t^*) \in V$ then $t^* > 0$ and $\gamma(t) \in V$ for all $t \in (t^* - \delta, t^*]$ for some $\delta > 0$, so no t in that interval has $\gamma(t) \in U$, and t^* is not the supremum. Both cases are impossible, since $\gamma(t^*) \in S = U \cup V$.

No partition $\Rightarrow$ polygonally connected. Fix $a \in S$, let U be the set of points of S that can be joined to a by a polygonal path in S , and $V := S \setminus U$. If $b \in S$ and $D(b, \varepsilon) \subseteq S$, then every $c \in D(b, \varepsilon)$ is joined to b by the segment $[b, c] \subseteq D(b, \varepsilon)$ (a disc is convex). Hence if $b \in U$ then $D(b, \varepsilon) \subseteq U$ (append $[b, c]$ to a path from a to b), and if $b \in V$ then $D(b, \varepsilon) \subseteq V$ (a path from a to some c in the disc followed by $[c, b]$ would put b in U). So U and V are open, disjoint, and $a \in U$; as S admits no partition, $V = \emptyset$, i.e. every point of S is joined to a , and therefore any two points of S are joined through a .

Remark 5.10 (Terminology). In these notes the word *domain* is used only for the domain of definition of a function (Definition 7.1); an open connected set is always called a *region*. Books differ here. Ahlfors and Stein–Shakarchi use *region* exactly as above, while Brown–Churchill call a nonempty connected open set a *domain* and reserve *region* for a domain together with none, some, or all of its boundary points.

Example 5.11. $D(0, 1)$, the half-plane $\{\operatorname{Re} z > 0\}$, the annulus $\{1 < |z| < 2\}$, and the slit plane $\mathbb{C} \setminus (-\infty, 0]$ are all regions. The set $\{\operatorname{Im} z \neq 0\}$ is open but not connected (any polygonal path from i to $-i$ must cross the real axis), so it is *not* a region; it is a disjoint union of two regions.

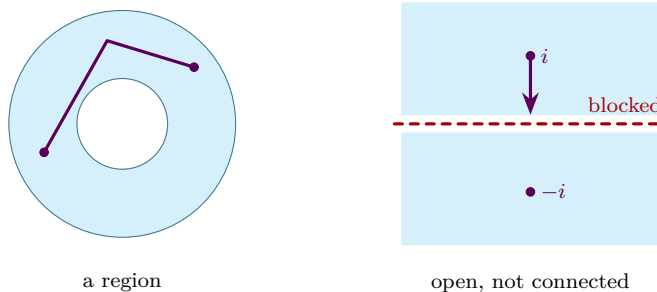


FIGURE 5.2. Connectedness in the sense of Definition 5.8. Left: any two points of the annulus are joined by a polygonal path staying inside it, so the annulus is a region even though it has a hole. Right: the set $\{\operatorname{Im} z \neq 0\}$ is open but *not* connected — every polygonal path from i to $-i$ must cross the forbidden real axis, so it is a disjoint union of two regions rather than one.

Definition 5.12 (Bounded, compact). A set $S \subseteq \mathbb{C}$ is *bounded* if there exists a real $R > 0$ with $S \subseteq D(0, R)$, i.e. $|z| < R$ for all $z \in S$; otherwise S is *unbounded*. A set that is both closed and bounded is called *compact*.

Theorem 5.13 (Heine–Borel / Bolzano–Weierstrass in $\mathbb{C}$). *Let $S \subseteq \mathbb{C}$. The following are equivalent:*

- (i) S is closed and bounded;
- (ii) every sequence $(z_n)_{n \in \mathbb{N}}$ in S has a subsequence converging to a point of S .

Proof. This is the corresponding theorem for $\mathbb{R}^2$, transported by the identification of Definition 2.1: by Proposition 2.2 the metric $|z - w|$ on $\mathbb{C}$ is the Euclidean metric on $\mathbb{R}^2$, so “closed”,

“bounded”, and “convergent” mean the same in both languages. We record the sequential form because it is what we use in Theorem 12.13. For completeness we deduce it from the one fact we quote from real analysis, the Bolzano–Weierstrass theorem on $\mathbb{R}$: every bounded real sequence has a convergent subsequence. Here $z_n \rightarrow z$ means $|z_n - z| \rightarrow 0$, which by $\max(|\operatorname{Re} w|, |\operatorname{Im} w|) \leq |w| \leq |\operatorname{Re} w| + |\operatorname{Im} w|$ holds if and only if $\operatorname{Re} z_n \rightarrow \operatorname{Re} z$ and $\operatorname{Im} z_n \rightarrow \operatorname{Im} z$.

(i) $\Rightarrow$ (ii). Let (z_n) lie in $S \subseteq D(0, R)$. The real sequence $(\operatorname{Re} z_n)$ is bounded by R , so it has a convergent subsequence $(\operatorname{Re} z_{n_j})$; the real sequence $(\operatorname{Im} z_{n_j})$ is bounded too, so it has a convergent subsequence. Along this sub-subsequence both real and imaginary parts converge, so it converges to some $z \in \mathbb{C}$. If $z \notin S$, then z lies in the open set $\mathbb{C} \setminus S$, so some disc $D(z, \varepsilon)$ misses S ; but the terms of the sub-subsequence lie in S and eventually in $D(z, \varepsilon)$, a contradiction. So $z \in S$.

(ii) $\Rightarrow$ (i). If S is unbounded, choose $z_n \in S$ with $|z_n| \geq n$; no subsequence converges, since a convergent sequence is bounded. If S is not closed, $\mathbb{C} \setminus S$ is not open, so there is $b \notin S$ such that every disc $D(b, 1/n)$ meets S ; choose $z_n \in S \cap D(b, 1/n)$. Then $z_n \rightarrow b$, so every subsequence converges to $b \notin S$ (limits are unique), and (ii) fails. $\square$

Definition 5.14 (Accumulation point). A point $z_0 \in \mathbb{C}$ is an *accumulation point* (or *limit point*) of $S \subseteq \mathbb{C}$ if every punctured neighbourhood $D'(z_0, \varepsilon)$ of (5.2) meets S . This is exactly the hypothesis under which the limit $\lim_{z \rightarrow z_0} f(z)$ is meaningful in Definition 12.1: it guarantees there are points of the domain of f arbitrarily close to z_0 other than z_0 itself.

5.3 Reading a set from an inequality

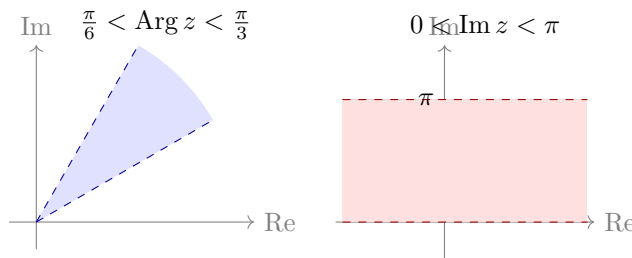
Most sets in this course are given by one or two inequalities involving $|\cdot|$, Re , Im , or Arg . The recipe is: rewrite until the condition is one of the four standard forms

$$|z - z_0| \leq \rho, \quad \operatorname{Re}(\gamma z) \leq c, \quad \operatorname{Arg} z \in (\alpha, \beta), \quad |z - a| \leq k|z - b|,$$

with $\rho, k > 0$, $c \in \mathbb{R}$, $\gamma \in \mathbb{C} \setminus \{0\}$ and $-\pi \leq \alpha < \beta \leq \pi$. These are, respectively: a disc or the exterior of a closed disc; a half-plane; a sector; and, for $k \neq 1$, an Apollonius disc or the exterior of a closed Apollonius disc (for $k = 1$ a half-plane, by Proposition 2.5). For instance $|z - a| < k|z - b|$ with $k > 1$ is the exterior of a closed disc: it contains all z of large modulus, since $|z - a|/|z - b| \rightarrow 1 < k$ as $|z| \rightarrow \infty$.

Example 5.15. The set $S = \{z \neq 0 : \frac{\pi}{6} < \operatorname{Arg} z < \frac{\pi}{3}\}$ is the open sector between the rays of angle $\pi/6$ and $\pi/3$, excluding the origin. It is open (each of its points has a small disc inside the sector) and connected, hence a region; it is unbounded; its boundary consists of the two closed rays together with 0.

Example 5.16. The set $T = \{z : 0 < \operatorname{Im} z < \pi\}$ is a horizontal open strip. It is a region, unbounded, with boundary the two lines $\operatorname{Im} z = 0$ and $\operatorname{Im} z = \pi$. It is half of the fundamental strip of the complex exponential: $\exp$ maps T one-to-one onto the upper half-plane $\{\operatorname{Im} w > 0\}$ (by Theorem 16.3(iii),(iv), which also describes the whole fundamental strip $\{-\pi < \operatorname{Im} z \leq \pi\}$).



5.4 Exercises

Exercise 5.1. For each set, determine $\text{int } S$, ∂S , $\overline{S}$; say whether S is open, closed, bounded, and, when S is open, whether it is connected (Definition 5.8) and a region:

$$|z - 1 + i| \leq 3, \quad \text{Re}(z^2) > 0, \quad 0 < |z| < 1, \quad \{1/n : n \in \mathbb{N}\}, \quad |\text{Im } z| \geq 1.$$

Exercise 5.2. Prove that an arbitrary union of open sets is open and that a finite intersection of open sets is open. Give an example showing that an infinite intersection of open sets need not be open.

Exercise 5.3. Prove $\overline{S}$ is closed, and that $\overline{S}$ is the smallest closed set containing S .

Exercise 5.4. (★) Let D be a region and let $f : D \rightarrow \mathbb{C}$ be continuous with $f(D) \subseteq \{0, 1\}$. Prove f is constant. (*This is the mechanism behind Theorem 14.13: connectedness converts a local statement into a global one.*)

Exercise 5.5. Show that the sector $\{z \neq 0 : \alpha < \text{Arg } z < \beta\}$ with $-\pi < \alpha < \beta \leq \pi$ is a region, and that it is convex if and only if $\beta - \alpha \leq \pi$.

Exercise 5.6. Determine the set of accumulation points (Definition 5.14) of each of

$$\{1/n : n \in \mathbb{N}\}, \quad \{i^n/n : n \in \mathbb{N}\}, \quad \{m + in : m, n \in \mathbb{Z}\}, \quad \{e^{in} : n \in \mathbb{N}\}.$$

Which of these sets are closed?

Exercise 5.7. Prove that a nonempty set $S \subseteq \mathbb{C}$ is both open and closed if and only if $S = \mathbb{C}$. (*This is the connectedness of $\mathbb{C}$, and it is the engine of Theorem 24.8 and Theorem 24.11.*)

Exercise 5.8. (★) Let K be compact and F closed with $K \cap F = \emptyset$. Prove that $\text{dist}(K, F) := \inf\{|z - w| : z \in K, w \in F\}$ is *strictly* positive and is attained. Give an example of two disjoint *closed* sets whose distance is 0, showing compactness cannot be dropped.

Chapter 6

Applications: Quadratics and Differential Equations

Complex numbers were forced on mathematics not by quadratics — one can simply declare $x^2+1=0$ unsolvable — but by *cubics*, where the real roots of a real cubic are produced by Cardano's formula only through detours into $\mathbb{C}$. This chapter gives two payoffs available immediately: a complete theory of the quadratic, and a clean derivation of the real solutions of linear constant-coefficient ODEs.

6.1 Quadratic equations over $\mathbb{C}$

Lemma 6.1 (Square roots always exist). *Every $w \in \mathbb{C}$ has exactly two square roots if $w \neq 0$, and exactly one (0) if $w = 0$. Explicitly, if $w = a + bi$ with $a, b \in \mathbb{R}$ and $b \neq 0$, the two square roots are $\pm s$ where*

$$s = \sqrt{\frac{|w|+a}{2}} + i \operatorname{sgn}(b) \sqrt{\frac{|w|-a}{2}}, \quad \operatorname{sgn}(b) := \frac{b}{|b|}. \quad (6.1)$$

Proof. Existence and the count are Theorem 4.5 with $n = 2$. For the formula, seek $s = x + iy$ with $x, y \in \mathbb{R}$ and $s^2 = w$. Comparing real and imaginary parts in $s^2 = (x^2 - y^2) + 2ixy$ gives

$$x^2 - y^2 = a, \quad 2xy = b. \quad (6.2)$$

Taking moduli in $s^2 = w$ and using (1.24) gives $x^2 + y^2 = |s|^2 = |w|$. Adding and subtracting this to the first equation of (6.2),

$$x^2 = \frac{|w|+a}{2}, \quad y^2 = \frac{|w|-a}{2},$$

both right-hand sides being ≥ 0 because $|w| \geq |a|$ by (1.26). Thus $x = \pm\sqrt{(|w|+a)/2}$ and $y = \pm\sqrt{(|w|-a)/2}$, and the second equation of (6.2) says xy has the sign of b , which selects the two sign combinations $(+, +\operatorname{sgn} b)$ and $(-, -\operatorname{sgn} b)$, i.e. $\pm s$ with s as in (6.1). $\square$

Theorem 6.2 (Quadratic formula in $\mathbb{C}$). *Let $a, b, c \in \mathbb{C}$ with $a \neq 0$. The equation*

$$az^2 + bz + c = 0 \quad (6.3)$$

has exactly the solutions

$$z = \frac{-b + s}{2a}, \quad z = \frac{-b - s}{2a}, \quad (6.4)$$

where s is either square root of the discriminant $\Delta := b^2 - 4ac \in \mathbb{C}$. These two solutions coincide if and only if $\Delta = 0$.

Proof. Since $a \neq 0$ and $\mathbb{C}$ is a field (Theorem 1.2), we may complete the square:

$$az^2 + bz + c = a \left[\left(z + \frac{b}{2a} \right)^2 - \frac{b^2 - 4ac}{4a^2} \right].$$

Hence (6.3) holds iff $\left(z + \frac{b}{2a} \right)^2 = \frac{\Delta}{4a^2}$. By Lemma 6.1 the square roots of $\Delta/(4a^2)$ are exactly $\pm s/(2a)$ where $s^2 = \Delta$, and solving for z gives (6.4). The two coincide iff $s = -s$, i.e. $s = 0$, i.e. $\Delta = 0$. $\square$

Remark 6.3. In (6.4) the symbol s is a *chosen* square root, not “the” square root: over $\mathbb{C}$ there is no order to pick a “positive” one. The familiar notation $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ is still correct provided $\sqrt{}$ is read as “either square root”, as Warning 1.16 demands.

Example 6.4. Solve $z^2 - (3+i)z + (2+2i) = 0$. Here $\Delta = (3+i)^2 - 4(2+2i) = (9-1+6i) - (8+8i) = -2i$. To find a square root of $-2i$ use (6.1) with $a = 0$, $b = -2$, $|w| = 2$:

$$s = \sqrt{\frac{2+0}{2}} + i(-1)\sqrt{\frac{2-0}{2}} = 1 - i.$$

Check: $(1-i)^2 = 1 - 2i + i^2 = -2i$. Hence

$$z = \frac{(3+i) \pm (1-i)}{2} \in \{2, 1+i\}.$$

Verification: $2^2 - (3+i)2 + (2+2i) = 4 - 6 - 2i + 2 + 2i = 0$, and $(1+i)^2 - (3+i)(1+i) + (2+2i) = 2i - (2+4i) + (2+2i) = 0$.

Theorem 6.5 (Fundamental Theorem of Algebra; stated). *Every polynomial $p(z) = a_n z^n + \cdots + a_1 z + a_0$ with $n \geq 1$ and $a_0, \dots, a_n \in \mathbb{C}$, $a_n \neq 0$, has at least one root in $\mathbb{C}$; consequently it factors completely,*

$$p(z) = a_n \prod_{j=1}^n (z - \zeta_j), \quad \zeta_j \in \mathbb{C}. \quad (6.5)$$

We do not prove Theorem 6.5 here. Every known proof needs a genuinely analytic (or topological) ingredient; in Chapter 22 (Theorem 22.4) it drops out of Liouville’s theorem in two lines. It is stated now because it explains the difference between $\mathbb{R}$ and $\mathbb{C}$: passing to $\mathbb{C}$ is done *once*, and no further extension is ever needed.

6.2 Linear ODEs with constant coefficients

Complex exponentials turn linear constant-coefficient differential equations into algebra. Throughout this section x denotes a real variable and y a function of x ; we allow y to take complex values and then extract real solutions at the end.

Definition 6.6 (Complex-valued function of a real variable). Let $I \subseteq \mathbb{R}$ be an interval. A function $y : I \rightarrow \mathbb{C}$ is written $y(x) = u(x) + iv(x)$ with $u = \operatorname{Re} y$ and $v = \operatorname{Im} y$ real-valued. We say y is differentiable at x if u and v are, and set

$$y'(x) := u'(x) + i v'(x). \quad (6.6)$$

The usual sum, product and quotient rules hold, with the same proofs, by separating real and imaginary parts.

Lemma 6.7 (Derivative of $e^{\lambda x}$). Let $\lambda = \alpha + i\beta \in \mathbb{C}$ with $\alpha, \beta \in \mathbb{R}$ and define $E_\lambda : \mathbb{R} \rightarrow \mathbb{C}$ by

$$E_\lambda(x) := e^{\alpha x}(\cos \beta x + i \sin \beta x) = e^{\alpha x} e^{i\beta x}, \quad (6.7)$$

the second expression using Definition 3.8. Then E_λ is differentiable on $\mathbb{R}$ and

$$E'_\lambda(x) = \lambda E_\lambda(x) \quad \text{for all } x \in \mathbb{R}. \quad (6.8)$$

Moreover $E_\lambda(x)E_\mu(x) = E_{\lambda+\mu}(x)$ and $E_\lambda(x) \neq 0$ for every x .

Proof. Differentiate real and imaginary parts of (6.7):

$$\begin{aligned} \operatorname{Re} E_\lambda &= e^{\alpha x} \cos \beta x, & (\operatorname{Re} E_\lambda)' &= \alpha e^{\alpha x} \cos \beta x - \beta e^{\alpha x} \sin \beta x, \\ \operatorname{Im} E_\lambda &= e^{\alpha x} \sin \beta x, & (\operatorname{Im} E_\lambda)' &= \alpha e^{\alpha x} \sin \beta x + \beta e^{\alpha x} \cos \beta x. \end{aligned}$$

On the other hand, by (1.11),

$$\lambda E_\lambda(x) = (\alpha + i\beta)e^{\alpha x}(\cos \beta x + i \sin \beta x) = e^{\alpha x}[(\alpha \cos \beta x - \beta \sin \beta x) + i(\alpha \sin \beta x + \beta \cos \beta x)],$$

which matches. The multiplicative property follows from $e^{\alpha_1 x} e^{\alpha_2 x} = e^{(\alpha_1 + \alpha_2)x}$ (real exponentials) together with (3.8). Non-vanishing follows from $|E_\lambda(x)| = e^{\alpha x} > 0$ and (1.22). $\square$

Proposition 6.8 (Real and imaginary parts of a solution). Let $a_0, \dots, a_m \in \mathbb{R}$ and let L be the differential operator

$$L[y] := a_m y^{(m)} + a_{m-1} y^{(m-1)} + \dots + a_1 y' + a_0 y. \quad (6.9)$$

If $y : I \rightarrow \mathbb{C}$ satisfies $L[y] = 0$ on I , then the real-valued functions $u = \operatorname{Re} y$ and $v = \operatorname{Im} y$ each satisfy $L[u] = 0$ and $L[v] = 0$ on I .

Proof. By (6.6) and induction, $y^{(k)} = u^{(k)} + i v^{(k)}$ for each k . Since the a_j are real,

$$0 = L[y] = \sum_j a_j (u^{(j)} + i v^{(j)}) = \underbrace{\sum_j a_j u^{(j)}}_{\text{real}} + i \underbrace{\sum_j a_j v^{(j)}}_{\text{real}} = L[u] + i L[v].$$

A complex number vanishes iff its real and imaginary parts vanish (1.5), so $L[u] = L[v] = 0$. $\square$

Proposition 6.8 fails if the a_j are non-real; the reality of the coefficients is exactly what makes the “take real parts” trick legal.

Theorem 6.9 (Second-order equation, complex roots). *Let $p, q \in \mathbb{R}$ and consider*

$$y'' + p y' + q y = 0 \quad \text{on } \mathbb{R}. \quad (6.10)$$

Let $\lambda_1, \lambda_2 \in \mathbb{C}$ be the roots of the characteristic equation

$$\lambda^2 + p\lambda + q = 0, \quad (6.11)$$

which exist by Theorem 6.2. Then:

- (i) E_{λ_1} and E_{λ_2} of (6.7) solve (6.10);
- (ii) if $p^2 - 4q < 0$, so that $\lambda_{1,2} = \alpha \pm i\beta$ with $\alpha = -p/2$ and $\beta = \frac{1}{2}\sqrt{4q - p^2} > 0$, then the general real solution of (6.10) is

$$y(x) = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x), \quad c_1, c_2 \in \mathbb{R}, \quad (6.12)$$

equivalently $y(x) = A e^{\alpha x} \cos(\beta x - \phi)$ with amplitude $A \geq 0$ and phase $\phi \in (-\pi, \pi]$.

Proof. (i) By Lemma 6.7, $E'_\lambda = \lambda E_\lambda$ and hence $E''_\lambda = \lambda^2 E_\lambda$. Therefore

$$E''_\lambda + p E'_\lambda + q E_\lambda = (\lambda^2 + p\lambda + q) E_\lambda,$$

which vanishes identically precisely when λ solves (6.11), because E_λ never vanishes.

(ii) Let $\Delta = p^2 - 4q < 0$. Since $(i\sqrt{-\Delta})^2 = \Delta$, where $\sqrt{-\Delta} > 0$ is the real square root of the positive number $-\Delta$, we may take $s = i\sqrt{-\Delta}$ in Theorem 6.2, so $\lambda_{1,2} = \frac{-p \pm i\sqrt{4q - p^2}}{2} = \alpha \pm i\beta$. By part (i) the complex function $E_{\lambda_1}(x) = e^{\alpha x} (\cos \beta x + i \sin \beta x)$ solves (6.10), so by Proposition 6.8 both

$$y_1(x) = e^{\alpha x} \cos \beta x, \quad y_2(x) = e^{\alpha x} \sin \beta x$$

are real solutions. They are linearly independent over $\mathbb{R}$: their Wronskian is

$$W(x) = y_1 y'_2 - y'_1 y_2 = e^{2\alpha x} [\cos \beta x (\alpha \sin \beta x + \beta \cos \beta x) - (\alpha \cos \beta x - \beta \sin \beta x) \sin \beta x] = \beta e^{2\alpha x},$$

which is nonzero since $\beta > 0$. By the standard existence and uniqueness theorem for second-order linear ODEs (which we quote from the theory of ordinary differential equations), the real solution space of (6.10) is two-dimensional, so $\{y_1, y_2\}$ is a basis and (6.12) is the general real solution. For the amplitude-phase form, set $A := \sqrt{c_1^2 + c_2^2}$ and let $\phi := \text{Arg}(c_1 + i c_2) \in (-\pi, \pi]$ if $(c_1, c_2) \neq (0, 0)$, and $\phi := 0$ (with $A = 0$) otherwise; in both cases $c_1 = A \cos \phi$ and $c_2 = A \sin \phi$, so $c_1 \cos \beta x + c_2 \sin \beta x = A(\cos \phi \cos \beta x + \sin \phi \sin \beta x) = A \cos(\beta x - \phi)$. $\square$

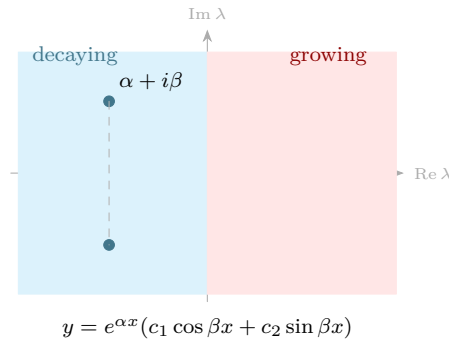


FIGURE 6.1. The characteristic root $\lambda = \alpha + i\beta$ of (6.11) packs the entire qualitative behaviour of the solution (6.12) into one point of the plane (the other root is its mirror image $\alpha - i\beta$, also drawn). Its real part is the exponential growth rate (negative means decay), its imaginary part the angular frequency. Left half-plane: decay. Right half-plane: growth. On the imaginary axis: undamped oscillation. This is why engineers speak of “poles in the left half-plane” (the roots are the poles of the system’s transfer function; poles are defined in Chapter 26).

Example 6.10 (Damped oscillator). Solve $y'' + 2y' + 5y = 0$ with $y(0) = 1$, $y'(0) = 0$. The characteristic equation $\lambda^2 + 2\lambda + 5 = 0$ has discriminant $4 - 20 = -16$, so $\lambda = \frac{-2 \pm 4i}{2} = -1 \pm 2i$, giving $\alpha = -1$, $\beta = 2$. By (6.12), $y(x) = e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$. From $y(0) = c_1 = 1$ and

$$y'(x) = e^{-x}[(-c_1 + 2c_2) \cos 2x + (-c_2 - 2c_1) \sin 2x], \quad y'(0) = -c_1 + 2c_2 = 0,$$

we get $c_2 = \frac{1}{2}$. Hence $y(x) = e^{-x}(\cos 2x + \frac{1}{2} \sin 2x) = \frac{\sqrt{5}}{2}e^{-x} \cos(2x - \phi)$ with $\phi = \arctan(1/2)$. The complex root $-1 + 2i$ encodes both the decay rate ($\operatorname{Re} \lambda = -1$) and the oscillation frequency ($\operatorname{Im} \lambda = 2$) in one number; this is the reason engineers speak of poles in the left half-plane.

Example 6.11 (Forced oscillation by phasors). Solve $y'' + \omega_0^2 y = F \cos \omega x$ with $\omega \neq \pm \omega_0$ and $F, \omega_0, \omega \in \mathbb{R}$, $\omega_0 > 0$. Consider the *complexified* equation

$$Y'' + \omega_0^2 Y = F e^{i\omega x}, \quad (6.13)$$

whose right-hand side has real part $F \cos \omega x$. Try $Y(x) = C e^{i\omega x}$ with $C \in \mathbb{C}$ constant. By Lemma 6.7, $Y'' = -\omega^2 C e^{i\omega x}$, so (6.13) becomes $C(\omega_0^2 - \omega^2) e^{i\omega x} = F e^{i\omega x}$, whence

$$C = \frac{F}{\omega_0^2 - \omega^2}. \quad (6.14)$$

Taking real parts, and using that $\operatorname{Re} L[Y] = L[\operatorname{Re} Y]$ because $L[y] = y'' + \omega_0^2 y$ has real coefficients (the computation in the proof of Proposition 6.8), we get $L[\operatorname{Re} Y] = \operatorname{Re}(F e^{i\omega x}) = F \cos \omega x$; this gives the real particular solution $y_p(x) = \frac{F}{\omega_0^2 - \omega^2} \cos \omega x$. The whole method — replace $\cos$ by $e^{i\omega x}$, do algebra instead of trigonometry, take real parts — is the phasor calculus of electrical engineering. With damping, $y'' + 2\gamma y' + \omega_0^2 y = F \cos \omega x$, the same computation gives the complex amplitude

$$C = \frac{F}{\omega_0^2 - \omega^2 + 2i\gamma\omega}, \quad (6.15)$$

and now $|C|$ is the gain and $-\operatorname{Arg} C$ the phase lag — two real quantities read off from one complex number by Definitions 1.12 and 3.3.

6.3 Exercises

Exercise 6.1. Solve over $\mathbb{C}$: $z^2 + z + 1 = 0$, $z^2 - 2iz - 2 = 0$, $z^2 - (1+i)z + i = 0$, $2z^2 + (3-i)z - 2 = 0$.

Exercise 6.2. Find a square root of $-5 + 12i$ using (6.1), and verify it.

Exercise 6.3. Solve the initial value problems and express the answer in amplitude–phase form:

$$(a) \ y'' + 4y = 0, \ y(0) = 0, \ y'(0) = 3; \quad (b) \ y'' + y' + y = 0, \ y(0) = 2, \ y'(0) = -1.$$

Exercise 6.4. (*) Using (6.15), show that for fixed $F, \gamma > 0$ the gain $|C|$ is maximised at $\omega = \sqrt{\omega_0^2 - 2\gamma^2}$ when $\omega_0^2 > 2\gamma^2$ (resonance), and is decreasing in ω otherwise.

Exercise 6.5. Show that if $\lambda_1 = \lambda_2 = \lambda$ (repeated root) then $x E_\lambda(x)$ also solves (6.10). (Differentiate $\lambda \mapsto E_\lambda(x)$, or substitute directly.)

Exercise 6.6. Show that if $p, q \in \mathbb{R}$ and the roots of (6.11) are $\alpha \pm i\beta$ with $\beta > 0$, then $\alpha = -p/2$ and $\alpha^2 + \beta^2 = q$. Deduce that the solution (6.12) decays for every initial condition if and only if $p > 0$ and $q > 0$.

Exercise 6.7. Use the phasor method of Example 6.11 to solve $y'' + 3y' + 2y = 5 \cos 2x$, giving the steady-state solution in amplitude-phase form. Identify $|C|$ and $\text{Arg } C$ from (6.15).

Exercise 6.8. ($\star$) (*Cardano and the birth of $\mathbb{C}$.*) The cubic $x^3 = 15x + 4$ has the obvious real root $x = 4$. Cardano's formula gives $x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}$, where $\sqrt{-121}$ is read as $11i$ and the two cube roots u of $2 + 11i$ and v of $2 - 11i$ must be chosen with $uv = 5$ (one third of the coefficient 15). Verify that $(2 + i)^3 = 2 + 11i$ and $(2 - i)^3 = 2 - 11i$, and hence that the choice $u = 2 + i$, $v = 2 - i$ makes the formula produce 4 — but only by passing through numbers that are not real. Using Theorem 4.5, find the other two admissible pairs (u, v) and the values $u + v$ they give. This is the historical reason complex numbers had to be taken seriously.

Exercise 6.9. An RLC series circuit satisfies $L\ddot{q} + R\dot{q} + \frac{1}{C}q = V_0 \cos \omega t$. Write the steady-state current $\dot{q}$ using a complex impedance $Z = R + i(\omega L - \frac{1}{\omega C})$, and identify $|Z|$ and $\text{Arg } Z$ physically. (Here $L, R, C > 0$ are the physical constants of the circuit, not the operator L of (6.9) or the amplitude C of (6.14).)

Part II

Complex Functions as Mappings

Chapter 7

Complex Functions as Mappings

7.1 What a complex function is, and why we cannot graph it

Definition 7.1 (Complex function). Let $S \subseteq \mathbb{C}$. A *complex function on S* is a rule f assigning to each $z \in S$ exactly one $w = f(z) \in \mathbb{C}$. The set S is the *domain of definition* of f (in these notes the word *domain* is used only in this sense; an open connected set is a *region*, Definition 5.8), and

$$f(S) := \{ f(z) : z \in S \} \quad (7.1)$$

is the *image* (or *range*) of f .

Definition 7.2 (Real and imaginary components). Given $f : S \rightarrow \mathbb{C}$, define real-valued functions u, v of two real variables by

$$u(x, y) := \operatorname{Re} f(x + iy), \quad v(x, y) := \operatorname{Im} f(x + iy), \quad (7.2)$$

for all $x, y \in \mathbb{R}$ with $x + iy \in S$. Then

$$f(z) = u(x, y) + i v(x, y), \quad z = x + iy. \quad (7.3)$$

We call u and v the *component functions* of f . Sometimes it is more convenient to use polar coordinates and write $f(z) = u(r, \theta) + i v(r, \theta)$ where $z = r e^{i\theta}$; the meaning is always clear from the names of the variables.

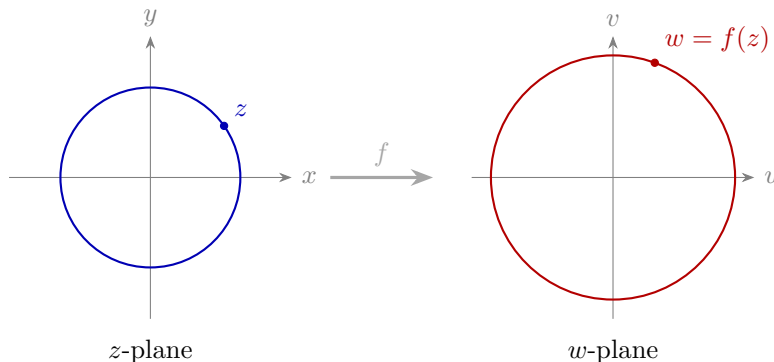
Example 7.3. For $f(z) = z^2$ we have $f(x + iy) = (x^2 - y^2) + i(2xy)$, so

$$u(x, y) = x^2 - y^2, \quad v(x, y) = 2xy. \quad (7.4)$$

For $f(z) = \bar{z}$ we have $u = x, v = -y$. For $f(z) = 1/z$ ($z \neq 0$), using (1.20),

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{x}{x^2 + y^2} + i \frac{-y}{x^2 + y^2}. \quad (7.5)$$

A real function $g : \mathbb{R} \rightarrow \mathbb{R}$ is visualised by its graph, a curve in $\mathbb{R}^2$. The graph of $f : S \rightarrow \mathbb{C}$ lives in $\mathbb{C} \times \mathbb{C} \cong \mathbb{R}^4$ and cannot be drawn. The standard replacement is the *two-plane picture*: draw the domain in one copy of $\mathbb{C}$ (the *z -plane*) and the image in a second copy (the *w -plane*), and study what f does to selected curves and regions.



7.2 Mapping curves, mapping regions

Definition 7.4 (Image of a curve). Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a continuous function of the real parameter t (a *parametrised curve*) with $\gamma([a, b]) \subseteq S$, and let $f : S \rightarrow \mathbb{C}$. The *image curve* is $f \circ \gamma : [a, b] \rightarrow \mathbb{C}$, $t \mapsto f(\gamma(t))$.

The practical technique for understanding a map f has three steps.

- (1) Choose a family of curves that fills the domain and is adapted to f : usually the Cartesian grid ($x = \text{const}$, $y = \text{const}$) or the polar grid ($r = \text{const}$, $\theta = \text{const}$).
- (2) Compute the image of each curve in the family, by eliminating the parameter from the pair u, v of (7.2).
- (3) Assemble: the image of a region bounded by such curves is the region bounded by their images, provided f is continuous and one-to-one on the closed region, and one checks which side is the image by testing a single interior point.

Step (3) is a heuristic, not a theorem of these notes: its proof rests on topological facts of Jordan-curve type, which we take as known and do not prove, and whenever we use it (for instance in Examples 9.5 and 11.5) the answer can be checked directly. Injectivity on the closed region cannot be dropped: z^2 is one-to-one on the open half-annulus $\{1 < |z| < 2, \text{Im } z > 0\}$ but not on its closure, and it maps it onto the annulus $\{1 < |w| < 4\}$ minus the segment $(1, 4)$, which is not “the region bounded by” the image curves.

Example 7.5 (The map $f(z) = z^2$ on the Cartesian grid). Use (7.4). On the vertical line $x = c$ ($c \neq 0$) we have $u = c^2 - y^2$ and $v = 2cy$, so $y = v/(2c)$ and

$$u = c^2 - \frac{v^2}{4c^2}, \quad (7.6)$$

a parabola opening in the $-u$ direction with vertex $(c^2, 0)$. On the horizontal line $y = d$ ($d \neq 0$), $u = x^2 - d^2$, $v = 2dx$, so

$$u = \frac{v^2}{4d^2} - d^2, \quad (7.7)$$

a parabola opening in the $+u$ direction. The two families are *confocal* (common focus 0) and meet at right angles — a first glimpse of conformality, proved in general in Chapter 13.

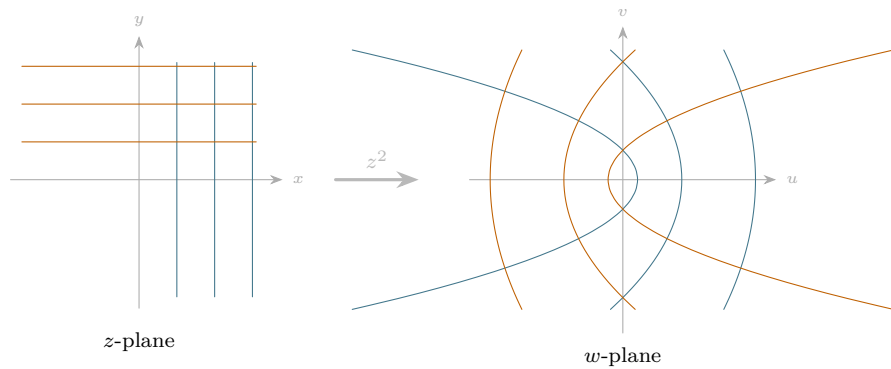


FIGURE 7.1. Example 7.5: the Cartesian grid under $w = z^2$. Vertical lines $x = c$ become the leftward parabolas (7.6); horizontal lines $y = d$ become the rightward parabolas (7.7). The two families are confocal, with common focus 0, and they meet at right angles — a fact we will explain in one line once conformality is available (Theorem 13.10).

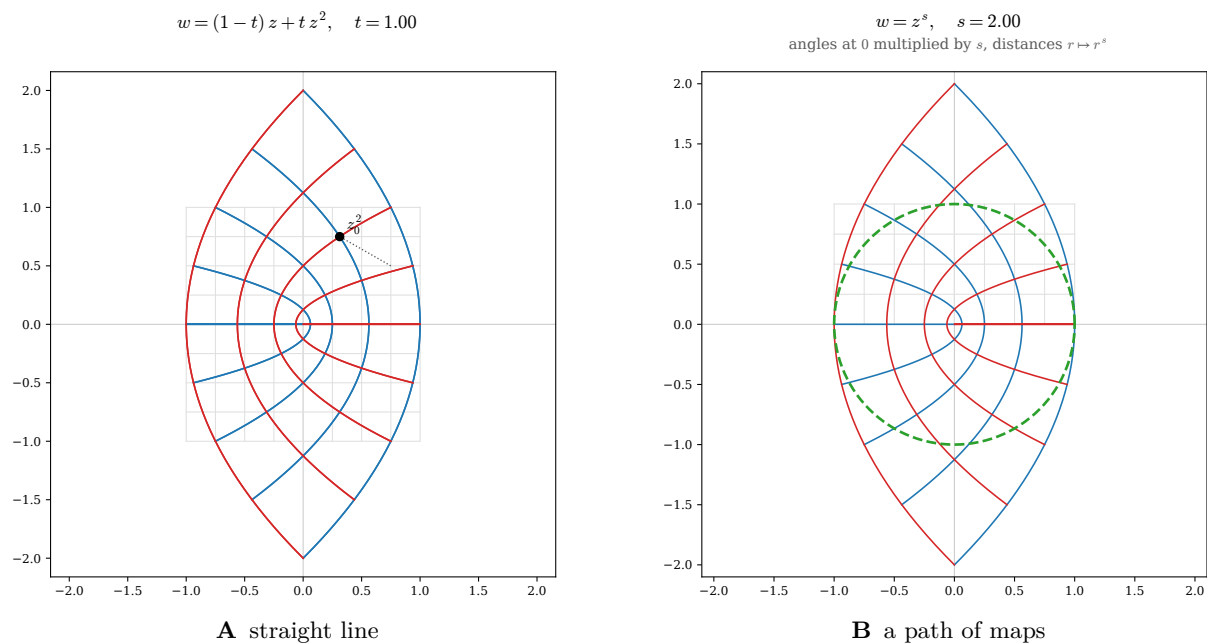
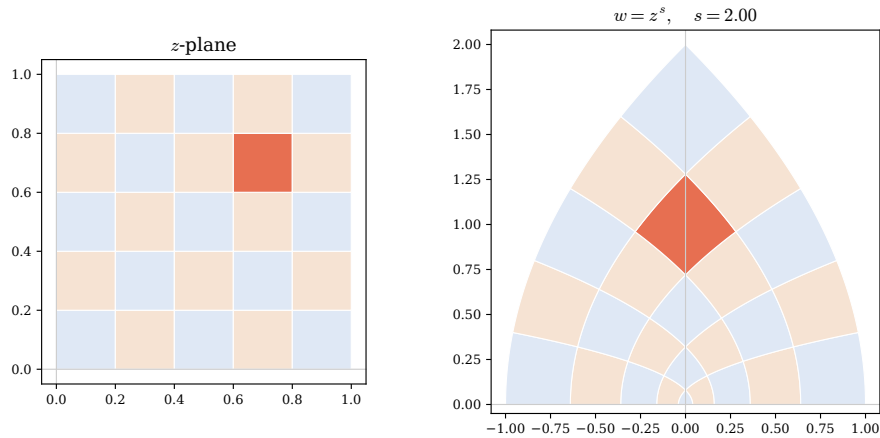


FIGURE 7.2 (animated). z^2 on the Cartesian grid. **A**: every point moves in a straight line from z to $f(z)$. **B**: the upper half of the square unfolds through z^s , $1 \leq s \leq 2$: angles at 0 are multiplied by s , the unit semicircle closes into the unit circle. **C**: the tiles on the left deform along the path of B.

small squares stay square; the quarter-plane opens to a half-plane (angles at 0 double)



C tiles (Figure 7.2, continued)

Example 7.6 (The same map on the polar grid). With $z = re^{i\theta}$, (4.2) gives $z^2 = r^2 e^{2i\theta}$. So the circle $r = \rho$ maps onto the circle of radius ρ^2 , traversed *twice*, and the ray $\theta = \theta_0$ maps onto the ray $\theta = 2\theta_0$. In polar coordinates the map is transparent; in Cartesian coordinates it produced the parabolas of Example 7.5. Choosing coordinates adapted to the map is most of the work.

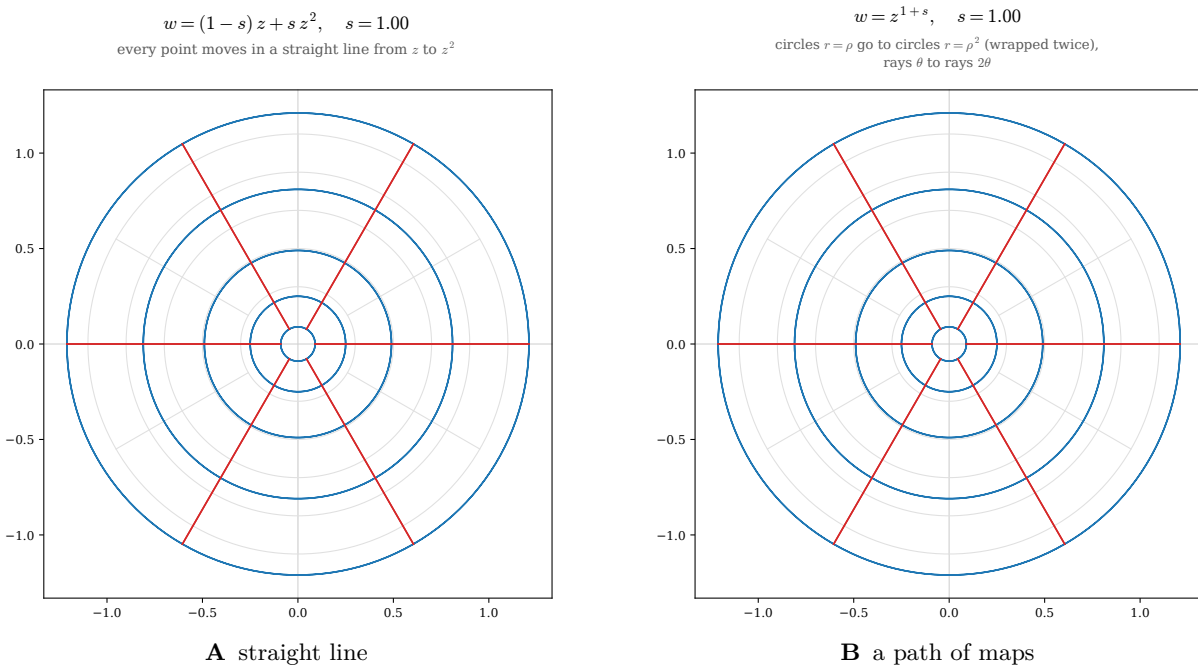
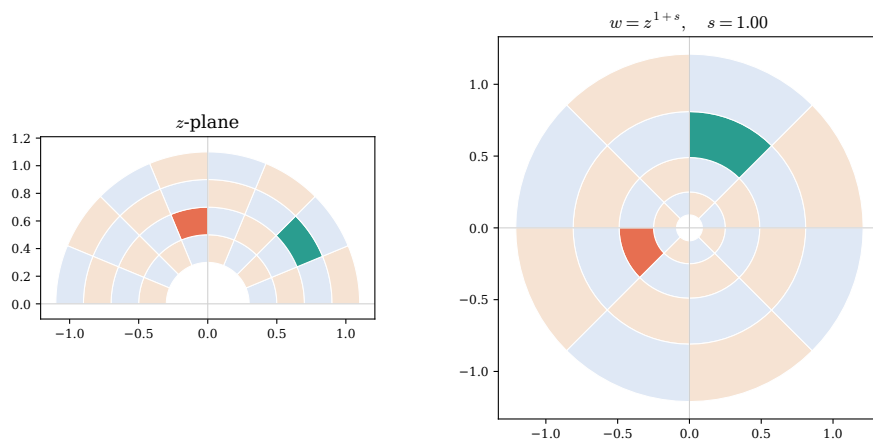


FIGURE 7.3 (animated). z^2 on the polar grid. **A:** every point moves in a straight line from z to $f(z)$. **B:** circles $r = \rho$ go to circles $r = \rho^2$ (wrapped twice), rays θ to rays 2θ . **C:** the tiles on the left deform along the path of B.

the upper half of the annulus wraps once around: angles double



C tiles (Figure 7.3, continued)

7.3 Injectivity, surjectivity, and inverse maps

Definition 7.7 (One-to-one; onto). Let $f : S \rightarrow \mathbb{C}$. We call f *one-to-one* (injective) on S if $f(z_1) = f(z_2)$ with $z_1, z_2 \in S$ implies $z_1 = z_2$, and *onto* T (surjective) if $f(S) = T$. If f is one-to-one on S then the *inverse function* $f^{-1} : f(S) \rightarrow S$ is well defined by

$$f^{-1}(w) = z \iff f(z) = w, \quad z \in S. \quad (7.8)$$

Example 7.8. $f(z) = z^2$ is not one-to-one on $\mathbb{C}$, since $f(z) = f(-z)$. It *is* one-to-one on the open upper half-plane $\{\text{Im } z > 0\}$: if $z_1^2 = z_2^2$ then $z_1 = \pm z_2$, and $z_1 = -z_2$ is impossible for two points both with positive imaginary part. More generally, by Example 7.6, z^2 is one-to-one on any open sector of opening angle at most π . This observation is the seed of the theory of branches in Chapter 10.

Remark 7.9 (Multiple-valued “functions”). Expressions such as $z^{1/2}$, $z^{1/n}$, $\arg z$, $\log z$ assign to each z a *set* of values, and are therefore *not* functions in the sense of Definition 7.1. We will call them *multiple-valued relations* and will always convert them into genuine functions by selecting a *branch*: a choice of one value at each point, made so that the result is continuous on a suitable region. Doing this well — and knowing where it is impossible — is one of the central techniques of the subject.

7.4 Exercises

Exercise 7.1. Find $u(x, y)$ and $v(x, y)$ for $f(z) = z^3$, $f(z) = \frac{z}{z+1}$, $f(z) = z\bar{z} + i \text{Re } z$, $f(z) = \frac{1}{\bar{z}}$.

Exercise 7.2. Let $f(z) = \bar{z}$. Describe the images under f of: the line $y = x$; the circle $|z - i| = 1$; the sector $0 < \text{Arg } z < \pi/4$. Is f one-to-one on $\mathbb{C}$?

Exercise 7.3. For $f(z) = z^2$, find the image of the square with vertices $0, 1, 1+i, i$. (Map each of the four edges using Example 7.5 and then locate the image of the interior point $\frac{1+i}{2}$.)

Exercise 7.4. Show that $f(z) = z + 1/z$ maps the unit circle $|z| = 1$ onto the segment $[-2, 2]$ of the real axis, two-to-one except at $z = \pm 1$. What is the image of the circle $|z| = \rho$ for $\rho > 1$? (*This is the Joukowski map.*)

Exercise 7.5. For $f(z) = z^2$, find and sketch the images of the polar grid curves $|z| = \frac{1}{2}$, $|z| = 1$, $|z| = 2$ and $\text{Arg } z = 0, \pi/4, \pi/2$. Compare with Example 7.6 and say which coordinate system makes the map easier to describe.

Exercise 7.6. Let $f(z) = \frac{1}{\bar{z}}$. Show f maps circles centred at 0 to circles centred at 0, and determine the image of the ray $\text{Arg } z = \theta_0$. Is f one-to-one on $\mathbb{C} \setminus \{0\}$? Compare with Proposition 11.2.

Exercise 7.7. Show that $f(z) = z + \bar{z}$ maps all of $\mathbb{C}$ onto the real axis, so its image has empty interior. Contrast with Theorem 31.10, and say which hypothesis of that theorem f violates.

Exercise 7.8. ($\star$) Let $f(z) = u + iv$ and suppose the image $f(S)$ lies on the line $au + bv = c$ with $a, b, c \in \mathbb{R}$ and $(a, b) \neq (0, 0)$. Show this says $\text{Re}(\bar{\alpha}f(z)) = c$ for all $z \in S$, where $\alpha = a + ib$. This reformulation is used in Theorem 14.13.

Chapter 8

Linear Mappings

8.1 The three elementary motions

Definition 8.1 (Translation, rotation, magnification). Let $b \in \mathbb{C}$, let $\alpha \in \mathbb{R}$, and let $\rho > 0$ be real. Define

$$T_b(z) := z + b \quad (\text{translation by } b), \quad (8.1)$$

$$R_\alpha(z) := e^{i\alpha} z \quad (\text{rotation through } \alpha \text{ about } 0), \quad (8.2)$$

$$M_\rho(z) := \rho z \quad (\text{magnification by factor } \rho). \quad (8.3)$$

Proposition 8.2 (Geometric description). For all $z, z_1, z_2 \in \mathbb{C}$:

- (i) $|T_b(z_1) - T_b(z_2)| = |z_1 - z_2|$: translation is an isometry, moving every point by the same vector b .
- (ii) $|R_\alpha(z)| = |z|$ and, for $z \neq 0$, $\arg R_\alpha(z) = \arg z + \alpha$: rotation preserves moduli and adds α to every argument. It is an isometry.
- (iii) $|M_\rho(z)| = \rho|z|$ and, for $z \neq 0$, $\arg M_\rho(z) = \arg z$: magnification preserves arguments and scales all distances by ρ , i.e. $|M_\rho(z_1) - M_\rho(z_2)| = \rho|z_1 - z_2|$.

Proof. (i) $T_b(z_1) - T_b(z_2) = (z_1 + b) - (z_2 + b) = z_1 - z_2$. (ii) and (iii) are Theorem 3.11(i) with $z_2 = e^{i\alpha}$ (modulus 1) and $z_2 = \rho$ (argument 0) respectively (for $z = 0$ only the statements about moduli are claimed, and they hold trivially); the isometry/similarity statements follow from $|az_1 - az_2| = |a||z_1 - z_2|$, which is (1.24). $\square$

8.2 The general linear map

Definition 8.3 (Linear mapping). Let $a, b \in \mathbb{C}$ with $a \neq 0$. The map

$$f(z) = az + b, \quad z \in \mathbb{C}, \quad (8.4)$$

is called a *(complex) linear mapping*. (It is an *affine* map in the language of linear algebra, since $f(0) = b$ need not be 0; the terminology is traditional.)

Theorem 8.4 (Decomposition). Let $f(z) = az + b$ with $a \neq 0$, and write $a = \rho e^{i\alpha}$ with $\rho = |a| > 0$ and $\alpha = \arg a$. Then

$$f = T_b \circ M_\rho \circ R_\alpha, \quad (8.5)$$

that is, f is: rotate through α , then magnify by ρ , then translate by b . Consequently:

- (i) f is a bijection of $\mathbb{C}$ onto $\mathbb{C}$, with inverse $f^{-1}(w) = \frac{w-b}{a}$, again linear;
- (ii) f is a similarity: $|f(z_1) - f(z_2)| = |a| |z_1 - z_2|$ for all z_1, z_2 ;
- (iii) f maps lines onto lines and circles onto circles; the image of $C(z_0, r)$ is $C(f(z_0), |a|r)$;
- (iv) f preserves angles, including their orientation.

Proof. For (8.5): $T_b(M_\rho(R_\alpha(z))) = \rho e^{i\alpha} z + b = az + b$.

(i) Solve $w = az + b$ for z : since $a \neq 0$ it has the unique solution $z = (w - b)/a$, which proves bijectivity and gives the inverse.

(ii) $f(z_1) - f(z_2) = a(z_1 - z_2)$; apply (1.24).

(iii) By (ii), $|z - z_0| = r$ if and only if $|f(z) - f(z_0)| = |a|r$; as f is onto, the image of the circle $C(z_0, r)$ is exactly $C(f(z_0), |a|r)$. For a line $\ell = \{z_0 + td : t \in \mathbb{R}\}$ with direction $d \neq 0$,

$$f(\ell) = \{az_0 + b + t(ad) : t \in \mathbb{R}\},$$

the line through $f(z_0)$ with direction $ad \neq 0$.

(iv) Let two curves meet at z_0 with tangent directions $d_1, d_2 \neq 0$; the (signed) angle between them is any element of $\arg(d_2/d_1)$. Under f the tangent directions become ad_1, ad_2 , and $\frac{ad_2}{ad_1} = \frac{d_2}{d_1}$, so the signed angle is unchanged. $\square$

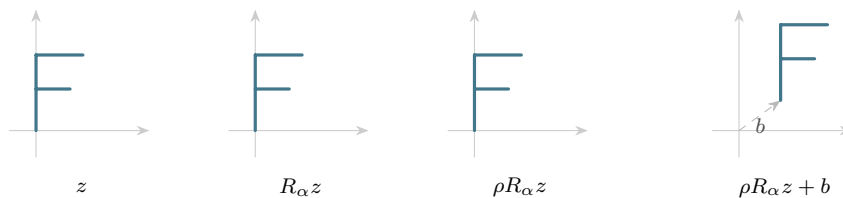


FIGURE 8.1. The decomposition (8.5) of $f(z) = az + b$ into rotate, magnify, translate, applied to a letter “F” so that orientation is visible. Nothing about the shape changes except its size, its tilt, and its position: that is what Theorem 8.4 means by *similarity*. Note the “F” is never mirrored — linear maps preserve orientation.

Example 8.5 (Reading off a linear map). Find the linear map taking $z = 0 \mapsto 1 + i$ and $z = 1 \mapsto 3 + 3i$. Writing $f(z) = az + b$: from $f(0) = b = 1 + i$ and $f(1) = a + b = 3 + 3i$ we get $a = 2 + 2i = 2\sqrt{2}e^{i\pi/4}$. So f rotates by 45° , magnifies by $2\sqrt{2}$, then translates by $1 + i$. Under f , the unit disc $|z| < 1$ maps onto the disc $|w - (1 + i)| < 2\sqrt{2}$, and the real axis maps onto the line through $1 + i$ in the direction $2 + 2i$. Since $(1 + i) + t(2 + 2i) = (1 + 2t)(1 + i)$, this is the line $v = u$ itself: the translation by $1 + i$ happens to move this line along itself.

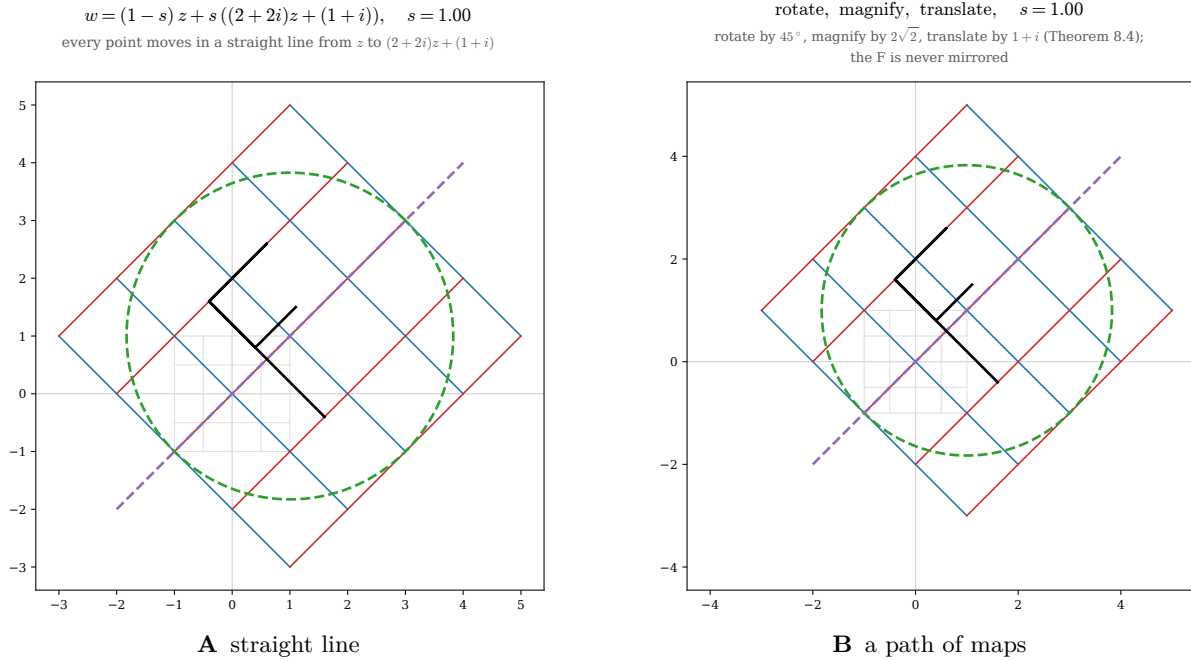
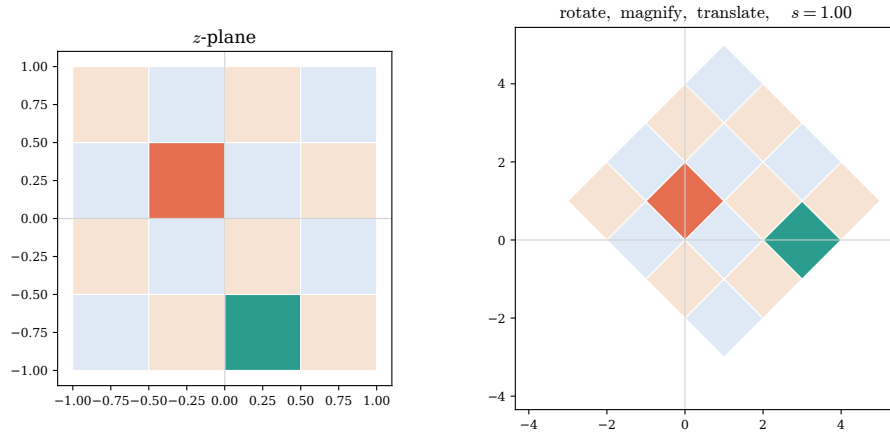


FIGURE 8.2 (animated). The linear map $(2+2i)z + (1+i)$. **A**: every point moves in a straight line from z to $f(z)$. **B**: rotate by 45° , magnify by $2\sqrt{2}$, translate by $1+i$ (Theorem 8.4); the F is never mirrored. **C**: the tiles on the left deform along the path of B.

a similarity: every tile stays a square, all scaled by $|a| = 2\sqrt{2}$



C tiles (Figure 8.2, continued)

Example 8.6 (Mapping one disc onto another). Given discs $D(z_0, r)$ and $D(w_0, R)$ with $r, R > 0$, the linear map

$$f(z) = \frac{R}{r} e^{i\alpha} (z - z_0) + w_0 \quad (8.6)$$

maps the first onto the second, for *any* choice of $\alpha \in \mathbb{R}$. The freedom in α reflects the rotational symmetry of a disc: linear maps between discs are never unique.

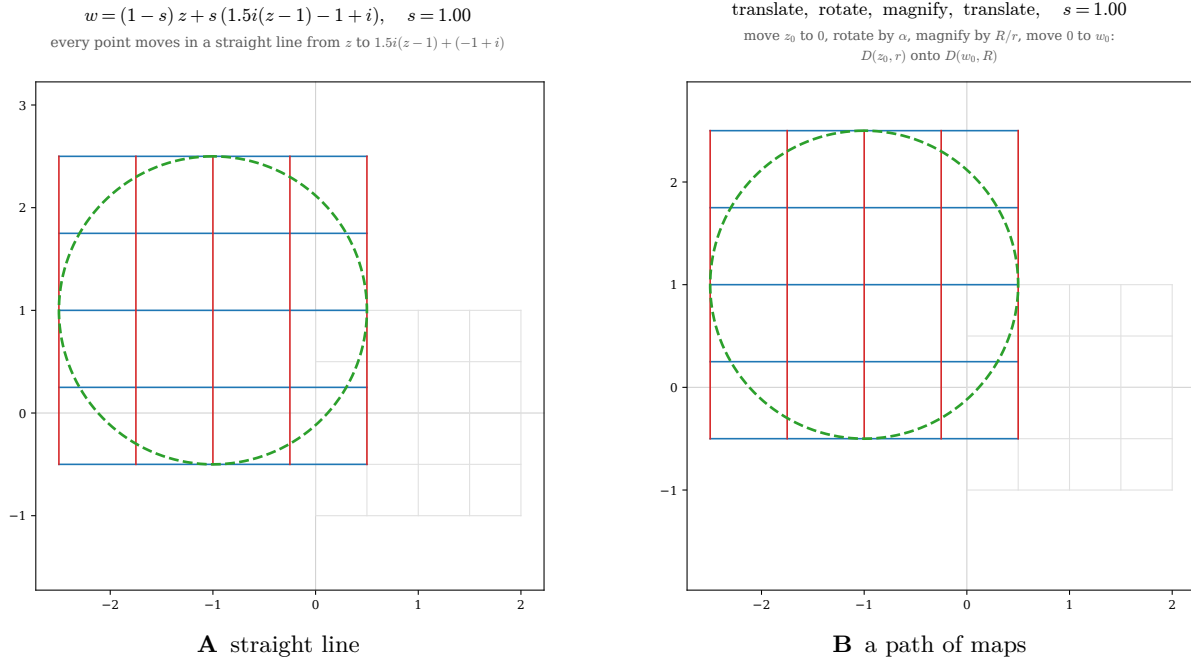
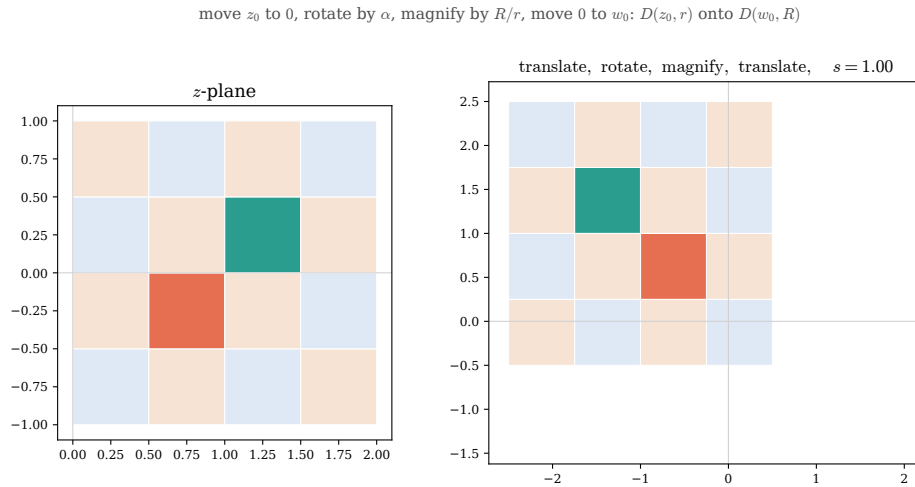


FIGURE 8.3 (animated). A disc onto a disc: $1.5i(z - 1) + (-1 + i)$. **A**: every point moves in a straight line from z to $f(z)$. **B**: move z_0 to 0, rotate by α , magnify by R/r , move 0 to w_0 : $D(z_0, r)$ onto $D(w_0, R)$. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 8.3, continued)

Example 8.7 (Mapping a strip onto a strip). Let $S = \{z : 0 < \text{Im } z < 1\}$ and $S' = \{w : 1 < \text{Re } w < 3\}$. Seek $f(z) = az + b$. The map must send horizontal lines to vertical lines, so a must rotate by $-\pi/2$ (or $+\pi/2$), i.e. $a = \rho e^{-i\pi/2} = -i\rho$; the width must go from 1 to 2, so $\rho = 2$ and $a = -2i$. Then $\text{Im } z = 0 \mapsto \text{Re } w = \text{Re}(b)$, so we need $\text{Re } b = 1$; take $b = 1$. Thus $f(z) = -2iz + 1$. Check: $z = iy$ with $0 < y < 1$ gives $w = -2i(iy) + 1 = 2y + 1 \in (1, 3)$ as required, and the whole line $\text{Im } z = y$ maps to $\text{Re } w = 2y + 1$.

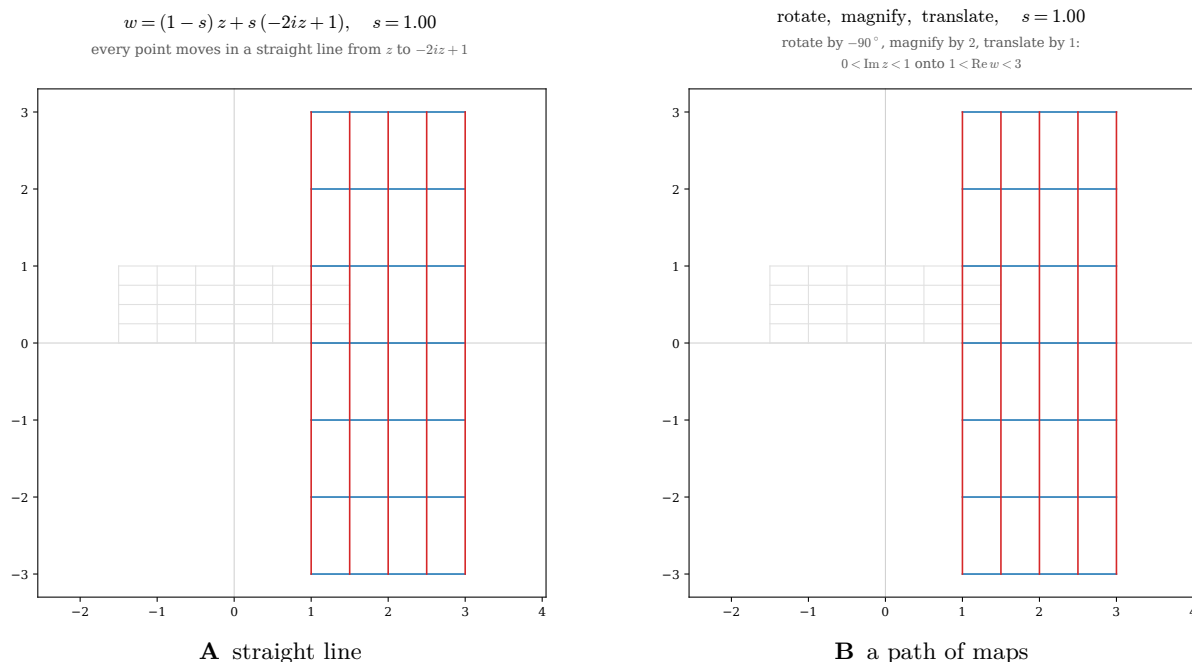
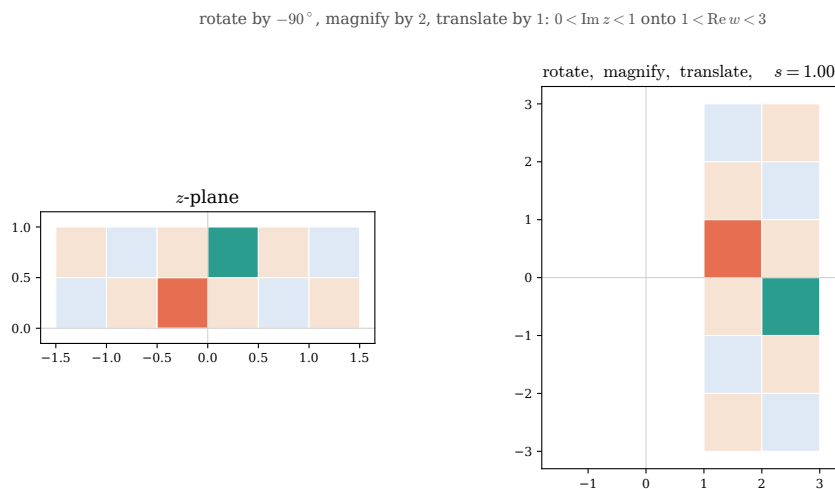


FIGURE 8.4 (animated). A strip onto a strip: $-2iz + 1$. **A**: every point moves in a straight line from z to $f(z)$. **B**: rotate by -90° , magnify by 2, translate by 1: $0 < \operatorname{Im} z < 1$ onto $1 < \operatorname{Re} w < 3$. **C**: the tiles on the left deform along the path of B.



8.3 Exercises

Exercise 8.1. Describe $f(z) = (1 - i)z + 2$ as a composition of a rotation, a magnification, and a translation, and find the image of the square with vertices $0, 1, 1 + i, i$.

Exercise 8.2. Find a linear map taking the disc $|z - 1| < 2$ onto the disc $|w + 3i| < 1$ and sending $1 \mapsto -3i$ and $3 \mapsto -3i + 1$.

Exercise 8.3. Find a linear map taking the horizontal strip $-1 < \operatorname{Im} z < 1$ onto the vertical strip $0 < \operatorname{Re} w < 4$ with $f(0) = 2$ and $f(i) = 0$. Is it unique?

Exercise 8.4. Prove that a map $f : \mathbb{C} \rightarrow \mathbb{C}$ satisfying $|f(z_1) - f(z_2)| = |z_1 - z_2|$ for all z_1, z_2 and $f(0) = 0$, $f(1) = 1$ is either the identity or $z \mapsto \bar{z}$. (*Hint: the conditions pin $|f(z)| = |z|$ and $|f(z) - 1| = |z - 1|$; intersect two circles.*)

Exercise 8.5. ($\star$) Show that $f(z) = a\bar{z} + b$ with $a \neq 0$ is also a similarity, but *reverses* the orientation of angles. Conclude that the orientation-preserving similarities of the plane are exactly the maps (8.4).

Exercise 8.6. (After Chapter 14.) Express $f(z) = az + b$ in the matrix language of Section 1.5: show that the real map $\tilde{f}$ of (14.8) is $\mathbf{x} \mapsto \Phi(a)\mathbf{x} + \mathbf{b}$ with Φ as in (1.35), and confirm $\det D\tilde{f} = |a|^2$ directly.

Exercise 8.7. Find the linear map carrying the triangle with vertices $0, 1, i$ onto the triangle with vertices $2i, 2i + 1 + i, 2i - 1 + i$, or show that none exists. (*Compare the two triangles' shapes first: a similarity cannot change the ratios of side lengths.*)

Exercise 8.8. ($\star$) Prove that the set of linear maps (8.4) forms a group under composition, and that it is isomorphic to the group of pairs $(a, b) \in \mathbb{C}^\times \times \mathbb{C}$ with the product $(a_1, b_1) \cdot (a_2, b_2) = (a_1 a_2, a_1 b_2 + b_1)$. Identify the subgroup of rotations about a fixed point z_* .

Exercise 8.9. Let $f(z) = az + b$ with $a \neq 1$. Prove that f has a unique fixed point $z_* = \frac{b}{1-a}$ and that $f(z) - z_* = a(z - z_*)$; hence f is a rotation-magnification about z_* . What happens when $a = 1$?

Chapter 9

The Power Function z^n

Throughout this chapter $n \in \mathbb{N}$ with $n \geq 2$ is fixed, and

$$p_n : \mathbb{C} \rightarrow \mathbb{C}, \quad p_n(z) := z^n. \quad (9.1)$$

9.1 Action on the polar grid

Theorem 9.1 (z^n in polar coordinates). *Let $z = re^{i\theta}$ with $r \geq 0$ and $\theta \in \mathbb{R}$. Then*

$$p_n(z) = r^n e^{in\theta}, \quad (9.2)$$

so that $|p_n(z)| = |z|^n$ and $\arg p_n(z) = n \arg z$ (as sets, in the sense of Convention 3.4). Consequently:

- (i) p_n maps the circle $C(0, \rho)$ onto $C(0, \rho^n)$, wrapping it n times;
- (ii) p_n maps the ray $\{te^{i\theta_0} : t > 0\}$ onto the ray $\{se^{in\theta_0} : s > 0\}$;
- (iii) p_n maps the sector $S(\alpha, \beta) := \{re^{i\theta} : r > 0, \alpha < \theta < \beta\}$ onto the sector $S(n\alpha, n\beta)$ when $\beta - \alpha \leq 2\pi/n$, and onto all of $\mathbb{C} \setminus \{0\}$ when $\beta - \alpha > 2\pi/n$.

Proof. Equation (9.2) is De Moivre, Theorem 4.2. (i) If $|z| = \rho$ then $|z^n| = \rho^n$; and as θ increases by 2π , $n\theta$ increases by $2n\pi$, so the image point traverses $C(0, \rho^n)$ exactly n times. Surjectivity onto $C(0, \rho^n)$ is Theorem 4.5: every w with $|w| = \rho^n$ has an n -th root of modulus ρ . (ii) is immediate from (9.2): $t \mapsto t^n$ is a bijection of $(0, \infty)$ onto itself. (iii) By (ii) the image is $\{se^{i\psi} : s > 0, \psi \in (n\alpha, n\beta)\}$, understood modulo 2π . If $n(\beta - \alpha) \leq 2\pi$ the interval $(n\alpha, n\beta)$ represents a genuine sector; if $n(\beta - \alpha) > 2\pi$ it covers all residues mod 2π , so the image is $\mathbb{C} \setminus \{0\}$. $\square$

Corollary 9.2 (Injectivity). *p_n is one-to-one on $S(\alpha, \beta)$ if and only if $\beta - \alpha \leq 2\pi/n$. In particular p_n is one-to-one on the open sector*

$$\Sigma_n := \left\{ z \neq 0 : -\frac{\pi}{n} < \operatorname{Arg} z < \frac{\pi}{n} \right\}, \quad (9.3)$$

which it maps bijectively onto the slit plane $\mathbb{C} \setminus (-\infty, 0]$.

Proof. If $\beta - \alpha \leq 2\pi/n$ and $z_1 = r_1 e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$ lie in $S(\alpha, \beta)$ with $z_1^n = z_2^n$, then $r_1^n = r_2^n$ gives $r_1 = r_2$, and $n\theta_1 - n\theta_2 \in 2\pi\mathbb{Z}$ gives $\theta_1 - \theta_2 \in \frac{2\pi}{n}\mathbb{Z}$. But $|\theta_1 - \theta_2| < \beta - \alpha \leq \frac{2\pi}{n}$, so $\theta_1 = \theta_2$. Conversely if $\beta - \alpha > 2\pi/n$ pick $\theta_1 \in (\alpha, \beta)$ with $\theta_2 := \theta_1 + \frac{2\pi}{n} \in (\alpha, \beta)$ as well; then $e^{i\theta_1} \neq e^{i\theta_2}$ but their n -th powers agree. For Σ_n : here $\beta - \alpha = 2\pi/n$ exactly, and the image is $S(-\pi, \pi) = \mathbb{C} \setminus (-\infty, 0]$ by Theorem 9.1(iii). $\square$

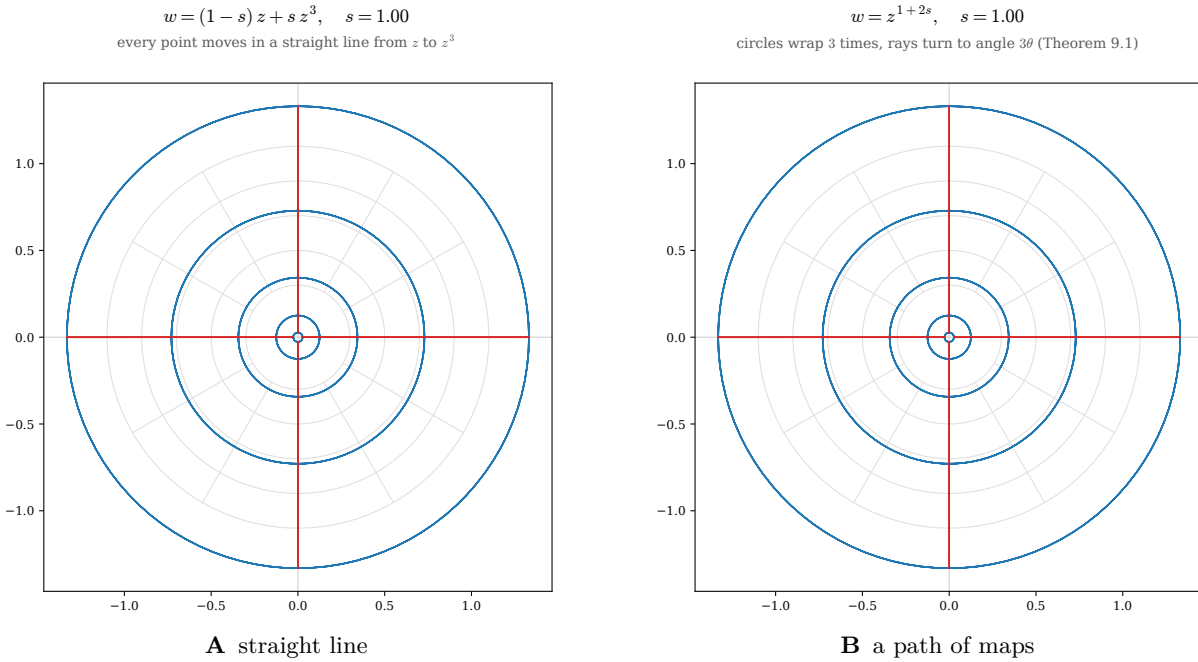
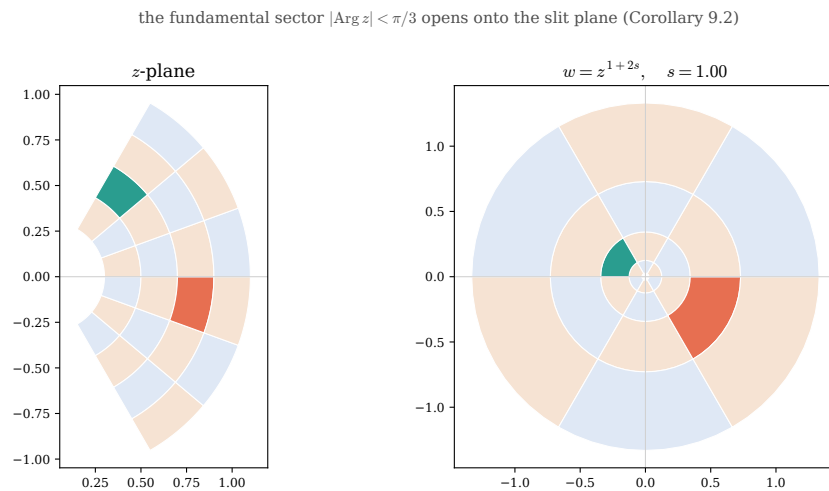


FIGURE 9.1 (animated). z^n for $n = 3$. **A**: every point moves in a straight line from z to $f(z)$. **B**: circles wrap 3 times, rays turn to angle 3θ (Theorem 9.1). **C**: the tiles on the left deform along the path of B.



C tiles (Figure 9.1, continued)

The sector Σ_n of (9.3) is the standard *fundamental sector* for p_n : by Corollary 9.2, p_n can be inverted on an open sector only if its opening is at most $2\pi/n$, and Σ_n has exactly this opening. The figure below shows p_2 on the smaller sector $|\text{Arg } z| < \pi/4$, which lies inside $\Sigma_2 = \{\text{Re } z > 0\}$;

its image is the right half-plane, whereas Σ_2 itself is mapped onto $\mathbb{C} \setminus (-\infty, 0]$.

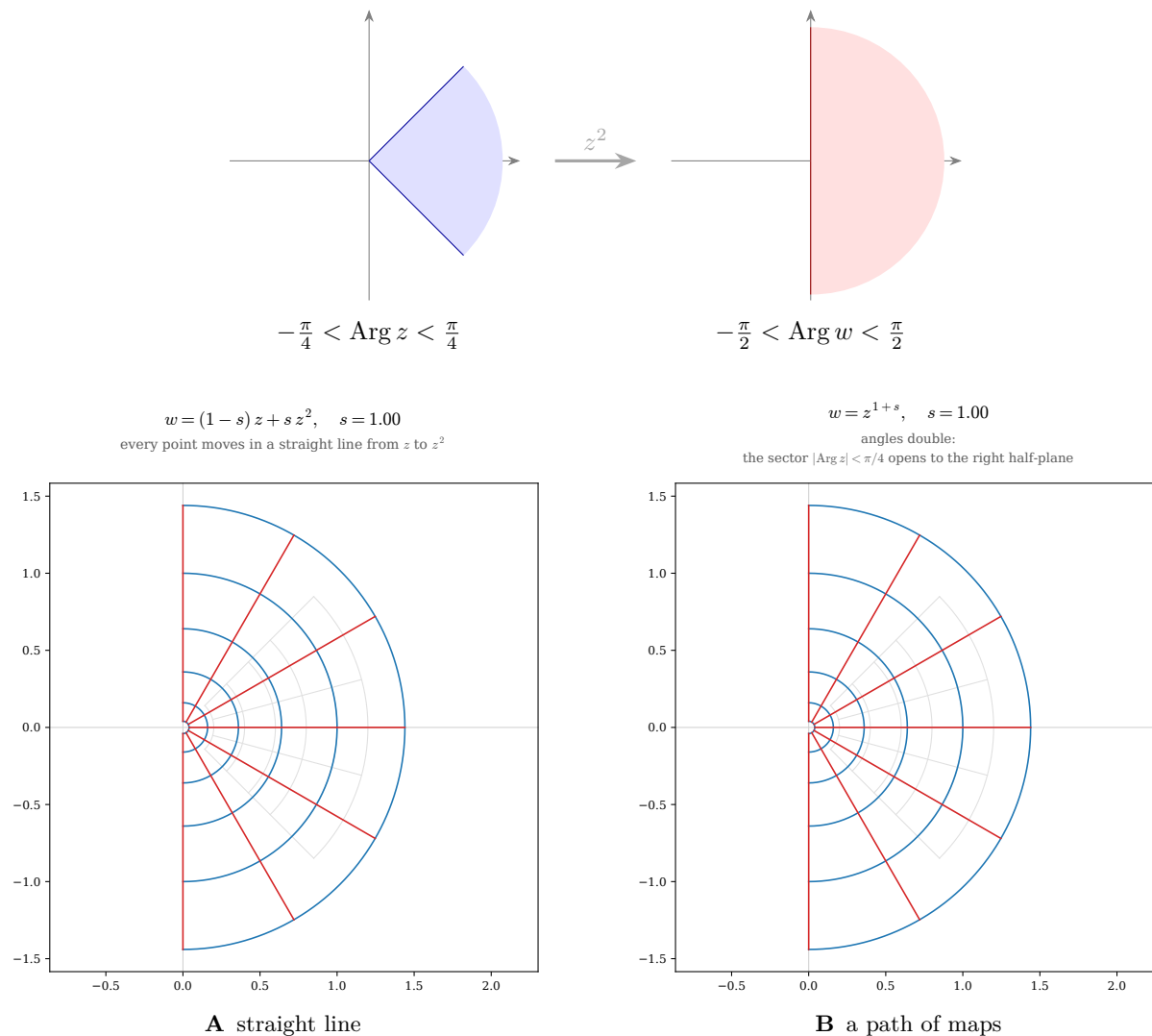
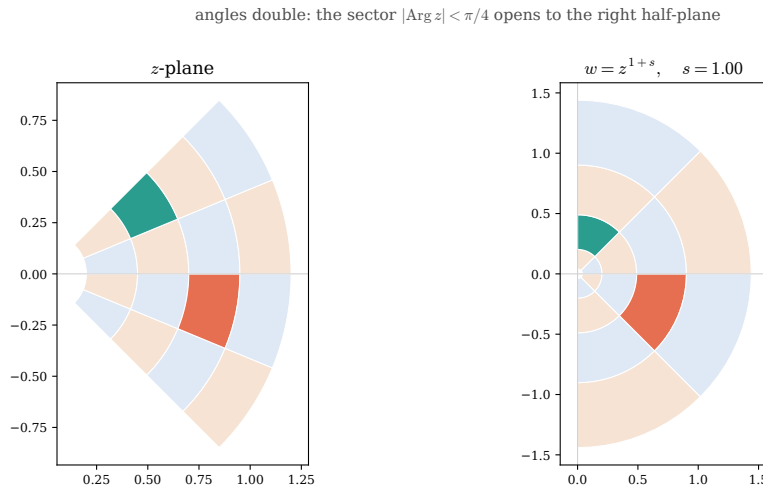


FIGURE 9.2 (animated). z^2 on the sector $|\text{Arg } z| < \pi/4$. **A**: every point moves in a straight line from z to $f(z)$. **B**: angles double: the sector $|\text{Arg } z| < \pi/4$ opens to the right half-plane. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 9.2, continued)

9.2 Action on the Cartesian grid; the case $n = 2$

For $n = 2$ the component functions were computed in (7.4): $u = x^2 - y^2$, $v = 2xy$. Two further families of curves are worth recording, because they are the level sets of u and v .

Proposition 9.3 (Level curves of z^2). *Let $c \in \mathbb{R} \setminus \{0\}$. Under p_2 :*

- (i) *the hyperbola $x^2 - y^2 = c$ maps into the vertical line $u = c$;*
- (ii) *the hyperbola $2xy = c$ maps into the horizontal line $v = c$;*
- (iii) *the vertical line $x = c$ maps onto the parabola (7.6), and the horizontal line $y = c$ onto the parabola (7.7).*

The two families of hyperbolas are mutually orthogonal, and so are the two families of parabolas.

Proof. (i) and (ii) restate (7.4). (iii) was Example 7.5. Orthogonality of the hyperbola families: the gradients are $\nabla(x^2 - y^2) = (2x, -2y)$ and $\nabla(2xy) = (2y, 2x)$, and their dot product is $4xy - 4yx = 0$. For the parabolas, differentiate (7.6) and (7.7) implicitly at a common point and check the slopes are negative reciprocals; alternatively, note that the parabola families are the images of the orthogonal Cartesian grid under a map that we will prove in Theorem 13.10 is angle-preserving away from $z = 0$. $\square$

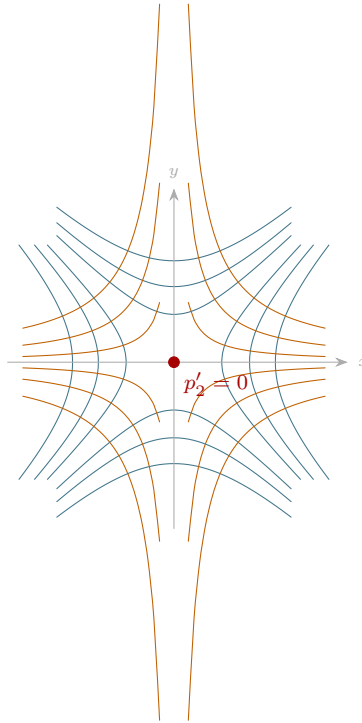


FIGURE 9.3. The level curves of Proposition 9.3: the hyperbolas $x^2 - y^2 = c$ (level sets of u) and $2xy = c$ (level sets of v) for $w = z^2$, $c \neq 0$. The two families cross at right angles wherever they meet; none of these curves passes through the origin. Only the degenerate level sets $c = 0$ (the axes and the diagonals $y = \pm x$, not drawn) pass through 0, where $p'_2 = 0$ and these four lines meet at 45° instead of 90° .

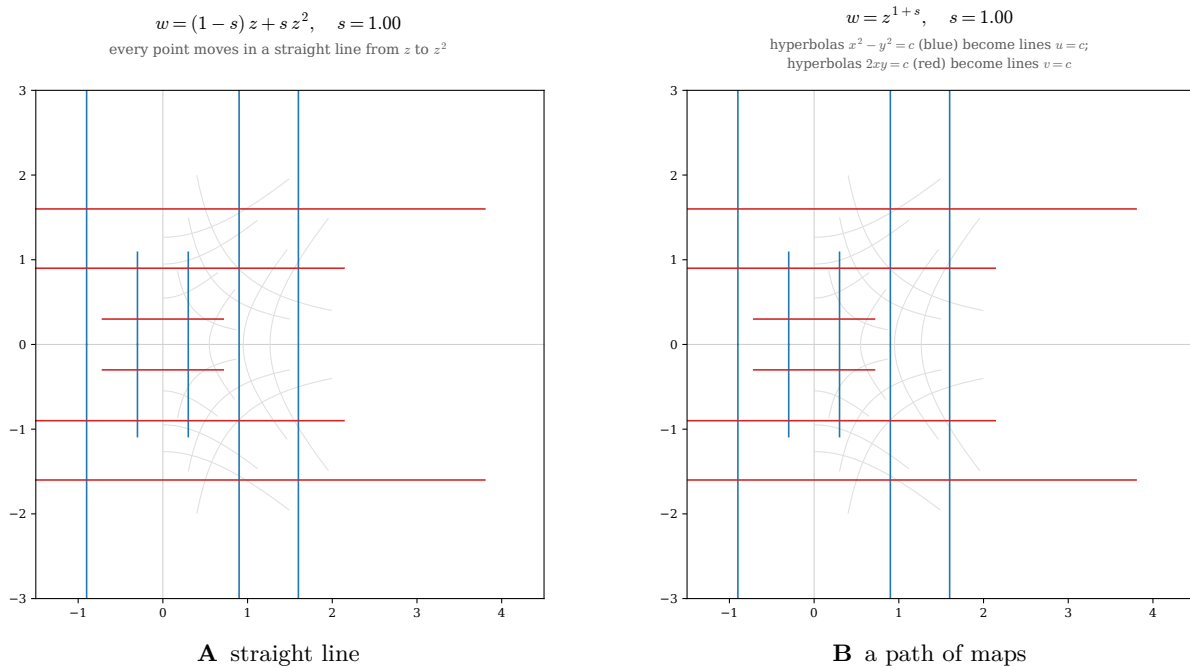
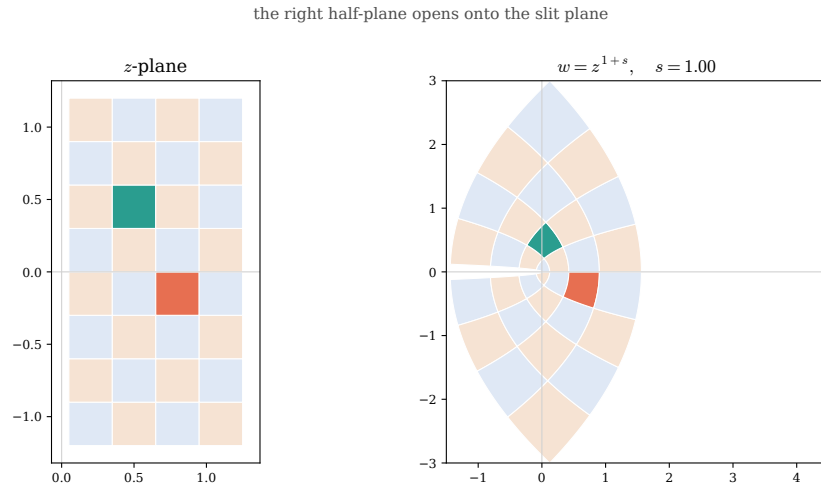


FIGURE 9.4 (animated). z^2 straightens the hyperbolas. **A**: every point moves in a straight line from z to $f(z)$. **B**: hyperbolas $x^2 - y^2 = c$ (blue) become lines $u = c$; hyperbolas $2xy = c$ (red) become lines $v = c$. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 9.4, continued)

Remark 9.4 (The critical point at the origin). Angles are *not* preserved at $z = 0$: by Theorem 9.1(ii) two rays meeting at 0 at angle φ map to rays meeting at angle $n\varphi$. The origin is the unique point where $p'_n(z) = nz^{n-1}$ vanishes, and this is exactly the general mechanism: conformality fails precisely at the zeros of the derivative. See Theorem 13.10.

Example 9.5 (Image of a rectangle). Let $R = \{x + iy : 1 \leq x \leq 2, 0 \leq y \leq 1\}$ and $f = p_2$. The four edges map as follows:

- $x = 1, 0 \leq y \leq 1$: by (7.6), $u = 1 - v^2/4$ with $v = 2y \in [0, 2]$ — an arc of a leftward parabola from $(1, 0)$ to $(0, 2)$;
- $x = 2, 0 \leq y \leq 1$: $u = 4 - v^2/16$ with $v = 4y \in [0, 4]$ — an arc from $(4, 0)$ to $(3, 4)$;
- $y = 0, 1 \leq x \leq 2$: $u = x^2 \in [1, 4], v = 0$ — the segment $[1, 4]$ of the real axis;
- $y = 1, 1 \leq x \leq 2$: by (7.7), $u = v^2/4 - 1$ with $v = 2x \in [2, 4]$ — an arc from $(0, 2)$ to $(3, 4)$.

The image of R is the curvilinear quadrilateral bounded by these four arcs; the interior point $z = \frac{3}{2} + \frac{i}{2}$ maps to $\frac{9}{4} - \frac{1}{4} + i\frac{3}{2} = 2 + \frac{3}{2}i$, which is inside it, confirming the orientation.

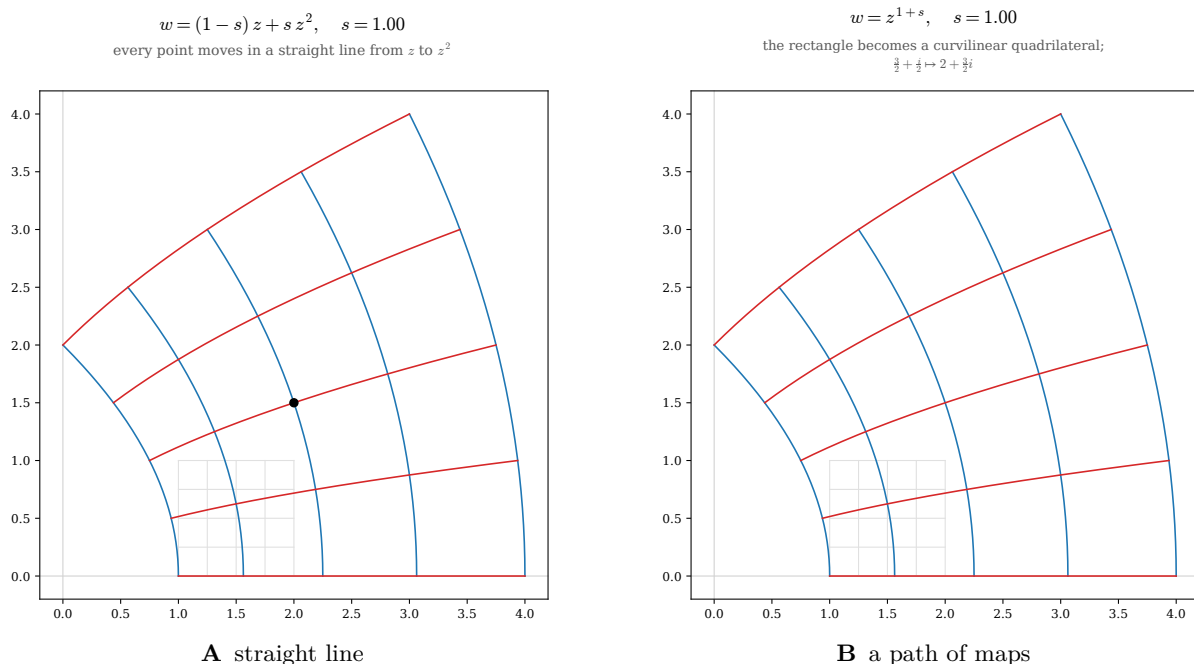
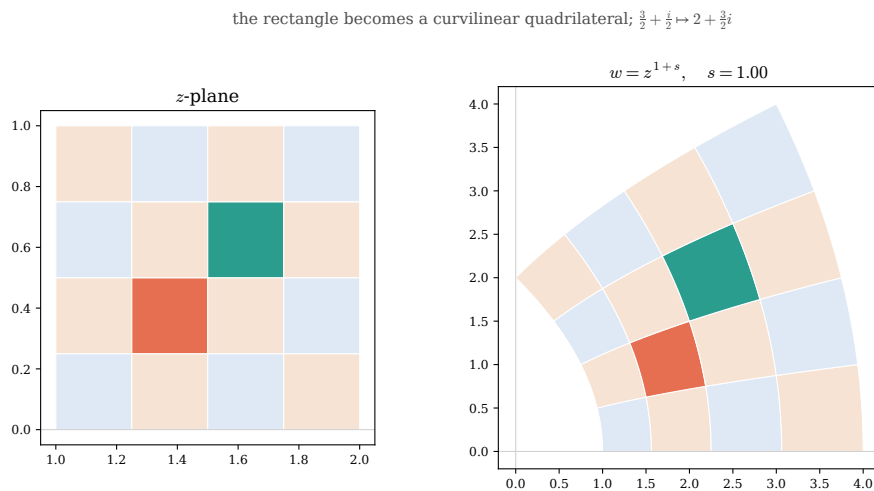


FIGURE 9.5 (animated). z^2 on the rectangle $1 \leq x \leq 2$, $0 \leq y \leq 1$. **A**: every point moves in a straight line from z to $f(z)$. **B**: the rectangle becomes a curvilinear quadrilateral; $\frac{3}{2} + \frac{i}{2} \mapsto 2 + \frac{3}{2}i$. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 9.5, continued)

9.3 Exercises

Exercise 9.1. Find the image under $z \mapsto z^3$ of: the sector $0 < \text{Arg } z < \pi/6$; the circle $|z| = 2$; the ray $\text{Arg } z = 3\pi/4$; the annulus $1 < |z| < 2$.

Exercise 9.2. Determine all sectors with vertex 0 on which $z \mapsto z^4$ is one-to-one, and describe the image in each case.

Exercise 9.3. Find the image of the half-disc $\{|z| < 1, \text{Im } z > 0\}$ under $z \mapsto z^2$. (Answer: the unit disc minus the segment $[0, 1)$.)

Exercise 9.4. Show that $z \mapsto z^2$ maps each line through the origin onto a closed ray from the origin, two-to-one except at 0, and that it maps the family of open rays $\{se^{i\theta} : s > 0\}$ from the origin onto itself, but not bijectively. Which rays are mapped into themselves?

Exercise 9.5. Let $n \geq 2$. Show that p_n maps the circle $|z - 1| = 1$ onto a closed curve passing through 0 and 2^n , and sketch it for $n = 2$. (*This image is the cardioid when $n = 2$.*)

Exercise 9.6. Using Theorem 9.1, determine the image under $z \mapsto z^5$ of the “pie slice” $\{re^{i\theta} : 1 < r < 2, \pi/6 < \theta < \pi/3\}$. Is the map one-to-one there?

Exercise 9.7. (After Chapter 14.) Compute $D\tilde{p}_2$, the Jacobian matrix (14.9) of p_2 , at a general point, and verify directly that it lies in $\overline{CO_+(2)}$ and equals $\Phi(2z)$. At which point does it fall out of $CO_+(2)$ into $\{0\}$, and what does Remark 9.4 say happens there?

Exercise 9.8. (★) Let $n \geq 2$. Prove that p_n maps $\mathbb{C}$ onto $\mathbb{C}$ and that every $w \neq 0$ has exactly n preimages, while $w = 0$ has exactly one. Deduce that p_n is an n -to-one map away from a single point — the model example of a *branched covering*.

Chapter 10

The Power Function $z^{1/n}$

By Theorem 4.5 each $w \neq 0$ has n distinct n -th roots, so “ $w^{1/n}$ ” names a set of n points and is not a function. This chapter converts it into functions. The construction — *branch*, *branch cut*, *branch point* — is the same one used later for $\log z$ and z^α , so it repays close attention.

10.1 Branches

Definition 10.1 (Multiple-valued relation and branch). Let F assign to each z in a set $S \subseteq \mathbb{C}$ a nonempty set $F(z) \subseteq \mathbb{C}$. A *branch* of F on a region $D \subseteq S$ is a function $f : D \rightarrow \mathbb{C}$ that is continuous on D and satisfies

$$f(z) \in F(z) \quad \text{for every } z \in D. \quad (10.1)$$

The two requirements pull against each other: picking a value at each point is trivial; picking them so that the result is *continuous* is not always possible on a given D , and the obstruction is the whole phenomenon.

Definition 10.2 (Principal n -th root function). For $n \in \mathbb{N}$ define

$$z_{\text{pr}}^{1/n} := |z|^{1/n} \exp\left(i \frac{\text{Arg } z}{n}\right), \quad z \in \mathbb{C} \setminus \{0\}, \quad (10.2)$$

with Arg as in Definition 3.3 and $|z|^{1/n} > 0$ the positive real n -th root. We also set $0_{\text{pr}}^{1/n} := 0$. When $n = 2$ we write $\sqrt{z} := z_{\text{pr}}^{1/2}$.

Theorem 10.3 (The principal branch). Let $n \geq 2$ and let $D_0 := \mathbb{C} \setminus (-\infty, 0]$ be the plane slit along the non-positive real axis. Then:

- (i) D_0 is a region, and $z_{\text{pr}}^{1/n}$ restricted to D_0 is a branch of $z \mapsto z^{1/n}$ in the sense of Definition 10.1;
- (ii) it maps D_0 bijectively onto the sector Σ_n of (9.3), and is the inverse of the restriction $p_n|_{\Sigma_n}$ of Corollary 9.2;
- (iii) $z_{\text{pr}}^{1/n}$ is discontinuous at every point of the slit $(-\infty, 0)$; precisely, for $x_0 < 0$,

$$\lim_{\substack{z \rightarrow x_0 \\ \text{Im } z > 0}} z_{\text{pr}}^{1/n} = |x_0|^{1/n} e^{i\pi/n}, \quad \lim_{\substack{z \rightarrow x_0 \\ \text{Im } z < 0}} z_{\text{pr}}^{1/n} = |x_0|^{1/n} e^{-i\pi/n}, \quad (10.3)$$

and these two values differ because $0 < 2\pi/n \leq \pi$.

Proof. (i) D_0 is open (its complement $(-\infty, 0]$ is closed) and any two of its points are joined by a polygonal path avoiding the slit (go around through the upper or lower half-plane), so D_0 is a region. That $z_{\text{pr}}^{1/n}$ takes a value in $z^{1/n}$ is Theorem 4.5 with $k = 0$, $\varphi = \text{Arg } z$. Continuity on D_0 : on D_0 we have $\text{Arg } z \in (-\pi, \pi)$, an open interval, and Arg is continuous there. Indeed, on D_0 the formula (3.4) of Proposition 3.5 reads $\arctan(y/x)$ on the open set $\{x > 0\}$, $\arctan(y/x) + \pi$ on the open set $\{x < 0, y > 0\}$ and $\arctan(y/x) - \pi$ on the open set $\{x < 0, y < 0\}$ (the points with $x < 0, y = 0$ are not in D_0), each continuous there; the remaining points of D_0 are those of the imaginary axis. Let $z_0 = iy_0$ with $y_0 > 0$, so $\text{Arg } z_0 = \pi/2$. As $z = x + iy \rightarrow z_0$ with $x > 0$ we have $y/x \rightarrow +\infty$ and $\arctan(y/x) \rightarrow \pi/2$; with $x < 0$ we have $y > 0$ near z_0 , $y/x \rightarrow -\infty$ and $\arctan(y/x) + \pi \rightarrow -\pi/2 + \pi = \pi/2$; with $x = 0$ the value is $\pi/2$. So $\text{Arg } z \rightarrow \text{Arg } z_0$. The case $y_0 < 0$ is the same with all signs reversed (the limit is $-\pi/2$ from both sides). Also $z \mapsto |z|^{1/n}$ is continuous and $t \mapsto e^{it}$ is continuous, so (10.2) is a composition of continuous functions.

(ii) By (10.2) the argument of $z_{\text{pr}}^{1/n}$ is $\text{Arg } z/n \in (-\pi/n, \pi/n)$, so the image lies in Σ_n ; conversely if $\zeta \in \Sigma_n$ then $z := \zeta^n$ lies in D_0 by Corollary 9.2 and $z_{\text{pr}}^{1/n} = \zeta$, since $|z|^{1/n} = |\zeta|$ and $\text{Arg } z = n \text{Arg } \zeta$ (legitimate because $n \text{Arg } \zeta \in (-\pi, \pi)$). Injectivity follows since $p_n \circ z_{\text{pr}}^{1/n} = \text{id}$ on D_0 .

(iii) Let $x_0 < 0$. If $z \rightarrow x_0$ with $\text{Im } z > 0$ then $\text{Arg } z \rightarrow \pi$, whereas if $\text{Im } z < 0$ then $\text{Arg } z \rightarrow -\pi$; substituting into (10.2) and using continuity of $|\cdot|$ and $t \mapsto e^{it}$ gives (10.3). The two limits are equal only if $e^{2\pi i/n} = 1$, i.e. $n \mid 1$, excluded by $n \geq 2$. $\square$

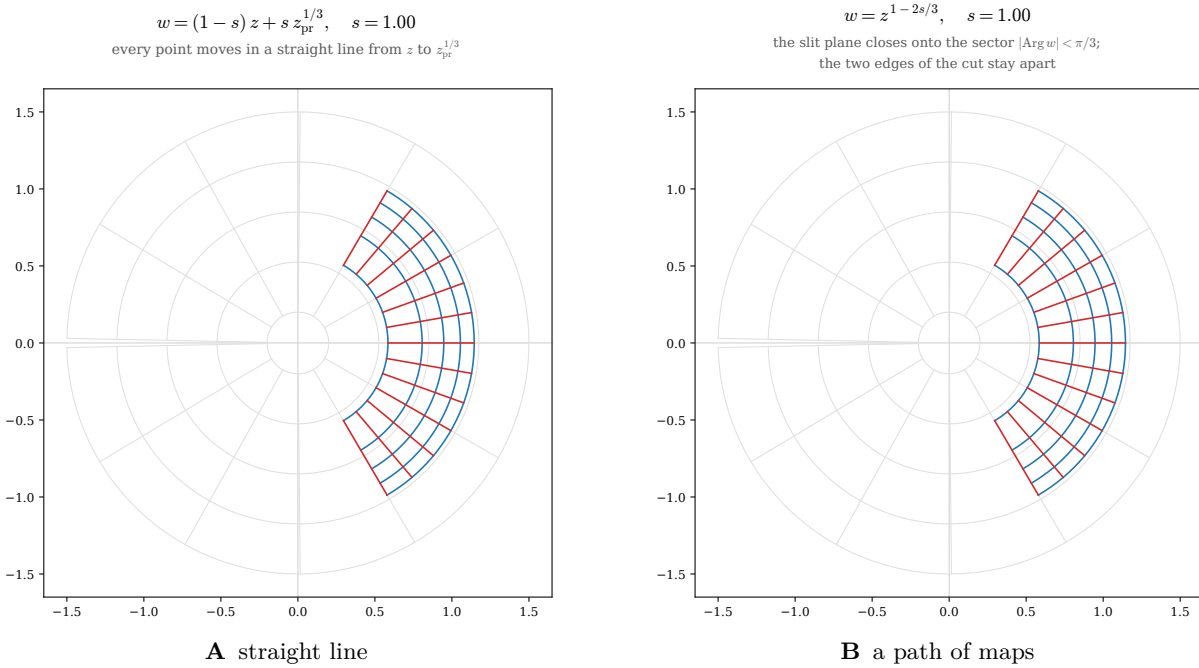
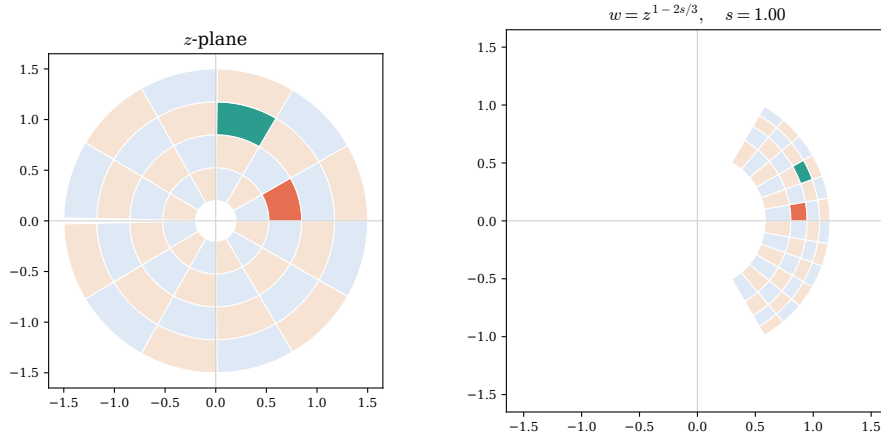


FIGURE 10.1 (animated). The principal cube root $z_{\text{pr}}^{1/3}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: the slit plane closes onto the sector $|\text{Arg } w| < \pi/3$; the two edges of the cut stay apart. **C**: the tiles on the left deform along the path of B.

the slit plane closes onto the sector $|\operatorname{Arg} w| < \pi/3$; the two edges of the cut stay apart



C tiles (Figure 10.1, continued)

Definition 10.4 (Branch cut, branch point). In the situation of Theorem 10.3, the ray $(-\infty, 0]$ removed from the plane is called the *branch cut* of the principal branch, and 0 is called a *branch point* of $z^{1/n}$.

The cut is a matter of choice; the branch point is not.

Proposition 10.5 (Other branches; the cut can be moved but not removed). *Let $n \geq 2$.*

- (i) *For each $\tau \in \mathbb{R}$, let $D_\tau := \mathbb{C} \setminus \{te^{i\tau} : t \geq 0\}$ be the plane cut along the ray of angle τ , and let $\arg_\tau z$ be the unique argument of z in $(\tau, \tau + 2\pi)$. (Caution: the slit plane $D_0 = \mathbb{C} \setminus (-\infty, 0]$ of Theorem 10.3 is D_π in this notation, with $\arg_\pi z \in (\pi, 3\pi)$; the set D_τ with $\tau = 0$ is the plane cut along $[0, \infty)$.) Then*

$$f_{\tau,k}(z) := |z|^{1/n} \exp\left(i \frac{\arg_\tau z + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1, \quad (10.4)$$

are exactly n distinct branches of $z^{1/n}$ on D_τ , and every branch on D_τ is one of them.

- (ii) *There is no branch of $z^{1/n}$ on any region D containing a circle $C(0, \rho)$, $\rho > 0$. In particular there is no branch on $\mathbb{C} \setminus \{0\}$.*

Proof. (i) Each $f_{\tau,k}$ is continuous on D_τ (same argument as in Theorem 10.3(i), since $\arg_\tau$ is continuous on D_τ : $\arg_\tau z = \operatorname{Arg}(e^{-i(\tau+\pi)}z) + \tau + \pi$, and $z \mapsto e^{-i(\tau+\pi)}z$ maps D_τ into $\mathbb{C} \setminus (-\infty, 0]$, where Arg is continuous) and takes values in $z^{1/n}$ by Theorem 4.5. They are distinct because their values at any fixed z differ by the distinct factors $e^{2\pi ik/n}$. Conversely, let f be a branch on D_τ . Then $g := f/f_{\tau,0}$ is continuous on D_τ (the denominator never vanishes) and, since both are n -th roots of the same number, $g(z)^n = 1$ for every z , i.e. g maps the *connected* set D_τ into the finite set μ_n of Proposition 4.9. A continuous map from a region to a finite set is constant (Exercise 5.4, applied to the function equal to 1 where g takes one given value and 0 elsewhere, which is continuous because the points of μ_n are at positive distance from each other), so $g \equiv e^{2\pi ik/n}$ for some $k \in \{0, \dots, n-1\}$, i.e. $f = f_{\tau,k}$.

(ii) Suppose f were a branch on a region $D \supseteq C(0, \rho)$. Consider $\gamma(t) := \rho e^{it}$ for $t \in [0, 2\pi]$ and $h(t) := f(\gamma(t))/(\rho^{1/n} e^{it/n})$. The denominator is a continuous nonvanishing n -th root of $\gamma(t)$

along the whole circle, so h is continuous on $[0, 2\pi]$ with $h(t)^n = 1$ for all t ; as $[0, 2\pi]$ is connected, h is a constant $c \in \mu_n$. Evaluating at $t = 0$ and $t = 2\pi$, where $\gamma(0) = \gamma(2\pi) = \rho$ and hence $f(\gamma(0)) = f(\gamma(2\pi))$, gives

$$c \rho^{1/n} e^{i \cdot 0} = f(\rho) = c \rho^{1/n} e^{2\pi i/n},$$

so $e^{2\pi i/n} = 1$, contradicting $n \geq 2$. $\square$

Part (ii) is the precise sense in which the multivaluedness of $z^{1/n}$ is not an artefact of bad bookkeeping: going once around the origin permutes the n roots cyclically, and no choice can survive the trip. That permutation is called the *monodromy* of the branch point.

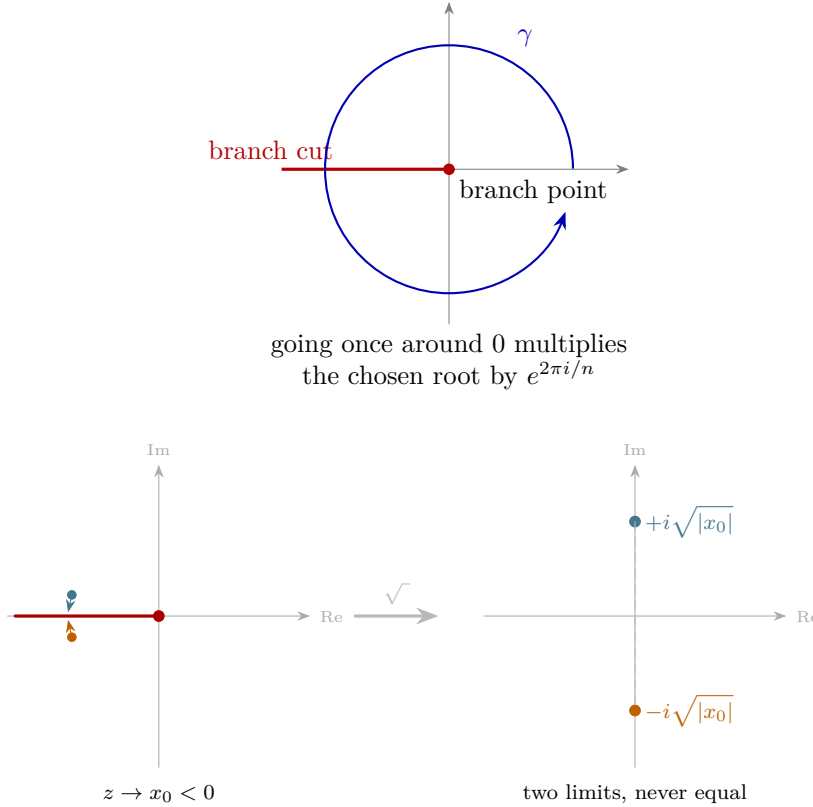


FIGURE 10.2. The jump (10.3) for $n = 2$. Approaching a point $x_0 < 0$ from above, $\text{Arg } z \rightarrow \pi$ and $\sqrt{z} \rightarrow |x_0|^{1/2} e^{i\pi/2}$; from below, $\text{Arg } z \rightarrow -\pi$ and $\sqrt{z} \rightarrow |x_0|^{1/2} e^{-i\pi/2}$. The two limits are negatives of each other, so no value assigned on the cut can make the function continuous there.

Warning 10.6 (Identities that fail). The familiar rule $\sqrt{z_1 z_2} = \sqrt{z_1} \sqrt{z_2}$ is *false* for the principal branch. Take $z_1 = z_2 = -1 + i\epsilon$ with small $\epsilon > 0$ and let $\epsilon \downarrow 0$: then $\text{Arg } z_j \rightarrow \pi$, $\sqrt{z_j} \rightarrow e^{i\pi/2} = i$, so $\sqrt{z_1} \sqrt{z_2} \rightarrow -1$; but $z_1 z_2 \rightarrow 1$ and $\sqrt{z_1 z_2} \rightarrow 1$. The correct statement is that $\sqrt{z_1} \sqrt{z_2}$ is *one of* the two square roots of $z_1 z_2$, and it is the principal one precisely when $\text{Arg } z_1 + \text{Arg } z_2 \in (-\pi, \pi]$ — exactly the correction term k of (3.9).

Example 10.7 (Mapping by $\sqrt{z}$). The principal square root maps:

- the slit plane $\mathbb{C} \setminus (-\infty, 0]$ onto the right half-plane $\{\text{Re } w > 0\}$, bijectively (Theorem 10.3(ii) with $n = 2$);
- the upper half-plane $\{\text{Im } z > 0\}$, i.e. $0 < \text{Arg } z < \pi$, onto the first-quadrant sector $0 < \text{Arg } w < \pi/2$;

- the punctured disc $0 < |z| < 1$ minus the slit, onto the half-disc $\{0 < |w| < 1, \operatorname{Re} w > 0\}$;
- the circle $|z| = \rho$ minus the point $-\rho$, onto the right-hand semicircle $|w| = \sqrt{\rho}$, $\operatorname{Re} w > 0$.

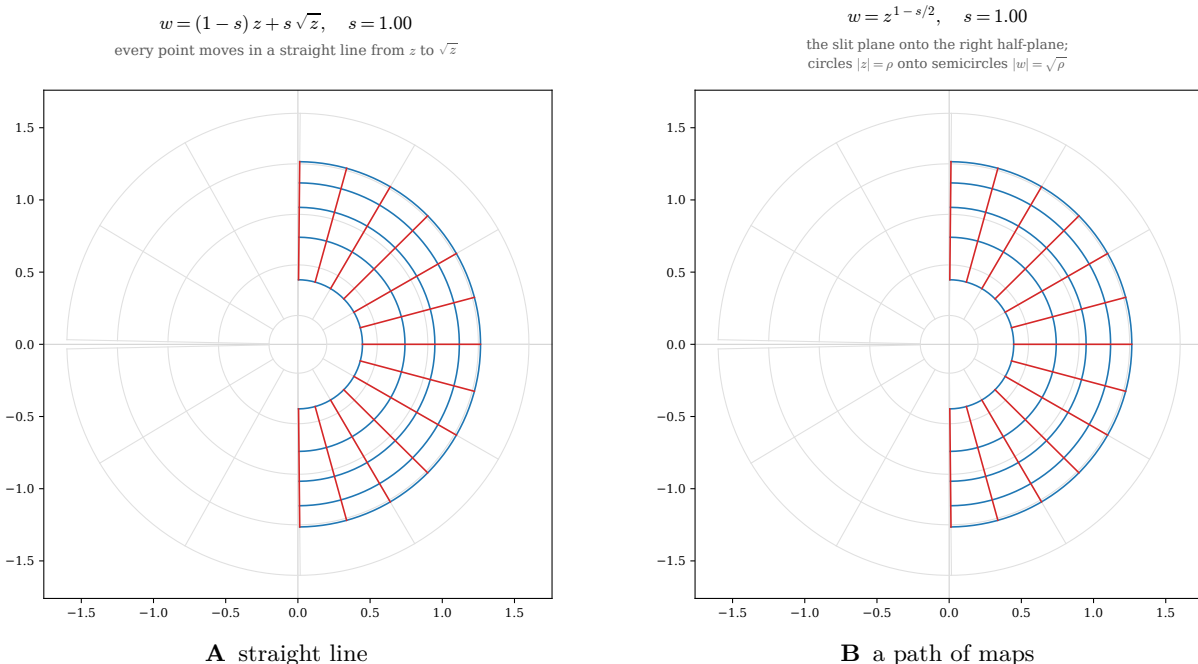
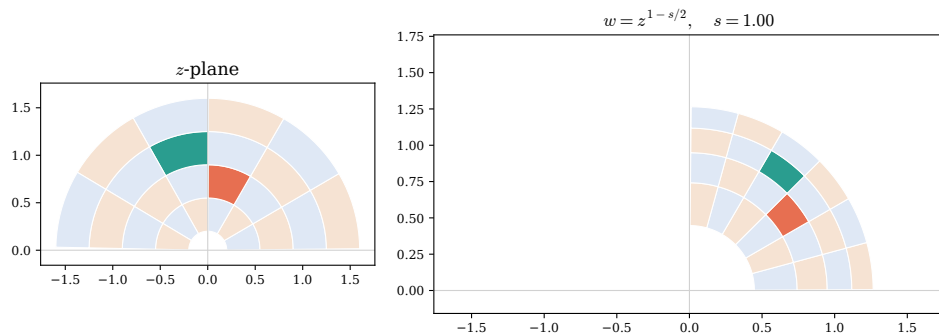


FIGURE 10.3 (animated). The principal square root $\sqrt{z}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: the slit plane onto the right half-plane; circles $|z| = \rho$ onto semicircles $|w| = \sqrt{\rho}$. **C**: the tiles on the left deform along the path of B.

the upper half-plane onto the first quadrant (Example 10.7)



C tiles (Figure 10.3, continued)

Example 10.8 (Choosing a cut to fit a problem). To define a continuous square root of $z^2 - 1$ one must exclude the two branch points ± 1 . Two standard choices are: cut along the segment $[-1, 1]$, or cut along $(-\infty, -1] \cup [1, \infty)$. The first gives a function continuous on $\mathbb{C} \setminus [-1, 1]$, with $\sqrt{z^2 - 1}$ behaving like z (or like $-z$, for the other branch) near infinity; this is the choice needed to invert the Joukowski map of aerodynamics (Exercise 10.5). The second keeps the segment $(-1, 1)$ inside

the domain, which is what one wants for $\sqrt{1-z^2}$ and, later, $\arcsin$. Neither is “the” answer: the cut is data supplied by the problem.

10.2 Exercises

Exercise 10.1. Compute the principal values: $(-8)_{\text{pr}}^{1/3}$, $\sqrt{i}$, $\sqrt{-4}$, $(2i)_{\text{pr}}^{1/4}$, and in each case list all the roots.

Exercise 10.2. Find the image of the sector $\pi/2 < \text{Arg } z < \pi$ under the principal cube root, and under the branch $f_{0,1}$ of (10.4) with $n = 3$, $\tau = 0$.

Exercise 10.3. Show that $\sqrt{\bar{z}} = \overline{\sqrt{z}}$ for all $z \in \mathbb{C} \setminus (-\infty, 0]$, and explain why the identity fails on the cut.

Exercise 10.4. (★) Let f be a branch of $z^{1/2}$ on the region D and suppose $a \in D$. Prove that f is one-to-one on any disc $D(a, \varepsilon) \subseteq D$. (Use $f(z_1) = f(z_2) \Rightarrow z_1 = z_2$ by squaring.)

Exercise 10.5. Prove that there is a branch of $(z^2 - 1)^{1/2}$ on $\mathbb{C} \setminus [-1, 1]$. (Hint: on that set $\frac{z-1}{z+1} \notin (-\infty, 0]$, so $F(z) := (z+1)\left(\frac{z-1}{z+1}\right)_{\text{pr}}^{1/2}$ is continuous, and $F(z)^2 = z^2 - 1$.)

Exercise 10.6. Explain, using Theorem 10.3(iii) and the jump (10.3), why $|z|^{1/2} e^{i \text{Arg } z/2}$ cannot be made continuous at any point of the negative real axis no matter how the value there is defined.

Exercise 10.7. Let f be the branch of $z^{1/3}$ on $D_0 = \mathbb{C} \setminus (-\infty, 0]$ with $f(1) = e^{2\pi i/3}$. Compute $f(i)$, $f(-1+i)$ and $f(8)$, and say which τ and which k in (10.4) this branch corresponds to. (Note that $D_0 = D_\pi$ in the notation of Proposition 10.5.)

Exercise 10.8. Show that the four maps $z \mapsto \pm\sqrt{z}$ and $z \mapsto \pm\sqrt{-z}$ (principal branches, on their natural cut planes) are pairwise distinct, and describe a region on which all four are simultaneously defined and continuous.

Exercise 10.9. (★) (Monodromy, counted.) Let f be a branch of $z^{1/n}$ near a point $z_1 \neq 0$, and continue it once counterclockwise around the circle through z_1 centred at 0 (that is, follow a continuous choice of n -th root along the circle, as in the proof of Proposition 10.5(ii)). Prove that the value returns multiplied by $e^{2\pi i/n}$, so that n circuits are needed to recover f . The order of a branch point is the least number of circuits needed to return to the starting value, minus one. Deduce that the branch point at 0 has order $n - 1$, and explain why $\log z$ (Chapter 16) never returns to its starting value however many circuits are made.

Chapter 11

The Reciprocal Function

11.1 Inversion and reflection

Definition 11.1 (Reciprocal map). Define

$$\iota(z) := \frac{1}{z}, \quad z \in \mathbb{C} \setminus \{0\}. \quad (11.1)$$

By (1.20),

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \overline{\left(\frac{z}{|z|^2}\right)}, \quad (11.2)$$

so ι is the composition of the *inversion in the unit circle*

$$\sigma(z) := \frac{z}{|z|^2} \quad \left(\text{same argument as } z, \text{ modulus } \frac{1}{|z|}\right) \quad (11.3)$$

with the reflection $z \mapsto \bar{z}$ in the real axis, in either order.

Proposition 11.2 (Basic properties). *For $z \in \mathbb{C} \setminus \{0\}$, write $z = re^{i\theta}$ with $r > 0$. Then*

$$\frac{1}{z} = \frac{1}{r} e^{-i\theta}, \quad \text{so} \quad \left|\frac{1}{z}\right| = \frac{1}{|z|}, \quad \arg \frac{1}{z} = -\arg z. \quad (11.4)$$

Consequently ι is a bijection of $\mathbb{C} \setminus \{0\}$ onto itself with $\iota \circ \iota = \text{id}$; it maps

- the unit circle $\mathbb{T}$ onto itself (with $z \mapsto \bar{z}$),
- the punctured open unit disc $0 < |z| < 1$ onto the exterior $|w| > 1$, and conversely,
- the ray $\text{Arg } z = \theta_0$ onto the ray $\text{Arg } w = -\theta_0$ (for $\theta_0 \neq \pi$).

Proof. Formula (11.4) is Theorem 3.11(i) applied to $1/z$ with $|1| = 1$, $\arg 1 = 2\pi\mathbb{Z}$. The listed consequences are immediate: $|z| = 1 \iff |1/z| = 1$; $|z| < 1 \iff |1/z| > 1$; and the argument statement is (11.4). Involutivity is $1/(1/z) = z$. $\square$

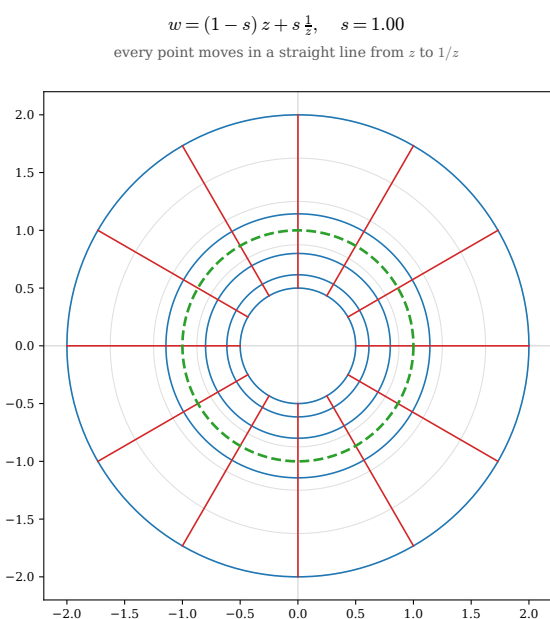
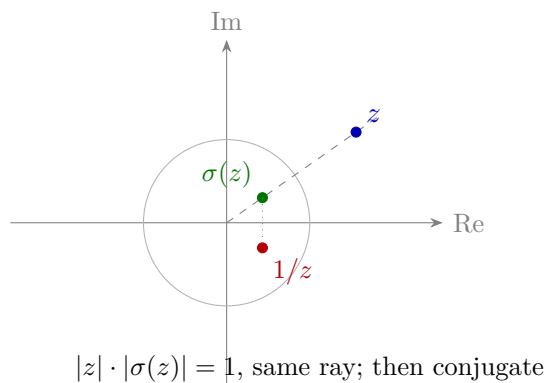
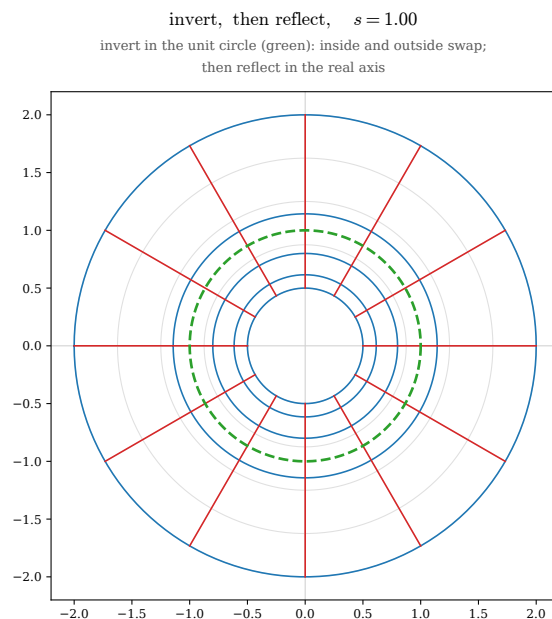
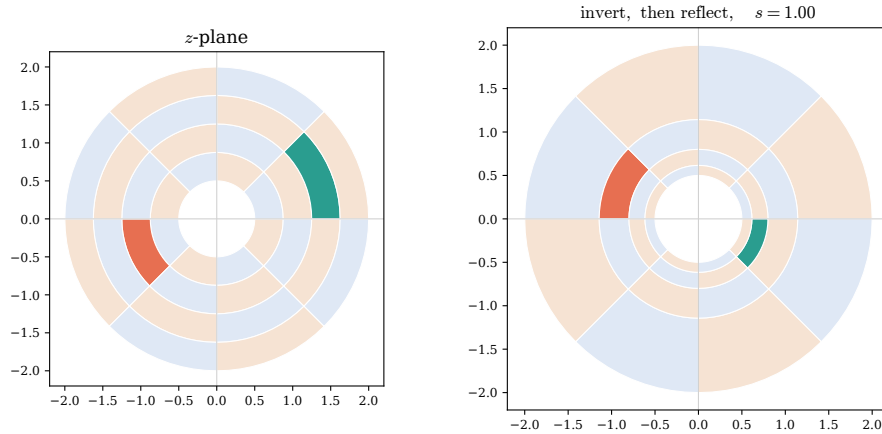
**A** straight line**B** a path of maps

FIGURE 11.1 (animated). $1/z$ as inversion, then reflection. **A**: every point moves in a straight line from z to $f(z)$. **B**: invert in the unit circle (green): inside and outside swap; then reflect in the real axis. **C**: the tiles on the left deform along the path of B.

invert in the unit circle (green): inside and outside swap; then reflect in the real axis



C tiles (Figure 11.1, continued)

11.2 Circles and lines are preserved

Theorem 11.3 ($1/z$ maps circles-or-lines to circles-or-lines). *Let S be a circle or a line in $\mathbb{C}$, described by (2.6):*

$$A(x^2 + y^2) + Bx + Cy + D = 0, \quad A, B, C, D \in \mathbb{R}, \quad (A, B, C) \neq (0, 0, 0). \quad (11.5)$$

Then $\iota(S \setminus \{0\})$ is the set T of points $w = u + iv$ with

$$D(u^2 + v^2) + Bu - Cv + A = 0, \quad (11.6)$$

with the point 0 removed if it lies on T ; and T is again a circle or a line. More precisely:

- (i) *a circle not through 0 ($A \neq 0, D \neq 0$) maps onto a circle not through 0;*
- (ii) *a circle through 0 ($A \neq 0, D = 0$) maps onto a line not through 0;*
- (iii) *a line not through 0 ($A = 0, D \neq 0$) maps onto a circle through 0, punctured at 0;*
- (iv) *a line through 0 ($A = 0, D = 0$) maps onto a line through 0, punctured at 0.*

Proof. Write $w = 1/z$, $w = u + iv$, so $z = 1/w$ and by (7.5)

$$x = \frac{u}{u^2 + v^2}, \quad y = \frac{-v}{u^2 + v^2}, \quad x^2 + y^2 = \frac{1}{u^2 + v^2}. \quad (11.7)$$

Substituting into (11.5),

$$\frac{A}{u^2 + v^2} + \frac{Bu}{u^2 + v^2} - \frac{Cv}{u^2 + v^2} + D = 0.$$

Multiplying by $u^2 + v^2$, which is nonzero since $w \neq 0$, gives exactly (11.6). The coefficient vector of (11.6) is $(D, B, -C, A)$, which is nonzero whenever (A, B, C, D) is; so by Proposition 2.6 the image equation defines a circle, a line, a point, or the empty set. This proves $\iota(S \setminus \{0\}) \subseteq T$.

Conversely, let $w = u + iv \in T \setminus \{0\}$ and put $z := 1/w = x + iy$. Then $u = x/(x^2 + y^2)$, $v = -y/(x^2 + y^2)$ and $u^2 + v^2 = 1/(x^2 + y^2)$ by (7.5); substituting into (11.6) and multiplying by

$x^2 + y^2 \neq 0$ gives (11.5), so $z \in S \setminus \{0\}$ and $w = \iota(z)$. (This is the same computation with the roles of z and w exchanged: $\iota^{-1} = \iota$, and $(A, B, C, D) \mapsto (D, B, -C, A)$ is an involution.) Hence

$$\iota(S \setminus \{0\}) = T \setminus \{0\}.$$

In particular T is neither a point nor empty, since $S \setminus \{0\}$ is infinite and ι is injective; so T is a circle or a line.

The four cases are read off from (11.6): the image is a line iff its $x^2 + y^2$ -coefficient D vanishes, i.e. iff S passes through the origin ($D = 0$ in (11.5) says $(0, 0) \in S$); and the image passes through the origin iff its constant term A vanishes, i.e. iff S is a line. In (i) and (ii) we have $A \neq 0$, so $0 \notin T$ and $\iota(S \setminus \{0\}) = T$; the punctures in (iii) and (iv) occur because 0 is not in the range of ι . $\square$

Example 11.4. Find the image of the vertical line $x = 1$ under $w = 1/z$. Here $A = 0, B = 1, C = 0, D = -1$, so (11.6) reads $-(u^2 + v^2) + u = 0$, i.e.

$$\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4},$$

the circle of centre $\frac{1}{2}$ and radius $\frac{1}{2}$, punctured at the origin. Sanity check: $z = 1 \mapsto 1$ (on the circle); $z = 1 + i \mapsto \frac{1-i}{2}$, and $\left|\frac{1}{2} - i\frac{1}{2} - \frac{1}{2}\right| = \frac{1}{2}$. As $|z| \rightarrow \infty$ along the line, $w \rightarrow 0$: the missing point.

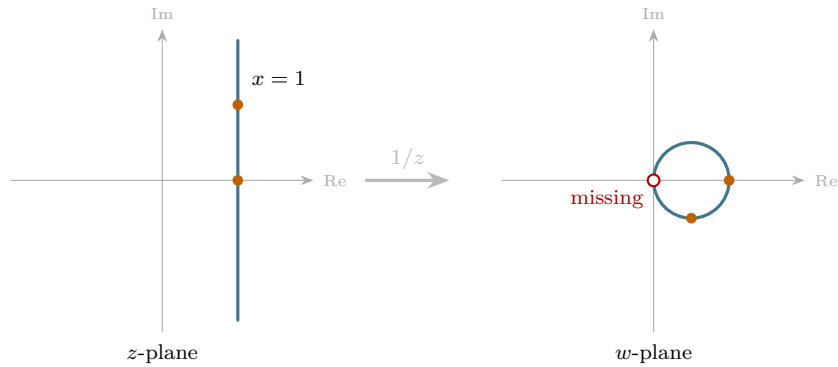


FIGURE 11.2. Example 11.4: the vertical line $x = 1$, a “circle through ∞ ”, maps under $w = 1/z$ onto the circle of centre $\frac{1}{2}$ and radius $\frac{1}{2}$, punctured at the origin. The puncture is exactly the image of the point at infinity, and adjoining ∞ to the plane (Definition 11.6) repairs it.

Example 11.5. The image of the half-plane $\operatorname{Re} z > 1$ under $w = 1/z$ is exactly the open disc $\left|w - \frac{1}{2}\right| < \frac{1}{2}$, the interior of the circle found in Example 11.4 (note that 0 lies on the boundary circle, not inside the disc). Heuristically: the boundary line maps to the boundary circle by Example 11.4, and the interior point $z = 2$ maps to $w = \frac{1}{2}$, which lies inside the circle. Proof: every point of the disc is nonzero, and for $w \neq 0$,

$$\left|w - \frac{1}{2}\right|^2 < \frac{1}{4} \iff |w|^2 < \operatorname{Re} w \iff \operatorname{Re} \frac{1}{w} = \frac{\operatorname{Re} w}{|w|^2} > 1;$$

since $z = 1/w$, the half-plane $\operatorname{Re} z > 1$ corresponds exactly to the disc.

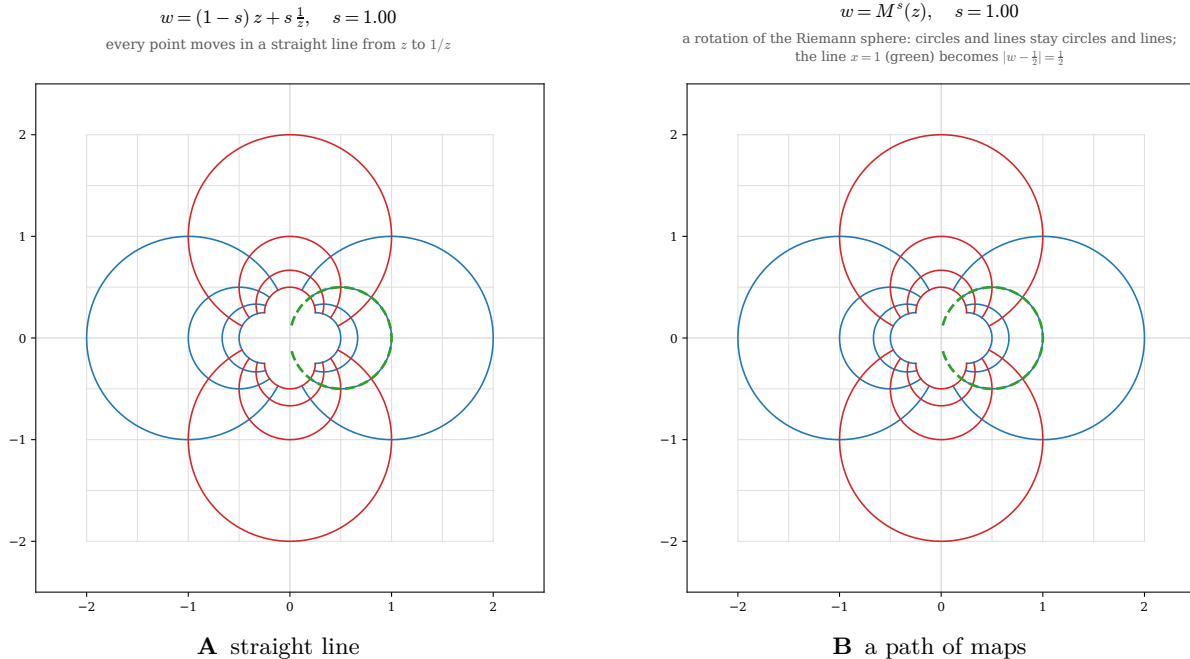
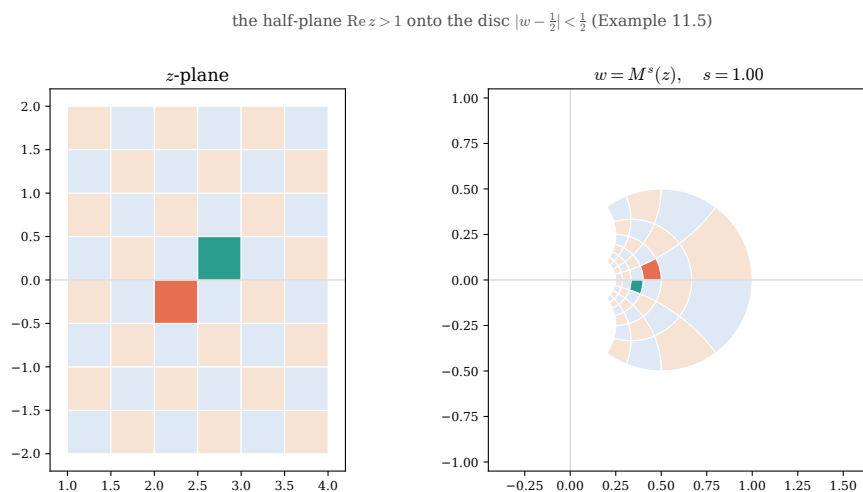


FIGURE 11.3 (animated). $1/z$: circles and lines. **A**: every point moves in a straight line from z to $f(z)$. **B**: a rotation of the Riemann sphere: circles and lines stay circles and lines; the line $x = 1$ (green) becomes $|w - \frac{1}{2}| = \frac{1}{2}$. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 11.3, continued)

11.3 The extended plane and the Riemann sphere

The punctures in Theorem 11.3 (and the fact that the point 0 on the boundary circle in Example 11.5 is the image of no point of the boundary line) are bookkeeping annoyances. They disappear if we add one point.

Definition 11.6 (Extended complex plane). Let ∞ be an object not in $\mathbb{C}$ and set

$$\widehat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}. \quad (11.8)$$

Extend ι to $\widehat{\mathbb{C}}$ by

$$\iota(0) := \infty, \quad \iota(\infty) := 0. \quad (11.9)$$

We declare a set $U \subseteq \widehat{\mathbb{C}}$ to be a *neighbourhood of ∞* if $U \supseteq \{z \in \mathbb{C} : |z| > R\} \cup \{\infty\}$ for some $R > 0$.

Theorem 11.7 (Stereographic projection). *Let*

$$\mathbb{S} := \{(\xi_1, \xi_2, \xi_3) \in \mathbb{R}^3 : \xi_1^2 + \xi_2^2 + \xi_3^2 = 1\} \quad (11.10)$$

be the unit sphere and $N := (0, 0, 1)$ its north pole. Define $\Pi : \mathbb{S} \setminus \{N\} \rightarrow \mathbb{C}$ by

$$\Pi(\xi_1, \xi_2, \xi_3) := \frac{\xi_1 + i\xi_2}{1 - \xi_3}. \quad (11.11)$$

Then Π is a bijection, with inverse

$$\Pi^{-1}(z) = \left(\frac{2 \operatorname{Re} z}{|z|^2 + 1}, \frac{2 \operatorname{Im} z}{|z|^2 + 1}, \frac{|z|^2 - 1}{|z|^2 + 1} \right), \quad (11.12)$$

and setting $\Pi(N) := \infty$ extends Π to a bijection $\mathbb{S} \rightarrow \widehat{\mathbb{C}}$ under which neighbourhoods of N correspond exactly to neighbourhoods of ∞ in the sense of Definition 11.6. Moreover Π carries circles on $\mathbb{S}$ (with N removed, if it lies on them) onto circles-or-lines in $\mathbb{C}$, and every circle or line in $\mathbb{C}$ arises in this way from exactly one circle on $\mathbb{S}$ — lines being precisely the images of circles through N .

Proof. Geometrically, $\Pi(\xi)$ is the point where the straight line from N through ξ meets the plane $\xi_3 = 0$, identified with $\mathbb{C}$; we verify the formulas algebraically.

Let $\xi \in \mathbb{S}$, $\xi \neq N$, and $z = \Pi(\xi)$ as in (11.11). Then

$$|z|^2 = \frac{\xi_1^2 + \xi_2^2}{(1 - \xi_3)^2} = \frac{1 - \xi_3^2}{(1 - \xi_3)^2} = \frac{1 + \xi_3}{1 - \xi_3},$$

using (11.10). Solving for ξ_3 gives $\xi_3 = \frac{|z|^2 - 1}{|z|^2 + 1}$, whence $1 - \xi_3 = \frac{2}{|z|^2 + 1}$, and then (11.11) gives $\xi_1 + i\xi_2 = (1 - \xi_3)z = \frac{2z}{|z|^2 + 1}$. This is (11.12), and it shows Π is injective with the stated inverse; conversely one checks directly that the right side of (11.12) lies on $\mathbb{S}$, so Π is onto.

As $|z| \rightarrow \infty$, (11.12) gives $\xi \rightarrow N$, and quantitatively $1 - \xi_3 = 2/(|z|^2 + 1)$, so $\{|z| > R\}$ corresponds to $\{\xi \in \mathbb{S} : 1 - \xi_3 < 2/(R^2 + 1)\}$, a spherical cap about N . These caps form a neighbourhood basis of N , which proves the correspondence of neighbourhoods.

For circles: a circle on $\mathbb{S}$ is the intersection of $\mathbb{S}$ with a plane $a\xi_1 + b\xi_2 + c\xi_3 = d$ with $a^2 + b^2 + c^2 > d^2$. Substituting (11.12) with $z = x + iy$ and multiplying by $|z|^2 + 1 = x^2 + y^2 + 1$:

$$2ax + 2by + c(x^2 + y^2 - 1) = d(x^2 + y^2 + 1),$$

that is,

$$(c - d)(x^2 + y^2) + 2ax + 2by - (c + d) = 0,$$

which is of the form (2.6); it is a line exactly when $c = d$, i.e. exactly when $N = (0, 0, 1)$ satisfies the plane equation, i.e. when the circle passes through N . Since $|z|^2 + 1 \neq 0$, these steps are

reversible: for $z \in \mathbb{C}$, the point $\Pi^{-1}(z)$ lies on the plane if and only if z satisfies the last equation. As Π is a bijection of $\mathbb{S} \setminus \{N\}$ onto $\mathbb{C}$, the image of the circle (minus N) is the *whole* set

$$\{x + iy : (c - d)(x^2 + y^2) + 2ax + 2by - (c + d) = 0\}.$$

Conversely, let a circle or line be given by (2.6), $A(x^2 + y^2) + Bx + Cy + D = 0$; by Proposition 2.6 (and its proof) it is a circle or a line exactly when $B^2 + C^2 > 4AD$. Put $a = \frac{B}{2}$, $b = \frac{C}{2}$, $c = \frac{A-D}{2}$, $d = -\frac{A+D}{2}$, so that $c - d = A$, $2a = B$, $2b = C$, $-(c + d) = D$. Then

$$a^2 + b^2 + c^2 - d^2 = \frac{1}{4}(B^2 + C^2 + (A - D)^2 - (A + D)^2) = \frac{1}{4}(B^2 + C^2 - 4AD) > 0,$$

so the plane $a\xi_1 + b\xi_2 + c\xi_3 = d$ is at distance $|d|/\sqrt{a^2 + b^2 + c^2} < 1$ from the origin and meets $\mathbb{S}$ in a circle, whose image under Π is the given circle or line by what we have just proved. It is the only such circle, since Π is a bijection and a circle on $\mathbb{S}$ is determined by any three of its points. $\square$

a great circle tilted 51° from the equator, and its stereographic image
image of the great circle $m \cdot \xi = 0$: the circle $|w + (m_1 + im_2)/m_3| = 1/|m_3|$, a line through 0 when $m_3 = 0$

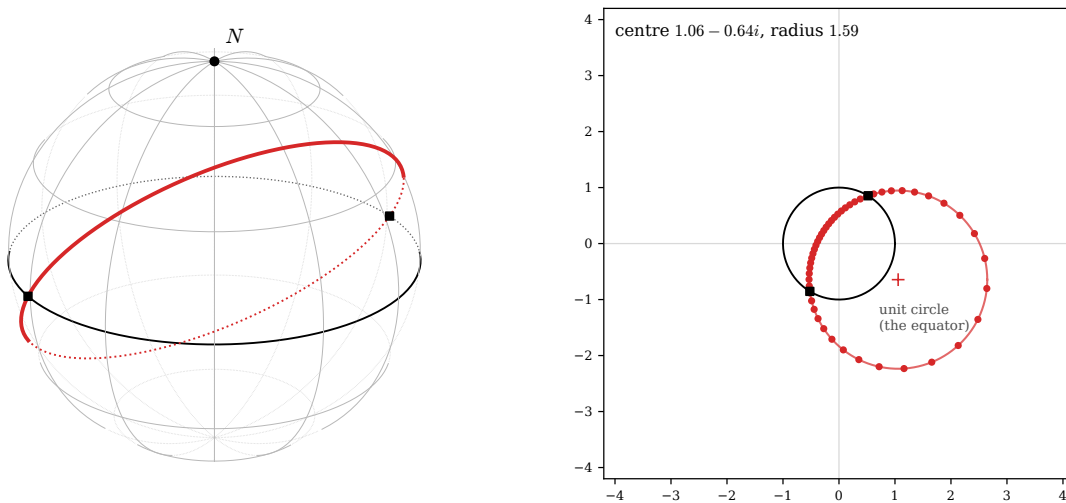


FIGURE 11.4 (animated). Theorem 11.7 for great circles. The great circle with unit normal $m = (m_1, m_2, m_3)$, i.e. $\{\xi \in \mathbb{S} : m \cdot \xi = 0\}$, projects to the circle $|w + (m_1 + im_2)/m_3| = 1/|m_3|$, or to a line through 0 when $m_3 = 0$ (the great circle passes through N); substitute (11.12) into $m \cdot \xi = 0$ and complete the square. Left: a great circle tilting from 8° to 90° . Right: its projected points (dots) on the circle given by this formula; it crosses the unit circle, the image of the equator, at the two opposite points where the great circle crosses the equator.

Corollary 11.8. *On $\widehat{\mathbb{C}}$ the extended map ι of (11.9) is a bijection that carries the family of circles-and-lines — equivalently, by Theorem 11.7, the family of circles on the Riemann sphere — onto itself, without exceptions or punctures.*

Proof. Combine Theorem 11.3 with the observation that adjoining ∞ repairs each of the four cases: in (ii) the point $0 \in S$ maps to ∞ , which lies on every line viewed in $\widehat{\mathbb{C}}$; in (iii) and (iv) the missing point 0 is the image of $\infty \in S$. $\square$

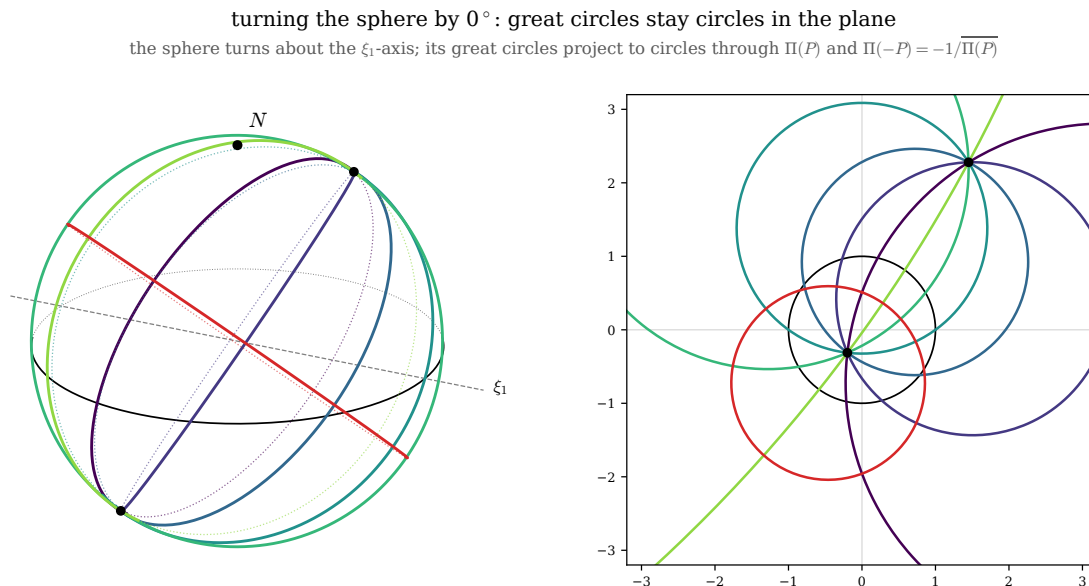


FIGURE 11.5 (animated). Rotations of the sphere keep circles circles. A net of great circles, six through two opposite points P and $-P$ and the one halfway between them, turns rigidly about the ξ_1 -axis; in the plane every one of them stays a circle or a line, the six all through $\Pi(P)$ and $\Pi(-P) = -1/\overline{\Pi(P)}$. Turning by 180° about the ξ_1 -axis is $z \mapsto 1/\bar{z}$ (Exercise 11.3); every rotation of the sphere acts on the plane as a linear fractional map (Chapter 34).

Remark 11.9 (Behaviour at 0 and at ∞). Under ι , small circles about 0 become large circles about 0: $|z| = \epsilon \mapsto |w| = 1/\epsilon$. In the language of Chapter 12 this reads $\lim_{z \rightarrow 0} 1/z = \infty$; in the language of Chapter 26, ι has a *pole* at 0. On the sphere there is nothing singular happening: the pole is just the point N , and ι is a rotation of $\mathbb{S}$ by π about the ξ_1 -axis (Exercise 11.3).

11.4 Exercises

Exercise 11.1. Find and sketch the images under $w = 1/z$ of: the line $y = 2$; the line $y = x$; the circle $|z - 1| = 1$; the circle $|z - 2| = 1$; the half-plane $\operatorname{Im} z > 1$.

Exercise 11.2. Show that $w = 1/z$ maps the annulus $1 < |z| < 2$ onto the annulus $\frac{1}{2} < |w| < 1$, and describe what it does to the rays inside.

Exercise 11.3. ($\star$) Using (11.12), show that $z \mapsto 1/\bar{z}$ corresponds on $\mathbb{S}$ to the reflection $(\xi_1, \xi_2, \xi_3) \mapsto (\xi_1, \xi_2, -\xi_3)$, and hence that $z \mapsto 1/z$ corresponds to $(\xi_1, \xi_2, \xi_3) \mapsto (\xi_1, -\xi_2, -\xi_3)$, a rotation of the sphere by π about the ξ_1 -axis.

Exercise 11.4. Let $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$. Prove that the *Möbius transformation* $T(z) = \frac{az + b}{cz + d}$ is, when $c \neq 0$, the composition

$$z \mapsto cz + d \mapsto \frac{1}{cz + d} \mapsto \frac{bc - ad}{c} \cdot \frac{1}{cz + d} + \frac{a}{c} = T(z),$$

of two linear maps and one reciprocal. Deduce from Theorems 8.4 and 11.3 that T maps circles-and-lines to circles-and-lines.

Exercise 11.5. Compute the image under stereographic projection of the equator, of the meridian $\xi_2 = 0$, and of the circle $\xi_3 = \frac{1}{2}$.

Exercise 11.6. Show that $\iota(z) = 1/z$ has exactly two fixed points in $\mathbb{C}$, and determine them. What are the fixed points of σ of (11.3), and how do the two answers differ?

Exercise 11.7. Using (11.12), compute the *chordal distance* $|\Pi^{-1}(z) - \Pi^{-1}(w)|$ between the sphere points corresponding to $z, w \in \mathbb{C}$, and show it equals

$$\frac{2|z - w|}{\sqrt{1 + |z|^2}\sqrt{1 + |w|^2}}.$$

Deduce that $z_n \rightarrow \infty$ in the sense of Definition 11.6 exactly when the corresponding sphere points converge to the north pole.

Exercise 11.8. (★) Prove that $1/z$ maps the family of circles-and-lines through two fixed points $a \neq b$ in $\mathbb{C} \setminus \{0\}$ onto the family of circles-and-lines through $1/a$ and $1/b$, and that it maps the orthogonal family (the Apollonius circles of Exercise 2.2) onto the corresponding orthogonal family. (Use Corollary 34.3 and conformality, Proposition 34.4.)

Part III

Limits, Continuity, and Analyticity

Chapter 12

Limits and Continuity

12.1 Limits

Definition 12.1 (Limit of a complex function). Let $S \subseteq \mathbb{C}$, let $f : S \rightarrow \mathbb{C}$, let $z_0 \in \mathbb{C}$ be an accumulation point of S in the sense of Definition 5.14, and let $L \in \mathbb{C}$. We say

$$\lim_{z \rightarrow z_0} f(z) = L \quad (12.1)$$

if for every real $\varepsilon > 0$ there exists a real $\delta > 0$ such that

$$z \in S \text{ and } 0 < |z - z_0| < \delta \implies |f(z) - L| < \varepsilon. \quad (12.2)$$

In the language of Definition 5.1: for every $\varepsilon > 0$ there is $\delta > 0$ with $f(S \cap D'(z_0, \delta)) \subseteq D(L, \varepsilon)$.

The definition is word-for-word the real one with $|\cdot|$ now meaning the modulus (1.21). What changes is the geometry: z may approach z_0 from infinitely many directions and along arbitrarily wild paths, and (12.2) demands the same L for all of them. This makes complex limits substantially more restrictive than limits of functions of one real variable.

Theorem 12.2 (Uniqueness of limits). *Under the hypotheses of Definition 12.1, if $\lim_{z \rightarrow z_0} f(z) = L_1$ and $\lim_{z \rightarrow z_0} f(z) = L_2$ then $L_1 = L_2$.*

Proof. Suppose $L_1 \neq L_2$ and put $\varepsilon := \frac{1}{2}|L_1 - L_2| > 0$. Choose $\delta_1, \delta_2 > 0$ as in (12.2) for L_1, L_2 respectively, and let $\delta = \min(\delta_1, \delta_2)$. Since z_0 is an accumulation point of S there exists $z \in S \cap D'(z_0, \delta)$. For that z ,

$$|L_1 - L_2| \leq |L_1 - f(z)| + |f(z) - L_2| < \varepsilon + \varepsilon = |L_1 - L_2|,$$

by the triangle inequality (1.27), a contradiction. $\square$

Theorem 12.3 (Componentwise criterion). *Let $f = u + iv$ as in (7.3), let $z_0 = x_0 + iy_0$ and $L = L_1 + iL_2$ with $x_0, y_0, L_1, L_2 \in \mathbb{R}$. Then (12.1) holds if and only if*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} u(x,y) = L_1 \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} v(x,y) = L_2, \quad (12.3)$$

these being limits of real functions of two real variables.

Proof. ($\Rightarrow$) By (1.26), $|u(x, y) - L_1| = |\operatorname{Re}(f(z) - L)| \leq |f(z) - L|$, so any δ that works for f works for u ; similarly for v .

($\Leftarrow$) Given $\varepsilon > 0$ choose δ_1 for u with tolerance $\varepsilon/2$ and δ_2 for v with tolerance $\varepsilon/2$, and set $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |z - z_0| < \delta$, by the triangle inequality,

$$|f(z) - L| = |(u - L_1) + i(v - L_2)| \leq |u - L_1| + |v - L_2| < \varepsilon. \quad \square$$

Theorem 12.4 (Algebra of limits). *Suppose $\lim_{z \rightarrow z_0} f(z) = L$ and $\lim_{z \rightarrow z_0} g(z) = M$, with f, g defined on a common set S having z_0 as accumulation point. Then*

$$\lim_{z \rightarrow z_0} (f(z) + g(z)) = L + M, \quad (12.4)$$

$$\lim_{z \rightarrow z_0} (f(z)g(z)) = LM, \quad (12.5)$$

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{L}{M} \quad \text{provided } M \neq 0. \quad (12.6)$$

Moreover $\lim_{z \rightarrow z_0} \overline{f(z)} = \overline{L}$ and $\lim_{z \rightarrow z_0} |f(z)| = |L|$.

Proof. (12.4): given $\varepsilon > 0$ pick δ_1, δ_2 for f, g with tolerance $\varepsilon/2$, and set $\delta = \min(\delta_1, \delta_2)$. Then

$$|(f + g) - (L + M)| \leq |f - L| + |g - M| < \varepsilon.$$

(12.5): write

$$f(z)g(z) - LM = (f(z) - L)g(z) + L(g(z) - M).$$

Choose δ_0 so that $|g(z) - M| < 1$, hence $|g(z)| < |M| + 1$, for $0 < |z - z_0| < \delta_0$ (possible by (1.28)). Given $\varepsilon > 0$, choose δ_1 with $|f(z) - L| < \frac{\varepsilon}{2(|M|+1)}$ and δ_2 with $|g(z) - M| < \frac{\varepsilon}{2(|L|+1)}$. For $\delta = \min(\delta_0, \delta_1, \delta_2)$ and $0 < |z - z_0| < \delta$, (1.24) and (1.27) give

$$|fg - LM| \leq |f - L||g| + |L||g - M| < \frac{\varepsilon}{2(|M|+1)}(|M|+1) + |L|\frac{\varepsilon}{2(|L|+1)} < \varepsilon.$$

(12.6): by (12.5) it suffices to show $\lim 1/g = 1/M$. Since $M \neq 0$, choose δ_0 with $|g(z) - M| < |M|/2$, hence $|g(z)| > |M|/2$ by (1.28), for $0 < |z - z_0| < \delta_0$. Then

$$\left| \frac{1}{g(z)} - \frac{1}{M} \right| = \frac{|M - g(z)|}{|g(z)||M|} \leq \frac{2}{|M|^2} |g(z) - M|,$$

which is $< \varepsilon$ once $|g(z) - M| < \varepsilon |M|^2/2$.

The last two statements follow from $|\overline{f} - \overline{L}| = |f - L|$ (by (1.23)) and $||f| - |L|| \leq |f - L|$ (by (1.28)). $\square$

Corollary 12.5 (Limits of polynomials and rational functions). *Let p, q be polynomials with complex coefficients. Then $\lim_{z \rightarrow z_0} p(z) = p(z_0)$ for every $z_0 \in \mathbb{C}$, and $\lim_{z \rightarrow z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q(z_0)}$ whenever $q(z_0) \neq 0$.*

Proof. $\lim_{z \rightarrow z_0} z = z_0$ (take $\delta = \varepsilon$) and $\lim_{z \rightarrow z_0} c = c$ for constants. Apply (12.5) repeatedly to get $\lim z^k = z_0^k$, then (12.4), then (12.6). $\square$

12.2 Showing a limit does not exist

Proposition 12.6 (Path test). *Let z_0 be an accumulation point of S and $f : S \rightarrow \mathbb{C}$. Suppose $\gamma_1, \gamma_2 : (0, 1] \rightarrow S$ satisfy $\gamma_j(t) \rightarrow z_0$ as $t \downarrow 0$ and $\gamma_j(t) \neq z_0$. If*

$$\lim_{t \downarrow 0} f(\gamma_1(t)) \neq \lim_{t \downarrow 0} f(\gamma_2(t)) \quad (12.7)$$

(both limits existing), then $\lim_{z \rightarrow z_0} f(z)$ does not exist.

Proof. If the limit existed and equalled L , then for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$. Since $\gamma_j(t) \rightarrow z_0$, for small t we have $0 < |\gamma_j(t) - z_0| < \delta$, so $|f(\gamma_j(t)) - L| < \varepsilon$; letting $t \downarrow 0$ gives both path limits equal to L , contradicting (12.7). $\square$

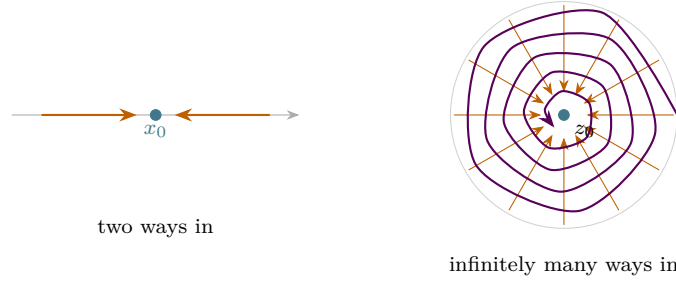


FIGURE 12.1. Why complex limits are harder than one-variable real ones. On the left, $x \rightarrow x_0$ along a line: there are two directions. On the right, $z \rightarrow z_0$ in the plane: there are infinitely many directions and, worse, infinitely many curved approaches. The definition (12.2) demands the *same* limit L for every one of them, which is why Proposition 12.6 is such an effective way to prove a limit does not exist.

Example 12.7 (The model non-existent limit). Let $f(z) = \frac{\bar{z}}{z}$ for $z \neq 0$. Along the real axis, $\gamma_1(t) = t$, $f(\gamma_1(t)) = t/t = 1$. Along the imaginary axis, $\gamma_2(t) = it$, $f(\gamma_2(t)) = \frac{-it}{it} = -1$. Since $1 \neq -1$, $\lim_{z \rightarrow 0} \bar{z}/z$ does not exist. In fact along the ray $\gamma_\theta(t) = te^{i\theta}$ one gets $f = e^{-2i\theta}$: every unimodular value is a directional limit.

Example 12.8. $\lim_{z \rightarrow 0} \frac{\operatorname{Re} z}{|z|}$ does not exist: along $te^{i\theta}$ the value is $\cos \theta$, which depends on θ . By contrast $\lim_{z \rightarrow 0} \frac{(\operatorname{Re} z)^2}{|z|} = 0$, since $\left| \frac{(\operatorname{Re} z)^2}{|z|} \right| \leq \frac{|z|^2}{|z|} = |z| \rightarrow 0$ by (1.26); here one $\delta = \varepsilon$ works.

Definition 12.9 (Limits involving ∞). Let $f : S \rightarrow \mathbb{C}$. In (12.8) let $z_0 \in \mathbb{C}$ be an accumulation point of S ; in (12.9) let S contain $\{|z| > R_0\}$ for some $R_0 > 0$. We write:

$$\lim_{z \rightarrow z_0} f(z) = \infty \iff \forall M > 0 \exists \delta > 0 : z \in S, 0 < |z - z_0| < \delta \Rightarrow |f(z)| > M; \quad (12.8)$$

$$\lim_{z \rightarrow \infty} f(z) = L \iff \forall \varepsilon > 0 \exists R > 0 : |z| > R \Rightarrow |f(z) - L| < \varepsilon. \quad (12.9)$$

Equivalently, using $\iota(z) = 1/z$: $\lim_{z \rightarrow z_0} f(z) = \infty$ iff $\lim_{z \rightarrow z_0} 1/f(z) = 0$, and $\lim_{z \rightarrow \infty} f(z) = L$ iff $\lim_{\zeta \rightarrow 0} f(1/\zeta) = L$. These are the ordinary limits of Definition 12.1 transported to the Riemann sphere of Theorem 11.7.

Example 12.10. For $f(z) = \frac{2z^3 - z + 1}{5z^3 + z^2}$ we have, substituting $z = 1/\zeta$ and simplifying, $f(1/\zeta) = \frac{2 - \zeta^2 + \zeta^3}{5 + \zeta} \rightarrow \frac{2}{5}$ as $\zeta \rightarrow 0$, so $\lim_{z \rightarrow \infty} f(z) = \frac{2}{5}$. And $\lim_{z \rightarrow 0} \frac{1}{z^2} = \infty$ since $|1/z^2| = 1/|z|^2 > M$ as soon as $|z| < 1/\sqrt{M}$.

12.3 Continuity

Definition 12.11 (Continuity). Let $f : S \rightarrow \mathbb{C}$ and $z_0 \in S$. Then f is *continuous at z_0* if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$z \in S, |z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon. \quad (12.10)$$

If z_0 is an accumulation point of S this is equivalent to

$$\lim_{z \rightarrow z_0} f(z) = f(z_0), \quad (12.11)$$

i.e. to the three requirements: $f(z_0)$ is defined, the limit exists, and the two agree. f is *continuous on S* if it is continuous at every point of S .

Theorem 12.12 (Continuity toolkit). (i) f is continuous at z_0 if and only if its components u, v of (7.2) are continuous at (x_0, y_0) .

(ii) Sums, products, and quotients (with nonvanishing denominator) of functions continuous at z_0 are continuous at z_0 .

(iii) If f is continuous at z_0 and g is continuous at $f(z_0)$, then $g \circ f$ is continuous at z_0 .

(iv) $z \mapsto z, z \mapsto \bar{z}, z \mapsto |z|, z \mapsto \operatorname{Re} z, z \mapsto \operatorname{Im} z$ and all constants are continuous on $\mathbb{C}$; hence every polynomial is continuous on $\mathbb{C}$ and every rational function is continuous off the zeros of its denominator.

(v) f is continuous at z_0 if and only if $f(z_n) \rightarrow f(z_0)$ for every sequence $z_n \in S$ with $z_n \rightarrow z_0$.

Proof. (i) is Theorem 12.3 with $L = f(z_0)$; (ii) is Theorem 12.4; (iv) follows from (ii), from $|\bar{z} - \bar{z}_0| = |z - z_0|$, from (1.28) $||z| - |z_0|| \leq |z - z_0|$, and from (1.26); each of these lets us take $\delta = \varepsilon$.

(iii): let $\varepsilon > 0$. Continuity of g at $w_0 := f(z_0)$ gives $\eta > 0$ with $|g(w) - g(w_0)| < \varepsilon$ whenever $|w - w_0| < \eta$. Continuity of f at z_0 gives $\delta > 0$ with $|f(z) - w_0| < \eta$ whenever $|z - z_0| < \delta$. Composing, $|g(f(z)) - g(f(z_0))| < \varepsilon$ for $|z - z_0| < \delta$.

(v) ($\implies$) Given $z_n \rightarrow z_0$ and $\varepsilon > 0$, take δ from (12.10) and then N with $|z_n - z_0| < \delta$ for $n \geq N$; then $|f(z_n) - f(z_0)| < \varepsilon$ for $n \geq N$. ($\impliedby$) If f is not continuous at z_0 , there is $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$ some $z_n \in S$ has $|z_n - z_0| < 1/n$ but $|f(z_n) - f(z_0)| \geq \varepsilon_0$. Then $z_n \rightarrow z_0$ while $f(z_n) \not\rightarrow f(z_0)$. $\square$

Theorem 12.13 (Extreme value theorem for $|f|$). Let $K \subseteq \mathbb{C}$ be compact (closed and bounded, Definition 5.12) and nonempty, and let $f : K \rightarrow \mathbb{C}$ be continuous. Then $|f|$ is bounded on K and attains a maximum and a minimum: there exist $z_{\max}, z_{\min} \in K$ with

$$|f(z_{\min})| \leq |f(z)| \leq |f(z_{\max})| \quad \text{for all } z \in K. \quad (12.12)$$

Proof. Let $M := \sup_{z \in K} |f(z)| \in [0, \infty]$ and choose $z_n \in K$ with $|f(z_n)| \rightarrow M$. By Theorem 5.13 there is a subsequence $z_{n_k} \rightarrow z_* \in K$. By Theorem 12.12(v) and continuity of $|\cdot|$, $|f(z_{n_k})| \rightarrow |f(z_*)|$, so $M = |f(z_*)|$ is finite and attained; take $z_{\max} = z_*$. The minimum is obtained the same way with $\inf$ in place of $\sup$. $\square$

Remark 12.14. Theorem 12.13 is used later to prove the maximum modulus principle and (together with Liouville's theorem, Theorem 22.2) the Fundamental Theorem of Algebra, Theorem 22.4. Note that the theorem concerns $|f|$, a real-valued function: by Warning 1.8 there is no such thing as “the maximum of f ” when f is complex-valued.

12.4 Exercises

Exercise 12.1. Evaluate or show the limit does not exist:

$$\lim_{z \rightarrow i} \frac{z^2 + 1}{z - i}, \quad \lim_{z \rightarrow 0} \frac{|z|^2}{z}, \quad \lim_{z \rightarrow 0} \frac{z}{\bar{z}}, \quad \lim_{z \rightarrow 0} \frac{\operatorname{Im}(z^2)}{|z|^2}, \quad \lim_{z \rightarrow \infty} \frac{z^2 + 1}{z^2 - 3z}.$$

Exercise 12.2. Prove directly from Definition 12.1 that $\lim_{z \rightarrow z_0} z^2 = z_0^2$, exhibiting δ explicitly in terms of ε and $|z_0|$.

Exercise 12.3. Let $f(z) = \operatorname{Arg} z$ on $\mathbb{C} \setminus \{0\}$. At which points is f continuous? Prove your claim using (10.3)-style one-sided limits.

Exercise 12.4. (★) Let f be continuous on the compact set K . Prove f is *uniformly* continuous: for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(z) - f(w)| < \varepsilon$ for *all* $z, w \in K$ with $|z - w| < \delta$.

Exercise 12.5. Show that $f(z) = \frac{\bar{z}^2}{|z|}$ (with $f(0) = 0$) is continuous at 0 but that $\frac{\bar{z}^2}{|z|^2}$ is not.

Exercise 12.6. Suppose f is continuous on a region D and $f(z) \neq 0$ for all $z \in D$. Prove that $1/f$ is continuous on D , and that $|f|$ has a positive lower bound on every compact subset of D .

Exercise 12.7. Show that $\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z) \operatorname{Im}(z)}{|z|^2}$ does not exist, but that $\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z) \operatorname{Im}(z)}{|z|} = 0$. In each case give either the two paths or the explicit $\delta(\varepsilon)$.

Exercise 12.8. Let $f(z) = \frac{az + b}{cz + d}$ with $ad - bc \neq 0$ and $c \neq 0$. Compute $\lim_{z \rightarrow -d/c} f(z)$ and $\lim_{z \rightarrow \infty} f(z)$ in the sense of Definition 12.9, and check the answers against (34.3).

Exercise 12.9. (★) Prove that if f is continuous on $\mathbb{C}$ and $\lim_{z \rightarrow \infty} f(z) = L$ exists in $\mathbb{C}$, then f is *uniformly* continuous on $\mathbb{C}$ and $f(\mathbb{C}) \cup \{L\}$ is compact. (Extend f to the Riemann sphere of Theorem 11.7 and use Exercise 12.4.)

Chapter 13

Differentiability and Analyticity

13.1 The complex derivative

Definition 13.1 (Derivative). Let f be defined on a set containing a neighbourhood $D(z_0, \varepsilon)$ of the point $z_0 \in \mathbb{C}$. We say f is *differentiable at z_0* if the limit

$$f'(z_0) := \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \quad (13.1)$$

exists in $\mathbb{C}$, in the sense of Definition 12.1 applied to the difference quotient as a function of the complex variable Δz on the punctured disc $D'(0, \varepsilon)$. The number $f'(z_0)$ is the *derivative* of f at z_0 . Equivalently, writing $z = z_0 + \Delta z$,

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}. \quad (13.2)$$

The formula is identical to the real one, but the requirement is far stronger: the quotient must tend to one and the same complex number $f'(z_0)$ *no matter how $\Delta z \rightarrow 0$ in the plane*. Chapter 14 extracts the exact consequence of that requirement.

Proposition 13.2 (Differentiable $\Rightarrow$ continuous). *If f is differentiable at z_0 then f is continuous at z_0 .*

Proof. For $z \neq z_0$ write $f(z) - f(z_0) = \frac{f(z) - f(z_0)}{z - z_0} \cdot (z - z_0)$. By (12.5) the right side tends to $f'(z_0) \cdot 0 = 0$, so $\lim_{z \rightarrow z_0} f(z) = f(z_0)$, which is (12.11). $\square$

Theorem 13.3 (Differentiation rules). *Let f, g be differentiable at z_0 and $c \in \mathbb{C}$. Then cf , $f + g$, fg are differentiable at z_0 , as is f/g if $g(z_0) \neq 0$, and*

$$(cf)'(z_0) = c f'(z_0), \quad (f + g)'(z_0) = f'(z_0) + g'(z_0), \quad (13.3)$$

$$(fg)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0), \quad (13.4)$$

$$\left(\frac{f}{g}\right)'(z_0) = \frac{f'(z_0)g(z_0) - f(z_0)g'(z_0)}{g(z_0)^2}. \quad (13.5)$$

If moreover g is differentiable at z_0 and f is differentiable at $g(z_0)$, then $f \circ g$ is differentiable at z_0 with

$$(f \circ g)'(z_0) = f'(g(z_0)) g'(z_0). \quad (13.6)$$

Finally, for every $n \in \mathbb{Z}$,

$$\frac{d}{dz} z^n = n z^{n-1} \quad (z \neq 0 \text{ if } n < 0). \quad (13.7)$$

Proof. All proofs are the real ones verbatim, since they use only the field axioms of $\mathbb{C}$ (Theorem 1.2) and the algebra of limits (Theorem 12.4). We give two.

Product rule. For $z \neq z_0$,

$$\frac{f(z)g(z) - f(z_0)g(z_0)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} g(z) + f(z_0) \frac{g(z) - g(z_0)}{z - z_0}.$$

Let $z \rightarrow z_0$; by Proposition 13.2 $g(z) \rightarrow g(z_0)$, and (12.4), (12.5) give (13.4).

Chain rule. Define $h : \mathbb{C} \rightarrow \mathbb{C}$ near $w_0 := g(z_0)$ by

$$h(w) := \begin{cases} \frac{f(w) - f(w_0)}{w - w_0}, & w \neq w_0, \\ f'(w_0), & w = w_0. \end{cases}$$

Then h is continuous at w_0 by (13.2), and $f(w) - f(w_0) = h(w)(w - w_0)$ holds for all w near w_0 including $w = w_0$. Substituting $w = g(z)$ and dividing by $z - z_0$,

$$\frac{f(g(z)) - f(g(z_0))}{z - z_0} = h(g(z)) \cdot \frac{g(z) - g(z_0)}{z - z_0}.$$

As $z \rightarrow z_0$, $g(z) \rightarrow g(z_0) = w_0$ (Proposition 13.2), so $h(g(z)) \rightarrow h(w_0) = f'(w_0)$ by Theorem 12.12(iii), and the second factor tends to $g'(z_0)$. This is (13.6).

Power rule. For $n \geq 1$ use induction with (13.4) starting from $\frac{d}{dz} z = 1$ (the difference quotient is identically 1). For $n = 0$ the function is constant with derivative 0. For $n < 0$ write $z^n = 1/z^{-n}$ and apply (13.5). $\square$

Example 13.4 (A nowhere differentiable function). Let $f(z) = \bar{z}$. For any z_0 and $\Delta z \neq 0$,

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \frac{\overline{z_0 + \Delta z} - \bar{z}_0}{\Delta z} = \frac{\bar{\Delta z}}{\Delta z}.$$

By Example 12.7 this quotient has no limit as $\Delta z \rightarrow 0$: it equals 1 along the real direction and -1 along the imaginary direction. Hence $\bar{z}$ is differentiable at *no* point of $\mathbb{C}$, even though its components $u = x$, $v = -y$ are as smooth as one could wish. This single example already shows that complex differentiability is not a coordinatewise notion.

Example 13.5 (Differentiable at one point only). Let $f(z) = |z|^2 = z\bar{z}$. Then for $z_0 \in \mathbb{C}$,

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \frac{(z_0 + \Delta z)(\overline{z_0 + \Delta z}) - z_0 \bar{z}_0}{\Delta z} = \bar{z}_0 + \overline{\Delta z} + z_0 \frac{\bar{\Delta z}}{\Delta z}.$$

As $\Delta z \rightarrow 0$ the first two terms tend to $\bar{z}_0$. The third has a limit only if $z_0 = 0$ (again Example 12.7). Hence f is differentiable exactly at $z_0 = 0$, with $f'(0) = 0$.

13.2 Analyticity

Example 13.5 shows that differentiability *at a point* is a weak condition. The useful notion demands differentiability throughout a neighbourhood.

Definition 13.6 (Analytic, entire, singular point). Let $D \subseteq \mathbb{C}$ be open and $f : D \rightarrow \mathbb{C}$.

- f is *analytic at z_0* (synonyms: *holomorphic*, *regular*) if there exists $\varepsilon > 0$ with $D(z_0, \varepsilon) \subseteq D$ such that f is differentiable at *every* point of $D(z_0, \varepsilon)$.
- f is *analytic on D* if it is analytic at every point of D ; equivalently (since D is open) if f is differentiable at every point of D .
- f is *entire* if it is analytic on all of $\mathbb{C}$.
- A point z_1 is a *singular point* (or *singularity*) of f if f is not analytic at z_1 but every neighbourhood of z_1 contains a point at which f is analytic.

Example 13.7. (a) Every polynomial is entire, by Theorem 13.3.

(b) A rational function p/q (with p, q polynomials, $q \neq 0$, and p/q in lowest terms) is analytic on $\mathbb{C}$ minus the zero set of q , and each zero of q is a singular point. (If p and q had a common zero, p/q as written would be undefined there, so that point would also be a singular point in the sense above; such *removable* singularities are studied in Chapter 26.) For instance $f(z) = \frac{z}{(z-i)(z+2)}$ is analytic on $\mathbb{C} \setminus \{i, -2\}$ with singular points i and -2 .

(c) $f(z) = |z|^2$ of Example 13.5 is differentiable at 0 but analytic *nowhere*: no neighbourhood of 0 consists of points of differentiability. This is exactly the distinction Definition 13.6 is designed to make.

(d) $f(z) = \bar{z}$ is nowhere differentiable, hence nowhere analytic, and has no singular points in the sense above.

Remark 13.8 (The word “analytic”). In real analysis “analytic” means “locally equal to the sum of its Taylor series”. One of the central theorems of the second part of this course states that for complex functions the two meanings coincide: differentiability on a disc implies infinite differentiability and convergence of the Taylor series there. Nothing remotely like this is true over $\mathbb{R}$ (consider e^{-1/x^2}). We use “analytic” in the sense of Definition 13.6 throughout Part I.

13.3 Conformality

Definition 13.9 (Angle between curves). Let γ_1, γ_2 be smooth curves through z_0 with nonzero tangent vectors $d_1, d_2 \in \mathbb{C} \setminus \{0\}$ at z_0 . The *signed angle from γ_1 to γ_2 at z_0* is any element of $\arg(d_2/d_1)$; it is well defined modulo 2π .

Theorem 13.10 (Analytic maps are conformal where $f' \neq 0$). *Let f be analytic on a neighbourhood of z_0 and suppose $f'(z_0) \neq 0$. Then for any two smooth curves through z_0 with tangents $d_1, d_2 \neq 0$, the image curves have tangents $f'(z_0)d_1$ and $f'(z_0)d_2$ at $f(z_0)$; consequently the signed angle between the curves is unchanged, and lengths near z_0 are scaled by the factor $|f'(z_0)|$.*

Proof. Let $\gamma : (-1, 1) \rightarrow \mathbb{C}$ be smooth with $\gamma(0) = z_0$ and $\gamma'(0) = d \neq 0$, where γ' is the derivative of Definition 6.6. The image curve is $f \circ \gamma$ and, by the chain rule (13.6) adapted to a real parameter (same proof, with Δt real),

$$(f \circ \gamma)'(0) = f'(\gamma(0))\gamma'(0) = f'(z_0) d \neq 0.$$

Hence the image tangents are $f'(z_0)d_1$ and $f'(z_0)d_2$, and

$$\frac{f'(z_0)d_2}{f'(z_0)d_1} = \frac{d_2}{d_1},$$

so by Definition 13.9 the signed angle is preserved. The scaling statement follows from $|f(z) - f(z_0)| = |f'(z_0)||z - z_0| + o(|z - z_0|)$, which is (13.2) multiplied out. $\square$

Theorem 13.10 explains, in one stroke, the orthogonality observed in Proposition 9.3: the Cartesian grid is orthogonal, p_2 is analytic with $p_2'(z) = 2z \neq 0$ off the origin, so the image grid of parabolas is orthogonal too. It also explains why the origin was exceptional: $p_2'(0) = 0$, and there angles double, in accordance with Remark 9.4.

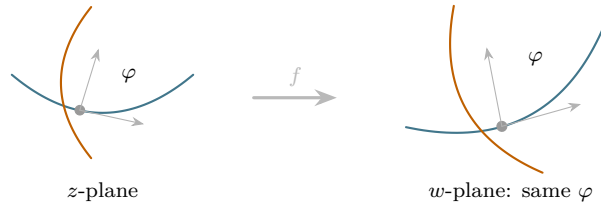


FIGURE 13.1. Conformality, Theorem 13.10. Two curves meeting at z_0 with tangents d_1, d_2 are carried to curves meeting at $f(z_0)$ with tangents $f'(z_0)d_1$ and $f'(z_0)d_2$. Both tangents are turned through the same angle $\arg f'(z_0)$ and stretched by the same factor $|f'(z_0)|$, so the angle φ between them is unchanged — in magnitude and in sign.

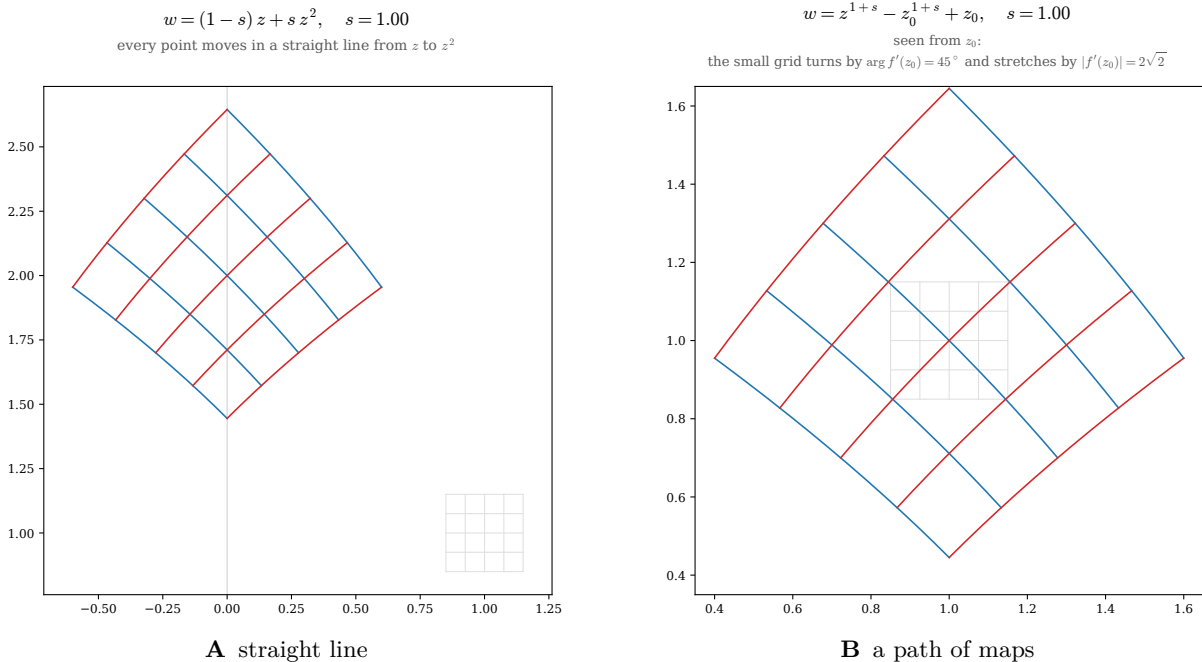
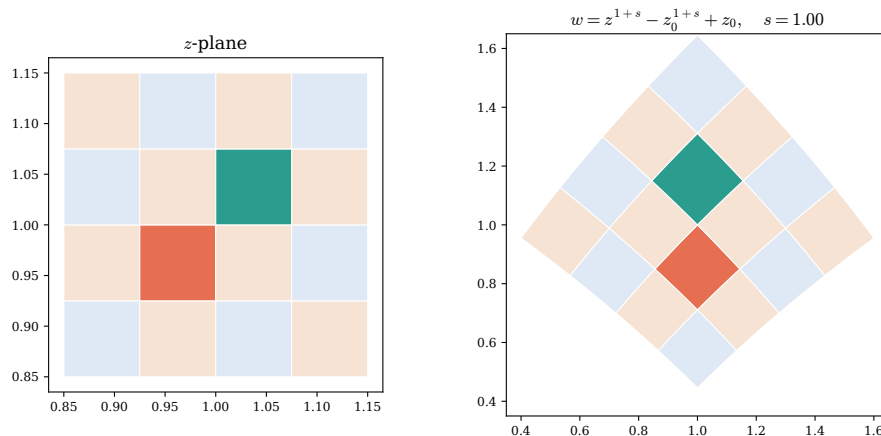


FIGURE 13.2 (animated). Conformality at a point: z^2 near $z_0 = 1 + i$. **A**: every point moves in a straight line from z to $f(z)$. **B**: seen from z_0 : the small grid turns by $\arg f'(z_0) = 45^\circ$ and stretches by $|f'(z_0)| = 2\sqrt{2}$. **C**: the tiles on the left deform along the path of B.

seen from z_0 : the small grid turns by $\arg f'(z_0) = 45^\circ$ and stretches by $|f'(z_0)| = 2\sqrt{2}$



C tiles (Figure 13.2, continued)

Proposition 13.11 (Behaviour at a critical point). *Suppose f is analytic near z_0 , $f'(z_0) = 0$, and f is not constant. If $k \geq 2$ is the least integer with $f^{(k)}(z_0) \neq 0$ (such k exists by the theory of Part II), then f multiplies angles at z_0 by the factor k . For $f = p_n$ and $z_0 = 0$ we have $k = n$, recovering Theorem 9.1(ii).*

Proof sketch. Near z_0 one has $f(z) - f(z_0) = c(z - z_0)^k(1 + o(1))$ with $c = f^{(k)}(z_0)/k! \neq 0$; a curve leaving z_0 in the direction $e^{i\varphi}$ therefore leaves $f(z_0)$ in the direction $\arg c + k\varphi$, so the difference of two such directions is multiplied by k . The existence of k and the local expansion require Taylor's theorem for analytic functions, proved in Part II. $\square$

13.4 Exercises

Exercise 13.1. Differentiate, stating where the result is valid: $f(z) = (z^2 + 3iz)^5$, $g(z) = \frac{z-1}{z^2+1}$, $h(z) = \left(\frac{2z+i}{z-3}\right)^4$.

Exercise 13.2. Using Definition 13.1 directly, determine all points at which $f(z) = z \operatorname{Re} z$ is differentiable, and compute f' there. (*Answer: only $z = 0$.*)

Exercise 13.3. Show that $f(z) = \operatorname{Re} z$ is nowhere differentiable, and that $f(z) = \operatorname{Im} z$ is nowhere differentiable, although $f = \operatorname{Re} z + i \operatorname{Im} z = z$ is entire.

Exercise 13.4. Let f be analytic on a region D and suppose $f(D) \subseteq \mathbb{R}$. Anticipating Theorem 14.13, guess what must be true of f and test your guess on $D = \mathbb{C}$, $f(z) = \operatorname{Re} z$ — why is this not a counterexample?

Exercise 13.5. ($\star$) Prove that if f is differentiable at z_0 then

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \eta(z)(z - z_0), \quad \lim_{z \rightarrow z_0} \eta(z) = 0,$$

and conversely. This “little-o” form is the cleanest statement of differentiability and is what makes Theorem 13.10 work.

Exercise 13.6. Find the points where $f(z) = z^3 - 3z$ fails to be conformal, and compute the angle multiplication factor there.

Exercise 13.7. Determine all points at which $f(z) = z^2\bar{z}$ is differentiable, and compute f' there. Is f analytic anywhere? (Write $f = |z|^2 z$ and use Example 13.5.)

Exercise 13.8. (After Chapter 14.) Let f be analytic at z_0 with $f'(z_0) = 2 + 2i$. By what factor are short segments through z_0 stretched, and through what angle are they turned? By what factor are small areas near z_0 magnified? (Use (14.11).)

Exercise 13.9. (★) Suppose f and g are analytic on a region D and $fg \equiv 0$ there. Prove that $f \equiv 0$ or $g \equiv 0$. (Zeros of a nonzero analytic function are isolated, Theorem 24.7; so the ring of analytic functions on a region has no zero divisors.) Show by example that this fails on a disconnected open set.

Chapter 14

The Cauchy–Riemann Equations

Complex differentiability at z_0 requires the difference quotient (13.1) to have the same limit along every approach. Comparing just two approaches — horizontal and vertical — already produces a pair of partial differential equations. Remarkably, those equations plus mild regularity are also *sufficient*.

14.1 Necessity

Theorem 14.1 (Cauchy–Riemann equations: necessity). *Let $f = u + iv$ be differentiable at $z_0 = x_0 + iy_0$. Then the four first-order partial derivatives u_x, u_y, v_x, v_y exist at (x_0, y_0) and satisfy the Cauchy–Riemann equations*

$$u_x(x_0, y_0) = v_y(x_0, y_0), \quad u_y(x_0, y_0) = -v_x(x_0, y_0). \quad (14.1)$$

Moreover the derivative is given by either of

$$f'(z_0) = u_x(x_0, y_0) + i v_x(x_0, y_0) = v_y(x_0, y_0) - i u_y(x_0, y_0). \quad (14.2)$$

Proof. By hypothesis the limit (13.1) exists; call it $f'(z_0)$. Since the limit is the same along every path, we compute it along two.

Horizontal approach. Take $\Delta z = h$ with $h \in \mathbb{R}$, $h \rightarrow 0$. Then $z_0 + \Delta z = (x_0 + h) + iy_0$ and

$$\begin{aligned} f'(z_0) &= \lim_{h \rightarrow 0} \frac{u(x_0 + h, y_0) + iv(x_0 + h, y_0) - u(x_0, y_0) - iv(x_0, y_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{u(x_0 + h, y_0) - u(x_0, y_0)}{h} + i \lim_{h \rightarrow 0} \frac{v(x_0 + h, y_0) - v(x_0, y_0)}{h}, \end{aligned}$$

where splitting the limit is justified by Theorem 12.3: since the complex limit exists, so do the limits of the real and imaginary parts. The two real limits are by definition $u_x(x_0, y_0)$ and $v_x(x_0, y_0)$; hence these partials exist and

$$f'(z_0) = u_x + i v_x \quad \text{at } (x_0, y_0). \quad (14.3)$$

Vertical approach. Take $\Delta z = ik$ with $k \in \mathbb{R}$, $k \rightarrow 0$. Then $z_0 + \Delta z = x_0 + i(y_0 + k)$ and, using $1/i = -i$,

$$f'(z_0) = \lim_{k \rightarrow 0} \frac{u(x_0, y_0 + k) + iv(x_0, y_0 + k) - u(x_0, y_0) - iv(x_0, y_0)}{ik}$$

$$= -i \lim_{k \rightarrow 0} \frac{u(x_0, y_0 + k) - u(x_0, y_0)}{k} + \lim_{k \rightarrow 0} \frac{v(x_0, y_0 + k) - v(x_0, y_0)}{k},$$

so u_y and v_y exist at (x_0, y_0) and

$$f'(z_0) = v_y - iu_y \quad \text{at } (x_0, y_0). \quad (14.4)$$

Equating real and imaginary parts of (14.3) and (14.4), which is legitimate by (1.5), gives exactly (14.1), and (14.2) is (14.3) together with (14.4). $\square$

Corollary 14.2 (A fast non-differentiability test). *If (14.1) fails at (x_0, y_0) , then f is not differentiable at z_0 , hence not analytic at z_0 . If (14.1) fails at every point of an open set, then f is analytic at no point of that set.*

Example 14.3. For $f(z) = \bar{z}$ we have $u = x$, $v = -y$, so $u_x = 1$ and $v_y = -1$. The first equation of (14.1) fails at every point, so $\bar{z}$ is nowhere differentiable — the conclusion of Example 13.4, now obtained without computing a single limit.

Example 14.4. For $f(z) = |z|^2$ we have $u = x^2 + y^2$, $v = 0$, so $u_x = 2x$, $u_y = 2y$, $v_x = v_y = 0$. The equations (14.1) read $2x = 0$ and $2y = 0$, satisfied only at the origin. Consistent with Example 13.5: differentiable at 0 only, analytic nowhere.

Warning 14.5. The Cauchy–Riemann equations alone do *not* imply differentiability. Define

$$f(z) := \begin{cases} \frac{\bar{z}^2}{z}, & z \neq 0, \\ 0, & z = 0. \end{cases} \quad (14.5)$$

Writing $z = x + iy$ one computes $u = \frac{x^3 - 3xy^2}{x^2 + y^2}$, $v = \frac{y^3 - 3x^2y}{x^2 + y^2}$ for $z \neq 0$. At the origin,

$$u_x(0, 0) = \lim_{h \rightarrow 0} \frac{u(h, 0) - 0}{h} = \lim_{h \rightarrow 0} \frac{h^3/h^2}{h} = 1, \quad v_y(0, 0) = \lim_{k \rightarrow 0} \frac{k^3/k^2}{k} = 1,$$

and similarly $u_y(0, 0) = v_x(0, 0) = 0$; so (14.1) holds at 0. Yet along the ray $z = te^{i\theta}$, $t \downarrow 0$,

$$\frac{f(z) - f(0)}{z} = \frac{\bar{z}^2}{z^2} = e^{-4i\theta},$$

which depends on θ ; so f is *not* differentiable at 0. The missing hypothesis is continuity of the partials, supplied next.

14.2 Sufficiency

Theorem 14.6 (Cauchy–Riemann equations: sufficiency). *Let D be a region and $f = u + iv$ defined on D . Suppose that on D the four partials u_x, u_y, v_x, v_y exist, are continuous, and satisfy (14.1) at every point of D . Then f is analytic on D , and $f' = u_x + iv_x$ there.*

Proof. Fix $z_0 = x_0 + iy_0 \in D$ and choose $r > 0$ with $D(z_0, r) \subseteq D$. Let $\Delta z = h + ik$ with $h, k \in \mathbb{R}$ and $0 < |\Delta z| < r$. Write

$$\Delta u := u(x_0 + h, y_0 + k) - u(x_0, y_0),$$

$$\Delta v := v(x_0 + h, y_0 + k) - v(x_0, y_0).$$

Because u has continuous first partials on the disc, it is (real-)differentiable there, and the increment can be written

$$\Delta u = u_x(x_0, y_0)h + u_y(x_0, y_0)k + \varepsilon_1(h, k)h + \varepsilon_2(h, k)k, \quad (14.6)$$

where $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as $(h, k) \rightarrow (0, 0)$. (This is the standard increment theorem of multivariable calculus: apply the mean value theorem in each variable separately,

$$\Delta u = [u(x_0 + h, y_0 + k) - u(x_0, y_0 + k)] + [u(x_0, y_0 + k) - u(x_0, y_0)] = u_x(\xi, y_0 + k)h + u_y(x_0, \eta)k$$

for some ξ between $x_0, x_0 + h$ and η between $y_0, y_0 + k$, and use continuity of the partials to write $u_x(\xi, y_0 + k) = u_x(x_0, y_0) + \varepsilon_1$, $u_y(x_0, \eta) = u_y(x_0, y_0) + \varepsilon_2$ with $\varepsilon_j \rightarrow 0$.) Similarly

$$\Delta v = v_x h + v_y k + \varepsilon_3 h + \varepsilon_4 k, \quad \varepsilon_3, \varepsilon_4 \rightarrow 0, \quad (14.7)$$

all partials evaluated at (x_0, y_0) .

Now use (14.1) to eliminate u_y and v_y : $u_y = -v_x$ and $v_y = u_x$. Then

$$\begin{aligned} \Delta u + i\Delta v &= (u_x h - v_x k) + i(v_x h + u_x k) + E \\ &= u_x(h + ik) + iv_x(h + ik) + E = (u_x + iv_x)\Delta z + E, \end{aligned}$$

where the error term is $E := (\varepsilon_1 + i\varepsilon_3)h + (\varepsilon_2 + i\varepsilon_4)k$. Since $|h| \leq |\Delta z|$ and $|k| \leq |\Delta z|$ by (1.26),

$$\left| \frac{E}{\Delta z} \right| \leq |\varepsilon_1 + i\varepsilon_3| + |\varepsilon_2 + i\varepsilon_4| \xrightarrow{\Delta z \rightarrow 0} 0.$$

Therefore

$$\lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \left[(u_x + iv_x) + \frac{E}{\Delta z} \right] = u_x(x_0, y_0) + iv_x(x_0, y_0),$$

so f is differentiable at z_0 with the stated derivative. As $z_0 \in D$ was arbitrary and D is open, f is analytic on D . $\square$

Example 14.7. Let $f(z) = x^2 - y^2 + 2ixy$. Then $u = x^2 - y^2$, $v = 2xy$, and

$$u_x = 2x, \quad v_y = 2x, \quad u_y = -2y, \quad v_x = 2y.$$

Both equations of (14.1) hold everywhere, and all four partials are continuous on $\mathbb{C}$; by Theorem 14.6, f is entire with $f' = u_x + iv_x = 2x + 2iy = 2z$. Of course $f(z) = z^2$, so this merely confirms (13.7); the point is that the test is mechanical.

Example 14.8 (Where is it analytic?). Let $f(z) = x^3 + 3xy^2 + i(y^3 + 3x^2y)$. Then

$$u_x = 3x^2 + 3y^2, \quad v_y = 3y^2 + 3x^2, \quad u_y = 6xy, \quad v_x = 6xy.$$

The first CR equation holds everywhere. The second requires $u_y = -v_x$, i.e. $6xy = -6xy$, i.e. $xy = 0$: the union of the two coordinate axes. That set has empty interior, so although f is *differentiable* at every point of the axes, it is analytic *nowhere*. (Differentiability on the axes does not follow from Theorem 14.6, which needs (14.1) on an open set; it follows from Theorem 14.10 below, (ii) $\Rightarrow$ (i), since the partials of u and v are continuous, so that $\tilde{f}$ is real-differentiable by the increment argument in the proof of Theorem 14.6.) This is another instance of the distinction in Definition 13.6.

14.3 The Cauchy–Riemann equations as a differential inclusion

We now restate everything above in the language of Section 1.5. The gain is not merely cosmetic: the reformulation is what makes it possible to say precisely why the theory stops in dimension two.

Definition 14.9 (The real map underlying f). Let $\Omega \subseteq \mathbb{C}$ be open and $f = u + iv : \Omega \rightarrow \mathbb{C}$. Identifying $\mathbb{C}$ with $\mathbb{R}^2$ by $x + iy \leftrightarrow (x, y)^\top$ as in Definition 2.1, write

$$\tilde{f} : \Omega \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \quad \tilde{f}(x, y) := (u(x, y), v(x, y)). \quad (14.8)$$

When u and v are (real-)differentiable at $p = (x_0, y_0)$, the *differential* of $\tilde{f}$ at p is the linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose matrix — the *Jacobian matrix* — is

$$D\tilde{f}(p) = \begin{pmatrix} u_x(p) & u_y(p) \\ v_x(p) & v_y(p) \end{pmatrix} \in \mathbb{R}^{2 \times 2}. \quad (14.9)$$

(Real differentiability at p means $\tilde{f}(p + \mathbf{h}) = \tilde{f}(p) + D\tilde{f}(p)\mathbf{h} + o(|\mathbf{h}|)$; see Appendix A.)

Theorem 14.10 (Cauchy–Riemann is the inclusion $D\tilde{f} \in \overline{CO_+(2)}$). Let $\Omega \subseteq \mathbb{C}$ be open, let $f = u + iv$ on Ω , let $z_0 = x_0 + iy_0 \in \Omega$ and suppose $\tilde{f}$ is real-differentiable at $p = (x_0, y_0)$. Then the following are equivalent:

- (i) f is complex-differentiable at z_0 in the sense of Definition 13.1;
- (ii) the Cauchy–Riemann equations (14.1) hold at p ;
- (iii) $D\tilde{f}(p) \in \mathcal{C}$, the space (1.33);
- (iv) $D\tilde{f}(p)$ satisfies the differential inclusion

$$D\tilde{f}(p) \in \overline{CO_+(2)} = CO_+(2) \cup \{0\}, \quad (14.10)$$

that is, $D\tilde{f}(p)$ is either 0 or an orientation-preserving similarity of $\mathbb{R}^2$.

When these hold,

$$\Phi(f'(z_0)) = D\tilde{f}(p), \quad \text{and consequently} \quad \det D\tilde{f}(p) = |f'(z_0)|^2. \quad (14.11)$$

Hence: f is analytic on Ω if and only if $\tilde{f}$ is differentiable on Ω and satisfies the pointwise inclusion (14.10) everywhere on Ω .

Proof. (ii) $\Leftrightarrow$ (iii). By (14.9), the matrix $D\tilde{f}(p)$ lies in $\mathcal{C}$ of (1.33) — i.e. has the shape $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ — exactly when its diagonal entries agree and its off-diagonal entries are negatives:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x,$$

which is (14.1) verbatim.

(iii) $\Leftrightarrow$ (iv) is (1.36) of Theorem 1.19.

(i) $\Rightarrow$ (ii) is Theorem 14.1.

(ii) $\Rightarrow$ (i). This is where real differentiability is used, and it is exactly the hypothesis that replaces the continuity of the partials in Theorem 14.6. Write $A := D\tilde{f}(p)$ and let $\mathbf{h} \leftrightarrow \Delta z$ under the identification. Real differentiability says

$$\tilde{f}(p + \mathbf{h}) - \tilde{f}(p) = A\mathbf{h} + o(|\mathbf{h}|).$$

By (iii), $A = \Phi(c)$ for some $c \in \mathbb{C}$, and by Theorem 1.19 (iv) the map $\mathbf{h} \mapsto A\mathbf{h}$ is $\Delta z \mapsto c\Delta z$. Translating back to complex notation,

$$f(z_0 + \Delta z) - f(z_0) = c\Delta z + o(|\Delta z|),$$

so the difference quotient $\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = c + \frac{o(|\Delta z|)}{\Delta z} \rightarrow c$, and $f'(z_0) = c$ exists.

Finally $\Phi(f'(z_0)) = \Phi(c) = A = D\tilde{f}(p)$, and taking determinants in (1.38) gives $\det D\tilde{f}(p) = |f'(z_0)|^2$.

For the last sentence of the theorem we also need that complex differentiability implies real differentiability, which (i)–(iv) take as a hypothesis. If f is complex-differentiable at z_0 , then $f(z_0 + \Delta z) - f(z_0) = f'(z_0)\Delta z + o(|\Delta z|)$ (Exercise 13.5), and by Theorem 1.19 (iv) the map $\Delta z \mapsto f'(z_0)\Delta z$ is $\mathbf{h} \mapsto \Phi(f'(z_0))\mathbf{h}$; so $\tilde{f}(p + \mathbf{h}) - \tilde{f}(p) = \Phi(f'(z_0))\mathbf{h} + o(|\mathbf{h}|)$, i.e. $\tilde{f}$ is real-differentiable at p . Hence, if f is analytic on Ω , then $\tilde{f}$ is differentiable on Ω and (i) $\Rightarrow$ (iv) gives the inclusion at every point; conversely, if $\tilde{f}$ is differentiable on Ω and satisfies the inclusion everywhere, (iv) $\Rightarrow$ (i) gives complex differentiability at every point of the open set Ω . $\square$

Remark 14.11 (Reading the inclusion). Three consequences are worth naming.

- (a) *Conformality is now a tautology.* Away from the zeros of f' the inclusion (14.10) says $D\tilde{f}(p)$ is a positive multiple of a rotation, and a similarity preserves angles: that is Theorem 13.10, with no computation.
- (b) *The Jacobian determinant is never negative.* By (14.11), $\det D\tilde{f} = |f'|^2 \geq 0$, so an analytic map never reverses orientation. Contrast $\bar{z}$, whose Jacobian is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ with determinant -1 — conformal but orientation-reversing, hence not in $CO_+(2)$, hence nowhere differentiable (Example 14.3).
- (c) *It is a genuine constraint, not a normalisation.* The set $\overline{CO_+(2)}$ is a two-dimensional subspace of the four-dimensional space $\mathbb{R}^{2 \times 2}$. Requiring $D\tilde{f}(p)$ to lie in it at *every* point is two scalar PDEs on two unknown functions — an elliptic, determined system. That balance is why solutions exist in abundance yet are rigid.

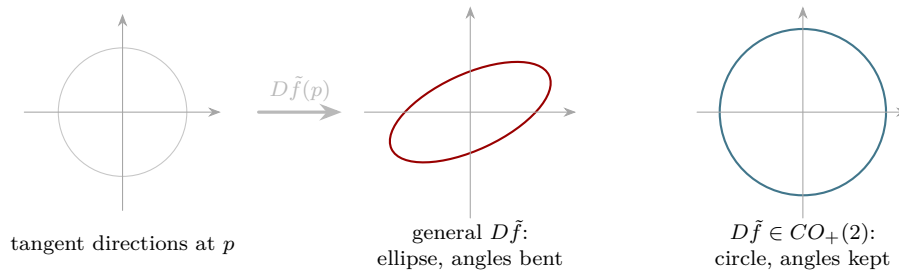


FIGURE 14.1. The inclusion, drawn on the unit circle of tangent directions. Left: a general Jacobian sends the circle to an ellipse — angles are distorted. Right: a Jacobian in $CO_+(2)$ sends it to a circle, so every angle survives and only the scale changes. The Cauchy–Riemann equations say precisely that the right-hand picture holds at every point.

Remark 14.12 (Liouville’s rigidity theorem: why there is no complex analysis in $\mathbb{R}^n$ for $n \geq 3$).

Definition 1.17 makes sense in every dimension. For $n \geq 1$ set

$$CO_+(n) := \{ \lambda R : \lambda > 0, R \in SO(n) \} \subseteq \mathbb{R}^{n \times n}, \quad (14.12)$$

the orientation-preserving similarities of $\mathbb{R}^n$; equivalently $M \in CO_+(n)$ iff $M^\top M = \lambda^2 I$ for some $\lambda > 0$ and $\det M > 0$, the proof of Lemma 1.18 (iii) $\Leftrightarrow$ (iv) being dimension-free. One may therefore pose, on a region $\Omega \subseteq \mathbb{R}^n$, the same differential inclusion as (14.10):

$$g : \Omega \rightarrow \mathbb{R}^n, \quad Dg(x) \in CO_+(n) \quad \text{for all } x \in \Omega. \quad (14.13)$$

For $n = 2$, Theorem 14.10 says the solutions of (14.13) are exactly the analytic functions with nowhere-vanishing derivative — an *infinite-dimensional* family, as rich as the subject we are studying. For $n \geq 3$ the situation collapses completely.

Theorem (Liouville, 1850). *Let $n \geq 3$, let $\Omega \subseteq \mathbb{R}^n$ be a region, and let $g : \Omega \rightarrow \mathbb{R}^n$ be a non-constant solution of (14.13). Then g is the restriction to Ω of a Möbius transformation of $\widehat{\mathbb{R}^n} = \mathbb{R}^n \cup \{\infty\}$: that is, g is a composition of translations $x \mapsto x + b$, positive dilations $x \mapsto \lambda x$, rotations $x \mapsto Rx$ with $R \in SO(n)$, and at most one reflected inversion*

$$x \mapsto \frac{\rho(x)}{|x|^2}, \quad (14.14)$$

with ρ the reflection in a hyperplane through 0. (The inversion $x \mapsto x/|x|^2$ alone reverses orientation: its differential $|x|^{-2}(I - 2\hat{x}\hat{x}^\top)$, $\hat{x} = x/|x|$, has negative determinant; composing with ρ restores it. Equivalently, g is an orientation-preserving Möbius transformation.)

Liouville proved this for g of class C^3 ; the hypothesis was weakened to C^1 by Hartman, and then to the Sobolev class $W_{\text{loc}}^{1,n}$ by Gehring and Rešetnyak (the natural class for quasiconformal maps). Iwaniec and Martin later lowered this to $W_{\text{loc}}^{1,n/2}$ in even dimensions, and showed that it fails below $n/2$; in odd dimensions the optimal exponent is, as far as we know, still open. The conclusion is the same in every case.

The contrast is stark, and it is the reason this course exists in the form it does.

- In dimension 2, the conformal maps form an infinite-dimensional family: for *every* analytic f with $f' \neq 0$ we get one, and Chapter 33 exploits exactly this abundance — the Riemann mapping theorem 33.10 says the family is large enough to carry *any* simply connected proper subregion onto the disc.
- In dimension $n \geq 3$, the conformal maps form a Lie group of finite dimension $\frac{1}{2}(n+1)(n+2)$. There is no Riemann mapping theorem, no residue calculus, no Taylor theory — because there are, essentially, no maps. The Möbius transformations of Chapter 34 are all that survives, and they survive in every dimension.

So “complex analysis in several variables” is not the study of conformal maps of $\mathbb{R}^{2n}$; it is a different subject, built on the Cauchy–Riemann system applied one variable at a time. The two-dimensional theory is genuinely exceptional, and (14.10) is the cleanest statement of what makes it so.

14.4 Consequences: rigidity

Theorem 14.13 (Vanishing derivative on a region). *Let D be a region and f analytic on D .*

- (i) *If $f'(z) = 0$ for all $z \in D$, then f is constant on D .*
- (ii) *If $|f|$ is constant on D , then f is constant on D .*
- (iii) *If $f(D) \subseteq \mathbb{R}$, then f is constant on D . More generally if $f(D)$ is contained in a line or a circle, f is constant.*

Proof. (i) By (14.2), $f' = u_x + iv_x = v_y - iu_y = 0$ forces $u_x = u_y = v_x = v_y = 0$ throughout D . Fix $a, b \in D$ and take a polygonal path in D from a to b (Definition 5.8). On each segment $t \mapsto z(t) = (1-t)p + tq$, the real function $t \mapsto u(z(t))$ has derivative $u_x \operatorname{Re}(q-p) + u_y \operatorname{Im}(q-p) = 0$, so u is constant on the segment by the mean value theorem; the same for v . Chaining the segments, $f(a) = f(b)$.

(ii) Say $|f| \equiv c \geq 0$. If $c = 0$ then $f \equiv 0$ by (1.22). Assume $c > 0$. From $u^2 + v^2 = c^2$, differentiate in x and in y :

$$uu_x + vv_x = 0, \quad uu_y + vv_y = 0. \quad (14.15)$$

Substituting (14.1) ($u_y = -v_x, v_y = u_x$) into the second equation gives $-uv_x + vu_x = 0$. Together with the first equation of (14.15) we get the linear system

$$\begin{pmatrix} u & v \\ v & -u \end{pmatrix} \begin{pmatrix} u_x \\ v_x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

whose coefficient matrix has determinant $-(u^2 + v^2) = -c^2 \neq 0$. Hence $u_x = v_x = 0$ on D , so $f' = 0$ by (14.2) and (i) applies.

(iii) If $f(D) \subseteq \mathbb{R}$ then $v \equiv 0$, so $v_x = v_y = 0$, and (14.1) forces $u_x = v_y = 0$ and $u_y = -v_x = 0$; apply (i). For a general line $\{w : \operatorname{Re}(\bar{\alpha}w) = c\}$ with $\alpha \neq 0$ (Exercise 7.8), the function $g := \bar{\alpha}f$ is analytic with $\operatorname{Re} g \equiv c$; then g_x -computations as above (or applying the case just proved to $i(g - c)$) give g constant, hence f constant. If $f(D)$ lies on a circle then $|f - \text{centre}|$ is constant and (ii) applies. $\square$

Theorem 14.13 has no analogue for smooth maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. The map $g(x, y) = (\cos(x + y), \sin(x + y))$ is real-analytic, non-constant, has constant norm, and takes values on a circle — so the hypotheses of (ii) and of (iii) can both be met by a non-constant smooth planar map. What is special is that the Cauchy–Riemann equations couple u and v so tightly that a single scalar constraint destroys all freedom. This rigidity is the recurring theme of the entire subject.

14.5 Exercises

Exercise 14.1. Verify the Cauchy–Riemann equations and find f' : $f(z) = z^3$; $f(z) = \frac{1}{z}$ ($z \neq 0$); $f(z) = e^x \cos y + ie^x \sin y$.

Exercise 14.2. Determine where each function is differentiable and where it is analytic:

$$f(z) = 2x^2 + y + i(y^2 - x), \quad g(z) = x^2 + y^2 - 2ixy, \quad h(z) = \bar{z}z^2.$$

Exercise 14.3. (★) (*Wirtinger form.*) Define $\partial_{\bar{z}} := \frac{1}{2}(\partial_x + i\partial_y)$. Show that (14.1) is equivalent to the single equation $\partial_{\bar{z}}f = 0$, and that then $f' = \partial_z f := \frac{1}{2}(\partial_x - i\partial_y)f$. Use it to re-derive Examples 14.3 and 14.4 in one line each.

Exercise 14.4. Suppose f and $\bar{f}$ are both analytic on a region D . Prove f is constant.

Exercise 14.5. Suppose f is analytic on a region D and $\operatorname{Re} f$ is constant. Prove f is constant, directly from (14.1).

Exercise 14.6. For each map below, write the Jacobian matrix $D\tilde{f}$ of (14.9) at a general point and decide, using Theorem 14.10 (iv), at which points it lies in $\overline{CO_+(2)}$:

$$f(z) = z^2, \quad f(z) = \bar{z}, \quad f(z) = |z|^2, \quad f(z) = \operatorname{Re} z.$$

In each case compute $\det D\tilde{f}$ and compare with $|f'|^2$ where the latter exists.

Exercise 14.7. Let f be analytic on a region D with f' never zero, and let $E \subseteq D$ be a region of finite area. Using (14.11) and the change-of-variables formula (Appendix A), show

$$\operatorname{area}(f(E)) = \iint_E |f'(x + iy)|^2 \, dx \, dy.$$

Evaluate it for $f(z) = z^2$ on the quarter-disc $\{|z| < 1, 0 < \operatorname{Arg} z < \pi/2\}$ and check your answer against Theorem 9.1.

Exercise 14.8. (★) (*The inclusion is not preserved by averaging.*) Exhibit two matrices $A, B \in CO_+(2)$ with $\frac{1}{2}(A + B) \notin \overline{CO_+(2)}$ — or prove no such pair exists. Then do the same for $CO_+(3)$ in place of $CO_+(2)$, and explain how your answers relate to Remark 14.12. (*Hint: $\overline{CO_+(2)} = \mathcal{C}$ is a linear subspace by (1.33). For $n = 3$, try $A = I$ and B a rotation by π about an axis.*)

Chapter 15

Harmonic Functions

15.1 Laplace's equation

Definition 15.1 (Harmonic function). Let $D \subseteq \mathbb{R}^2$ be a region (equivalently, a region in $\mathbb{C}$ via Definition 2.1). A function $\phi : D \rightarrow \mathbb{R}$ is *harmonic on D* if ϕ has continuous second-order partial derivatives on D and satisfies *Laplace's equation*

$$\nabla^2 \phi := \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad \text{at every point of } D. \quad (15.1)$$

Theorem 15.2 (Components of an analytic function are harmonic). *Let $f = u + iv$ be analytic on a region D . Then u and v are harmonic on D .*

Proof. We use the following fact, proved in Chapter 21 (Corollary 21.5): an analytic function has derivatives of all orders, and consequently u and v have continuous partial derivatives of all orders on D . (This is genuinely a theorem, not a convention; there is no elementary proof, and it is one of the most striking differences between real and complex analysis.)

Granting that, differentiate the first Cauchy–Riemann equation (14.1) with respect to x and the second with respect to y :

$$u_{xx} = v_{yx}, \quad u_{yy} = -v_{xy}.$$

Since the second partials are continuous, Clairaut's theorem gives $v_{yx} = v_{xy}$, so

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0.$$

Similarly, differentiating the first CR equation with respect to y and the second with respect to x ,

$$u_{xy} = v_{yy}, \quad u_{yx} = -v_{xx} \implies v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0. \quad \square$$

Example 15.3. Since $z^2 = (x^2 - y^2) + i(2xy)$ is entire, both $x^2 - y^2$ and $2xy$ are harmonic on $\mathbb{R}^2$; check: $\nabla^2(x^2 - y^2) = 2 - 2 = 0$ and $\nabla^2(2xy) = 0 + 0 = 0$. Since $1/z = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$ is analytic on $\mathbb{C} \setminus \{0\}$, the function $\frac{x}{x^2+y^2}$ is harmonic there. And $\ln|z| = \frac{1}{2}\ln(x^2 + y^2)$ is harmonic on $\mathbb{C} \setminus \{0\}$, being the real part of the principal logarithm (Chapter 16); direct check:

$$\partial_x \frac{1}{2} \ln(x^2 + y^2) = \frac{x}{x^2 + y^2}, \quad \partial_{xx} = \frac{y^2 - x^2}{(x^2 + y^2)^2},$$

and by symmetry $\partial_{yy} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$; the sum is 0.

15.2 Harmonic conjugates

Definition 15.4 (Harmonic conjugate). Let u be harmonic on a region D . A function v harmonic on D is called a *harmonic conjugate of u on D* if u and v satisfy the Cauchy–Riemann equations (14.1) throughout D , i.e. if $f = u + iv$ is analytic on D .

Warning 15.5. The relation is *not* symmetric: if v is a harmonic conjugate of u then $-u$ is a harmonic conjugate of v , not u . (Because $-if = -iu + v = v + i(-u)$ is analytic.) The order matters.

Theorem 15.6 (Existence of a harmonic conjugate). *Let u be harmonic on a disc $D = D(z_0, \rho)$, or on $D = \mathbb{C}$. Then u has a harmonic conjugate v on D , unique up to an additive real constant. The same holds on any simply connected region; that case is proved in Chapter 20 (Corollary 20.13) and is not used before then.*

Proof. Without loss of generality $z_0 = 0$ (for $D = \mathbb{C}$ take $z_0 = 0$); write points of D as (x, y) . For $(x, y) \in D$ the horizontal segment from $(0, 0)$ to $(x, 0)$ and the vertical segment from $(x, 0)$ to (x, y) lie in D , so the following integrals are over subsets of D . Define

$$v(x, y) := \int_0^y u_x(x, t) dt - \int_0^x u_y(s, 0) ds. \quad (15.2)$$

(All integrands are continuous, so the integrals exist and may be differentiated under the integral sign, the integrands having continuous partials.)

Differentiating (15.2) in y gives immediately

$$v_y(x, y) = u_x(x, y), \quad (15.3)$$

which is the first CR equation. Differentiating in x ,

$$\begin{aligned} v_x(x, y) &= \int_0^y u_{xx}(x, t) dt - u_y(x, 0) \stackrel{(15.1)}{=} - \int_0^y u_{tt}(x, t) dt - u_y(x, 0) \\ &= -[u_y(x, y) - u_y(x, 0)] - u_y(x, 0), \end{aligned}$$

so

$$v_x(x, y) = -u_y(x, y), \quad (15.4)$$

the second CR equation. Since $v_x = -u_y$ and $v_y = u_x$ are continuous, $f := u + iv$ satisfies the hypotheses of Theorem 14.6 and is analytic on D . Moreover v_x and v_y have continuous first partials, because u has continuous second partials; so v has continuous second partials and, by Clairaut's theorem for u , $v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0$. Thus v is harmonic (without appeal to the fact from Chapter 21 used in Theorem 15.2).

Uniqueness: if v_1, v_2 are two harmonic conjugates then $f_1 - f_2 = i(v_1 - v_2)$ is analytic with purely imaginary values, hence constant by Theorem 14.13(iii) applied to $-i(f_1 - f_2)$. So $v_1 - v_2$ is a real constant.

For a general simply connected region the same construction works with (15.2) replaced by a line integral along any path from a base point, the CR equations guaranteeing path-independence; this uses material from Part II, and is carried out in Corollary 20.13. $\square$

Remark 15.7 (Where it fails). Simple connectivity is essential. On the punctured plane $D = \mathbb{C} \setminus \{0\}$ the function $u = \ln |z|$ is harmonic (Example 15.3) but has *no* harmonic conjugate on all of D .

Indeed, suppose v were one. On the upper half-plane $\{y > 0\}$ we have $\operatorname{Arg} z = \frac{\pi}{2} - \arctan(x/y)$, and on the lower half-plane $\operatorname{Arg} z = -\frac{\pi}{2} - \arctan(x/y)$ (check the quadrants against (3.4)); in both cases

$$\partial_x \operatorname{Arg} z = \frac{-y}{x^2 + y^2} = -u_y, \quad \partial_y \operatorname{Arg} z = \frac{x}{x^2 + y^2} = u_x.$$

So $v - \operatorname{Arg}$ has vanishing gradient on each half-plane (both satisfy $v_x = -u_y$, $v_y = u_x$), and is therefore a constant c_+ on $\{y > 0\}$ and a constant c_- on $\{y < 0\}$ (as in the proof of Theorem 14.13(i)). Let $x_0 \neq 0$ be real. Letting $z \rightarrow x_0$ from above and from below and using continuity of v at x_0 : for $x_0 > 0$, $\operatorname{Arg} z \rightarrow 0$ from both sides, so $c_+ = c_-$; for $x_0 < 0$, $\operatorname{Arg} z \rightarrow \pi$ from above and $\rightarrow -\pi$ from below, so $\pi + c_+ = -\pi + c_-$. These contradict each other. (This is the same obstruction as in Proposition 10.5(ii): a conjugate would be a continuous argument function on $\mathbb{C} \setminus \{0\}$.) On any disc contained in D a conjugate exists by Theorem 15.6, and on the slit plane $\mathbb{C} \setminus (-\infty, 0]$ the function Arg is one (Chapter 16).

Example 15.8 (Finding a conjugate, method of partial integration). Let $u(x, y) = x^3 - 3xy^2 + 2y$. First check harmonicity: $u_{xx} = 6x$ and $u_{yy} = -6x$, sum 0; so u is harmonic on $\mathbb{R}^2$. Now solve the CR equations for v . From $v_y = u_x = 3x^2 - 3y^2$, integrate in y holding x fixed:

$$v(x, y) = 3x^2y - y^3 + \psi(x) \tag{15.5}$$

for some differentiable function ψ of x alone. Differentiate (15.5) in x and use the second CR equation $v_x = -u_y$:

$$6xy + \psi'(x) = -(-6xy + 2) = 6xy - 2 \implies \psi'(x) = -2 \implies \psi(x) = -2x + c.$$

Hence $v(x, y) = 3x^2y - y^3 - 2x + c$ with $c \in \mathbb{R}$ arbitrary, and

$$f(z) = u + iv = (x^3 - 3xy^2 + 2y) + i(3x^2y - y^3 - 2x + c).$$

One recognises $f(z) = z^3 - 2iz + ic$: indeed $z^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$ and $-2iz = -2i(x + iy) = 2y - 2ix$.

Proposition 15.9 (Recovering f from u directly). *Let u be harmonic on a disc D about z_0 and let $f = u + iv$ be an analytic function on D with real part u . Then*

$$f'(z) = u_x(x, y) - iu_y(x, y), \tag{15.6}$$

and f may be recovered by antidifferentiating (15.6). In practice one uses Milne's method: substitute $x \rightarrow z$, $y \rightarrow 0$ to get $f'(z) = u_x(z, 0) - iu_y(z, 0)$. This is valid when D meets the real axis and the function $g(z) := u_x(z, 0) - iu_y(z, 0)$, obtained by extending the real functions $x \mapsto u_x(x, 0)$ and $x \mapsto u_y(x, 0)$ to complex arguments by their formulas, is analytic on D (for instance when these formulas are polynomials, or built from e^x , $\cos x$, $\sin x$ by sums and products).

Proof. Equation (15.6) is (14.2) combined with $v_x = -u_y$. For the substitution step: f' is analytic on D (by Part II, Corollary 21.5), and for real x with $x \in D$, (15.6) gives $f'(x) = u_x(x, 0) - iu_y(x, 0) = g(x)$. So the analytic functions f' and g agree on the segment $D \cap \mathbb{R}$, which has accumulation points in D , and hence on the whole disc by the identity theorem of Part II (Theorem 24.8). $\square$

Example 15.10. For $u = x^3 - 3xy^2 + 2y$ as above, $u_x = 3x^2 - 3y^2$ and $u_y = -6xy + 2$. Milne: $f'(z) = u_x(z, 0) - iu_y(z, 0) = 3z^2 - 2i$, so $f(z) = z^3 - 2iz + C$ — the same answer as Example 15.8, in two lines.

15.3 Level curves and physical interpretation

Theorem 15.11 (Orthogonality of level curves). *Let $f = u + iv$ be analytic on a region D and let $z_0 \in D$ with $f'(z_0) \neq 0$. Then the level curve of u through z_0 and the level curve of v through z_0 are smooth near z_0 and meet orthogonally there.*

Proof. By (14.2), $f'(z_0) \neq 0$ means $(u_x, v_x) \neq (0, 0)$ at z_0 ; combined with (14.1) this gives $\nabla u = (u_x, u_y) \neq (0, 0)$ and $\nabla v = (v_x, v_y) = (-u_y, u_x) \neq (0, 0)$. By the implicit function theorem both level curves are smooth near z_0 , with normals ∇u and ∇v . Their dot product is

$$\nabla u \cdot \nabla v = u_x v_x + u_y v_y = u_x(-u_y) + u_y(u_x) = 0,$$

so the normals, and hence the curves, are orthogonal. $\square$

Remark 15.12 (Potentials and flows). If ϕ is a harmonic function modelling a steady-state physical potential — temperature in a plate with no internal sources, the electrostatic potential in a charge-free region, or the velocity potential of an incompressible irrotational planar flow — then the *complex potential*

$$W(z) = \phi(x, y) + i\psi(x, y), \quad (15.7)$$

with ψ a harmonic conjugate of ϕ , packages the whole solution in a single analytic function. The level curves $\phi = \text{const}$ are the *equipotentials* (isotherms, field-equipotentials), and by Theorem 15.11 the level curves $\psi = \text{const}$ — called *stream lines* or *lines of flux* — cross them at right angles, which is exactly the physical statement that heat and field lines flow along the steepest descent of the potential. Moreover $W' = \phi_x - i\phi_y$ by (15.6), so the conjugate of W' is the gradient field itself.

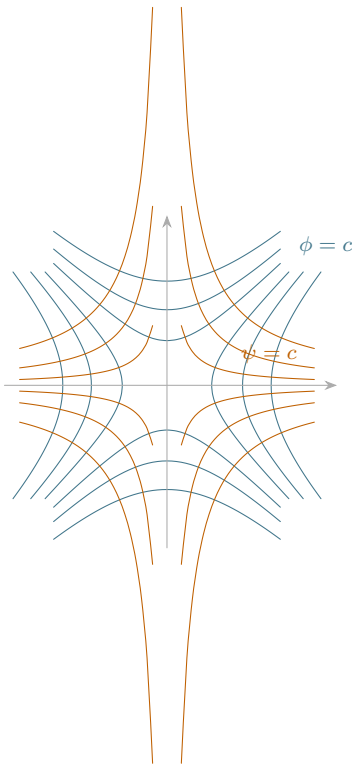
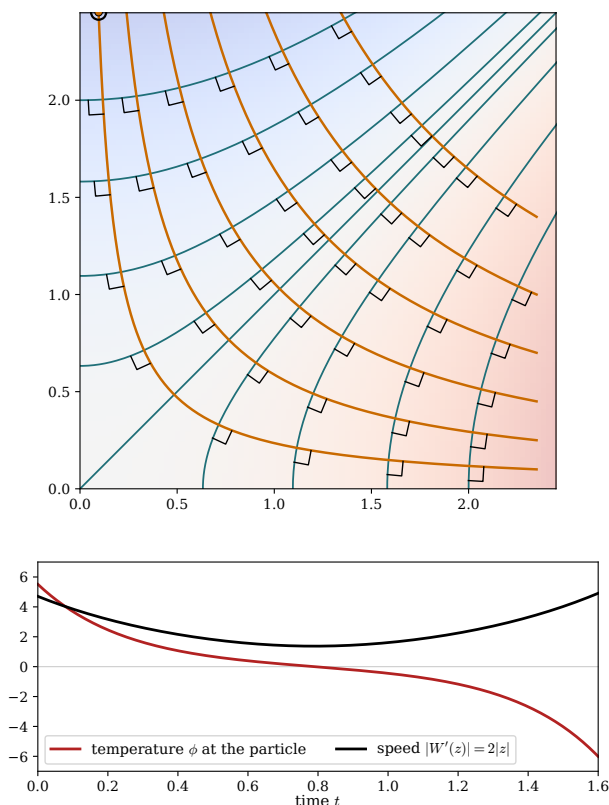


FIGURE 15.1. Theorem 15.11 for the complex potential $W(z) = z^2$ of Example 15.13. The equipotentials $\phi = x^2 - y^2 = c$ and the stream lines $\psi = 2xy = c$ are two families of hyperbolas crossing at right angles. Physically: heat flows along the stream lines, always perpendicular to the isotherms, which is exactly what “down the steepest gradient” means.

Example 15.13 (A steady temperature field). Let $W(z) = z^2$, so $\phi = x^2 - y^2$ and $\psi = 2xy$. The isotherms $x^2 - y^2 = c$ and the flux lines $2xy = c$ are the two orthogonal families of hyperbolas of Proposition 9.3 — the classical picture of heat flow near a corner. In the first quadrant with the two axes held at temperature 0, the function $\phi = 2xy$ (real part of $-iz^2$) is a harmonic function vanishing on both walls (the simplest one; it is not unique, since for instance every multiple of $2xy$, and $\text{Im } z^4 = 4x^3y - 4xy^3$, also vanish there), giving the temperature distribution in a right-angled corner.

Stream lines cross isotherms at right angles (Theorem 15.11)

$W = z^2$ (Fig. 15.1). Teal: isotherms $\phi = x^2 - y^2 = c$ (red warm, blue cold). Heat moves with velocity $-\nabla\phi = -\overline{W'}$ (Remark 15.12), along $\psi = 2xy = c$



ringed particle: $t = 1.60$, $z = 0.096 + 2.453i$, temperature -6.009, speed 4.910
angle between trace and isotherm, measured at the latest crossing: 89.96°

FIGURE 15.2 (animated). Theorem 15.11 through motion: heat particles move with velocity $-\nabla\phi$ along the stream lines $\psi = \text{const}$, and cross the fixed isotherms $\phi = \text{const}$ at right angles (the marked crossings are within 0.11° of 90°). First $W = z^2$: the particles turn round the corner, slowest near it, where $|W'| = 2|z|$ is small. Then $W = e^{-i\theta} z^2$ turns until $W = -iz^2$, $\phi = 2xy$ of Example 15.13: now the particles flow into the walls. Below: temperature and speed of the ringed particle.

15.4 Bonus background: harmonic functions as steady states of heat flow

Bonus background. This section explains where harmonic functions come from physically and why they are “optimal”. Nothing later in the notes depends on it, and it can be skipped.

Remark 15.12 read a harmonic function as a steady temperature. This section explains where that comes from: harmonic functions are exactly the temperatures that heat flow no longer changes, and among all temperatures with the same boundary values they are the most economical way to carry heat from the hot part of the boundary to the cold part.

The heat equation. Let D be a bounded region, a thin plate, and let $\varphi(x, y, t)$ be its temperature at the point (x, y) at time t . Heat flows from hot to cold at a rate proportional to the temperature gradient (Fourier’s law: the heat flux is $\mathbf{q} = -k\nabla\varphi$ with a conductivity $k > 0$), and heat is conserved: the rate at which a small piece of the plate warms up is the net heat flowing into it, $-\operatorname{div} \mathbf{q} = k\nabla^2\varphi$ per unit area. Choosing units of time in which the material constants equal 1, this is the *heat equation*

$$\frac{\partial\varphi}{\partial t} = \nabla^2\varphi = \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2}. \quad (15.8)$$

We hold the edge of the plate at fixed temperatures, $\varphi(x, y, t) = g(x, y)$ for $(x, y) \in \partial D$ and all $t \geq 0$, and start from any initial temperature $\varphi(x, y, 0)$. The model is physics, not a theorem; what follows is mathematics about it.

A *steady state* is a temperature that does not change: $\partial\varphi/\partial t = 0$. By (15.8) this says exactly $\nabla^2\varphi = 0$:

the steady states of heat flow are the harmonic functions with the given boundary values.

The animations of this section run (15.8) from an arbitrary initial temperature until nothing changes any more, and the picture that remains is harmonic. That the solution always does settle down to the harmonic function with boundary values g is a theorem about the heat equation that we do not prove here; the next result explains why it should.

Energy. Throughout, D is a bounded region whose boundary is a finite union of simple closed piecewise- C^1 curves, as in Green’s theorem (Theorem A.17), and the functions are C^2 on a neighbourhood of $\bar{D}$. The *Dirichlet energy* of a function v on D is

$$E(v) = \iint_D |\nabla v|^2 \, dA = \iint_D (v_x^2 + v_y^2) \, dA. \quad (15.9)$$

It measures how steep v is, in total. Everything rests on *Green’s first identity*: for w of class C^1 and φ of class C^2 ,

$$\iint_D (\nabla w \cdot \nabla \varphi + w \nabla^2 \varphi) \, dA = \oint_{\partial D} w \frac{\partial \varphi}{\partial n} \, ds, \quad (15.10)$$

where $\partial\varphi/\partial n = \nabla\varphi \cdot \mathbf{n}$ is the derivative along the outward unit normal $\mathbf{n}$. To prove it, apply Green’s theorem (A.16) to $P = -w\varphi_y$, $Q = w\varphi_x$: then $Q_x - P_y = w_x\varphi_x + w\varphi_{xx} + w_y\varphi_y + w\varphi_{yy}$, the left side of (15.10), while $P \, dx + Q \, dy = w(\varphi_x \, dy - \varphi_y \, dx) = w \nabla\varphi \cdot \mathbf{n} \, ds$, because $(dy, -dx) = \mathbf{n} \, ds$ when D lies to the left of ∂D .

Proposition 15.14 (Heat flow lowers the energy). *Let φ solve (15.8) with $\varphi = g$ on ∂D for all t . Then*

$$\frac{d}{dt}E(\varphi(\cdot, t)) = -2 \iint_D (\nabla^2 \varphi)^2 dA \leq 0,$$

with equality exactly when $\varphi(\cdot, t)$ is harmonic.

Proof. Differentiating under the integral sign, $\frac{d}{dt}E = 2 \iint_D \nabla \varphi \cdot \nabla \varphi_t dA$. Apply (15.10) with $w = \varphi_t$: the boundary term vanishes, because the edge temperature does not change, so $\varphi_t = 0$ on ∂D . Hence $\frac{d}{dt}E = -2 \iint_D \varphi_t \nabla^2 \varphi dA = -2 \iint_D (\nabla^2 \varphi)^2 dA$ by (15.8). The integrand is continuous and ≥ 0 , so the integral is 0 exactly when $\nabla^2 \varphi \equiv 0$. $\square$

So heat flow always runs *downhill* in energy, and it only stops where the ground is flat: at a harmonic function. (In fact (15.8) is the steepest descent of E : by the proof, $\frac{d}{dt}E = -2 \iint \varphi_t \nabla^2 \varphi$, so among all ways of changing φ at a given speed $\iint \varphi_t^2$, the Cauchy–Schwarz inequality shows that $\varphi_t = \nabla^2 \varphi$ lowers E fastest.) The next theorem says that the bottom it reaches is the lowest point there is.

Theorem 15.15 (Dirichlet’s principle). *Let φ be harmonic on D with $\varphi = g$ on ∂D , and let v be any function with the same boundary values, $v = g$ on ∂D . Then*

$$E(v) \geq E(\varphi),$$

with equality only if $v = \varphi$. In particular there is at most one harmonic function with given boundary values.

Proof. Put $w = v - \varphi$, so $w = 0$ on ∂D . Expanding $|\nabla \varphi + \nabla w|^2$,

$$E(v) = E(\varphi) + 2 \iint_D \nabla w \cdot \nabla \varphi dA + E(w).$$

By (15.10) the middle integral is $\oint_{\partial D} w \partial_n \varphi ds - \iint_D w \nabla^2 \varphi dA = 0 - 0$, because $w = 0$ on ∂D and $\nabla^2 \varphi = 0$. So $E(v) = E(\varphi) + E(w) \geq E(\varphi)$. Equality forces $E(w) = 0$, i.e. $\nabla w \equiv 0$ on D . Then $f = w + 0i$ satisfies the Cauchy–Riemann equations with continuous partials, so it is analytic with $f' = w_x = 0$ (Theorem 14.6), hence constant on the region D (Theorem 14.13 (i)); being 0 on ∂D , $w \equiv 0$. $\square$

Remark 15.16 (The most efficient way to carry heat). Suppose ∂D is the union of a hot part Γ_1 held at temperature T_1 and a cold part Γ_0 held at $T_0 < T_1$, each of Γ_0, Γ_1 a union of whole boundary curves of D (for example the inner and the outer circle of an annulus). This matters: if a hot and a cold arc of the same boundary curve met, the boundary temperature would jump there, φ could not be C^1 up to the boundary, and in fact $E(\varphi) = \infty$, as Remark 15.17 shows for the wedge. Applying (15.10) with $w = \varphi$ to the harmonic φ gives

$$E(\varphi) = \oint_{\partial D} \varphi \frac{\partial \varphi}{\partial n} ds = T_1 \oint_{\Gamma_1} \frac{\partial \varphi}{\partial n} ds + T_0 \oint_{\Gamma_0} \frac{\partial \varphi}{\partial n} ds = (T_1 - T_0) Q,$$

where $Q = \oint_{\Gamma_1} \partial_n \varphi ds$ is the heat current entering through the hot edge (here $\oint_{\partial D} \partial_n \varphi ds = 0$, by (15.10) with $w = 1$: in the steady state what comes in goes out). So the energy is *heat current* $\times$ *temperature drop*, exactly as power = current $\times$ voltage for an electric current. By Fourier’s law it is also the rate at which conduction dissipates, $k \iint |\nabla \varphi|^2 = \iint \mathbf{q} \cdot (-\nabla \varphi)$; and when the temperature differences are small compared with the absolute temperature T , the rate at which

the plate produces entropy is $\approx (k/T^2) E(\varphi)$. Dirichlet's principle therefore says: *the steady state carries heat from hot to cold with the least waste*: no other temperature distribution with the same edge temperatures does it with less dissipation. It has no hot spots and no cold pockets to feed, and its isotherms and flux lines cross at right angles (Theorem 15.11), so the heat takes no detours.

Remark 15.17. The wedge of Example 33.8 is held at 0 on one edge and at 100 on the other, so its boundary temperature jumps at the vertex 0. There the smoothness assumed above fails, and indeed $|\nabla\varphi| = \frac{400}{\pi r}$ makes $E(\varphi) = \infty$ near 0. The first animation below therefore cuts the vertex off, using the truncated wedge $\{0.2 < |z| < 1, 0 < \text{Arg } z < \pi/4\}$, and holds the two arcs at the steady values; its boundary temperature is then continuous and everything above applies.

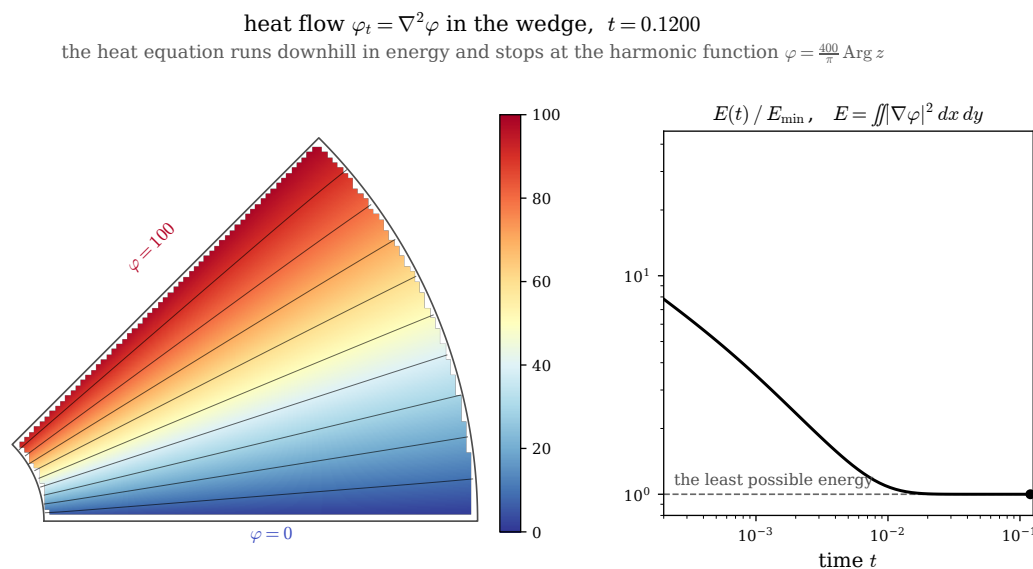


FIGURE 15.3 (animated). Heat flow in the wedge of Example 33.8, cut off at $|z| = 0.2$, starting from the constant temperature 45 plus a hot and a cold spot. Left: the temperature and its isotherms, which end as the rays of Example 33.8. Right: the energy $E(t)/E_{\min}$ falls monotonically to 1 (Proposition 15.14, Theorem 15.15).

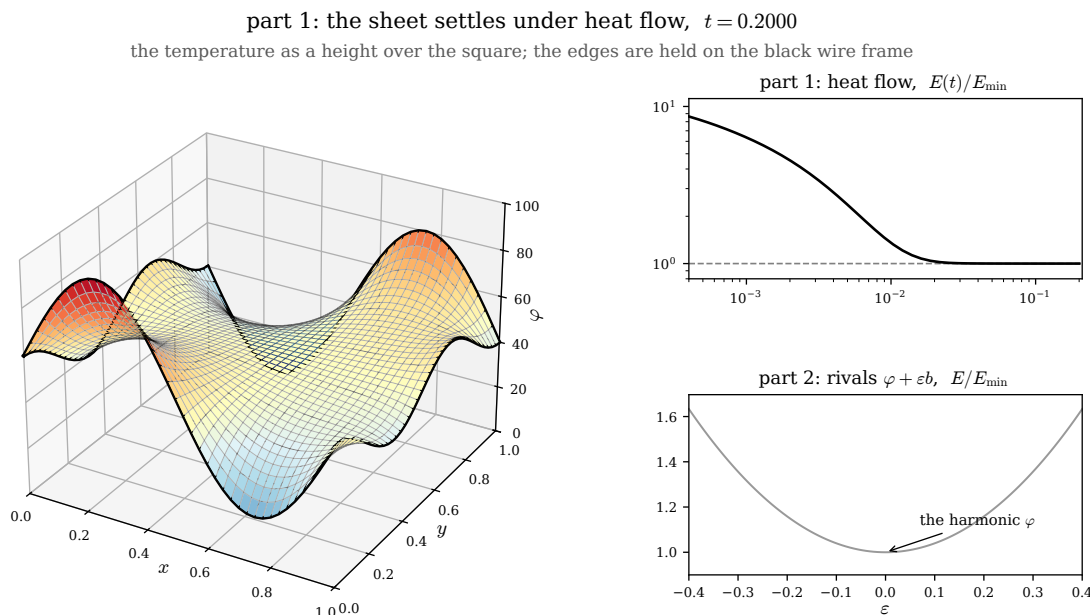


FIGURE 15.4 (animated). The temperature on a square drawn as a height: a “rubber sheet” on a wavy wire frame. Part 1: under heat flow the bumpy sheet settles as E falls to its minimum. Part 2, Dirichlet’s principle made visible: other sheets on the same frame, $\varphi + \varepsilon b$ with a bump b vanishing on the edges, all have more energy, $E(\varphi + \varepsilon b) = E(\varphi) + \varepsilon^2 E(b)$, a parabola with its minimum at the harmonic φ .

15.5 Exercises

Exercise 15.1. Show that each is harmonic and find a harmonic conjugate:

$$u = 2x - x^3 + 3xy^2, \quad u = e^{-x}(x \sin y - y \cos y), \quad u = \ln(x^2 + y^2) \text{ on } \{x > 0\}.$$

Exercise 15.2. Show $u = x^2 + y^2$ is *not* harmonic, and explain why it therefore cannot be the real part of any analytic function.

Exercise 15.3. Use Milne’s method (Proposition 15.9) to find the analytic f with $\operatorname{Re} f = e^x \cos y$ and $f(0) = 1$.

Exercise 15.4. (★) Prove that if u and v are both harmonic on D and v is a harmonic conjugate of u , then uv is harmonic. (*Hint:* $uv = \frac{1}{2} \operatorname{Im}(f^2)$.) Is u^2 harmonic?

Exercise 15.5. Let u be harmonic on $\mathbb{C}$ and bounded above. Anticipating Liouville’s theorem, argue that u must be constant. (Consider e^f where $f = u + iv$; you may use Theorem 15.6 on $\mathbb{C}$.)

Exercise 15.6. Decide which of the following are harmonic on the indicated set, and find a conjugate where one exists:

$$u = x^3 - 3xy^2 \text{ on } \mathbb{R}^2; \quad u = \frac{y}{x^2 + y^2} \text{ on } \mathbb{C} \setminus \{0\}; \quad u = xy + x \text{ on } \mathbb{R}^2; \quad u = e^{x^2 - y^2} \text{ on } \mathbb{R}^2.$$

Exercise 15.7. Prove that if u is harmonic then so are u_x and u_y . (You may use Corollary 21.5 to justify differentiating Laplace’s equation.)

Exercise 15.8. (★) (*Mean value property for harmonic functions.*) Let u be harmonic on a neighbourhood of $\overline{D}(z_0, \rho)$. Prove

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + \rho e^{i\theta}) d\theta,$$

and deduce a maximum principle: a harmonic function on a region that attains an interior maximum is constant. (*Take real parts in Exercise 21.4, and imitate Theorem 24.11.*)

Exercise 15.9. Verify Theorem 15.11 explicitly for $f(z) = 1/z$ by computing the level curves of $u = \frac{x}{x^2+y^2}$ and $v = \frac{-y}{x^2+y^2}$ and showing that they are two orthogonal families of circles through the origin.

Part IV

The Elementary Functions

Chapter 16

Exponential and Logarithmic Functions

16.1 The complex exponential

Definition 16.1 (Complex exponential). For $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$ define

$$e^z := e^x (\cos y + i \sin y) = e^x e^{iy}, \quad (16.1)$$

where e^x is the real exponential and e^{iy} is the Euler abbreviation (3.6). We also write $\exp z$ for e^z .

Definition 16.1 is forced: any extension of e^x that is entire and satisfies $f' = f$ must be (16.1), as Exercise 16.5 shows. It is consistent with the earlier notation: setting $x = 0$ recovers (3.6), and setting $y = 0$ recovers the real exponential.

Theorem 16.2 (Properties of e^z). For all $z, w \in \mathbb{C}$:

$$|e^z| = e^{\operatorname{Re} z} > 0, \quad \arg e^z = \operatorname{Im} z + 2\pi\mathbb{Z}; \quad (16.2)$$

$$e^z \neq 0 \text{ for every } z \in \mathbb{C}; \quad (16.3)$$

$$e^z e^w = e^{z+w}, \quad e^{-z} = 1/e^z, \quad (e^z)^n = e^{nz} \quad (n \in \mathbb{Z}); \quad (16.4)$$

$$\overline{e^z} = e^{\bar{z}}; \quad (16.5)$$

$$e^{z+2\pi i} = e^z \quad (\text{periodicity}). \quad (16.6)$$

Moreover e^z is entire and

$$\frac{d}{dz} e^z = e^z. \quad (16.7)$$

Proof. (16.2): by (1.24) and $|e^{iy}| = 1$ (Proposition 3.9), $|e^z| = e^x |e^{iy}| = e^x$, which is > 0 ; and (16.1) exhibits y as an argument.

(16.3): immediate from $|e^z| = e^x > 0$ and (1.22).

(16.4): with $z = x + iy$, $w = s + it$,

$$e^z e^w = e^x e^{iy} e^s e^{it} = e^{x+s} e^{i(y+t)} = e^{(x+s)+i(y+t)} = e^{z+w},$$

using the real law of exponents and Proposition 3.9. Setting $w = -z$ gives $e^z e^{-z} = e^0 = 1$. The integer power law follows by induction and the inverse rule.

(16.5): $\overline{e^x(\cos y + i \sin y)} = e^x(\cos y - i \sin y) = e^x(\cos(-y) + i \sin(-y)) = e^{x-iy}$.

(16.6): $e^{2\pi i} = \cos 2\pi + i \sin 2\pi = 1$; apply (16.4).

Entirety and (16.7): here $u = e^x \cos y$ and $v = e^x \sin y$, so

$$u_x = e^x \cos y, \quad v_x = e^x \cos y, \quad u_y = -e^x \sin y, \quad v_y = e^x \sin y.$$

Both Cauchy–Riemann equations (14.1) hold on all of $\mathbb{R}^2$ and the four partials are continuous, so Theorem 14.6 gives analyticity on $\mathbb{C}$ with $f' = u_x + iv_x = e^x \cos y + ie^x \sin y = e^z$. $\square$

Theorem 16.3 (Periodicity and the fundamental strip). (i) *The set of periods of $\exp$ is exactly $2\pi i\mathbb{Z}$: $e^{z+p} = e^z$ for all z if and only if $p = 2k\pi i$ for some $k \in \mathbb{Z}$.*

(ii) $e^{z_1} = e^{z_2}$ if and only if $z_1 - z_2 \in 2\pi i\mathbb{Z}$.

(iii) $\exp$ maps the fundamental strip

$$\Lambda := \{z \in \mathbb{C} : -\pi < \operatorname{Im} z \leq \pi\} \quad (16.8)$$

bijectionally onto $\mathbb{C} \setminus \{0\}$; it maps the open strip $\{-\pi < \operatorname{Im} z < \pi\}$ bijectionally onto the slit plane $\mathbb{C} \setminus (-\infty, 0]$.

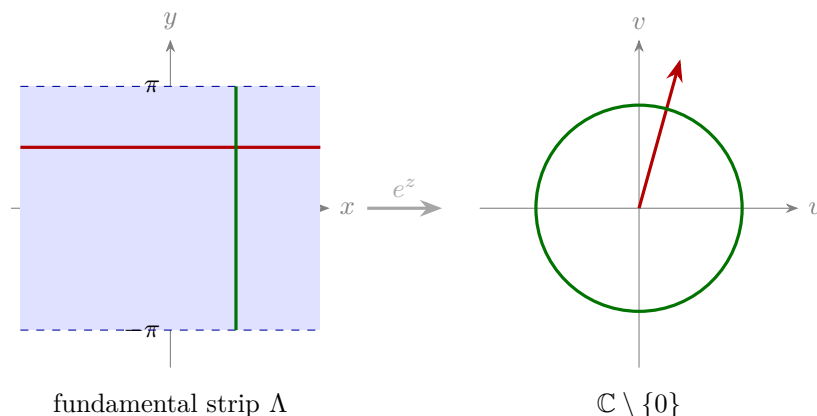
(iv) $\exp$ maps the horizontal line $\operatorname{Im} z = b$ onto the open ray $\{re^{ib} : r > 0\}$, and the vertical segment $\{a + it : -\pi < t \leq \pi\}$ onto the circle $C(0, e^a)$. A rectangle $\{a_1 < x < a_2, b_1 < y < b_2\}$ with $b_2 - b_1 \leq 2\pi$ maps onto the annular sector $\{re^{i\theta} : e^{a_1} < r < e^{a_2}, b_1 < \theta < b_2\}$.

Proof. (ii) By (16.4), $e^{z_1} = e^{z_2}$ iff $e^{z_1 - z_2} = 1$. Write $z_1 - z_2 = x + iy$. By (16.2), $e^{x+iy} = 1$ forces $e^x = |1| = 1$, so $x = 0$; then $\cos y + i \sin y = 1$ forces $\cos y = 1, \sin y = 0$, i.e. $y \in 2\pi\mathbb{Z}$.

(i) is (ii) with $z_1 = z + p, z_2 = z$.

(iii) *Injectivity on Λ :* if $z_1, z_2 \in \Lambda$ and $e^{z_1} = e^{z_2}$, then by (ii) $\operatorname{Im} z_1 - \operatorname{Im} z_2 \in 2\pi\mathbb{Z}$; but both imaginary parts lie in the half-open interval $(-\pi, \pi]$ of length 2π , so the difference is 0, and then $\operatorname{Re} z_1 = \operatorname{Re} z_2$ by (16.2). *Surjectivity onto $\mathbb{C} \setminus \{0\}$:* given $w \neq 0$, put $x := \ln |w|$ (real logarithm of a positive number) and $y := \operatorname{Arg} w \in (-\pi, \pi]$; then $z = x + iy \in \Lambda$ and $e^z = |w|e^{i \operatorname{Arg} w} = w$ by Theorem 3.7. That e^z never takes the value 0 is (16.3). Restricting to the open strip removes exactly the points with $\operatorname{Arg} w = \pi$, i.e. the negative reals.

(iv) On $\operatorname{Im} z = b$, $e^z = e^x e^{ib}$ with e^x ranging over $(0, \infty)$ as x ranges over $\mathbb{R}$; that is the stated ray. On $\operatorname{Re} z = a$, $e^z = e^a e^{it}$ traces the circle of radius e^a once as t runs over an interval of length 2π . The rectangle statement follows by combining the two. $\square$



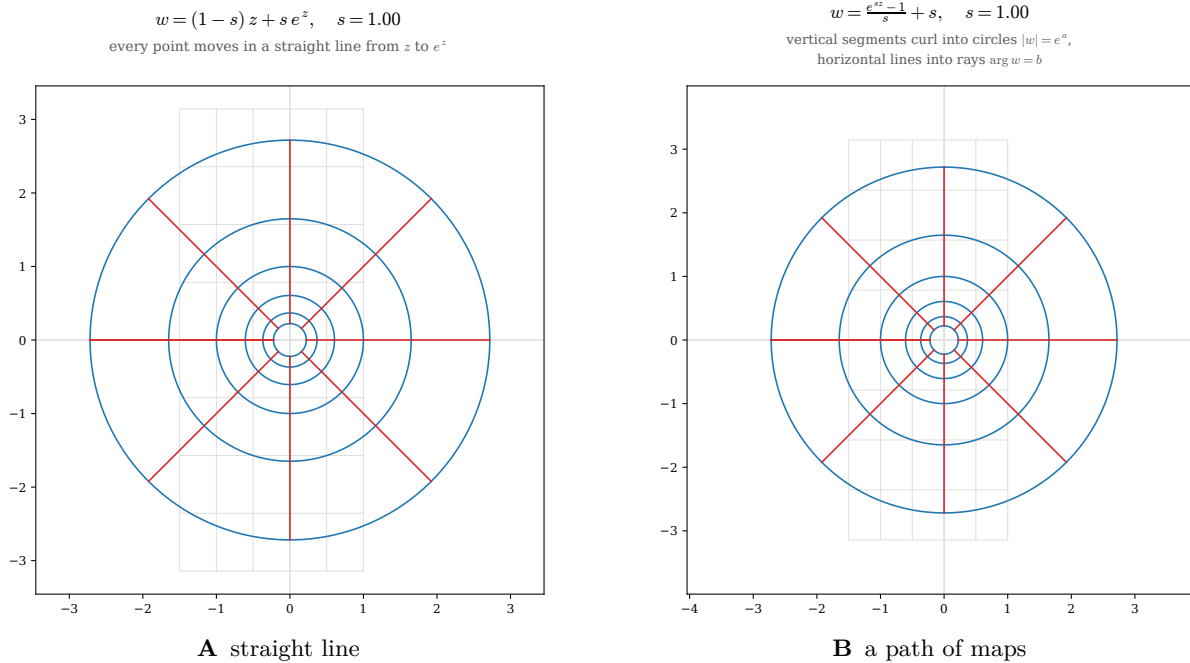
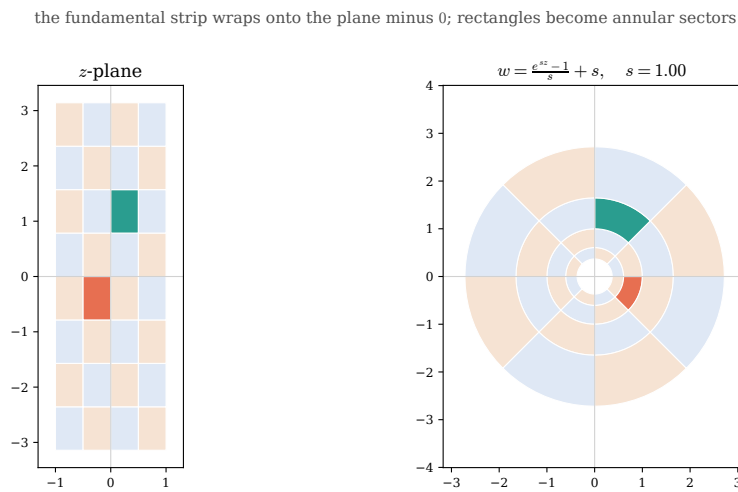


FIGURE 16.1 (animated). The exponential e^z . **A**: every point moves in a straight line from z to $f(z)$. **B**: vertical segments curl into circles $|w| = e^a$, horizontal lines into rays $\arg w = b$. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 16.1, continued)

Example 16.4 (Solving $e^z = w$). Solve $e^z = -1 - i$. Here $|w| = \sqrt{2}$ and $\text{Arg } w = -\frac{3\pi}{4}$ (third quadrant, by Proposition 3.5). By the proof of Theorem 16.3(iii), the solutions are

$$z = \ln \sqrt{2} + i \left(-\frac{3\pi}{4} + 2k\pi \right), \quad k \in \mathbb{Z}.$$

There are infinitely many, differing by $2\pi i$; exactly one lies in Λ , namely $\frac{1}{2} \ln 2 - \frac{3\pi i}{4}$.

16.2 The complex logarithm

Definition 16.5 (Logarithm). For $z \in \mathbb{C} \setminus \{0\}$ define the set

$$\log z := \{w \in \mathbb{C} : e^w = z\} = \{\ln |z| + i\theta : \theta \in \arg z\}, \quad (16.9)$$

where $\ln$ is the real natural logarithm on $(0, \infty)$. The *principal logarithm* is

$$\operatorname{Log} z := \ln |z| + i \operatorname{Arg} z \in \log z, \quad (16.10)$$

so that

$$\log z = \{\operatorname{Log} z + 2k\pi i : k \in \mathbb{Z}\}. \quad (16.11)$$

$\log 0$ is undefined, since e^w never vanishes.

Justification of (16.9). $e^w = z$ with $w = s + it$ means $e^s = |z|$ and $t \in \arg z$, by (16.2) and Theorem 3.7; the first forces $s = \ln |z|$. $\square$

Example 16.6. $\log 1 = \{2k\pi i\}$, so $\operatorname{Log} 1 = 0$. $\log(-1) = \{(2k+1)\pi i\}$, so $\operatorname{Log}(-1) = i\pi$ — a value with no real analogue. $\log i = \{i(\frac{\pi}{2} + 2k\pi)\}$, $\operatorname{Log} i = \frac{i\pi}{2}$. $\log(-2) = \{\ln 2 + i(\pi + 2k\pi)\}$, $\operatorname{Log}(-2) = \ln 2 + i\pi$.

Theorem 16.7 (Algebra of the logarithm). For $z_1, z_2 \in \mathbb{C} \setminus \{0\}$, as an identity of sets,

$$\log(z_1 z_2) = \log z_1 + \log z_2, \quad \log \frac{z_1}{z_2} = \log z_1 - \log z_2. \quad (16.12)$$

For the principal branch there exists $k \in \{-1, 0, 1\}$ with

$$\operatorname{Log}(z_1 z_2) = \operatorname{Log} z_1 + \operatorname{Log} z_2 + 2k\pi i, \quad (16.13)$$

and all three values of k occur. Also $e^{\operatorname{Log} z} = z$ for all $z \neq 0$, but $\operatorname{Log}(e^z) = z$ only when $\operatorname{Im} z \in (-\pi, \pi]$.

Proof. (16.12) follows from (16.9) and Theorem 3.11(ii), since $\ln |z_1 z_2| = \ln |z_1| + \ln |z_2|$ by (1.24). (16.13) is Theorem 3.11(iii) multiplied by i and added to the modulus identity; the examples given there transfer verbatim. $e^{\operatorname{Log} z} = e^{\ln |z|} e^{i \operatorname{Arg} z} = |z| e^{i \operatorname{Arg} z} = z$. Finally $\operatorname{Log}(e^z) = \ln |e^z| + i \operatorname{Arg}(e^z) = \operatorname{Re} z + i \operatorname{Arg}(e^z)$, and $\operatorname{Arg}(e^z) = \operatorname{Im} z$ precisely when $\operatorname{Im} z \in (-\pi, \pi]$ by (16.2) and Definition 3.3. $\square$

Warning 16.8. Take $z_1 = z_2 = -1$. Then $\operatorname{Log} z_1 + \operatorname{Log} z_2 = 2\pi i$ while $\operatorname{Log}(z_1 z_2) = \operatorname{Log} 1 = 0$: the naive identity fails by $2\pi i$. Likewise $\operatorname{Log}(z^2) \neq 2 \operatorname{Log} z$ in general, and $\operatorname{Log}(1/z) \neq -\operatorname{Log} z$ when z is a negative real. Every logarithm identity must be checked, or stated for the multivalued log.

Theorem 16.9 (Analyticity of the principal logarithm). Let $D_0 = \mathbb{C} \setminus (-\infty, 0]$ be the principal slit plane (as in Theorem 10.3). Then Log is analytic on D_0 and

$$\frac{d}{dz} \operatorname{Log} z = \frac{1}{z}, \quad z \in D_0. \quad (16.14)$$

Log is discontinuous at every point of $(-\infty, 0)$, with

$$\lim_{\substack{z \rightarrow x_0 \\ \operatorname{Im} z > 0}} \operatorname{Log} z = \ln |x_0| + i\pi, \quad \lim_{\substack{z \rightarrow x_0 \\ \operatorname{Im} z < 0}} \operatorname{Log} z = \ln |x_0| - i\pi, \quad x_0 < 0, \quad (16.15)$$

a jump of $2\pi i$.

Proof. Write $z = x + iy \in D_0$, so $\text{Log } z = u + iv$ with $u = \frac{1}{2} \ln(x^2 + y^2)$ and $v = \text{Arg } z$. On D_0 we have $x + |z| > 0$, and the half-angle formula gives $v = \text{Arg } z = 2 \arctan \frac{y}{x + \sqrt{x^2 + y^2}}$, a smooth function there. Differentiating,

$$u_x = \frac{x}{x^2 + y^2}, \quad u_y = \frac{y}{x^2 + y^2}, \quad v_x = \frac{-y}{x^2 + y^2}, \quad v_y = \frac{x}{x^2 + y^2}.$$

These four partials are continuous on D_0 and satisfy $u_x = v_y$, $u_y = -v_x$, so Theorem 14.6 gives analyticity and

$$\frac{d}{dz} \text{Log } z = u_x + iv_x = \frac{x - iy}{x^2 + y^2} = \frac{\bar{z}}{|z|^2} = \frac{1}{z}.$$

The jump (16.15) follows from the corresponding statement for Arg , exactly as in Theorem 10.3(iii). $\square$

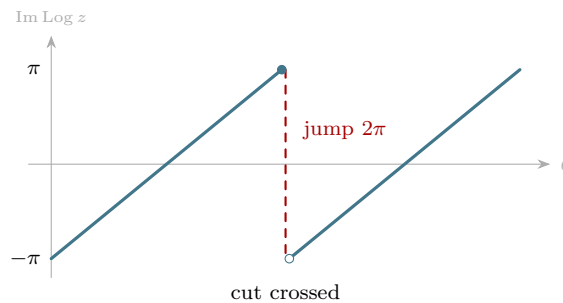


FIGURE 16.2. The jump (16.15). Following $\text{Log } z$ once counterclockwise around the origin, the imaginary part $\text{Arg } z$ climbs steadily from $-\pi$ to π and then drops by 2π as the cut is crossed. The real part $\ln |z|$ returns to its starting value; only the argument fails to. This single picture explains why $\oint_{|z|=1} dz/z = 2\pi i \neq 0$ and why $\ln |z|$ has no global harmonic conjugate (Remark 15.7).

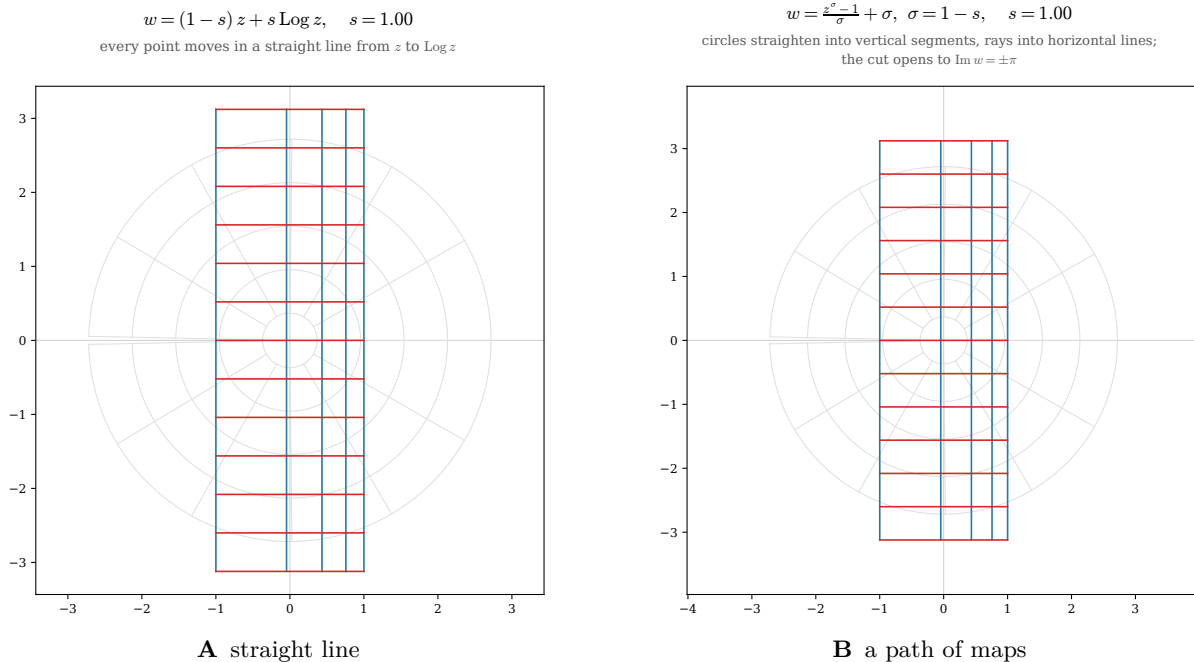
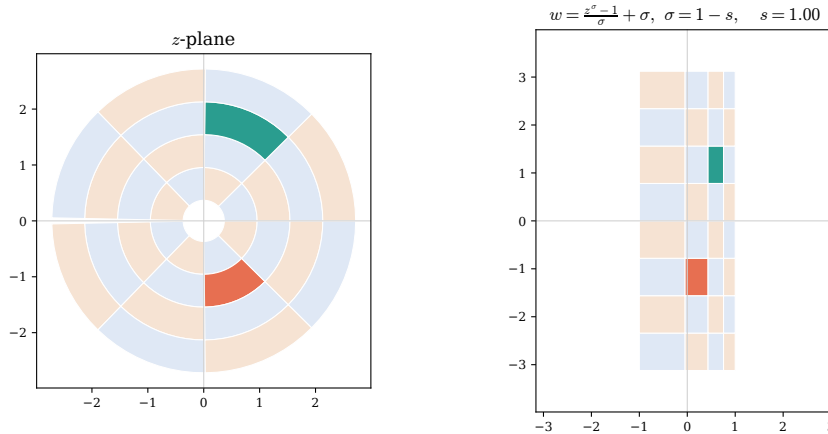


FIGURE 16.3 (animated). The principal logarithm $\text{Log } z$. **A**: every point moves in a straight line from z to $f(z)$. **B**: circles straighten into vertical segments, rays into horizontal lines; the cut opens to $\text{Im } w = \pm\pi$. **C**: the tiles on the left deform along the path of B.

circles straighten into vertical segments, rays into horizontal lines; the cut opens to $\operatorname{Im} w = \pm\pi$



C tiles (Figure 16.3, continued)

Definition 16.10 (Branches of the logarithm). Let $\tau \in \mathbb{R}$ and let $D_\tau = \mathbb{C} \setminus \{te^{i\tau} : t \geq 0\}$ be the plane cut along the ray of angle τ , and $\arg_\tau$ the argument in $(\tau, \tau + 2\pi)$ as in Proposition 10.5. Then

$$L_\tau(z) := \ln |z| + i \arg_\tau z, \quad z \in D_\tau, \quad (16.16)$$

is a branch of $\log$ on D_τ , analytic there with $L'_\tau(z) = 1/z$; and every branch of $\log$ on D_τ equals $L_\tau + 2k\pi i$ for some fixed $k \in \mathbb{Z}$. There is no branch of $\log$ on any region containing a circle around 0.

Caution on notation. The principal slit plane $D_0 = \mathbb{C} \setminus (-\infty, 0]$ of Theorem 16.9 is the member $D_{-\pi}$ (equivalently D_π , the same set) of this family, and on it $L_{-\pi} = \operatorname{Log}$. The member with $\tau = 0$ is the plane cut along $[0, \infty)$; whenever it is meant we write $\mathbb{C} \setminus [0, \infty)$ and $\arg_0$, L_0 explicitly, and D_0 always denotes the principal slit plane.

Justification of the claims in Definition 16.10. Let $z \in D_\tau$ and put $w := e^{-i(\tau+\pi)}z$. Then $|w| = |z|$ and $\arg_\tau z - (\tau + \pi) \in (-\pi, \pi)$ is an argument of w , so $w \in D_0$ and $\operatorname{Arg} w = \arg_\tau z - \tau - \pi$. Hence

$$L_\tau(z) = \operatorname{Log}(e^{-i(\tau+\pi)}z) + i(\tau + \pi), \quad z \in D_\tau.$$

By Theorem 16.9 and the chain rule (13.6) L_τ is analytic on D_τ with $L'_\tau(z) = e^{-i(\tau+\pi)}/(e^{-i(\tau+\pi)}z) = 1/z$; and $e^{L_\tau(z)} = |z|e^{i\arg_\tau z} = z$, so L_τ is a branch of $\log$.

If L is any branch of $\log$ on D_τ , then $e^L = e^{L_\tau}$, so by Theorem 16.3(ii) the function $k := (L - L_\tau)/(2\pi i)$ is continuous on D_τ with integer values. A continuous integer-valued function on the connected set D_τ is constant (as in the proof of Proposition 10.5(i): the integers are at distance ≥ 1 from each other), so $L = L_\tau + 2k\pi i$ with $k \in \mathbb{Z}$ fixed.

Finally suppose L were a branch of $\log$ on a region $D \supseteq C(0, \rho)$, $\rho > 0$. Put $\gamma(t) := \rho e^{it}$, $t \in [0, 2\pi]$, and $h(t) := (L(\gamma(t)) - \ln \rho - it)/(2\pi i)$. Since $\ln \rho + it \in \log \gamma(t)$, the function h is continuous on $[0, 2\pi]$ with integer values (Theorem 16.3(ii)), hence constant, say $h \equiv k$. As $\gamma(0) = \gamma(2\pi) = \rho$, evaluating at $t = 0$ and $t = 2\pi$ gives $L(\rho) = \ln \rho + 2k\pi i$ and $L(\rho) = \ln \rho + 2\pi i + 2k\pi i$, a contradiction. This is the proof of Proposition 10.5(ii), with the quotient by $\rho^{1/n}e^{it/n}$ replaced by the difference with $\ln \rho + it$. $\square$

16.3 Complex powers

Definition 16.11 (Complex power). Let $z \in \mathbb{C} \setminus \{0\}$ and $\alpha \in \mathbb{C}$. Define the set

$$z^\alpha := \{e^{\alpha w} : w \in \log z\} = \{e^{\alpha(\operatorname{Log} z + 2k\pi i)} : k \in \mathbb{Z}\}, \quad (16.17)$$

and the *principal value*

$$z_{\text{pr}}^\alpha := e^{\alpha \operatorname{Log} z}. \quad (16.18)$$

Proposition 16.12 (How many values?). Let $z \neq 0$ and $\alpha \in \mathbb{C}$. The set z^α of (16.17) has:

- (i) exactly one element, if $\alpha \in \mathbb{Z}$; and then it is the ordinary power of Definition 4.1;
- (ii) exactly q elements, if $\alpha = p/q \in \mathbb{Q}$ in lowest terms with $q \geq 2$; and then it is the set of q -th roots of z^p , in agreement with Theorem 4.5;
- (iii) infinitely many elements, if $\alpha \in \mathbb{C} \setminus \mathbb{Q}$ (in particular whenever $\operatorname{Im} \alpha \neq 0$).

Proof. Two values $e^{\alpha(\operatorname{Log} z + 2k\pi i)}$ and $e^{\alpha(\operatorname{Log} z + 2l\pi i)}$ coincide iff $e^{2\pi i \alpha(k-l)} = 1$ iff $\alpha(k-l) \in \mathbb{Z}$, by Theorem 16.3(ii). If $\alpha = m \in \mathbb{Z}$ this holds for all k, l : one value, and it equals $e^{m \operatorname{Log} z} = (e^{\operatorname{Log} z})^m = z^m$ by (16.4). If $\alpha = p/q$ in lowest terms, $\alpha(k-l) \in \mathbb{Z}$ iff $q \mid k-l$: exactly q values, and each satisfies $(z^\alpha)^q = z^p$. Otherwise $\alpha(k-l) \in \mathbb{Z}$ forces $k=l$ (if α is irrational real) or is impossible for $k \neq l$ (if $\operatorname{Im} \alpha \neq 0$, since then $\alpha(k-l) \notin \mathbb{R}$): infinitely many values. $\square$

Example 16.13. i^i : here $\operatorname{Log} i = \frac{i\pi}{2}$ and $\log i = \{i(\frac{\pi}{2} + 2k\pi)\}$, so

$$i^i = \{e^{i \cdot i(\pi/2 + 2k\pi)}\} = \{e^{-\pi/2 - 2k\pi} : k \in \mathbb{Z}\} \subseteq (0, \infty).$$

Every value is a positive *real* number; the principal value is $i_{\text{pr}}^i = e^{-\pi/2} \approx 0.20788$. This is the classical example of the phenomenon that “imaginary to the imaginary power” is real.

Example 16.14. $(1+i)_{\text{pr}}^{2i} = e^{2i \operatorname{Log}(1+i)} = e^{2i(\frac{1}{2} \ln 2 + i\pi/4)} = e^{-\pi/2} e^{i \ln 2}$, of modulus $e^{-\pi/2}$ and argument $\ln 2$. $(-1)^{1/2} = \{e^{(1/2)(i\pi + 2k\pi i)}\} = \{i, -i\}$, matching Theorem 4.5. $1^{\sqrt{2}} = \{e^{\sqrt{2} \cdot 2k\pi i}\}$ is an infinite dense subset of the unit circle — so even 1 has infinitely many “ $\sqrt{2}$ -th powers”.

Proposition 16.15 (Derivative of a power). Fix $\alpha \in \mathbb{C}$. On $D_0 = \mathbb{C} \setminus (-\infty, 0]$ the principal power $z \mapsto z_{\text{pr}}^\alpha = e^{\alpha \operatorname{Log} z}$ is analytic, and

$$\frac{d}{dz} z_{\text{pr}}^\alpha = \alpha z_{\text{pr}}^{\alpha-1}. \quad (16.19)$$

Proof. By Theorems 16.2 and 16.9 and the chain rule (13.6),

$$\frac{d}{dz} e^{\alpha \operatorname{Log} z} = e^{\alpha \operatorname{Log} z} \cdot \frac{\alpha}{z} = \alpha e^{\alpha \operatorname{Log} z} e^{-\operatorname{Log} z} = \alpha e^{(\alpha-1) \operatorname{Log} z} = \alpha z_{\text{pr}}^{\alpha-1},$$

using $z = e^{\operatorname{Log} z}$ and (16.4). $\square$

Warning 16.16. $z^\alpha w^\alpha = (zw)^\alpha$ and $(z^\alpha)^\beta = z^{\alpha\beta}$ are false in general for principal values, because they inherit the failure (16.13). For instance with $z = w = -1$ and $\alpha = \frac{1}{2}$: $z_{\text{pr}}^{1/2} w_{\text{pr}}^{1/2} = i \cdot i = -1$ while $(zw)_{\text{pr}}^{1/2} = 1^{1/2} = 1$. Compare Warning 10.6: it is the same phenomenon, since (16.18) with $\alpha = 1/n$ reproduces Definition 10.2.

16.4 Exercises

Exercise 16.1. Evaluate in the form $x + iy$: $e^{2-i\pi/3}$, $e^{1+i\pi}$, $e^{\ln 2 + i\pi/4}$, $|e^{3+4i}|$, $\text{Arg}(e^{2+7i})$.

Exercise 16.2. Solve for all z : $e^z = 2i$, $e^z = -3$, $e^{2z} + e^z = 0$, $e^z = 1 + i\sqrt{3}$.

Exercise 16.3. Compute Log and the full set $\log$ for $z = -i$, $z = 1 - i$, $z = -e$, $z = 3$.

Exercise 16.4. Compute all values, and the principal value, of $(-1)^i$, 2^{1+i} , $(1+i)^{1-i}$, $i^{2/3}$.

Exercise 16.5. ($\star$) Suppose f is entire, $f' = f$, and $f(x) = e^x$ for all real x . Prove $f(z) = e^z$ as in (16.1). (*Hint: consider $g(z) = f(z)e^{-z}$ and use Theorem 14.13(i).*)

Exercise 16.6. Find the image under $w = e^z$ of: the rectangle $0 < x < 1$, $0 < y < \pi/2$; the line $y = x$; the strip $0 < y < \pi$.

Exercise 16.7. Show that Log is *not* the inverse of $\exp$ as a function on $\mathbb{C}$, by exhibiting z with $\text{Log}(e^z) \neq z$. On which set is it a genuine inverse?

Exercise 16.8. Prove that $|z_{\text{pr}}^\alpha| = |z|^{\text{Re } \alpha} e^{-\text{Im } \alpha \text{ Arg } z}$ for $z \in D_0$, and deduce that $|z_{\text{pr}}^\alpha|$ depends on $\text{Arg } z$ unless α is real.

Exercise 16.9. Prove that e^z takes every value in $\mathbb{C} \setminus \{0\}$ infinitely often, and that the solution set of $e^z = w$ is a translate of the vertical lattice $2\pi i\mathbb{Z}$, namely $\text{Log } w + 2\pi i\mathbb{Z}$. Sketch the solution set of $e^z = -1$ in the plane.

Exercise 16.10. Show that $|e^z| \leq e^{|z|}$ for all z , with equality if and only if z is a non-negative real. Deduce that e^z is bounded on every left half-plane $\{\text{Re } z \leq c\}$ but unbounded on every right half-plane.

Exercise 16.11. ($\star$) Let L be any branch of $\log$ on a region D not containing 0. Prove $L'(z) = 1/z$ on D , and deduce that any two branches on the same region differ by a constant in $2\pi i\mathbb{Z}$. Then (after Chapter 20; use Theorem 20.4) show that D admits a branch of $\log$ if and only if $\oint_\gamma \frac{dz}{z} = 0$ for every closed contour γ in D — so the existence of a logarithm is a purely topological condition on D .

Chapter 17

Trigonometric and Hyperbolic Functions

17.1 Definitions

For real θ , Euler's formula (3.6) gives $e^{i\theta} = \cos \theta + i \sin \theta$ and $e^{-i\theta} = \cos \theta - i \sin \theta$; adding and subtracting,

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad (\theta \in \mathbb{R}). \quad (17.1)$$

The right-hand sides make sense for complex arguments, which dictates the definition.

Definition 17.1 (Complex trigonometric functions). For $z \in \mathbb{C}$ define

$$\sin z := \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z := \frac{e^{iz} + e^{-iz}}{2}, \quad (17.2)$$

and, where the denominators do not vanish,

$$\tan z := \frac{\sin z}{\cos z}, \quad \cot z := \frac{\cos z}{\sin z}, \quad \sec z := \frac{1}{\cos z}, \quad \csc z := \frac{1}{\sin z}. \quad (17.3)$$

Definition 17.2 (Complex hyperbolic functions). For $z \in \mathbb{C}$ define

$$\sinh z := \frac{e^z - e^{-z}}{2}, \quad \cosh z := \frac{e^z + e^{-z}}{2}, \quad (17.4)$$

with $\tanh z := \sinh z / \cosh z$, $\coth z := \cosh z / \sinh z$, $\operatorname{sech} z := 1 / \cosh z$, $\operatorname{csch} z := 1 / \sinh z$ where defined.

By (17.1) these agree with the real functions on $\mathbb{R}$.

Theorem 17.3 (Elementary properties). For all $z, w \in \mathbb{C}$:

$$\sin(-z) = -\sin z, \quad \cos(-z) = \cos z; \quad (17.5)$$

$$\sin^2 z + \cos^2 z = 1; \quad (17.6)$$

$$\sin(z + w) = \sin z \cos w + \cos z \sin w, \quad (17.7)$$

$$\cos(z + w) = \cos z \cos w - \sin z \sin w; \quad (17.8)$$

$$\sin(z + 2\pi) = \sin z, \quad \cos(z + 2\pi) = \cos z, \quad \tan(z + \pi) = \tan z; \quad (17.9)$$

$\sin$ and $\cos$ are entire, with

$$\frac{d}{dz} \sin z = \cos z, \quad \frac{d}{dz} \cos z = -\sin z, \quad \frac{d}{dz} \tan z = \sec^2 z. \quad (17.10)$$

The analogous statements hold for the hyperbolic functions (listed at the end of the proof; the periods become $2\pi i$ and πi), with

$$\cosh^2 z - \sinh^2 z = 1, \quad \frac{d}{dz} \sinh z = \cosh z, \quad \frac{d}{dz} \cosh z = \sinh z. \quad (17.11)$$

Proof. All of these are algebraic identities in $e^{\pm iz}$ and follow from Theorem 16.2. We verify three. (17.6): writing $E := e^{iz}$, so $e^{-iz} = 1/E$ by (16.4),

$$\sin^2 z + \cos^2 z = \frac{(E - E^{-1})^2}{(2i)^2} + \frac{(E + E^{-1})^2}{4} = \frac{-(E^2 - 2 + E^{-2}) + (E^2 + 2 + E^{-2})}{4} = 1.$$

(17.8): expand the right-hand side with $A := e^{iz}$, $B := e^{iw}$:

$$\begin{aligned} & \frac{(A + A^{-1})(B + B^{-1})}{4} - \frac{(A - A^{-1})(B - B^{-1})}{(2i)^2} \\ &= \frac{AB + AB^{-1} + A^{-1}B + A^{-1}B^{-1}}{4} + \frac{AB - AB^{-1} - A^{-1}B + A^{-1}B^{-1}}{4} \\ &= \frac{2AB + 2A^{-1}B^{-1}}{4} = \frac{e^{i(z+w)} + e^{-i(z+w)}}{2} = \cos(z + w). \end{aligned}$$

Identity (17.7) is the same computation.

(17.10): by (16.7) and the chain rule (13.6), $\frac{d}{dz} e^{\pm iz} = \pm i e^{\pm iz}$, so

$$\frac{d}{dz} \sin z = \frac{ie^{iz} + ie^{-iz}}{2i} = \frac{e^{iz} + e^{-iz}}{2} = \cos z,$$

and similarly for $\cos$. Both are entire as linear combinations of entire functions. The formula for $\tan$ is the quotient rule (13.5) with (17.6).

Periodicity (17.9) follows from $e^{i(z+2\pi)} = e^{iz} e^{2\pi i} = e^{iz}$. Likewise $e^{i(z+\pi)} = e^{iz} e^{i\pi} = -e^{iz}$ and $e^{-i(z+\pi)} = -e^{-iz}$ give $\sin(z + \pi) = -\sin z$ and $\cos(z + \pi) = -\cos z$, hence $\tan(z + \pi) = \tan z$.

The hyperbolic statements are obtained in the same way from (17.4), or from the trigonometric ones via the dictionary (17.12) of Proposition 17.4 below. In particular $\sinh$ and $\cosh$ are entire with period $2\pi i$, $\tanh(z + \pi i) = \tanh z$, $\sinh$ is odd and $\cosh$ even, and the addition laws read $\sinh(z + w) = \sinh z \cosh w + \cosh z \sinh w$ and $\cosh(z + w) = \cosh z \cosh w + \sinh z \sinh w$ (note the sign). $\square$

Proposition 17.4 (The trigonometric–hyperbolic dictionary). *For all $z \in \mathbb{C}$,*

$$\sin(iz) = i \sinh z, \quad \cos(iz) = \cosh z, \quad \sinh(iz) = i \sin z, \quad \cosh(iz) = \cos z. \quad (17.12)$$

Consequently every hyperbolic identity is a trigonometric identity in disguise and vice versa; over $\mathbb{C}$ the two families are one family.

Proof. $\sin(iz) = \frac{e^{i(iz)} - e^{-i(iz)}}{2i} = \frac{e^{-z} - e^z}{2i} = \frac{-(e^z - e^{-z})}{2i} = i \cdot \frac{e^z - e^{-z}}{2} = i \sinh z$, using $1/i = -i$. The other three are identical computations. $\square$

17.2 Real and imaginary parts; size

Theorem 17.5 (Component formulas). *Let $z = x + iy$ with $x, y \in \mathbb{R}$. Then*

$$\sin z = \sin x \cosh y + i \cos x \sinh y, \quad (17.13)$$

$$\cos z = \cos x \cosh y - i \sin x \sinh y, \quad (17.14)$$

$$\sinh z = \sinh x \cos y + i \cosh x \sin y, \quad (17.15)$$

$$\cosh z = \cosh x \cos y + i \sinh x \sin y, \quad (17.16)$$

and consequently

$$|\sin z|^2 = \sin^2 x + \sinh^2 y, \quad |\cos z|^2 = \cos^2 x + \sinh^2 y. \quad (17.17)$$

Proof. Apply (17.7) with $z = x$ and $w = iy$, then (17.12):

$$\sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy) = \sin x \cosh y + i \cos x \sinh y.$$

The others are the same. For (17.17),

$$|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sin^2 x (1 + \sinh^2 y) + (1 - \sin^2 x) \sinh^2 y = \sin^2 x + \sinh^2 y,$$

using $\cosh^2 = 1 + \sinh^2$ from (17.11). The computation for $\cos$ is identical. $\square$

Corollary 17.6 (Zeros). *$\sin z = 0$ if and only if $z = k\pi$ with $k \in \mathbb{Z}$; $\cos z = 0$ if and only if $z = \frac{\pi}{2} + k\pi$ with $k \in \mathbb{Z}$. In particular the complex sine and cosine have exactly the same zeros as their real counterparts, all of them real. Consequently $\tan z$ and $\sec z$ are analytic on $\mathbb{C} \setminus \{\frac{\pi}{2} + k\pi\}$, and $\cot z, \csc z$ on $\mathbb{C} \setminus \{k\pi\}$.*

Proof. By (17.17), $|\sin z|^2 = \sin^2 x + \sinh^2 y$ is a sum of two non-negative real numbers, so it vanishes iff both vanish: $\sin x = 0$ (i.e. $x = k\pi$) and $\sinh y = 0$ (i.e. $y = 0$, since $\sinh$ is strictly increasing on $\mathbb{R}$ with $\sinh 0 = 0$). Same for $\cos$. Analyticity of the quotients follows from Theorem 13.3. $\square$

Corollary 17.7 (The complex sine is unbounded). *$\sin$ and $\cos$ are unbounded on $\mathbb{C}$; indeed*

$$|\sin(x + iy)| \geq |\sinh y| \xrightarrow{|y| \rightarrow \infty} \infty. \quad (17.18)$$

For example $|\cos(10i)| = \cosh 10 > 11000$.

Proof. Immediate from (17.17) and $\sinh y = \frac{1}{2}(e^y - e^{-y}) \rightarrow \pm\infty$. $\square$

Remark 17.8. Corollary 17.7 is a genuine surprise for students arriving from real calculus, where $|\sin| \leq 1$ is the first fact one learns. The inequality is an accident of restricting to $\mathbb{R}$. In fact by Liouville's theorem (Theorem 22.2) a bounded entire function must be constant, so no non-constant entire extension of $\sin$ could possibly be bounded. Equations such as $\sin z = 5$ therefore do have solutions; see Example 17.9.

Example 17.9 (Solving $\sin z = 5$). Put $E = e^{iz}$. By (17.2) the equation reads $\frac{E - E^{-1}}{2i} = 5$, i.e. $E^2 - 10iE - 1 = 0$. By Theorem 6.2,

$$E = \frac{10i \pm \sqrt{-100 + 4}}{2} = (5 \pm 2\sqrt{6})i.$$

Both values are purely imaginary with positive imaginary part, so $\arg E = \frac{\pi}{2} + 2k\pi$ and $|E| = 5 \pm 2\sqrt{6}$. From $E = e^{iz}$ we get $iz \in \log E$, i.e.

$$z = -i \log E = -i \left[\ln(5 \pm 2\sqrt{6}) + i \left(\frac{\pi}{2} + 2k\pi \right) \right] = \frac{\pi}{2} + 2k\pi - i \ln(5 \pm 2\sqrt{6}), \quad k \in \mathbb{Z}.$$

Since $(5 + 2\sqrt{6})(5 - 2\sqrt{6}) = 25 - 24 = 1$, the two logarithms are negatives of each other, so the solutions come in conjugate pairs about the real axis, as they must because $\sin$ has real coefficients. Numerically $\ln(5 + 2\sqrt{6}) \approx 2.2924$, so $z \approx \frac{\pi}{2} \pm 2.2924i$ modulo 2π .

17.3 Mapping properties of $\sin$

Theorem 17.10 (sin on the fundamental strip). *Let $\Lambda_{\sin} := \{z = x + iy : -\frac{\pi}{2} < x < \frac{\pi}{2}\}$. Then $\sin$ maps $\Lambda_{\sin}$ bijectively onto $\mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$. Moreover, for $0 < c < \frac{\pi}{2}$ and $d > 0$:*

(i) *the vertical line $x = c$ maps onto the branch of the hyperbola*

$$\frac{u^2}{\sin^2 c} - \frac{v^2}{\cos^2 c} = 1 \tag{17.19}$$

lying in $u > 0$;

(ii) *the horizontal segment $y = d$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, maps onto the upper half of the ellipse*

$$\frac{u^2}{\cosh^2 d} + \frac{v^2}{\sinh^2 d} = 1; \tag{17.20}$$

(iii) *the real segment $y = 0$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, maps onto $[-1, 1]$.*

Proof. By (17.13), $u = \sin x \cosh y$ and $v = \cos x \sinh y$.

(i) On $x = c$: $\frac{u}{\sin c} = \cosh y$ and $\frac{v}{\cos c} = \sinh y$; subtract the squares and use $\cosh^2 - \sinh^2 = 1$ to get (17.19). Since $\cosh y \geq 1$ and $\sin c > 0$ we have $u \geq \sin c > 0$, the right branch.

(ii) On $y = d > 0$: $\frac{u}{\cosh d} = \sin x$ and $\frac{v}{\sinh d} = \cos x$; add the squares and use $\sin^2 + \cos^2 = 1$ to get (17.20). For $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ we have $\cos x > 0$, so $v > 0$: the upper half.

(iii) $y = 0$ gives $u = \sin x \in [-1, 1]$, $v = 0$.

Bijectivity: for $z = x + iy \in \Lambda_{\sin}$ we have $\cos x > 0$, so $v = \cos x \sinh y$ has the sign of y . Let (u, v) with $v > 0$ be given. The function $d \mapsto u^2/\cosh^2 d + v^2/\sinh^2 d$ is continuous and strictly decreasing on $(0, \infty)$ (the second term strictly, the first weakly), tends to $+\infty$ as $d \downarrow 0$ and to 0 as $d \rightarrow \infty$; so it equals 1 for exactly one $d > 0$. For that d the numbers $u/\cosh d$ and $v/\sinh d > 0$ are the sine and cosine of exactly one $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, and by (17.13) the point $x + id$ is the only point of $\Lambda_{\sin}$ with $\sin z = u + iv$ (any preimage must have $y > 0$ and, by (ii), $y = d$). Thus the half-ellipses (17.20), $d > 0$, foliate the open upper half-plane, each covered exactly once. The case $v < 0$ is symmetric ($y < 0$). Finally $v = 0$ forces $y = 0$, and then (iii) shows that $u = \sin x$ takes each value in $(-1, 1)$ exactly once for $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and never a value with $|u| \geq 1$. So the image is $\mathbb{C}$ minus the two real rays $|u| \geq 1$, $v = 0$, and each of its points is attained exactly once. $\square$

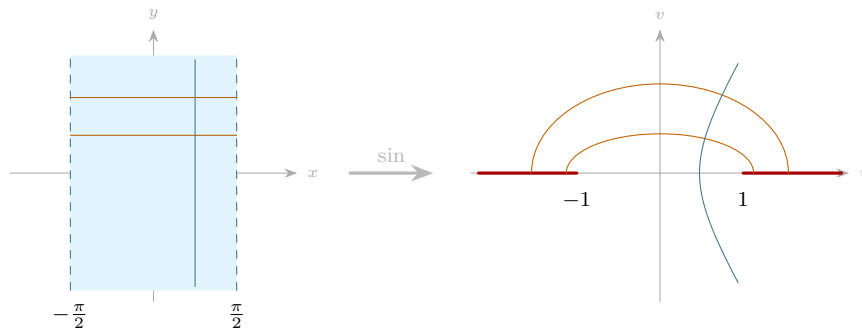


FIGURE 17.1. Theorem 17.10: $\sin$ carries the vertical strip $|\operatorname{Re} z| < \pi/2$ onto the plane cut along $(-\infty, -1] \cup [1, \infty)$. Horizontal lines $y = d$ become the half-ellipses (17.20), which shrink to the segment $[-1, 1]$ as $d \rightarrow 0$; vertical lines $x = c$ become the hyperbola branches (17.19). The two families are confocal, with foci ± 1 .

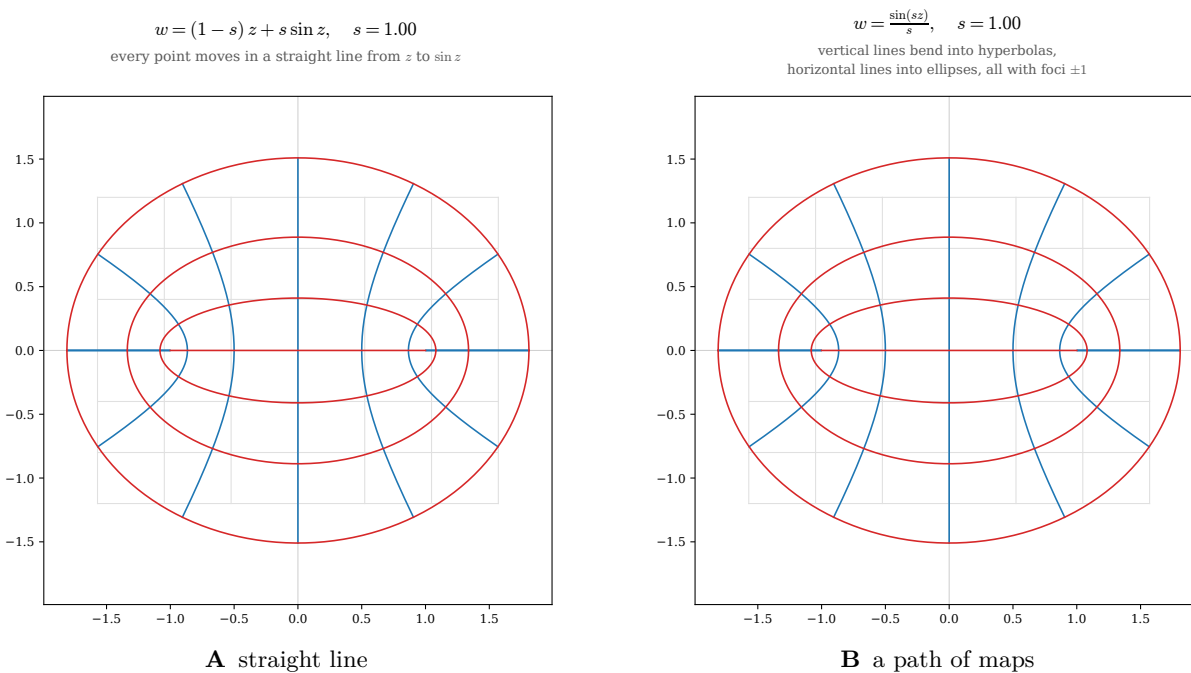
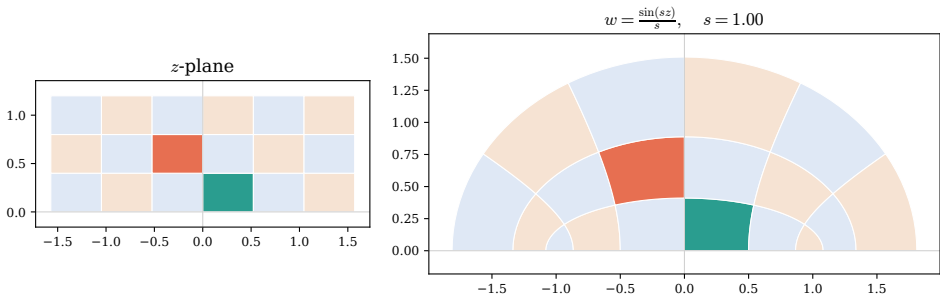


FIGURE 17.2 (animated). The sine $\sin z$. **A**: every point moves in a straight line from z to $f(z)$. **B**: vertical lines bend into hyperbolas, horizontal lines into ellipses, all with foci ± 1 . **C**: the tiles on the left deform along the path of B.

the half-strip $|\operatorname{Re} z| < \pi/2$, $\operatorname{Im} z > 0$ onto the upper half-plane



C tiles (Figure 17.2, continued)

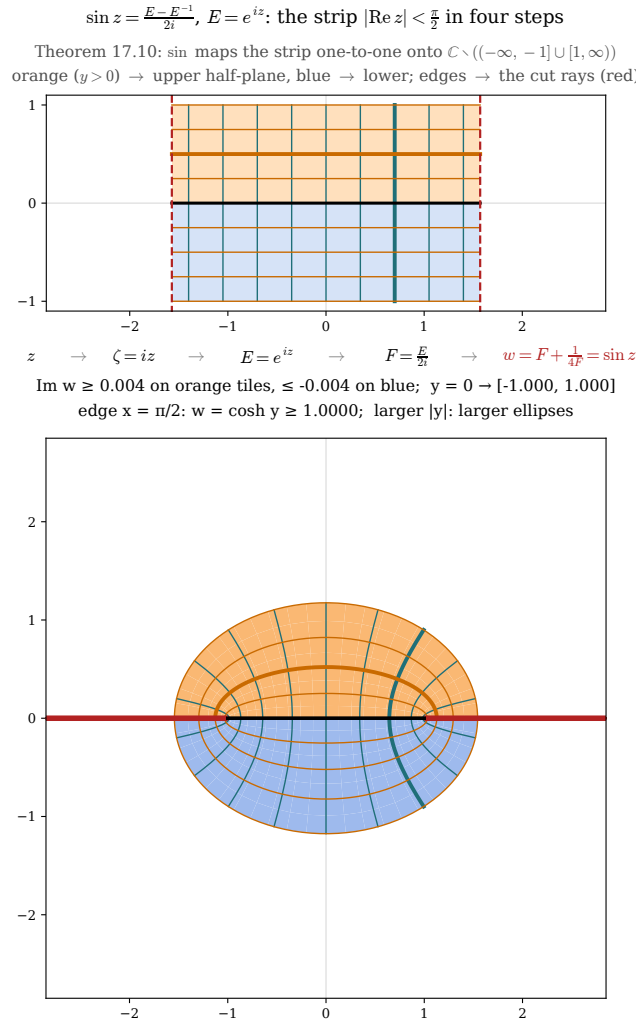


FIGURE 17.3 (animated). Why Theorem 17.10 holds: $\sin z = \frac{E-E^{-1}}{2i}$ with $E = e^{iz}$ (Definition 17.1), done in steps. The strip $|\operatorname{Re} z| < \frac{\pi}{2}$ is turned by $z \mapsto iz$; then $E = e^{iz}$ sends it onto the right half-plane, vertical lines $x = c$ to rays $\arg E = c$ and horizontal lines $y = d$ to circles $|E| = e^{-d}$; then $F = E/(2i)$; and finally $w = F + \frac{1}{4F} = \sin z$, a Joukowski map with foci ± 1 , bends the circles into the ellipses (17.20) and the rays into the hyperbolas (17.19), and folds the two edges $x = \pm \frac{\pi}{2}$ onto the cut rays $(-\infty, -1]$ and $[1, \infty)$. Each intermediate picture is the image under a genuine map. The upper half of the strip (orange) fills the upper half-plane, the lower half (blue) the lower one.

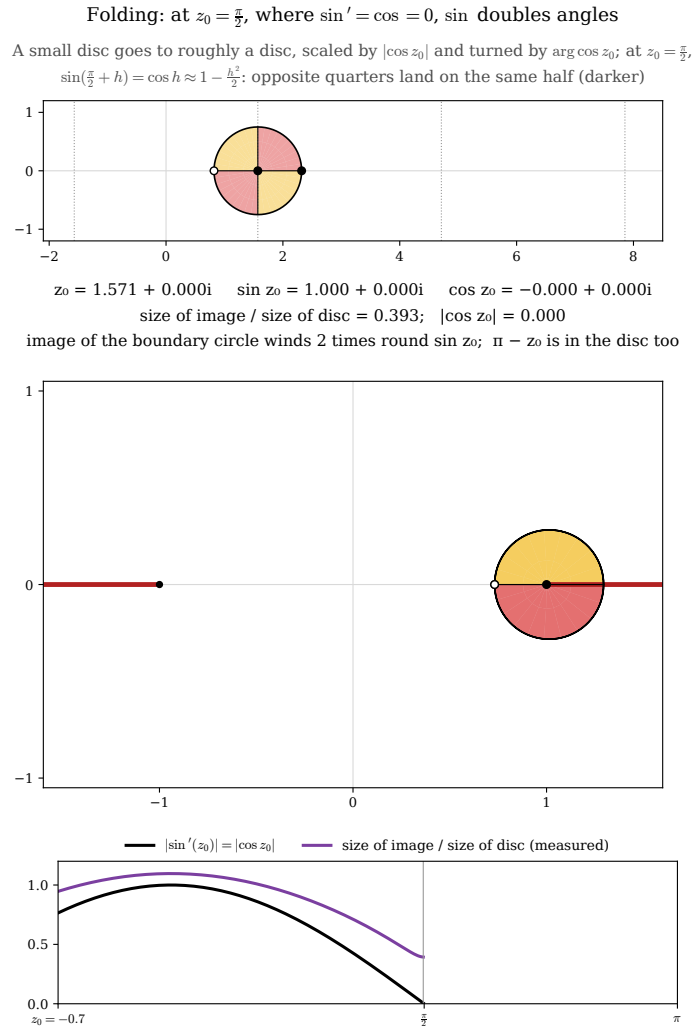


FIGURE 17.4 (animated). Where $\sin$ stops being one-to-one. A disc slides along the real axis; its image is roughly a disc, scaled by $|\sin' z_0| = |\cos z_0|$ (graph below). At $z_0 = \frac{\pi}{2}$, where $\cos z_0 = 0$, $\sin(\frac{\pi}{2} + h) = \cos h \approx 1 - \frac{h^2}{2}$ doubles angles: opposite quarters of the disc land on the same half-disc, and the image of the boundary winds twice round 1. Then a window $a < x < a + \pi$ slides across the plane: $\sin$ is one-to-one on it only when it lines up with a strip $-\frac{\pi}{2} + k\pi < x < \frac{\pi}{2} + k\pi$, since $\sin(\pi - z) = \sin z$; otherwise it contains two solutions of $\sin z = 0.5 + 0.4i$ (counted on screen). The strips next to $\Lambda_{\sin}$ cover the cut plane again, their upper halves going to the lower half-plane.

Remark 17.11 (Inverse trigonometric functions). Solving $\sin w = z$ as in Example 17.9 gives, in general,

$$\sin^{-1} z = -i \log \left(iz + (1 - z^2)^{1/2} \right), \quad (17.21)$$

a multiple-valued relation with two sources of multivaluedness: the square root (2 values) and the logarithm (infinitely many). A branch of $\sin^{-1}$ is obtained by choosing a branch of each; the principal branch uses the principal square root and Log, and inverts Theorem 17.10. Similarly

$$\tan^{-1} z = \frac{i}{2} \log \frac{i + z}{i - z}, \quad \sinh^{-1} z = \log \left(z + (z^2 + 1)^{1/2} \right). \quad (17.22)$$

Differentiating the principal branches gives the expected formulas, e.g. $\frac{d}{dz} \sin^{-1} z = (1 - z^2)^{-1/2}$

and $\frac{d}{dz} \tan^{-1} z = \frac{1}{1+z^2}$ on suitable cut planes.

Derivation of (17.21). Set $E = e^{iw}$. Then $\sin w = z$ reads $E - E^{-1} = 2iz$, i.e. $E^2 - 2izE - 1 = 0$, so by Theorem 6.2 $E = iz + s$ where s is either square root of $(iz)^2 + 1 = 1 - z^2$. Then $iw \in \log E$, i.e. $w = -i \log(iz + (1 - z^2)^{1/2})$. $\square$

17.4 Exercises

Exercise 17.1. Express in the form $x + iy$: $\sin(1 + i)$, $\cos(2i)$, $\tan(i \ln 2)$, $\sinh(1 + i\pi/2)$.

Exercise 17.2. Find all solutions: $\cos z = 2$, $\sin z = i$, $\cosh z = 0$, $e^z = \sin z$ (numerically: a pair of roots near $0.363 \pm 1.134i$, and real roots near -3.183 , -6.281 , $-9.425, \dots$).

Exercise 17.3. Prove $|\sinh y| \leq |\sin z| \leq \cosh y$ for $z = x + iy$, and identify where each inequality is an equality.

Exercise 17.4. (★) Prove that $\tan z$ maps the strip $-\frac{\pi}{4} < \operatorname{Re} z < \frac{\pi}{4}$ bijectively onto the unit disc. (Write $\tan z = \frac{1}{i} \cdot \frac{e^{2iz} - 1}{e^{2iz} + 1}$ and compose known maps.)

Exercise 17.5. Verify the Cauchy–Riemann equations directly for $\sin z = \sin x \cosh y + i \cos x \sinh y$, and confirm $\frac{d}{dz} \sin z = \cos z$ using (14.2).

Exercise 17.6. Show that $\sin z$ and $\cos z$ take every complex value infinitely often, whereas e^z takes every complex value *except* 0 infinitely often. (These are instances of Picard's theorem.)

Exercise 17.7. Find all z with $\sin z = \cos z$, and all z with $\tan z = i$. (The second has no solutions — explain why, using (17.2).)

Exercise 17.8. Locate all zeros and all poles of $\tan z$, $\cot z$, $\tanh z$ and $\coth z$, giving the order of each. (Use Corollary 17.6 and the dictionary (17.12).)

Exercise 17.9. (★) Prove that $\cos z$ maps the half-strip $\{0 < \operatorname{Re} z < \pi, \operatorname{Im} z > 0\}$ conformally onto the lower half-plane. (Use $\cos z = \sin(\frac{\pi}{2} - z)$ and Theorem 17.10.) Where does the map fail to be conformal on the boundary?

Exercise 17.10. Prove $\overline{\sin z} = \sin \bar{z}$ and $\overline{\cos z} = \cos \bar{z}$, and deduce that the solutions of $\sin z = r$ with $r \in \mathbb{R}$ are symmetric about the real axis.

Chapter 18

Complex Powers, Continued

The multiple-valued relation z^α was defined in (16.17) and its principal value in (16.18), and Proposition 16.15 computed the derivative of the principal branch. This short chapter finishes the topic: it describes *all* the branches, records exactly which algebraic laws survive, and examines the power functions as mappings. It also states the binomial series, whose proof must wait for Chapter 24.

18.1 All branches of z^α

Recall the definitions. For $z \neq 0$ and $\alpha \in \mathbb{C}$,

$$z^\alpha = \{ e^{\alpha w} : w \in \log z \}, \quad z_{\text{pr}}^\alpha = e^{\alpha \text{Log } z}, \quad (18.1)$$

with $\log$ and Log as in (16.9) and (16.10).

Theorem 18.1 (Branches of the power function). *Let $\alpha \in \mathbb{C}$ and let $\tau \in \mathbb{R}$. Let $D_\tau = \mathbb{C} \setminus \{te^{i\tau} : t \geq 0\}$ and let L_τ be the logarithm branch (16.16) on D_τ . Then for each $k \in \mathbb{Z}$ the function*

$$P_{\tau,k}(z) := e^{\alpha(L_\tau(z) + 2k\pi i)} = e^{2k\pi i\alpha} e^{\alpha L_\tau(z)} \quad (18.2)$$

is analytic on D_τ , takes values in z^α , and satisfies

$$P'_{\tau,k}(z) = \alpha \frac{P_{\tau,k}(z)}{z}, \quad z \in D_\tau. \quad (18.3)$$

Every branch of z^α on D_τ is one of the $P_{\tau,k}$, and the number of distinct functions among them is the number of values of z^α counted in Proposition 16.12.

Proof. Analyticity: L_τ is analytic on D_τ with $L'_\tau = 1/z$ (Definition 16.10 and its justification), and $\exp$ is entire (16.7), so the chain rule (13.6) gives both analyticity and

$$P'_{\tau,k}(z) = e^{\alpha(L_\tau(z) + 2k\pi i)} \cdot \alpha L'_\tau(z) = \alpha \frac{P_{\tau,k}(z)}{z},$$

which is (18.3). The value lies in z^α because $L_\tau(z) + 2k\pi i \in \log z$ by (16.11).

Conversely let f be any branch of z^α on D_τ , i.e. f continuous with $f(z) \in z^\alpha$ for all z . Then $g := f/P_{\tau,0}$ is continuous on the connected set D_τ (the denominator never vanishes, by (16.3)) and takes values in the set

$$E := \{ e^{2k\pi i\alpha} : k \in \mathbb{Z} \}, \quad (18.4)$$

which is *countable*. Now $g(D_\tau)$ is a connected subset of E , hence countable; and a connected metric space with more than one point is uncountable. Therefore $g(D_\tau)$ is a single point, i.e. g is constant, and $f = P_{\tau,k}$ for some k .

Warning 18.2. It is tempting to argue instead that E of (18.4) is *discrete* and quote the fact that a continuous map from a connected set to a discrete set is constant. That argument is *false*: for α real and irrational, E is a dense subset of the unit circle by Weyl's equidistribution theorem, and has no isolated points at all. Countability, not discreteness, is what does the work above. $\square$

Remark 18.3 (When the exponent is real and the branch is invisible). If $\alpha = n \in \mathbb{Z}$ then $e^{2k\pi in} = 1$ for all k , so there is one branch, defined on all of $\mathbb{C} \setminus \{0\}$ and in fact entire when $n \geq 0$: the cut is an artefact one may discard. This is the only case in which the cut can be dispensed with. For $\alpha = p/q$ in lowest terms with $q \geq 2$ one gets exactly q branches — the q determinations of the q -th root of z^p — and for all other α infinitely many.

Proposition 18.4 (Which power laws survive). *Let $z, w \in \mathbb{C} \setminus \{0\}$ and $\alpha, \beta \in \mathbb{C}$. As identities of sets,*

$$z^\alpha z^\beta = z^{\alpha+\beta} \quad \text{is false in general,} \quad \frac{1}{z^\alpha} = z^{-\alpha} \quad \text{is true,} \quad (18.5)$$

where the left-hand side of the first means $\{ab : a \in z^\alpha, b \in z^\beta\}$. For principal values the following hold for all $z \in \mathbb{C} \setminus \{0\}$ (the cut plays no role here):

$$z_{\text{pr}}^\alpha z_{\text{pr}}^\beta = z_{\text{pr}}^{\alpha+\beta}, \quad (18.6)$$

$$(z_{\text{pr}}^\alpha)^{-1} = z_{\text{pr}}^{-\alpha}, \quad (18.7)$$

$$|z_{\text{pr}}^\alpha| = |z|^{\text{Re } \alpha} e^{-\text{Im } \alpha \text{ Arg } z}. \quad (18.8)$$

By contrast $(zw)_{\text{pr}}^\alpha = z_{\text{pr}}^\alpha w_{\text{pr}}^\alpha$ and $(z_{\text{pr}}^\alpha)_{\text{pr}}^\beta = z_{\text{pr}}^{\alpha\beta}$ both fail in general.

Proof. Same base: $e^{\alpha \text{Log } z} e^{\beta \text{Log } z} = e^{(\alpha+\beta) \text{Log } z}$ by (16.4); this needs no correction because the same logarithm is used twice, which is exactly why (18.6) is safe while the product rule in the base is not. (18.7) is (18.6) with $\beta = -\alpha$. For (18.8), write $\alpha = a + ib$ and $\text{Log } z = \ln |z| + i \text{Arg } z$; then

$$\alpha \text{Log } z = (a \ln |z| - b \text{Arg } z) + i(a \text{Arg } z + b \ln |z|),$$

and (16.2) gives $|e^{\alpha \text{Log } z}| = e^{a \ln |z| - b \text{Arg } z}$, which is (18.8).

For the set identity $z^\alpha z^\beta = z^{\alpha+\beta}$: the left side is $\{e^{\alpha \text{Log } z + 2k\pi i \alpha + \beta \text{Log } z + 2l\pi i \beta}\}$ over $k, l \in \mathbb{Z}$, the right side has only the diagonal $k = l$. Taking $z = 1$, $\alpha = \beta = \frac{1}{2}$ gives left side $\{1, -1\} \cdot \{1, -1\} = \{1, -1\}$ and right side $1^1 = \{1\}$: not equal. The failure of $(zw)_{\text{pr}}^\alpha$ is Warning 16.16. For the iterated power, take $z = e$, $\alpha = 2\pi i$, $\beta = i$: then $z_{\text{pr}}^\alpha = e^{2\pi i} = 1$, so $(z_{\text{pr}}^\alpha)_{\text{pr}}^\beta = 1_{\text{pr}}^i = 1$, whereas $z_{\text{pr}}^{\alpha\beta} = e^{-2\pi} \neq 1$. $\square$

18.2 Power functions as mappings

Proposition 18.5 (z^α on a sector, $\alpha > 0$ real). *Let $\alpha > 0$ be real and let $S(\theta_1, \theta_2) = \{re^{i\theta} : r > 0, \theta_1 < \theta < \theta_2\}$ with $-\pi \leq \theta_1 < \theta_2 \leq \pi$. Then the principal branch z_{pr}^α maps $S(\theta_1, \theta_2)$ bijectively onto $S(\alpha\theta_1, \alpha\theta_2)$, provided $\alpha(\theta_2 - \theta_1) \leq 2\pi$; it is conformal there, since by (18.3) its derivative $\alpha z_{\text{pr}}^{\alpha-1}$ never vanishes.*

Proof. By (16.18) with real α ,

$$z_{\text{pr}}^\alpha = e^{\alpha \ln|z|} e^{i\alpha \text{Arg } z} = |z|^\alpha e^{i\alpha \text{Arg } z},$$

so the modulus map $r \mapsto r^\alpha$ is a bijection of $(0, \infty)$ onto itself and the argument map $\theta \mapsto \alpha\theta$ is a bijection of (θ_1, θ_2) onto $(\alpha\theta_1, \alpha\theta_2)$. Injectivity requires that the image interval have length at most 2π , which is the stated hypothesis. Non-vanishing of the derivative is (18.3) together with (16.3), and Theorem 13.10 then gives conformality. $\square$

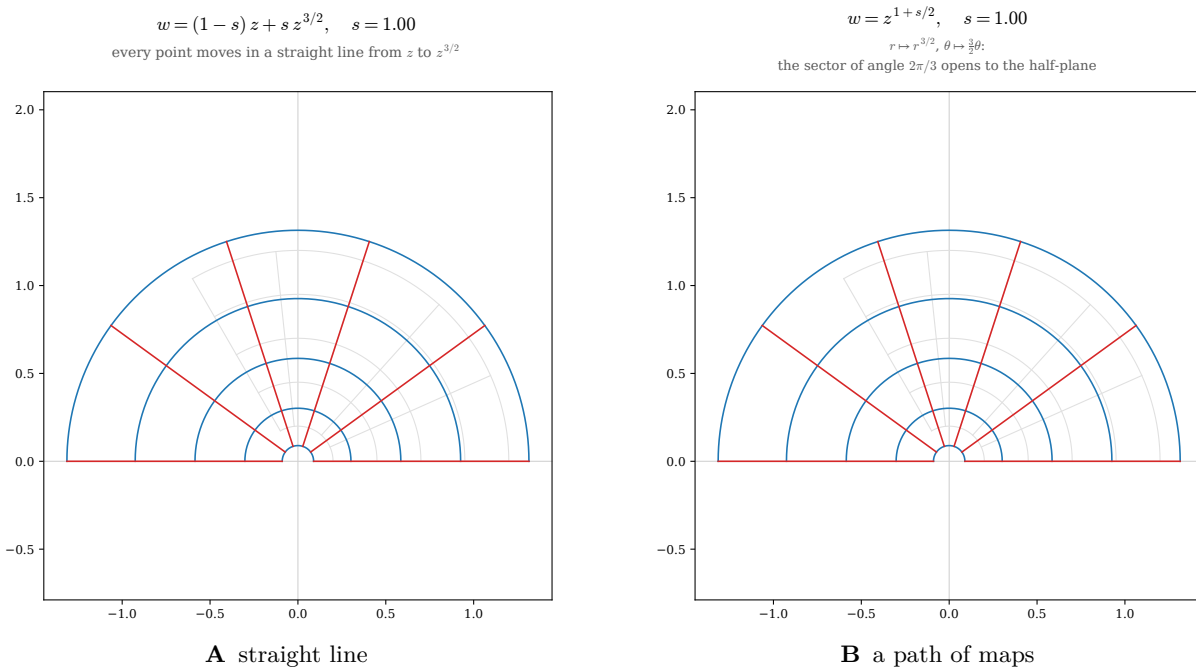
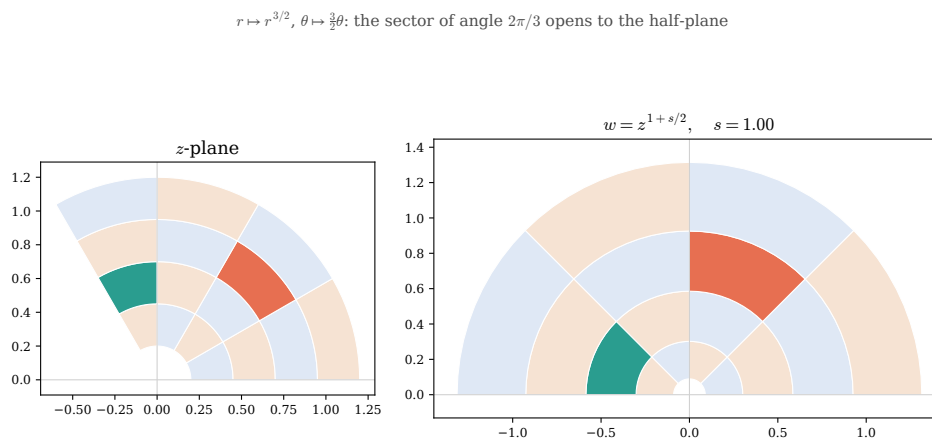


FIGURE 18.1 (animated). $z_{\text{pr}}^{3/2}$ on a sector of angle $2\pi/3$. **A**: every point moves in a straight line from z to $f(z)$. **B**: $r \mapsto r^{3/2}, \theta \mapsto \frac{3}{2}\theta$: the sector of angle $2\pi/3$ opens to the half-plane. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 18.1, continued)

Example 18.6 (Opening a corner). The map $w = z_{\text{pr}}^{1/2}$ halves angles at 0: it takes the slit plane $-\pi < \text{Arg } z < \pi$ onto the right half-plane $-\pi/2 < \text{Arg } w < \pi/2$, as recorded in Theorem 10.3(ii). Conversely $w = z^2$ doubles angles and opens a quarter-plane into a half-plane. Choosing $\alpha = \pi/\varphi$ opens a wedge of angle φ into a half-plane; this is the single most useful move in Chapter 33, where it is used to reduce flow and temperature problems in corners to problems in a half-plane.

Example 18.7 (z^i takes circles to spirals). Let $\alpha = i$ and consider $w = z_{\text{pr}}^i = e^{i \text{Log } z}$. By (18.8), $|w| = e^{-\text{Arg } z}$, while

$$\text{Arg } w \equiv \text{Re}(\text{Log } z) = \ln |z| \pmod{2\pi}.$$

So the modulus of w is controlled by the *argument* of z and the argument of w by the *modulus* of z : the roles are exchanged. The circle $|z| = \rho$ maps into the ray $\{te^{i \ln \rho} : t > 0\}$ (that is, $\ln \rho \in \arg w$; this is $\text{Arg } w = \ln \rho$ only when $-\pi < \ln \rho \leq \pi$), and in fact onto its segment $e^{-\pi} \leq t < e^\pi$, since $|w| = e^{-\text{Arg } z}$ with $\text{Arg } z \in (-\pi, \pi]$; the ray $\text{Arg } z = \theta$ maps onto the circle $|w| = e^{-\theta}$. The image of a logarithmic spiral is therefore another logarithmic spiral.

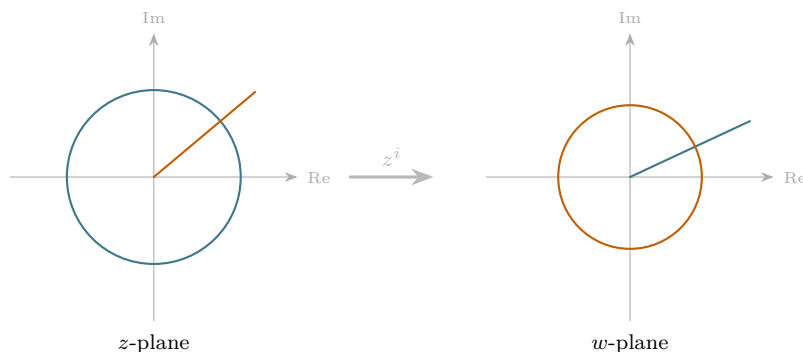


FIGURE 18.2. Example 18.7: under $w = z^i$ the roles of modulus and argument are exchanged. A circle $|z| = \rho$ (left, cyan) has constant modulus and varying argument, and its image lies on a *ray* $\{te^{i \ln \rho} : t > 0\}$ of constant argument and varying modulus; a ray $\text{Arg } z = \theta$ (left, amber) maps to a *circle* $|w| = e^{-\theta}$. Consequently logarithmic spirals map to logarithmic spirals.

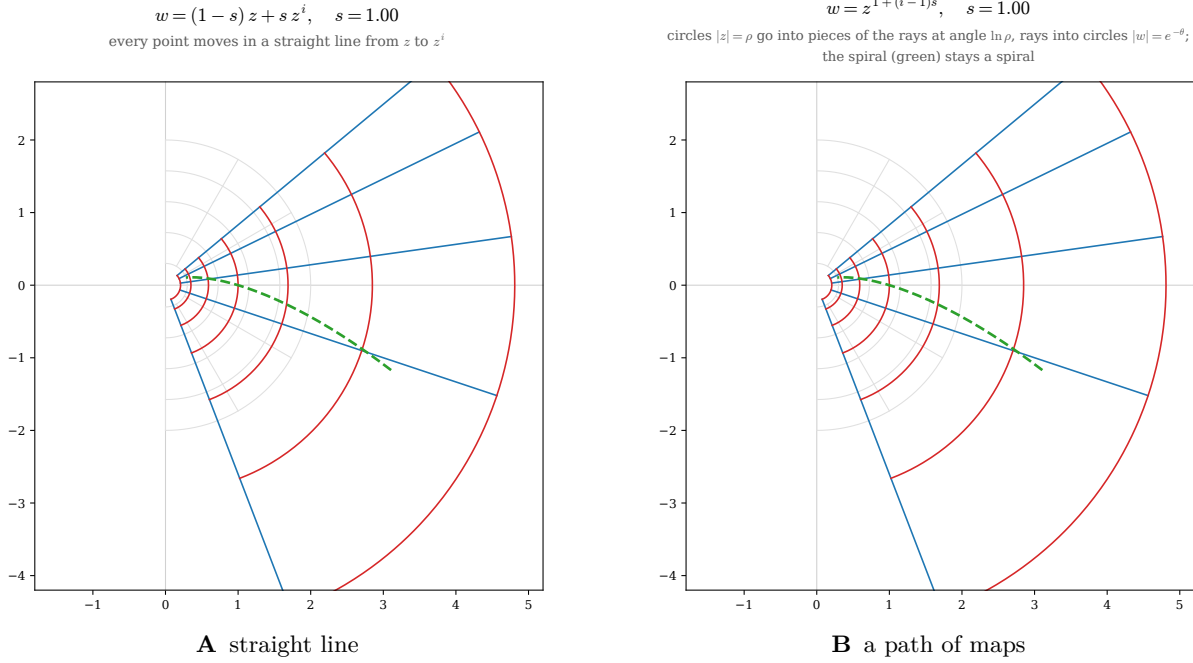
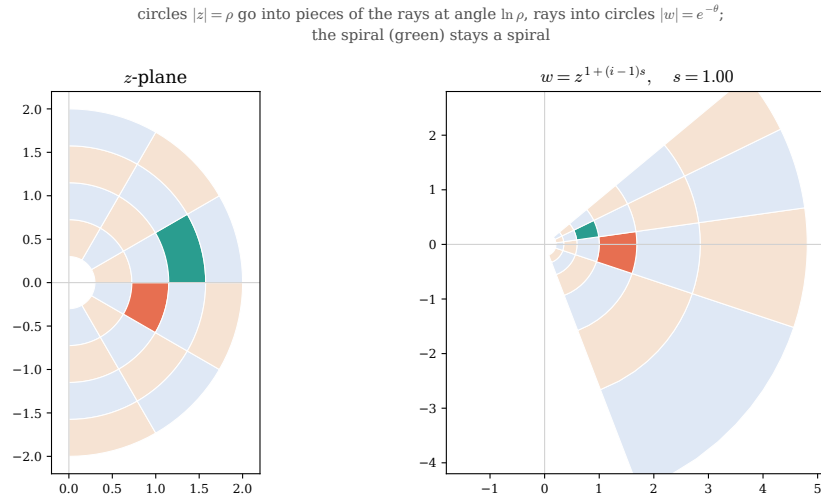


FIGURE 18.3 (animated). z_{pr}^i : circles to rays, rays to circles. **A**: every point moves in a straight line from z to $f(z)$. **B**: circles $|z| = \rho$ go into pieces of the rays at angle $\ln \rho$, rays into circles $|w| = e^{-\theta}$; the spiral (green) stays a spiral. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 18.3, continued)

18.3 The binomial series

Theorem 18.8 (Binomial series; proof deferred). *Let $\alpha \in \mathbb{C}$. Define the generalised binomial coefficients*

$$\binom{\alpha}{0} := 1, \quad \binom{\alpha}{n} := \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} \quad (n \geq 1). \quad (18.9)$$

Then the principal branch of $(1+z)^\alpha$ satisfies

$$(1+z)_{\text{pr}}^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} z^n, \quad |z| < 1, \quad (18.10)$$

the series converging absolutely on $|z| < 1$ and uniformly on $|z| \leq \rho$ for each $\rho < 1$. If $\alpha = m \in \{0, 1, 2, \dots\}$ the coefficients vanish for $n > m$ and (18.10) is the ordinary binomial theorem, valid for all z .

The proof is postponed to Example 24.5, once Taylor's theorem is available. What can be said now is that the radius of convergence is exactly 1 when $\alpha \notin \{0, 1, 2, \dots\}$: then no coefficient (18.9) vanishes, and

$$\left| \binom{\alpha}{n+1} / \binom{\alpha}{n} \right| = \frac{|\alpha - n|}{n+1} \rightarrow 1 \quad (n \rightarrow \infty),$$

so the ratio formula of Theorem 23.10 gives radius 1. The reason behind it is the singularity of $(1+z)^\alpha$ at $z = -1$ (a branch point, or a pole when α is a negative integer).

18.4 Exercises

Exercise 18.1. Find all values, and the principal value, of $(1+i)^i$, $(-8)^{2/3}$, $2^{\sqrt{2}}$, $(2i)^{1-i}$.

Exercise 18.2. Verify (18.8) numerically for $z = -1+i$ and $\alpha = 2-3i$, by computing both sides to four decimals.

Exercise 18.3. Give explicit z, w, α with $(zw)_{\text{pr}}^\alpha \neq z_{\text{pr}}^\alpha w_{\text{pr}}^\alpha$, and compute the ratio of the two sides; show that the ratio is always of the form $e^{2\pi i k \alpha}$ with $k \in \{-1, 0, 1\}$.

Exercise 18.4. Find the image of the sector $0 < \text{Arg } z < \pi/3$ under $z_{\text{pr}}^{3/2}$ and under z^3 . Is either map conformal at 0? Explain, and describe what each does to angles at 0.

Exercise 18.5. (★) Show that z_{pr}^α is real-valued for all $z > 0$ if and only if $\alpha \in \mathbb{R}$, and that it is bounded on the punctured unit disc $0 < |z| < 1$ if and only if $\text{Re } \alpha \geq 0$.

Exercise 18.6. Write out the first four terms of (18.10) for $\alpha = \frac{1}{2}$ and for $\alpha = -1$, and identify the second series.

Exercise 18.7. Determine, for each α below, how many distinct values z^α has and describe the set: $\alpha = 5$, $\alpha = \frac{3}{4}$, $\alpha = -\frac{7}{6}$, $\alpha = \sqrt{3}$, $\alpha = 1+i$. (Use Proposition 16.12.)

Exercise 18.8. Show that $z \mapsto z_{\text{pr}}^\alpha$ maps the unit circle minus -1 into the unit circle if and only if α is real, and describe the image when $\alpha = \frac{1}{2}$ and when $\alpha = 3$.

Exercise 18.9. (★) Prove that the set $\{i^{i^i}\}$ — computed by taking all values at each stage — is an infinite subset of the unit circle, namely $\{e^{ic_k(\pi/2+2m\pi)} : k, m \in \mathbb{Z}\}$ with $c_k = e^{-\pi/2-2k\pi}$. (Work out i^i first, from Example 16.13, then raise i to each of those powers.) Contrast this with $(i^i)^i$, and show that the two bracketings give different sets.

Part V

Integration

Chapter 19

Real Integrals and Contour Integrals

Two different objects are both called “integral” in this subject, and confusing them is the commonest early error. The first integrates a complex-valued function over an interval of the *real* line; it is elementary, being two real integrals side by side. The second integrates a complex-valued function along a *curve in the plane*; it is the object on which the whole theory turns. This chapter defines both and proves the estimates that everything later depends on.

19.1 Integrals of a complex-valued function of a real variable

Definition 19.1 (Real-variable integral). Let $a, b \in \mathbb{R}$ with $a \leq b$ and let $h : [a, b] \rightarrow \mathbb{C}$ be written $h(t) = u(t) + iv(t)$ with $u = \operatorname{Re} h$ and $v = \operatorname{Im} h$ real-valued. If u and v are Riemann integrable on $[a, b]$ we say h is integrable and define

$$\int_a^b h(t) dt := \int_a^b u(t) dt + i \int_a^b v(t) dt \in \mathbb{C}. \quad (19.1)$$

Proposition 19.2 (Elementary properties). Let $h, g : [a, b] \rightarrow \mathbb{C}$ be integrable and let $c \in \mathbb{C}$. Then

$$\int_a^b (h + g) = \int_a^b h + \int_a^b g, \quad \int_a^b c h = c \int_a^b h, \quad (19.2)$$

$$\operatorname{Re} \int_a^b h = \int_a^b \operatorname{Re} h, \quad \operatorname{Im} \int_a^b h = \int_a^b \operatorname{Im} h, \quad (19.3)$$

$$\int_a^b h = \int_a^c h + \int_c^b h \quad (a \leq c \leq b). \quad (19.4)$$

If h is continuous and $H' = h$ on $[a, b]$ in the sense of (6.6), then

$$\int_a^b h(t) dt = H(b) - H(a). \quad (19.5)$$

Proof. Additivity and (19.3) and (19.4) are immediate from (19.1) and the corresponding real statements. For $\int c h$ write $c = \lambda + i\mu$ and $h = u + iv$; then $ch = (\lambda u - \mu v) + i(\lambda v + \mu u)$, so

$$\int ch = \left(\lambda \int u - \mu \int v \right) + i \left(\lambda \int v + \mu \int u \right) = (\lambda + i\mu) \left(\int u + i \int v \right) = c \int h.$$

For (19.5), apply the real fundamental theorem of calculus to $\operatorname{Re} H$ and $\operatorname{Im} H$ separately and recombine using (6.6). $\square$

Lemma 19.3 (The modulus inequality). *Let $h : [a, b] \rightarrow \mathbb{C}$ be integrable, $a \leq b$. Then*

$$\left| \int_a^b h(t) dt \right| \leq \int_a^b |h(t)| dt. \quad (19.6)$$

Proof. The inequality is trivial if $I := \int_a^b h = 0$, so assume $I \neq 0$ and write $I = |I| e^{i\varphi}$ using Theorem 3.7. Put $c := e^{-i\varphi}$, so that $|c| = 1$ and $cI = |I|$ is real and positive. Then, using (19.2) and (19.3),

$$\left| \int_a^b h \right| = cI = \int_a^b c h(t) dt = \operatorname{Re} \int_a^b c h(t) dt = \int_a^b \operatorname{Re}(c h(t)) dt,$$

the third equality because the quantity is real. Now $\operatorname{Re}(ch) \leq |ch| = |c||h| = |h|$ pointwise by (1.26) and (1.24), and the real integral is monotone, so the last expression is $\leq \int_a^b |h|$. $\square$

Lemma 19.3 is the reason the *ML*-inequality below is true, and through it the reason almost every estimate in this course works. Note that the proof had to rotate I onto the positive real axis first: there is no order on $\mathbb{C}$ (Warning 1.8), so one cannot argue directly.

19.2 Contours

Definition 19.4 (Smooth curve, contour). A *smooth curve* in $\mathbb{C}$ is a function $\gamma : [a, b] \rightarrow \mathbb{C}$ with γ' continuous on $[a, b]$ and $\gamma'(t) \neq 0$ for all t , where γ' is as in (6.6). A *contour* (or *piecewise smooth curve*) is a continuous $\gamma : [a, b] \rightarrow \mathbb{C}$ for which there is a partition $a = t_0 < t_1 < \dots < t_m = b$ such that each restriction $\gamma|_{[t_{j-1}, t_j]}$ is a smooth curve.

The point $\gamma(a)$ is the *initial point* and $\gamma(b)$ the *terminal point*. The contour is *closed* if $\gamma(a) = \gamma(b)$, and *simple* if γ is injective on $[a, b]$ or, when γ is closed, injective on $[a, b)$ — that is, it does not cross itself. A *simple closed contour* is both.

Definition 19.5 (Standard contours). Two parametrisations are used constantly.

$$\text{Segment from } z_1 \text{ to } z_2 : \quad \gamma(t) = (1-t)z_1 + tz_2, \quad t \in [0, 1]; \quad (19.7)$$

$$\text{Circle } C(z_0, \rho), \text{ counterclockwise: } \gamma(t) = z_0 + \rho e^{it}, \quad t \in [0, 2\pi]. \quad (19.8)$$

Given a contour $\gamma : [a, b] \rightarrow \mathbb{C}$ we write $-\gamma$ for the *reversed* contour $(-\gamma)(t) := \gamma(a+b-t)$, and $\gamma_1 + \gamma_2$ for the concatenation of two contours when the terminal point of γ_1 is the initial point of γ_2 .

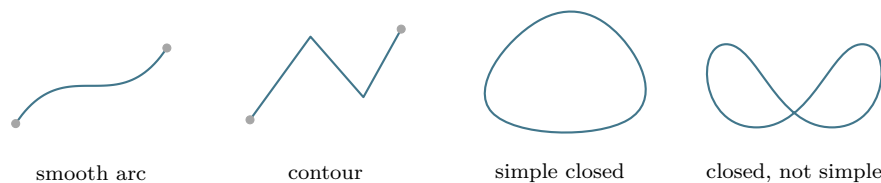


FIGURE 19.1. The vocabulary of Definition 19.4. From the left: a smooth arc; a contour, smooth in pieces with corners allowed; a simple closed contour, which does not cross itself; and a closed contour that is *not* simple. Cauchy's theorems are stated for the third kind, and the residue theorem in its general form (27.7) handles the fourth by counting winding numbers.

Definition 19.6 (Length). The *length* of a contour $\gamma : [a, b] \rightarrow \mathbb{C}$ is

$$\text{length}(\gamma) := \int_a^b |\gamma'(t)| dt = \int_a^b \sqrt{(\operatorname{Re} \gamma'(t))^2 + (\operatorname{Im} \gamma'(t))^2} dt, \quad (19.9)$$

the usual arclength of the plane curve $t \mapsto (\operatorname{Re} \gamma, \operatorname{Im} \gamma)$. For the circle (19.8), $\gamma'(t) = i\rho e^{it}$ so $|\gamma'| = \rho$ and $\text{length} = 2\pi\rho$, as it should be.

19.3 The contour integral

Definition 19.7 (Contour integral). Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a contour and let f be continuous on the image $\gamma([a, b])$. The *contour integral of f along γ* is

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt, \quad (19.10)$$

the right-hand side being a real-variable integral in the sense of (19.1). (For a piecewise smooth γ , apply (19.10) on each smooth piece and add.) When γ is closed we often write $\oint_{\gamma} f(z) dz$.

Proposition 19.8 (Well-definedness). *The value of (19.10) is unchanged under an orientation-preserving reparametrisation: if $\varphi : [c, d] \rightarrow [a, b]$ is continuously differentiable with $\varphi' > 0$, $\varphi(c) = a$, $\varphi(d) = b$, then $\int_{\gamma \circ \varphi} f = \int_{\gamma} f$.*

Proof. By the chain rule $(\gamma \circ \varphi)'(s) = \gamma'(\varphi(s))\varphi'(s)$, so

$$\int_c^d f(\gamma(\varphi(s))) \gamma'(\varphi(s)) \varphi'(s) ds = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

by the real substitution rule applied to real and imaginary parts. $\square$

Theorem 19.9 (Properties of the contour integral). *Let f, g be continuous on the contour γ and $c \in \mathbb{C}$. Then*

$$\int_{\gamma} (f + g) = \int_{\gamma} f + \int_{\gamma} g, \quad \int_{\gamma} cf = c \int_{\gamma} f; \quad (19.11)$$

$$\int_{-\gamma} f = - \int_{\gamma} f; \quad (19.12)$$

$$\int_{\gamma_1 + \gamma_2} f = \int_{\gamma_1} f + \int_{\gamma_2} f. \quad (19.13)$$

Proof. Linearity is (19.2). For (19.12), put $\sigma(t) = \gamma(a + b - t)$, so $\sigma'(t) = -\gamma'(a + b - t)$ and

$$\int_{-\gamma} f = \int_a^b f(\gamma(a + b - t)) (-\gamma'(a + b - t)) dt \stackrel{s=a+b-t}{=} - \int_a^b f(\gamma(s)) \gamma'(s) ds.$$

(19.13) is (19.4). $\square$

Theorem 19.10 (The ML -inequality). *Let γ be a contour of length $L = \text{length}(\gamma)$ and let f be continuous on γ with*

$$|f(z)| \leq M \quad \text{for all } z \text{ on } \gamma. \quad (19.14)$$

Then

$$\left| \int_{\gamma} f(z) dz \right| \leq ML. \quad (19.15)$$

Proof. By (19.10), Lemma 19.3, and (1.24),

$$\left| \int_{\gamma} f \right| = \left| \int_a^b f(\gamma(t)) \gamma'(t) dt \right| \leq \int_a^b |f(\gamma(t))| |\gamma'(t)| dt \leq M \int_a^b |\gamma'(t)| dt = ML. \quad \square$$

The *ML*-inequality is crude — it discards all cancellation — and yet it proves Liouville's theorem and Cauchy's inequalities, it proves Jordan's lemma (together with Lemma 19.3 and Jordan's inequality $\sin \theta \geq 2\theta/\pi$), and it gives the vanishing of nearly every auxiliary arc in Chapters 29 and 30. Its power comes from being applied on contours whose length or whose bound is being sent to a limit.

19.4 Two computations that decide the theory

Example 19.11 (The key integral). Let $z_0 \in \mathbb{C}$, $\rho > 0$, and $n \in \mathbb{Z}$. Let C be the circle $C(z_0, \rho)$ traversed once counterclockwise, parametrised by (19.8). Then

$$\oint_C (z - z_0)^n dz = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \in \mathbb{Z}, n \neq -1. \end{cases} \quad (19.16)$$

Proof. With $\gamma(t) = z_0 + \rho e^{it}$ we have $\gamma(t) - z_0 = \rho e^{it}$ and $\gamma'(t) = i\rho e^{it}$, so by (19.10) and De Moivre (4.1),

$$\oint_C (z - z_0)^n dz = \int_0^{2\pi} \rho^n e^{int} \cdot i\rho e^{it} dt = i\rho^{n+1} \int_0^{2\pi} e^{i(n+1)t} dt.$$

If $n = -1$ the integrand is 1 and the value is $i\rho^0 \cdot 2\pi = 2\pi i$. If $n \neq -1$, put $m := n + 1 \neq 0$; then by (19.5) with $H(t) = e^{imt}/(im)$,

$$\int_0^{2\pi} e^{imt} dt = \frac{e^{2\pi im} - 1}{im} = 0,$$

since $e^{2\pi im} = 1$ for integer m by Theorem 16.3(ii). $\square$

Two features deserve emphasis. First, the answer does not depend on ρ or on z_0 : the integral is a topological quantity, not a metric one. Second, the exceptional exponent is $n = -1$, and -1 is exactly the exponent for which $(z - z_0)^n$ has no antiderivative that is single-valued on a punctured disc — the antiderivative would have to be a branch of $\log(z - z_0)$, and the last claim of Definition 16.10 (proved just after it) forbids one around a circle. The entire residue calculus is the systematic exploitation of (19.16).

Example 19.12 (Path dependence for a non-analytic integrand). Compute $\int_\gamma \bar{z} dz$ from 0 to $1+i$ along two paths.

Path 1: the segment. By (19.7), $\gamma(t) = t(1+i)$, $\gamma'(t) = 1+i$, $\overline{\gamma(t)} = t(1-i)$, so

$$\int_{\gamma_1} \bar{z} dz = \int_0^1 t(1-i)(1+i) dt = 2 \int_0^1 t dt = 1.$$

Path 2: along the real axis to 1, then vertically to $1+i$. On the first leg $\gamma(t) = t$, $\gamma' = 1$, $\bar{z} = t$, giving $\int_0^1 t dt = \frac{1}{2}$. On the second, $\gamma(t) = 1+it$, $\gamma' = i$, $\bar{z} = 1-it$, giving

$$\int_0^1 (1-it)i dt = i \int_0^1 dt + \int_0^1 t dt = i + \frac{1}{2}.$$

Total: $1+i$. Since $1 \neq 1+i$, the integral of $\bar{z}$ is *path dependent*. Compare Example 13.4: $\bar{z}$ is nowhere analytic. Theorem 20.4 will show that this is no coincidence.

Example 19.13 (A path-independent one). Compute $\int_\gamma z dz$ along the same two paths. Along the segment, $\int_0^1 t(1+i) \cdot (1+i) dt = (1+i)^2/2 = i$. Along the bent path, $\int_0^1 t dt + \int_0^1 (1+it)i dt = \frac{1}{2} + (i - \frac{1}{2}) = i$. Equal — and indeed $\frac{1}{2}z^2$ is an antiderivative of z on all of $\mathbb{C}$, so Theorem 20.2 below applies.

19.5 Exercises

Exercise 19.1. Evaluate $\int_0^\pi (e^{2it} + it^2) dt$ and $\int_0^1 \frac{dt}{1+it}$, writing each answer in the form $x + iy$.

Exercise 19.2. Compute $\int_\gamma \operatorname{Re}(z) dz$ where γ is (a) the segment from 0 to $2 + i$, (b) the arc of the parabola $y = x^2/4$ from 0 to $2 + i$. Are they equal? What does this tell you about $\operatorname{Re} z$?

Exercise 19.3. Evaluate $\oint_C \frac{dz}{z-2}$ where C is (a) $|z-2| = 3$, (b) $|z| = 1$, both counterclockwise. (For (b) you may use Theorem 20.8 once you have read the next chapter, or find an antiderivative on a disc containing $|z| \leq 1$ and use Theorem 20.2.)

Exercise 19.4. Use the ML -inequality to show $\left| \oint_{|z|=2} \frac{dz}{z^2+1} \right| \leq \frac{4\pi}{3}$.

Exercise 19.5. (*) Let γ be the circle $|z| = \rho$ counterclockwise. Prove that $\oint_\gamma \bar{z} dz = 2\pi i \rho^2$, and note that this is *nonzero* even though γ is closed. Reconcile this with Theorem 20.9.

Exercise 19.6. Evaluate $\int_\gamma |z| dz$ and $\int_\gamma |z|^2 dz$ where γ is the upper half of $|z| = 1$ from 1 to -1 . Are either of these path-independent? (Compare Example 19.12.)

Exercise 19.7. Using (A.17), write $\oint_{|z|=1} \bar{z} dz$ as a pair of real line integrals and evaluate each. Confirm the total against Exercise 19.5.

Exercise 19.8. (*) Let f be continuous on the contour γ and suppose $|\int_\gamma f| = M \operatorname{length}(\gamma)$ with $M = \max_\gamma |f|$, i.e. equality holds in the ML -inequality (19.15). Prove that $|f| \equiv M$ on γ and that $f(\gamma(t))\gamma'(t)$ has constant argument. (Trace the two inequalities in the proof of Lemma 19.3.)

Exercise 19.9. Prove that if f is continuous on γ and $|f| \leq M$, and if γ is the arc of $|z| = \rho$ subtending angle θ , then $|\int_\gamma f| \leq M\rho\theta$. Deduce a bound for $|\int_\gamma \frac{e^{iz}}{z} dz|$ on the upper half of $|z| = \rho$.

Chapter 20

The Cauchy–Goursat Theorem and Independence of Path

Example 19.12 showed that $\int_{\gamma} \bar{z} dz$ depends on the path and Example 19.13 that $\int_{\gamma} z dz$ does not. The difference is analyticity. This chapter proves the theorem that makes that precise, in the sharp form due to Goursat, in which *no* continuity assumption on f' is made — a point that matters, because it is what eventually lets us prove that analytic functions are automatically infinitely differentiable without circularity.

20.1 Antiderivatives and the fundamental theorem

Definition 20.1 (Antiderivative). Let $D \subseteq \mathbb{C}$ be a region and $f : D \rightarrow \mathbb{C}$. A function $F : D \rightarrow \mathbb{C}$ is an *antiderivative* (or *primitive*) of f on D if F is analytic on D and

$$F'(z) = f(z) \quad \text{for every } z \in D. \quad (20.1)$$

Theorem 20.2 (Fundamental theorem for contour integrals). *Let D be a region, let f be continuous on D , and suppose f has an antiderivative F on D . Then for every contour γ in D with initial point z_1 and terminal point z_2 ,*

$$\int_{\gamma} f(z) dz = F(z_2) - F(z_1). \quad (20.2)$$

In particular the integral depends only on the endpoints, and $\oint_{\gamma} f = 0$ for every closed contour γ in D .

Proof. Assume first that $\gamma : [a, b] \rightarrow D$ is smooth. Define $h(t) := F(\gamma(t))$. By the chain rule — in the mixed form used in Theorem 13.10, where the outer function is complex analytic and the inner variable is real —

$$h'(t) = F'(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t) \quad (t \in [a, b]),$$

with one-sided derivatives at a and b ; this is continuous on $[a, b]$. For completeness, here is the proof of the mixed chain rule. Fix $t \in [a, b]$, put $w_0 := \gamma(t)$, and let $\eta(w) := \frac{F(w) - F(w_0)}{w - w_0} - F'(w_0)$ for $w \neq w_0$, $\eta(w_0) := 0$; then η is continuous at w_0 and $F(w) - F(w_0) = (F'(w_0) + \eta(w))(w - w_0)$ for all $w \in D$. Hence for real $s \neq 0$ with $t + s \in [a, b]$,

$$\frac{h(t+s) - h(t)}{s} = \left(F'(w_0) + \eta(\gamma(t+s)) \right) \frac{\gamma(t+s) - \gamma(t)}{s} \rightarrow F'(w_0)\gamma'(t) \quad (s \rightarrow 0),$$

because γ is continuous, so $\eta(\gamma(t+s)) \rightarrow \eta(w_0) = 0$. Hence by (19.10) and (19.5),

$$\int_{\gamma} f(z) dz = \int_a^b h'(t) dt = h(b) - h(a) = F(z_2) - F(z_1).$$

For a piecewise smooth contour, apply this on each smooth piece and add: the intermediate values of F cancel telescopically. If γ is closed then $z_1 = z_2$ and the right side of (20.2) is 0. $\square$

Example 20.3. $\int_{\gamma} z^n dz = \frac{z_2^{n+1} - z_1^{n+1}}{n+1}$ for $n \geq 0$ and any contour, since $z^{n+1}/(n+1)$ is entire. For $n \leq -2$ the same formula holds on $\mathbb{C} \setminus \{0\}$. For $n = -1$ there is no antiderivative on $\mathbb{C} \setminus \{0\}$ — and indeed (19.16) gives $2\pi i \neq 0$ around a circle. The absence of a global antiderivative and the nonvanishing of a closed integral are the same fact.

Theorem 20.4 (Converse: path independence gives an antiderivative). *Let D be a region and f continuous on D . The following are equivalent.*

- (i) f has an antiderivative on D .
- (ii) $\oint_{\gamma} f = 0$ for every closed contour γ in D .
- (iii) $\int_{\gamma} f$ depends only on the initial and terminal points of γ , for contours γ in D .

Proof. (i) $\Rightarrow$ (ii) is Theorem 20.2.

(ii) $\Leftrightarrow$ (iii): if γ_1, γ_2 share endpoints then $\gamma_1 + (-\gamma_2)$ is closed, and by (19.12) and (19.13), $\oint_{\gamma_1 - \gamma_2} f = \int_{\gamma_1} f - \int_{\gamma_2} f$; this vanishes iff the two agree. Conversely every closed contour splits into two contours with common endpoints.

(iii) $\Rightarrow$ (i): fix $z_0 \in D$ and define

$$F(z) := \int_{\gamma_z} f(\zeta) d\zeta, \quad (20.3)$$

where γ_z is any contour in D from z_0 to z ; such a contour exists because D is a region (Definition 5.8), and by (iii) the value does not depend on the choice, so F is well defined. Fix $z \in D$ and choose $\varepsilon > 0$ with $D(z, \varepsilon) \subseteq D$. For $0 < |\Delta z| < \varepsilon$ the segment $[z, z + \Delta z]$ lies in $D(z, \varepsilon)$, so we may compute $F(z + \Delta z)$ by appending it to γ_z :

$$F(z + \Delta z) - F(z) = \int_{[z, z + \Delta z]} f(\zeta) d\zeta.$$

Since $\int_{[z, z + \Delta z]} d\zeta = \Delta z$ (take $f \equiv 1$, $F(\zeta) = \zeta$ in Theorem 20.2), we may write

$$\frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) = \frac{1}{\Delta z} \int_{[z, z + \Delta z]} (f(\zeta) - f(z)) d\zeta.$$

Given $\eta > 0$, continuity of f at z gives $\delta \in (0, \varepsilon)$ with $|f(\zeta) - f(z)| < \eta$ whenever $|\zeta - z| < \delta$. For $|\Delta z| < \delta$ every ζ on the segment satisfies this, so the ML -inequality (19.15), with $M = \eta$ and $L = |\Delta z|$, bounds the right side by $\frac{1}{|\Delta z|} \cdot \eta |\Delta z| = \eta$. Hence the difference quotient tends to $f(z)$: F is differentiable at z with $F'(z) = f(z)$. As z was arbitrary, F is an antiderivative. $\square$

20.2 Goursat's lemma

Theorem 20.5 (Goursat). *Let D be an open set and f analytic on D . Let $\Delta \subseteq D$ be a closed triangle (the convex hull of three points), and let $\partial\Delta$ denote its boundary traversed once. Then*

$$\oint_{\partial\Delta} f(z) \, dz = 0. \quad (20.4)$$

More generally, (20.4) holds if f is only assumed continuous on D and analytic on $D \setminus \{p\}$ for one point $p \in D$.

Proof. Write $I(\Delta) := \oint_{\partial\Delta} f$. Bisect the sides of Δ and thereby divide it into four closed subtriangles $\Delta^{(1)}, \dots, \Delta^{(4)}$, each similar to Δ with ratio $\frac{1}{2}$. Orienting each boundary consistently, every interior edge is traversed twice in opposite directions, so those contributions cancel by (19.12) and

$$I(\Delta) = \sum_{j=1}^4 I(\Delta^{(j)}). \quad (20.5)$$

Hence $|I(\Delta)| \leq 4 \max_j |I(\Delta^{(j)})|$, so at least one j has $|I(\Delta^{(j)})| \geq \frac{1}{4} |I(\Delta)|$. Choose such a subtriangle and call it Δ_1 . Repeating, we obtain a nested sequence

$$\Delta = \Delta_0 \supseteq \Delta_1 \supseteq \Delta_2 \supseteq \dots$$

with

$$|I(\Delta_n)| \geq 4^{-n} |I(\Delta)|, \quad \text{diam}(\Delta_n) = 2^{-n} \text{diam}(\Delta), \quad \text{length}(\partial\Delta_n) = 2^{-n} L, \quad (20.6)$$

where $L = \text{length}(\partial\Delta)$.

The Δ_n are nonempty nested compact sets with diameters tending to 0, so by Theorem 5.13 their intersection is a single point $z_* \in \Delta \subseteq D$. (Pick $w_n \in \Delta_n$; the sequence is Cauchy by the diameter bound, and its limit lies in every Δ_n since each is closed.)

Now use differentiability at z_* , in the little- o form of Exercise 13.5; the direction needed here is immediate, since $\frac{f(z) - f(z_*)}{z - z_*} - f'(z_*) \rightarrow 0$ as $z \rightarrow z_*$ (and for $z = z_*$ both sides below vanish). Thus given $\eta > 0$ there is $\delta > 0$ with $D(z_*, \delta) \subseteq D$ and

$$|f(z) - f(z_*) - f'(z_*)(z - z_*)| \leq \eta |z - z_*| \quad \text{for } |z - z_*| < \delta. \quad (20.7)$$

The function $z \mapsto f(z_*) + f'(z_*)(z - z_*)$ is a polynomial, hence has an antiderivative on $\mathbb{C}$, hence integrates to zero over the *closed* contour $\partial\Delta_n$ by Theorem 20.2. Therefore

$$I(\Delta_n) = \oint_{\partial\Delta_n} [f(z) - f(z_*) - f'(z_*)(z - z_*)] \, dz.$$

Choose n so large that $\text{diam}(\Delta_n) < \delta$; since $z_* \in \Delta_n$, every $z \in \partial\Delta_n$ satisfies $|z - z_*| \leq \text{diam}(\Delta_n) = 2^{-n} \text{diam}(\Delta)$, and (20.7) applies. The ML -inequality (19.15) with $M = \eta 2^{-n} \text{diam}(\Delta)$ and length $2^{-n} L$ gives

$$|I(\Delta_n)| \leq \eta 2^{-n} \text{diam}(\Delta) \cdot 2^{-n} L = \eta 4^{-n} \text{diam}(\Delta) L.$$

Combining with the first inequality of (20.6),

$$4^{-n} |I(\Delta)| \leq |I(\Delta_n)| \leq \eta 4^{-n} \text{diam}(\Delta) L \implies |I(\Delta)| \leq \eta \text{diam}(\Delta) L.$$

Since $\eta > 0$ was arbitrary and $\text{diam}(\Delta)L$ is a fixed finite number, $I(\Delta) = 0$.

One exceptional point. Now let f be continuous on D and analytic on $D \setminus \{p\}$. For points u, v, w write

$$J(u, v, w) := \int_{[u,v]} f + \int_{[v,w]} f + \int_{[w,u]} f,$$

the integral over the boundary of the triangle with vertices u, v, w , traversed $u \rightarrow v \rightarrow w \rightarrow u$ (an edge with equal endpoints contributes 0). Since $J(u, v, w) = J(v, w, u)$ we may take $\partial\Delta$ to start at any vertex. If w lies on the segment $[u, v]$, then $\int_{[u,v]} f = \int_{[u,w]} f + \int_{[w,v]} f$ (split the parameter interval and use (19.4) and Proposition 19.8), and the four resulting edge integrals cancel in pairs by (19.12). Hence $J = 0$ for a *degenerate* triangle (three collinear vertices), for every continuous f . Also, for any four points u, v, w, q ,

$$J(u, v, w) = J(u, v, q) + J(v, w, q) + J(w, u, q),$$

because the added edges $[v, q]$ and $[q, v]$, $[w, q]$ and $[q, w]$, $[u, q]$ and $[q, u]$ cancel in pairs by (19.12).

If $p \notin \Delta$, then Δ lies in the open set $D \setminus \{p\}$, on which f is analytic, and the case already proved applies. If Δ is degenerate, $I(\Delta) = 0$ as just shown. So let Δ be non-degenerate with $p \in \Delta$. If p is not a vertex, write Δ with vertices u, v, w and apply the last identity with $q = p$: each of the three triangles on the right lies in Δ (which is convex), has p as a vertex, and is either degenerate or non-degenerate. So it suffices to treat a non-degenerate Δ with vertices p, b, c . For $0 < s \leq 1$ put $b_s := p + s(b - p)$ and $c_s := p + s(c - p)$. Splitting $[p, b]$ at b_s and $[c, p]$ at c_s as above, and cancelling the edges $[b_s, c_s]$, $[c_s, b_s]$ and $[b_s, c]$, $[c, b_s]$,

$$J(p, b, c) = J(p, b_s, c_s) + J(b_s, b, c) + J(b_s, c, c_s).$$

Every point of Δ is uniquely $\lambda p + \mu b + \nu c$ with $\lambda, \mu, \nu \geq 0$ and $\lambda + \mu + \nu = 1$ (because Δ is non-degenerate). The point p has $\lambda = 1$, while b_s, c_s have $\lambda = 1 - s$ and b, c have $\lambda = 0$; so every point of the two closed triangles with vertices b_s, b, c and b_s, c, c_s has $\lambda \leq 1 - s < 1$. These triangles therefore lie in $D \setminus \{p\}$, and their integrals vanish by the case already proved. The triangle with vertices p, b_s, c_s is Δ scaled by the factor s about p , so its boundary has length sL ; and $M := \max_{\Delta} |f| < \infty$ by Theorem 12.13, since f is continuous on the compact set Δ . The ML -inequality gives

$$|I(\Delta)| = |J(p, b_s, c_s)| \leq M s L \quad \text{for every } s \in (0, 1],$$

and letting $s \downarrow 0$ gives $I(\Delta) = 0$. □

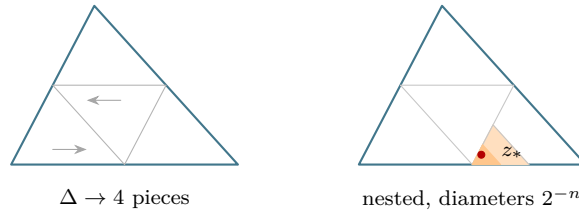


FIGURE 20.1. The bisection in Goursat's proof. Joining the midpoints of the sides splits Δ into four similar triangles; each interior edge is traversed once in each direction, so those contributions cancel and (20.5) holds. One of the four carries at least a quarter of the integral; iterating produces the nested sequence (20.6), whose diameters halve each time and whose intersection is the single point z_* where differentiability is finally used.

On every inside edge two arrows point opposite ways
so those parts of the four integrals cancel

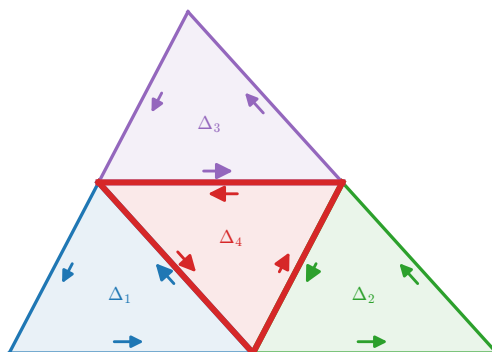


FIGURE 20.2 (animated). Why (20.5) holds. A point runs once round $\partial\Delta$, counterclockwise. The four subtriangles are pulled apart and a point runs round each, counterclockwise, all at once, leaving an arrow on each edge. The pieces slide back: along every interior edge two arrows now point in opposite directions, so by (19.12) those two contributions cancel, and the arrows left over are exactly those of $\partial\Delta$. This uses nothing about f beyond continuity.

Goursat, stage 1: zooming into Δ_1 , diameter 1.500, $4^n |I(\Delta_n)| = 1.965$
 darker: a larger $|I_j|$ among the four pieces of Δ_n ; the outlined piece carries at least a quarter of $I(\Delta_n)$, so $4^n |I(\Delta_n)|$ never decreases

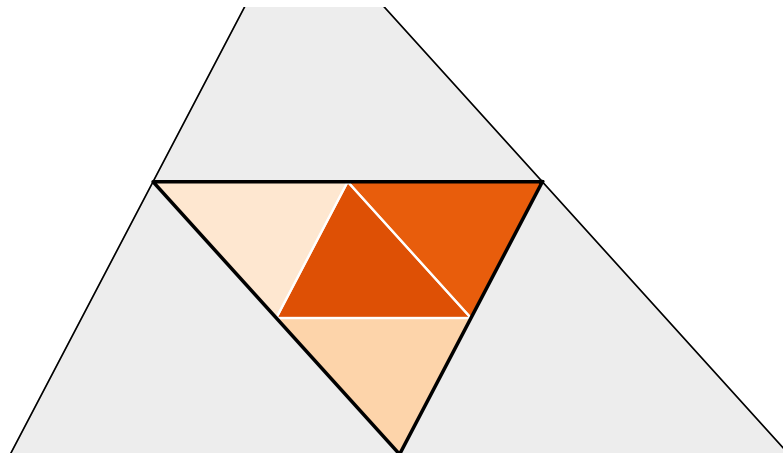


FIGURE 20.3 (animated). The bisection in the proof of Goursat's theorem, as a zoom through the nested triangles $\Delta_0 \supseteq \Delta_1 \supseteq \dots$. At each level the four pieces of Δ_n are shaded by $|I(\Delta_n^{(j)})|$, darker for larger; by (20.5) one of them has $|I(\Delta_n^{(j)})| \geq \frac{1}{4} |I(\Delta_n)|$, and it is outlined and becomes Δ_{n+1} ; the pieces not chosen turn grey. The title shows $\text{diam}(\Delta_n)$, which halves at each step, and $4^n |I(\Delta_n)|$, which therefore never decreases: if $I(\Delta) \neq 0$ it stays at least $|I(\Delta)| > 0$ while the triangles shrink to z_* , which the rest of the proof shows is impossible. Since the argument works by showing that $|I(\Delta)|$ is smaller than every positive number, the animation needs a function whose integrals are not 0: it uses the non-analytic $f(z) = \overline{(z-p)} e^{-|z-p|^2/s} e^{2.2i \operatorname{Re} z}$. For an analytic f every $I(\Delta_n)$ is 0; the bookkeeping is the same for every continuous f .

Remark 20.6 (Why Goursat, and not Green). If one *assumes* f' continuous, (20.4) follows in three lines from Green's theorem: writing $f = u + iv$ and $dz = dx + i dy$,

$$\oint_{\partial\Delta} f dz = \oint (u dx - v dy) + i \oint (v dx + u dy) = \iint_{\Delta} (-v_x - u_y) dA + i \iint_{\Delta} (u_x - v_y) dA,$$

and both double integrals vanish by the Cauchy–Riemann equations (14.1). Goursat's achievement is to avoid the continuity hypothesis. This is not fastidiousness: in Chapter 21 we deduce from the Cauchy–Goursat theorem that f' is continuous — indeed that f has derivatives of all orders — and that deduction would be circular if we had assumed it here.

20.3 The Cauchy–Goursat theorem

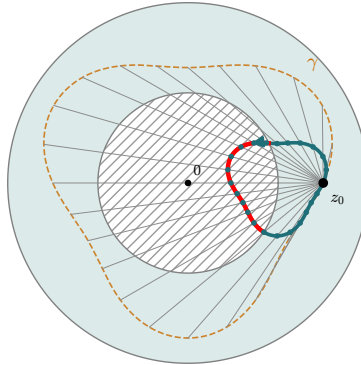
Definition 20.7 (Star-shaped and simply connected). A region D is *star-shaped with respect to* $z_0 \in D$ if for every $z \in D$ the segment $[z_0, z]$ of (19.7) lies in D . It is *convex* if it is star-shaped with respect to each of its points. A region D is *simply connected* if every closed contour in D can be continuously deformed within D to a point; precisely, for every closed contour $\gamma : [a, b] \rightarrow D$ there is a continuous map $H : [a, b] \times [0, 1] \rightarrow D$ (a *homotopy* from γ to a constant) with

$$H(t, 0) = \gamma(t), \quad H(a, s) = H(b, s), \quad H(t, 1) = H(a, 1) \quad (t \in [a, b], s \in [0, 1]).$$

Informally, D has no holes.

An annulus is not star-shaped: the same homotopy leaves D

red: the part of $H(\cdot, s)$ in the hole $|z| \leq 1$, which is not in D



$D = \{1 < |z| < 2\}$; the hole $|z| \leq 1$ is not in D

$s = 0.659$

is $H(\cdot, s)$ inside D ? **NO (red part)**

$$\frac{1}{2\pi i} \oint_{H(\cdot, s)} dz/z = 0.0000$$

$\frac{1}{2\pi i} \oint_{H(\cdot, s)} dz/z$ against the parameter

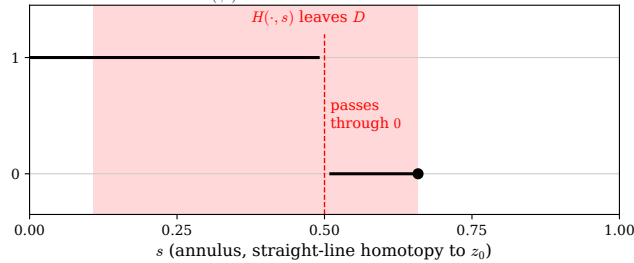


FIGURE 20.4 (animated). Definition 20.7 as a motion. In a star-shaped sector every point of the loop slides along its segment to z_0 , the loop stays in the region and collapses, and $\frac{1}{2\pi i} \oint dz/z$ stays 0. In the annulus $1 < |z| < 2$ the same homotopy leaves the region (red) and the value jumps from 1 to 0 where the curve passes through 0; deformations that stay inside the annulus leave it at 1, even when the loop is pressed against the hole. (That no deformation inside the annulus shrinks the loop is the argument of the text; the animation shows some deformations, not all.)

Every star-shaped region D , with star centre z_0 , is simply connected: $H(t, s) := (1-s)\gamma(t) + sz_0$ is continuous, lies in D because the segment $[z_0, \gamma(t)]$ does, and has the three properties required. Discs, half-planes and convex sets are star-shaped. So is every sector $\{re^{i\theta} : r > 0, \theta_1 < \theta < \theta_2\}$ with $0 < \theta_2 - \theta_1 \leq 2\pi$, with respect to any point q on its bisecting ray: if the opening is at most π the sector is convex; otherwise its complement K is a closed convex cone containing $-q$, and if the segment from q to a point z of the sector met K at $w = (1-t)q + tz$ with $0 < t < 1$, then $z = \frac{1}{t}w + \frac{1-t}{t}(-q) \in K$, which is absurd. In particular the slit plane $\mathbb{C} \setminus (-\infty, 0]$ (star-shaped with respect to 1) is simply connected. The punctured plane $\mathbb{C} \setminus \{0\}$ and any annulus $\{r < |z - c| < R\}$ are not: $1/(z - c)$ is analytic there, but its integral over a circle $C(c, \rho)$ inside is $2\pi i \neq 0$ by (19.16), contradicting Theorem 20.9 (so this argument uses Theorem 20.9, whose proof below is only sketched).

Theorem 20.8 (Cauchy–Goursat, star-shaped case). *Let D be a star-shaped region and f analytic*

on D . Then f has an antiderivative on D , and consequently

$$\oint_{\gamma} f(z) \, dz = 0 \quad (20.8)$$

for every closed contour γ in D . The same conclusions hold if f is only assumed continuous on D and analytic on $D \setminus \{p\}$ for one point $p \in D$.

Proof. Let z_0 be a star centre and define

$$F(z) := \int_{[z_0, z]} f(\zeta) \, d\zeta, \quad (20.9)$$

legitimate because the segment lies in D . Fix $z \in D$ and take $\varepsilon > 0$ with $D(z, \varepsilon) \subseteq D$. For $|\Delta z| < \varepsilon$ the triangle Δ with vertices $z_0, z, z + \Delta z$ lies in D : its edges $[z_0, z]$ and $[z_0, z + \Delta z]$ lie in D by star-shapedness, the edge $[z, z + \Delta z]$ lies in $D(z, \varepsilon)$, and the interior is covered by the segments from z_0 to points of $[z, z + \Delta z]$. Goursat's Theorem 20.5 gives $\oint_{\partial\Delta} f = 0$, i.e.

$$\int_{[z_0, z]} f + \int_{[z, z + \Delta z]} f + \int_{[z + \Delta z, z_0]} f = 0,$$

which by (19.12) rearranges to

$$F(z + \Delta z) - F(z) = \int_{[z, z + \Delta z]} f(\zeta) \, d\zeta.$$

This is exactly the identity established in the proof of Theorem 20.4, and the same estimate shows $F'(z) = f(z)$. Now (20.8) follows from Theorem 20.2.

This proof used analyticity of f only through Goursat's theorem, applied to closed triangles in D , and otherwise only continuity of f . If f is merely continuous on D and analytic on $D \setminus \{p\}$, Goursat's theorem still applies, by its last assertion, so the proof goes through unchanged. $\square$

Theorem 20.9 (Cauchy–Goursat, simply connected case). *Let D be a simply connected region and f analytic on D . Then*

$$\oint_{\gamma} f(z) \, dz = 0 \quad (20.10)$$

for every closed contour γ in D ; equivalently (by Theorem 20.4) f has an antiderivative on D and $\int_{\gamma} f$ depends only on the endpoints of γ .

Proof sketch. Cover the compact image of γ by finitely many discs contained in D ; on each disc f has an antiderivative by Theorem 20.8, and these local antiderivatives agree up to constants on overlaps. Following γ through the chain of discs produces a continuous choice of antiderivative along γ , whose net change is $\oint_{\gamma} f$. Simple connectivity says γ is homotopic in D to a constant path; the net change varies continuously under the homotopy while taking values in a set determined by the local antiderivatives, hence is constant, hence equals its value for the constant path, namely 0. The full argument (the “homotopy form of Cauchy's theorem”, in which a Lebesgue-number argument subdivides $[a, b] \times [0, 1]$ into small rectangles, each mapped by the homotopy into a disc contained in D , and Theorem 20.8 is applied on each disc) is standard but longer, and we take it as known; see for instance Stein and Shakarchi, *Complex Analysis*, Chapter 3, Section 5.

What depends on this sketch. Theorem 20.9 is used in the proof of the deformation Theorem 20.10, and through it in Examples 20.11 and 20.12 and in Cauchy's integral formula for

general simple closed contours (Theorem 21.1, Step 2 of the proof); it is used directly in Corollary 20.13. It is *not* needed for circles: Cauchy’s integral formula and the formulas for derivatives on a circle, hence the infinite differentiability of analytic functions (Corollary 21.5), Cauchy’s inequalities and Liouville’s theorem, rest only on Goursat’s Theorem 20.5 and the star-shaped case, Theorem 20.8. $\square$

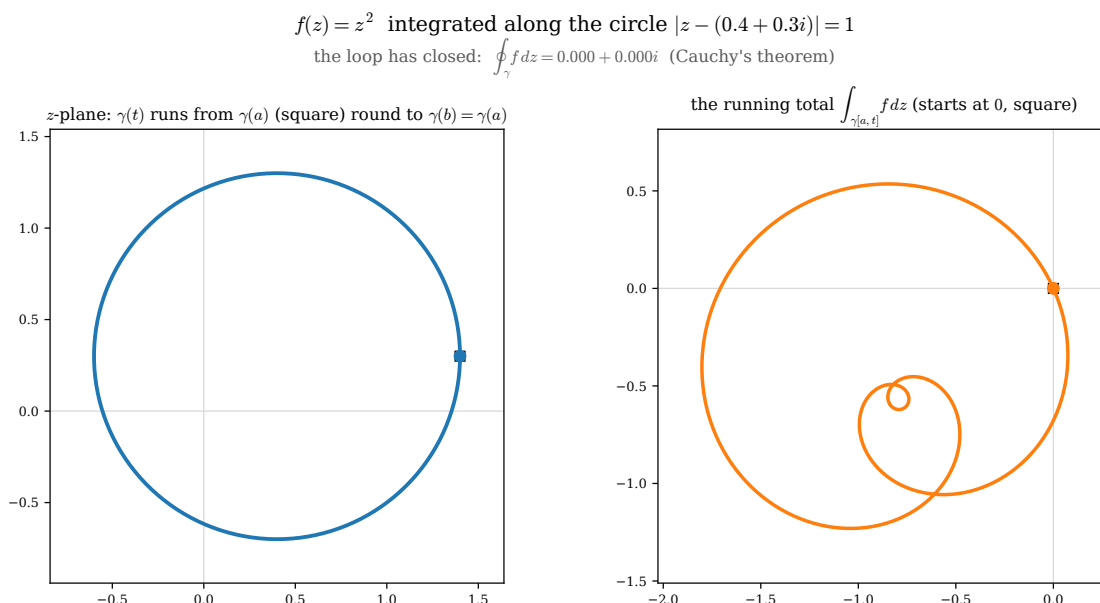


FIGURE 20.5 (animated). The Cauchy–Goursat theorem watched as t runs from a to b : the point $\gamma(t)$ goes round a closed contour (left) while the running integral $\int_{\gamma[a,t]} f dz$ moves in its own plane (right).

It starts at 0 and returns to 0 exactly when the loop closes: z^2 on a circle, e^z on a square, and $1/z$ on the circle $|z - 2| = 1$, whose singularity 0 lies outside the loop. Last, the counterexample $1/z$ on the unit circle: 0 is inside, f is not analytic there, and the integral ends at $2\pi i$ (Example 19.11).

Background on simple closed contours. The deformation theorem below, and Cauchy’s integral formula for general contours in Chapter 21, speak of the *interior* of a simple closed contour and of its *counterclockwise* orientation. These rest on the *Jordan curve theorem*, which we state and take as known: the complement of a simple closed contour C consists of exactly two regions, a bounded one, the *interior* of C , and an unbounded one, the *exterior*, and C is the boundary of each. The contour is traversed *counterclockwise* if its interior lies to the left of the direction of travel; for the circle (19.8) this is the usual meaning, and for circles, rectangles, triangles and ellipses these facts can be checked by hand. We use them without proof.

Theorem 20.10 (Deformation of contours). *Let C and C' be simple closed contours, both traversed counterclockwise, with C' lying in the interior of C . Suppose f is analytic on an open set containing C , C' and the region between them (the interior of C minus C' and its interior). Then*

$$\oint_C f(z) dz = \oint_{C'} f(z) dz. \quad (20.11)$$

More generally, if $C'_1, \dots, C'_m$ are simple closed contours, counterclockwise, lying in the interior of C with pairwise disjoint interiors, and f is analytic on an open set containing C , every C'_k ,

and all points inside C and outside every C'_k , then

$$\oint_C f(z) \, dz = \sum_{k=1}^m \oint_{C'_k} f(z) \, dz. \quad (20.12)$$

Proof. Introduce two disjoint “cut” arcs σ_1, σ_2 joining C to C' inside the region. They divide the region between the curves into two pieces, each bounded by a closed contour lying in a simply connected region on which f is analytic. (That such cuts exist and that the pieces have this property is evident in a picture; for general C and C' it is a statement of Jordan–Schoenflies type, which we do not prove and take as known together with the Jordan curve theorem. The proof also uses Theorem 20.9, whose proof was only sketched.) By Theorem 20.9 the integral of f around each piece is 0. Adding the two, the contributions of σ_1 and σ_2 cancel (each is traversed once in each direction), leaving

$$\oint_C f - \oint_{C'} f = 0,$$

the sign arising because the inner curve is traversed clockwise as part of the boundary of the region. The many-hole version is the same argument with $m + 1$ cuts. $\square$

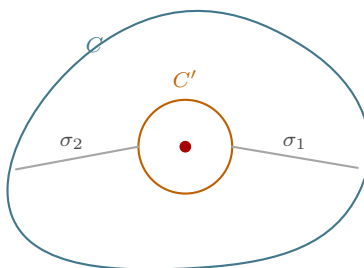


FIGURE 20.6. The deformation theorem 20.10. Two cuts σ_1, σ_2 divide the region between C and C' into two pieces, each lying in a simply connected subregion where Cauchy–Goursat applies. Adding the two vanishing integrals, the cuts cancel (each is traversed once in each direction) and only C and $-C'$ survive — which is (20.11).

Each cut is run twice, in opposite directions
so the cuts cancel

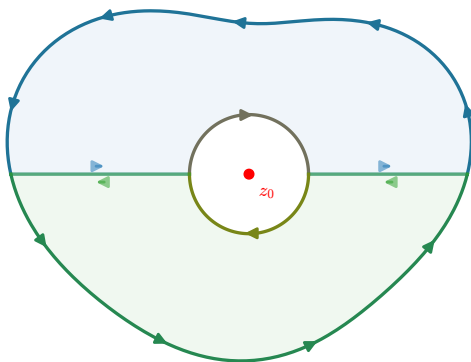


FIGURE 20.7 (animated). The proof of Theorem 20.10, with the two cuts σ_1, σ_2 of the figure above. f is analytic except at z_0 , inside C' . The two pieces of the region between C and C' are pulled apart and run round counterclockwise; neither contains a singularity, so each integral is 0 by Theorem 20.9. When the pieces are put back, each cut carries two arrows pointing in opposite directions, and the cuts cancel by (19.12). What is left is C counterclockwise and C' clockwise, so $\oint_C f - \oint_{C'} f = 0$; running C' counterclockwise instead changes the sign of its integral and gives (20.11).

Example 20.11 (Deformation in action). Let C be any simple closed contour, counterclockwise, whose interior contains z_0 . Then

$$\oint_C \frac{dz}{z - z_0} = 2\pi i. \quad (20.13)$$

Indeed, choose $\rho > 0$ so small that $C(z_0, \rho)$ lies inside C . The integrand is analytic on $\mathbb{C} \setminus \{z_0\}$, hence on a neighbourhood of the region between the two curves, so Theorem 20.10 applies and reduces the integral to the one computed in (19.16). The shape of C is irrelevant: only the fact that z_0 is inside matters.

Example 20.12 (A rational integrand). Evaluate $\oint_{|z|=3} \frac{2z+1}{z^2+z} dz$. Partial fractions give $\frac{2z+1}{z(z+1)} = \frac{1}{z} + \frac{1}{z+1}$. Both singularities 0 and -1 lie inside $|z| = 3$, so by (20.13) applied twice the answer is $2\pi i + 2\pi i = 4\pi i$. If instead the contour were $|z+1| = \frac{1}{2}$, only -1 would be enclosed and the value would be $2\pi i$; the term $1/z$ would contribute 0 by Theorem 20.9, being analytic inside.

Each cut is run twice, in opposite directions
so the cuts cancel

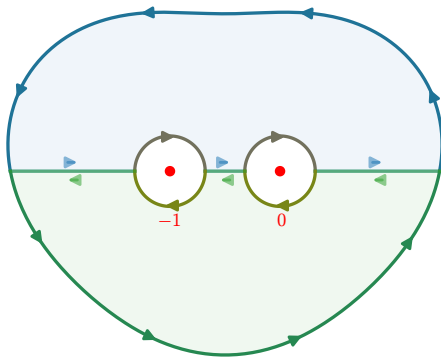


FIGURE 20.8 (animated). The many-hole form (20.12) for Example 20.12: $f(z) = \frac{2z+1}{z^2+z} = \frac{1}{z} + \frac{1}{z+1}$ is analytic except at 0 and -1 , inside the circles C'_1 and C'_2 . Three cuts along the real axis ($m+1$ cuts for $m=2$ holes) divide the region inside C and outside both circles into two pieces with no singularity inside. Pull apart, run round, put back: the cuts cancel, leaving C counterclockwise and both circles clockwise, so $\oint_C f = \oint_{C'_1} f + \oint_{C'_2} f = 2\pi i + 2\pi i = 4\pi i$.

20.4 Consequences for harmonic conjugates

We can now close a gap left in Theorem 15.6, whose last sentence was not proved there. (The proof below uses Theorem 20.9, whose proof was only sketched.)

Corollary 20.13 (Harmonic conjugates on simply connected regions). *Let D be a simply connected region and u harmonic on D . Then u has a harmonic conjugate on D , unique up to an additive real constant; equivalently there is an analytic f on D with $\operatorname{Re} f = u$.*

Proof. Set $g := u_x - i u_y$ on D . Its components are $U = u_x$ and $V = -u_y$, and

$$U_x = u_{xx} \stackrel{(15.1)}{=} -u_{yy} = V_y, \quad U_y = u_{xy} = u_{yx} = -V_x,$$

using Clairaut's theorem, valid since u has continuous second partials. So g satisfies the Cauchy–Riemann equations (14.1) with continuous partials and is analytic on D by Theorem 14.6. By Theorem 20.9, g has an antiderivative F on D . By (14.2), $F' = (\operatorname{Re} F)_x + i(\operatorname{Im} F)_x$, and comparing with $g = u_x - i u_y$ gives $(\operatorname{Re} F)_x = u_x$ and, using (14.1) for F , $(\operatorname{Re} F)_y = -(\operatorname{Im} F)_x = u_y$. Hence $\operatorname{Re} F - u$ has vanishing gradient on the region D and is therefore constant; adjusting F by that real constant gives $\operatorname{Re} F = u$, and $v := \operatorname{Im} F$ is the required conjugate. Uniqueness was proved in Theorem 15.6. $\square$

Remark 15.7 showed that simple connectivity cannot be dropped: $\ln|z|$ on $\mathbb{C} \setminus \{0\}$ has no global conjugate. The obstruction is precisely that $\oint_{|z|=1} \frac{dz}{z} = 2\pi i \neq 0$.

20.5 Exercises

Exercise 20.1. Evaluate, with C the circle $|z| = 2$ counterclockwise:

$$\oint_C \frac{dz}{z-1}, \quad \oint_C \frac{dz}{z-5}, \quad \oint_C \frac{z \, dz}{z^2-9}, \quad \oint_C \frac{e^z}{z^2+16} \, dz.$$

Exercise 20.2. Evaluate $\oint_C \frac{3z+2}{z^2-z} \, dz$ for (a) $C : |z| = \frac{1}{2}$, (b) $C : |z-1| = \frac{1}{2}$, (c) $C : |z| = 4$.

Exercise 20.3. Show that $\int_\gamma \frac{dz}{z} = \text{Log } z_2 - \text{Log } z_1$ whenever γ is a contour in the slit plane $\mathbb{C} \setminus (-\infty, 0]$ from z_1 to z_2 , and explain why the formula fails if γ is allowed to cross the cut.

Exercise 20.4. ($\star$) Let f be analytic on a region containing the closed annulus $1 \leq |z| \leq 2$. Prove $\oint_{|z|=2} f = \oint_{|z|=1} f$ directly from Theorem 20.10, and deduce that $\oint_{|z|=\rho} \frac{e^z}{z} \, dz$ is independent of $\rho > 0$.

Exercise 20.5. Give an example of a region D , an f analytic on D , and a closed contour γ in D with $\oint_\gamma f \neq 0$. Which hypothesis of Theorem 20.9 does your example violate?

Exercise 20.6. Prove that if f is analytic on a region D and γ is a closed contour in D , then $\oint_\gamma f'(z) \, dz = 0$, with no hypothesis on D beyond being a region. (*This is easier than Cauchy-Goursat: f itself is the antiderivative.*)

Exercise 20.7. Decide, for each region, whether every analytic function on it has an antiderivative, and justify by Theorem 20.9 or by a counterexample:

$$\mathbb{D}; \quad \mathbb{C} \setminus \{0\}; \quad \mathbb{C} \setminus (-\infty, 0]; \quad \{1 < |z| < 2\}; \quad \mathbb{C} \setminus \{1, -1\}.$$

Exercise 20.8. Evaluate $\oint_C \left(\frac{2}{z-i} + \frac{3}{z+2} \right) \, dz$ where C is the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ counterclockwise, using (20.13) twice.

Exercise 20.9. ($\star$) Let f be analytic on the punctured disc $0 < |z| < 1$ and suppose $\oint_{|z|=\rho} f = 0$ for one (hence every) $\rho \in (0, 1)$. Prove f has an antiderivative on the punctured disc. (*The slit discs $\mathbb{D} \setminus (-1, 0]$ and $\mathbb{D} \setminus [0, 1)$ are star-shaped, with respect to $\frac{1}{2}$ and $-\frac{1}{2}$, so by Theorem 20.8 f has antiderivatives F_1, F_2 on them. On the upper and on the lower half-disc $F_1 - F_2$ is constant, say c_+ and c_- . Integrate over the right half of $|z| = \rho$ using F_1 and over the left half using F_2 to show $\oint_{|z|=\rho} f = c_+ - c_-$ for every ρ ; then glue F_1 and $F_2 + c_+$.)*

Chapter 21

Cauchy's Integral Formulas

Cauchy–Goursat says that the integral of an analytic function around a closed contour is zero. The integral formulas say something far stronger and stranger: the values of an analytic function *inside* a contour are completely determined by its values *on* the contour, by an explicit formula. Nothing of the kind is true for real-differentiable functions.

21.1 The first formula

Theorem 21.1 (Cauchy's integral formula). *Let f be analytic on an open set D , let C be a simple closed contour traversed counterclockwise such that C and its interior lie in D , and let z_0 be a point in the interior of C . Then*

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz. \quad (21.1)$$

This applies in particular when D is a simply connected region containing C , since then the interior of C lies in D (a fact that rests on the Jordan curve theorem and that we take as known).

Proof. Step 1: C is a circle. Let $C = C(c, r)$, whose interior is the open disc $D(c, r)$, so that $\overline{D(c, r)} \subseteq D$, and let $z_0 \in D(c, r)$. There is $R > r$ with $D(c, R) \subseteq D$: otherwise there are points $w_n \notin D$ with $|w_n - c| < r + 1/n$, by Theorem 5.13 a subsequence converges to some w with $|w - c| \leq r$, and $w \notin D$ because $\mathbb{C} \setminus D$ is closed — contradicting $\overline{D(c, r)} \subseteq D$. Define

$$g(z) := \frac{f(z) - f(z_0)}{z - z_0} \quad (z \in D(c, R), z \neq z_0), \quad g(z_0) := f'(z_0).$$

Then g is analytic on $D(c, R) \setminus \{z_0\}$ by Theorem 13.3, and continuous at z_0 by the definition of $f'(z_0)$. The disc $D(c, R)$ is convex, hence star-shaped, so Theorem 20.8, in its form with one exceptional point, gives $\oint_C g = 0$. Since $z \neq z_0$ on C , this says

$$\oint_C \frac{f(z)}{z - z_0} dz = f(z_0) \oint_C \frac{dz}{z - z_0}.$$

It remains to show that the last integral is $2\pi i$. Put $A := z - c$ and $w := z_0 - c$, so that $|A| = r > |w|$ on C . For every $N \geq 1$, multiplying out a finite geometric sum gives

$$\frac{1}{z - z_0} = \frac{1}{A - w} = \sum_{k=0}^{N-1} \frac{w^k}{A^{k+1}} + \frac{w^N}{A^N(A - w)}.$$

Integrate over C : by (19.16) (with centre c) the term $k = 0$ gives $2\pi i$ and the terms $k \geq 1$ give 0, while the remainder has modulus at most $|w|^N / (r^N(r - |w|))$ on C , so by the ML -inequality (19.15) its integral has modulus at most $2\pi r (|w|/r)^N / (r - |w|) \rightarrow 0$ as $N \rightarrow \infty$. Hence $\oint_C \frac{dz}{z - z_0} = 2\pi i$, and (21.1) follows.

Step 2: general C . The integrand $\frac{f(z)}{z - z_0}$ is analytic on the open set $D \setminus \{z_0\}$. Choose $\rho > 0$ small enough that the closed disc $\overline{D}(z_0, \rho)$ lies inside C , and let C_ρ be the circle $C(z_0, \rho)$ counterclockwise. By the deformation theorem 20.10,

$$\oint_C \frac{f(z)}{z - z_0} dz = \oint_{C_\rho} \frac{f(z)}{z - z_0} dz \quad \text{for every such } \rho. \quad (21.2)$$

Now split the numerator:

$$\oint_{C_\rho} \frac{f(z)}{z - z_0} dz = f(z_0) \oint_{C_\rho} \frac{dz}{z - z_0} + \oint_{C_\rho} \frac{f(z) - f(z_0)}{z - z_0} dz.$$

The first integral is $2\pi i$ by (19.16). For the second, let $\varepsilon > 0$. Since f is continuous at z_0 (Proposition 13.2) there is $\delta > 0$ with $|f(z) - f(z_0)| < \varepsilon$ whenever $|z - z_0| < \delta$. Take $\rho < \delta$. On C_ρ we have $|z - z_0| = \rho$, so the integrand is bounded in modulus by ε/ρ , and $\text{length}(C_\rho) = 2\pi\rho$; the ML -inequality (19.15) gives

$$\left| \oint_{C_\rho} \frac{f(z) - f(z_0)}{z - z_0} dz \right| \leq \frac{\varepsilon}{\rho} \cdot 2\pi\rho = 2\pi\varepsilon.$$

Therefore

$$\left| \oint_C \frac{f(z)}{z - z_0} dz - 2\pi i f(z_0) \right| \leq 2\pi\varepsilon$$

by (21.2), and the left-hand side does not depend on ε . Letting $\varepsilon \downarrow 0$ gives (21.1). (Alternatively, Step 1 applied to the circle C_ρ evaluates the right-hand side of (21.2) at once.)

What the proof uses. Step 1 uses only Goursat's Theorem 20.5 (with its exceptional point), the star-shaped case Theorem 20.8, and the computation (19.16). Step 2 uses in addition the deformation Theorem 20.10, and so the Jordan curve facts stated before it and Theorem 20.9, whose proof was only sketched. $\square$

Shrink the circle: each piece tends to $f(z_0) \Delta\theta/2\pi$

$\Delta z \approx i\rho e^{i\theta} \Delta\theta$ is small, $1/(z - z_0) = e^{-i\theta}/\rho$ is large: the product $i \Delta\theta$ does not depend on ρ

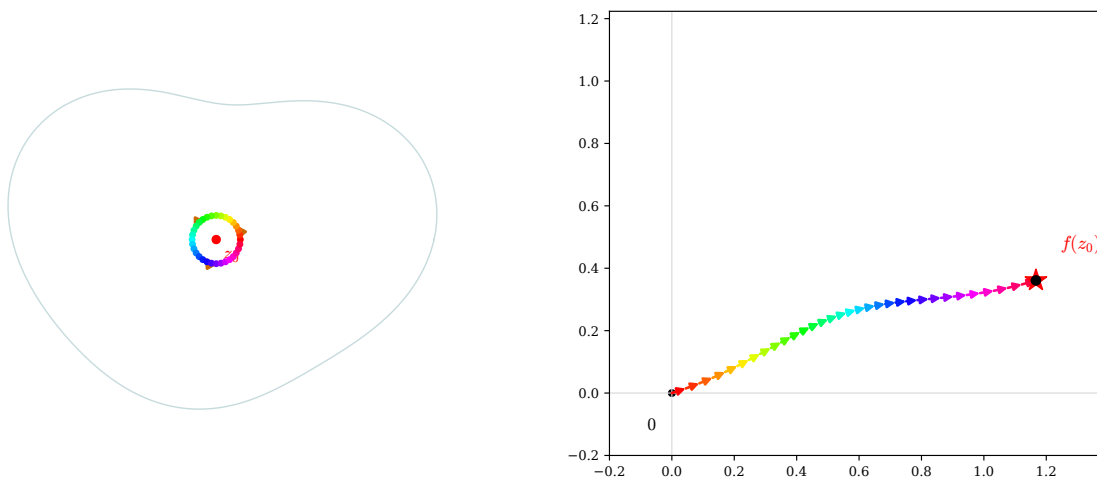
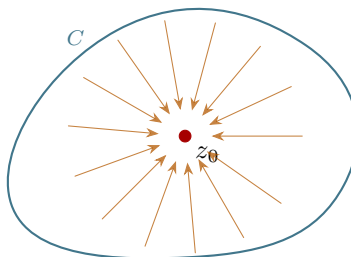


FIGURE 21.1 (animated). The proof of Theorem 21.1 for $f(z) = e^z$ and $z_0 = 0.2 + 0.3i$. The curve (left) is cut into 32 pieces, and their contributions $\frac{1}{2\pi i} \int \frac{f(z)}{z - z_0} dz$ over each piece (roughly $\frac{1}{2\pi i} \frac{f(z)}{z - z_0} \Delta z$) are added tip to tail (right); the chain ends at $f(z_0)$ (the star). Deforming C into the circle C_ρ changes the chain but not its end, by (21.2). On C_ρ , $\Delta z \approx i\rho e^{i\theta} \Delta\theta$ is small but $1/(z - z_0) = e^{-i\theta}/\rho$ is large: the product $i \Delta\theta$ does not depend on ρ . As the circle shrinks, f is nearly $f(z_0)$ on it, so each piece tends to $f(z_0) \Delta\theta/2\pi$ and the chain straightens into 32 equal steps from 0 to $f(z_0)$.

Remark 21.2 (Reading the formula). Written as

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta$$

— which is (21.1) for the circle C_ρ , after substituting $z = z_0 + \rho e^{i\theta}$, $dz = i\rho e^{i\theta} d\theta$ — the formula says that *the value at the centre is the average of the values on any circle $C(z_0, \rho)$ about it with $\overline{D}(z_0, \rho) \subseteq D$* . This is the *mean value property*, and it is the source of the maximum modulus principle (Theorem 24.11 in Chapter 24).



$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

FIGURE 21.2. Cauchy's integral formula (21.1), and the mean value form of Remark 21.2. The value at the centre is the *average* of the boundary values: the data on the curve determines the function everywhere inside, with no freedom left. Nothing remotely like this holds for real differentiable functions, where interior values can be altered without touching the boundary.

Example 21.3. Evaluate $\oint_{|z|=2} \frac{e^z}{z-1} dz$. Here $f(z) = e^z$ is entire and $z_0 = 1$ lies inside $|z| = 2$, so (21.1) gives the value $2\pi i e^1 = 2\pi e i$.

Evaluate $\oint_{|z|=2} \frac{\cos z}{z^2 + 9} dz$. The singularities $\pm 3i$ lie *outside* $|z| = 2$, so the integrand is analytic inside and on the contour, and Theorem 20.9 gives 0.

Evaluate $\oint_{|z-i|=1} \frac{z}{z^2 + 1} dz$. Write $\frac{z}{(z-i)(z+i)} = \frac{f(z)}{z-i}$ with $f(z) = \frac{z}{z+i}$, which is analytic on and inside $|z-i| = 1$ (the other singularity $-i$ is at distance 2 from the centre). By (21.1) the value is $2\pi i f(i) = 2\pi i \cdot \frac{i}{2i} = \pi i$.

21.2 The second formula: derivatives

Theorem 21.4 (Cauchy's integral formula for derivatives). *Under the hypotheses of Theorem 21.1, f has derivatives of all orders at every point z_0 interior to C , and*

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz, \quad n = 0, 1, 2, \dots \quad (21.3)$$

Proof. We prove by induction on n the statement

$$P(n) : f^{(n)} \text{ exists at every interior point and satisfies (21.3).}$$

$P(0)$ is Theorem 21.1.

Assume $P(n)$. Fix z_0 interior to C and choose $d > 0$ with $\overline{D}(z_0, 2d)$ inside C . Let Δz satisfy $0 < |\Delta z| < d$, so $z_0 + \Delta z$ is also interior. Abbreviate $a := z - z_0$, $b := z - z_0 - \Delta z$. From (21.3),

$$\frac{f^{(n)}(z_0 + \Delta z) - f^{(n)}(z_0)}{\Delta z} = \frac{n!}{2\pi i} \oint_C f(z) \frac{1}{\Delta z} \left(\frac{1}{b^{n+1}} - \frac{1}{a^{n+1}} \right) dz. \quad (21.4)$$

Use the algebraic identity

$$\frac{1}{b^{n+1}} - \frac{1}{a^{n+1}} = \frac{a^{n+1} - b^{n+1}}{a^{n+1}b^{n+1}} = \frac{(a-b) \sum_{k=0}^n a^k b^{n-k}}{a^{n+1}b^{n+1}}, \quad (21.5)$$

valid for $a, b \neq 0$, together with $a - b = \Delta z$. Substituting into (21.4) cancels Δz :

$$\frac{f^{(n)}(z_0 + \Delta z) - f^{(n)}(z_0)}{\Delta z} = \frac{n!}{2\pi i} \oint_C f(z) \frac{\sum_{k=0}^n a^k b^{n-k}}{a^{n+1}b^{n+1}} dz. \quad (21.6)$$

Now let $\Delta z \rightarrow 0$. On C we have $|a| = |z - z_0| \geq 2d$ and $|b| \geq |a| - |\Delta z| \geq 2d - d = d$, so the integrand is bounded uniformly in Δz ; moreover $b \rightarrow a$ pointwise on C , uniformly, and hence

$$\frac{\sum_{k=0}^n a^k b^{n-k}}{a^{n+1}b^{n+1}} \rightarrow \frac{(n+1)a^n}{a^{2n+2}} = \frac{n+1}{a^{n+2}} \quad \text{uniformly on } C.$$

Uniform convergence of continuous integrands on a contour of finite length permits passing to the limit under the integral sign (apply the ML -inequality (19.15) to the difference). Therefore the limit in (21.6) exists and equals

$$\frac{n!}{2\pi i} \oint_C \frac{(n+1)f(z)}{(z - z_0)^{n+2}} dz = \frac{(n+1)!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+2}} dz,$$

which is $P(n+1)$. In particular $f^{(n+1)}(z_0)$ exists. $\square$

Corollary 21.5 (Analytic implies infinitely differentiable). *If f is analytic on an open set D , then f has derivatives of all orders on D , and each $f^{(n)}$ is analytic on D . Writing $f = u + iv$, the real functions u and v have continuous partial derivatives of all orders on D .*

Proof. Let $z_0 \in D$ and choose $\rho > 0$ with $\overline{D}(z_0, \rho) \subseteq D$. Theorem 21.4 applies with $C = \overline{D}(z_0, \rho/2)$, since C and its interior lie in D , and shows that all derivatives of f exist at every point of $D(z_0, \rho/2)$. As C is a circle, only Step 1 of the proof of Theorem 21.1 is involved. Since $f^{(n)}$ is differentiable (its derivative $f^{(n+1)}$ exists), $f^{(n)}$ is analytic. Finally $u_x = \operatorname{Re} f'$, $v_x = \operatorname{Im} f'$ and so on by (14.2), so all partials of u, v of all orders exist and are continuous, being real and imaginary parts of analytic — hence continuous — functions. $\square$

Corollary 21.5 closes the gap flagged in the proof of Theorem 15.2: the real and imaginary parts of an analytic function really do have the continuous second partials that Laplace's equation requires. Note the logical order — Goursat (with one exceptional point allowed), then the star-shaped Cauchy–Goursat theorem, then the integral formula on circles, then C^∞ — which uses neither the sketched Theorem 20.9 nor the deformation theorem; and note that Goursat's refusal to assume f' continuous (Remark 20.6) is what keeps this chain free of circularity.

Example 21.6. Evaluate $\oint_{|z|=1} \frac{e^{2z}}{z^4} dz$. Take $f(z) = e^{2z}$, $z_0 = 0$, $n = 3$ in (21.3):

$$\oint_{|z|=1} \frac{e^{2z}}{z^4} dz = \frac{2\pi i}{3!} f'''(0) = \frac{2\pi i}{6} \cdot 8e^0 = \frac{8\pi i}{3}.$$

Evaluate $\oint_C \frac{z+1}{z^2(z-2)} dz$ where C is $|z| = 1$. Only $z_0 = 0$ is enclosed, and there the factor z appears squared (a *double pole*, in the language of Chapter 26), so write $\frac{z+1}{z^2(z-2)} = \frac{f(z)}{z^2}$ with $f(z) = \frac{z+1}{z-2}$, analytic on and inside C . Then $n = 1$ and

$$\oint_C \frac{f(z)}{z^2} dz = 2\pi i f'(0), \quad f'(z) = \frac{(z-2) - (z+1)}{(z-2)^2} = \frac{-3}{(z-2)^2},$$

so $f'(0) = -3/4$ and the integral equals $-3\pi i/2$.

21.3 Morera's theorem

Theorem 21.7 (Morera). *Let D be a region and let f be continuous on D . If*

$$\oint_{\partial\Delta} f(z) dz = 0 \quad \text{for every closed triangle } \Delta \subseteq D, \quad (21.7)$$

then f is analytic on D .

Proof. Analyticity is local, so fix $z_0 \in D$ and a disc $D(z_0, \rho) \subseteq D$; it suffices to prove f analytic there. The disc is convex, hence star-shaped with respect to z_0 . Define $F(z) := \int_{[z_0, z]} f(\zeta) d\zeta$ as in (20.9). The proof of Theorem 20.8 used analyticity of f only through Goursat's Theorem 20.5, i.e. only through the vanishing of f 's integral over triangles — which is now the hypothesis (21.7). That proof therefore applies verbatim and gives $F' = f$ on $D(z_0, \rho)$. So F is analytic there, and by Corollary 21.5 its derivative $f = F'$ is analytic too. $\square$

Morera's theorem is the converse of Cauchy–Goursat and is the standard tool for proving that a function *is* analytic when no formula for its derivative is available — for instance a limit of analytic functions, or a function defined by an integral.

Corollary 21.8 (Analyticity of a uniform limit). *Let f_n be analytic on a region D and suppose $f_n \rightarrow f$ uniformly on every compact subset of D . Then f is analytic on D , and $f'_n \rightarrow f'$ uniformly on compact subsets.*

Proof. f is continuous as a locally uniform limit of continuous functions. For any closed triangle $\Delta \subseteq D$, the boundary $\partial\Delta$ is compact, so $f_n \rightarrow f$ uniformly on it and hence

$$\oint_{\partial\Delta} f = \lim_n \oint_{\partial\Delta} f_n = 0$$

by Theorem 20.5, the interchange being justified by the *ML*-inequality. Morera's Theorem 21.7 gives analyticity. For the derivatives, fix a compact $K \subseteq D$ and $\rho > 0$ with $K_\rho := \{z : \text{dist}(z, K) \leq \rho\} \subseteq D$. (Such ρ exists. If $D = \mathbb{C}$ any ρ will do. Otherwise $\delta(z) := \text{dist}(z, \mathbb{C} \setminus D)$ satisfies $|\delta(z) - \delta(z')| \leq |z - z'|$, so it is continuous, and it is positive on D because D is open; by Theorem 12.13 it has a positive minimum m on K , and $\rho := m/2$ works, since a point within ρ of K is at distance at least $m - \rho > 0$ from $\mathbb{C} \setminus D$.) For $z_0 \in K$, (21.3) with $n = 1$ on $C(z_0, \rho)$ gives

$$|f'_n(z_0) - f'(z_0)| = \left| \frac{1}{2\pi i} \oint_{C(z_0, \rho)} \frac{f_n(z) - f(z)}{(z - z_0)^2} dz \right| \leq \frac{1}{2\pi} \cdot \frac{\sup_{K_\rho} |f_n - f|}{\rho^2} \cdot 2\pi\rho = \frac{\sup_{K_\rho} |f_n - f|}{\rho},$$

which tends to 0 uniformly for $z_0 \in K$. $\square$

21.4 Exercises

Exercise 21.1. Evaluate, all contours counterclockwise:

$$\oint_{|z|=3} \frac{\sin z}{z - \pi/2} dz, \quad \oint_{|z|=1} \frac{\cos z}{z^3} dz, \quad \oint_{|z-1|=1} \frac{z^2}{z^2 - 1} dz.$$

Exercise 21.2. Evaluate $\oint_C \frac{e^z}{z(z-3)} dz$ for (a) $C : |z| = 1$, (b) $C : |z-3| = 1$, (c) $C : |z| = 5$. Check that (c) equals (a)+(b) and explain why, using Theorem 20.10.

Exercise 21.3. Evaluate $\oint_{|z|=2} \frac{z^3 + 2z}{(z-i)^3} dz$.

Exercise 21.4. $(\star)$ (*Gauss mean value property.*) Let f be analytic on a neighbourhood of $\overline{D}(z_0, \rho)$. Prove

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta,$$

and deduce the same statement for $u = \text{Re } f$; conclude that a harmonic function equals its average over every circle $C(z_0, \rho)$ with $\overline{D}(z_0, \rho)$ contained in its domain. (Use Theorem 15.6 on a slightly larger disc.)

Exercise 21.5. Suppose f is entire and $\oint_{|z|=1} \frac{f(z)}{z^{n+1}} dz = 0$ for all $n \geq 0$. Prove $f \equiv 0$. (You may use Theorem 24.1.)

Exercise 21.6. Evaluate $\oint_{|z|=2} \frac{dz}{(z-1)^2(z+1)}$ two ways: by (21.3) after splitting into partial fractions, and by the residue theorem of Chapter 27 once you have read it.

Exercise 21.7. Let f be entire with $|f(z)| \leq e^{|z|}$ for all z . Use (21.3) and the ML -inequality on the circle $|z| = n$ (this is Cauchy's inequality (22.2) of the next chapter) to show $|f^{(n)}(0)| \leq n! e^n / n^n$, and estimate this for large n using Stirling's formula.

Exercise 21.8. ($\star$) (*Cauchy's formula on an annulus.*) Let f be analytic on a neighbourhood of $\{r \leq |z - z_0| \leq R\}$ and let z lie in the open annulus. Prove

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = R} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

(This is (25.5), the first step of Laurent's theorem.)

Exercise 21.9. Use Corollary 21.8 to prove that $\zeta(s) = \sum_{n \geq 1} n^{-s}$ is analytic on $\{\operatorname{Re} s > 1\}$. (Show the partial sums converge uniformly on $\{\operatorname{Re} s \geq 1 + \delta\}$ for each $\delta > 0$, using $|n^{-s}| = n^{-\operatorname{Re} s}$ from (18.8).)

Chapter 22

Consequences of the Integral Formulas

Everything in this chapter follows from (21.3) by applying the ML -inequality to a large or a small circle. The results are out of all proportion to the effort. The maximum modulus principle and Schwarz's lemma, which are also consequences of the integral formulas, are deferred to Chapter 24: their proofs need the identity theorem.

22.1 Cauchy's inequalities and Liouville's theorem

Theorem 22.1 (Cauchy's inequalities). *Let f be analytic on an open set containing the closed disc $\overline{D}(z_0, \rho)$ and suppose*

$$|f(z)| \leq M \quad \text{for all } z \text{ with } |z - z_0| = \rho. \quad (22.1)$$

Then

$$|f^{(n)}(z_0)| \leq \frac{n! M}{\rho^n}, \quad n = 0, 1, 2, \dots \quad (22.2)$$

Proof. Apply (21.3) with $C = C(z_0, \rho)$. On C the integrand has modulus $|f(z)|/\rho^{n+1} \leq M/\rho^{n+1}$, and $\text{length}(C) = 2\pi\rho$, so the ML -inequality (19.15) gives

$$|f^{(n)}(z_0)| \leq \frac{n!}{2\pi} \cdot \frac{M}{\rho^{n+1}} \cdot 2\pi\rho = \frac{n!M}{\rho^n}. \quad \square$$

Theorem 22.2 (Liouville). *Let f be entire and bounded: $|f(z)| \leq M$ for all $z \in \mathbb{C}$ and some real M . Then f is constant.*

Proof. Fix $z_0 \in \mathbb{C}$. For every $\rho > 0$, hypothesis (22.1) holds with the same M , so (22.2) with $n = 1$ gives

$$|f'(z_0)| \leq \frac{M}{\rho}.$$

Letting $\rho \rightarrow \infty$ forces $f'(z_0) = 0$. As z_0 was arbitrary, $f' \equiv 0$ on the region $\mathbb{C}$, and Theorem 14.13(i) gives f constant. $\square$

Corollary 22.3 (Extended Liouville). *Let f be entire and suppose there are constants $A > 0$, $\rho_0 > 0$ and an integer $m \geq 0$ with $|f(z)| \leq A|z|^m$ for $|z| \geq \rho_0$. Then f is a polynomial of degree at most m .*

Proof. For $n > m$ and $\rho > \rho_0$, (22.2) at $z_0 = 0$ with $M = A\rho^m$ gives $|f^{(n)}(0)| \leq n!A\rho^{m-n} \rightarrow 0$ as $\rho \rightarrow \infty$, so $f^{(n)}(0) = 0$ for all $n > m$. The same argument at any z_0 (with $M = A(|z_0| + \rho)^m$) gives $f^{(n)} \equiv 0$ for $n > m$. Hence $f^{(m+1)} \equiv 0$, and integrating $m+1$ times using Theorem 14.13(i) shows f is a polynomial of degree $\leq m$. $\square$

22.2 The Fundamental Theorem of Algebra

We can now supply the proof promised at Theorem 6.5.

Theorem 22.4 (Fundamental Theorem of Algebra). *Every polynomial $p(z) = a_n z^n + \cdots + a_0$ with $n \geq 1$ and $a_n \neq 0$ has a root in $\mathbb{C}$.*

Proof. Suppose not, so $p(z) \neq 0$ for all z . Then $g := 1/p$ is entire, being a quotient of entire functions with nonvanishing denominator (Theorem 13.3).

We show g is bounded. By Proposition 2.7 there is $R_0 > 0$ with $|p(z)| \geq \frac{1}{2}|a_n||z|^n$ for $|z| \geq R_0$; hence for such z ,

$$|g(z)| \leq \frac{2}{|a_n|R_0^n}.$$

On the compact disc $|z| \leq R_0$, $|g|$ is continuous and therefore bounded by Theorem 12.13. So g is bounded on $\mathbb{C}$.

By Liouville's Theorem 22.2, g is constant, hence p is constant, contradicting $n \geq 1$ and $a_n \neq 0$.

The factorisation (6.5) now follows by induction: given a root ζ_1 , polynomial division gives $p(z) = (z - \zeta_1)q(z)$ with $\deg q = n - 1$, and one repeats. $\square$

22.3 Exercises

Exercise 22.1. Let f be entire with $|f(z)| \leq 7|z|^3 + 5$ for all z . What can you conclude? Prove it.

Exercise 22.2. Prove that a nonconstant entire function has dense image in $\mathbb{C}$. (Suppose the image misses a disc $D(w_0, \varepsilon)$ and apply Liouville to $1/(f - w_0)$.) This is the elementary ancestor of Picard's theorem.

Exercise 22.3. Suppose f is entire and $\operatorname{Re} f(z) \leq 100$ for all z . Prove f is constant. (Consider e^f and use (16.2).)

Exercise 22.4. Let f be entire with $f(z + 2\pi i) = f(z)$ for all z and $|f(z)| \leq C$ on the strip $|\operatorname{Im} z| \leq \pi$. Prove f is constant. (Periodicity plus the strip bound gives a global bound.)

Exercise 22.5. Suppose f is entire and $|f(z)| \geq 1$ for all z . Prove f is constant. (Apply Liouville to $1/f$.) Where does the argument break if the hypothesis is weakened to $|f(z)| \geq 1$ for $|z| \geq 1$?

Part VI

Series

Chapter 23

Sequences and Series

23.1 Sequences

Definition 23.1 (Convergent sequence). A *sequence* in $\mathbb{C}$ is a function $\mathbb{N} \rightarrow \mathbb{C}$, written $(z_n)_{n \geq 1}$. It *converges* to $L \in \mathbb{C}$, written $z_n \rightarrow L$ or $\lim_{n \rightarrow \infty} z_n = L$, if for every real $\varepsilon > 0$ there is $N \in \mathbb{N}$ with

$$n \geq N \implies |z_n - L| < \varepsilon. \quad (23.1)$$

A sequence that does not converge *diverges*.

Proposition 23.2 (Componentwise convergence). Write $z_n = x_n + iy_n$ and $L = a + ib$ with all of x_n, y_n, a, b real. Then $z_n \rightarrow L$ if and only if $x_n \rightarrow a$ and $y_n \rightarrow b$ in $\mathbb{R}$. Moreover convergent sequences are bounded, and the algebra of limits holds: if $z_n \rightarrow L$ and $w_n \rightarrow M$ then $z_n + w_n \rightarrow L + M$, $z_n w_n \rightarrow LM$, and $z_n/w_n \rightarrow L/M$ when $M \neq 0$.

Proof. The equivalence is (1.26) and (1.27) exactly as in Theorem 12.3. Boundedness: take $\varepsilon = 1$ to get $|z_n| \leq |L| + 1$ for $n \geq N$, and enlarge the bound by the finitely many earlier terms. The algebra of limits is Theorem 12.4 with the same proofs, reading $n \geq N$ for $0 < |z - z_0| < \delta$. $\square$

Definition 23.3 (Cauchy sequence). (z_n) is *Cauchy* if for every $\varepsilon > 0$ there is N with $|z_n - z_m| < \varepsilon$ for all $n, m \geq N$.

Theorem 23.4 (Completeness of $\mathbb{C}$). A sequence in $\mathbb{C}$ converges if and only if it is Cauchy.

Proof. If $z_n \rightarrow L$ then $|z_n - z_m| \leq |z_n - L| + |L - z_m| < \varepsilon$ for n, m large. Conversely, if (z_n) is Cauchy then by (1.26) so are (x_n) and (y_n) in $\mathbb{R}$; completeness of $\mathbb{R}$ gives $x_n \rightarrow a$, $y_n \rightarrow b$, and Proposition 23.2 gives $z_n \rightarrow a + ib$. $\square$

23.2 Series

Definition 23.5 (Series, partial sums). Given $(a_k)_{k \geq 1}$ in $\mathbb{C}$, the n -th *partial sum* is $S_n := \sum_{k=1}^n a_k$. The *series* $\sum_{k=1}^{\infty} a_k$ *converges* to S if $S_n \rightarrow S$, and then we write $\sum_{k=1}^{\infty} a_k = S$. It *converges absolutely* if the real series $\sum_{k=1}^{\infty} |a_k|$ converges.

Theorem 23.6 (Basic tests). Let $\sum a_k$ be a series in $\mathbb{C}$.

- (i) If $\sum a_k$ converges then $a_k \rightarrow 0$. Equivalently, if $a_k \not\rightarrow 0$ the series diverges.

(ii) *Absolute convergence implies convergence.*

(iii) (Geometric series.) *For $z \in \mathbb{C}$,*

$$\sum_{k=0}^{\infty} z^k = \frac{1}{1-z} \quad \text{if } |z| < 1, \quad \text{and the series diverges if } |z| \geq 1. \quad (23.2)$$

(iv) (Ratio test.) *If $a_k \neq 0$ for large k and $\lim_k |a_{k+1}/a_k| = L$ exists, then $\sum a_k$ converges absolutely if $L < 1$ and diverges if $L > 1$.*

(v) (Root test.) *If $\limsup_k |a_k|^{1/k} = L$, the same conclusions hold.*

Proof. (i) $a_n = S_n - S_{n-1} \rightarrow S - S = 0$.

(ii) If $\sum |a_k|$ converges, its partial sums are Cauchy, and by the triangle inequality (1.27), $|S_n - S_m| = |\sum_{k=m+1}^n a_k| \leq \sum_{k=m+1}^n |a_k|$, so (S_n) is Cauchy; apply Theorem 23.4.

(iii) For $z \neq 1$ the identity $(1-z) \sum_{k=0}^n z^k = 1 - z^{n+1}$ gives $S_n = \frac{1-z^{n+1}}{1-z}$. If $|z| < 1$ then $|z^{n+1}| = |z|^{n+1} \rightarrow 0$, so $S_n \rightarrow \frac{1}{1-z}$. If $|z| \geq 1$ then $|z^k| \geq 1 \not\rightarrow 0$ and (i) applies.

(iv) and (v) are the real tests applied to $\sum |a_k|$ (for convergence), together with (i) and $|a_k| \not\rightarrow 0$ (for divergence). $\square$

Definition 23.7 (Uniform convergence). Let $f_n, f : S \rightarrow \mathbb{C}$ with $S \subseteq \mathbb{C}$. We say $f_n \rightarrow f$ *uniformly on S* if

$$\sup_{z \in S} |f_n(z) - f(z)| \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (23.3)$$

A series $\sum_k g_k$ of functions converges uniformly on S if its partial sums do.

Theorem 23.8 (Weierstrass M -test). Let $g_k : S \rightarrow \mathbb{C}$ and suppose there are real constants $M_k \geq 0$ with

$$|g_k(z)| \leq M_k \quad \text{for all } z \in S, \quad \text{and} \quad \sum_{k=1}^{\infty} M_k < \infty. \quad (23.4)$$

Then $\sum_k g_k$ converges absolutely and uniformly on S .

Proof. Absolute convergence at each z is comparison with $\sum M_k$. Let S_n denote the partial sums and F the pointwise limit. For $n \geq m$ and all $z \in S$,

$$|F(z) - S_n(z)| = \left| \sum_{k>n} g_k(z) \right| \leq \sum_{k>n} M_k,$$

a tail of a convergent series of non-negative reals, hence $\rightarrow 0$ independently of z . $\square$

23.3 Power series

Definition 23.9 (Power series). Let $z_0 \in \mathbb{C}$ and $(a_k)_{k \geq 0}$ in $\mathbb{C}$. The *power series* centred at z_0 with coefficients a_k is

$$\sum_{k=0}^{\infty} a_k (z - z_0)^k. \quad (23.5)$$

Theorem 23.10 (Radius of convergence). *For the series (23.5) define*

$$\frac{1}{R} := \limsup_{k \rightarrow \infty} |a_k|^{1/k} \in [0, \infty], \quad (23.6)$$

with the conventions $1/0 = \infty$ and $1/\infty = 0$. Then:

- (i) *the series converges absolutely for $|z - z_0| < R$ and diverges for $|z - z_0| > R$;*
- (ii) *the convergence is uniform on every closed disc $|z - z_0| \leq r$ with $r < R$;*
- (iii) *nothing general can be said on the circle $|z - z_0| = R$.*

The number R is the radius of convergence and $D(z_0, R)$ the disc of convergence. If $a_k \neq 0$ for large k and $\lim_k |a_{k+1}/a_k|$ exists in $[0, \infty]$, it equals $1/R$.

Proof. (i) Apply the root test, Theorem 23.6(v), to $a_k(z - z_0)^k$:

$$\limsup_k |a_k(z - z_0)^k|^{1/k} = |z - z_0| \limsup_k |a_k|^{1/k} = \frac{|z - z_0|}{R},$$

which is < 1 exactly when $|z - z_0| < R$ and > 1 exactly when $|z - z_0| > R$.

(ii) Fix $r < \rho < R$. Since $\limsup_k |a_k|^{1/k} = 1/R < 1/\rho$, there is K with $|a_k| \leq \rho^{-k}$ for $k \geq K$. For $|z - z_0| \leq r$ this gives $|a_k(z - z_0)^k| \leq M_k$ for every k , where $M_k := (r/\rho)^k$ for $k \geq K$ and $M_k := |a_k| r^k$ for the finitely many $k < K$; and $\sum M_k < \infty$ because $r/\rho < 1$. The M -test 23.8 applies.

(iii) Examples: $\sum z^k$ ($R = 1$, diverges everywhere on $|z| = 1$), $\sum z^k/k^2$ ($R = 1$, converges everywhere on $|z| = 1$), $\sum z^k/k$ ($R = 1$, diverges at $z = 1$, converges at $z = -1$).

(Ratio formula.) Let $L := \lim_k |a_{k+1}/a_k| \in [0, \infty]$. For $z \neq z_0$ the terms $b_k := a_k(z - z_0)^k$ are nonzero for large k and $|b_{k+1}/b_k| \rightarrow L|z - z_0|$. By the ratio test, Theorem 23.6(iv), the series converges for $|z - z_0| < 1/L$ and diverges for $|z - z_0| > 1/L$. If $R < 1/L$, a point with $R < |z - z_0| < 1/L$ would contradict (i); if $R > 1/L$, a point with $1/L < |z - z_0| < R$ would. Hence $R = 1/L$. $\square$

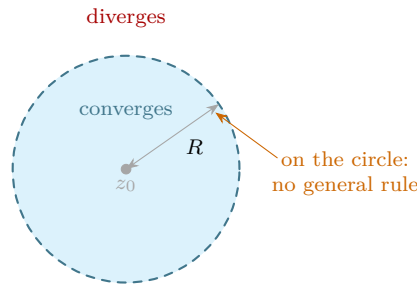


FIGURE 23.1. The disc of convergence, Theorem 23.10. Inside $|z - z_0| < R$ the series converges absolutely, and uniformly on each closed subdisc; outside it diverges. On the boundary circle itself anything can happen, and all three behaviours occur among $\sum z^k$, $\sum z^k/k$ and $\sum z^k/k^2$ — which is why 23.10(iii) makes no claim there.

Theorem 23.11 (A power series is analytic inside its disc). *Let $R > 0$ be the radius of convergence of (23.5) and let*

$$f(z) := \sum_{k=0}^{\infty} a_k(z - z_0)^k, \quad z \in D(z_0, R). \quad (23.7)$$

Then f is analytic on $D(z_0, R)$; it may be differentiated term by term,

$$f'(z) = \sum_{k=1}^{\infty} k a_k (z - z_0)^{k-1}, \quad (23.8)$$

and this series has the same radius of convergence R . Consequently f has derivatives of all orders on $D(z_0, R)$ and

$$a_k = \frac{f^{(k)}(z_0)}{k!}, \quad k = 0, 1, 2, \dots \quad (23.9)$$

so the coefficients are determined by f . Term-by-term integration is likewise valid on any contour in $D(z_0, R)$.

Proof. Same radius. $\limsup_k (k |a_k|)^{1/k} = \limsup_k k^{1/k} |a_k|^{1/k} = \limsup_k |a_k|^{1/k}$, since $k^{1/k} \rightarrow 1$.

Analyticity. The partial sums S_n are polynomials, hence entire, and by Theorem 23.10(ii) $S_n \rightarrow f$ uniformly on each closed disc $|z - z_0| \leq r < R$; every compact subset of $D(z_0, R)$ lies in such a disc. Corollary 21.8 gives that f is analytic and $S'_n \rightarrow f'$ locally uniformly, and S'_n is exactly the n -th partial sum of (23.8).

Coefficients. Differentiating (23.7) k times and setting $z = z_0$ kills every term but the k -th, whose value is $k! a_k$.

Integration. Uniform convergence on the (compact) contour allows interchange of sum and integral by the *ML*-inequality. $\square$

Example 23.12 (Manufacturing series from the geometric series). From (23.2), for $|z| < 1$,

$$\frac{1}{1-z} = \sum_{k=0}^{\infty} z^k, \quad \frac{1}{1+z} = \sum_{k=0}^{\infty} (-1)^k z^k, \quad (23.10)$$

$$\frac{1}{(1-z)^2} = \sum_{k=1}^{\infty} k z^{k-1} \quad (\text{differentiate}), \quad \frac{1}{1+z^2} = \sum_{k=0}^{\infty} (-1)^k z^{2k} \quad (|z| < 1). \quad (23.11)$$

Integrating the last from 0 to z term by term gives the principal branch of $\tan^{-1}$: $\tan^{-1} z = \sum_{k \geq 0} \frac{(-1)^k z^{2k+1}}{2k+1}$ on $|z| < 1$. Similarly, writing $\frac{1}{a-z} = \frac{1}{a-z_0} \cdot \frac{1}{1-w}$ with $w = (z - z_0)/(a - z_0)$ and applying (23.10) gives the expansion of $\frac{1}{a-z}$ about any $z_0 \neq a$:

$$\frac{1}{a-z} = \sum_{k=0}^{\infty} \frac{(z - z_0)^k}{(a - z_0)^{k+1}}, \quad |z - z_0| < |a - z_0|. \quad (23.12)$$

Identity (23.12) is the whole proof of Taylor's theorem in the next chapter.

23.4 Exercises

Exercise 23.1. Determine convergence or divergence: $\sum_{k \geq 1} \frac{i^k}{k}$, $\sum_{k \geq 1} \frac{(1+i)^k}{k!}$, $\sum_{k \geq 1} \left(\frac{2+i}{2}\right)^k$, $\sum_{k \geq 1} \frac{k!}{k^k} i^k$.

Exercise 23.2. Find the radius of convergence:

$$\sum_{k \geq 0} \frac{(z - 2i)^k}{3^k}, \quad \sum_{k \geq 1} \frac{k! z^k}{k^k}, \quad \sum_{k \geq 0} \frac{z^k}{k!}, \quad \sum_{k \geq 0} 2^k (z + 1)^k, \quad \sum_{k \geq 1} \frac{(-1)^k z^{2k}}{k}.$$

Exercise 23.3. Prove that $\lim_n z^n$ exists if and only if $|z| < 1$ or $z = 1$, and identify the limit.

Exercise 23.4. ($\star$) Let $\sum a_k z^k$ have radius R . Show that $\sum a_k^2 z^k$ has radius R^2 and $\sum \frac{a_k}{k!} z^k$ has radius ∞ when $R > 0$.

Exercise 23.5. Use the M -test to prove that $\sum_{k \geq 1} \frac{z^k}{k^2}$ converges uniformly on $\overline{D}(0, 1)$, and that $\sum_{k \geq 1} \frac{1}{k^2 + z^2}$ converges uniformly on $|z| \leq \frac{1}{2}$.

Exercise 23.6. Expand $\frac{1}{3-z}$ in powers of $z-1$ and give the radius of convergence; do the same in powers of $z+2$.

Exercise 23.7. Determine the behaviour of $\sum_{k \geq 1} z^k/k$ at each of $z = 1$, $z = -1$ and $z = i$ (divergence, conditional convergence, absolute convergence?), and show that it converges absolutely at no point of $|z| = 1$. Then determine the behaviour of $\sum_{k \geq 1} z^k/k^2$ on $|z| = 1$, and conclude that divergence, conditional convergence and absolute convergence can all occur on a circle of convergence (Theorem 23.10(iii)).

Exercise 23.8. Let $\sum a_k z^k$ and $\sum b_k z^k$ have radii R_1 and R_2 . What can be said about the radius of $\sum (a_k + b_k) z^k$? Show it is $\min(R_1, R_2)$ when $R_1 \neq R_2$, and give an example where it is strictly larger when $R_1 = R_2$.

Exercise 23.9. ($\star$) (*Abel's theorem, one direction.*) Suppose $\sum a_k$ converges. Prove that $\lim_{r \uparrow 1} \sum a_k r^k = \sum a_k$. (*Summation by parts.*) Use it to evaluate $\sum_{k \geq 1} (-1)^{k-1}/k$ from (24.7).

Chapter 24

Taylor Series

Theorem 23.11 showed that every convergent power series defines an analytic function. This chapter proves the converse, which is false over $\mathbb{R}$ and is the single deepest fact in elementary complex analysis: *every* analytic function is locally the sum of its Taylor series. From it we derive the identity theorem, and then two results whose natural home this is because their proofs use it: the maximum modulus principle and Schwarz's lemma.

24.1 Taylor's theorem

Theorem 24.1 (Taylor). *Let f be analytic on a region D , let $z_0 \in D$, and let $R := \sup\{\rho > 0 : D(z_0, \rho) \subseteq D\} \in (0, \infty]$ (so $R = \infty$ if $D = \mathbb{C}$; in every case $D(z_0, R) \subseteq D$). Then*

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k \quad \text{for every } z \in D(z_0, R), \quad (24.1)$$

the series converging absolutely on $D(z_0, R)$ and uniformly on each closed subdisc. The representation is unique: if $f(z) = \sum_k a_k (z - z_0)^k$ on a neighbourhood of z_0 , then $a_k = f^{(k)}(z_0)/k!$ for all k .

Proof. Fix z with $|z - z_0| < R$ and choose r with $|z - z_0| < r < R$. Let $C := C(z_0, r)$ counter-clockwise; then $\overline{D}(z_0, r) \subseteq D$, so Cauchy's integral formula (21.1) gives

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (24.2)$$

Expand the kernel by (23.12) with $a = \zeta$: for ζ on C we have $|\zeta - z_0| = r > |z - z_0|$, so

$$\frac{1}{\zeta - z} = \sum_{k=0}^{\infty} \frac{(z - z_0)^k}{(\zeta - z_0)^{k+1}}, \quad (24.3)$$

and the series converges uniformly in $\zeta \in C$: its k -th term has modulus $\frac{|z - z_0|^k}{r^{k+1}} = \frac{1}{r} \left(\frac{|z - z_0|}{r}\right)^k$, summable by the M -test 23.8 since $|z - z_0|/r < 1$.

Multiply (24.3) by the bounded continuous function $f(\zeta)$ — bounded on the compact set C by Theorem 12.13 — so uniform convergence persists, and integrate term by term over C (legitimate by the ML -inequality applied to the tail). From (24.2),

$$f(z) = \sum_{k=0}^{\infty} \left(\frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta \right) (z - z_0)^k = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k,$$

the last step by the integral formula for derivatives (21.3). This proves (24.1) at z , and z was an arbitrary point of the disc. Absolute and uniform convergence follow from Theorem 23.10, since we have exhibited a power series representation valid on $D(z_0, R)$, forcing its radius of convergence to be at least R .

Uniqueness is (23.9). $\square$

Corollary 24.2 (Radius of convergence is the distance to the nearest singularity). *Let f be analytic on a region D and $z_0 \in D$, and let R_T be the radius of convergence of the Taylor series (24.1).*

- (i) $R_T \geq \sup\{\rho > 0 : D(z_0, \rho) \subseteq D\}$; in particular $R_T = \infty$ if f is entire.
- (ii) Let $w \in \mathbb{C}$ with $D(z_0, |w - z_0|) \subseteq D$, and suppose f is unbounded on $D(z_0, |w - z_0|) \cap D(w, \varepsilon)$ for every $\varepsilon > 0$ (for example, w is a pole of f , or the branch point of a logarithm). Then $R_T = |w - z_0|$.

So when the only obstructions to analyticity are points near which f is unbounded, the radius is exactly the distance from z_0 to the nearest of them. Some such hypothesis is needed: “the nearest point where f is not analytic” is not a well-defined notion for a function given only on D . (For instance Log on $\mathbb{C} \setminus (-\infty, 0]$ about $z_0 = -1 + i$ has radius $|z_0| = \sqrt{2}$, by (ii) with $w = 0$, although the cut is at distance 1 from z_0 .)

Proof. (i) is Taylor’s theorem: the series converges on $D(z_0, \rho)$ whenever $D(z_0, \rho) \subseteq D$. (ii) Put $d := |w - z_0|$. By (i), $R_T \geq d$, and the series has sum f on $D(z_0, d)$ by Theorem 24.1. If $R_T > d$, the sum S of the series is continuous on $D(z_0, R_T)$ (Theorem 23.11), hence bounded on the compact set $\overline{D}(z_0, d)$ (Theorem 12.13); since $f = S$ on $D(z_0, d)$, f would be bounded there, contrary to the hypothesis. So $R_T = d$. $\square$

Example 24.3 (Why $\frac{1}{1+x^2}$ has radius 1 over $\mathbb{R}$). The real function $x \mapsto \frac{1}{1+x^2}$ is infinitely differentiable on all of $\mathbb{R}$, yet its Maclaurin series $\sum (-1)^k x^{2k}$ converges only for $|x| < 1$. From the real side this is unexplained. From the complex side it is obvious: $f(z) = \frac{1}{1+z^2}$ has singularities at $\pm i$, at distance 1 from the origin, and Corollary 24.2 says the radius must be exactly 1. This example alone justifies studying real problems in the complex plane.

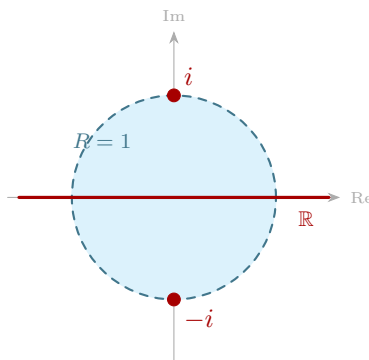


FIGURE 24.1. Corollary 24.2 explains Example 24.3. The real function $1/(1+x^2)$ is smooth on all of $\mathbb{R}$, yet its Maclaurin series stops converging at $|x| = 1$. Seen in the plane the reason is visible: the singularities $\pm i$ sit at distance 1 from the origin, and the disc of convergence must stop when it reaches them. The real line simply passes underneath the obstruction without meeting it.

24.2 The standard expansions

Proposition 24.4 (Maclaurin series of the elementary functions). *For all $z \in \mathbb{C}$,*

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}, \quad (24.4)$$

$$\sin z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{(2k+1)!}, \quad \cos z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k}}{(2k)!}, \quad (24.5)$$

$$\sinh z = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!}, \quad \cosh z = \sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!}, \quad (24.6)$$

each with $R = \infty$, and for $|z| < 1$,

$$\operatorname{Log}(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} z^k}{k}, \quad R = 1. \quad (24.7)$$

Proof. $\exp$ is entire with $\exp^{(k)} = \exp$ by (16.7), so $f^{(k)}(0) = 1$ and (24.1) gives (24.4) with $R = \infty$ by Corollary 24.2. The trigonometric series follow by substituting $\pm iz$ into (24.4) and using (17.2):

$$\sin z = \frac{1}{2i} \sum_k \frac{(iz)^k - (-iz)^k}{k!} = \frac{1}{2i} \sum_{k \text{ odd}} \frac{2(iz)^k}{k!} = \sum_{m=0}^{\infty} \frac{i^{2m} z^{2m+1}}{(2m+1)!},$$

and $i^{2m} = (-1)^m$. The hyperbolic series are the same computation without the i 's, or via (17.12). For (24.7), $\operatorname{Log}(1+z)$ is analytic on $|z| < 1$ (the branch point is at $z = -1$) with derivative $\frac{1}{1+z} = \sum_{k \geq 0} (-1)^k z^k$ by Theorem 16.9 and (23.10); integrating term by term from 0 (Theorem 23.11) and using $\operatorname{Log} 1 = 0$ gives the result, and $R = 1$ by Corollary 24.2. $\square$

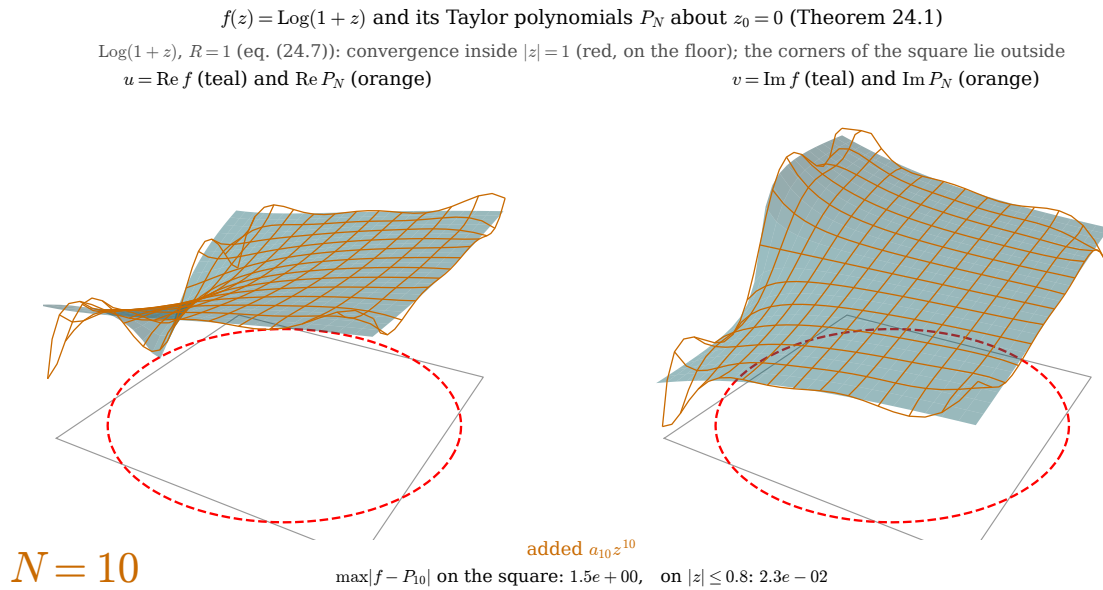


FIGURE 24.2 (animated). Taylor's theorem 24.1 for real-valued functions: the real part $u = \text{Re } f$ (left) and the imaginary part $v = \text{Im } f$ (right) of $f = e^z$, $\sin z$ and $\text{Log}(1+z)$ over a square about 0 (teal), with $\text{Re } P_N$ and $\text{Im } P_N$ for the Taylor polynomials P_N of Proposition 24.4 (orange mesh). Each polynomial blends into the next, the term being added is shown, and at a few degrees the camera turns once round and zooms in on the point where $|f - P_N|$ is largest (red: the gap). For e^z and $\sin z$ ($R = \infty$) the meshes settle onto the surfaces on the whole square. For $\text{Log}(1+z)$ ($R = 1$, (24.7)) they settle inside the circle $|z| = 1$ (red, on the floor) but not at the corners of the square, which lie outside it; heights there are cut off at the box.

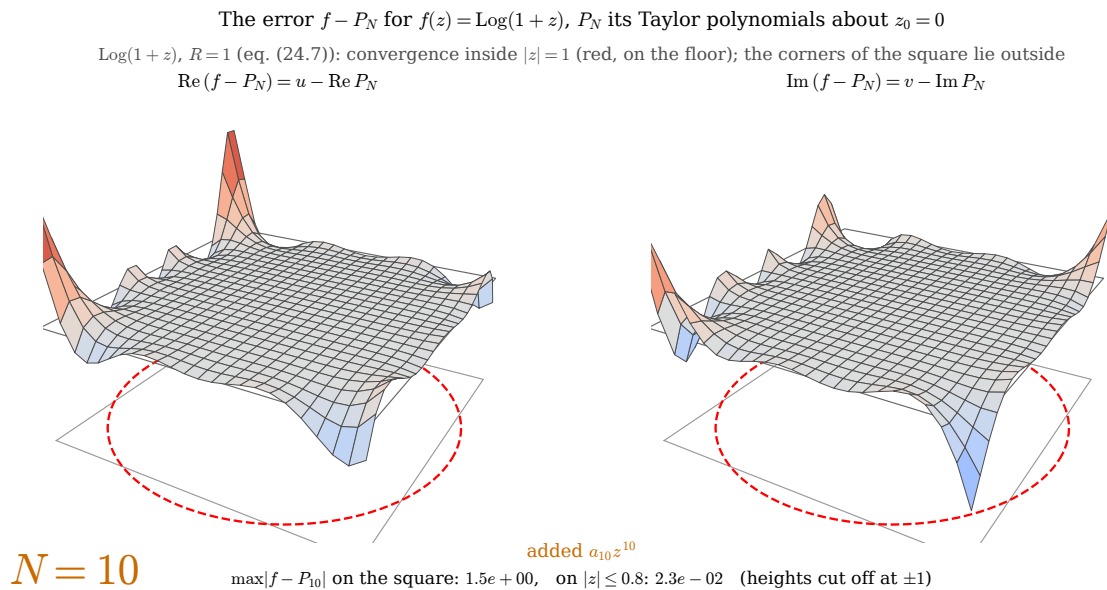


FIGURE 24.3 (animated). The same three functions, showing the errors $\text{Re}(f - P_N)$ and $\text{Im}(f - P_N)$ on a fixed vertical scale (cut off at ± 1). For e^z and $\sin z$ the error surfaces flatten onto 0 over the whole square. For $\text{Log}(1+z)$ they flatten inside $|z| = 1$, slowly near the singular point -1 , and grow spikes at the four corners, outside the disc of convergence.

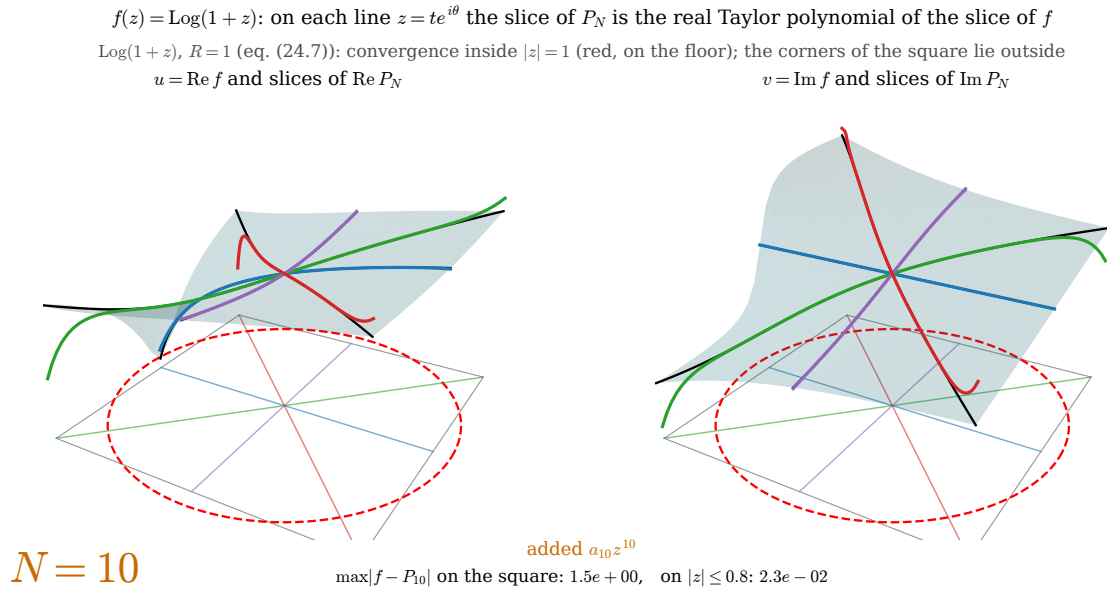


FIGURE 24.4 (animated). The surfaces of Figure 24.2 cut along four lines $z = te^{i\theta}$ through 0. On each line the slice of $\text{Re } P_N$ (or $\text{Im } P_N$) is the real Taylor polynomial of degree N of the slice of u (or v), since $\text{Re } P_N(te^{i\theta}) = \sum_{k \leq N} \text{Re}(a_k e^{ik\theta}) t^k$: on every line the real Taylor polynomials converge, and for $\text{Log}(1+z)$ they fail where the line leaves the disc $|z| < 1$.

Example 24.5 (Proof of the binomial series). We can now prove Theorem 18.8. Let $\alpha \in \mathbb{C}$ and $f(z) := (1+z)^\alpha_{\text{pr}} = e^{\alpha \text{Log}(1+z)}$, analytic on $|z| < 1$. By (18.3) applied to $1+z$,

$$f'(z) = \frac{\alpha f(z)}{1+z}, \quad \text{hence} \quad f^{(k)}(z) = \alpha(\alpha-1) \cdots (\alpha-k+1)(1+z)^{\alpha-k}_{\text{pr}}$$

by induction. Setting $z = 0$ gives $f^{(k)}(0) = \alpha(\alpha-1) \cdots (\alpha-k+1) = k! \binom{\alpha}{k}$, and Theorem 24.1 yields (18.10). The radius is 1 unless $\alpha \in \{0, 1, 2, \dots\}$. Indeed, for any other α , $\binom{\alpha}{k} \neq 0$ for all k and $|\binom{\alpha}{k+1}/\binom{\alpha}{k}| = |\alpha-k|/(k+1) \rightarrow 1$, so Theorem 23.10 gives radius 1. If $\alpha \in \{0, 1, 2, \dots\}$, all coefficients beyond the α -th vanish and the (finite) series is valid everywhere. $\square$

Example 24.6 (Series by manipulation, not differentiation). Expand $f(z) = \frac{1}{z^2 - 3z + 2}$ about $z_0 = 0$. Partial fractions: $\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{1-z} - \frac{1}{2} \cdot \frac{1}{1-z/2}$. By (23.10),

$$f(z) = \sum_{k=0}^{\infty} z^k - \frac{1}{2} \sum_{k=0}^{\infty} \frac{z^k}{2^k} = \sum_{k=0}^{\infty} \left(1 - \frac{1}{2^{k+1}}\right) z^k,$$

valid for $|z| < 1$ — the distance to the nearest singularity 1, confirming Corollary 24.2. Computing $f^{(k)}(0)$ directly would be miserable.

24.3 The identity theorem

Theorem 24.7 (Zeros of an analytic function are isolated). *Let f be analytic on a region D , not identically zero, and suppose $f(z_0) = 0$ for some $z_0 \in D$. Then there exist a unique integer $m \geq 1$ and a function g analytic on a neighbourhood of z_0 with*

$$f(z) = (z - z_0)^m g(z), \quad g(z_0) \neq 0. \quad (24.8)$$

Consequently z_0 is an isolated zero: there is $\varepsilon > 0$ with $f(z) \neq 0$ for $0 < |z - z_0| < \varepsilon$. The integer m is called the order (or multiplicity) of the zero, written $m = \text{ord}_{z_0} f$.

Proof. By Theorem 24.1, $f(z) = \sum_{k \geq 0} a_k(z - z_0)^k$ on some $D(z_0, R)$ with $a_k = f^{(k)}(z_0)/k!$.

Not all a_k vanish. Let $A := \{z \in D : f^{(k)}(z) = 0 \text{ for all } k \geq 0\}$. A is closed in D , being an intersection of zero sets of continuous functions (Corollary 21.5), and open, since if $z_1 \in A$ all Taylor coefficients of f at z_1 vanish, so $f \equiv 0$ on a disc about z_1 by Theorem 24.1 and that disc lies in A . As D is connected, $A = \emptyset$ or $A = D$; and $A \neq D$ because $f \neq 0$. So $z_0 \notin A$, i.e. some $a_k \neq 0$.

Factorisation. Let m be the least index with $a_m \neq 0$; since $a_0 = f(z_0) = 0$ we have $m \geq 1$. Then

$$f(z) = (z - z_0)^m \underbrace{\sum_{j=0}^{\infty} a_{m+j}(z - z_0)^j}_{=: g(z)},$$

the series for g having the same radius of convergence and hence defining an analytic function with $g(z_0) = a_m \neq 0$. By continuity there is $\varepsilon > 0$ with $g \neq 0$ on $D(z_0, \varepsilon)$, and then $f(z) = (z - z_0)^m g(z) \neq 0$ for $0 < |z - z_0| < \varepsilon$. Uniqueness of m is clear from (23.9). $\square$

Theorem 24.8 (Identity theorem). *Let f and h be analytic on a region D and suppose the set $\{z \in D : f(z) = h(z)\}$ has an accumulation point in D . Then $f \equiv h$ on D .*

Proof. Put $F := f - h$, analytic on D , and let $z_0 \in D$ be an accumulation point of the zero set $Z := \{F = 0\}$; note $F(z_0) = 0$ by continuity. Let

$$A := \{z \in D : F^{(k)}(z) = 0 \text{ for all } k \geq 0\}.$$

As in the proof of Theorem 24.7, A is open and closed in D .

A is nonempty: we claim $z_0 \in A$. If not, some Taylor coefficient of F at z_0 is nonzero, and the factorisation step of the proof of Theorem 24.7 (which uses only the Taylor expansion at z_0) lets us write $F(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$ and g continuous, so F would be nonzero on a punctured disc about z_0 — contradicting that z_0 is an accumulation point of Z .

So A is a nonempty subset of the connected set D that is both open and closed in D ; therefore $A = D$ and $F \equiv 0$. $\square$

Remark 24.9 (What the identity theorem buys). Three uses recur.

- (a) *Extension of identities from $\mathbb{R}$ to $\mathbb{C}$.* Since $\sin^2 x + \cos^2 x = 1$ on $\mathbb{R}$, and both sides are entire, they agree on $\mathbb{C}$: the real line has accumulation points. This is the legitimate version of “the same algebra works”.
- (b) *Milne’s method.* Proposition 15.9 substituted $y = 0$, $x \rightarrow z$; the identity theorem is exactly what licenses that substitution.
- (c) *Rigidity.* An analytic function on a region is determined by its values on any sequence of distinct points converging to a point of the region — for example by its values at $1/n$, $n \in \mathbb{N}$, if 0 is in the region. The limit must lie in the region: $\sin(\pi/z)$ is analytic on $\mathbb{C} \setminus \{0\}$ and vanishes at every $1/n$ without being $\equiv 0$. Nothing like this holds for C^∞ real functions.

Example 24.10. There is no function f analytic on a neighbourhood of 0 with $f(1/n) = (-1)^n/n$ for all $n \geq 2$. Indeed such an f would satisfy $f(1/(2n)) = 1/(2n)$, so $f(z) = z$ on a sequence accumulating at 0, whence $f(z) \equiv z$ by Theorem 24.8; but then $f(1/(2n+1)) = 1/(2n+1) \neq -1/(2n+1)$.

24.4 The maximum modulus principle

Theorem 24.11 (Maximum modulus principle). *Let D be a region and f analytic on D . If $|f|$ attains a local maximum at some $z_0 \in D$ — that is, if there is $\varepsilon > 0$ with $|f(z)| \leq |f(z_0)|$ for all $z \in D(z_0, \varepsilon) \subseteq D$ — then f is constant on D .*

Proof. *Step 1: f is constant near z_0 .* For $0 < \rho < \varepsilon$ the mean value property (Remark 21.2) gives

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta,$$

whence by Lemma 19.3

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \rho e^{i\theta})| d\theta \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0)| d\theta = |f(z_0)|, \quad (24.9)$$

the second inequality by the local maximum hypothesis. So both inequalities are equalities. Equality in the second forces

$$\int_0^{2\pi} (|f(z_0)| - |f(z_0 + \rho e^{i\theta})|) d\theta = 0$$

with a continuous, non-negative integrand, hence $|f(z_0 + \rho e^{i\theta})| = |f(z_0)|$ for all θ and all $\rho < \varepsilon$. Thus $|f|$ is constant on $D(z_0, \varepsilon)$, and Theorem 14.13(ii) makes f constant there.

Step 2: f is constant on D . Let $c := f(z_0)$ and let $A := \{z \in D : f \equiv c \text{ on a neighbourhood of } z\}$. A is open by definition and nonempty by Step 1. It is also closed in D : if $z_1 \in D$ is a limit of points of A , then all derivatives $f^{(n)}(z_1)$ vanish for $n \geq 1$ (they vanish on A and are continuous by Corollary 21.5), and $f(z_1) = c$; the identity theorem (Theorem 24.8) then gives $f \equiv c$ near z_1 , so $z_1 \in A$. Since D is connected and A is nonempty, open and closed in D , we get $A = D$. $\square$

Corollary 24.12 (Maximum on the boundary). *Let D be a bounded region and let f be continuous on $\overline{D}$ and analytic on D . Then $|f|$ attains its maximum over $\overline{D}$ on the boundary ∂D . If f is nonconstant, the maximum is attained only on ∂D . If moreover f has no zero in D , the same holds for the minimum of $|f|$.*

Proof. $\overline{D}$ is compact, so by Theorem 12.13 $|f|$ attains a maximum at some $z_* \in \overline{D}$. If $z_* \in D$ then Theorem 24.11 makes f constant on D , hence on $\overline{D}$ by continuity, and the maximum is attained on ∂D as well. If f is nonconstant, no interior point can be a maximum, so $z_* \in \partial D$. For the minimum: if f vanishes at a point of ∂D , the minimum of $|f|$ over $\overline{D}$ is 0 and is attained there, and (as f has no zero in D) only on ∂D . Otherwise f has no zero on $\overline{D}$, and we apply the result to $1/f$, which is analytic on D and continuous on $\overline{D}$. $\square$

Example 24.13 (Estimating without computing). Let $f(z) = z^2 + 3z$ on $\overline{D}(0, 1)$. By Corollary 24.12, $\max_{|z| \leq 1} |f|$ occurs on $|z| = 1$, where $|f| = |z||z + 3| = |z + 3| \leq |z| + 3 = 4$, with equality at $z = 1$. So the maximum is 4, attained only at $z = 1$ — established without differentiating anything.

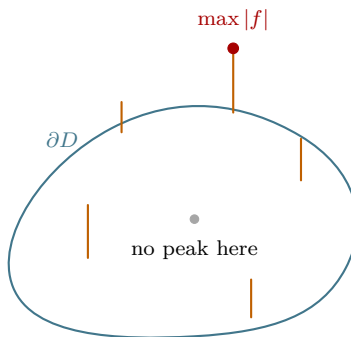


FIGURE 24.5. The maximum modulus principle, Corollary 24.12. Think of $|f|$ as the height of a surface over the region. For a nonconstant analytic f that surface has no interior peak: it can only reach its highest point on the boundary. The mean value inequality (24.9) is why — a value at the centre that exceeded every value on a surrounding circle could not also be their average.

Theorem 24.14 (Schwarz’s lemma). *Let f be analytic on the open unit disc $\mathbb{D}$ with $f(0) = 0$ and $|f(z)| \leq 1$ for all $z \in \mathbb{D}$. Then*

$$|f(z)| \leq |z| \quad \text{for all } z \in \mathbb{D}, \quad \text{and} \quad |f'(0)| \leq 1. \quad (24.10)$$

If equality holds in either place — $|f(z_1)| = |z_1|$ for some $z_1 \neq 0$, or $|f'(0)| = 1$ — then $f(z) = \lambda z$ for some constant λ with $|\lambda| = 1$.

Proof. Define $g : \mathbb{D} \rightarrow \mathbb{C}$ by $g(z) := f(z)/z$ for $z \neq 0$ and $g(0) := f'(0)$. Then g is analytic on $\mathbb{D}$. Indeed, by Theorem 24.1 f is the sum of its Taylor series $\sum_{n \geq 0} a_n z^n$ on $\mathbb{D}$, and $a_0 = f(0) = 0$; hence $f(z) = z \sum_{n \geq 1} a_n z^{n-1}$, the series on the right converging on all of $\mathbb{D}$. That series is g , and $g(0) = a_1 = f'(0)$.

Fix r with $0 < r < 1$. On $|z| = r$ we have $|g(z)| = |f(z)|/r \leq 1/r$, so by Corollary 24.12 $|g| \leq 1/r$ on $\overline{D}(0, r)$. Fix $z \in \mathbb{D}$ and let $r \uparrow 1$ through values $> |z|$: we obtain $|g(z)| \leq 1$, i.e. $|f(z)| \leq |z|$, and at $z = 0$, $|f'(0)| = |g(0)| \leq 1$.

If equality holds at some point of $\mathbb{D}$, then $|g|$ attains the value 1, its maximum, at an interior point; by Theorem 24.11 g is constant, say $g \equiv \lambda$ with $|\lambda| = 1$, and $f(z) = \lambda z$. $\square$

Schwarz’s lemma is the engine of Chapter 34: it pins down all the analytic bijections of the disc.

24.5 Underneath: the maximum principle for harmonic functions

The heart of the proof of Theorem 24.11 was the mean value property of f . That property is really about *real-valued* functions, and so is the maximum principle. Think of a real function u as the height of a landscape. A mountain top at z_0 is a point higher than everything around it, so on every small circle about z_0 the height is at most $u(z_0)$; a harmonic function, however, *equals* its average on every circle whose closed disc lies in its region. So a harmonic function cannot have a mountain top.

Theorem 24.15 (Maximum principle for harmonic functions). *Let D be a region and $u : D \rightarrow \mathbb{R}$ harmonic. If u attains a local maximum at some $z_0 \in D$, then u is constant on D . Consequently, if D is bounded and u is continuous on $\overline{D}$ and harmonic on D , then the maximum and the minimum of u over $\overline{D}$ are attained on ∂D .*

Proof. Step 1: the mean value property. Let $D(z_0, \rho') \subseteq D$. By Theorem 15.6, u has a harmonic conjugate v on this disc, so $u = \operatorname{Re} f$ with $f = u + iv$ analytic there. For $0 < \rho < \rho'$, the mean value form of Cauchy's integral formula (Remark 21.2) gives $f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta$; taking real parts,

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + \rho e^{i\theta}) d\theta. \quad (24.11)$$

Step 2: no mountain top. Suppose $u(z) \leq u(z_0)$ for $z \in D(z_0, \varepsilon) \subseteq D$. For $0 < \rho < \varepsilon$, (24.11) says

$$\int_0^{2\pi} (u(z_0) - u(z_0 + \rho e^{i\theta})) d\theta = 0,$$

and the integrand is continuous and non-negative, so it vanishes: $u = u(z_0)$ on every circle about z_0 of radius less than ε . Thus u is constant on $D(z_0, \varepsilon)$. (A genuine mountain top, higher than the points around it, would make this integral strictly positive: the average over every small circle would be below the peak. Figures 24.6–24.8 show this.)

Step 3: from near z_0 to all of D . Let $M = u(z_0)$ and $A = \{z \in D : u \equiv M \text{ on a neighbourhood of } z\}$. A is open, and nonempty by Step 2. It is closed in D : let $z_1 \in D$ be a limit of points of A and $D(z_1, \delta) \subseteq D$. On this disc $u = \operatorname{Re} g$ with g analytic (Theorem 15.6 again). The disc contains a point of A , hence a smaller open disc on which $\operatorname{Re} g \equiv M$. There g takes values in the line $\operatorname{Re} w = M$, so g is constant there (Theorem 14.13(iii)), and by the identity theorem 24.8 g is constant on $D(z_1, \delta)$. So $u \equiv M$ near z_1 , i.e. $z_1 \in A$. Since D is connected, $A = D$.

For the consequence, $\overline{D}$ is compact, so u attains its maximum at some $z_* \in \overline{D}$ (Theorem 12.13). If $z_* \in D$, then u is constant on D , hence on $\overline{D}$ by continuity, and the maximum is attained on ∂D too. For the minimum apply this to the harmonic function $-u$. $\square$

Theorem 24.15 is Exercise 15.8, and Step 1 is Exercise 21.4.

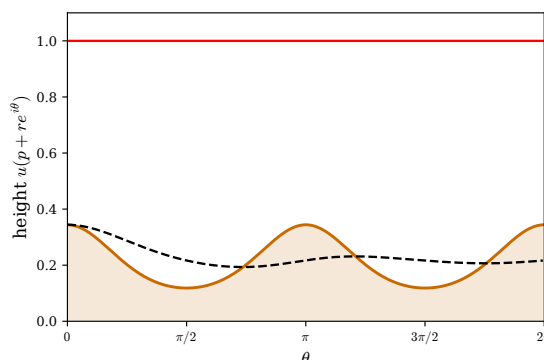
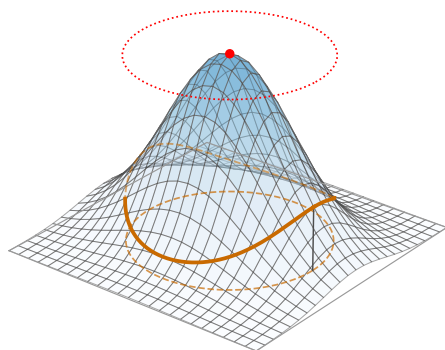
The average height round the circle is below the peak

every point of the circle is lower than the peak $u(p)$, so their average is lower too

$u(p) = 1.000$ (red: the peak)

average over the circle = 0.21651 (dashed)

$u(p) - \text{average} = +0.78349$; $r = 0.800$



$u = e^{-x^2/0.6 - y^2/0.3}$; made up, NOT harmonic ($\nabla^2 u(0) = -10$)

FIGURE 24.6 (animated). A mountain top has no mean value property. The surface is the height $u = e^{-x^2/0.6 - y^2/0.3}$, with its peak $u(p) = 1$ at $p = 0$; it is made up and is *not* harmonic ($\nabla^2 u(0) = -10$). A circle of radius r about p is drawn on the surface, with its shadow below. On the right the height $u(p + re^{i\theta})$ is graphed as θ runs from 0 to 2π , together with the running average $\frac{1}{\theta} \int_0^\theta u(p + re^{it}) dt$ (dashed) and the peak height (red). Every point of the circle lies below the peak, so the average does too: 0.2165 for $r = 0.8$, 0.7381 for $r = 0.35$, and still $0.9960 < 1$ for $r = 0.04$. A harmonic function would equal its average, (24.11).

For every radius: the mountain's average is below, the harmonic one is equal

the harmonic saddle rises and falls round the circle by exactly equal amounts
made-up mountain, NOT harmonic harmonic: $u = \text{Re}(1 + 0.8z^2)$

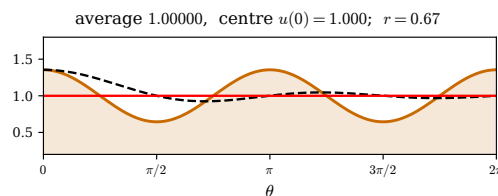
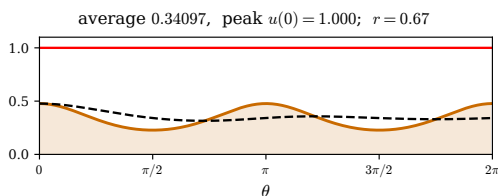
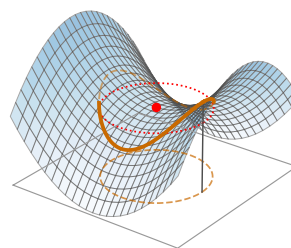
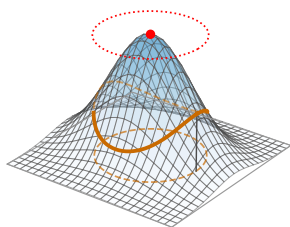


FIGURE 24.7 (animated). The mountain of Figure 24.6 beside the harmonic saddle $u = 1 + 0.8(x^2 - y^2) = \text{Re}(1 + 0.8z^2)$; both have height 1 at the centre. The same average is computed round circles of every radius. The mountain's average is always below its peak. The saddle rises and falls round each circle by exactly equal amounts, and its average is 1, the centre value, for every radius, as (24.11) says. So a harmonic function has no room for a peak.

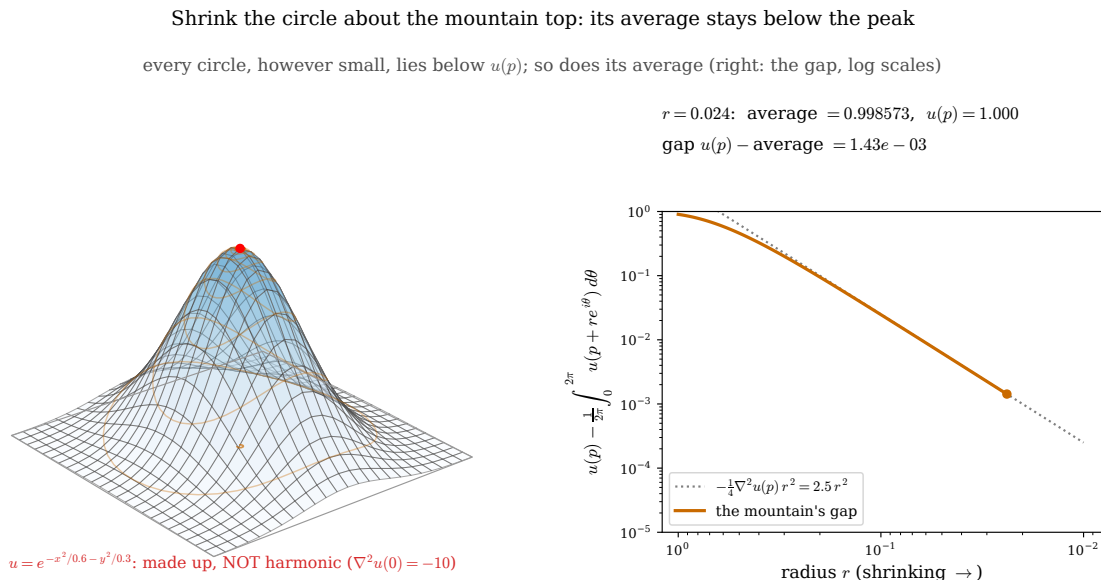


FIGURE 24.8 (animated). However small the circle, the average stays below the peak. Circles about the mountain top shrink from $r = 1$ to $r = 0.01$; on the right the gap $u(p) - \frac{1}{2\pi} \int_0^{2\pi} u(p + re^{i\theta}) d\theta$ is plotted on logarithmic scales. It shrinks like $2.5r^2$ but stays positive (for this u the gap is about $-\frac{1}{4}\nabla^2 u(p)r^2$, a fact not needed here). For the harmonic saddle the computed gap is 0 for every r . This is Step 2 of the proof of Theorem 24.15: at an interior maximum of a harmonic u the gap must be 0, which forces u to be constant on the small circles.

Remark 24.16 (This is what lies underneath the maximum modulus principle). Theorem 24.11 follows from Theorem 24.15. Let f be analytic on the region D , and suppose $|f|$ has a local maximum at z_0 .

- If $f(z_0) = 0$, then $|f| \equiv 0$ near z_0 , so $f \equiv 0$ near z_0 , and on D by the identity theorem 24.8.
- Otherwise write $f(z_0) = |f(z_0)|e^{i\alpha}$ and turn f so that its value at z_0 is positive: put $g = e^{-i\alpha}f$ and $u = \operatorname{Re} g$. Then u is harmonic (Theorem 15.2), and near z_0

$$u(z) \leq |g(z)| = |f(z)| \leq |f(z_0)| = g(z_0) = u(z_0).$$

So the *real-valued harmonic* function u has a local maximum at z_0 . By Theorem 24.15, u is constant on D . Then $g(D)$ lies in a vertical line, so g is constant (Theorem 14.13(iii)), and so is $f = e^{i\alpha}g$.

The same turn explains the first inequality in (24.9): $|f(z_0)| = u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u d\theta \leq \frac{1}{2\pi} \int_0^{2\pi} |f| d\theta$.

So the landscape $|f|$ has no peak because the harmonic landscape $u = \operatorname{Re}(e^{-i\alpha}f)$, which lies below it and touches it at z_0 , has none. The complex principle is the scalar principle for harmonic functions, applied to the real part of a suitably turned f .

24.6 Exercises

Exercise 24.1. Find the Maclaurin series and its radius of convergence:

$$\frac{z}{1+z^2}, \quad e^{z^2}, \quad \sin(2z), \quad \frac{1}{(1-z)^3}, \quad \operatorname{Log}(1-z).$$

Exercise 24.2. Expand $f(z) = \frac{1}{z}$ in powers of $z - 1$; in powers of $z + 2$. Give the radius of convergence in each case and check against Corollary 24.2.

Exercise 24.3. Determine the order of the zero of each function at the stated point: $z^2 \sin z$ at 0; $1 - \cos z$ at 0; $(e^z - 1)^3$ at 0; $z^4 + 2z^2 + 1$ at i .

Exercise 24.4. Without computing many derivatives, find the first four nonzero terms of the Maclaurin series of $\tan z$. (*Divide the series for $\sin$ by that for $\cos$.*)

Exercise 24.5. (★) Let f be entire with $f(z) = f(z+1) = f(z+i)$ for all z . Prove f is constant. (*Doubly periodic $\Rightarrow$ determined by a compact fundamental square $\Rightarrow$ bounded $\Rightarrow$ Liouville.*)

Exercise 24.6. Suppose f is analytic on $D(0, 1)$ and $f(1/n) = 0$ for all $n \geq 2$. Prove $f \equiv 0$. Why does the hypothesis $f(n) = 0$ for all $n \in \mathbb{N}$ not force an entire f to vanish?

Exercise 24.7. Give the radius of convergence of the Taylor series about the stated centre, *without* computing any coefficients, by locating the nearest singularity:

$$\frac{1}{z^2 + 4} \text{ about } 0; \quad \frac{e^z}{z - 3} \text{ about } 1; \quad \text{Log } z \text{ about } 2; \quad \tan z \text{ about } 0.$$

Exercise 24.8. Prove that if f is analytic at z_0 and $f^{(k)}(z_0) = 0$ for all $k \geq 1$ but $f(z_0) \neq 0$, then f is constant near z_0 . Is the same true for real C^∞ functions? Give the standard counterexample.

Exercise 24.9. (★) Let f be analytic on a region D and suppose that for each $z \in D$ there is some $k = k(z) \geq 0$ with $f^{(k)}(z) = 0$. Prove that f is a polynomial. (*Baire: one of the closed sets $E_k = \{z : f^{(k)}(z) = 0\}$ has nonempty interior; then apply Theorem 24.8.*)

Exercise 24.10. Find $\max_{|z| \leq 2} |z^2 - z|$ and the points where it is attained.

Exercise 24.11. (★) Let f be analytic on $\mathbb{D}$ with $f(0) = 0$ and $|f| \leq 1$. Prove $|f(z) + f(-z)| \leq 2|z|^2$ for $|z| < 1$. (*Apply Schwarz to $g(z) = \frac{1}{2}(f(z) + f(-z))$ suitably normalised.*)

Exercise 24.12. Let f be analytic and nonconstant on a bounded region D , continuous on $\overline{D}$, with $|f|$ constant on ∂D . Prove that f has at least one zero in D .

Exercise 24.13. (★) (*Schwarz–Pick.*) Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be analytic. Prove that for all $a \in \mathbb{D}$,

$$\frac{|f'(a)|}{1 - |f(a)|^2} \leq \frac{1}{1 - |a|^2}.$$

(*Apply Theorem 24.14 to $\varphi_{f(a),0} \circ f \circ \varphi_{a,0}^{-1}$, with φ as in (34.9).*) Deduce that every analytic self map of $\mathbb{D}$ is a contraction for the hyperbolic metric $\frac{2|dz|}{1-|z|^2}$.

Chapter 25

Laurent Series

Taylor series expand a function analytic on a *disc*. Many of the functions we care about — $1/z$, $e^{1/z}$, $\frac{\sin z}{z^2}$ — fail to be analytic at the centre. Laurent's theorem expands such a function on an *annulus*, at the cost of admitting negative powers. Those negative powers turn out to carry all the information the residue calculus needs.

25.1 Laurent's theorem

Definition 25.1 (Annulus). For $z_0 \in \mathbb{C}$ and $0 \leq r < R \leq \infty$ let

$$A(z_0; r, R) := \{ z \in \mathbb{C} : r < |z - z_0| < R \}. \quad (25.1)$$

The cases $r = 0$ (a punctured disc) and $R = \infty$ (the exterior of a disc) are allowed and are the ones that occur most often.

Definition 25.2 (Laurent series). A *Laurent series* centred at z_0 is a doubly infinite expression

$$\sum_{k=-\infty}^{\infty} a_k(z - z_0)^k := \underbrace{\sum_{k=1}^{\infty} \frac{a_{-k}}{(z - z_0)^k}}_{\text{principal part}} + \underbrace{\sum_{k=0}^{\infty} a_k(z - z_0)^k}_{\text{analytic part}}, \quad (25.2)$$

and it *converges* at z precisely when both displayed series converge there.

Theorem 25.3 (Laurent). Let f be analytic on the annulus $A = A(z_0; r, R)$ of (25.1). Then

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k, \quad z \in A, \quad (25.3)$$

where

$$a_k = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta, \quad k \in \mathbb{Z}, \quad (25.4)$$

and C is any circle $C(z_0, \rho)$ with $r < \rho < R$, traversed counterclockwise. The series converges absolutely on A and uniformly on every closed sub-annulus $\{r < r_1 \leq |z - z_0| \leq r_2 < R\}$. The coefficients are uniquely determined by f and the annulus.

Proof. Fix $z \in A$ and choose r_1, r_2 with $r < r_1 < |z - z_0| < r_2 < R$. Let $C_1 = C(z_0, r_1)$ and $C_2 = C(z_0, r_2)$, both counterclockwise.

Step 1: a two-circle Cauchy formula. The function $\zeta \mapsto \frac{f(\zeta)}{\zeta - z}$ is analytic on $A \setminus \{z\}$. Apply the many-hole deformation theorem 20.10 to the region between C_2 and the two curves C_1 and a small circle C_ε around z :

$$\oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta = \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta + \oint_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta,$$

and the last integral equals $2\pi i f(z)$ by the computation in the proof of Theorem 21.1. Hence

$$f(z) = \frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (25.5)$$

Step 2: expand the outer kernel. For $\zeta \in C_2$ we have $|z - z_0| < r_2 = |\zeta - z_0|$, so (24.3) applies and converges uniformly on C_2 ; multiplying by $f(\zeta)$ and integrating term by term,

$$\frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{k=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta \right) (z - z_0)^k. \quad (25.6)$$

Step 3: expand the inner kernel the other way. For $\zeta \in C_1$ we have $|\zeta - z_0| = r_1 < |z - z_0|$, so now we factor out $z - z_0$ instead:

$$\frac{-1}{\zeta - z} = \frac{1}{z - \zeta} = \frac{1}{(z - z_0) - (\zeta - z_0)} = \frac{1}{z - z_0} \cdot \frac{1}{1 - \frac{\zeta - z_0}{z - z_0}} = \sum_{j=0}^{\infty} \frac{(\zeta - z_0)^j}{(z - z_0)^{j+1}},$$

using (23.2) with ratio of modulus $r_1/|z - z_0| < 1$; the convergence is uniform in $\zeta \in C_1$. Multiplying by $f(\zeta)$ and integrating,

$$-\frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{j=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{C_1} f(\zeta) (\zeta - z_0)^j d\zeta \right) \frac{1}{(z - z_0)^{j+1}}. \quad (25.7)$$

Setting $k = -(j + 1)$, i.e. $j = -k - 1$, the bracket becomes $\frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta$, which is (25.4) computed on C_1 .

Step 4: one contour suffices. For each fixed k , the integrand $\frac{f(\zeta)}{(\zeta - z_0)^{k+1}}$ is analytic on all of A , so by the deformation theorem 20.10 its integral is the same over every circle $C(z_0, \rho)$ with $r < \rho < R$. Hence the coefficients in (25.6) and (25.7) are both given by (25.4) with a common C . Adding the two expansions and using (25.5) yields (25.3).

Absolute and uniform convergence. This part holds for *any* Laurent series $\sum_k c_k (z - z_0)^k$ that converges at every point of A . The analytic part $\sum_{k \geq 0} c_k (z - z_0)^k$ is a power series converging at every z with $r < |z - z_0| < R$, so by Theorem 23.10(i) its radius of convergence is at least each such $|z - z_0|$, hence at least R ; by Theorem 23.10(i),(ii) it converges absolutely on $|z - z_0| < R$ and uniformly on $|z - z_0| \leq r_2$ for every $r_2 < R$. The principal part is the power series $\sum_{k \geq 1} c_{-k} w^k$ in $w := 1/(z - z_0)$, which converges for every w with $1/R < |w| < 1/r$ (read $1/0 = \infty$, $1/\infty = 0$); so its radius is at least $1/r$, and it converges absolutely for $|w| < 1/r$ and uniformly on $|w| \leq 1/r_1$, i.e. on $|z - z_0| \geq r_1$, for every $r_1 > r$. Adding, the series converges absolutely on A and uniformly on every closed sub-annulus $r_1 \leq |z - z_0| \leq r_2$.

Uniqueness. Suppose $f(z) = \sum_k b_k (z - z_0)^k$ at every point of A . By the previous paragraph, applied to the b_k , the convergence is uniform on every circle $C(z_0, \rho)$, $r < \rho < R$. Fix $n \in \mathbb{Z}$,

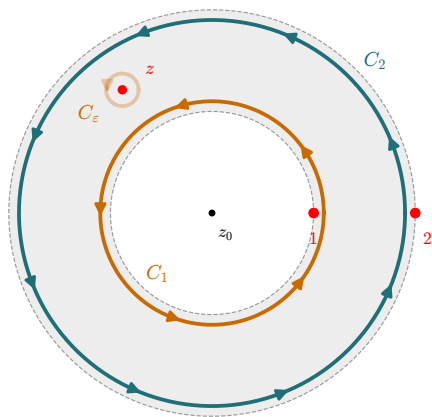
multiply by $(z - z_0)^{-n-1}$ and integrate over $C = C(z_0, \rho)$; uniform convergence permits term-by-term integration, and by (19.16)

$$\oint_C (z - z_0)^{k-n-1} dz = \begin{cases} 2\pi i, & k = n, \\ 0, & k \neq n, \end{cases}$$

so $\oint_C f(z)(z - z_0)^{-n-1} dz = 2\pi i b_n$, i.e. $b_n = a_n$. $\square$

The two-circle Cauchy formula (25.5)

$$f(z) = \frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta$$



$$\frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta = -0.2947 - 0.1240 i$$

$$\frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta = -0.3754 - 0.2418 i$$

$$\frac{1}{2\pi i} \oint_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta = 0.0807 + 0.1178 i$$

(each counterclockwise)

$$C_\varepsilon \text{ gives } f(z) = 0.0807 + 0.1178 i$$

(Cauchy's integral formula, Theorem 21.1)

$$\frac{1}{2\pi i} \oint_{C_2} - \frac{1}{2\pi i} \oint_{C_1} = 0.0807 + 0.1178 i$$

$= f(z)$, for z between the two circles

FIGURE 25.1 (animated). Step 1 of the proof of Theorem 25.3, the two-circle Cauchy formula (25.5), for $f(z) = \frac{1}{(z-1)(z-2)}$ of Example 25.6 on the annulus $1 < |z| < 2$, with $C_1 = C(0, 1.1)$, $C_2 = C(0, 1.9)$ and the point $z = 1.5e^{2.2i}$. The integrand $\frac{f(\zeta)}{\zeta - z}$ is analytic on the annulus except at $\zeta = z$, so a small circle C_ε about z is cut out too. Three cuts divide what is left into two pieces; the integral round each is 0 (right: computed as the points run), and when the pieces are put back the cuts cancel. What remains is C_2 counterclockwise and C_1 , C_ε clockwise, which is the many-hole deformation theorem 20.10. The small circle gives $f(z)$, as in the proof of Theorem 21.1, and so $f(z) = \frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta$ (computed values on the right).

Remark 25.4 (The coefficient that matters). Taking $k = -1$ in (25.4) gives

$$a_{-1} = \frac{1}{2\pi i} \oint_C f(\zeta) d\zeta, \quad \text{equivalently} \quad \oint_C f(z) dz = 2\pi i a_{-1}. \quad (25.8)$$

So *one* Laurent coefficient computes the integral. When $r = 0$ (a punctured disc) that coefficient is the residue of Chapter 27 (Definition 27.1), and the whole of residue calculus is the business of extracting a_{-1} without computing the integral.

Warning 25.5 (The expansion depends on the annulus). A function analytic on several disjoint annuli about the same centre has a *different* Laurent series on each. There is no such thing as “the Laurent series of f at z_0 ” without naming the region; see Example 25.6.

25.2 Computing Laurent series

In practice one never uses (25.4). One manipulates known series — almost always the geometric series (23.2) and the exponential series (24.4) — and appeals to uniqueness.

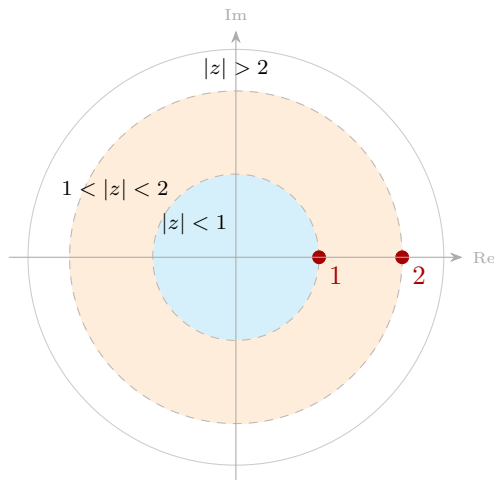


FIGURE 25.2. Warning 25.5, drawn for $f(z) = \frac{1}{(z-1)(z-2)}$ of Example 25.6. The two singularities cut the plane into three annuli centred at 0, and f has a *different* Laurent expansion on each. There is no such thing as “the Laurent series at 0” until the region is named.

Example 25.6 (Three annuli, three series). Let $f(z) = \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$. It is analytic on three regions centred at 0: $|z| < 1$, $1 < |z| < 2$, $|z| > 2$.

On $|z| < 1$: expand both terms in powers of z , as in Example 24.6:

$$f(z) = \sum_{k=0}^{\infty} \left(1 - \frac{1}{2^{k+1}}\right) z^k \quad (\text{a Taylor series; no negative powers}).$$

On $1 < |z| < 2$: now $|1/z| < 1$ and $|z/2| < 1$, so

$$-\frac{1}{z-1} = -\frac{1}{z} \cdot \frac{1}{1-1/z} = -\sum_{k=1}^{\infty} \frac{1}{z^k}, \quad \frac{1}{z-2} = -\frac{1}{2} \cdot \frac{1}{1-z/2} = -\sum_{k=0}^{\infty} \frac{z^k}{2^{k+1}},$$

so

$$f(z) = -\sum_{k=1}^{\infty} \frac{1}{z^k} - \sum_{k=0}^{\infty} \frac{z^k}{2^{k+1}}.$$

On $|z| > 2$: both $|1/z| < 1$ and $|2/z| < 1$, so

$$\frac{1}{z-2} = \frac{1}{z} \cdot \frac{1}{1-2/z} = \sum_{k=0}^{\infty} \frac{2^k}{z^{k+1}}, \quad -\frac{1}{z-1} = -\sum_{k=0}^{\infty} \frac{1}{z^{k+1}},$$

giving $f(z) = \sum_{k=0}^{\infty} \frac{2^k-1}{z^{k+1}}$ — only negative powers, as it must be, since $f(z) \rightarrow 0$ as $z \rightarrow \infty$.

Note the coefficient a_{-1} of $1/z$: it is 0 on the first region (there is no $1/z$ term), -1 on the second, and $2^0 - 1 = 0$ on the third. The value depends on the annulus, and only the first is a residue: the residue at 0 — Definition 27.1 — refers to the expansion on a punctured disc, here $0 < |z| < 1$, where f is in fact analytic at 0 and $\text{Res}(f, 0) = 0$. On the other two annuli a_{-1} is still $\frac{1}{2\pi i} \oint_C f$ by (25.8), for a circle C in that annulus.

Example 25.7 (An essential singularity). Let $f(z) = e^{1/z}$, analytic on $0 < |z| < \infty$. Substituting $1/z$ for z in (24.4),

$$e^{1/z} = \sum_{k=0}^{\infty} \frac{1}{k!} z^{-k} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \cdots, \quad z \neq 0. \quad (25.9)$$

The principal part has infinitely many terms. Its residue is $a_{-1} = 1$, so $\oint_{|z|=1} e^{1/z} dz = 2\pi i$ by (25.8) — an integral one would not care to attempt directly.

Example 25.8 (Products and quotients). Expand $f(z) = \frac{\sin z}{z^4}$ about 0. By (24.5),

$$\frac{\sin z}{z^4} = \frac{1}{z^4} \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots \right) = \frac{1}{z^3} - \frac{1}{3!} \frac{1}{z} + \frac{z}{5!} - \cdots, \quad 0 < |z| < \infty.$$

So $a_{-1} = -\frac{1}{6}$ and $\oint_{|z|=1} \frac{\sin z}{z^4} dz = -\frac{\pi i}{3}$. Compare Example 21.6: the integral formula for derivatives would give the same answer, since $-1/3!$ is the coefficient of z^{-1} , i.e. $\frac{(\sin)'''(0)}{3!}$.

Expand $g(z) = \frac{e^z}{z^2(z-1)}$ on $0 < |z| < 1$. Write $\frac{1}{z-1} = -\sum_{k \geq 0} z^k$ and $e^z = 1 + z + \frac{z^2}{2} + \cdots$; then

$$e^z \cdot \frac{-1}{1-z} \cdot \frac{1}{z^2} = -\frac{1}{z^2} (1 + z + \frac{z^2}{2} + \cdots) (1 + z + z^2 + \cdots) = -\frac{1}{z^2} (1 + 2z + \frac{5}{2}z^2 + \cdots),$$

so $g(z) = -z^{-2} - 2z^{-1} - \frac{5}{2} - \cdots$ and $a_{-1} = -2$.

25.3 Exercises

Exercise 25.1. Expand in a Laurent series on the indicated region:

$$\frac{1}{z(z-3)} \quad \text{on } 0 < |z| < 3 \quad \text{and on } |z| > 3; \quad \frac{z}{(z+1)(z-2)} \quad \text{on } 1 < |z| < 2.$$

Exercise 25.2. Find the Laurent expansion about 0 and give a_{-1} :

$$z^3 e^{1/z}, \quad \frac{\cos z - 1}{z^5}, \quad \frac{1}{z^2 \sin z}, \quad \frac{e^z - 1}{z^3}.$$

Exercise 25.3. Expand $\frac{1}{z^2}$ in powers of $z-1$, valid on $|z-1| < 1$. (Differentiate the expansion of $-1/z$.)

Exercise 25.4. (*) Let f be analytic on $0 < |z - z_0| < R$ and bounded there. Prove that $a_k = 0$ for all $k < 0$. (Estimate (25.4) on $C(z_0, \rho)$ with the ML-inequality and let $\rho \downarrow 0$.) This is Riemann's removable singularity theorem, restated in Theorem 26.4.

Exercise 25.5. Use (25.8) to evaluate $\oint_{|z|=2} z^2 e^{1/z} dz$ and $\oint_{|z|=1} \frac{\cosh z}{z^3} dz$.

Exercise 25.6. Show that if f is analytic on $|z - z_0| < R$ then its Laurent series on $0 < |z - z_0| < R$ coincides with its Taylor series, i.e. all negative coefficients vanish. (Use uniqueness.)

Exercise 25.7. Find all three Laurent expansions of $f(z) = \frac{1}{z(z-1)(z-2)}$ about 0, naming the annulus for each, and give $\text{Res}(f, 0)$.

Exercise 25.8. Expand $\frac{1}{z^2 + 1}$ in a Laurent series valid for $|z| > 1$, and use it to compute $\oint_{|z|=2} \frac{dz}{z^2 + 1}$ from (25.8) alone. Check against Theorem 27.5.

Exercise 25.9. (*) (Cauchy estimates for Laurent coefficients.) Let f be analytic on $A(z_0; r, R)$ with $|f| \leq M(\rho)$ on $|z - z_0| = \rho$. Prove $|a_k| \leq M(\rho)\rho^{-k}$ for every $k \in \mathbb{Z}$ and every $\rho \in (r, R)$. Deduce Exercise 25.4 and, taking $R = \infty$, that an entire function bounded by $A|z|^m$ at infinity is a polynomial of degree $\leq m$.

Chapter 26

Zeros and Poles

26.1 Isolated singularities and their classification

Definition 26.1 (Isolated singularity). A point $z_0 \in \mathbb{C}$ is an *isolated singularity* of f if there is $R > 0$ such that f is analytic on the punctured disc $D'(z_0, R) = A(z_0; 0, R)$ but not analytic at z_0 itself (either f is undefined there, or defined but not differentiable).

By Theorem 25.3, such an f has a Laurent expansion

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k, \quad 0 < |z - z_0| < R, \quad (26.1)$$

and the shape of the principal part $\sum_{k \geq 1} a_{-k}(z - z_0)^{-k}$ gives the classification.

Definition 26.2 (The three types). Let z_0 be an isolated singularity of f with Laurent expansion (26.1). Then z_0 is

- *removable* if $a_k = 0$ for all $k < 0$ (no principal part);
- a *pole of order m* ($m \geq 1$) if $a_{-m} \neq 0$ and $a_k = 0$ for all $k < -m$ (finite principal part, longest term $(z - z_0)^{-m}$); a pole of order 1 is called *simple*;
- an *essential singularity* if $a_k \neq 0$ for infinitely many $k < 0$.

Example 26.3. $\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \dots$ has a removable singularity at 0. $\frac{\sin z}{z^4}$ has, by Example 25.8, principal part $z^{-3} - \frac{1}{6}z^{-1}$: a pole of order 3. And $e^{1/z}$ has an essential singularity at 0 by (25.9).

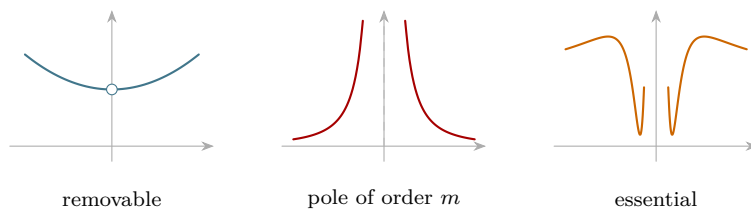


FIGURE 26.1. The trichotomy of Definition 26.2, read off from the behaviour of $|f|$ near the singularity. Removable: $|f|$ stays bounded and the graph fills in. Pole: $|f| \rightarrow \infty$, and the rate ($|z - z_0|^{-m}$) is the order. Essential: $|f|$ does neither — by Casorati–Weierstrass 26.8 f comes arbitrarily close to *every* complex value in every punctured neighbourhood, so $|f|$ has no limit in $[0, \infty]$.

26.2 Removable singularities

Theorem 26.4 (Riemann's removable singularity theorem). *Let z_0 be an isolated singularity of f . The following are equivalent:*

- (i) z_0 is removable;
- (ii) $\lim_{z \rightarrow z_0} f(z)$ exists in $\mathbb{C}$;
- (iii) f is bounded on some punctured disc $D'(z_0, \varepsilon)$;
- (iv) $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$.

In that case, defining $f(z_0) := a_0 = \lim_{z \rightarrow z_0} f(z)$ makes f analytic on the full disc $D(z_0, R)$.

Proof. (i) $\Rightarrow$ (ii): with no principal part, (26.1) is a power series, convergent on $D(z_0, R)$ and hence continuous there by Theorem 23.11; its value at z_0 is a_0 , and the extended function is analytic. (ii) $\Rightarrow$ (iii) is immediate, and (iii) $\Rightarrow$ (iv) because $|(z - z_0)f(z)| \leq M|z - z_0| \rightarrow 0$.

(iv) $\Rightarrow$ (i): let $\eta > 0$ and choose $\delta > 0$ with $|(z - z_0)f(z)| < \eta$ for $0 < |z - z_0| < \delta$, i.e. $|f(z)| < \eta/|z - z_0|$ there. For $k \leq -1$ and $0 < \rho < \delta$, estimate (25.4) on $C = C(z_0, \rho)$ with the ML -inequality (19.15):

$$|a_k| \leq \frac{1}{2\pi} \cdot \frac{\eta/\rho}{\rho^{k+1}} \cdot 2\pi\rho = \eta\rho^{-k-1}.$$

Since $k \leq -1$ we have $-k - 1 \geq 0$, so letting $\rho \downarrow 0$ gives $|a_k| \leq \eta \cdot 0 = 0$ when $k \leq -2$, and $|a_{-1}| \leq \eta$ for every $\eta > 0$, hence $a_{-1} = 0$ as well. $\square$

26.3 Poles

Theorem 26.5 (Characterisations of a pole). *Let z_0 be an isolated singularity of f and $m \geq 1$ an integer. The following are equivalent:*

- (i) z_0 is a pole of order m ;
- (ii) there is g analytic on a neighbourhood of z_0 with $g(z_0) \neq 0$ and

$$f(z) = \frac{g(z)}{(z - z_0)^m} \quad \text{for } z \neq z_0 \text{ near } z_0; \quad (26.2)$$

- (iii) $\lim_{z \rightarrow z_0} (z - z_0)^m f(z)$ exists, is finite, and is nonzero;
- (iv) $1/f$ has a removable singularity at z_0 and, once removed, a zero of order exactly m there.

In all cases $\lim_{z \rightarrow z_0} |f(z)| = \infty$.

Proof. (i) $\Rightarrow$ (ii): from (26.1) with $a_k = 0$ for $k < -m$,

$$f(z) = (z - z_0)^{-m} \underbrace{\sum_{j=0}^{\infty} a_{j-m} (z - z_0)^j}_{=: g(z)},$$

and g is a convergent power series with $g(z_0) = a_{-m} \neq 0$.

(ii) $\Rightarrow$ (iii): $(z - z_0)^m f(z) = g(z) \rightarrow g(z_0) \neq 0$.

(iii) $\Rightarrow$ (i): let $L \neq 0$ be the limit. Then $h(z) := (z - z_0)^m f(z)$ has a removable singularity at z_0 by Theorem 26.4, with $h(z_0) = L \neq 0$; expanding h in a Taylor series and dividing by $(z - z_0)^m$ produces a Laurent series for f whose most negative term is $L(z - z_0)^{-m}$.

(ii) $\Leftrightarrow$ (iv): from (26.2), $1/f(z) = (z - z_0)^m/g(z)$ near z_0 , and $1/g$ is analytic and nonzero near z_0 ; this is exactly the factorisation (24.8) of a zero of order m . Conversely if $1/f$ has a zero of order m , write $1/f = (z - z_0)^m h$ with $h(z_0) \neq 0$ and invert.

Finally, from (26.2), $|f(z)| = |g(z)|/|z - z_0|^m \rightarrow \infty$ since the numerator tends to $|g(z_0)| > 0$. $\square$

Corollary 26.6 (Poles of a quotient). *Let p, q be analytic near z_0 with $p \not\equiv 0$ and $q \not\equiv 0$, and set $f = p/q$. Let $\mu := \text{ord}_{z_0} p$ and $\nu := \text{ord}_{z_0} q$ (with $\text{ord} = 0$ meaning the function is nonzero at z_0). Then f has at z_0*

- a removable singularity, if $\mu \geq \nu$ (or no singularity at all, if $\nu = 0$: then f is analytic at z_0);
- a pole of order $\nu - \mu$, if $\mu < \nu$.

Proof. Write $p = (z - z_0)^\mu P$ and $q = (z - z_0)^\nu Q$ with P, Q analytic and nonzero at z_0 (Theorem 24.7). Then $f = (z - z_0)^{\mu-\nu} \frac{P}{Q}$ with P/Q analytic and nonzero at z_0 , and Theorem 26.5(ii) applies when $\mu - \nu < 0$; when $\mu \geq \nu$ the function extends analytically. $\square$

Example 26.7. (a) $f(z) = \frac{z+1}{(z-2)^3(z^2+1)}$ has a pole of order 3 at 2 and simple poles at $\pm i$.

(b) $f(z) = \frac{1}{\sin z}$ has simple poles exactly at $z = k\pi$, $k \in \mathbb{Z}$, by Corollary 17.6 and Theorem 26.5(iv), since each zero of $\sin$ is simple ($\cos k\pi = \pm 1 \neq 0$).

(c) $f(z) = \frac{1 - \cos z}{z^4}$: here $1 - \cos z = \frac{z^2}{2} - \frac{z^4}{24} + \cdots$ has a zero of order 2 at 0, so by Corollary 26.6 f has a pole of order $4 - 2 = 2$.

(d) $f(z) = \frac{z}{e^z - 1}$: the denominator has a simple zero at 0 (its derivative e^z is 1 there) and the numerator a simple zero, so the singularity at 0 is *removable*. The other singularities $z = 2k\pi i$, $k \neq 0$, are simple poles.

(e) $f(z) = z e^{1/z}$ has an essential singularity at 0: multiplying (25.9) by z still leaves infinitely many negative powers.

26.4 Essential singularities

Theorem 26.8 (Casorati–Weierstrass). *Let z_0 be an essential singularity of f , analytic on $D'(z_0, R)$. Then for every $\varepsilon \in (0, R)$ the image $f(D'(z_0, \varepsilon))$ is dense in $\mathbb{C}$: for every $w_0 \in \mathbb{C}$ and every $\eta > 0$ there is z with $0 < |z - z_0| < \varepsilon$ and $|f(z) - w_0| < \eta$.*

Proof. Suppose not: there are $w_0 \in \mathbb{C}$, $\eta > 0$, $\varepsilon > 0$ with

$$|f(z) - w_0| \geq \eta \quad \text{for } 0 < |z - z_0| < \varepsilon. \quad (26.3)$$

Define $h(z) := \frac{1}{f(z) - w_0}$ on $D'(z_0, \varepsilon)$. By (26.3) h is analytic there and $|h| \leq 1/\eta$, so by Theorem 26.4 h extends analytically to z_0 .

If $h(z_0) \neq 0$ then $f = w_0 + 1/h$ is analytic at z_0 after removing the singularity, so z_0 is removable — not essential. If $h(z_0) = 0$, say with a zero of order $m \geq 1$, then Theorem 26.5(iv) says $f - w_0$, hence f , has a pole of order m at z_0 — again not essential. Either way we contradict the hypothesis. $\square$

Remark 26.9 (Picard). Picard's great theorem sharpens Casorati–Weierstrass enormously: near an essential singularity f takes every complex value, with at most one exception, *infinitely often*. For $e^{1/z}$ at 0 the exceptional value is 0, consistent with (16.3). We do not prove this.

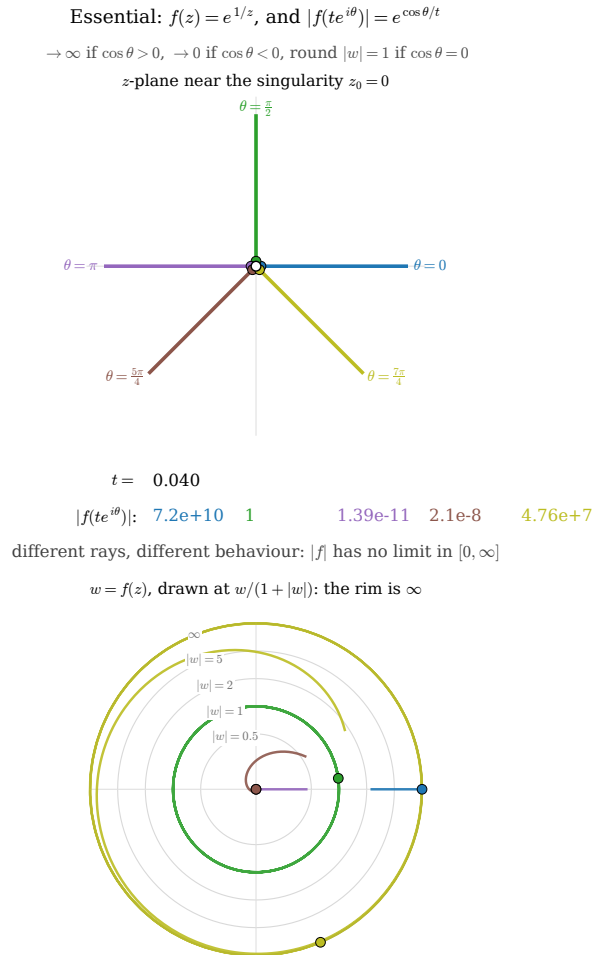


FIGURE 26.2 (animated). The three kinds of isolated singularity (Definition 26.2) seen along five rays into 0; the images are drawn at $w/(1 + |w|)$, so the rim is ∞ . For $\frac{\sin z}{z}$ all images tend to 1 (removable); for $\frac{\sin z}{z^4}$ all run out to ∞ at the rate t^{-3} , each arriving at its own angle (pole of order 3); for $e^{1/z}$ the rays behave differently: $|f| = e^{\cos \theta/t}$ goes to ∞ , stays 1, goes to 0 (essential). Then Casorati–Weierstrass (Theorem 26.8): in every punctured disc $D'(0, \varepsilon)$ lie infinitely many solutions of $e^{1/z} = w_0$, and they stay there as w_0 moves; this is the stronger fact of Remark 26.9.

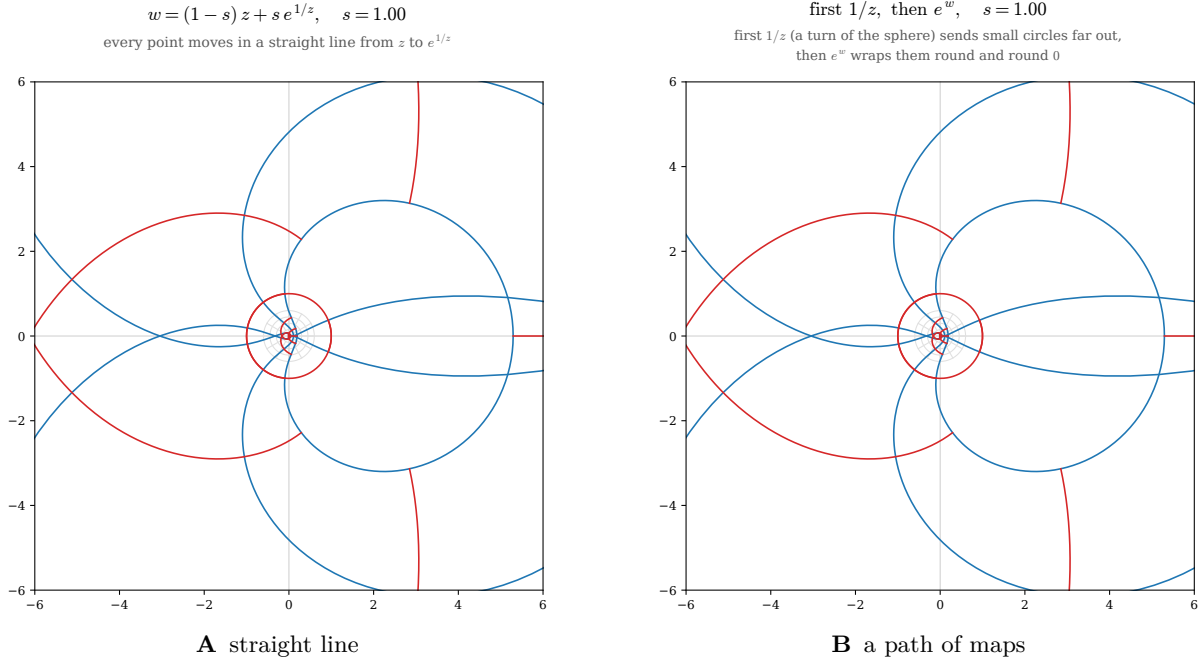
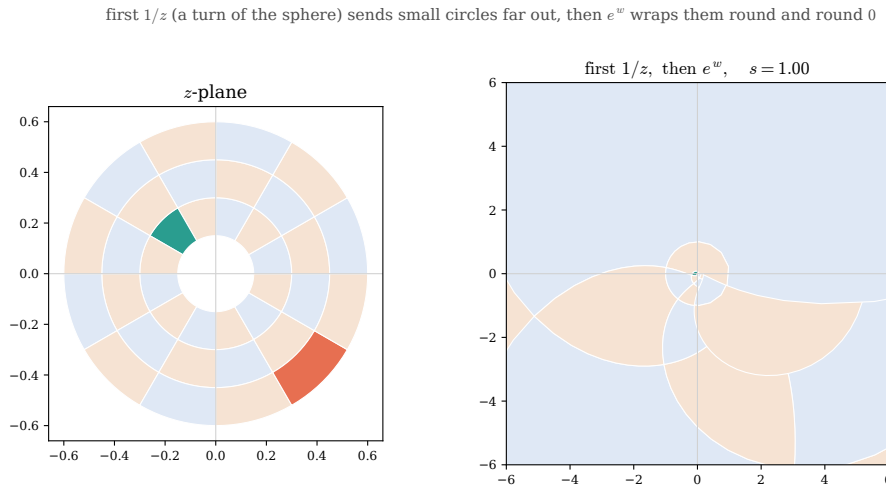


FIGURE 26.3 (animated). $e^{1/z}$ near its essential singularity. **A**: every point moves in a straight line from z to $f(z)$. **B**: first $1/z$ (a turn of the sphere) sends small circles far out, then e^w wraps them round and round 0. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 26.3, continued)

Proposition 26.10 (Behaviour at ∞). *Let f be analytic on $|z| > R_0$. Define $g(w) := f(1/w)$ on $0 < |w| < 1/R_0$. We say f has a removable singularity, a pole of order m , or an essential singularity at ∞ according as g has one at 0. Then:*

- (i) f is a polynomial of degree $m \geq 1$ if and only if f is entire with a pole of order m at ∞ ;
- (ii) f is constant if and only if f is entire with a removable singularity at ∞ ;
- (iii) e^z is entire with an essential singularity at ∞ .

Proof. (ii) A removable singularity at ∞ means f is bounded near ∞ ; being entire, it is bounded on compact sets too, hence bounded on $\mathbb{C}$, hence constant by Liouville 22.2. (i) A pole of order m at ∞ means $|f(1/w)| \sim c|w|^{-m}$, i.e. $|f(z)| \leq A|z|^m$ for large $|z|$; Corollary 22.3 makes f a polynomial of degree $\leq m$, and the degree is exactly m because the pole order is. Conversely, if $f(z) = \sum_{j=0}^m c_j z^j$ with $c_m \neq 0$ and $m \geq 1$, then f is entire and $f(1/w) = w^{-m}(c_m + c_{m-1}w + \cdots + c_0 w^m)$ for $w \neq 0$, the bracket being a polynomial in w that is nonzero at $w = 0$; so $f(1/w)$ has a pole of order m at 0 by Theorem 26.5(ii). (iii) $e^{1/w}$ has an essential singularity at 0 by (25.9). $\square$

26.5 Exercises

Exercise 26.1. Locate and classify all isolated singularities:

$$\frac{z^2}{z^2 + 4}, \quad \frac{\sin z}{z^3}, \quad \frac{e^z}{(z-1)^2}, \quad \frac{1}{e^z - 1}, \quad z^2 \sin \frac{1}{z}, \quad \frac{\cos z}{z^2 - \pi^2/4}.$$

Exercise 26.2. Show that $f(z) = \frac{1 - \cos z}{z^2}$ has a removable singularity at 0 and compute the value that removes it.

Exercise 26.3. Determine the order of the pole of $\frac{1}{(e^z - 1 - z)}$ at $z = 0$.

Exercise 26.4. (*) Let f be analytic on $\mathbb{C}$ except for finitely many poles, and suppose f has a pole or removable singularity at ∞ . Prove f is a rational function. (*Subtract the principal parts at each pole and apply Proposition 26.10.*)

Exercise 26.5. Verify Casorati–Weierstrass for $e^{1/z}$ at 0 directly: given $w_0 \neq 0$ and $\eta > 0$, exhibit z near 0 with $|e^{1/z} - w_0| < \eta$. (*Solve $e^{1/z} = w_0$ exactly, using (16.11).*)

Exercise 26.6. Classify the singularity of $\sin(\frac{1}{1-z})$ at $z = 1$, and of $\frac{1}{\sin(1/z)}$ at $z = 0$. (*The second is not isolated — explain why.*)

Exercise 26.7. Let f have a pole of order m at z_0 and g a zero of order k there. Determine the type and order of the singularity of fg , of f/g and of $f + g$ at z_0 , treating all cases of m versus k .

Exercise 26.8. Classify the singularity at ∞ (Proposition 26.10) of $p(z) = z^3 - z$, of $\frac{1}{z^2+1}$, of $\sin z$, and of $\frac{z^2}{z-1}$.

Exercise 26.9. (*) Prove that f has a pole at z_0 if and only if $\lim_{z \rightarrow z_0} |f(z)| = \infty$, and that f has an essential singularity if and only if $\lim_{z \rightarrow z_0} |f(z)|$ fails to exist in $[0, \infty]$. (*Combine Theorems 26.4, 26.5 and 26.8: the three types are exactly the three possible behaviours of $|f|$.*)

Part VII

Residues

Chapter 27

Residues and the Residue Theorem

27.1 The residue

Definition 27.1 (Residue). Let z_0 be an isolated singularity of f , with Laurent expansion (26.1) on $D'(z_0, R)$. The *residue of f at z_0* is the coefficient of $(z - z_0)^{-1}$:

$$\operatorname{Res}(f, z_0) := a_{-1} = \frac{1}{2\pi i} \oint_C f(z) \, dz, \quad (27.1)$$

where $C = C(z_0, \rho)$ counterclockwise with $0 < \rho < R$. If f is analytic at z_0 we set $\operatorname{Res}(f, z_0) = 0$, consistent with (27.1).

The second equality in (27.1) is (25.8), and it is the point of the whole construction: a single number extracted from a series computes a contour integral.

Theorem 27.2 (Computing residues at poles). *Let z_0 be a pole of f .*

(i) *If the pole is simple,*

$$\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z). \quad (27.2)$$

(ii) *If the pole has order $m \geq 1$,*

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]. \quad (27.3)$$

(iii) (Quotient rule.) *If $f = p/q$ with p, q analytic at z_0 , $p(z_0) \neq 0$, $q(z_0) = 0$ and $q'(z_0) \neq 0$ (so the pole is simple), then*

$$\operatorname{Res}\left(\frac{p}{q}, z_0\right) = \frac{p(z_0)}{q'(z_0)}. \quad (27.4)$$

Proof. (ii) By Theorem 26.5(ii), $h(z) := (z - z_0)^m f(z)$ extends analytically to z_0 , with Taylor expansion $h(z) = \sum_{j \geq 0} c_j (z - z_0)^j$ where, comparing with (26.1), $c_j = a_{j-m}$. We want $a_{-1} = c_{m-1}$, and by (23.9) $c_{m-1} = h^{(m-1)}(z_0)/(m-1)!$, which is (27.3).

(i) is (ii) with $m = 1$.

(iii) Since $q(z_0) = 0$ and $q'(z_0) \neq 0$, q has a simple zero, so p/q has a simple pole by Corollary 26.6. By (27.2),

$$\operatorname{Res}\left(\frac{p}{q}, z_0\right) = \lim_{z \rightarrow z_0} \frac{(z - z_0)p(z)}{q(z)} = \lim_{z \rightarrow z_0} \frac{p(z)}{\frac{q(z) - q(z_0)}{z - z_0}} = \frac{p(z_0)}{q'(z_0)}. \quad \square$$

Warning 27.3. Formulas (27.2)–(27.4) apply *only at poles*. At an essential singularity there is no shortcut: one must produce the Laurent series and read off a_{-1} , as in Example 25.7.

Example 27.4 (Residue computations). (a) $f(z) = \frac{z+1}{z^2-4}$ at $z_0 = 2$: simple pole, and by (27.4) with $p = z+1$, $q = z^2-4$, $q' = 2z$: $\text{Res} = \frac{3}{4}$.

(b) $f(z) = \frac{e^z}{(z-1)^3}$ at $z_0 = 1$: pole of order 3, so by (27.3), $\text{Res} = \frac{1}{2!} \lim_{z \rightarrow 1} \frac{d^2}{dz^2} e^z = \frac{e}{2}$.

(c) $f(z) = \cot z = \frac{\cos z}{\sin z}$ at $z_0 = k\pi$: by (27.4), $\text{Res} = \frac{\cos k\pi}{\cos k\pi} = 1$ for every integer k . This uniformity is what makes $\cot$ the summation kernel of Chapter 32.

(d) $f(z) = z^2 e^{1/z}$ at 0: essential. From (25.9), $z^2 e^{1/z} = \sum_{k \geq 0} \frac{z^{2-k}}{k!}$, and $2-k = -1$ when $k = 3$, so $\text{Res} = \frac{1}{3!} = \frac{1}{6}$.

(e) $f(z) = \frac{1}{z^2(z-1)}$ at 0: pole of order 2, so $\text{Res} = \lim_{z \rightarrow 0} \frac{d}{dz} \frac{1}{z-1} = \lim_{z \rightarrow 0} \frac{-1}{(z-1)^2} = -1$.

27.2 The residue theorem

Theorem 27.5 (Cauchy's residue theorem). *Let C be a simple closed contour traversed counterclockwise, let U be an open set containing C and all points inside C , and let f be analytic on U except for finitely many isolated singularities $z_1, \dots, z_n$, none of which lies on C . (This covers the usual situation: C in a simply connected region D , and f analytic on D except for finitely many isolated singularities, none on C .) Then*

$$\oint_C f(z) dz = 2\pi i \sum_{\substack{k=1 \\ z_k \text{ inside } C}}^n \text{Res}(f, z_k). \quad (27.5)$$

Proof. Relabel so that $z_1, \dots, z_p$ are the singularities inside C and the rest lie outside. Choose $\rho > 0$ small enough that the closed discs $\overline{D}(z_k, \rho)$, $k = 1, \dots, p$, are pairwise disjoint and all lie inside C ; let $C_k := C(z_k, \rho)$ counterclockwise. The function f is analytic on an open set containing C , every C_k , and all points inside C and outside every C_k , so the many-hole deformation theorem 20.10 gives

$$\oint_C f(z) dz = \sum_{k=1}^p \oint_{C_k} f(z) dz.$$

By the definition of the residue (27.1), each term equals $2\pi i \text{Res}(f, z_k)$. □

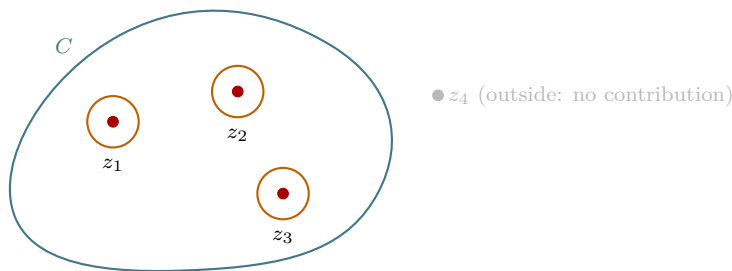


FIGURE 27.1. The residue theorem 27.5. Shrinking C onto small circles around each enclosed singularity — legitimate by the deformation theorem 20.10 — replaces one hard integral by a sum of local ones, and each of those is $2\pi i$ times a single Laurent coefficient. Singularities outside C contribute nothing.

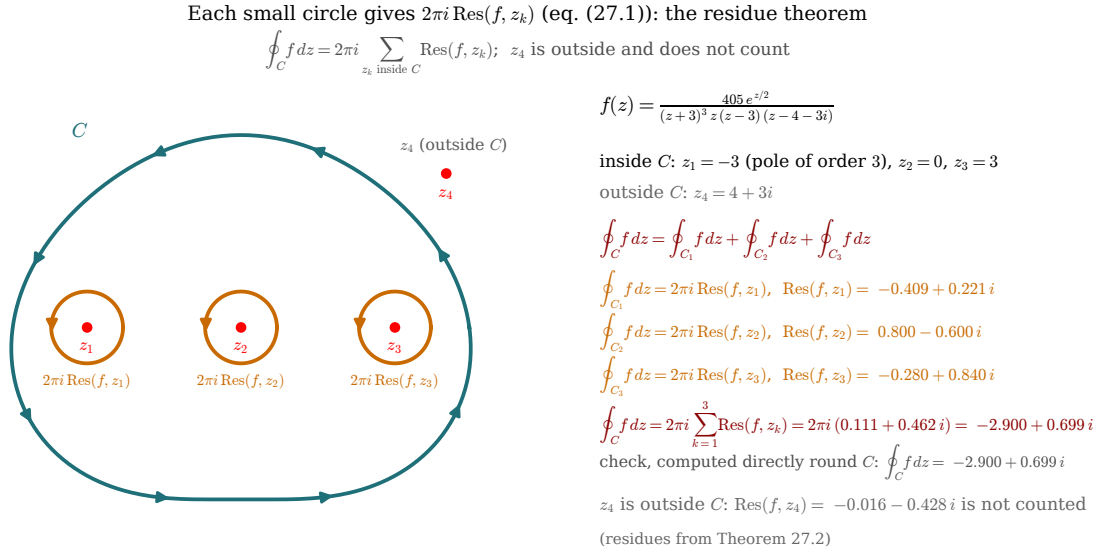


FIGURE 27.2 (animated). The proof of Theorem 27.5 for $f(z) = \frac{405 e^{z/2}}{(z+3)^3 z (z-3)(z-4-3i)}$, which has a pole of order 3 at $z_1 = -3$ and simple poles at $z_2 = 0$, $z_3 = 3$ inside C and at $z_4 = 4 + 3i$ outside. Small circles C_k are drawn about z_1, z_2, z_3 , and cuts along the real axis divide the region inside C and outside the circles into two pieces. No piece contains a singularity, so each integral round a piece is 0 (Theorem 20.9). When the pieces are put back, each cut is run once in each direction and cancels; what is left is C counterclockwise and the C_k clockwise, which is the many-hole deformation theorem 20.10. Each C_k gives $2\pi i \operatorname{Res}(f, z_k)$ by (27.1), with the residues computed by Theorem 27.2; their sum agrees with $\oint_C f dz$ computed directly. The pole z_4 outside C does not count.

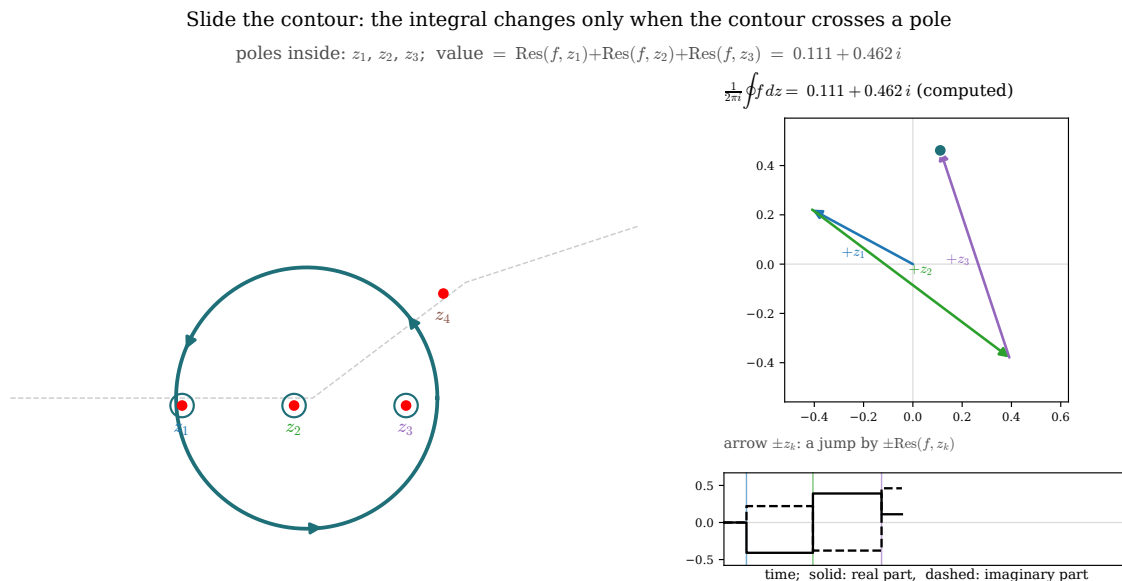


FIGURE 27.3 (animated). The residue theorem seen another way, for the same f . A circle slides across the plane, and $\frac{1}{2\pi i} \oint f dz$ round it is computed in every frame (right: as a point, and its real and imaginary parts against time). The value stays constant while no pole crosses the circle, and jumps by exactly $+\text{Res}(f, z_k)$ when z_k enters and $-\text{Res}(f, z_k)$ when it leaves: at every moment it is the sum of the residues inside, as (27.5) says.

Example 27.6. Evaluate $\oint_{|z|=3} \frac{4z-5}{z(z-1)^2} dz$. The singularities are 0 (simple) and 1 (double), both inside.

$$\text{Res}(f, 0) = \lim_{z \rightarrow 0} \frac{4z-5}{(z-1)^2} = -5, \quad \text{Res}(f, 1) = \lim_{z \rightarrow 1} \frac{d}{dz} \frac{4z-5}{z} = \lim_{z \rightarrow 1} \frac{5}{z^2} = 5.$$

By (27.5) the integral is $2\pi i(-5 + 5) = 0$.

Evaluate $\oint_{|z|=2} \tan z dz$. Here $\tan z = \frac{\sin z}{\cos z}$ has simple poles where $\cos z = 0$, i.e. $z = \pm\pi/2$ (both inside $|z| = 2$, since $\pi/2 \approx 1.571$), with residues $\frac{\sin z}{-\sin z} = -1$ each by (27.4). So the integral is $2\pi i(-1 - 1) = -4\pi i$.

Evaluate $\oint_{|z|=1} \frac{e^{1/z}}{z} dz$. The only singularity is the essential one at 0. Multiplying (25.9) by $1/z$, $\frac{e^{1/z}}{z} = \sum_{k \geq 0} \frac{1}{k! z^{k+1}}$, whose z^{-1} -coefficient is the $k = 0$ term, namely 1. So the value is $2\pi i$.

27.3 The winding number and the general form

For contours that are not simple, the correct statement counts how many times the contour encircles each singularity.

Definition 27.7 (Winding number). Let γ be a closed contour and $z_0 \in \mathbb{C}$ a point not on γ . The *winding number* (or *index*) of γ about z_0 is

$$\text{Ind}(\gamma; z_0) := \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z - z_0}. \quad (27.6)$$

Proposition 27.8 (The index is an integer). $\text{Ind}(\gamma; z_0) \in \mathbb{Z}$ for every closed contour γ and every $z_0 \notin \gamma$. It is constant on each connected component of $\mathbb{C} \setminus \gamma$ and is 0 on the unbounded component. For the circle traversed n times counterclockwise, $\gamma(t) = z_0 + \rho e^{it}$, $t \in [0, 2\pi n]$ ($n \geq 1$), the index about its centre z_0 is n .

Proof. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ and define

$$\varphi(t) := \int_a^t \frac{\gamma'(s)}{\gamma(s) - z_0} ds.$$

Then φ is continuous, piecewise differentiable, with $\varphi'(t) = \frac{\gamma'(t)}{\gamma(t) - z_0}$ where defined. Consider $\psi(t) := e^{-\varphi(t)}(\gamma(t) - z_0)$. Wherever differentiable,

$$\psi'(t) = e^{-\varphi}[-\varphi'(\gamma - z_0) + \gamma'] = e^{-\varphi}[-\gamma' + \gamma'] = 0,$$

so ψ is constant on $[a, b]$ (it is continuous and piecewise constant). Hence $\psi(b) = \psi(a)$, i.e.

$$e^{-\varphi(b)}(\gamma(b) - z_0) = e^{-\varphi(a)}(\gamma(a) - z_0) = \gamma(a) - z_0,$$

using $\varphi(a) = 0$. Since γ is closed, $\gamma(b) = \gamma(a)$, and $\gamma(a) - z_0 \neq 0$, so $e^{-\varphi(b)} = 1$ and therefore $\varphi(b) \in 2\pi i\mathbb{Z}$ by Theorem 16.3(ii). But $\varphi(b) = \oint_{\gamma} \frac{dz}{z - z_0} = 2\pi i \text{Ind}(\gamma; z_0)$.

Local constancy: $z_0 \mapsto \text{Ind}(\gamma; z_0)$ is continuous on $\mathbb{C} \setminus \gamma$ (differentiate under the integral, or estimate directly) and integer-valued, hence locally constant. On the unbounded component, the *ML*-inequality (19.15) gives $|\text{Ind}| \leq \frac{\text{length}(\gamma)}{2\pi \text{dist}(z_0, \gamma)} \rightarrow 0$ as $|z_0| \rightarrow \infty$, so the constant value is 0. For the circle traversed n times, $\oint_{\gamma} \frac{dz}{z - z_0} = \int_0^{2\pi n} \frac{i\rho e^{it}}{\rho e^{it}} dt = 2\pi i n$, so the index is n (for $n = 1$ this is (19.16)). $\square$

Theorem 27.9 (Residue theorem, general form). Let D be a simply connected region, f analytic on D except at isolated singularities $z_1, \dots, z_n$, and γ a closed contour in D avoiding all z_k . Then

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Ind}(\gamma; z_k) \text{Res}(f, z_k). \quad (27.7)$$

Proof. For each k let $P_k(z) := \sum_{j \geq 1} a_{-j}^{(k)}(z - z_k)^{-j}$ be the principal part of f at z_k , where f is analytic on $A(z_k; 0, R_k)$. It converges on $\mathbb{C} \setminus \{z_k\}$: by Theorem 25.3 it converges for $0 < |z - z_k| < R_k$, i.e. the power series $\sum_{j \geq 1} a_{-j}^{(k)} w^j$ converges for every w with $|w| > 1/R_k$, so by Theorem 23.10(i) its radius of convergence is ∞ . Hence P_k is the composition of an entire function of w (Theorem 23.11) with $w = 1/(z - z_k)$, and is analytic on $\mathbb{C} \setminus \{z_k\}$. Then $F := f - \sum_k P_k$ has removable singularities at every z_k and so extends analytically to D ; by Theorem 20.9, $\oint_{\gamma} F = 0$. Hence

$$\oint_{\gamma} f = \sum_{k=1}^n \oint_{\gamma} P_k.$$

We may integrate P_k term by term over γ : γ is compact and avoids z_k , so $\delta := \min_{z \in \gamma} |z - z_k| > 0$, and on γ we have $|w| = 1/|z - z_k| \leq 1/\delta$, where the series in w converges uniformly by Theorem 23.10(ii). (This matters when z_k is essential and P_k has infinitely many terms.) Term by term, $\oint_{\gamma} (z - z_k)^{-j} dz = 0$ for $j \geq 2$ because $(z - z_k)^{-j}$ has the antiderivative $\frac{(z - z_k)^{1-j}}{1-j}$ on $\mathbb{C} \setminus \{z_k\}$ and Theorem 20.2 applies to the closed contour γ ; and for $j = 1$ the integral is $2\pi i \text{Ind}(\gamma; z_k)$ by (27.6). Only $a_{-1}^{(k)} = \text{Res}(f, z_k)$ survives. $\square$

Example 27.10. Let γ be the circle $|z| = 2$ traversed *twice* clockwise. Then $\text{Ind}(\gamma; 0) = -2$, and

$$\oint_{\gamma} \frac{e^z}{z} dz = 2\pi i \cdot (-2) \cdot \text{Res}\left(\frac{e^z}{z}, 0\right) = 2\pi i \cdot (-2) \cdot 1 = -4\pi i.$$

Move z_0 : the index is constant on each region, jumps by ± 1 across γ
 z_0 in a region of index -1 : the ray turns once, clockwise

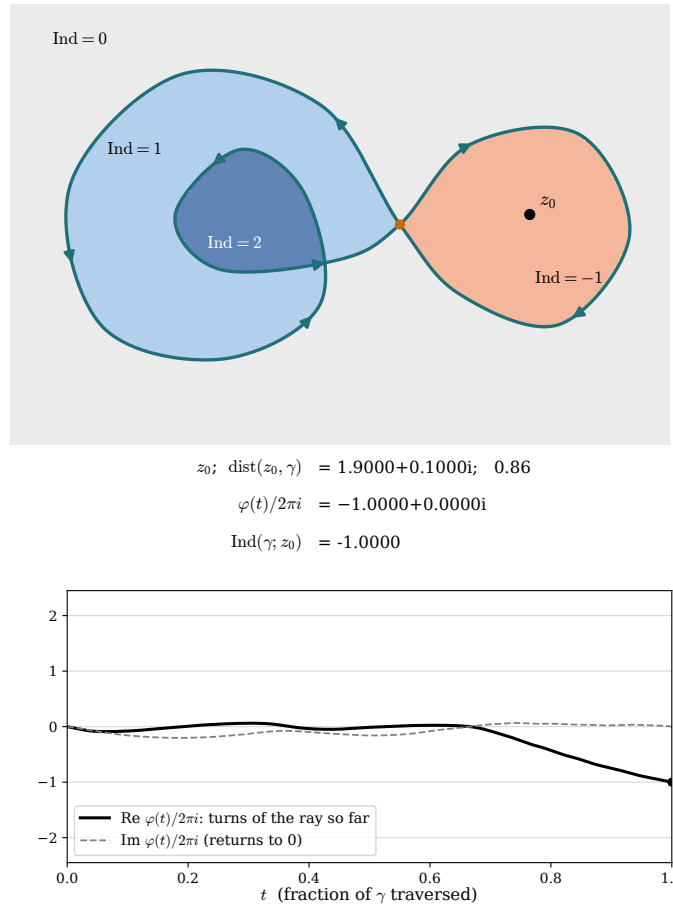


FIGURE 27.4 (animated). The index $\text{Ind}_{\gamma}(z_0)$ as a function of the point (Definition 27.7), for a closed curve with regions of index 2, 1, 0, -1 . As a point runs round γ , the ray from z_0 to $\gamma(t)$ turns, and below, $\varphi(t)/2\pi i$ from the proof of Proposition 27.8 is drawn: its real part counts the turns so far. Then z_0 moves across γ : the index is constant on each region and jumps by ± 1 at each crossing. Last, Theorem 27.9 for a function with poles in regions of index 2, -1 , 0: $\frac{1}{2\pi i} \oint_{\gamma} f dz = 1.5000 = 2 \cdot 1 + (-1) \cdot 0.5 + 0 \cdot 3$, each residue weighted by its index; and Example 27.10, $\oint e^z/z dz = -4\pi i$.

27.4 Exercises

Exercise 27.1. Compute all residues:

$$\frac{z}{z^2 + 1}, \quad \frac{e^z}{z^2(z - 2)}, \quad \frac{1}{z^4 + 1}, \quad \frac{\cos z}{z^5}, \quad z^3 e^{2/z}.$$

Exercise 27.2. Evaluate by residues, all contours counterclockwise:

$$\oint_{|z|=2} \frac{dz}{z^2-1}, \quad \oint_{|z|=4} \frac{z}{(z-1)(z-3)} dz, \quad \oint_{|z|=1} \frac{\sin z}{z^4} dz, \quad \oint_{|z|=1} \cot z dz.$$

Exercise 27.3. Let $f(z) = \frac{1}{z^5-1}$. Show $\text{Res}(f, \omega) = \frac{\omega}{5}$ at each fifth root of unity ω , and deduce $\sum_{\omega^5=1} \text{Res}(f, \omega) = 0$ using (4.6). Interpret the result via $\oint_{|z|=2} f$.

Exercise 27.4. ($\star$) (*Residue at infinity.*) Define $\text{Res}(f, \infty) := -\frac{1}{2\pi i} \oint_{|z|=R} f(z) dz$ for R large. Prove that if f is analytic on $\mathbb{C}$ except for finitely many singularities, then the sum of *all* residues, including the one at ∞ , is 0. Use it to evaluate $\oint_{|z|=5} \frac{z^3}{z^4-1} dz$ in one line.

Exercise 27.5. Give an example of f with $\text{Res}(f, z_0) = 0$ at a pole z_0 of order 2, and explain why this does not contradict Theorem 26.5.

Exercise 27.6. Let γ be the figure-eight contour that encircles 1 once counterclockwise and -1 once clockwise. Compute $\oint_{\gamma} \frac{dz}{z^2-1}$ using Theorem 27.9.

Exercise 27.7. Compute $\text{Res}(\frac{1}{\sin z}, k\pi)$ for every $k \in \mathbb{Z}$, and evaluate $\oint_{|z|=4} \csc z dz$.

Exercise 27.8. Let f have a pole of order 2 at z_0 . Show $\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} \frac{d}{dz} [(z - z_0)^2 f(z)]$, and use it to compute the residue of $\frac{e^z}{(z-1)^2(z-3)}$ at 1.

Exercise 27.9. ($\star$) Let f be analytic and nonzero at z_0 and let g have a simple zero there. Prove $\text{Res}(\frac{f}{g}, z_0) = \frac{f(z_0)}{g'(z_0)}$ and, more generally, that if g has a zero of order 2 then

$$\text{Res}\left(\frac{f}{g}, z_0\right) = \frac{2f'(z_0)}{g''(z_0)} - \frac{2f(z_0)g'''(z_0)}{3(g''(z_0))^2}.$$

(Expand both f and g in Taylor series about z_0 and divide.)

Chapter 28

Evaluation of Real Trigonometric Integrals

The residue theorem evaluates definite *real* integrals that elementary calculus cannot touch. This chapter treats the first and easiest family: integrals of a rational expression in $\cos \theta$ and $\sin \theta$ over a full period. The idea is that such an integral *is already* a contour integral over the unit circle, once one notices what $e^{i\theta}$ is doing.

28.1 The substitution

Theorem 28.1 (Trigonometric integrals by residues). *Let $F(x, y)$ be a rational function of two variables, with complex coefficients, whose denominator does not vanish on the unit circle $x^2 + y^2 = 1$. Then*

$$\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta = \oint_{|z|=1} F\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2i}\right) \frac{dz}{iz} = 2\pi i \sum_{|z_k|<1} \text{Res}(G, z_k), \quad (28.1)$$

where

$$G(z) := \frac{1}{iz} F\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2i}\right) \quad (28.2)$$

is a rational function of z , and the sum runs over its poles inside the unit circle.

Proof. Parametrise the unit circle by $z = e^{i\theta}$, $\theta \in [0, 2\pi]$, as in (19.8). Then $dz = ie^{i\theta} d\theta = iz d\theta$, so

$$d\theta = \frac{dz}{iz}, \quad (28.3)$$

and by (17.1) together with $z^{-1} = e^{-i\theta}$ (valid since $|z| = 1$, using (3.8)),

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + z^{-1}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{z - z^{-1}}{2i}. \quad (28.4)$$

Substituting (28.3) and (28.4) into the left side of (28.1) converts it into $\oint_{|z|=1} G(z) dz$ by Definition 19.7. The integrand G is rational in z : F is a quotient of polynomials in $\cos \theta, \sin \theta$, and both are Laurent polynomials in z . Its poles are the zeros of the denominator; by hypothesis none lies on $|z| = 1$, and the residue theorem 27.5 gives the last equality. $\square$

Remark 28.2 (Why a full period). The substitution requires θ to traverse $[0, 2\pi]$ so that z goes once around the circle. Integrals over $[0, \pi]$ or $[0, \pi/2]$ must first be extended to a full period using symmetry — see Example 28.6.

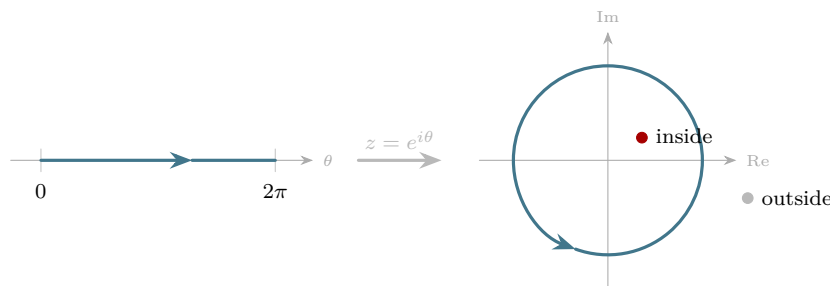


FIGURE 28.1. The substitution $z = e^{i\theta}$ of Theorem 28.1. As θ runs once over $[0, 2\pi]$ the point z runs once counterclockwise around the unit circle, so a real integral over a period is a contour integral. The rational integrand acquires poles; only those *inside* the circle contribute, which is what makes the method finite.

28.2 Worked examples

Example 28.3 (The basic integral). Evaluate $I = \int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$ for real $a > 1$.

By (28.1),

$$I = \oint_{|z|=1} \frac{1}{a + \frac{z+z^{-1}}{2}} \cdot \frac{dz}{iz} = \oint_{|z|=1} \frac{2 dz}{i(z^2 + 2az + 1)}.$$

The denominator vanishes at

$$z_{\pm} = -a \pm \sqrt{a^2 - 1},$$

both real and negative, with $z_+ z_- = 1$ (product of the roots). Since $a > 1$, $|z_-| = a + \sqrt{a^2 - 1} > 1$, so $|z_+| < 1$: exactly one pole is inside. It is simple, and by (27.4) with $p = \frac{2}{i}$, $q = z^2 + 2az + 1$, $q' = 2z + 2a$:

$$\text{Res}(G, z_+) = \frac{2/i}{2z_+ + 2a} = \frac{2/i}{2\sqrt{a^2 - 1}} = \frac{1}{i\sqrt{a^2 - 1}}.$$

Hence

$$\int_0^{2\pi} \frac{d\theta}{a + \cos \theta} = 2\pi i \cdot \frac{1}{i\sqrt{a^2 - 1}} = \frac{2\pi}{\sqrt{a^2 - 1}}, \quad a > 1. \quad (28.5)$$

Sanity checks: the answer is positive, as it must be since the integrand is; it blows up as $a \downarrow 1$, when the integrand develops a non-integrable singularity at $\theta = \pi$; and it behaves like $2\pi/a$ for large a , matching the crude approximation $\frac{1}{a + \cos \theta} \approx \frac{1}{a}$.

Example 28.4 (A higher-order pole). Evaluate $I = \int_0^{2\pi} \frac{d\theta}{(2 + \cos \theta)^2}$.

By the same substitution,

$$I = \oint_{|z|=1} \frac{1}{(2 + \frac{z+z^{-1}}{2})^2} \cdot \frac{dz}{iz} = \oint_{|z|=1} \frac{4z dz}{i(z^2 + 4z + 1)^2}.$$

The roots of $z^2 + 4z + 1$ are $z_{\pm} = -2 \pm \sqrt{3}$, with $|z_+| = 2 - \sqrt{3} \approx 0.268 < 1$ inside and $|z_-| = 2 + \sqrt{3} > 1$ outside. Inside, z_+ is a pole of order 2. Write $z^2 + 4z + 1 = (z - z_+)(z - z_-)$, so

$$G(z) = \frac{4z}{i(z - z_+)^2(z - z_-)^2}, \quad (z - z_+)^2 G(z) = \frac{4z}{i(z - z_-)^2}.$$

By (27.3) with $m = 2$,

$$\operatorname{Res}(G, z_+) = \frac{4}{i} \frac{d}{dz} \left[\frac{z}{(z - z_-)^2} \right]_{z=z_+},$$

and the derivative is

$$\frac{d}{dz} \frac{z}{(z - z_-)^2} = \frac{(z - z_-)^2 - z \cdot 2(z - z_-)}{(z - z_-)^4} = \frac{(z - z_-) - 2z}{(z - z_-)^3} = \frac{-z - z_-}{(z - z_-)^3}.$$

At $z = z_+$: numerator $-(z_+ + z_-) = 4$ (sum of roots is -4), and $z_+ - z_- = 2\sqrt{3}$, so the derivative equals $\frac{4}{(2\sqrt{3})^3} = \frac{4}{24\sqrt{3}} = \frac{1}{6\sqrt{3}}$. Therefore $\operatorname{Res}(G, z_+) = \frac{4}{i} \cdot \frac{1}{6\sqrt{3}} = \frac{2}{3i\sqrt{3}}$, and

$$I = 2\pi i \cdot \frac{2}{3i\sqrt{3}} = \frac{4\pi}{3\sqrt{3}} = \frac{4\pi\sqrt{3}}{9} \approx 2.4184. \quad (28.6)$$

Example 28.5 (Sine in the numerator). Evaluate $I = \int_0^{2\pi} \frac{\sin^2 \theta}{5 + 4 \cos \theta} d\theta$.

Here $\sin^2 \theta = \left(\frac{z - z^{-1}}{2i}\right)^2 = -\frac{(z^2 - 1)^2}{4z^2}$ and $5 + 4 \cos \theta = 5 + 2(z + z^{-1}) = \frac{2z^2 + 5z + 2}{z}$, so

$$G(z) = \frac{1}{iz} \cdot \frac{-\frac{(z^2 - 1)^2}{4z^2}}{\frac{2z^2 + 5z + 2}{z}} = \frac{-(z^2 - 1)^2}{4i z^2 (2z^2 + 5z + 2)}.$$

Now $2z^2 + 5z + 2 = (2z + 1)(z + 2)$, so the poles are $z = 0$ (order 2), $z = -\frac{1}{2}$ (simple, inside), and $z = -2$ (outside).

At $z = -\frac{1}{2}$, by (27.2), writing $2z^2 + 5z + 2 = 2(z + \frac{1}{2})(z + 2)$:

$$\operatorname{Res}(G, -\frac{1}{2}) = \frac{-\left(\frac{1}{4} - 1\right)^2}{4i \cdot \frac{1}{4} \cdot 2 \cdot \left(-\frac{1}{2} + 2\right)} = \frac{-\frac{9}{16}}{3i} = \frac{-3}{16i}.$$

At $z = 0$, order 2: with $H(z) := z^2 G(z) = \frac{-(z^2 - 1)^2}{4i(2z^2 + 5z + 2)}$,

$$H'(z) = \frac{-1}{4i} \cdot \frac{4z(z^2 - 1)(2z^2 + 5z + 2) - (z^2 - 1)^2(4z + 5)}{(2z^2 + 5z + 2)^2},$$

and at $z = 0$ the first term of the numerator vanishes, leaving $\frac{-1}{4i} \cdot \frac{-(1)(5)}{4} = \frac{5}{16i}$.

Summing, $I = 2\pi i \left(\frac{-3}{16i} + \frac{5}{16i}\right) = 2\pi i \cdot \frac{2}{16i} = \frac{\pi}{4}$.

Check. Since $\sin^2 \theta \leq 1$ and $5 + 4 \cos \theta \geq 1$, we have $0 \leq I \leq 2\pi$; and $\pi/4 \approx 0.785$ is plausible. A second check: expanding $\frac{1}{5 + 4 \cos \theta}$ in a Fourier series and using $\int_0^{2\pi} \sin^2 \theta d\theta = \pi$ gives the same value.

Example 28.6 (Half a period, by symmetry). Evaluate $J = \int_0^\pi \frac{d\theta}{3 + 2 \cos \theta}$. The integrand is invariant under $\theta \mapsto 2\pi - \theta$ (since $\cos$ is), so

$$\int_0^{2\pi} \frac{d\theta}{3 + 2 \cos \theta} = 2J.$$

Applying (28.5) with $a + \cos \theta$ replaced by $3 + 2 \cos \theta = 2(\frac{3}{2} + \cos \theta)$,

$$2J = \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{\frac{3}{2} + \cos \theta} = \frac{1}{2} \cdot \frac{2\pi}{\sqrt{\frac{9}{4} - 1}} = \frac{\pi}{\sqrt{5/2}} = \frac{2\pi}{\sqrt{5}},$$

so $J = \frac{\pi}{\sqrt{5}}$.

28.3 A general formula

Proposition 28.7. For real $a > |b| > 0$ and $n \geq 0$ an integer,

$$\int_0^{2\pi} \frac{\cos n\theta}{a + b \cos \theta} d\theta = \frac{2\pi}{\sqrt{a^2 - b^2}} \left(\frac{\sqrt{a^2 - b^2} - a}{b} \right)^n. \quad (28.7)$$

Proof. Write $\cos n\theta = \operatorname{Re}(z^n)$ on $|z| = 1$ and apply (28.1):

$$\int_0^{2\pi} \frac{e^{in\theta}}{a + b \cos \theta} d\theta = \oint_{|z|=1} \frac{z^n}{a + \frac{b}{2}(z + z^{-1})} \frac{dz}{iz} = \frac{2}{ib} \oint_{|z|=1} \frac{z^n dz}{z^2 + \frac{2a}{b}z + 1}.$$

The roots are $z_{\pm} = \frac{-a \pm \sqrt{a^2 - b^2}}{b}$ with $z_+ z_- = 1$, and $|z_+| < 1 < |z_-|$ because $a > |b|$. Only z_+ contributes, and by (27.4)

$$\operatorname{Res} = \frac{(2/ib)z_+^n}{2z_+ + 2a/b} = \frac{z_+^n}{i\sqrt{a^2 - b^2}}.$$

Multiplying by $2\pi i$ gives the right side of (28.7), which is real; taking real parts recovers the $\cos n\theta$ integral (the $\sin n\theta$ integral vanishes by oddness). $\square$

28.4 Exercises

Exercise 28.1. Evaluate:

$$\int_0^{2\pi} \frac{d\theta}{5 - 3 \sin \theta}, \quad \int_0^{2\pi} \frac{d\theta}{1 + \frac{1}{2} \cos \theta}, \quad \int_0^{2\pi} \frac{\cos \theta}{3 + \cos \theta} d\theta.$$

Exercise 28.2. Evaluate $\int_0^{2\pi} \frac{d\theta}{(5 - 4 \cos \theta)^2}$.

Exercise 28.3. Evaluate $\int_0^{\pi} \frac{\cos 2\theta}{5 - 4 \cos \theta} d\theta$ using (28.7) and a symmetry argument.

Exercise 28.4. ($\star$) Prove that for $|r| < 1$,

$$\int_0^{2\pi} \frac{1 - r^2}{1 - 2r \cos \theta + r^2} d\theta = 2\pi.$$

(This integrand is the Poisson kernel; the identity says it has total mass 2π , which is why it solves the Dirichlet problem for the disc.)

Exercise 28.5. Show that $\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta + c \sin \theta} = \frac{2\pi}{\sqrt{a^2 - b^2 - c^2}}$ when $a > \sqrt{b^2 + c^2} > 0$. (Write $b \cos \theta + c \sin \theta = \operatorname{Re}((b - ic)e^{i\theta})$ and rotate.)

Exercise 28.6. Explain what goes wrong in Theorem 28.1 if the denominator of F vanishes at some point of the unit circle, and illustrate with $\int_0^{2\pi} \frac{d\theta}{1 + \cos \theta}$.

Exercise 28.7. Evaluate $\int_0^{2\pi} \frac{\sin^2 \theta}{2 - \cos \theta} d\theta$ and $\int_0^{2\pi} \frac{d\theta}{3 + 2 \sin \theta}$.

Exercise 28.8. Show that $\int_0^{2\pi} \cos^{2n} \theta \, d\theta = \frac{2\pi}{4^n} \binom{2n}{n}$ by writing $\cos \theta = \frac{z+z^{-1}}{2}$ and picking out the coefficient of z^{-1} . Check the cases $n = 1, 2$ against elementary calculus.

Exercise 28.9. (★) Prove that for $0 < a < 1$,

$$\int_0^{2\pi} \ln(a^2 - 2a \cos \theta + 1) \, d\theta = 0.$$

(Recognise the integrand as $2 \ln |1 - ae^{i\theta}|$ and use the mean value property of the harmonic function $\ln |1 - az|$ on the disc $|z| < 1/a$, Exercise 15.8.) What is the value when $a > 1$?

Chapter 29

Evaluation of Real Improper Integrals

29.1 Principal values and the semicircular contour

Definition 29.1 (Cauchy principal value). Let $g : \mathbb{R} \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be integrable on every bounded interval. The *Cauchy principal value* of $\int_{-\infty}^{\infty} g$ is

$$\text{p.v.} \int_{-\infty}^{\infty} g(x) dx := \lim_{R \rightarrow \infty} \int_{-R}^R g(x) dx, \quad (29.1)$$

when the limit exists. If the improper integral converges in the ordinary sense — that is, if $\lim_{R_1 \rightarrow \infty} \int_{-R_1}^0$ and $\lim_{R_2 \rightarrow \infty} \int_0^{R_2}$ both exist separately — then the principal value exists and equals it. The converse is false: $\text{p.v.} \int_{-\infty}^{\infty} x dx = 0$, while the ordinary integral diverges.

More generally, let $x_1 < \dots < x_k$ be finitely many real points and let g be integrable on every bounded interval that stays a positive distance away from all of them. Then

$$\text{p.v.} \int_{-\infty}^{\infty} g(x) dx := \lim_{\substack{R \rightarrow \infty \\ \varepsilon \downarrow 0}} \int_{[-R, R] \setminus \bigcup_{j=1}^k (x_j - \varepsilon, x_j + \varepsilon)} g(x) dx,$$

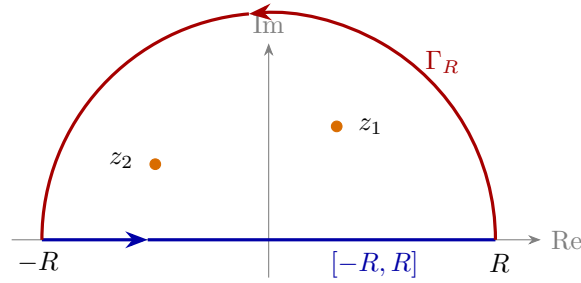
when the limit exists: the singular points are excised *symmetrically*. For instance $\text{p.v.} \int_{-1}^1 \frac{dx}{x} = 0$ in this sense, although $\int_0^1 \frac{dx}{x}$ diverges.

Convention 29.2. In the theorems below the ordinary convergence is guaranteed by decay hypotheses, so the principal value is the honest value; we still write p.v. where the distinction matters, as in Section 29.4.

Definition 29.3 (The standard contours). For $R > 0$ let

- $[-R, R]$ denote the segment of the real axis, left to right;
- Γ_R denote the upper semicircle $\Gamma_R(t) = Re^{it}$, $t \in [0, \pi]$.

Their concatenation $C_R := [-R, R] + \Gamma_R$ is a simple closed contour, counterclockwise, enclosing the part of the upper half-plane inside $|z| < R$.



29.2 Rational integrands

Lemma 29.4 (Vanishing of the large semicircle). *Let f be continuous on $\{|z| \geq R_0, \operatorname{Im} z \geq 0\}$ and suppose there are constants $A > 0$ and $p > 1$ with*

$$|f(z)| \leq \frac{A}{|z|^p} \quad \text{for } |z| \text{ large, } \operatorname{Im} z \geq 0. \quad (29.2)$$

Then $\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z) dz = 0$.

Proof. $\operatorname{length}(\Gamma_R) = \pi R$ and $|f| \leq AR^{-p}$ on Γ_R , so the *ML*-inequality (19.15) gives

$$\left| \int_{\Gamma_R} f \right| \leq \frac{A}{R^p} \cdot \pi R = \frac{\pi A}{R^{p-1}} \rightarrow 0$$

because $p - 1 > 0$. □

Theorem 29.5 (Rational improper integrals). *Let p, q be polynomials with real coefficients such that*

(a) q has no real zeros, and

(b) $\deg q \geq \deg p + 2$.

Then $\int_{-\infty}^{\infty} \frac{p(x)}{q(x)} dx$ converges and

$$\int_{-\infty}^{\infty} \frac{p(x)}{q(x)} dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}\left(\frac{p}{q}, z_k\right), \quad (29.3)$$

the sum being over the poles of p/q in the open upper half-plane.

Proof. Convergence: by (b) there are A, R_0 with $|p(x)/q(x)| \leq A|x|^{-2}$ for $|x| \geq R_0$ (Proposition 2.7), and $\int_{-\infty}^{\infty} x^{-2} dx$ converges; (a) makes the integrand continuous on $\mathbb{R}$.

Let R be large enough that all poles of p/q in the upper half-plane lie inside $|z| < R$; there are finitely many, being zeros of a polynomial. Apply the residue theorem 27.5 to C_R :

$$\int_{-R}^R \frac{p(x)}{q(x)} dx + \int_{\Gamma_R} \frac{p(z)}{q(z)} dz = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}\left(\frac{p}{q}, z_k\right).$$

(The first term is the contour integral over the real segment, which by (19.10) with $\gamma(t) = t$, $\gamma' = 1$, is the ordinary real integral.) By (b) and Lemma 29.4 (with exponent 2 in (29.2)) the second term tends to 0 as $R \rightarrow \infty$, and the right side does not depend on R . Letting $R \rightarrow \infty$ gives (29.3). □

Example 29.6. Evaluate $\int_{-\infty}^{\infty} \frac{dx}{x^4 + 1}$.

Here $\deg q - \deg p = 4 \geq 2$ and q has no real zeros. The poles are the fourth roots of -1 , namely $e^{i\pi/4}, e^{3i\pi/4}, e^{5i\pi/4}, e^{7i\pi/4}$ by Theorem 4.5; the first two lie in the upper half-plane. Each is a simple zero of $q(z) = z^4 + 1$, so by (27.4) with $q' = 4z^3$,

$$\text{Res} = \frac{1}{4z_k^3} = \frac{z_k}{4z_k^4} = \frac{z_k}{4(-1)} = -\frac{z_k}{4},$$

using $z_k^4 = -1$. Hence

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + 1} = 2\pi i \left(-\frac{e^{i\pi/4} + e^{3i\pi/4}}{4} \right) = -\frac{\pi i}{2} \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \right) = -\frac{\pi i}{2} \cdot i\sqrt{2} = \frac{\pi}{\sqrt{2}}.$$

The answer is a positive real number, as it must be.

Example 29.7. Evaluate $\int_0^{\infty} \frac{x^2}{(x^2 + 1)(x^2 + 4)} dx$. The integrand is even, so the integral equals half of the integral over $\mathbb{R}$. In the upper half-plane the poles are i and $2i$, both simple. By (27.2),

$$\text{Res}(f, i) = \frac{i^2}{(i + i)(i^2 + 4)} = \frac{-1}{2i \cdot 3} = \frac{-1}{6i}, \quad \text{Res}(f, 2i) = \frac{(2i)^2}{((2i)^2 + 1)(2i + 2i)} = \frac{-4}{(-3)(4i)} = \frac{1}{3i}.$$

So $\int_{-\infty}^{\infty} f = 2\pi i \left(-\frac{1}{6i} + \frac{1}{3i} \right) = 2\pi i \cdot \frac{1}{6i} = \frac{\pi}{3}$, and the required integral is $\pi/6$.

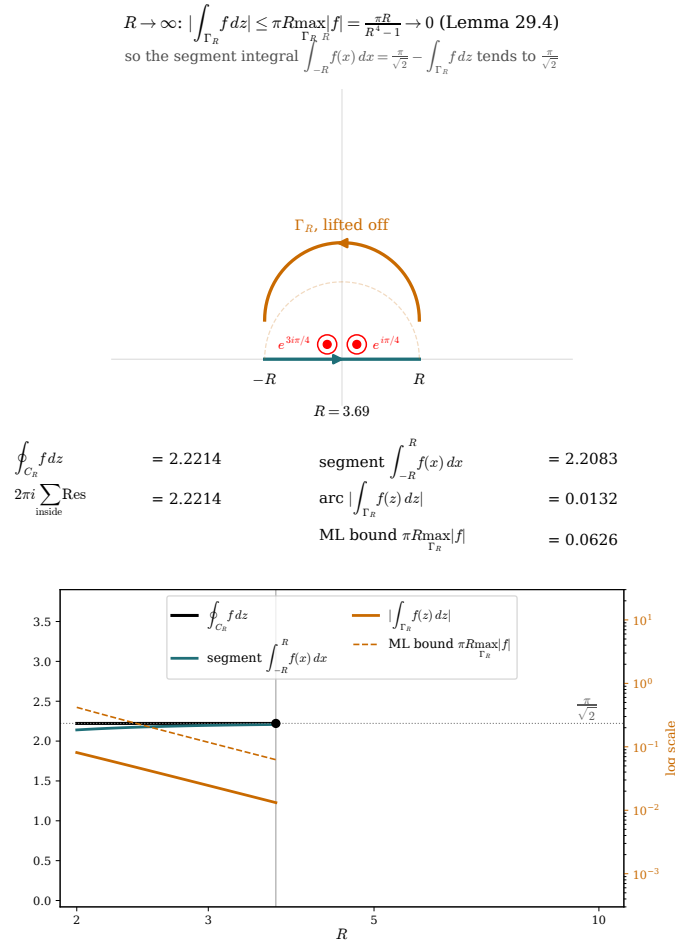


FIGURE 29.1 (animated). The proof of Theorem 29.5, in two parts. First, for $\frac{1}{z^4+1}$ (Example 29.6), $\frac{z^2}{(z^2+1)(z^2+4)}$ (Example 29.7) and $\frac{1}{(z^2+1)^2}$, the semicircular contour C_R grows past the poles in the upper half-plane. The closed integral $\oint_{C_R} f dz$ (below, against R) jumps by $2\pi i \text{Res}$ each time a pole enters and, once all of them are inside, stays constant however large R becomes. Then, for $\frac{1}{z^4+1}$, the arc Γ_R is lifted off: $\left| \int_{\Gamma_R} f dz \right|$ and its *ML* bound $\pi R \max_{\Gamma_R} |f|$ (Lemma 29.4; logarithmic scale on the right) fall to 0, so the segment integral $\int_{-R}^R f dx$ rises to the constant: in the limit only the real line is left, $\int_{-\infty}^{\infty} \frac{dx}{x^4+1} = \frac{\pi}{\sqrt{2}}$.

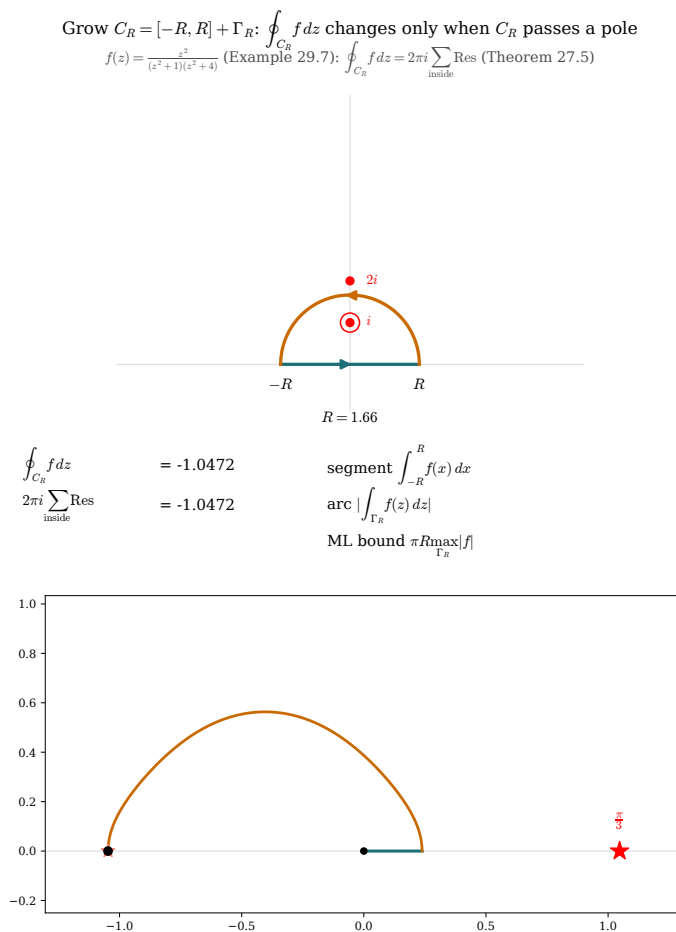


FIGURE 29.2 (animated). The same story with the running integral drawn as a path in its own plane, as z runs along $[-R, R]$ (teal) and back along Γ_R (orange). Once all poles are inside C_R , the path changes shape as R grows but always ends at the same point $2\pi i \sum \text{Res}$ (the star). When the arc is lifted off, its own running integral (inset, on a fixed scale) shrinks to a point, and the teal segment alone reaches the star.

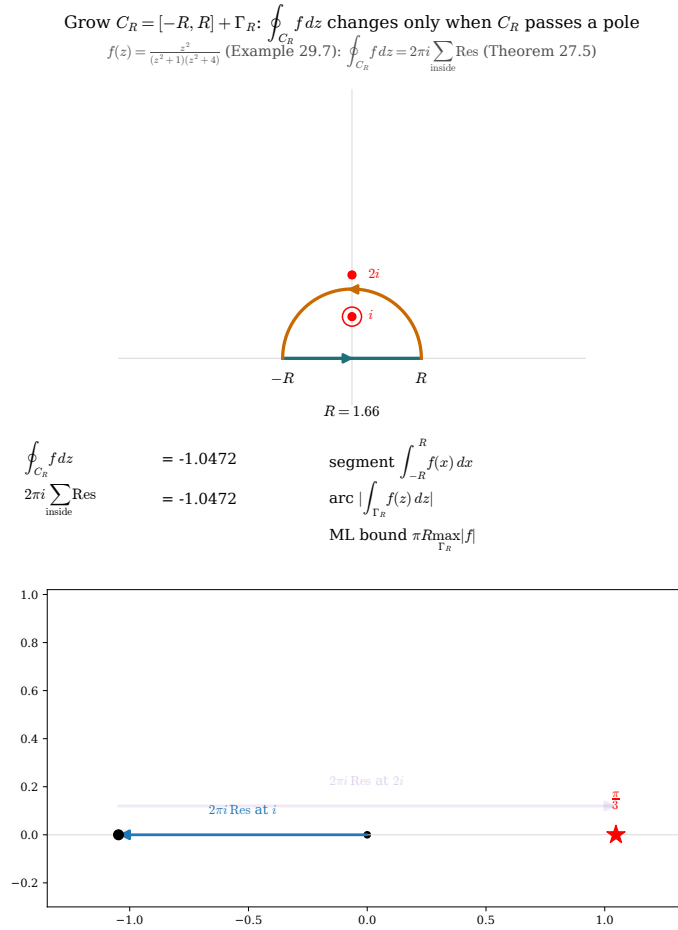


FIGURE 29.3 (animated). The same story with the residues as arrows: each pole that enters C_R adds its arrow $2\pi i \text{Res}(f, z_k)$, tip to tail, and the computed $\oint_{C_R} f dz$ sits at the tip. After the arc is lifted off it is drawn thick where $|f|$ is large, and below it the graph of $R|f(Re^{it})|$ over $[0, \pi]$: the area under the graph bounds $|\int_{\Gamma_R} f dz|$, the dashed rectangle is the ML bound, and both shrink to 0 as $R \rightarrow \infty$.

29.3 Fourier-type integrands and Jordan's lemma

For integrands like $\frac{\cos x}{x^2+1}$ the decay (29.2) fails on the semicircle: $|\cos z|$ grows like $e^{|\text{Im } z|}$ (Corollary 17.7). The fix is to work with e^{iaz} , whose modulus *decays* in the upper half-plane, and take real or imaginary parts at the end.

Lemma 29.8 (Jordan). *Let $a > 0$ and let f be continuous on $\{|z| \geq R_0, \text{Im } z \geq 0\}$ with*

$$M(R) := \max_{z \in \Gamma_R} |f(z)| \longrightarrow 0 \quad \text{as } R \rightarrow \infty. \quad (29.4)$$

Then

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z) e^{iaz} dz = 0. \quad (29.5)$$

Proof. On Γ_R write $z = Re^{i\theta}$, $\theta \in [0, \pi]$, so $dz = iRe^{i\theta} d\theta$ and, by (16.2),

$$|e^{iaz}| = e^{-a \text{Im } z} = e^{-aR \sin \theta}.$$

Hence by Lemma 19.3,

$$\left| \int_{\Gamma_R} f(z) e^{iaz} dz \right| \leq M(R) R \int_0^\pi e^{-aR \sin \theta} d\theta = 2M(R) R \int_0^{\pi/2} e^{-aR \sin \theta} d\theta,$$

by symmetry of $\sin$ about $\pi/2$. Now use *Jordan's inequality*: $\sin \theta \geq \frac{2\theta}{\pi}$ for $\theta \in [0, \pi/2]$, which holds because $\sin$ is concave there and the two sides agree at the endpoints. Therefore

$$\int_0^{\pi/2} e^{-aR \sin \theta} d\theta \leq \int_0^{\pi/2} e^{-2aR\theta/\pi} d\theta = \frac{\pi}{2aR} (1 - e^{-aR}) \leq \frac{\pi}{2aR}.$$

Combining,

$$\left| \int_{\Gamma_R} f(z) e^{iaz} dz \right| \leq 2M(R) R \cdot \frac{\pi}{2aR} = \frac{\pi M(R)}{a} \rightarrow 0. \quad \square$$

Note how weak the hypothesis (29.4) is: f need only tend to 0, with no rate. The factor e^{iaz} supplies the rest.

Theorem 29.9 (Fourier-type integrals). *Let $a > 0$ and let f be analytic on $\mathbb{C}$ except for finitely many poles, none on the real axis, with $|f(z)| \rightarrow 0$ as $|z| \rightarrow \infty$ in the closed upper half-plane, i.e.*

$$M^*(R) := \sup\{|f(z)| : |z| \geq R, \operatorname{Im} z \geq 0\} \rightarrow 0 \quad (R \rightarrow \infty).$$

Then the improper integral $\int_{-\infty}^{\infty} f(x) e^{iax} dx$ converges (in the ordinary sense) and

$$\int_{-\infty}^{\infty} f(x) e^{iax} dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}(f(z) e^{iaz}, z_k). \quad (29.6)$$

If moreover f is real-valued on $\mathbb{R}$, then

$$\int_{-\infty}^{\infty} f(x) \cos ax dx = \operatorname{Re}(RHS), \quad \int_{-\infty}^{\infty} f(x) \sin ax dx = \operatorname{Im}(RHS). \quad (29.7)$$

Proof. Convergence. Let R_1 exceed the modulus of every pole of f . For $R_1 \leq R < R'$ and $Y > 0$ integrate $f(z) e^{iaz}$ around the boundary of the rectangle with vertices $R, R', R' + iY, R + iY$, counterclockwise. The rectangle lies in the half-plane $\{\operatorname{Re} z > R_1\}$ where there are no poles; this half-plane is convex, hence star-shaped, so the integral is 0 by Theorem 20.8. On the two vertical sides $z = R + iy$ and $z = R' + iy$ ($0 \leq y \leq Y$) we have $|f(z) e^{iaz}| \leq M^*(R) e^{-ay}$, so by Lemma 19.3 each contributes at most $M^*(R) \int_0^Y e^{-ay} dy \leq M^*(R)/a$; on the top side $|f(z) e^{iaz}| \leq M^*(R) e^{-aY}$, so it contributes at most $(R' - R) M^*(R) e^{-aY}$. Hence

$$\left| \int_R^{R'} f(x) e^{iax} dx \right| \leq \frac{2M^*(R)}{a} + (R' - R) M^*(R) e^{-aY} \xrightarrow{Y \rightarrow \infty} \frac{2M^*(R)}{a},$$

which tends to 0 as $R \rightarrow \infty$, uniformly in $R' > R$. By the Cauchy criterion, $\lim_{R_2 \rightarrow \infty} \int_0^{R_2} f(x) e^{iax} dx$ exists. The rectangle with vertices $-R', -R, -R + iY, -R' + iY$ (in the convex half-plane $\{\operatorname{Re} z < -R_1\}$) gives the same bound for $\int_{-R'}^{-R}$, so $\int_{-\infty}^0$ converges too. (Note that f is continuous on $\mathbb{R}$, having no poles there.)

Value. Since the integral converges, it equals the limit of $\int_{-R}^R$. Apply the residue theorem to $f(z) e^{iaz}$ on C_R and let $R \rightarrow \infty$, using Lemma 29.8 to kill the arc (its hypothesis holds since $\max_{\Gamma_R} |f| \leq M^*(R)$).

Real and imaginary parts. If f is real on $\mathbb{R}$, then $f(x) \cos ax$ and $f(x) \sin ax$ are the real and imaginary parts of $f(x) e^{iax} = f(x) \cos ax + i f(x) \sin ax$, and (29.7) follows by taking real and imaginary parts of (29.6). Without that hypothesis (29.7) can fail: for $f(z) = \frac{1}{z+i}$, which has no poles in the upper half-plane, the right side of (29.6) is 0, yet $\int_{-\infty}^{\infty} \frac{\cos x}{x+i} dx = -\frac{i\pi}{e} \neq 0$. $\square$

Example 29.10. Evaluate $\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx$.

Take $f(z) = \frac{1}{z^2 + 1}$, $a = 1$. In the upper half-plane the only pole is $z = i$, simple, and by (27.4)

$$\operatorname{Res}\left(\frac{e^{iz}}{z^2 + 1}, i\right) = \frac{e^{i \cdot i}}{2i} = \frac{e^{-1}}{2i}.$$

By (29.6), $\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + 1} dx = 2\pi i \cdot \frac{e^{-1}}{2i} = \frac{\pi}{e}$, which is real; taking real and imaginary parts,

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx = \frac{\pi}{e}, \quad \int_{-\infty}^{\infty} \frac{\sin x}{x^2 + 1} dx = 0,$$

the second also being evident from oddness.

29.4 Indented contours and simple poles on the real axis

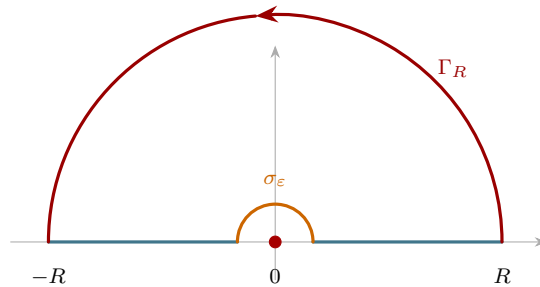


FIGURE 29.4. The indented contour of Example 29.12. The pole sitting *on* the path of integration is stepped over by a small semicircle σ_ε , traversed clockwise; by Lemma 29.11 it contributes $-\pi i$ times the residue — exactly half of a full circuit, with the sign of the direction taken. The large arc dies by Jordan's Lemma 29.8, and what remains is the real integral.

Lemma 29.11 (Small arc at a simple pole). *Let f have a simple pole at $x_0 \in \mathbb{R}$ with residue $\operatorname{Res}(f, x_0)$, and for $\varepsilon > 0$ let σ_ε be the semicircular arc $x_0 + \varepsilon e^{it}$, t from π to 0 (i.e. traversed clockwise, indenting above the pole). Then*

$$\lim_{\varepsilon \downarrow 0} \int_{\sigma_\varepsilon} f(z) dz = -\pi i \operatorname{Res}(f, x_0). \quad (29.8)$$

More generally, over an arc of opening angle α traversed counterclockwise the limit is $i\alpha \operatorname{Res}(f, x_0)$.

Proof. Write $f(z) = \frac{c}{z - x_0} + h(z)$ with $c = \operatorname{Res}(f, x_0)$ and h analytic near x_0 (Theorem 26.5). For the second piece, h is bounded near x_0 , say $|h| \leq M$, and $\operatorname{length}(\sigma_\varepsilon) = \pi\varepsilon$, so by (19.15) its contribution is at most $\pi M\varepsilon \rightarrow 0$. For the first piece, parametrise $z = x_0 + \varepsilon e^{it}$ with t running from π to 0 ; then $dz = i\varepsilon e^{it} dt$ and

$$\int_{\sigma_\varepsilon} \frac{c dz}{z - x_0} = \int_{\pi}^0 \frac{c i \varepsilon e^{it}}{\varepsilon e^{it}} dt = ci \int_{\pi}^0 dt = -\pi ic,$$

independently of ε . The general-angle statement is the same computation with limits 0 to α . $\square$

Example 29.12 (The Dirichlet integral). Prove

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}. \quad (29.9)$$

The function $\frac{\sin z}{z}$ has a removable singularity at 0, so we cannot see a residue; the trick is to work with $\frac{e^{iz}}{z}$, which has a *simple pole* at 0 with residue 1, and to indent.

Let $0 < \varepsilon < R$ and let C be the closed contour consisting of

$$[-R, -\varepsilon], \quad \sigma_\varepsilon \text{ (clockwise upper semicircle of radius } \varepsilon \text{ about } 0), \quad [\varepsilon, R], \quad \Gamma_R.$$

The integrand $\frac{e^{iz}}{z}$ is analytic inside and on C — the pole at 0 has been excluded by the indentation. More precisely, it is analytic on $D := \mathbb{C} \setminus \{iy : y \leq 0\}$, which contains C and is star-shaped about i (the segment from i to any $z \in D$ meets the imaginary axis only at i , or runs along the positive imaginary axis), so Theorem 20.8 gives

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx + \int_{\sigma_\varepsilon} \frac{e^{iz}}{z} dz + \int_{\varepsilon}^R \frac{e^{ix}}{x} dx + \int_{\Gamma_R} \frac{e^{iz}}{z} dz = 0. \quad (29.10)$$

As $R \rightarrow \infty$ the last term vanishes by Jordan's Lemma 29.8, since $|1/z| = 1/R \rightarrow 0$. As $\varepsilon \downarrow 0$ the second term tends to $-\pi i \cdot 1$ by (29.8). Combining the two real integrals by substituting $x \mapsto -x$ in the first,

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx + \int_{\varepsilon}^R \frac{e^{ix}}{x} dx = \int_{\varepsilon}^R \frac{e^{ix} - e^{-ix}}{x} dx = 2i \int_{\varepsilon}^R \frac{\sin x}{x} dx,$$

using (17.2). Therefore (29.10) becomes, in the limit,

$$2i \int_0^\infty \frac{\sin x}{x} dx - \pi i = 0,$$

which is (29.9). The *symmetric* excision $[-R, -\varepsilon] \cup [\varepsilon, R]$ was essential: each of $\int_{\varepsilon}^R \frac{e^{ix}}{x} dx$ and $\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx$ diverges as $\varepsilon \downarrow 0$ (their real parts involve $\frac{\cos x}{x}$, which is not integrable at 0), and only the sum of the two halves produces the integrable function $\frac{\sin x}{x}$. (Separately, the integral in (29.9) converges only conditionally at ∞ : $\int_0^\infty \frac{|\sin x|}{x} dx = \infty$.)

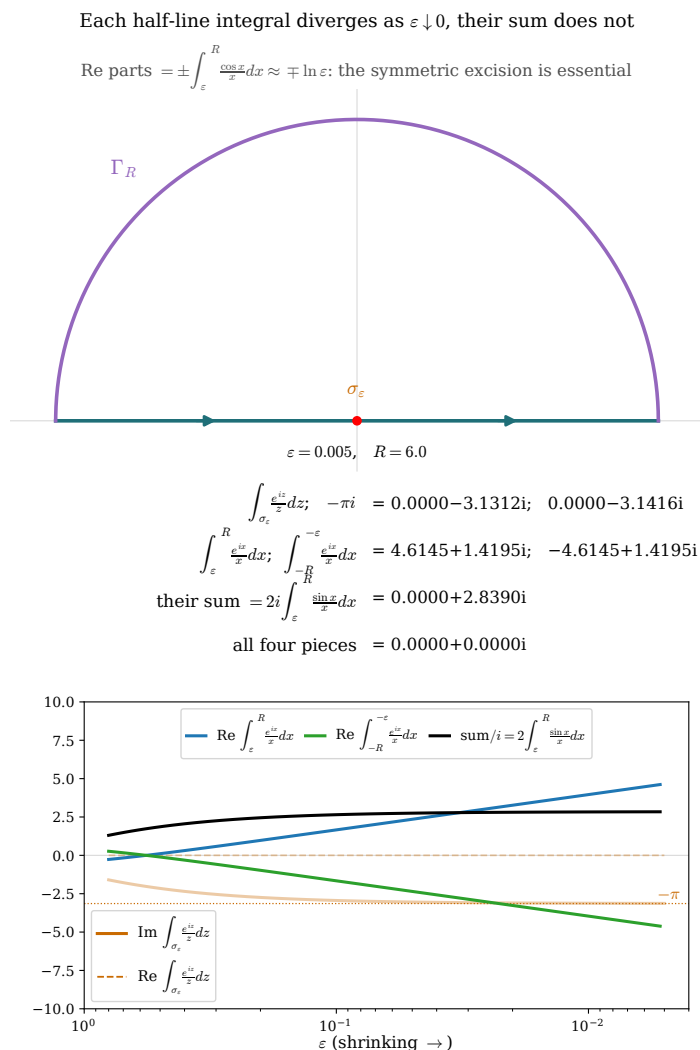


FIGURE 29.5 (animated). The indented contour of Example 29.12 for e^{iz}/z . First, with $\varepsilon = 0.8$ and $R = 6$, the running integral round the contour traces a path that closes up at 0 (Cauchy's theorem). As $\varepsilon \downarrow 0$, the small arc gives $-\pi i$, half a clockwise turn (Lemma 29.11); each half-line integral diverges like $\mp \ln \varepsilon$, but their sum $2i \int_{\varepsilon}^R \frac{\sin x}{x} dx$ stays bounded. Then R grows to 100: the big arc dies under Jordan's bound π/R (Lemma 29.8) and $\int_0^R \frac{\sin x}{x} dx$ oscillates into $\frac{\pi}{2}$.

Example 29.13 (A principal value). Evaluate p.v. $\int_{-\infty}^{\infty} \frac{dx}{x^2 - 1}$. The integrand has simple poles at ± 1 on the contour, with $\text{Res}(f, 1) = \frac{1}{2}$ and $\text{Res}(f, -1) = -\frac{1}{2}$ by (27.4). Indenting above both and closing with Γ_R (which contributes 0 by Lemma 29.4), and noting there are no poles in the open upper half-plane,

$$\text{p.v.} \int_{-\infty}^{\infty} \frac{dx}{x^2 - 1} - \pi i \left(\frac{1}{2} \right) - \pi i \left(-\frac{1}{2} \right) = 0,$$

so the principal value (in the sense of Definition 29.1, excising $(\pm 1 - \varepsilon, \pm 1 + \varepsilon)$) is 0: the two half-residue contributions cancel because the residues at ± 1 are opposite. A direct check: $F(x) =$

$\frac{1}{2} \ln \left| \frac{x-1}{x+1} \right|$ is an antiderivative on each of the three intervals, $F(\pm R) \rightarrow 0$ as $R \rightarrow \infty$, and

$$\int_{[-R, R] \setminus ((-1-\varepsilon, -1+\varepsilon) \cup (1-\varepsilon, 1+\varepsilon))} \frac{dx}{x^2 - 1} = F(R) - F(-R) + \ln \frac{2+\varepsilon}{2-\varepsilon} \rightarrow 0.$$

29.5 Exercises

Exercise 29.1. Evaluate:

$$\int_{-\infty}^{\infty} \frac{dx}{x^2 + 4}, \quad \int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)^2}, \quad \int_0^{\infty} \frac{x^2 dx}{x^6 + 1}, \quad \int_{-\infty}^{\infty} \frac{x^2 - 1}{x^4 + 1} dx.$$

Exercise 29.2. Evaluate $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 4} dx$ and $\int_0^{\infty} \frac{\cos 3x}{(x^2 + 1)^2} dx$.

Exercise 29.3. Show $\int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$. (Write $\sin^2 x = \frac{1 - \cos 2x}{2}$ and indent at 0.)

Exercise 29.4. (*) Prove $\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2 + 1)} dx = \pi \left(1 - \frac{1}{e}\right)$. (The integrand has a removable singularity at 0 and the integral converges absolutely; integrate $\frac{e^{iz}}{z(z^2 + 1)}$ over a contour indented at 0.)

Exercise 29.5. Explain why Theorem 29.5 fails for $\int_{-\infty}^{\infty} \frac{x}{x^2 + 1} dx$, and compute the principal value.

Exercise 29.6. Let $a < 0$, and assume now that $|f(z)| \rightarrow 0$ as $|z| \rightarrow \infty$ in the closed lower half-plane (the other hypotheses being as in Theorem 29.9). Show that closing in the lower half-plane gives $\int_{-\infty}^{\infty} f(x)e^{iax} dx = -2\pi i \sum_{\text{Im } z_k < 0} \text{Res}(f(z)e^{iaz}, z_k)$, and explain the sign.

Exercise 29.7. Evaluate $\int_0^{\infty} \frac{dx}{x^4 + x^2 + 1}$ and $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2 + 1)(x^2 + 9)}$.

Exercise 29.8. Prove Jordan's inequality $\sin \theta \geq \frac{2\theta}{\pi}$ on $[0, \pi/2]$ carefully, by showing $\theta \mapsto \sin \theta / \theta$ is decreasing there. Where exactly is it used in Lemma 29.8, and what would fail without it?

Exercise 29.9. (*) (A Fresnel integral.) By integrating e^{iz^2} around the circular sector $\{0 \leq |z| \leq R, 0 \leq \text{Arg } z \leq \pi/4\}$ and letting $R \rightarrow \infty$, prove

$$\int_0^{\infty} \cos(x^2) dx = \int_0^{\infty} \sin(x^2) dx = \frac{1}{2} \sqrt{\frac{\pi}{2}}.$$

(You will need $\int_0^{\infty} e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$ and a Jordan-type estimate on the arc.)

Chapter 30

Integration along a Branch Cut

Integrals such as $\int_0^\infty \frac{x^{a-1}}{1+x} dx$ involve a non-integer power, hence a branch point at 0 and a branch cut (Chapter 10). The residue theorem still applies, provided the contour is chosen to stay on one side of the cut. The resulting *keyhole* contour is the standard device, and the mechanism by which it works is worth stating plainly: the integral along the top of the cut and the integral along the bottom do *not* cancel, because the integrand jumps across the cut, and the surviving difference is the integral we want.

30.1 The keyhole contour

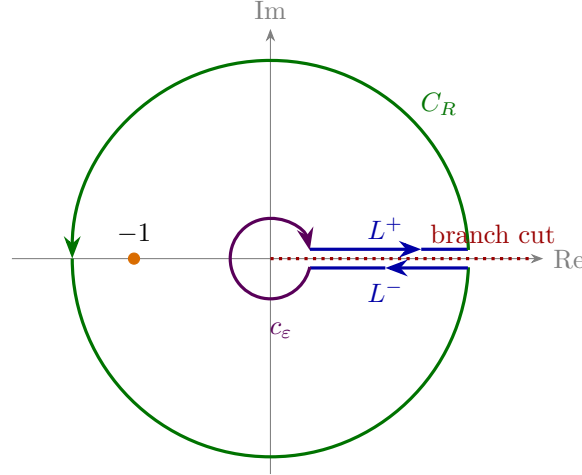
Definition 30.1 (Keyhole contour). Fix $0 < \varepsilon < R$. Cut the plane along the non-negative real axis $[0, \infty)$ and use the branch of z^{a-1} with argument in $(0, 2\pi)$:

$$z^{a-1} := e^{(a-1)(\ln|z|+i\theta)}, \quad \theta = \arg_0 z \in (0, 2\pi), \quad (30.1)$$

in the notation of (10.4) with $\tau = 0$. The *keyhole contour* $K_{\varepsilon,R}$ consists of

- L^+ : the segment from ε to R just *above* the cut, where $\theta = 0^+$;
- C_R : the circle $|z| = R$ counterclockwise, from $\theta = 0^+$ round to $\theta = 2\pi^-$;
- L^- : the segment from R back to ε just *below* the cut, where $\theta = 2\pi^-$;
- c_ε : the circle $|z| = \varepsilon$ clockwise, back to the start.

The values of z^{a-1} “on” $L^\pm$ are the limits of (30.1) from above and from below the cut. Since $K_{\varepsilon,R}$ runs along the cut, where the branch is not analytic, integrals over it are understood as limits of integrals over nearby contours that avoid the cut; this is made precise in the proof of Theorem 30.3.



Lemma 30.2 (The jump across the cut). *With the branch (30.1), for $x > 0$*

$$z^{a-1}|_{L^+} = x^{a-1}, \quad z^{a-1}|_{L^-} = x^{a-1}e^{2\pi i(a-1)} = x^{a-1}e^{2\pi ia}. \quad (30.2)$$

Proof. On L^+ we have $|z| = x$ and $\theta \rightarrow 0^+$, so (30.1) gives $e^{(a-1)\ln x} = x^{a-1}$ (the ordinary real power, since $x > 0$). On L^- , $\theta \rightarrow 2\pi^-$, so we pick up the extra factor $e^{(a-1)2\pi i}$, and $e^{-2\pi i} = 1$ turns it into $e^{2\pi ia}$. $\square$

For an integrand $z^{a-1}g(z)$ with g continuous across the cut, the two segments therefore contribute $(1 - e^{2\pi ia}) \int_\varepsilon^R x^{a-1}g(x) dx$ rather than 0: this is the whole mechanism.

30.2 The model integral

Theorem 30.3. *Let $a \in \mathbb{R}$ with $0 < a < 1$. Then*

$$\int_0^\infty \frac{x^{a-1}}{1+x} dx = \frac{\pi}{\sin \pi a}. \quad (30.3)$$

Proof. Let $f(z) = \frac{z^{a-1}}{1+z}$ with the branch (30.1). Its only singularity off the cut is a simple pole at $z = -1$, which in our branch has $|-1| = 1$ and $\theta = \pi$, so by (30.1) and (27.2)

$$\text{Res}(f, -1) = \lim_{z \rightarrow -1} z^{a-1} = e^{(a-1)(\ln 1 + i\pi)} = e^{i\pi(a-1)} = -e^{i\pi a}. \quad (30.4)$$

Fix $\varepsilon < 1 < R$. The function f is analytic on the slit plane $D := \mathbb{C} \setminus [0, \infty)$ except for the pole at -1 , and D is simply connected (it is star-shaped about -1 : a segment from -1 to a point of D meets the real axis only at -1 , or stays in $(-\infty, 0)$). The keyhole itself touches the cut, so we first work with a nearby contour. For $0 < \delta < \varepsilon$ put $x_r := \sqrt{r^2 - \delta^2}$ and let K^δ consist of the horizontal segment L_δ^+ from $x_\varepsilon + i\delta$ to $x_R + i\delta$, the arc C_R^δ of $|z| = R$ counterclockwise from $x_R + i\delta$ to $x_R - i\delta$, the segment L_δ^- from $x_R - i\delta$ to $x_\varepsilon - i\delta$, and the arc C_ε^δ of $|z| = \varepsilon$ clockwise from $x_\varepsilon - i\delta$ back to $x_\varepsilon + i\delta$. This is a simple closed counterclockwise contour in D with -1 inside, so the residue theorem 27.5 gives

$$\int_{L_\delta^+} f + \int_{C_R^\delta} f + \int_{L_\delta^-} f + \int_{C_\varepsilon^\delta} f = 2\pi i \text{Res}(f, -1).$$

Now let $\delta \downarrow 0$. Formula (30.1) with $\theta \in [0, \pi]$ defines a continuous function on the compact set $\{\frac{\varepsilon}{2} \leq |z| \leq 2R, \operatorname{Im} z \geq 0\}$, equal to z^{a-1} off the real axis and to x^{a-1} at $x > 0$; similarly with $\theta \in [\pi, 2\pi]$ on the lower half, with value $x^{a-1}e^{2\pi ia}$ at $x > 0$. By uniform continuity on these compact sets, $f(x \pm i\delta)$ converges uniformly on $[\frac{\varepsilon}{2}, 2R]$ to the boundary values of Lemma 30.2 divided by $1+x$, and $x_\varepsilon \rightarrow \varepsilon$, $x_R \rightarrow R$; so $\int_{L^\pm_\delta} f$ converge to the integrals over $L^\pm$ computed below. The arc integrals converge as well: C_R^δ is $\theta \mapsto Re^{i\theta}$ for $\theta \in [\theta_\delta, 2\pi - \theta_\delta]$ with $\theta_\delta \rightarrow 0$, and $\theta \mapsto f(Re^{i\theta})$ is continuous and bounded on $(0, 2\pi)$, so $\int_{C_R^\delta} f$ tends to $\int_0^{2\pi} f(Re^{i\theta}) iRe^{i\theta} d\theta$, which is what $\int_{C_R} f$ means; likewise for c_ε^δ and c_ε . The bounds used below for the circles are of the form $\max |f| \times \text{length}$. With this meaning,

$$\int_{L^+} f + \int_{C_R} f + \int_{L^-} f + \int_{c_\varepsilon} f = 2\pi i \operatorname{Res}(f, -1) = -2\pi i e^{i\pi a}. \quad (30.5)$$

The big circle. On C_R , $|z^{a-1}| = |e^{(a-1)\ln R}| |e^{i(a-1)\theta}| = R^{a-1}$ by (30.1), since $a-1$ is real (so no exponential factor), and $|1+z| \geq R-1$, so by the *ML*-inequality (19.15)

$$\left| \int_{C_R} f \right| \leq \frac{R^{a-1}}{R-1} \cdot 2\pi R = \frac{2\pi R^a}{R-1} \rightarrow 0 \quad \text{as } R \rightarrow \infty,$$

because $a < 1$.

The small circle. On c_ε , $|z^{a-1}| = \varepsilon^{a-1}$ and $|1+z| \geq 1-\varepsilon$, so

$$\left| \int_{c_\varepsilon} f \right| \leq \frac{\varepsilon^{a-1}}{1-\varepsilon} \cdot 2\pi \varepsilon = \frac{2\pi \varepsilon^a}{1-\varepsilon} \rightarrow 0 \quad \text{as } \varepsilon \downarrow 0,$$

because $a > 0$. Both inequalities $0 < a < 1$ are used, one at each end — this is why the hypothesis is exactly $0 < a < 1$.

The segments. By Lemma 30.2, and remembering that L^- is traversed from R to ε so contributes with a minus sign (19.12),

$$\int_{L^+} f + \int_{L^-} f = (1 - e^{2\pi ia}) \int_\varepsilon^R \frac{x^{a-1}}{1+x} dx.$$

Letting $\varepsilon \downarrow 0$ and $R \rightarrow \infty$ in (30.5),

$$(1 - e^{2\pi ia}) \int_0^\infty \frac{x^{a-1}}{1+x} dx = -2\pi i e^{i\pi a}.$$

Divide, and simplify by factoring $e^{i\pi a}$ out of the bracket:

$$\int_0^\infty \frac{x^{a-1}}{1+x} dx = \frac{-2\pi i e^{i\pi a}}{1 - e^{2\pi ia}} = \frac{-2\pi i e^{i\pi a}}{e^{i\pi a}(e^{-i\pi a} - e^{i\pi a})} = \frac{-2\pi i}{-2i \sin \pi a} = \frac{\pi}{\sin \pi a},$$

where $e^{i\pi a} - e^{-i\pi a} = 2i \sin \pi a$ by (17.2). The convergence of the original integral is guaranteed by $0 < a < 1$: near 0 the integrand is $\sim x^{a-1}$ with $a-1 > -1$, and near ∞ it is $\sim x^{a-2}$ with $a-2 < -1$. $\square$

Remark 30.4 (A sanity check and a corollary). At $a = \frac{1}{2}$, (30.3) gives $\int_0^\infty \frac{dx}{\sqrt{x(1+x)}} = \pi$, which one can verify elementarily by substituting $x = u^2$: $\int_0^\infty \frac{2du}{1+u^2} = 2 \cdot \frac{\pi}{2} = \pi$. Formula (30.3) is also, via the substitution $x = t/(1-t)$, the reflection formula $\Gamma(a)\Gamma(1-a) = \frac{\pi}{\sin \pi a}$ for the gamma function.

30.3 Logarithms in the integrand

A second use of the cut: inserting $\log z$ produces integrals of $\ln x$, and inserting $(\log z)^2$ produces integrals with no logarithm at all.

Example 30.5. Evaluate $\int_0^\infty \frac{dx}{x^2+1}$ and $\int_0^\infty \frac{\ln x}{x^2+1} dx$ simultaneously.

Take $f(z) = \frac{(\log z)^2}{z^2+1}$ with the branch $\log z = \ln|z| + i\theta$, $\theta = \arg_0 z \in (0, 2\pi)$, of Definition 30.1, cut along $[0, \infty)$, and integrate over the keyhole $K_{\varepsilon, R}$ (with $\varepsilon < 1 < R$, understood as in the proof of Theorem 30.3). On L^+ , $\log z = \ln x$; on L^- , $\log z = \ln x + 2\pi i$. Hence

$$\int_{L^+} f + \int_{L^-} f = \int_0^\infty \frac{(\ln x)^2 - (\ln x + 2\pi i)^2}{x^2+1} dx = \int_0^\infty \frac{-4\pi i \ln x + 4\pi^2}{x^2+1} dx.$$

The circles vanish as before (the log growth is beaten by the ε, R powers). The poles are $z = \pm i$; in our branch i has $\theta = \frac{\pi}{2}$ and $-i$ has $\theta = \frac{3\pi}{2}$, so

$$\operatorname{Res}(f, i) = \frac{(i\pi/2)^2}{2i} = \frac{-\pi^2/4}{2i}, \quad \operatorname{Res}(f, -i) = \frac{(3i\pi/2)^2}{-2i} = \frac{-9\pi^2/4}{-2i}.$$

Their sum is $\frac{1}{2i}(-\frac{\pi^2}{4} + \frac{9\pi^2}{4}) = \frac{2\pi^2}{2i} = \frac{\pi^2}{i}$, so the residue theorem gives

$$-4\pi i \int_0^\infty \frac{\ln x}{x^2+1} dx + 4\pi^2 \int_0^\infty \frac{dx}{x^2+1} = 2\pi i \cdot \frac{\pi^2}{i} = 2\pi^3.$$

Both integrals are real, so equating real and imaginary parts:

$$\int_0^\infty \frac{\ln x}{x^2+1} dx = 0, \quad \int_0^\infty \frac{dx}{x^2+1} = \frac{2\pi^3}{4\pi^2} = \frac{\pi}{2}.$$

Both answers are correct and the first — which is not obvious — also follows from the substitution $x \mapsto 1/x$, which maps the integral to its own negative.

30.4 Exercises

Exercise 30.1. Use (30.3) to evaluate $\int_0^\infty \frac{dx}{x^{1/3}(1+x)}$ and $\int_0^\infty \frac{x^{-3/4}}{1+x} dx$.

Exercise 30.2. Prove $\int_0^\infty \frac{x^{a-1}}{1+x^2} dx = \frac{\pi}{2 \sin(\pi a/2)}$ for $0 < a < 2$. (Substitute $x^2 = t$ in (30.3), or run the keyhole argument again.)

Exercise 30.3. Evaluate $\int_0^\infty \frac{\ln x}{(x+1)^3} dx$ using a keyhole contour with $f(z) = \frac{(\log z)^2}{(z+1)^3}$. (Why would $\frac{\log z}{(z+1)^3}$ not do? Compare Example 30.5.)

Exercise 30.4. (*) Show $\int_0^\infty \frac{\ln x}{x^2+a^2} dx = \frac{\pi \ln a}{2a}$ for $a > 0$, and check it against Example 30.5 at $a = 1$.

Exercise 30.5. Where exactly does the proof of Theorem 30.3 break if $a \geq 1$? If $a \leq 0$? Give the divergent integral in each case.

Exercise 30.6. Evaluate $\int_0^\infty \frac{\sqrt{x}}{x^2+1} dx$.

Exercise 30.7. Verify (30.3) numerically at $a = \frac{1}{4}$ and $a = \frac{3}{4}$ by evaluating both sides to three decimals, and confirm the symmetry $a \leftrightarrow 1-a$ that the formula predicts. What identity of $\sin$ is that symmetry?

Exercise 30.8. Evaluate $\int_0^\infty \frac{x^{a-1}}{(1+x)^2} dx$ for $0 < a < 2$, $a \neq 1$, by the keyhole method, and treat $a = 1$ directly. (*The pole at -1 is now double, so use (27.3).*)

Exercise 30.9. (★) Let $0 < a < 1$. Show that the substitution $x = e^t$ turns (30.3) into

$$\int_{-\infty}^\infty \frac{e^{at}}{1+e^t} dt = \frac{\pi}{\sin \pi a},$$

and prove this directly by integrating $\frac{e^{az}}{1+e^z}$ around the rectangle with vertices $\pm R$, $\pm R + 2\pi i$. (*Only one pole, at $z = i\pi$, is enclosed; the top edge reproduces the bottom edge times $e^{2\pi ia}$, which is the same cancellation mechanism as Lemma 30.2.*)

Chapter 31

The Argument Principle and Rouché's Theorem

The residue theorem applied to the particular function f'/f counts zeros and poles. That single observation yields a practical method for locating roots of equations, a two-line proof of the Fundamental Theorem of Algebra, and the open mapping theorem.

31.1 The argument principle

Definition 31.1 (Meromorphic). A function f is *meromorphic* on a region D if there is a set $P \subseteq D$ with no accumulation point in D such that f is analytic on $D \setminus P$ and has a pole at each point of P .

Lemma 31.2 (The logarithmic derivative). *Let f be meromorphic near z_0 and not identically 0.*

- (i) *If f has a zero of order m at z_0 , then f'/f has a simple pole at z_0 with $\text{Res}(f'/f, z_0) = m$.*
- (ii) *If f has a pole of order m at z_0 , then f'/f has a simple pole at z_0 with $\text{Res}(f'/f, z_0) = -m$.*
- (iii) *Otherwise f'/f is analytic at z_0 .*

Proof. (i) By (24.8), $f(z) = (z - z_0)^m g(z)$ with g analytic and $g(z_0) \neq 0$. Differentiating, $f'(z) = m(z - z_0)^{m-1}g(z) + (z - z_0)^m g'(z)$, so

$$\frac{f'(z)}{f(z)} = \frac{m}{z - z_0} + \frac{g'(z)}{g(z)}. \quad (31.1)$$

The second term is analytic near z_0 because $g(z_0) \neq 0$. Hence the pole is simple with residue m .

(ii) By Theorem 26.5(ii), $f(z) = (z - z_0)^{-m}g(z)$ with g analytic and nonzero at z_0 ; the same computation gives $\frac{f'}{f} = \frac{-m}{z - z_0} + \frac{g'}{g}$.

(iii) If $f(z_0) \neq 0$ and f is analytic there, f'/f is a quotient of analytic functions with nonvanishing denominator. $\square$

Theorem 31.3 (Argument principle). *Let C be a simple closed contour, counterclockwise, and let f be meromorphic on an open set containing C and its interior, with no zeros and no poles*

on C . Let N be the number of zeros and P the number of poles of f inside C , each counted with multiplicity. Then

$$\frac{1}{2\pi i} \oint_C \frac{f'(z)}{f(z)} dz = N - P. \quad (31.2)$$

Equivalently, $N - P$ equals the winding number of the image curve $f \circ C$ about the origin:

$$N - P = \text{Ind}(f \circ C; 0) = \frac{1}{2\pi} \left[\text{net change in } \arg f(z) \text{ as } z \text{ traverses } C \right]. \quad (31.3)$$

Proof. The zeros and poles inside C are finite in number: they are isolated (Theorem 24.7 and Definition 31.1) and lie in a compact set, so an infinite family would have an accumulation point, forcing $f \equiv 0$ or contradicting Definition 31.1 (which we use with “region” replaced by “open set”). Apply the residue theorem 27.5 to f'/f and use Lemma 31.2. Here we need the residue theorem in the form its proof actually gives: that proof (deformation onto small circles, Theorem 20.10) uses only that the function is analytic on an open set containing C and its interior, except for finitely many isolated singularities inside C . This holds for f'/f : if U is the given open set and Z the set of zeros and poles of f in U (closed in U , having no accumulation point in U), then f'/f is analytic on the open set obtained by removing from U the points of Z that are not inside C , which still contains C and its interior. So: each zero of order m contributes m , each pole of order m contributes $-m$, and nothing else contributes. That is (31.2).

For (31.3), parametrise C by $\gamma : [a, b] \rightarrow \mathbb{C}$ and substitute $w = f(z)$:

$$\oint_C \frac{f'(z)}{f(z)} dz = \int_a^b \frac{f'(\gamma(t))\gamma'(t)}{f(\gamma(t))} dt = \int_a^b \frac{(f \circ \gamma)'(t)}{(f \circ \gamma)(t)} dt = \oint_{f \circ C} \frac{dw}{w},$$

which is $2\pi i \text{Ind}(f \circ C; 0)$ by (27.6). The interpretation as net change of argument is the identity $\frac{d}{dt} \log(f \circ \gamma) = \frac{(f \circ \gamma)'}{f \circ \gamma}$ along a continuous branch of $\log$, whose imaginary part is a continuous branch of the argument; the real part $\ln |f \circ \gamma|$ returns to its starting value because the contour is closed, so only the argument change survives. $\square$

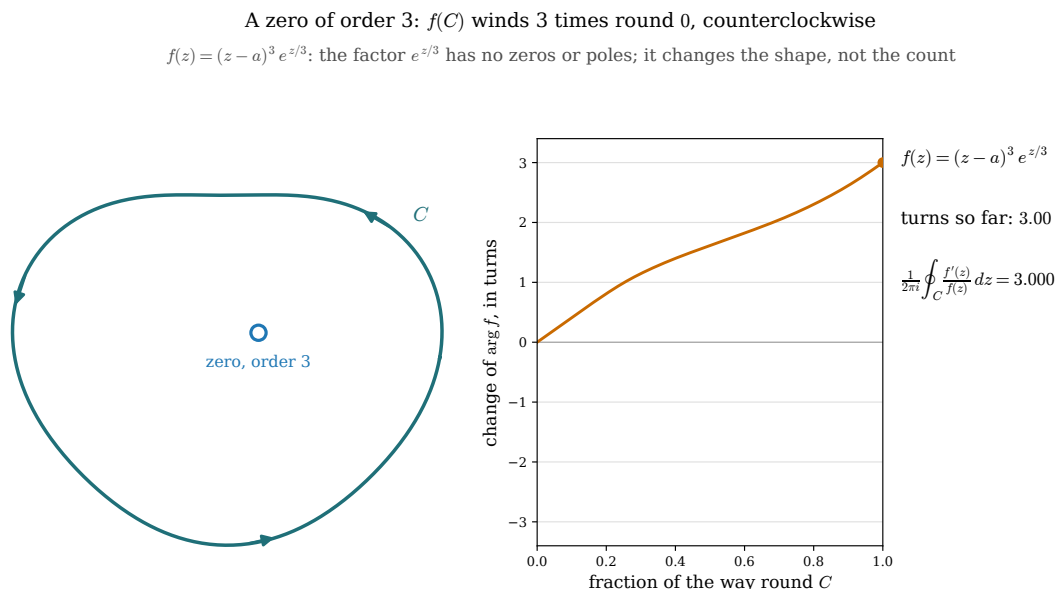


FIGURE 31.1 (animated). Zeros and poles of different orders, for $f(z) = (z-a)^m e^{z/3}$ with $m = 1, 2, 3$ (zeros) and $m = -1, -2, -3$ (poles). As z runs round C (left), the change of $\arg f(z)$, measured in turns, is graphed (right); it ends at m . Indeed $\frac{f'}{f} = \frac{m}{z-a} + \frac{1}{3}$, so $\frac{1}{2\pi i} \oint_C \frac{f'}{f} dz = m$ by (20.13) and Theorem 20.9: a zero of order m counts m and a pole of order k counts $-k$. The factor $e^{z/3}$, which has no zeros or poles, changes the shape of the graph but not its end point.

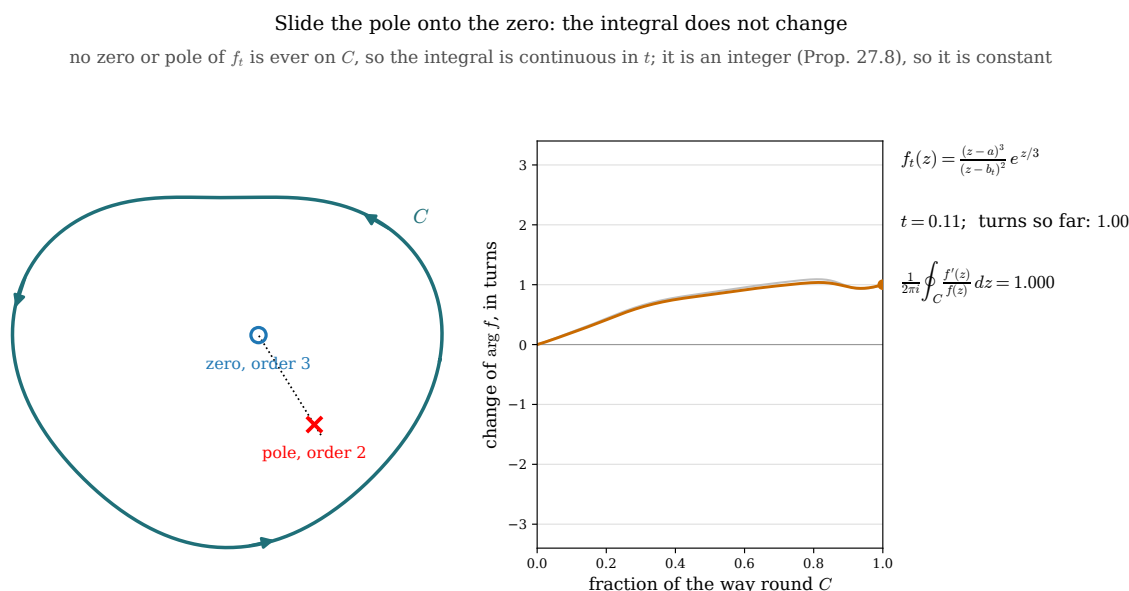


FIGURE 31.2 (animated). A proof of (31.2) by merging. First a simple zero and a simple pole, then a zero of order 3 and a pole of order 2, lie inside C , and the pole slides onto the zero: $f_t(z) = \frac{(z-a)^3}{(z-b_t)^2} e^{z/3}$. No zero or pole of f_t is ever on C , so $N(t) = \frac{1}{2\pi i} \oint_C \frac{f'_t}{f_t} dz$ is continuous in t . It is also an integer, the index of $f_t(C)$ about 0 (Proposition 27.8), so it is constant. The graph of $\arg f_t$ changes (orange; grey is $t = 0$) but its end point does not. After the merge $f_1 = (z-a)^{m-k} e^{z/3}$, whose count is $m-k$ by the previous figure: here $0 = 1 - 1$ and $1 = 3 - 2$. Sliding every zero and pole inside C to one point in this way proves (31.2); the notes' proof above uses the residue theorem instead.

Example 31.4. Let $f(z) = \frac{z^3(z-2)}{(z-i)^2}$ and let C be $|z| = 3$. Inside C : zeros at 0 (order 3) and 2 (order 1), so $N = 4$; a pole at i of order 2, so $P = 2$. Therefore $\frac{1}{2\pi i} \oint_C \frac{f'}{f} = 2$: the image curve winds twice around the origin.

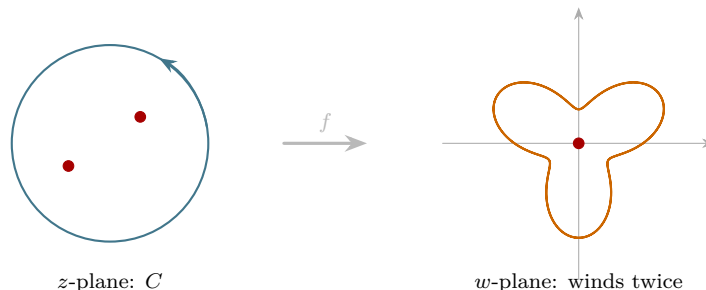


FIGURE 31.3. The argument principle (31.3) as bookkeeping on the image curve. As z traverses C once, the image $f(z)$ traces a closed curve in the w -plane; the number of times that curve winds around the origin is $N - P$. Here it winds twice, so C encloses two more zeros than poles, counted with multiplicity. No root finding is involved — one only has to watch $\arg f$.

Example 31.5 (Counting roots in a region). How many zeros does $f(z) = z^4 + z^3 + 1$ have in the first quadrant? Traverse the boundary of the quarter-disc of radius R (large) and track the argument.

- On $[0, R]$: $f(x) = x^4 + x^3 + 1 > 0$, so $\arg f$ stays at 0.
- On the arc $Re^{i\theta}$, $\theta : 0 \rightarrow \pi/2$: write $f(z) = z^4(1 + z^{-1} + z^{-4})$. The argument of z^4 increases by $4 \cdot \frac{\pi}{2} = 2\pi$, while $|z^{-1} + z^{-4}| \leq 2/R$, so the second factor stays in the disc $D(1, 2/R)$ and its argument changes by at most $2 \arcsin(2/R) = o(1)$. Net change $2\pi + o(1)$ as $R \rightarrow \infty$.
- On the segment from iR to 0: $f(iy) = y^4 - iy^3 + 1$, whose real part $y^4 + 1 > 0$ always. So $\arg f$ stays in $(-\frac{\pi}{2}, \frac{\pi}{2})$; it starts at $-\arctan \frac{R^3}{R^4+1} = o(1)$ and ends at 0, so the net change is $o(1)$.

There are no zeros on the boundary (on the positive axes, f and $\operatorname{Re} f$ are positive), so by (31.3) the total change is $2\pi N$, an integer multiple of 2π . It equals $2\pi + o(1)$, hence for R large it is exactly 2π . Taking R larger than the modulus of every zero, there is exactly $N = 1$ zero in the first quadrant.

31.2 Rouché's theorem

Theorem 31.6 (Rouché). *Let C be a simple closed contour and let f, g be analytic on an open set containing C and its interior. If*

$$|g(z)| < |f(z)| \quad \text{for every } z \text{ on } C, \quad (31.4)$$

then f and $f + g$ have the same number of zeros inside C , counted with multiplicity.

Proof. First, (31.4) forces $f \neq 0$ on C and, by the reverse triangle inequality (1.28), $|f + g| \geq |f| - |g| > 0$ on C , so $f + g \neq 0$ on C either. Both functions therefore satisfy the hypotheses of Theorem 31.3 with $P = 0$.

For $t \in [0, 1]$ put $h_t := f + tg$. By the same estimate $|h_t| \geq |f| - t|g| > 0$ on C , so

$$N(t) := \frac{1}{2\pi i} \oint_C \frac{h'_t(z)}{h_t(z)} dz$$

is defined for every $t \in [0, 1]$ and, by Theorem 31.3, is a non-negative *integer* — the number of zeros of h_t inside C . But $t \mapsto N(t)$ is continuous: the integrand $\frac{f'+tg'}{f+tg}$ depends continuously on t , uniformly on the compact contour C (the denominator is bounded away from 0), so the integral does too. A continuous integer-valued function on the connected interval $[0, 1]$ is constant. Hence $N(0) = N(1)$, i.e. f and $f + g$ have the same zero count. $\square$

Rouché's theorem as a dog on a leash, $C: |z| = 2$

person at $f(z) = z^4$ (blue), dog at $f(z) + g(z) = z^4 - 6z + 3$ (green), leash $g(z) = 3 - 6z$ (brown)

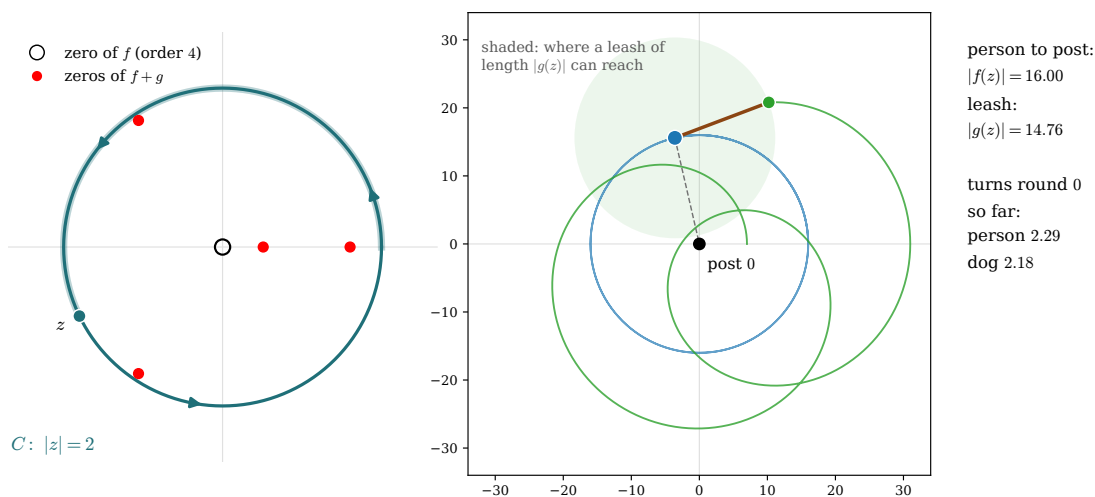


FIGURE 31.4 (animated). Rouché's theorem as a dog on a leash, for $f(z) = z^4$ and $g(z) = 3 - 6z$ on $C: |z| = 2$. As z runs round C (left), a person walks along $f(C)$ and a dog along $(f + g)(C)$ (right), joined by a leash of length $|g(z)| \leq 15$. The post 0 is always at distance $|f(z)| = 16$ from the person, farther than the leash reaches, so the dog never reaches the post. Both go round the post the same number of times, 4, and by (31.3) f and $f + g = z^4 - 6z + 3$ have the same number of zeros inside C . This picture illustrates the statement; the proof above uses the homotopy of the next figure.

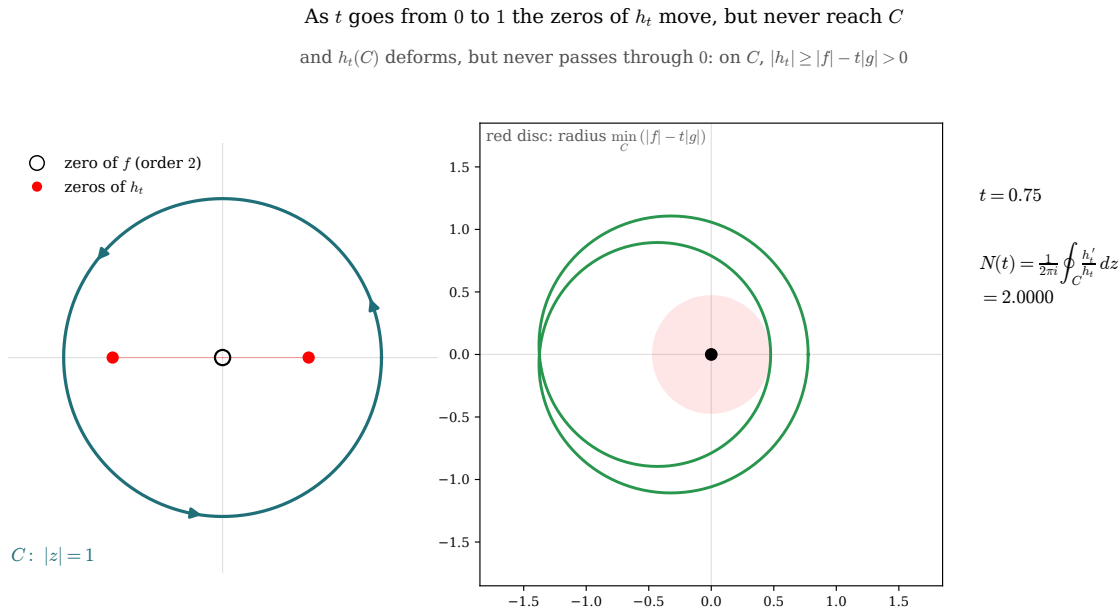


FIGURE 31.5 (animated). The proof of Theorem 31.6, for a smaller example: $f(z) = z^2$ and $g(z) = z/5 - 1/2$ on $C: |z| = 1$, where $|g| \leq 0.7 < 1 = |f|$. As t runs from 0 to 1 the zeros of $h_t = f + tg$ (left) leave the double zero of f at 0 and move apart, but never reach C . The curve $h_t(C)$ (right) deforms, but stays outside the red disc of radius $\min_C (|f| - t|g|) = 1 - 0.7t > 0$ about 0, so it never passes through 0. Hence $N(t) = \frac{1}{2\pi i} \oint_C \frac{h'_t}{h_t} dz$ is defined, continuous and integer-valued, so it is constant: $N(1) = N(0) = 2$.

Remark 31.7 (How to use it). Given a function F whose zeros you want to count inside C , split $F = f + g$ where f is a *dominant* term whose zeros you know, and verify $|g| < |f|$ on C . The art is entirely in the split.

Example 31.8 (The Fundamental Theorem of Algebra again). Let $p(z) = a_n z^n + \cdots + a_0$ with $a_n \neq 0$, $n \geq 1$. Put $f(z) = a_n z^n$ and $g(z) = a_{n-1} z^{n-1} + \cdots + a_0$. On $|z| = R$,

$$|g(z)| \leq \sum_{k < n} |a_k| R^k \leq \left(\sum_{k < n} |a_k| \right) R^{n-1} < |a_n| R^n = |f(z)|$$

as soon as $R > \max\left(1, \frac{\sum_{k < n} |a_k|}{|a_n|}\right)$. By Rouché, $p = f + g$ has as many zeros in $|z| < R$ as $f(z) = a_n z^n$ does, namely n (all at the origin, counted with multiplicity). So a degree- n polynomial has exactly n roots. This proves Theorem 6.5 again, and this time it delivers the count, not merely existence.

Example 31.9 (Locating roots). How many zeros does $F(z) = z^7 - 5z^3 + 12$ have in the annulus $1 \leq |z| < 2$?

On $|z| = 1$: take $f = 12$, $g = z^7 - 5z^3$. Then $|g| \leq 1 + 5 = 6 < 12 = |f|$, so F has the same number of zeros in $|z| < 1$ as the constant 12 — namely 0.

On $|z| = 2$: take $f = z^7$, $g = -5z^3 + 12$. Then $|g| \leq 5 \cdot 8 + 12 = 52 < 128 = 2^7 = |f|$, so F has as many zeros in $|z| < 2$ as z^7 — namely 7.

Hence all 7 zeros lie in $1 \leq |z| < 2$, and in fact in $1 < |z| < 2$ since F has no zero on $|z| = 1$ (the first estimate was strict).

31.3 The open mapping theorem

Theorem 31.10 (Open mapping theorem). *Let D be a region and f analytic and nonconstant on D . Then f is an open map: $f(U)$ is open in $\mathbb{C}$ for every open $U \subseteq D$.*

Proof. Let $U \subseteq D$ be open, $z_0 \in U$, $w_0 := f(z_0)$. We find $\eta > 0$ with $D(w_0, \eta) \subseteq f(U)$.

Since f is nonconstant, the identity theorem 24.8 makes the zeros of $f - w_0$ isolated, so there is $\rho > 0$ with $\overline{D}(z_0, \rho) \subseteq U$ and $f(z) \neq w_0$ for $0 < |z - z_0| \leq \rho$. Let

$$\eta := \frac{1}{2} \min_{|z-z_0|=\rho} |f(z) - w_0| > 0,$$

positive by Theorem 12.13 applied on the compact circle.

Fix $w \in D(w_0, \eta)$ and write

$$f(z) - w = \underbrace{(f(z) - w_0)}_{=:F(z)} + \underbrace{(w_0 - w)}_{=:G(z)}.$$

On $|z - z_0| = \rho$ we have $|G| = |w - w_0| < \eta \leq \frac{1}{2}|F| < |F|$, so Rouché's Theorem 31.6 says $f - w$ has the same number of zeros in $D(z_0, \rho)$ as $f - w_0$, which has at least one (namely z_0). Hence there is $z \in D(z_0, \rho) \subseteq U$ with $f(z) = w$, i.e. $w \in f(U)$. $\square$

Corollary 31.11. *The maximum modulus principle 24.11 follows: if $|f|$ had a local maximum at an interior point z_0 and f were nonconstant, then $f(U)$ would be an open neighbourhood of $f(z_0)$ for small U , and every neighbourhood of $f(z_0)$ contains points of larger modulus — contradiction.*

31.4 Exercises

Exercise 31.1. Compute $\frac{1}{2\pi i} \oint_{|z|=4} \frac{f'}{f} dz$ for $f(z) = \frac{(z-1)^2(z+3)}{z^3(z-2)}$.

Exercise 31.2. Use Rouché to show that all five zeros of $z^5 + 3z + 1$ lie in $|z| < 2$, and that exactly one lies in $|z| < 1$.

Exercise 31.3. How many zeros does $z^4 - 6z + 3$ have in the annulus $1 < |z| < 2$?

Exercise 31.4. (*) Prove that $e^z = 3z^n$ has exactly n solutions in $|z| < 1$ for every $n \geq 1$. (Compare $3z^n$ with $-e^z$ on $|z| = 1$, using $|e^z| = e^{\operatorname{Re} z} \leq e$.)

Exercise 31.5. Let f be analytic on $\overline{\mathbb{D}}$ (i.e. on an open set containing it) with $|f(z)| < 1$ for $|z| = 1$. Prove f has exactly one fixed point in $\mathbb{D}$. (Apply Rouché to $f(z) - z$ against $-z$.)

Exercise 31.6. Show that Theorem 31.10 fails for real-differentiable maps by exhibiting a smooth nonconstant $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose image of an open set is not open.

Exercise 31.7. How many zeros does $2z^5 - 6z^2 + z + 1$ have in $1 \leq |z| < 2$? (Two applications of Rouché, as in Example 31.9.)

Exercise 31.8. Determine the number of zeros of $z^4 + 2z^3 + 3z^2 + 4z + 5$ in each quadrant (first check that none lies on the real or imaginary axis), by tracking $\arg f$ along the boundary of a large quarter-disc as in Example 31.5.

Exercise 31.9. (*) (Hurwitz.) Let f_n be analytic and nowhere zero on a region D , and suppose $f_n \rightarrow f$ locally uniformly with $f \neq 0$. Prove f is nowhere zero on D . (If $f(z_0) = 0$, apply (31.2) to f on a small circle where $f \neq 0$, and use $f'_n/f_n \rightarrow f'/f$ uniformly there, from Corollary 21.8.) Deduce that a locally uniform limit of injective analytic functions is injective or constant.

Chapter 32

Summing Infinite Series

Example 27.4(c) recorded that $\cot z$ has a simple pole with residue 1 at every integer multiple of π . Consequently $\pi \cot \pi z$ has a simple pole of residue 1 at every *integer*. Multiplying it by a function f and integrating over a large contour therefore sums the values $f(n)$ over all $n \in \mathbb{Z}$ — and the residue theorem evaluates the result. This is the residue calculus applied to series rather than integrals, and it produces closed forms that are otherwise very hard to obtain.

32.1 The summation kernels

Lemma 32.1 (Poles and residues of the kernels). *Define*

$$\kappa(z) := \pi \cot \pi z = \pi \frac{\cos \pi z}{\sin \pi z}, \quad \lambda(z) := \pi \csc \pi z = \frac{\pi}{\sin \pi z}. \quad (32.1)$$

Both are meromorphic on $\mathbb{C}$ with simple poles exactly at the integers, and

$$\operatorname{Res}(\kappa, n) = 1, \quad \operatorname{Res}(\lambda, n) = (-1)^n, \quad n \in \mathbb{Z}. \quad (32.2)$$

Proof. By Corollary 17.6, $\sin \pi z$ vanishes exactly at $z = n \in \mathbb{Z}$, and the zeros are simple since $\frac{d}{dz} \sin \pi z = \pi \cos \pi z$ equals $\pi(-1)^n \neq 0$ there. Corollary 26.6 makes the poles simple, and (27.4) gives

$$\operatorname{Res}(\kappa, n) = \frac{\pi \cos \pi n}{\pi \cos \pi n} = 1, \quad \operatorname{Res}(\lambda, n) = \frac{\pi}{\pi \cos \pi n} = (-1)^n. \quad \square$$

Lemma 32.2 (The kernels are bounded on good contours). *For $N \in \mathbb{N}$ let S_N be the boundary of the square with vertices $(N + \frac{1}{2})(\pm 1 \pm i)$, traversed counterclockwise. There is a constant A , independent of N , with*

$$|\kappa(z)| \leq A \quad \text{and} \quad |\lambda(z)| \leq A \quad \text{for all } z \in S_N \text{ and all } N. \quad (32.3)$$

Proof. Write $z = x + iy$. By (17.17), $|\sin \pi z|^2 = \sin^2 \pi x + \sinh^2 \pi y$.

On the horizontal sides $|y| = N + \frac{1}{2}$, so $|\sin \pi z| \geq |\sinh \pi y| \geq \sinh \frac{\pi}{2} > 2$, whence $|\lambda| \leq \pi/2$. Also $|\cos \pi z|^2 = \cos^2 \pi x + \sinh^2 \pi y \leq 1 + \sinh^2 \pi y = \cosh^2 \pi y$, so $|\kappa| \leq \pi \coth \pi |y| \leq \pi \coth \frac{\pi}{2}$.

On the vertical sides $x = \pm(N + \frac{1}{2})$, so $\sin^2 \pi x = \sin^2(\pm(N + \frac{1}{2})\pi) = 1$ and $|\sin \pi z| \geq 1$, giving $|\lambda| \leq \pi$; and $\cos^2 \pi x = 0$, so $|\kappa| = \pi \frac{|\sinh \pi y|}{\sqrt{1 + \sinh^2 \pi y}} \leq \pi$.

Take $A := \pi \coth \frac{\pi}{2}$, which dominates all four bounds. $\square$

The choice of half-integer sides is what makes this work: the squares stay uniformly far from the poles.

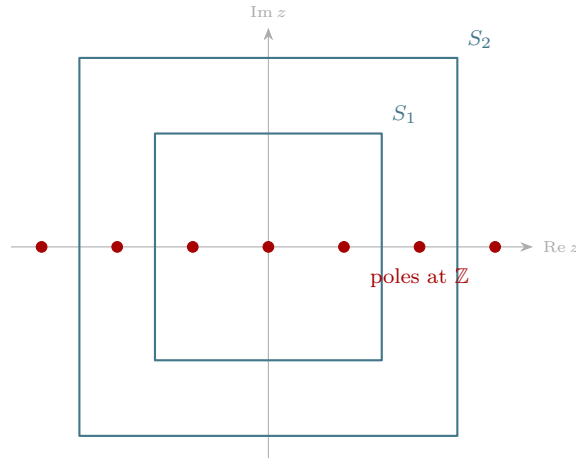


FIGURE 32.1. The squares S_N of Lemma 32.2, with half-integer sides. Their edges stay a fixed distance from every pole of $\pi \cot \pi z$ (the integers, marked), which is exactly why the kernel is bounded on them *uniformly in N* . Squares through the integers would fail: the bound would blow up and the argument of Theorem 32.3 would collapse.

32.2 The summation theorems

Theorem 32.3 (Summation by residues). *Let f be meromorphic on $\mathbb{C}$ with finitely many poles $w_1, \dots, w_r$, none of them an integer, and suppose there are constants $B > 0$, $p > 1$, $\rho_0 > 0$ with*

$$|f(z)| \leq \frac{B}{|z|^p} \quad \text{for } |z| \geq \rho_0. \quad (32.4)$$

Then

$$\sum_{n=-\infty}^{\infty} f(n) = - \sum_{k=1}^r \text{Res}(\pi \cot(\pi z) f(z), w_k), \quad (32.5)$$

$$\sum_{n=-\infty}^{\infty} (-1)^n f(n) = - \sum_{k=1}^r \text{Res}(\pi \csc(\pi z) f(z), w_k). \quad (32.6)$$

Both series converge absolutely.

Proof. We prove (32.5); the other is identical with λ in place of κ and (32.2) supplying the sign $(-1)^n$.

Absolute convergence of $\sum f(n)$ is comparison with $\sum |n|^{-p}$, $p > 1$.

Take N large enough that all the w_k and all of $\{|z| \leq \rho_0\}$ lie inside S_N . The function κf is meromorphic inside S_N with poles at the integers n with $|n| \leq N$ (simple, from κ) and at the w_k (from f). At an integer n , f is analytic and (32.2) with Theorem 27.2(i) gives

$$\text{Res}(\kappa f, n) = \lim_{z \rightarrow n} (z - n) \kappa(z) f(z) = 1 \cdot f(n) = f(n).$$

The residue theorem 27.5, applied on the open square Q_N with vertices $(N + \frac{3}{4})(\pm 1 \pm i)$ — a convex, hence simply connected, region containing S_N , on which κf has only the finitely many

poles just listed — therefore gives

$$\frac{1}{2\pi i} \oint_{S_N} \kappa(z) f(z) dz = \sum_{|n| \leq N} f(n) + \sum_{k=1}^r \text{Res}(\kappa f, w_k). \quad (32.7)$$

Now estimate the left side. The square S_N has perimeter $8(N + \frac{1}{2})$ and every point on it has $|z| \geq N + \frac{1}{2}$, so by (32.4) and Lemma 32.2 the ML -inequality (19.15) gives

$$\left| \oint_{S_N} \kappa f \right| \leq A \cdot \frac{B}{(N + \frac{1}{2})^p} \cdot 8(N + \frac{1}{2}) = \frac{8AB}{(N + \frac{1}{2})^{p-1}} \rightarrow 0$$

as $N \rightarrow \infty$, because $p > 1$. Letting $N \rightarrow \infty$ in (32.7) yields (32.5). $\square$

32.3 The Basel problem and its relatives

Example 32.4 ($\sum 1/n^2$). Take $f(z) = 1/z^2$. It satisfies (32.4) with $p = 2$, but its pole $w_1 = 0$ is an integer, so Theorem 32.3 does not apply verbatim. The remedy is to run the same contour argument directly: the function $\kappa(z)/z^2$ has a pole of order 3 at 0 and simple poles at each nonzero integer, with residue $1/n^2$ there. The estimate on S_N is unchanged, so

$$0 = \lim_{N \rightarrow \infty} \frac{1}{2\pi i} \oint_{S_N} \frac{\pi \cot \pi z}{z^2} dz = \sum_{n \neq 0} \frac{1}{n^2} + \text{Res}\left(\frac{\pi \cot \pi z}{z^2}, 0\right). \quad (32.8)$$

It remains to compute the residue at 0. From (24.5),

$$\pi \cot \pi z = \pi \frac{\cos \pi z}{\sin \pi z} = \pi \cdot \frac{1 - \frac{(\pi z)^2}{2} + \dots}{\pi z \left(1 - \frac{(\pi z)^2}{6} + \dots\right)} = \frac{1}{z} \left(1 - \frac{\pi^2 z^2}{2}\right) \left(1 + \frac{\pi^2 z^2}{6} + \dots\right) = \frac{1}{z} - \frac{\pi^2 z}{3} + O(z^3),$$

where we used $\frac{1}{1-u} = 1 + u + \dots$ with $u = \frac{(\pi z)^2}{6}$. Dividing by z^2 ,

$$\frac{\pi \cot \pi z}{z^2} = \frac{1}{z^3} - \frac{\pi^2}{3z} + O(z),$$

so the residue — the coefficient of z^{-1} — is $-\frac{\pi^2}{3}$. Substituting into (32.8),

$$\sum_{n \neq 0} \frac{1}{n^2} = \frac{\pi^2}{3}, \quad \text{and by symmetry} \quad \boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}}$$

— Euler's solution of the Basel problem, obtained here in half a page.

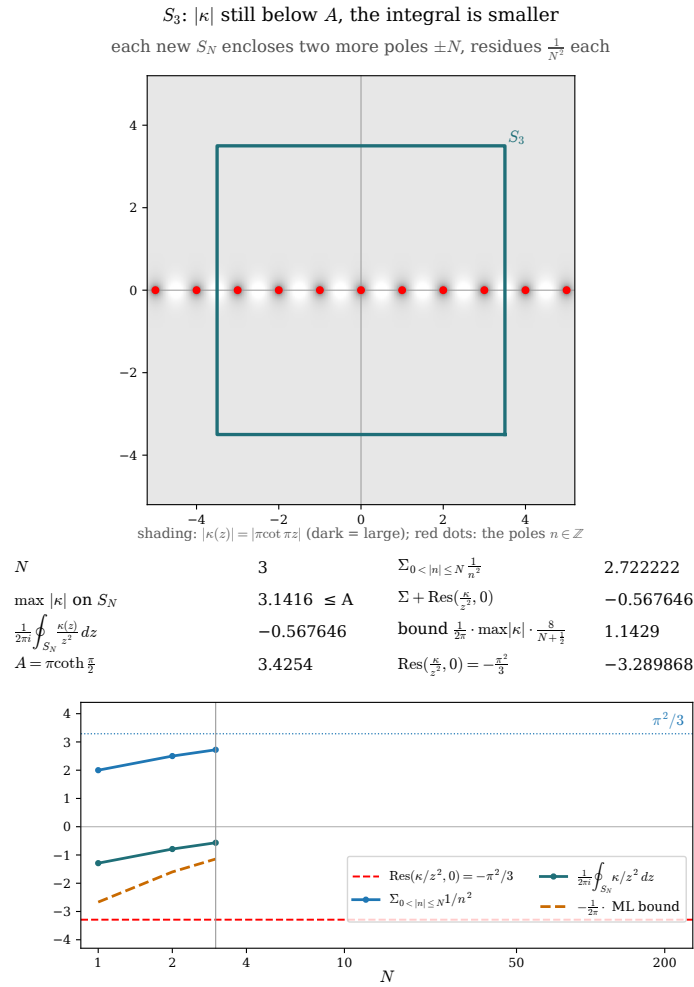


FIGURE 32.2 (animated). The squares S_N with sides on $x, y = \pm(N + \frac{1}{2})$ over a map of $|\pi \cot \pi z|$ (dark near the poles n , about π elsewhere). Each S_N passes halfway between two poles, so $\max_{S_N} |\pi \cot \pi z|$ stays below the bound A of Lemma 32.2; while the square grows from one S_N to the next it passes through the poles $\pm(N+1)$, where the maximum is infinite, and a square with sides $x = \pm 4.05$ misses the poles but has maximum 19.84. Below, against N : the contour integral of $\pi \cot(\pi z)/z^2$ tends to 0, and the partial sums $\sum_{0 < |n| \leq N} n^{-2}$ approach $\pi^2/3$, that is, $\sum_{n \geq 1} n^{-2} \rightarrow \pi^2/6$ (Example 32.4; 1.639947 at $N = 200$ against 1.644934).

Example 32.5 (The alternating version). The same computation with $\lambda = \pi \csc \pi z$ in place of κ requires the expansion

$$\pi \csc \pi z = \frac{\pi}{\pi z (1 - \frac{(\pi z)^2}{6} + \dots)} = \frac{1}{z} + \frac{\pi^2 z}{6} + O(z^3),$$

so $\text{Res}(\frac{\pi \csc \pi z}{z^2}, 0) = \frac{\pi^2}{6}$ and

$$\sum_{n \neq 0} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{6}, \quad \text{i.e.} \quad \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}.$$

Example 32.6 (A sum with poles off the integers). Evaluate $\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2}$ for real $a > 0$.

Here $f(z) = \frac{1}{z^2 + a^2}$ satisfies (32.4) with $p = 2$ and has simple poles at $\pm ia$, neither an integer. By (32.5) and (27.4),

$$\sum_{n \in \mathbb{Z}} \frac{1}{n^2 + a^2} = -\left[\frac{\pi \cot(i\pi a)}{2ia} + \frac{\pi \cot(-i\pi a)}{-2ia} \right] = -\frac{\pi \cot(i\pi a)}{ia},$$

the two terms being equal since $\cot$ is odd. Now $\cot(i\pi a) = \frac{\cos(i\pi a)}{\sin(i\pi a)} = \frac{\cosh \pi a}{i \sinh \pi a}$ by (17.12), so

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} = -\frac{\pi}{ia} \cdot \frac{\cosh \pi a}{i \sinh \pi a} = \frac{\pi}{a} \coth \pi a. \quad (32.9)$$

Two checks. As $a \rightarrow \infty$ the right side is asymptotic to π/a (the ratio tends to 1), matching the integral approximation $\int_{-\infty}^{\infty} \frac{dx}{x^2 + a^2} = \pi/a$. As $a \downarrow 0$, $\coth \pi a \sim \frac{1}{\pi a}$, so the right side $\sim \frac{1}{a^2}$ — the $n = 0$ term, which does indeed dominate. Subtracting it and letting $a \rightarrow 0$ recovers $2 \sum_{n \geq 1} n^{-2} = \pi^2/3$, consistent with Example 32.4.

Remark 32.7 (Higher even zeta values). Repeating Example 32.4 with $f(z) = z^{-2m}$ gives $\zeta(2m) = \sum_{n \geq 1} n^{-2m}$ in closed form for every $m \geq 1$: one needs the coefficient of z^{2m-1} in the expansion of $\pi \cot \pi z$, which is a Bernoulli number. Thus $\zeta(4) = \frac{\pi^4}{90}$, $\zeta(6) = \frac{\pi^6}{945}$, and so on. No comparable formula is known for $\zeta(3)$ — the method produces *even* arguments only, because $\cot$ is odd.

32.4 Exercises

Exercise 32.1. Evaluate $\sum_{n=1}^{\infty} \frac{1}{n^4}$ by the method of Example 32.4. (You will need the z^3 term of $\pi \cot \pi z$.)

Exercise 32.2. Show $\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{n^2 + a^2} = \frac{\pi}{a} \operatorname{csch} \pi a$ for $a > 0$, and deduce $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2 + 1}$.

Exercise 32.3. Evaluate $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$ from Examples 32.4 and 32.5 without any new contour work.

Exercise 32.4. (★) Prove the partial fraction expansion

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}, \quad z \notin \mathbb{Z},$$

by applying Theorem 32.3 to $f(w) = \frac{1}{w-z}$ with z fixed and non-integral. (Handle the failure of (32.4) with $p = 1$ by pairing n with $-n$.)

Exercise 32.5. Explain why the method cannot produce $\sum_{n \geq 1} \frac{1}{n}$, and what goes wrong in the estimate.

Exercise 32.6. Evaluate $\sum_{n=-\infty}^{\infty} \frac{1}{(n + \frac{1}{3})^2}$. (Take $f(z) = (z + \frac{1}{3})^{-2}$; the pole is not an integer, so (32.5) applies directly.)

Exercise 32.7. Evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2 + n}$ two ways: by telescoping, and by (32.5) applied to $f(z) = \frac{1}{z^2 + z}$ with the integer poles handled separately.

Exercise 32.8. Show $\sum_{n=-\infty}^{\infty} \frac{1}{(n-a)^2} = \frac{\pi^2}{\sin^2 \pi a}$ for $a \notin \mathbb{Z}$, and recover Example 32.4 by letting $a \rightarrow 0$ after subtracting the $n = 0$ term.

Exercise 32.9. (★) Prove that $\sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}$. (You need the coefficient of z^5 in $\pi \cot \pi z$; get it by dividing the series (24.5) far enough, or by the Bernoulli-number formula of Remark 32.7.)

Part VIII

Conformal Mapping

Chapter 33

Conformal Mapping

Theorem 13.10 established that an analytic map preserves angles wherever its derivative is nonzero. This chapter takes that fact as a design principle: to solve a boundary value problem on an awkward region, map the region conformally onto a standard one, solve there, and transport the answer back.

33.1 Conformal maps

Definition 33.1 (Conformal map). Let D be a region. A function f is *conformal at* $z_0 \in D$ if f is analytic at z_0 and $f'(z_0) \neq 0$. It is *conformal on* D if it is conformal at every point of D , and a *conformal equivalence* of D onto D' if in addition it is a bijection of D onto D' .

Theorem 33.2 (Local behaviour). *Let f be conformal at z_0 . Then:*

- (i) f preserves the magnitude and the sense of angles at z_0 (Theorem 13.10);
- (ii) f is one-to-one on some neighbourhood of z_0 ;
- (iii) the local inverse f^{-1} is analytic near $w_0 = f(z_0)$, with

$$(f^{-1})'(w_0) = \frac{1}{f'(z_0)}. \quad (33.1)$$

Proof. (ii) Since $f'(z_0) \neq 0$, the function $f - f(z_0)$ has a zero of order exactly 1 at z_0 (Theorem 24.7 with $m = 1$). Choose $\rho > 0$ with $f(z) \neq f(z_0)$ on $0 < |z - z_0| \leq \rho$ and $f' \neq 0$ on $\overline{D}(z_0, \rho)$. Put $w_0 := f(z_0)$. There is $\eta > 0$ such that for every $w \in D(w_0, \eta)$, Rouché's Theorem 31.6 (as in the proof of Theorem 31.10) shows $f - w$ has exactly one zero in $D(z_0, \rho)$, counted with multiplicity (as many as $f - w_0$, whose only zero there is z_0 , of order 1). So f is injective on the neighbourhood $D(z_0, \rho) \cap f^{-1}(D(w_0, \eta))$ of z_0 .

(iii) f is an open map (Theorem 31.10), so f^{-1} is continuous; then

$$\frac{f^{-1}(w) - f^{-1}(w_0)}{w - w_0} = \frac{z - z_0}{f(z) - f(z_0)} \longrightarrow \frac{1}{f'(z_0)}$$

as $w \rightarrow w_0$, since $z = f^{-1}(w) \rightarrow z_0$ and the denominator's difference quotient tends to $f'(z_0) \neq 0$. $\square$

Warning 33.3. Conformality on D does *not* imply injectivity on D . The map $f(z) = z^2$ is conformal on $\mathbb{C} \setminus \{0\}$ but two-to-one there; $f(z) = e^z$ is conformal on $\mathbb{C}$ but infinitely many-to-one. Global injectivity must be checked separately, usually by the arguments of Corollary 9.2 and Theorem 16.3.

33.2 A catalogue of standard maps

The following table collects the conformal equivalences established earlier in these notes. In each case the reference gives the proof.

Map	Domain	Image
$w = az + b, a \neq 0$	$\mathbb{C}$	$\mathbb{C}$ (Thm 8.4)
$w = z^n$	sector $ \operatorname{Arg} z < \pi/n$	slit plane $\mathbb{C} \setminus (-\infty, 0]$ (Cor 9.2)
$w = z_{\text{pr}}^{1/n}$	$\mathbb{C} \setminus (-\infty, 0]$	sector $ \operatorname{Arg} w < \pi/n$ (Thm 10.3)
$w = z_{\text{pr}}^\alpha, \alpha > 0$	sector $S(\theta_1, \theta_2), -\pi \leq \theta_1 < \theta_2 \leq \pi$, of angle $\varphi = \theta_2 - \theta_1 \leq 2\pi/\alpha$	sector $S(\alpha\theta_1, \alpha\theta_2)$ of angle $\alpha\varphi$ (Prop 18.5)
$w = 1/z$	$\mathbb{C} \setminus \{0\}$	$\mathbb{C} \setminus \{0\}$ (Prop 11.2)
$w = e^z$	strip $-\pi < \operatorname{Im} z < \pi$	slit plane $\mathbb{C} \setminus (-\infty, 0]$ (Thm 16.3)
$w = \operatorname{Log} z$	$\mathbb{C} \setminus (-\infty, 0]$	strip $ \operatorname{Im} w < \pi$ (Thm 16.9)
$w = \sin z$	strip $ \operatorname{Re} z < \pi/2$	$\mathbb{C}$ minus $(-\infty, -1] \cup [1, \infty)$ (Thm 17.10)
$w = \frac{z-a}{1-\bar{a}z}, a < 1$	unit disc $\mathbb{D}$	unit disc $\mathbb{D}$ (Thm 34.8)

Example 33.4 (Composing the catalogue). Map the half-disc $H = \{|z| < 1, \operatorname{Im} z > 0\}$ conformally onto the upper half-plane.

Step 1. $w_1 = \frac{1+z}{1-z}$ maps $\mathbb{D}$ onto the right half-plane $\{\operatorname{Re} w_1 > 0\}$ and the real diameter $(-1, 1)$ onto the positive real axis. Indeed $w_1 = -i\psi^{-1}(z)$ with ψ^{-1} the inverse Cayley transform (34.11), which maps $\mathbb{D}$ onto the upper half-plane (Theorem 34.9), and multiplication by $-i$ rotates the upper half-plane onto the right half-plane; for real $x \in (-1, 1)$, $\frac{1+x}{1-x} \in (0, \infty)$. Moreover

$$\operatorname{Im} w_1 = \operatorname{Im} \frac{(1+z)(1-\bar{z})}{|1-z|^2} = \frac{2 \operatorname{Im} z}{|1-z|^2},$$

so w_1 maps $\{\operatorname{Im} z > 0\}$ into $\{\operatorname{Im} w_1 > 0\}$ and $\{\operatorname{Im} z \leq 0\} \setminus \{1\}$ into $\{\operatorname{Im} w_1 \leq 0\}$. Since w_1 is a bijection of $\mathbb{D}$ onto the right half-plane, it therefore maps $H = \mathbb{D} \cap \{\operatorname{Im} z > 0\}$ onto the first quadrant $\{\operatorname{Re} w_1 > 0, \operatorname{Im} w_1 > 0\}$. (On the boundary, the upper semicircle $e^{i\theta}$, $0 < \theta < \pi$, goes to $\frac{1+e^{i\theta}}{1-e^{i\theta}} = i \cot \frac{\theta}{2}$, the positive imaginary axis.)

Step 2. $w_2 = w_1^2$ doubles the opening angle (Theorem 9.1), mapping the first quadrant onto the upper half-plane.

Hence $w = \left(\frac{1+z}{1-z}\right)^2$ maps H conformally onto $\{\operatorname{Im} w > 0\}$. Both steps are bijections onto their images, so the composition is too.

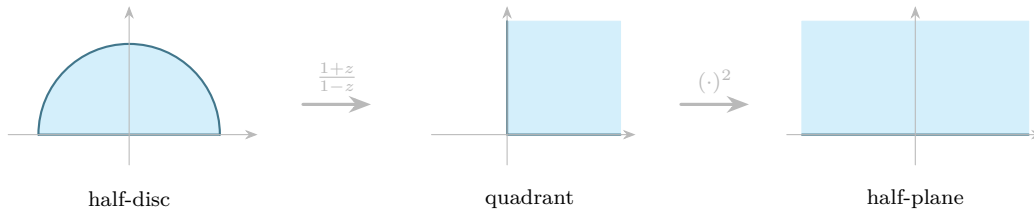


FIGURE 33.1. Example 33.4: a conformal map built by composing two entries of the catalogue. The half-disc goes to a quarter plane under $w_1 = \frac{1+z}{1-z}$, and the quarter plane opens to a half-plane under $w_2 = w_1^2$. Building maps by composition is almost always how explicit conformal maps are found; the Riemann mapping theorem 33.10 guarantees one exists but does not produce it.

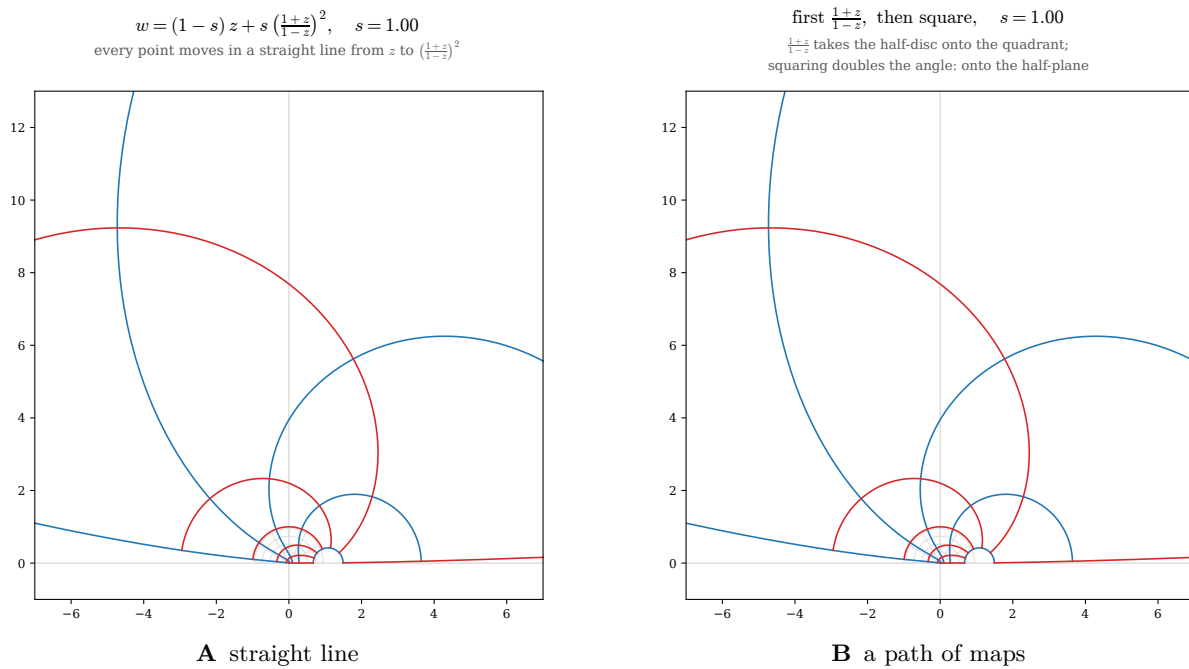
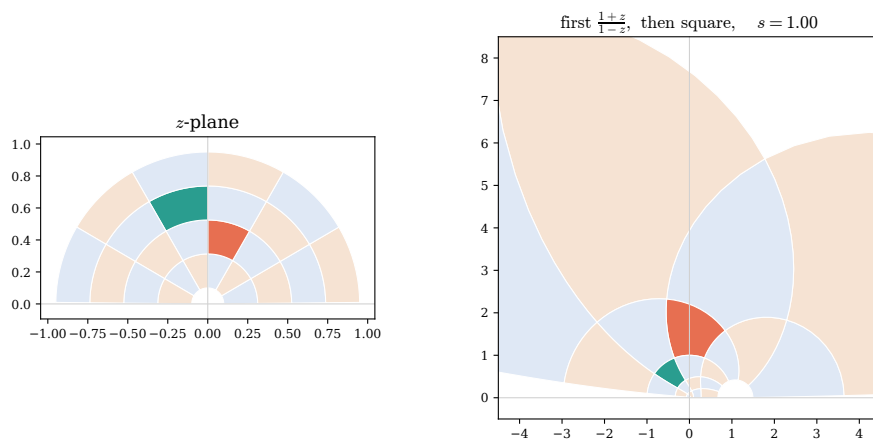


FIGURE 33.2 (animated). Composing: $\left(\frac{1+z}{1-z}\right)^2$. **A**: every point moves in a straight line from z to $f(z)$. **B**: $\frac{1+z}{1-z}$ takes the half-disc onto the quadrant; squaring doubles the angle: onto the half-plane. **C**: the tiles on the left deform along the path of B.

$\frac{1+z}{1-z}$ takes the half-disc onto the quadrant; squaring doubles the angle: onto the half-plane



C tiles (Figure 33.2, continued)

Example 33.5 (Opening a corner). Map the wedge $W = \{0 < \text{Arg } z < \pi/4\}$ onto the upper half-plane. By Proposition 18.5 with $\alpha = 4$, the map $w = z^4$ sends W onto $\{0 < \text{Arg } w < \pi\}$, the upper half-plane, conformally (and $z = 0$, where $f' = 0$, is a boundary point, so conformality inside is unaffected).

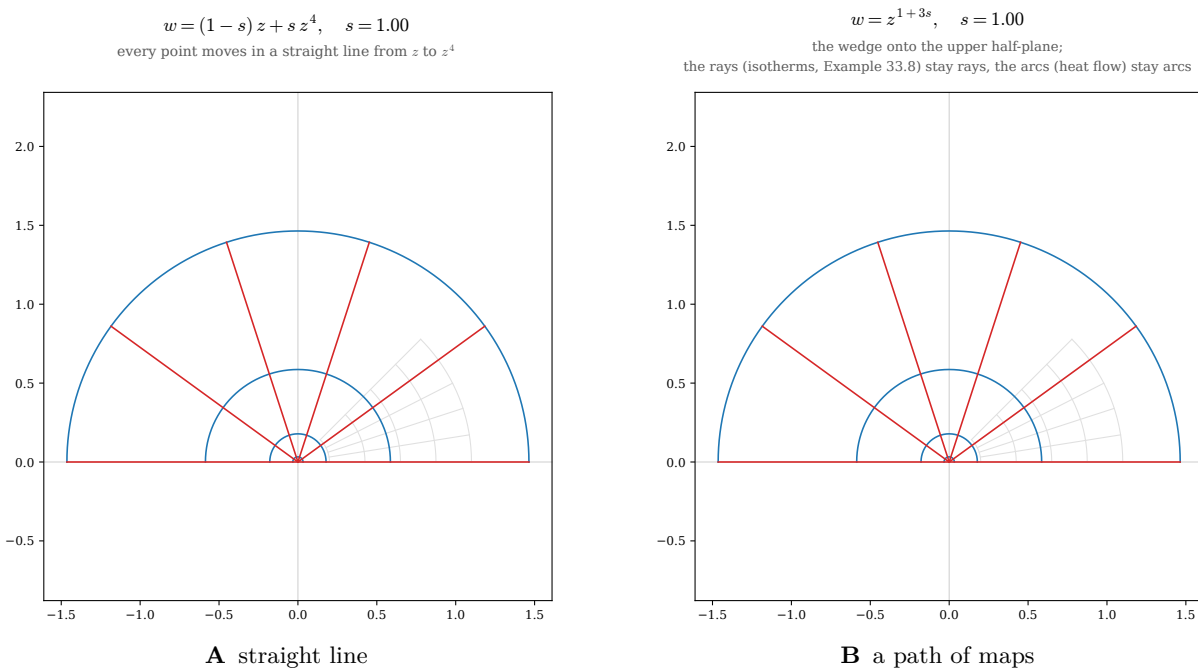
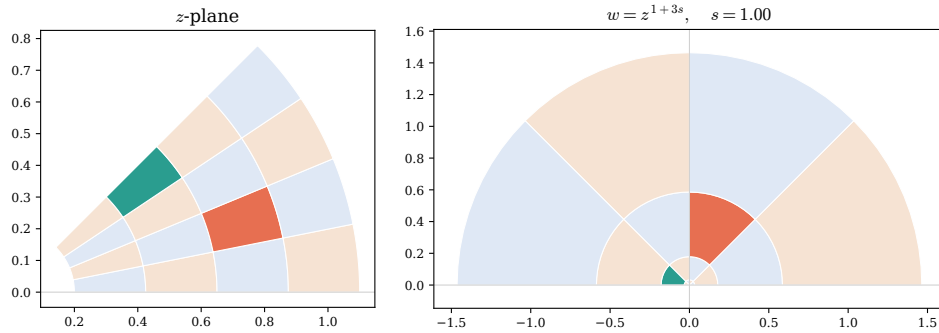


FIGURE 33.3 (animated). z^4 opens the wedge $0 < \text{Arg } z < \pi/4$. **A**: every point moves in a straight line from z to $f(z)$. **B**: the wedge onto the upper half-plane; the rays (isotherms, Example 33.8) stay rays, the arcs (heat flow) stay arcs. **C**: the tiles on the left deform along the path of B.

the wedge onto the upper half-plane; the rays (isotherms, Example 33.8) stay rays, the arcs (heat flow) stay arcs



C tiles (Figure 33.3, continued)

33.3 Transporting harmonic functions

The reason conformal maps matter to physics is the following invariance.

Theorem 33.6 (Conformal invariance of Laplace's equation). *Let $f : D \rightarrow D'$ be analytic and let Φ be harmonic on D' . Then*

$$\phi := \Phi \circ f \quad (33.2)$$

is harmonic on D .

Proof. Fix $z_0 \in D$ and a disc $D(w_0, \rho) \subseteq D'$ about $w_0 = f(z_0)$. By Theorem 15.6 there is F analytic on that disc with $\operatorname{Re} F = \Phi$. Then $F \circ f$ is analytic on $f^{-1}(D(w_0, \rho))$, a neighbourhood of z_0 (continuity of f), and

$$\operatorname{Re}(F \circ f) = \Phi \circ f = \phi.$$

By Theorem 15.2 the real part of an analytic function is harmonic, so ϕ is harmonic near z_0 . Harmonicity is a local property, so ϕ is harmonic on D . $\square$

Theorem 33.7 (Solving a Dirichlet problem by transport). *Terminology: let D' be a region, $E' \subseteq \partial D'$ a finite (possibly empty) set of exceptional points, and g a function on $\partial D' \setminus E'$. We say that Φ solves the Dirichlet problem on D' with boundary data g if Φ is harmonic on D' , bounded, continuous on $\overline{D'} \setminus E'$, and*

$$\nabla^2 \Phi = 0 \text{ on } D', \quad \Phi = g \text{ on } \partial D' \setminus E'.$$

Now let $f : D \rightarrow D'$ be a conformal equivalence which extends continuously to a map $\overline{D} \rightarrow \overline{D'}$ with $f(\partial D) \subseteq \partial D'$, and suppose the set $E := \{z \in \partial D : f(z) \in E'\}$ is finite. If Φ solves the Dirichlet problem on D' with boundary data g , then $\phi = \Phi \circ f$ solves the Dirichlet problem on D with exceptional set E and boundary data $g \circ f$.

Proof. Harmonicity is Theorem 33.6, and ϕ is bounded because Φ is. The extended f maps $\overline{D} \setminus E$ into $\overline{D'} \setminus E'$: points of D go into D' , which is disjoint from $E' \subseteq \partial D'$, and points of $\partial D \setminus E$ go outside E' by the definition of E . So ϕ is continuous on $\overline{D} \setminus E$ as a composition of continuous maps, and for $z \in \partial D \setminus E$, $\phi(z) = \Phi(f(z)) = g(f(z))$. $\square$

Boundedness is part of the definition because on unbounded regions the solution would otherwise not be unique: on the upper half-plane, $\operatorname{Im} w$ is harmonic, vanishes on $\mathbb{R}$, and could be added to any solution. (That a bounded solution with finitely many exceptional points is unique is true, but we do not prove it here.)

Example 33.8 (Temperature in a wedge). Find the steady temperature ϕ in the wedge $W = \{0 < \operatorname{Arg} z < \pi/4\}$ whose two edges are held at fixed temperatures: $\phi = 0$ on the ray $\operatorname{Arg} z = 0$ and $\phi = 100$ on the ray $\operatorname{Arg} z = \pi/4$.

In the half-plane. On $D' = \{\operatorname{Im} w > 0\}$, the function

$$\Phi(w) = \frac{100}{\pi} \operatorname{Arg} w \quad (33.3)$$

is harmonic (it is $\frac{100}{\pi} \operatorname{Im} \operatorname{Log} w$, the imaginary part of an analytic function by Theorem 16.9), equals 0 on the positive real axis and 100 on the negative real axis. It is bounded, and continuous on the closed half-plane except at the single exceptional point $E' = \{0\}$, where the data jump.

Transport. By Example 33.5, $w = z^4$ maps W onto D' , sending $\operatorname{Arg} z = 0$ to $\operatorname{Arg} w = 0$ and $\operatorname{Arg} z = \pi/4$ to $\operatorname{Arg} w = \pi$; it extends continuously to the closed wedge, and the only boundary point mapped to $E' = \{0\}$ is $z = 0$. Theorem 33.7 gives

$$\phi(z) = \Phi(z^4) = \frac{100}{\pi} \operatorname{Arg}(z^4) = \frac{400}{\pi} \operatorname{Arg} z = \frac{400}{\pi} \arctan \frac{y}{x} \quad (x > 0). \quad (33.4)$$

Check: $\phi = 0$ when $\operatorname{Arg} z = 0$ and $\phi = \frac{400}{\pi} \cdot \frac{\pi}{4} = 100$ when $\operatorname{Arg} z = \pi/4$, as required; and ϕ is harmonic, being a constant multiple of $\operatorname{Arg} z$.

The isotherms are the rays $\operatorname{Arg} z = \text{const}$, and by Theorem 15.11 the heat-flow lines are their orthogonal trajectories, the circles $|z| = \text{const}$.

Example 33.9 (The Poisson kernel). On the unit disc the Dirichlet problem is solved explicitly by

$$\phi(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1-r^2}{1-2r \cos(\theta-t) + r^2} g(e^{it}) dt, \quad 0 \leq r < 1, \quad (33.5)$$

for continuous boundary data g on $|z| = 1$. That the kernel has total mass 1 is Exercise 28.4; that (33.5) is harmonic follows because the kernel is $\operatorname{Re} \frac{e^{it}+z}{e^{it}-z}$, the real part of an analytic function of z for each fixed t , together with differentiation under the integral sign (legitimate because the integrand and its partial derivatives in x, y are continuous in (z, t) on $\{|z| \leq r_0\} \times [0, 2\pi]$ for each $r_0 < 1$). The remaining, and main, point is the boundary behaviour: $\phi(re^{i\theta}) \rightarrow g(e^{i\theta})$ as $r \uparrow 1$, uniformly in θ , so that ϕ extends continuously to $\overline{\mathbb{D}}$ with boundary values g . This follows from the fact that the kernel is positive, has total mass 1, and concentrates at $t = \theta$ as $r \uparrow 1$ (an “approximate identity”); we state it without proof. Combining (33.5) with Theorem 33.7 and a conformal map onto the disc solves the Dirichlet problem on any region for which such a map is known.

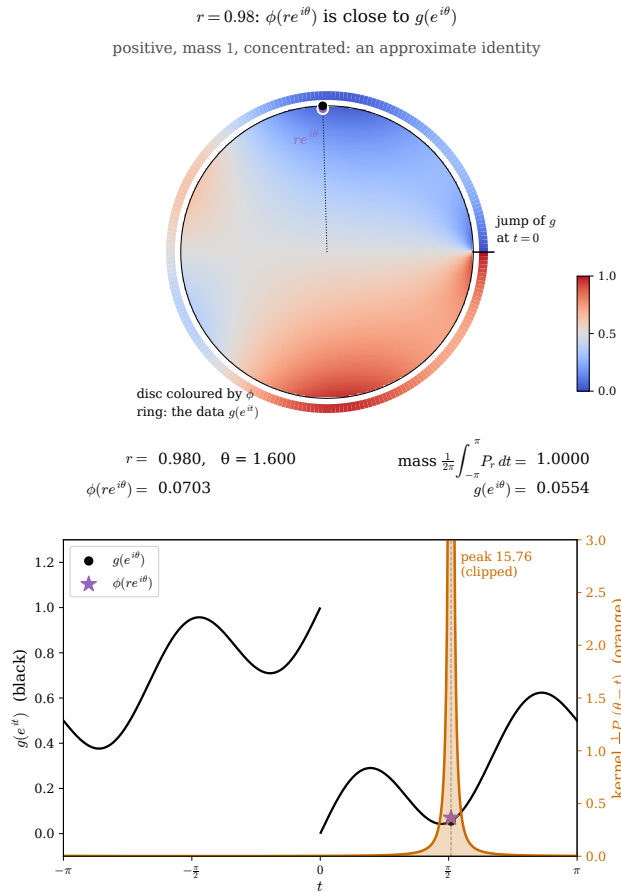


FIGURE 33.4 (animated). The Poisson kernel of Example 33.9 at work, for boundary data $g(e^{it}) = \frac{t}{2\pi} + 0.2 \sin 3t$, $0 < t < 2\pi$. The disc is coloured by ϕ ; below, the kernel $\frac{1}{2\pi} P_r(\theta - t)$ is drawn over the graph of g . As the probe $re^{i\theta}$ moves toward the boundary the kernel sharpens (peak 15.76 at $r = 0.98$) while its mass stays 1, so $\phi(re^{i\theta})$, the kernel-weighted average of g , approaches $g(e^{i\theta})$. At the point $t = 0$, where this g jumps from 1 to 0 (beyond the continuous g of the notes), the kernel puts mass $\frac{1}{2}$ on each side and ϕ stays at the midpoint $\frac{1}{2}$.

33.4 The Riemann mapping theorem

Theorem 33.10 (Riemann mapping theorem; stated without proof). *Let $D \subsetneq \mathbb{C}$ be a simply connected region that is not all of $\mathbb{C}$. Then there exists a conformal equivalence $f : D \rightarrow \mathbb{D}$ of D onto the open unit disc. If one fixes $z_0 \in D$ and demands $f(z_0) = 0$ and $f'(z_0) > 0$, the map is unique.*

Remark 33.11. Three comments.

- (a) The hypothesis $D \neq \mathbb{C}$ cannot be dropped: a conformal equivalence $\mathbb{C} \rightarrow \mathbb{D}$ would be a bounded entire function, hence constant by Liouville 22.2.
- (b) The theorem is an existence statement and is famously non-constructive; for explicit regions one still builds the map by composing catalogue entries as in Example 33.4, or by the Schwarz–Christoffel formula for polygons.

- (c) Simple connectivity is necessary: an annulus is not conformally equivalent to a disc, and two annuli are equivalent only when their radius ratios agree — so, unlike simply connected regions, annuli carry a genuine modulus.

Theorem 33.12 (Schwarz–Christoffel, stated). *Let P be a polygon with interior angles $\alpha_1\pi, \dots, \alpha_n\pi$ ($0 < \alpha_k < 2$, $\sum(\alpha_k - 1) = -2$). There are real $x_1 < \dots < x_n$ and complex constants A, B such that*

$$f(z) = A \int^z (\zeta - x_1)^{\alpha_1-1} \dots (\zeta - x_n)^{\alpha_n-1} d\zeta + B \quad (33.6)$$

maps the upper half-plane conformally onto the interior of P , with $x_k \mapsto$ the k -th vertex. The powers use branches analytic on the upper half-plane.

Each factor $(\zeta - x_k)^{\alpha_k-1}$ contributes a turn of $(1 - \alpha_k)\pi$ to $\arg f'$ as ζ crosses x_k from left to right along the real axis (there $\arg(\zeta - x_k)$ drops from π to 0), which is exactly the exterior angle $\pi - \alpha_k\pi$ of the polygon at the k -th vertex; that is the whole idea of (33.6).

33.5 Exercises

Exercise 33.1. Find a conformal map of the quarter-disc $\{|z| < 1, 0 < \arg z < \pi/2\}$ onto the upper half-plane.

Exercise 33.2. Find a conformal map of the strip $0 < \operatorname{Im} z < \pi$ onto the unit disc. (Compose e^z with a map of the upper half-plane onto $\mathbb{D}$ from Chapter 34.)

Exercise 33.3. Solve the Dirichlet problem on the upper half-plane with boundary data $g(x) = 0$ for $x < 0$ and $g(x) = 1$ for $x > 0$, and then transport your answer to the first quadrant using $w = z^2$.

Exercise 33.4. (★) Find the steady temperature in the half-disc $\{|z| < 1, \operatorname{Im} z > 0\}$ with the semicircular boundary held at 0 and the diameter at 50. (Use Example 33.4.)

Exercise 33.5. Show that $w = z + 1/z$ maps $\{|z| > 1, \operatorname{Im} z > 0\}$ conformally onto the upper half-plane, and find where the derivative vanishes.

Exercise 33.6. Explain why there is no conformal equivalence of the punctured disc $\{0 < |z| < 1\}$ onto the annulus $\{1 < |z| < 2\}$.

Exercise 33.7. Find a conformal map of the half-strip $\{0 < \operatorname{Re} z < \pi, \operatorname{Im} z > 0\}$ onto the upper half-plane, and one of the sector $\{0 < \arg z < \pi/6\}$ onto the unit disc. In each case say which catalogue entries you composed.

Exercise 33.8. Solve the Dirichlet problem on the strip $0 < \operatorname{Im} z < \pi$ with boundary data 0 on the lower edge and T on the upper. (The answer is linear in $\operatorname{Im} z$; verify it is harmonic. Transport it by $\operatorname{Log} z$, which maps the upper half-plane onto the strip (its inverse is e^z), to get (33.3); and by $4\operatorname{Log} z$, which maps the wedge $\{0 < \arg z < \pi/4\}$ onto the strip (its inverse is $e^{z/4}$), to get the wedge solution of Example 33.8 when $T = 100$.)

Exercise 33.9. (★) Let f be a conformal equivalence of a region D onto D' , and let u be harmonic and bounded on D' . Prove that $u \circ f$ is harmonic and bounded on D . Show that if u has a harmonic conjugate v on D' , then $v \circ f$ is a harmonic conjugate of $u \circ f$ on D , so this property always transfers. Show, by $u = \ln |z|$ on an annulus, that a harmonic function on a region that is not simply connected need not have a harmonic conjugate at all. (See Remark 15.7.)

Chapter 34

Linear Fractional Transformations

The maps $az + b$ and $1/z$ were studied separately in Chapters 8 and 11. Their compositions form a group, they are bijections of the Riemann sphere onto itself that are conformal wherever they are finite and finite-valued (in fact they are exactly the conformal self-maps of the sphere, a fact we do not prove), and they supply almost every explicit conformal map used in practice.

34.1 Definition and basic structure

Definition 34.1 (Linear fractional transformation). Let $a, b, c, d \in \mathbb{C}$ with

$$ad - bc \neq 0. \quad (34.1)$$

The *linear fractional transformation* (or *Möbius transformation*) with these coefficients is

$$T(z) = \frac{az + b}{cz + d}. \quad (34.2)$$

When $c \neq 0$ we extend T to the Riemann sphere $\hat{\mathbb{C}}$ of (11.8) by

$$T\left(-\frac{d}{c}\right) := \infty, \quad T(\infty) := \frac{a}{c}; \quad (34.3)$$

when $c = 0$ we set $T(\infty) := \infty$, and T is the linear map $\frac{a}{d}z + \frac{b}{d}$ of Definition 8.3.

The condition (34.1) is exactly what excludes the constant maps: if $ad - bc = 0$ then (a, b) and (c, d) are proportional and T is constant wherever defined.

Theorem 34.2 (The group structure). (i) *Every T as in (34.2) is a bijection of $\hat{\mathbb{C}}$ onto $\hat{\mathbb{C}}$, with inverse*

$$T^{-1}(w) = \frac{dw - b}{-cw + a}, \quad (34.4)$$

again a linear fractional transformation.

- (ii) *The composition of two linear fractional transformations is one, and the coefficient matrices multiply: if $T \leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $S \leftrightarrow \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ then $T \circ S \leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix}$.*
- (iii) *Consequently the linear fractional transformations form a group under composition, and T determines its matrix up to a nonzero scalar multiple.*

- (iv) Every T is a composition of translations, rotations, magnifications, and at most one inversion $z \mapsto 1/z$.

Proof. (i) Solve $w = \frac{az+b}{cz+d}$: cross-multiplying, $w(cz+d) = az+b$, so $z(wc-a) = b-wd$ and $z = \frac{b-wd}{wc-a} = \frac{dw-b}{-cw+a}$, whose determinant is $da - (-b)(-c) = ad - bc \neq 0$. The extensions (34.3) make this a bijection of $\widehat{\mathbb{C}}$ (the excluded point $-d/c$ maps to ∞ and back).

- (ii) Direct computation:

$$T(S(z)) = \frac{a \frac{pz+q}{rz+s} + b}{c \frac{pz+q}{rz+s} + d} = \frac{(ap+br)z + (aq+bs)}{(cp+dr)z + (cq+ds)},$$

whose coefficient array is the matrix product; the determinant is the product of the determinants, hence nonzero.

(iii) is (i) and (ii). For the scalar ambiguity, note that replacing (a, b, c, d) by $(\lambda a, \lambda b, \lambda c, \lambda d)$ with $\lambda \neq 0$ leaves (34.2) unchanged; conversely if two matrices give the same T , applying T to $0, 1, \infty$ forces proportionality.

- (iv) If $c = 0$, T is linear. If $c \neq 0$, divide out:

$$T(z) = \frac{az+b}{cz+d} = \frac{a}{c} + \frac{bc-ad}{c} \cdot \frac{1}{cz+d}, \quad (34.5)$$

verified by putting the right side over a common denominator. So T is $z \mapsto cz+d$, then $z \mapsto 1/z$, then multiplication by $\frac{bc-ad}{c}$ and translation by $\frac{a}{c}$. $\square$

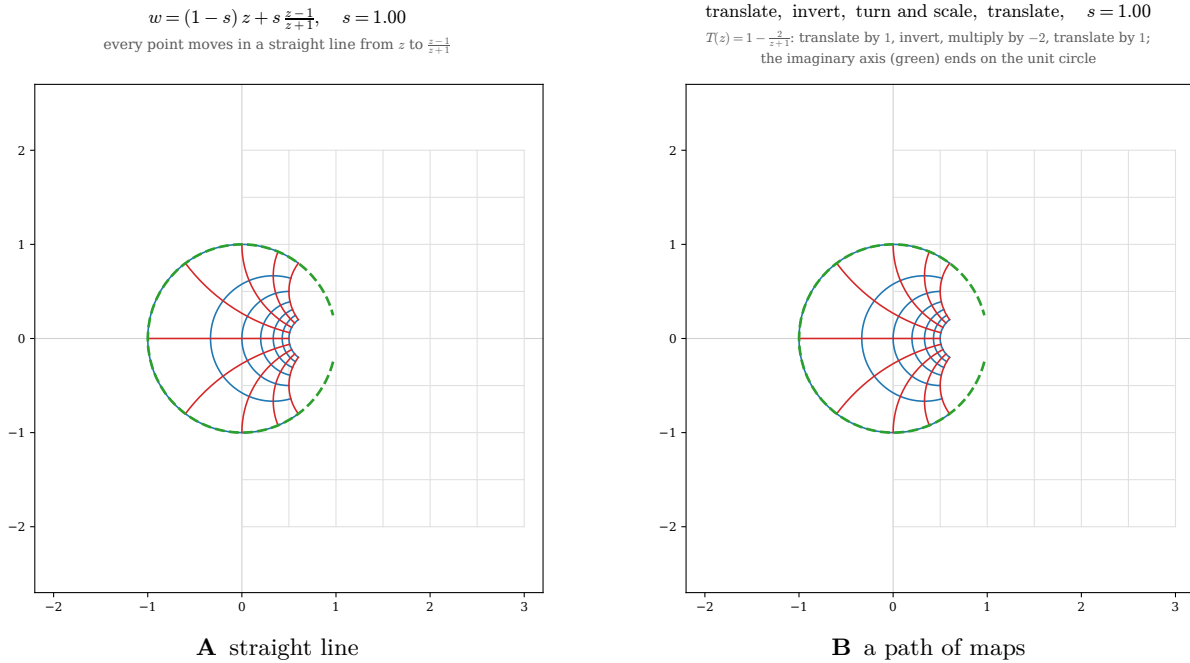
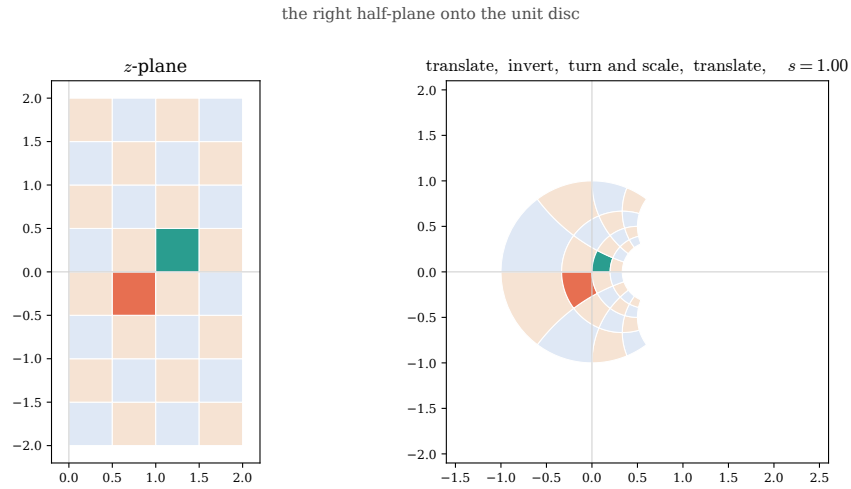


FIGURE 34.1 (animated). A linear fractional map, step by step: $\frac{z-1}{z+1}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: $T(z) = 1 - \frac{2}{z+1}$: translate by 1, invert, multiply by -2, translate by 1; the imaginary axis (green) ends on the unit circle. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 34.1, continued)

Corollary 34.3 (Circles and lines again). *Every linear fractional transformation maps the family of circles and lines in $\mathbb{C}$ onto itself. Viewed on the Riemann sphere, it maps circles to circles (Theorem 11.7).*

Proof. Combine Theorem 34.2(iv) with Theorem 8.4(iii) for the linear pieces and Theorem 11.3 (with Corollary 11.8) for the inversion. $\square$

Proposition 34.4 (Conformality). *T is conformal at every $z \in \mathbb{C}$ with $cz + d \neq 0$; explicitly, for such z ,*

$$T'(z) = \frac{ad - bc}{(cz + d)^2} \neq 0. \quad (34.6)$$

Proof. The quotient rule (13.5) gives $T' = \frac{a(cz+d) - c(az+b)}{(cz+d)^2} = \frac{ad-bc}{(cz+d)^2}$, and the numerator is nonzero by (34.1). $\square$

34.2 Three points determine the map

Theorem 34.5 (Three-point theorem). *Let z_1, z_2, z_3 be distinct points of $\hat{\mathbb{C}}$ and w_1, w_2, w_3 distinct points of $\hat{\mathbb{C}}$. Then there is exactly one linear fractional transformation T with $T(z_k) = w_k$ for $k = 1, 2, 3$. It is obtained by solving*

$$\frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)} \quad (34.7)$$

for w in terms of z . If one of the six points is ∞ , delete the two factors containing it.

Proof. Existence. The left side of (34.7), call it $S(w)$, is a linear fractional transformation of w with $S(w_1) = 0$, $S(w_2) = 1$, $S(w_3) = \infty$; its determinant is nonzero because w_1, w_2, w_3 are distinct. Likewise the right side is a linear fractional transformation $R(z)$ with $R(z_1) = 0$, $R(z_2) = 1$, $R(z_3) = \infty$. Then $T := S^{-1} \circ R$ is a linear fractional transformation by Theorem 34.2 and satisfies $T(z_k) = w_k$.

Uniqueness. Suppose T_1, T_2 both work. Then $U := T_2^{-1} \circ T_1$ fixes z_1, z_2, z_3 . A linear fractional transformation with three fixed points is the identity: $U(z) = z$ means $cz^2 + (d - a)z - b = 0$, a

polynomial of degree at most 2 with three distinct roots (interpreting $z = \infty$ as forcing $c = 0$), hence identically zero, so $c = b = 0$ and $a = d$, i.e. $U = \text{id}$. Therefore $T_1 = T_2$. $\square$

Definition 34.6 (Cross ratio). The quantity

$$(z, z_1, z_2, z_3) := \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)} \quad (34.8)$$

is the *cross ratio*; as a function of z it is the linear fractional transformation R with $R(z_1) = 0$, $R(z_2) = 1$, $R(z_3) = \infty$. The uniqueness part of Theorem 34.5 shows that the cross ratio is *invariant* under every linear fractional transformation T : $(Tz, Tz_1, Tz_2, Tz_3) = (z, z_1, z_2, z_3)$. Indeed, if $R'(\zeta) := (\zeta, Tz_1, Tz_2, Tz_3)$, then $R' \circ T$ and R both send z_1, z_2, z_3 to $0, 1, \infty$, so $R' \circ T = R$.

Example 34.7 (Building a map to order). Find T with $T(-1) = 0$, $T(0) = i$, $T(1) = \infty$.

Take $z_1 = -1$, $z_2 = 0$, $z_3 = 1$ and $w_1 = 0$, $w_2 = i$, $w_3 = \infty$. Since $w_3 = \infty$, delete the factors containing it on the left of (34.7):

$$\frac{w - 0}{i - 0} = \frac{(z + 1)(0 - 1)}{(z - 1)(0 + 1)} = \frac{-(z + 1)}{z - 1},$$

so

$$T(z) = \frac{-i(z + 1)}{z - 1} = \frac{i(z + 1)}{1 - z}.$$

Check: $T(-1) = 0$; $T(0) = i$; $T(1) = \infty$. Note that T maps the real axis (a circle through ∞) to the circle through $0, i, \infty$, i.e. the imaginary axis — consistent with Corollary 34.3.

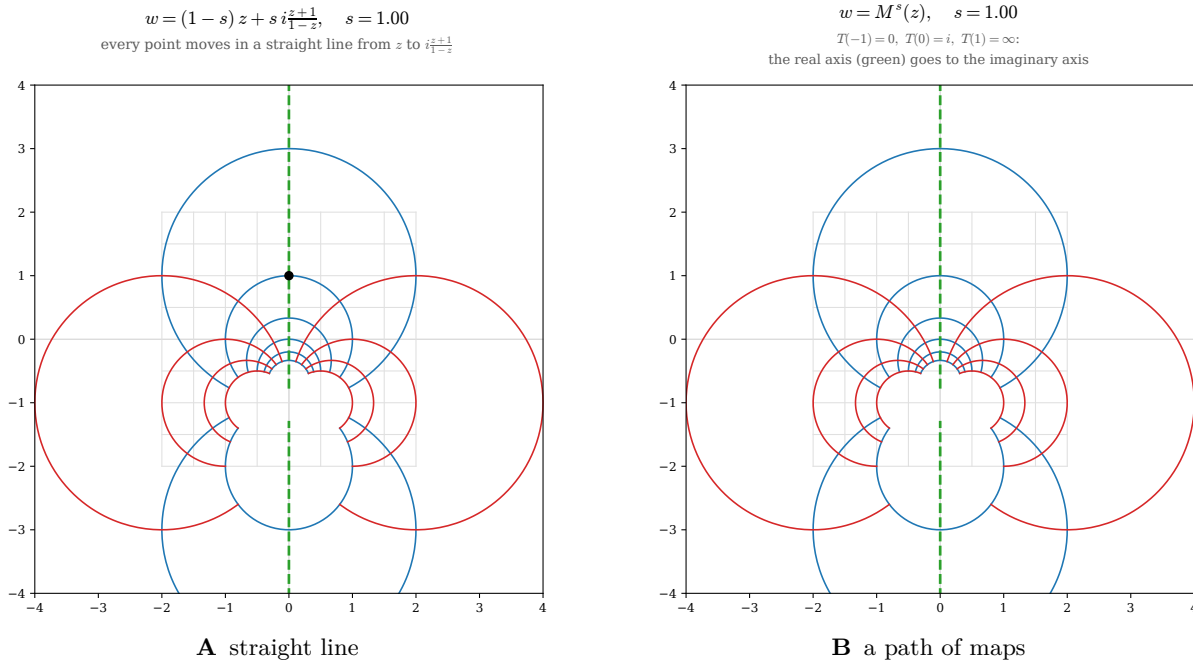
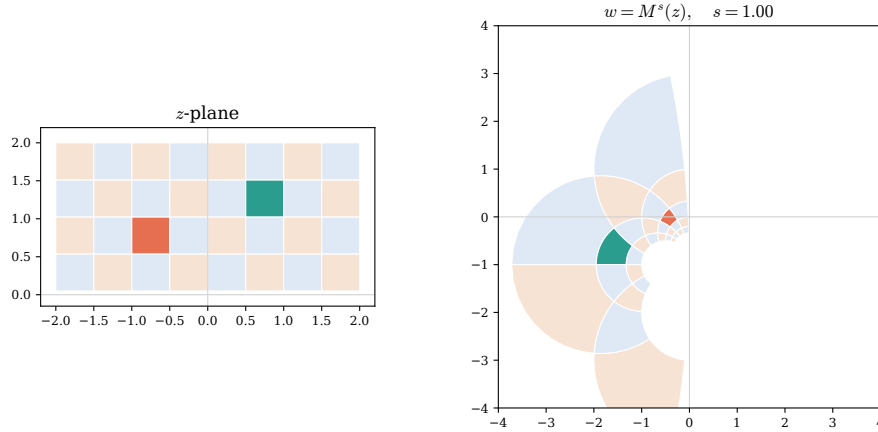


FIGURE 34.2 (animated). Three points: $T(z) = i \frac{z+1}{1-z}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: $T(-1) = 0$, $T(0) = i$, $T(1) = \infty$: the real axis (green) goes to the imaginary axis. **C**: the tiles on the left deform along the path of B.

$T(-1) = 0$, $T(0) = i$, $T(1) = \infty$: the real axis (green) goes to the imaginary axis



C tiles (Figure 34.2, continued)

34.3 The disc and the half-plane

Theorem 34.8 (Automorphisms of the disc). *For $a \in \mathbb{D}$ and $\theta \in \mathbb{R}$ define*

$$\varphi_{a,\theta}(z) := e^{i\theta} \frac{z - a}{1 - \bar{a}z}. \quad (34.9)$$

Then:

- (i) $\varphi_{a,\theta}$ is a conformal equivalence of $\mathbb{D}$ onto $\mathbb{D}$, with $\varphi_{a,\theta}(a) = 0$;
- (ii) its inverse is $\varphi_{-ae^{i\theta}, -\theta}$, again of the same form;
- (iii) every conformal equivalence of $\mathbb{D}$ onto $\mathbb{D}$ is of the form (34.9).

Proof. (i) The pole is at $z = 1/\bar{a}$, of modulus $1/|a| > 1$, so $\varphi_{a,\theta}$ is analytic on a neighbourhood of $\mathbb{D}$. For $|z| = 1$ we have $\bar{z} = 1/z$ (Exercise 1.5), so

$$|1 - \bar{a}z| = |z| \left| \frac{1}{z} - \bar{a} \right| = |\bar{z} - \bar{a}| = |\overline{z - a}| = |z - a|,$$

whence $|\varphi_{a,\theta}(z)| = 1$: the unit circle maps into itself. By Corollary 34.3 its image is a circle or a line; a line is not contained in the unit circle, and a circle contained in the unit circle equals it, so the unit circle is mapped *onto* itself. Since $\varphi_{a,\theta}$ is injective on $\hat{\mathbb{C}}$ (Theorem 34.2), no point of $\mathbb{D}$ is mapped onto the unit circle. Now $\varphi_{a,\theta}(\mathbb{D})$ is connected (the continuous image of a connected set; $\varphi_{a,\theta}$ is finite on $\mathbb{D}$), contains $\varphi_{a,\theta}(a) = 0$, and misses the unit circle; if it contained a point w with $|w| > 1$, then $|\cdot|$ would take on the connected set $\varphi_{a,\theta}(\mathbb{D})$ the values 0 and $|w|$, hence (its image being an interval) the value 1 — impossible. So $\varphi_{a,\theta}(\mathbb{D}) \subseteq \mathbb{D}$. The same argument applied to the inverse, which is of the same form by (ii), gives surjectivity. Conformality is Proposition 34.4.

- (ii) Solve $w = e^{i\theta} \frac{z - a}{1 - \bar{a}z}$ for z : $e^{-i\theta} w(1 - \bar{a}z) = z - a$, so $z(1 + \bar{a}e^{-i\theta}w) = e^{-i\theta}w + a$ and

$$z = e^{-i\theta} \frac{w + ae^{i\theta}}{1 + \bar{a}e^{-i\theta}w} = \varphi_{-ae^{i\theta}, -\theta}(w),$$

since $\overline{-ae^{i\theta}} = -\bar{a}e^{-i\theta}$; and $|-ae^{i\theta}| = |a| < 1$.

(iii) Let $F : \mathbb{D} \rightarrow \mathbb{D}$ be a conformal equivalence, put $a := F^{-1}(0)$, and set $G := F \circ \varphi_{a,0}^{-1}$. Then G is a conformal equivalence of $\mathbb{D}$ onto $\mathbb{D}$ with $G(0) = 0$. By Schwarz's lemma 24.14, $|G(z)| \leq |z|$ on $\mathbb{D}$; applying the lemma to G^{-1} likewise gives $|G^{-1}(w)| \leq |w|$, i.e. $|z| \leq |G(z)|$. Hence $|G(z)| = |z|$ throughout, and the equality case of Schwarz's lemma forces $G(z) = e^{i\theta}z$ for some real θ . Therefore $F = G \circ \varphi_{a,0} = e^{i\theta} \varphi_{a,0} = \varphi_{a,\theta}$. $\square$

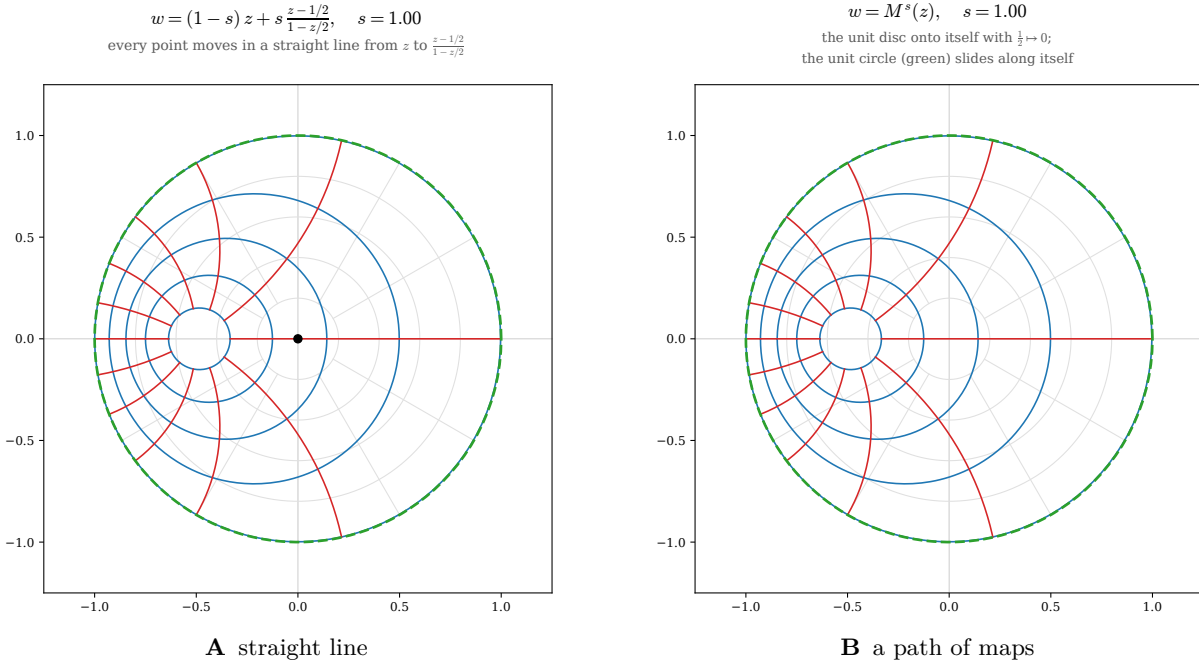
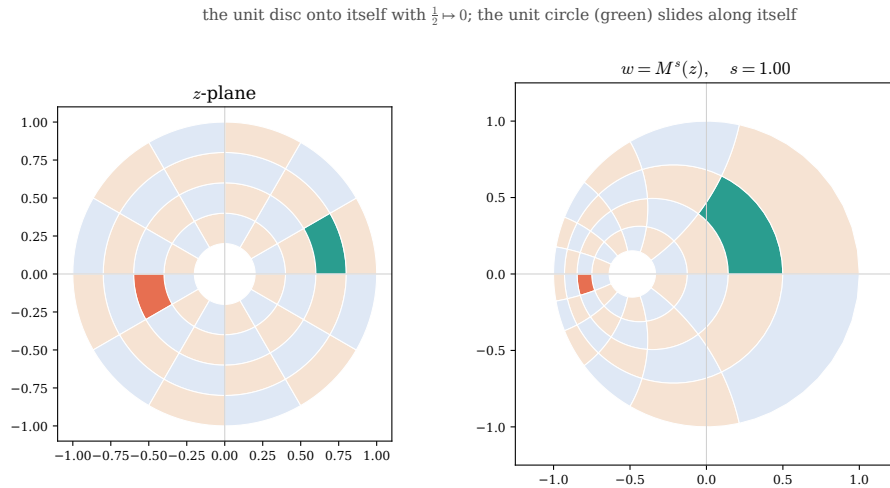


FIGURE 34.3 (animated). A disc automorphism: $\frac{z-1/2}{1-\bar{z}/2}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: the unit disc onto itself with $\frac{1}{2} \mapsto 0$; the unit circle (green) slides along itself. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 34.3, continued)

Theorem 34.9 (Half-plane to disc). *For $\operatorname{Im} \alpha > 0$ and $\theta \in \mathbb{R}$, the map*

$$\psi(z) = e^{i\theta} \frac{z - \alpha}{z - \bar{\alpha}} \quad (34.10)$$

is a conformal equivalence of the upper half-plane $\mathbb{H} = \{\operatorname{Im} z > 0\}$ onto $\mathbb{D}$, with $\psi(\alpha) = 0$; and every such equivalence has this form. The special case $\alpha = i$, $\theta = 0$ gives the Cayley transform

$$\psi(z) = \frac{z - i}{z + i}, \quad \text{with inverse} \quad \psi^{-1}(w) = i \frac{1 + w}{1 - w}. \quad (34.11)$$

Proof. For real x , the points α and $\bar{\alpha}$ are mirror images across the real axis, so $|x - \alpha| = |x - \bar{\alpha}|$ and hence $|\psi(x)| = 1$; also $\psi(\infty) = e^{i\theta}$. So the image of $\mathbb{R} \cup \{\infty\}$ (a circle through ∞) is, by Corollary 34.3, a circle contained in, hence equal to, the unit circle. Since ψ is a bijection of $\hat{\mathbb{C}}$ (Theorem 34.2), ψ maps $\mathbb{H}$ and the lower half-plane into the complement of the unit circle. Each half-plane is connected and ψ is continuous and finite on $\mathbb{H}$ (its pole $\bar{\alpha}$ lies in the lower half-plane), so, as in the proof of Theorem 34.8(i), $\psi(\mathbb{H})$ is a connected set containing $\psi(\alpha) = 0$ and missing the unit circle, hence $\psi(\mathbb{H}) \subseteq \mathbb{D}$. Likewise the lower half-plane minus $\bar{\alpha}$ is connected and its image contains points of arbitrarily large modulus (near $\psi(\bar{\alpha}) = \infty$), so it lies in $\{|w| > 1\}$. As ψ is onto $\hat{\mathbb{C}}$, it follows that $\psi(\mathbb{H}) = \mathbb{D}$. Conformality is Proposition 34.4. For the converse, compose with the Cayley transform and apply Theorem 34.8(iii). Formula (34.11) follows from (34.4). $\square$

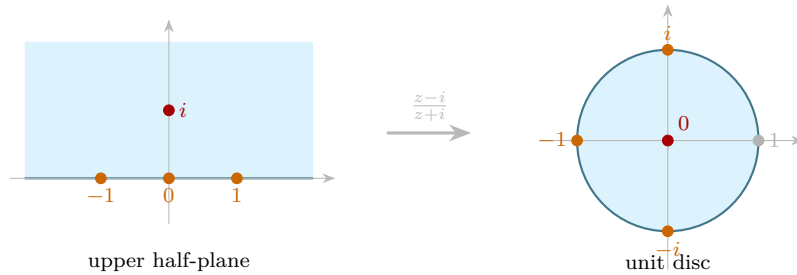


FIGURE 34.4. The Cayley transform (34.11). The real axis — a circle through ∞ — goes to the unit circle, and the upper half-plane goes to the disc, because $i \mapsto 0$ lands inside. The labelled points track the correspondence: $0 \mapsto -1$, $1 \mapsto -i$, $-1 \mapsto i$, $\infty \mapsto 1$.

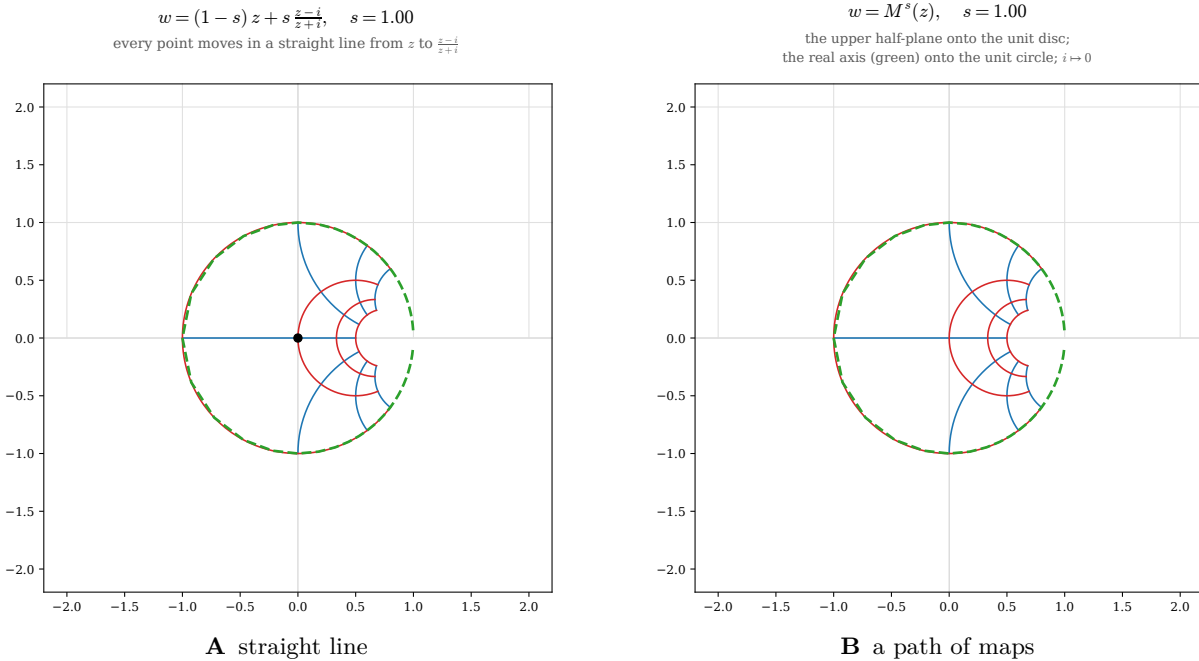
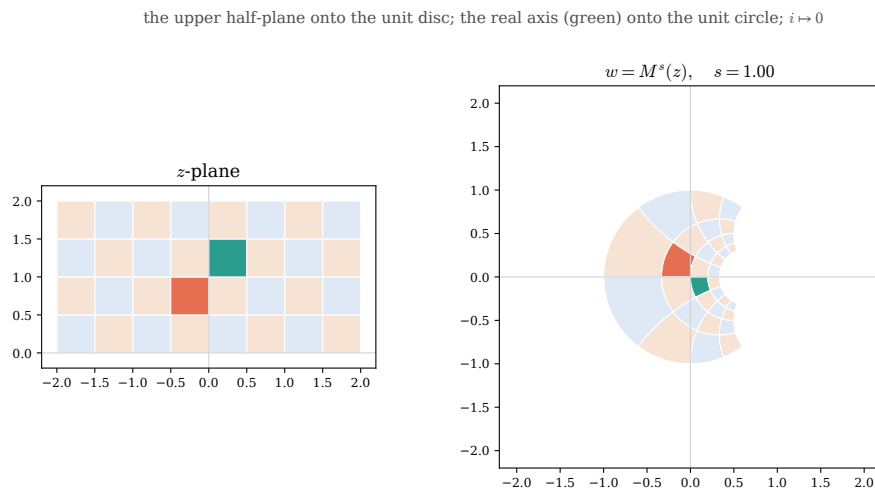


FIGURE 34.5 (animated). The Cayley map $\frac{z-i}{z+i}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: the upper half-plane onto the unit disc; the real axis (green) onto the unit circle; $i \mapsto 0$. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 34.5, continued)

Example 34.10. Map the upper half-plane onto the unit disc so that $2i \mapsto 0$ and $0 \mapsto 1$. By (34.10) with $\alpha = 2i$, $\psi(z) = e^{i\theta} \frac{z-2i}{z+2i}$. The condition $\psi(0) = 1$ gives $e^{i\theta} \frac{-2i}{2i} = -e^{i\theta} = 1$, so $e^{i\theta} = -1$ and

$$\psi(z) = \frac{2i - z}{2i + z}.$$

Check: $\psi(2i) = 0$, $\psi(0) = 1$, and $|\psi(1)| = \left| \frac{-1+2i}{1+2i} \right| = \frac{\sqrt{5}}{\sqrt{5}} = 1$, as required for a boundary point.

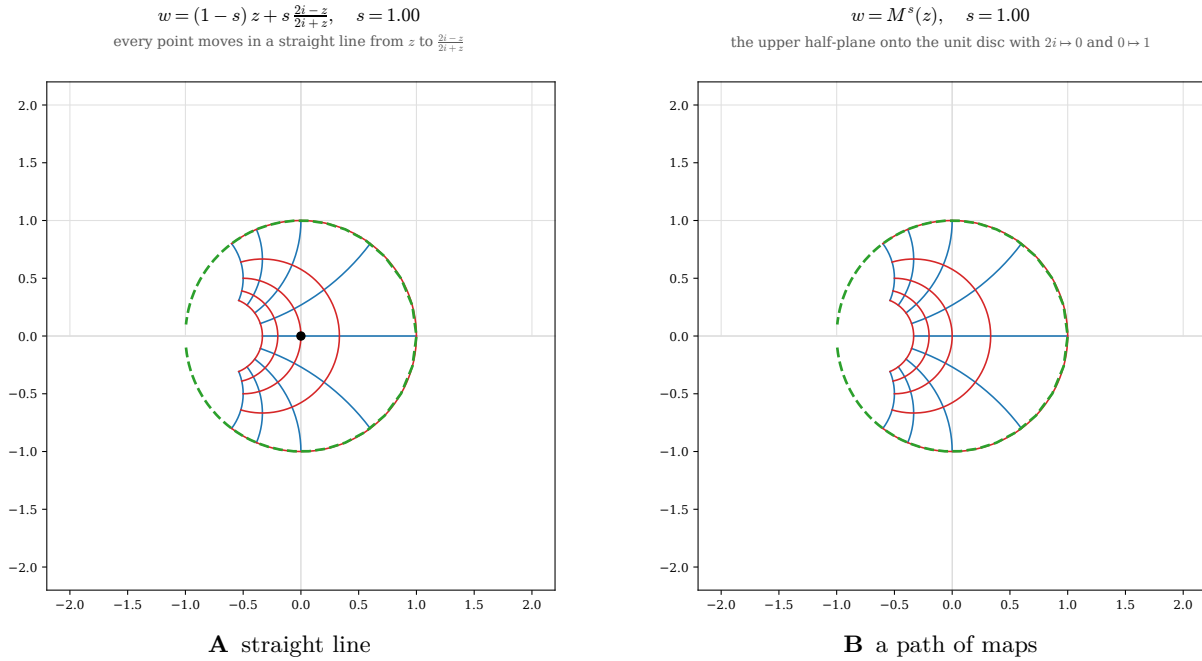
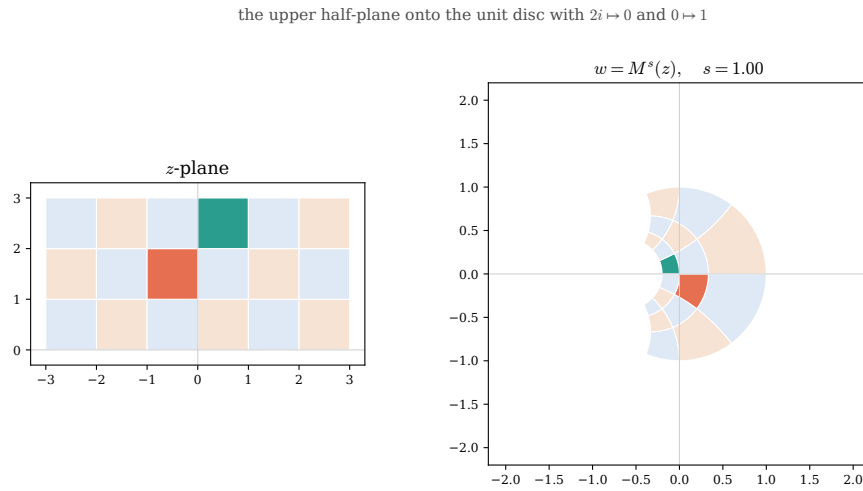


FIGURE 34.6 (animated). Half-plane to disc with $2i \mapsto 0$: $\frac{2i-z}{2i+z}$. **A**: every point moves in a straight line from z to $f(z)$. **B**: the upper half-plane onto the unit disc with $2i \mapsto 0$ and $0 \mapsto 1$. **C**: the tiles on the left deform along the path of B.



C tiles (Figure 34.6, continued)

34.4 Symmetry and fixed points

Definition 34.11 (Symmetric points). Two points z, z^* are *symmetric with respect to* a circle $C(z_0, \rho)$ if

$$z^* = z_0 + \frac{\rho^2}{\overline{z - z_0}}, \quad (34.12)$$

that is, if they lie on the same ray from z_0 with $|z - z_0| |z^* - z_0| = \rho^2$. By convention the centre z_0 and ∞ are symmetric with respect to $C(z_0, \rho)$. For a line, symmetry means ordinary mirror

reflection (and ∞ is symmetric to itself).

Theorem 34.12 (Symmetry principle). *Let T be a linear fractional transformation and C a circle or line. If z and z^* are symmetric with respect to C , then $T(z)$ and $T(z^*)$ are symmetric with respect to $T(C)$.*

Proof. Step 1: a cross-ratio characterisation. Let z_1, z_2, z_3 be distinct finite points on C . We claim that z^* is symmetric to z with respect to C if and only if

$$(z^*, z_1, z_2, z_3) = \overline{(z, z_1, z_2, z_3)}. \quad (\dagger)$$

Since $\zeta \mapsto (\zeta, z_1, z_2, z_3)$ is injective on $\widehat{\mathbb{C}}$, for a given z at most one z^* satisfies $(\dagger)$; so it suffices to check that the symmetric point does. Suppose first $C = C(z_0, \rho)$ and z, z^* finite and different from z_0 . Put $u := z - z_0$, $u_k := z_k - z_0$ and $u^* := z^* - z_0 = \rho^2/\bar{u}$. On C , $\bar{u}_k = \rho^2/u_k$, so

$$\bar{z} - \bar{z}_k = \frac{\rho^2}{u^*} - \frac{\rho^2}{u_k} = \frac{\rho^2(u_k - u^*)}{u^*u_k}, \quad \bar{z}_2 - \bar{z}_3 = \frac{\rho^2(u_3 - u_2)}{u_2u_3}, \quad \bar{z}_2 - \bar{z}_1 = \frac{\rho^2(u_1 - u_2)}{u_1u_2}.$$

Substituting into the conjugate of (34.8), all the factors ρ^2 , u^* and u_k cancel, and

$$\overline{(z, z_1, z_2, z_3)} = \frac{(u_1 - u^*)(u_3 - u_2)}{(u_3 - u^*)(u_1 - u_2)} = \frac{(z^* - z_1)(z_2 - z_3)}{(z^* - z_3)(z_2 - z_1)} = (z^*, z_1, z_2, z_3).$$

For the pair (z_0, ∞) : taking $u = 0$ and deleting the factors containing ∞ as in Theorem 34.5, $\overline{(z_0, z_1, z_2, z_3)} = \overline{\left(\frac{u_1(u_2 - u_3)}{u_3(u_2 - u_1)}\right)} = \frac{u_1^{-1}(u_2^{-1} - u_3^{-1})}{u_3^{-1}(u_2^{-1} - u_1^{-1})} = \frac{z_2 - z_3}{z_2 - z_1} = (\infty, z_1, z_2, z_3)$, and $(\dagger)$ with z, z^* exchanged is its conjugate. If C is a line $\{p + te^{i\beta} : t \in \mathbb{R}\}$, reflection is $z^* = p + e^{2i\beta}\overline{(z - p)}$, and for $z_k \in C$, $\overline{(z_k - p)} = e^{-2i\beta}(z_k - p)$; hence $\bar{z} - \bar{z}_k = e^{-2i\beta}(z^* - z_k)$ and $\bar{z}_j - \bar{z}_k = e^{-2i\beta}(z_j - z_k)$, and in the conjugated cross ratio the four factors $e^{-2i\beta}$ cancel, giving $(\dagger)$; for $z = z^* = \infty$, $(\infty, z_1, z_2, z_3) = \frac{z_2 - z_3}{z_2 - z_1}$ is real because z_1, z_2, z_3 are collinear.

Step 2. Now let z, z^* be symmetric with respect to C . By Corollary 34.3, $T(C)$ is a circle or a line. Choose distinct $z_1, z_2, z_3 \in C$, all finite and with Tz_k finite (possible, since at most one point of C is ∞ and at most one is sent to ∞). By Step 1 and the invariance of the cross ratio (Definition 34.6),

$$(Tz^*, Tz_1, Tz_2, Tz_3) = (z^*, z_1, z_2, z_3) = \overline{(z, z_1, z_2, z_3)} = \overline{(Tz, Tz_1, Tz_2, Tz_3)},$$

and since Tz_1, Tz_2, Tz_3 are distinct points of $T(C)$, Step 1 applied to $T(C)$ shows that Tz^* is symmetric to Tz with respect to $T(C)$. $\square$

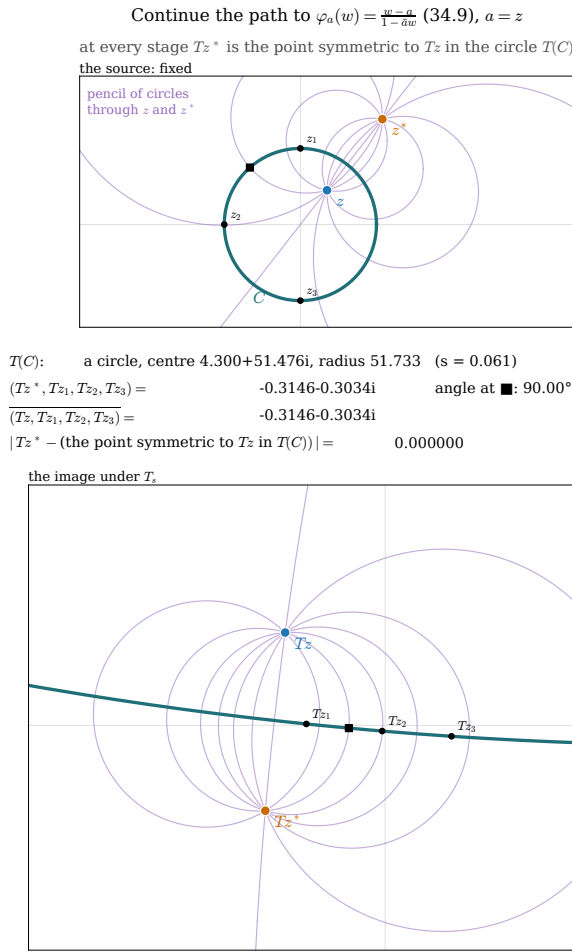


FIGURE 34.7 (animated). The symmetry principle, Theorem 34.12. Top (fixed): the unit circle C , the symmetric points $z = 0.35 + 0.45i$ and $z^* = 1/\bar{z}$ (Definition 34.11), and the circles through z and z^* , all orthogonal to C . Below: their images under a path of LFTs, through the inverse Cayley map (34.11) (the image of C is a line and Tz, Tz^* are mirror images) and through φ_a of (34.9) ($Tz = 0$ and $Tz^* = \infty$). The readouts, computed in every frame, show that Tz and Tz^* stay symmetric in $T(C)$ and that the image circles stay orthogonal to $T(C)$.

The symmetry principle is the fastest route to maps like (34.9): to send $\mathbb{D}$ to $\mathbb{D}$ with $a \mapsto 0$, note that 0 and ∞ are symmetric in the unit circle, so the point symmetric to a , namely $1/\bar{a}$, must go to ∞ — which forces the pole to sit at $1/\bar{a}$, giving (34.9) up to a constant.

Proposition 34.13 (Fixed points). *A linear fractional transformation $T \neq \text{id}$ has exactly one or two fixed points in $\hat{\mathbb{C}}$. They are the solutions of*

$$cz^2 + (d - a)z - b = 0 \quad (34.13)$$

together with $z = \infty$ when $c = 0$.

Proof. $T(z) = z$ with $cz + d \neq 0$ rearranges to (34.13), a polynomial of degree ≤ 2 ; it vanishes identically only for $c = b = 0$, $a = d$, i.e. $T = \text{id}$. (A root z of (34.13) automatically has $cz + d \neq 0$: otherwise $cz = -d$ and (34.13) gives $-az - b = 0$, so (a, b) and (c, d) would both annihilate $(z, 1)$, contradicting $ad - bc \neq 0$.) If $c \neq 0$, (34.13) has degree 2, hence 1 or 2 roots, and ∞ is not fixed

since $T(\infty) = a/c \neq \infty$. If $c = 0$, ∞ is fixed by (34.3), and (34.13) reads $(d - a)z = b$: it has exactly one root if $a \neq d$ (two fixed points in all), and none if $a = d$, $b \neq 0$ (a translation, whose only fixed point is ∞). $\square$

34.5 Exercises

Exercise 34.1. Find the linear fractional transformation with (a) $0 \mapsto 1$, $1 \mapsto \infty$, $\infty \mapsto -1$; (b) $1 \mapsto i$, $i \mapsto -1$, $-1 \mapsto 1$.

Exercise 34.2. Find a conformal map of $\mathbb{D}$ onto $\mathbb{D}$ with $\frac{1}{2} \mapsto 0$ and $0 \mapsto -\frac{1}{2}$.

Exercise 34.3. Find the image of the disc $|z - 1| < 1$ under $w = \frac{1}{z}$, and verify it against Theorem 11.3.

Exercise 34.4. Map the half-plane $\operatorname{Re} z > 0$ onto $\mathbb{D}$ with $1 \mapsto 0$.

Exercise 34.5. ($\star$) Prove that a linear fractional transformation maps $\mathbb{H}$ onto $\mathbb{H}$ if and only if it can be written with real coefficients a, b, c, d satisfying $ad - bc > 0$.

Exercise 34.6. Let $T(z) = \frac{z}{2 - z}$. Find its fixed points, and determine the image of the unit circle and of the unit disc.

Exercise 34.7. Combine Theorem 34.9 with Example 33.9 to write down the solution of the Dirichlet problem in the upper half-plane, and check that your kernel is $\frac{1}{\pi} \frac{y}{(x - t)^2 + y^2}$.

Exercise 34.8. Compute the cross ratio $(z, 0, 1, \infty)$, and deduce that z lies on the circle through $0, 1, \infty$ (the real line) precisely when it is real. Then compute $(-1, 0, 1, i)$ and use invariance (Definition 34.6) to decide whether $0, 1, i, -1$ lie on a common circle or line. (*Four points are concyclic-or-collinear precisely when their cross ratio is real.*)

Exercise 34.9. Show that the matrix correspondence of Theorem 34.2(ii) gives a surjective group homomorphism $GL_2(\mathbb{C}) \rightarrow \{\text{linear fractional transformations}\}$ with kernel the nonzero scalar matrices, so the group is $PGL_2(\mathbb{C}) = GL_2(\mathbb{C})/\mathbb{C}^\times$. What is the corresponding statement for the maps preserving $\mathbb{H}$ (Exercise 34.5)?

Exercise 34.10. ($\star$) Classify the linear fractional transformations by their fixed points: show that $T \neq \text{id}$ is conjugate, by another linear fractional transformation, either to $z \mapsto z + 1$ (one fixed point, *parabolic*) or to $z \mapsto \kappa z$ for some $\kappa \neq 0, 1$ (two fixed points). In the second case relate $|\kappa| = 1$, $|\kappa| \neq 1$ to the names *elliptic*, *hyperbolic* and *loxodromic*, and sketch the orbits of a point in each case.

Part IX

Appendix

Appendix A

Background from Multivariable Calculus

This appendix collects, with precise statements but without proofs, the facts from calculus of several real variables that the main text uses. Nothing here is complex analysis; everything here is assumed known, and the appendix exists so that “assumed known” has a definite meaning. Throughout, $n, m \in \mathbb{N}$, points of $\mathbb{R}^n$ are written $\mathbf{x} = (x_1, \dots, x_n)$ and treated as column vectors when matrices act on them, and $|\mathbf{x}| = (\sum_k x_k^2)^{1/2}$ is the Euclidean norm.

A.1 Open sets, limits, continuity

Definition A.1 (Balls and open sets). For $\mathbf{a} \in \mathbb{R}^n$ and $\varepsilon > 0$ the *open ball* is

$$B(\mathbf{a}, \varepsilon) := \{ \mathbf{x} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{a}| < \varepsilon \}. \quad (\text{A.1})$$

A set $\Omega \subseteq \mathbb{R}^n$ is *open* if every $\mathbf{a} \in \Omega$ has some $B(\mathbf{a}, \varepsilon) \subseteq \Omega$; it is a *region* if it is additionally nonempty and connected. For $n = 2$ these are Definitions 5.3 and 5.8 transcribed.

Definition A.2 (Limit and continuity). Let $\Omega \subseteq \mathbb{R}^n$, $\mathbf{F} : \Omega \rightarrow \mathbb{R}^m$, $\mathbf{a}$ an accumulation point of Ω , and $\mathbf{L} \in \mathbb{R}^m$. Then $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{F}(\mathbf{x}) = \mathbf{L}$ means: for every $\varepsilon > 0$ there is $\delta > 0$ with $|\mathbf{F}(\mathbf{x}) - \mathbf{L}| < \varepsilon$ whenever $\mathbf{x} \in \Omega$ and $0 < |\mathbf{x} - \mathbf{a}| < \delta$. $\mathbf{F}$ is *continuous at* $\mathbf{a} \in \Omega$ if $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{F}(\mathbf{x}) = \mathbf{F}(\mathbf{a})$. Writing $\mathbf{F} = (F_1, \dots, F_m)$ in components, $\mathbf{F}$ is continuous iff each $F_j : \Omega \rightarrow \mathbb{R}$ is.

A.2 Partial derivatives, the gradient, the Jacobian

Definition A.3 (Partial derivative). Let $\Omega \subseteq \mathbb{R}^n$ be open, $F : \Omega \rightarrow \mathbb{R}$, and $\mathbf{a} \in \Omega$. The *k-th partial derivative* of F at $\mathbf{a}$ is

$$\frac{\partial F}{\partial x_k}(\mathbf{a}) := \lim_{t \rightarrow 0} \frac{F(\mathbf{a} + t\mathbf{e}_k) - F(\mathbf{a})}{t}, \quad (\text{A.2})$$

when the limit exists, where $\mathbf{e}_k$ is the k -th standard basis vector. We also write F_{x_k} or $\partial_k F$. In two variables the notation of the main text is u_x, u_y .

Definition A.4 (Gradient). For a *scalar-valued* $F : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ whose partials exist at $\mathbf{a}$, the *gradient* is the row of partials, regarded as a vector,

$$\nabla F(\mathbf{a}) := \left(\frac{\partial F}{\partial x_1}(\mathbf{a}), \dots, \frac{\partial F}{\partial x_n}(\mathbf{a}) \right) \in \mathbb{R}^n. \quad (\text{A.3})$$

It is defined only when the target is $\mathbb{R}$ ($m = 1$). For $m > 1$ the correct object is the Jacobian matrix below, whose rows are the gradients of the component functions.

Definition A.5 (Differentiability and the Jacobian matrix). Let $\Omega \subseteq \mathbb{R}^n$ be open and $\mathbf{F} = (F_1, \dots, F_m) : \Omega \rightarrow \mathbb{R}^m$. We say $\mathbf{F}$ is *differentiable at* $\mathbf{a} \in \Omega$ if there is a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with

$$\mathbf{F}(\mathbf{a} + \mathbf{h}) = \mathbf{F}(\mathbf{a}) + A\mathbf{h} + \mathbf{r}(\mathbf{h}), \quad \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|\mathbf{r}(\mathbf{h})|}{|\mathbf{h}|} = 0. \quad (\text{A.4})$$

The map A is then unique, is written $D\mathbf{F}(\mathbf{a})$, and its matrix in the standard bases is the *Jacobian matrix*

$$D\mathbf{F}(\mathbf{a}) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}_{\mathbf{a}} \in \mathbb{R}^{m \times n}, \quad (\text{A.5})$$

whose j -th row is $\nabla F_j(\mathbf{a})$. When $m = n$ the number $\det D\mathbf{F}(\mathbf{a})$ is the *Jacobian determinant*, often written $J_{\mathbf{F}}(\mathbf{a})$ or $\frac{\partial(F_1, \dots, F_n)}{\partial(x_1, \dots, x_n)}$.



FIGURE A.1. Differentiability is the existence of a good linear approximation. Near $\mathbf{a}$ the map $\mathbf{F}$ agrees with the affine map $\mathbf{h} \mapsto \mathbf{F}(\mathbf{a}) + D\mathbf{F}(\mathbf{a})\mathbf{h}$ to first order: a small disc about $\mathbf{a}$ goes approximately to an ellipse about $\mathbf{F}(\mathbf{a})$, the error being $o(|\mathbf{h}|)$. Theorem 14.10 is the statement that for an analytic map the ellipse is always a circle.

Theorem A.6 (Sufficient condition; the two implications). *Let $\Omega \subseteq \mathbb{R}^n$ be open and $\mathbf{F} : \Omega \rightarrow \mathbb{R}^m$.*

- (i) *If $\mathbf{F}$ is differentiable at $\mathbf{a}$ then all partials $\partial F_j / \partial x_k(\mathbf{a})$ exist and $\mathbf{F}$ is continuous at $\mathbf{a}$.*
- (ii) *If all partials exist on a neighbourhood of $\mathbf{a}$ and are continuous at $\mathbf{a}$, then $\mathbf{F}$ is differentiable at $\mathbf{a}$. (If the partials exist and are continuous at every point of an open set, $\mathbf{F}$ is called continuously differentiable, or C^1 , there, and is then differentiable at every point of it.)*

Neither implication reverses: existence of all partials at a point does not give differentiability, nor even continuity; and a differentiable function need not have continuous partials (for instance $f(x) = x^2 \sin(1/x)$, $f(0) = 0$, is differentiable on $\mathbb{R}$ but f' is not continuous at 0).

Statement (ii) is exactly the ingredient used in the proof of Theorem 14.6, where it appears in the explicit increment form (14.6).

Theorem A.7 (Chain rule). *Let $\mathbf{G} : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $\mathbf{a}$ and $\mathbf{F} : \Omega' \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^p$ differentiable at $\mathbf{G}(\mathbf{a})$, with $\mathbf{G}(\Omega) \subseteq \Omega'$. Then $\mathbf{F} \circ \mathbf{G}$ is differentiable at $\mathbf{a}$ and*

$$D(\mathbf{F} \circ \mathbf{G})(\mathbf{a}) = D\mathbf{F}(\mathbf{G}(\mathbf{a})) D\mathbf{G}(\mathbf{a}) \quad (\text{A.6})$$

— a product of matrices, of sizes $(p \times m)(m \times n)$. In the special case of a curve, $\mathbf{G} = \gamma : [a, b] \rightarrow \mathbb{R}^n$ and $F : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\frac{d}{dt}F(\gamma(t)) = \nabla F(\gamma(t)) \cdot \gamma'(t). \quad (\text{A.7})$$

Theorem A.8 (Equality of mixed partials; Clairaut–Schwarz). *If $F : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ has second partials $\partial^2 F / \partial x_j \partial x_k$ existing on a neighbourhood of $\mathbf{a}$ and continuous at $\mathbf{a}$, then*

$$\frac{\partial^2 F}{\partial x_j \partial x_k}(\mathbf{a}) = \frac{\partial^2 F}{\partial x_k \partial x_j}(\mathbf{a}). \quad (\text{A.8})$$

This is the fact used in Theorem 15.2 to show that u and v are harmonic, legitimate there because Corollary 21.5 supplies continuity of *all* partials.

Theorem A.9 (Change of variables). *Let $\Phi : \Omega \rightarrow \Phi(\Omega)$ be a C^1 bijection between open subsets of $\mathbb{R}^n$ with $\det D\Phi \neq 0$ on Ω , and let g be integrable on $\Phi(\Omega)$. Then*

$$\int_{\Phi(\Omega)} g(\mathbf{y}) \, d\mathbf{y} = \int_{\Omega} g(\Phi(\mathbf{x})) |\det D\Phi(\mathbf{x})| \, d\mathbf{x}. \quad (\text{A.9})$$

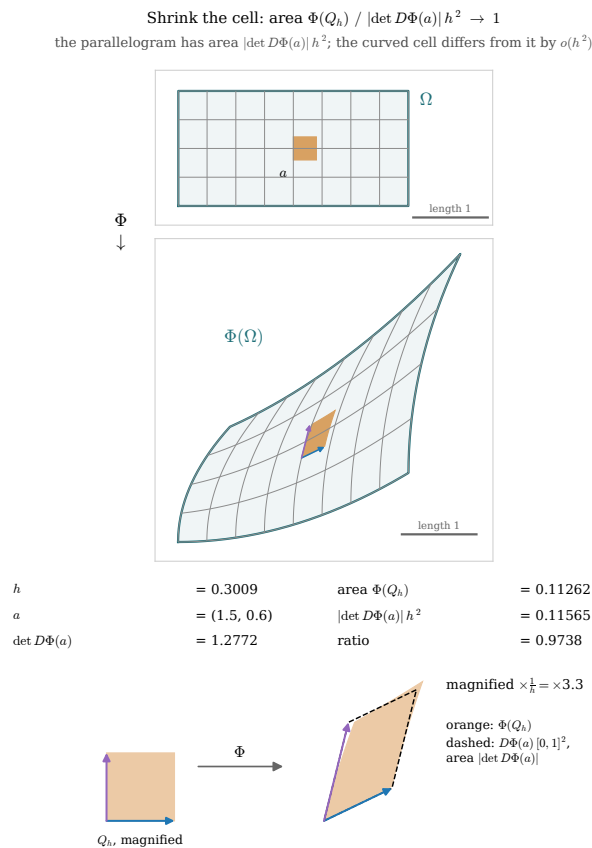


FIGURE A.2 (animated). The idea behind Theorem A.9 with $g \equiv 1$: $\text{area } \Phi(\Omega) = \int_{\Omega} |\det D\Phi|$, for $\Phi(x, y) = (x + 0.3y^2, y + 0.3xy + 0.1x^2)$ on $\Omega = [0, 3] \times [0, 1.5]$ (top) and its image (below). A small square Q_h at a is mapped to a curved cell that is nearly the parallelogram spanned by $D\Phi(a)(h, 0)$ and $D\Phi(a)(0, h)$, of area $|\det D\Phi(a)| h^2$ (bottom, magnified): the ratio of the two areas tends to 1 as $h \rightarrow 0$. Adding up over a grid, $\sum |\det D\Phi(x_{ij})| h^2$ tends to the area of $\Phi(\Omega)$, 5.31 (computed independently from the boundary curve). This is the standard heuristic; the notes state the theorem without proof.

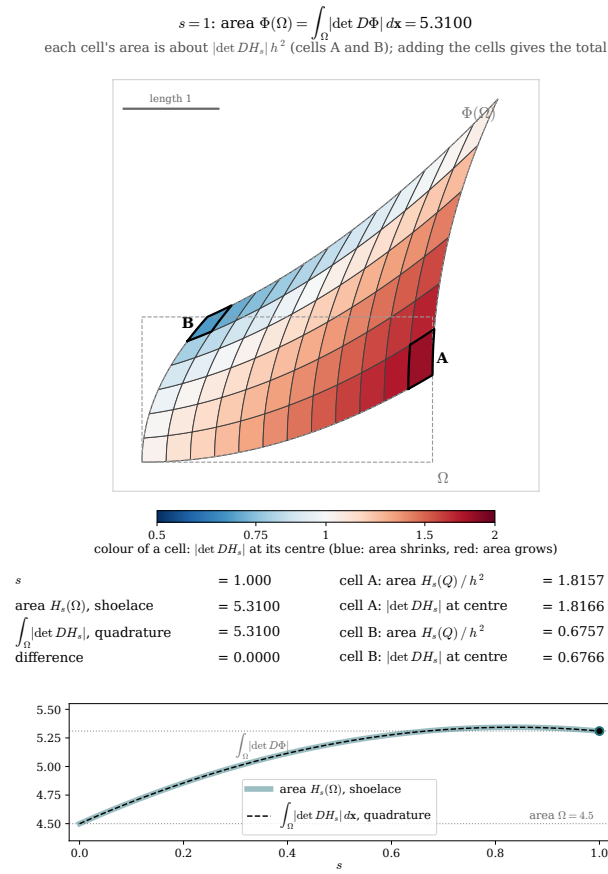


FIGURE A.3 (animated). The straight-line deformation $H_s(x) = (1-s)x + s\Phi(x)$ carries Ω onto $\Phi(\Omega)$ (same Φ). Cells are coloured by the local area factor $|\det DH_s|$ (blue: area shrinks, red: area grows). At every s the area of $H_s(\Omega)$, measured from its boundary, equals $\int_{\Omega} |\det DH_s|$ (graph below), and each outlined cell has area close to $|\det DH_s|$ at its centre times h^2 .

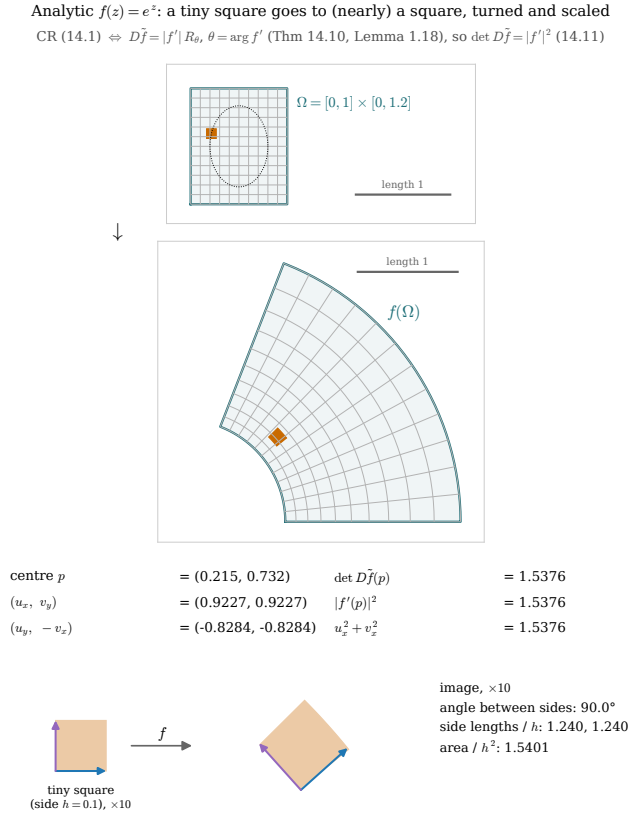


FIGURE A.4 (animated). For an analytic map the area factor is $|f'|^2$. A tiny square travels round $\Omega = [0, 1] \times [0, 1.2]$, first under the Φ of the previous figures, whose image is a skewed parallelogram, then under $f(z) = e^z$, whose image is (nearly) a square, turned by $\arg f'$ and scaled by $|f'|$: by the Cauchy–Riemann equations (14.1), Df is a conformal matrix and $\det Df = |f'|^2$ (Theorem 14.10). Tiling $f(\Omega)$ by such squares, $\sum |f'(z_{ij})|^2 h^2$ tends to $\text{area } f(\Omega) = \iint_{\Omega} |f'|^2 dA = 0.6(e^2 - 1)$.

A.3 Curves

Definition A.10 (Parametrised curve, arclength). A C^1 curve in $\mathbb{R}^n$ is a map $\gamma : [a, b] \rightarrow \mathbb{R}^n$ with continuous derivative $\gamma'(t) = (\gamma'_1(t), \dots, \gamma'_n(t))$. It is *regular* if $\gamma'(t) \neq \mathbf{0}$ for all t , in which case $\gamma'(t)$ is the tangent vector at $\gamma(t)$. Its *length* is

$$\text{length}(\gamma) := \int_a^b |\gamma'(t)| dt, \quad (\text{A.10})$$

and the *arclength element* is $ds = |\gamma'(t)| dt$. For $n = 2$ this is Definition 19.6.

A.4 Line integrals of scalar fields

Definition A.11 (Line integral of a scalar field). Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be a C^1 curve and let φ be continuous on its image. The *line integral of φ with respect to arclength* is

$$\int_{\gamma} \varphi ds := \int_a^b \varphi(\gamma(t)) |\gamma'(t)| dt. \quad (\text{A.11})$$

Proposition A.12 (Properties). *The integral (A.11) is linear in φ , additive under concatenation of curves, and independent of the parametrisation — including its orientation:*

$$\int_{-\gamma} \varphi \, ds = \int_{\gamma} \varphi \, ds. \quad (\text{A.12})$$

Taking $\varphi \equiv 1$ recovers (A.10). If $|\varphi| \leq M$ on the curve then $|\int_{\gamma} \varphi \, ds| \leq M \text{length}(\gamma)$.

The orientation-independence in (A.12) is the sharpest difference from the vector case below, and from the contour integral of Definition 19.7, both of which change sign.

A.5 Line integrals of vector fields

Definition A.13 (Line integral of a vector field; work). Let $\mathbf{V} = (V_1, \dots, V_n)$ be a continuous vector field on a neighbourhood of the image of a C^1 curve $\gamma : [a, b] \rightarrow \mathbb{R}^n$. The *line integral of $\mathbf{V}$ along γ* is

$$\int_{\gamma} \mathbf{V} \cdot d\mathbf{r} := \int_a^b \mathbf{V}(\gamma(t)) \cdot \gamma'(t) \, dt = \int_{\gamma} (V_1 \, dx_1 + \dots + V_n \, dx_n). \quad (\text{A.13})$$

Physically it is the *work* done by the force field $\mathbf{V}$ along the path.

Proposition A.14 (Properties). *The integral (A.13) is linear in $\mathbf{V}$, additive under concatenation, unchanged by orientation-preserving reparametrisation, and reverses sign under reversal:*

$$\int_{-\gamma} \mathbf{V} \cdot d\mathbf{r} = - \int_{\gamma} \mathbf{V} \cdot d\mathbf{r}. \quad (\text{A.14})$$

Moreover $|\int_{\gamma} \mathbf{V} \cdot d\mathbf{r}| \leq (\max |\mathbf{V}|) \text{length}(\gamma)$.

Compare Theorem 19.9 and the *ML*-inequality (19.15): the contour integral of Chapter 19 behaves like a vector line integral, not like a scalar one.

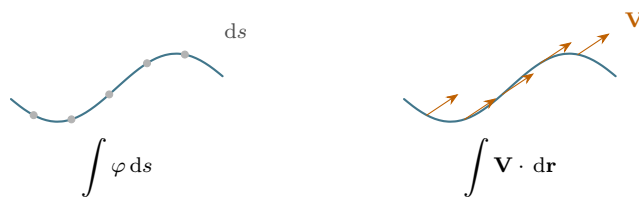


FIGURE A.5. The two line integrals. Left: $\int \varphi \, ds$ weights each element of arclength by the scalar φ — reversing the curve changes nothing. Right: $\int \mathbf{V} \cdot d\mathbf{r}$ records only the component of $\mathbf{V}$ along the tangent — reversing the curve flips every tangent, hence the sign.

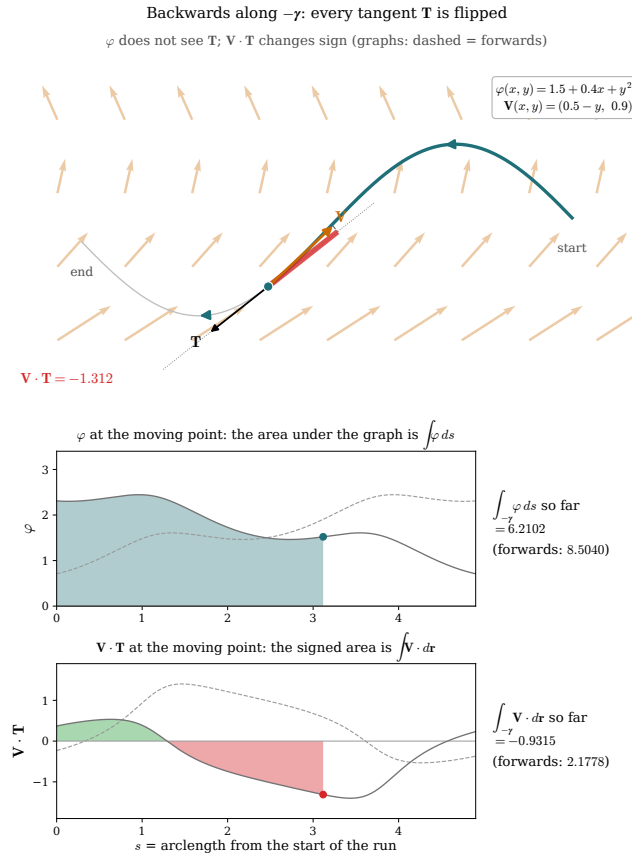


FIGURE A.6 (animated). The two line integrals on the curve $y = 0.7 \sin 1.5x$, $-2 \leq x \leq 2$, of the figure above, with $\varphi = 1.5 + 0.4x + y^2$ and $\mathbf{V} = (0.5 - y, 0.9)$. A point runs along the curve with its unit tangent $\mathbf{T}$; the component $\mathbf{V} \cdot \mathbf{T}$ is drawn green where positive and red where negative. Below, φ and $\mathbf{V} \cdot \mathbf{T}$ are graphed against arclength s , and the areas $\int \varphi ds$ and $\int \mathbf{V} \cdot \mathbf{T} ds = \int \mathbf{V} \cdot d\mathbf{r}$ fill up. Run backwards, the scalar integral is unchanged, 8.5040 (A.12), while the vector one changes sign, $2.1778 \mapsto -2.1778$ (A.14).

Theorem A.15 (Fundamental theorem for line integrals). *Let $\Omega \subseteq \mathbb{R}^n$ be a region and $F : \Omega \rightarrow \mathbb{R}$ be C^1 . Then for every C^1 curve γ in Ω from $\mathbf{p}$ to $\mathbf{q}$,*

$$\int_{\gamma} \nabla F \cdot d\mathbf{r} = F(\mathbf{q}) - F(\mathbf{p}). \quad (\text{A.15})$$

Consequently $\int_{\gamma} \nabla F \cdot d\mathbf{r} = 0$ whenever γ is closed.

Definition A.16 (Conservative field, potential). A continuous vector field $\mathbf{V}$ on a region Ω is *conservative* if $\mathbf{V} = \nabla F$ for some F , called a *potential*. By Theorem A.15, conservative fields have path-independent line integrals; the converse also holds, by the construction $F(\mathbf{x}) = \int_{\gamma_{\mathbf{x}}} \mathbf{V} \cdot d\mathbf{r}$ along any path from a fixed base point.

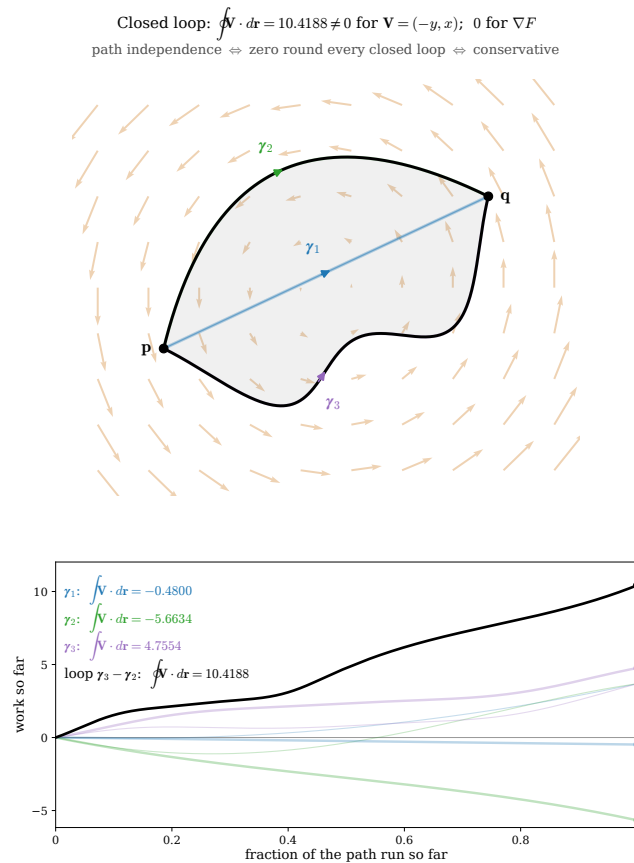


FIGURE A.7 (animated). Three paths $\gamma_1, \gamma_2, \gamma_3$ from $\mathbf{p}$ to $\mathbf{q}$, with the work along each graphed below as it accumulates. For the gradient field $\mathbf{V} = \nabla F$, $F = xy + x$ (level curves drawn), all three give $F(\mathbf{q}) - F(\mathbf{p}) = 3.68$ and the closed loop $\gamma_3 - \gamma_2$ gives 0 (Theorem A.15). For the rotating field $\mathbf{V} = (-y, x)$ the same paths give $-0.48, -5.66$ and 4.76 , and the loop gives 10.42 , twice the area it encloses: this field is not conservative (Definition A.16).

Theorem 20.4 is precisely this equivalence, transcribed to contour integrals, with “antiderivative” in place of “potential”.

Theorem A.17 (Green’s theorem in the plane). *Let $D \subseteq \mathbb{R}^2$ be a bounded region whose boundary ∂D is a finite union of simple closed piecewise- C^1 curves, oriented so that D lies to the left. If P, Q are C^1 on a neighbourhood of $\overline{D}$, then*

$$\oint_{\partial D} (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA. \quad (\text{A.16})$$

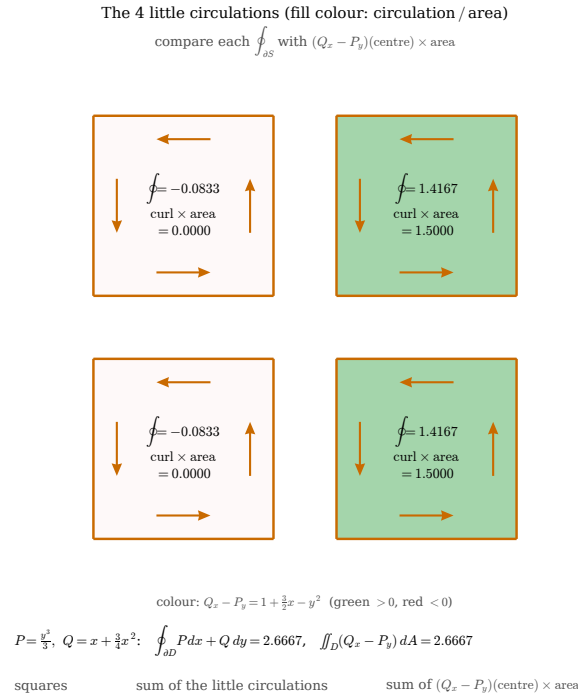


FIGURE A.8 (animated). Green's theorem A.17 by cancellation, for $P = y^3/3$, $Q = x + \frac{3}{4}x^2$ on $D = [-1, 1]^2$, coloured by $Q_x - P_y$ (green positive, red negative). D is cut into 2×2 , then 4×4 , then 8×8 squares; the pieces are pulled apart and run round counterclockwise, each circulation being about $(Q_x - P_y) \times \text{area}$. Put back, every inside edge is run twice in opposite directions and cancels, leaving ∂D : the little circulations always add up to $\oint_{\partial D} P dx + Q dy = \iint_D (Q_x - P_y) dA = \frac{8}{3}$. Applied to the two fields in (A.17), the Cauchy–Riemann equations make both double integrals vanish: the argument of Remark 20.6.

A.6 Dictionary: contour integrals in these terms

Let γ be a contour in $\mathbb{C} \cong \mathbb{R}^2$ and $f = u + iv$ continuous on it. Writing $dz = dx + i dy$ and multiplying out,

$$\int_{\gamma} f(z) dz = \underbrace{\int_{\gamma} (u dx - v dy)}_{\text{a real vector line integral}} + i \underbrace{\int_{\gamma} (v dx + u dy)}_{\text{another one}}. \quad (\text{A.17})$$

So a contour integral is a pair of vector line integrals, one for each of the vector fields $(u, -v)$ and (v, u) . Applying Green's theorem (A.16) to each and using the Cauchy–Riemann equations (14.1) makes both double integrals vanish — which is the short proof of the Cauchy–Goursat theorem sketched in Remark 20.6, valid when f' is assumed continuous. Goursat's Theorem 20.5 avoids that assumption and therefore avoids Green's theorem entirely.

A.7 Exercises

Exercise A.1. Compute ∇F and, where $m > 1$, the Jacobian matrix $D\mathbf{F}$:

$$F(x, y) = x^2y - y^3; \quad F(x, y, z) = e^{xy} \sin z; \quad \mathbf{F}(x, y) = (x^2 - y^2, 2xy).$$

For the last one compute $\det D\mathbf{F}$ and compare with $|f'(z)|^2$ for $f(z) = z^2$, as (14.11) predicts.

Exercise A.2. Let $\gamma(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$. Compute

$$\int_{\gamma} (x^2 + y^2) \, ds, \quad \int_{\gamma} \mathbf{V} \cdot d\mathbf{r} \text{ for } \mathbf{V} = (-y, x), \quad \int_{\gamma} \mathbf{W} \cdot d\mathbf{r} \text{ for } \mathbf{W} = (x, y).$$

Which of $\mathbf{V}, \mathbf{W}$ is conservative? Find a potential for the one that is.

Exercise A.3. Verify (A.17) directly for $f(z) = z$ and γ the unit circle, by computing the two real line integrals separately and checking the total against (19.16).

Exercise A.4. Show that $\mathbf{V}(x, y) = (\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2})$ satisfies $\partial_x V_2 = \partial_y V_1$ on $\mathbb{R}^2 \setminus \{0\}$, yet is *not* conservative there. (*Integrate around the unit circle.*) Relate this to Remark 15.7 and to the failure of $\log z$ to have a global branch.

Exercise A.5. State and verify Green's theorem (A.16) for $P = -y$, $Q = x$ on the unit disc, and observe that it computes twice the area. What is the corresponding statement about $\oint_{|z|=1} \bar{z} \, dz$ (Exercise 19.5)?